JEE MAIN 2024 APRIL SESSION

OFFICIAL QUESTION PAPERS WITH DETAILED SOLUTIONS

10 sets having 900 questions with detailed solutions of Physics, Chemistry and Mathematics

CAREERS 360

Contents

04 Apr - Morning Shift (Questions with detailed solutions) 3

04 Apr - Evening Shift (Questions with detailed solutions) 70

05 Apr - Morning Shift (Questions with detailed solutions) 143

05 Apr - Evening Shift (Questions with detailed solutions) 223

06 Apr - Morning Shift (Questions with detailed solutions) 299

06 Apr - Evening Shift (Questions with detailed solutions) 386

08 Apr - Morning Shift (Questions with detailed solutions) 445

08 Apr - Evening Shift (Questions with detailed solutions) 516

09 Apr - Morning Shift (Questions with detailed solutions) 596

09 Apr - Evening Shift (Questions with detailed solutions) 679

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4 Apr - Morning Shift

Physics

Q. 1 An electron is projected with uniform velocity along the axis inside a current carrying long solenoid. Then :

Option 1:
the electron will be accelerated along the axis.

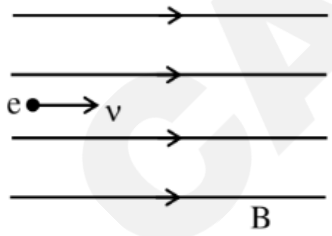
Option 2:
the electron will continue to move with uniform velocity along the axis of the solenoid.

Option 3:
the electron path will be circular about the axis.

Option 4:
the electron will experience a force at 45° to the axis and execute a helical path.

Correct Answer:
the electron will continue to move with uniform velocity along the axis of the solenoid.

Solution:



Since $\vec{v} \| \vec{B}$ so force on electron due to magnetic field is zero. So it will move along axis with uniform velocity.

Q. 2 The electric field in an electromagnetic wave is given by $\overrightarrow{\mathrm{E}} = \hat{\mathrm{i}} 40 \cos \omega \left(\mathrm{t} - \frac{\mathrm{z}}{\mathrm{c}}\right) \mathrm{NC}^{-1}$. The magnetic field induction of this wave is (in SI unit):

Option 1:
$\overrightarrow{\mathrm{B}}=\hat{\mathrm{i}}\frac{40}{\mathrm{c}}\cos\omega\left(\mathrm{t}-\frac{\mathrm{z}}{\mathrm{c}}\right)$

Option 2:
$\overrightarrow{\mathrm{B}}=\hat{\mathrm{j}}40\cos\omega\left(\mathrm{t}-\frac{\mathrm{z}}{\mathrm{c}}\right)$

Option 3:
$\overrightarrow{\mathrm{B}}=\hat{\mathrm{k}}\frac{40}{\mathrm{c}}\cos\omega\left(\mathrm{t}-\frac{\mathrm{z}}{\mathrm{c}}\right)$

Option 4:
$\overrightarrow{\mathrm{B}}=\hat{\mathrm{j}}\frac{40}{\mathrm{c}}\cos\omega\left(\mathrm{t}-\frac{\mathrm{z}}{\mathrm{c}}\right)$

Correct Answer:
$\overrightarrow{\mathrm{B}}=\hat{\mathrm{j}}\frac{40}{\mathrm{c}}\cos\omega\left(\mathrm{t}-\frac{\mathrm{z}}{\mathrm{c}}\right)$

Solution:
$\overrightarrow{\mathrm{E}}=\hat{\mathrm{i}}40\cos\omega\left(\mathrm{t}-\frac{\mathrm{z}}{\mathrm{c}}\right)$
$\overrightarrow{\mathrm{E}}$ is along $+\mathrm{x}$ direction

$\overrightarrow{\mathrm{v}}$ is along $+\mathrm{z}$ direction

So direction of $\overrightarrow{\mathrm{B}}$ will be along $+\mathrm{y}$ and magnitude of B will be $\frac{E}{c}$
So answer is $\frac{40}{c}\cos\omega\left(t-\frac{z}{c}\right)\hat{j}$

Hence, the answer is the option 4.

Q. 3 Which of the following nuclear fragments corresponding to nuclear fission between neutron $\left({}_{0}^{1}\mathrm{n}\right)$ and uranium isotope $\left({}_{92}^{235}\mathrm{U}\right)$ is correct:

Option 1:
${}_{56}^{144}\mathrm{Ba}+{}_{36}^{89}\mathrm{Kr}+4{}_{0}^{1}\mathrm{n}$

Option 2:
${}_{56}^{140}\mathrm{Xe}+{}_{38}^{94}\mathrm{Sr}+3{}_{0}^{1}\mathrm{n}$

Option 3:
${}_{51}^{153}\mathrm{Sb}+{}_{41}^{99}\mathrm{Nb}+3{}_{0}^{1}\mathrm{n}$

Option 4:
${}_{56}^{144}\mathrm{Ba}+{}_{36}^{89}\mathrm{Kr}+3{}_{0}^{1}\mathrm{n}$

Correct Answer:
${}_{56}^{144}\mathrm{Ba}+{}_{36}^{89}\mathrm{Kr}+3{}_{0}^{1}\mathrm{n}$

Solution:
Balancing mass number and atomic number
$_{92}^{235}\mathrm{U} + _{0}^{1}\mathrm{n} \rightarrow _{56}^{144}\mathrm{Ba} + _{36}^{89}\mathrm{Kr} + 3_{0}^{1}\mathrm{n}$

Hence, the answer is the option 4.

Q. 4 In an experiment to measure focal length (f) of convex lens, the least counts of the measuring scales for the position of object (u) and for the position of image (v) are Δu and Δv, respectively. The error in the measurement of the focal length of the convex lens will be :

Option 1:
$\frac{\Delta \mathrm{u}}{\mathrm{u}} + \frac{\Delta \mathrm{v}}{\mathrm{v}}$

Option 2:
$\mathrm{f}^2\left[\frac{\Delta \mathrm{u}}{\mathrm{u}^2} + \frac{\Delta \mathrm{v}}{\mathrm{v}^2}\right]$

Option 3:
$2\mathrm{f}\left[\frac{\Delta \mathrm{u}}{\mathrm{u}} + \frac{\Delta \mathrm{v}}{\mathrm{v}}\right]$

Option 4:
$\mathrm{f}\left[\frac{\Delta \mathrm{u}}{\mathrm{u}} + \frac{\Delta \mathrm{v}}{\mathrm{v}}\right]$

Correct Answer:
$\mathrm{f}^2\left[\frac{\Delta \mathrm{u}}{\mathrm{u}^2} + \frac{\Delta \mathrm{v}}{\mathrm{v}^2}\right]$

Solution:
$\mathrm{f}^{-1} = \mathrm{v}^{-1} - \mathrm{u}^{-1}$

$-\mathrm{f}^{-2}\mathrm{df} = -\mathrm{v}^{-2}\mathrm{dv} - \mathrm{u}^{-2}\mathrm{du}$

$\frac{\mathrm{df}}{\mathrm{f}^2} = \frac{\mathrm{dv}}{\mathrm{v}^2} + \frac{\mathrm{du}}{\mathrm{u}^2}$

$\mathrm{df} = \mathrm{f}^2\left[\frac{\mathrm{dv}}{\mathrm{v}^2} + \frac{\mathrm{du}}{\mathrm{u}^2}\right]$

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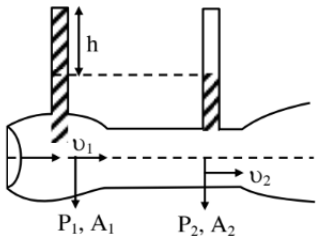
Q. 5 Given below are two statements :

Statement I : When speed of liquid is zero everywhere, pressure difference at any two points depends on equation $\mathrm{P_1 - P_2 = \rho g\,(h_2 - h_1)}$

Statement II : In ventury tube shown $\mathrm{2gh = v_1^2 - v_2^2}$



In the light of the above statements, choose the most appropriate answer from the options given below.

Option 1:

Both Statement I and Statement II are correct.

Option 2:

Statement I is incorrect but Statement II is correct.

Option 3:

Both Statement I and Statement II are incorrect.

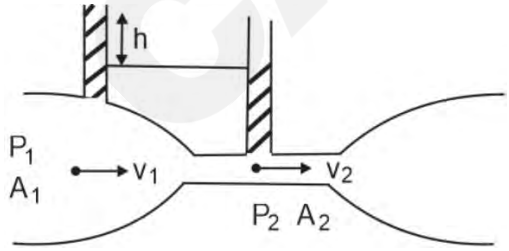
Option 4:

Statement I is correct but Statement II is incorrect.

Correct Answer:

Statement I is correct but Statement II is incorrect.

Solution:



Applying Bernoulli's equation

$\mathrm{P_1 + \rho g h_1 + \frac{1}{2}\rho v_1^2 = P_2 + \rho g h_2 + \frac{1}{2}\rho v_2^2}$

[$h_1 \& h_2$ are height of point from any reference level]

Given $V_1 = V_2 = 0$(for statement -1)

$\therefore \mathrm{P_1 - P_2 = \rho g\,(h_2 - h_2)}\$$

For statement-2

$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$

$P_1 - P_2 = \rho gh$

$P_1 - P_2 = \frac{1}{2}\rho v_2^2 - \frac{1}{2}\rho v_1^2$

$\rho \mathrm{gh} = \frac{1}{2}\rho v_2^2 - \frac{1}{2}\rho v_1^2$\$

$2\mathrm{gh} = v_2^2 - v_1^2$

Hence answer (4)

Q. 6 The resistances of the platinum wire of a platinum resistance thermometer at the ice point and steam point are 8Ω and 10Ω respectively. After inserting in a hot bath of temperature $400^{\circ}\mathrm{C}$, the resistance of platinum wire is :

Option 1:
2Ω

Option 2:
16Ω

Option 3:
8Ω

Option 4:
10Ω

Correct Answer:
16Ω

Solution:
$\text{Given } \mathrm{R}_0 = 8\Omega, \mathrm{R}_{100} = 10\Omega$
$\therefore \mathrm{R}_{100} = \mathrm{R}_0(1 + \alpha\Delta\mathrm{T})$

Also, $R_{400} = R_0\left(1 + \alpha\Delta\mathrm{T}^1\right)$
$\therefore 10 = 8(1 + \alpha \times 100) \Rightarrow 100\alpha = \frac{1}{4}$\$

$\therefore \mathrm{R}_{400} = 8(1 + 400\alpha) = 8(1 + 1) = 16\Omega$\$

Hence, the answer is the option 2.

Q. 7 A metal wire of uniform mass density having length L and mass M is bent to form a semicircular arc and a particle of mass m is placed at the centre of the arc. The gravitational force on the particle by the wire is:

Option 1:
$\frac{\mathrm{GMm}\pi}{2\,\mathrm{L}^2}$

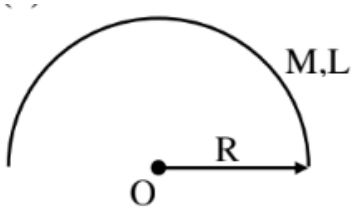
Option 2:
0

Option 3:
$\frac{\mathrm{GmM}\pi^2}{\mathrm{L}^2}$

Option 4:
$\frac{2\mathrm{GmM}\pi}{\mathrm{L}^2}$

Correct Answer:
$\frac{2\mathrm{GmM}\pi}{\mathrm{L}^2}$

Solution:



We have $\mathrm{R} = \frac{\mathrm{L}}{\pi}$

$\mathrm{g_0} = \frac{2\mathrm{G}\frac{\mathrm{M}}{\mathrm{L}}}{\mathrm{R}} = \frac{2\mathrm{GM}\pi}{\mathrm{L}^2}$

$\therefore \mathrm{F_m} = \mathrm{mg_0} = \frac{2\mathrm{GM}\pi\mathrm{m}}{\mathrm{L}^2}$

Q. 8 On the Celsius scale, the body's temperature increases by $40^{\circ}\mathrm{C}$. The increase in temperature on the Fahrenheit scale is:

Option 1:
$70^{\circ}\mathrm{F}$

Option 2:
$68^{\circ}\mathrm{F}$

Option 3:
$72^{\circ}\mathrm{F}$

Option 4:
$75^{\circ}\mathrm{F}$

Correct Answer:
$72^{\circ}\mathrm{F}$

Solution:
We know that per $^{\circ}\mathrm{C}$ change is equivalent to 1.8° change in $^{\circ}\mathrm{F}$.
$\therefore 40^{\circ}$ change on celcius scale will corresponds to 72° change on Fahrenheit scale.
Hence, the answer is the option 3.

Q. 9 An effective power of a combination of 5 identical convex lenses which are kept in contact along the principal axis is $25\mathrm{D}$. The focal length of each convex lens is :

Option 1:
20cm

Option 2:
50cm

Option 3:
500cm

Option 4:
25cm

Correct Answer:
20cm

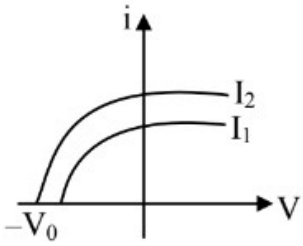
Solution:
We know that $\mathrm{P_{eq}} = \Sigma \mathrm{P_i}$
$\therefore$ given all lenses are identical
$\therefore 5P = 25D$ $
$\therefore P = 5D$

$\therefore \frac{1}{\mathrm{f}} = 5 \Rightarrow \mathrm{f} = \frac{1}{5}\ \mathrm{m} = 20\ \mathrm{cm}$

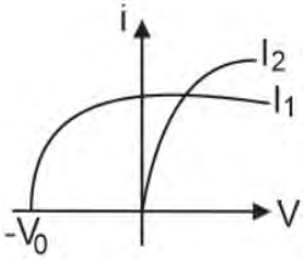
Hence, the answer is the option 1.

Q. 10 Which figure shows the correct variation of applied potential difference (V) with photoelectric current (I) at two different intensities of light $(\mathrm{I_1} < \mathrm{I_2})$ of same wavelengths :

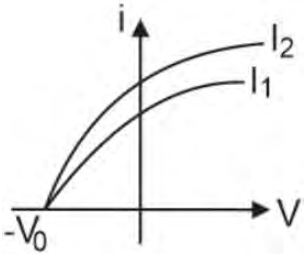
Option 1:



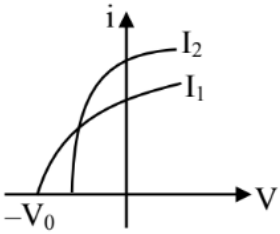
Option 2:



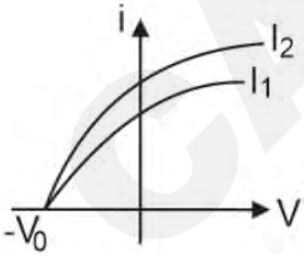
Option 3:



Option 4:



Correct Answer:



Solution:

Given lights are of same wavelength.

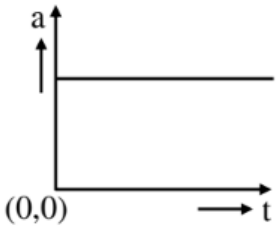
Hence stopping potential will remain same.

Since $I_2 > I_1$, hence saturation current corresponding to I_2 will be greater than that corresponding to I_1.

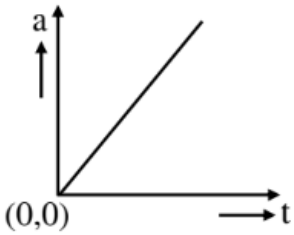
Hence option (3)

Q. 11 A wooden block, initially at rest on the ground, is pushed by a force which increases linearly with time t. Which of the following curve best describes acceleration of the block with time :

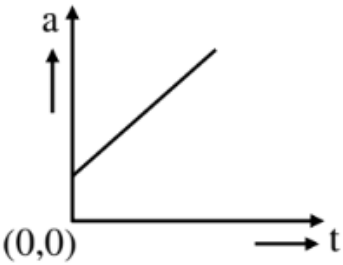
Option 1:



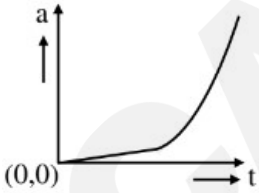
Option 2:



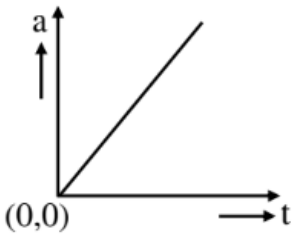
Option 3:



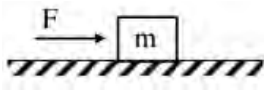
Option 4:



Correct Answer:



Solution:



$F = ma \Rightarrow a = \frac{F}{m} = \frac{kt}{m}$

a vs t will be straight line passing through origin. Since option (2).

Q. 12 If a rubber ball falls from a height of h and rebounds up to the height of $h/2$. The percentage loss of total energy of the initial system as well as the velocity ball before it strikes the ground, respectively, are :

Option 1:
$50\%, \sqrt{\frac{gh}{2}}$

Option 2:
$50\%, \sqrt{gh}$

Option 3:
$40\%, \sqrt{2gh}$
(4) $50\%, \sqrt{2gh}$

Option 4:
$50\%, \sqrt{2gh}$

Correct Answer:
$50\%, \sqrt{2gh}$

Solution:
Velocity just before collision $= \sqrt{2gh}$
Velocity just after collision $= \sqrt{2g\left(\frac{h}{2}\right)}$

$\therefore \Delta KE = \frac{1}{2}\, m(2gh) - \frac{1}{2}mgh$
$= \frac{1}{2}mgh$
$\therefore \%$ loss in energy

$= \frac{\Delta KE}{KE_i} \times 100 = \frac{\frac{1}{2}mgh}{\frac{1}{2}mg2\,h} \times 100 = 50\%$

Hence, the answer is the option 4.

Q. 13 The equation of stationary wave is :
$y = 2a \sin\left(\frac{2\pi nt}{\lambda}\right) \cos\left(\frac{2\pi x}{\lambda}\right)\$$

Which of the following is NOT correct

Option 1:
The dimensions of nt is [L]

Option 2:
The dimensions of n is $\left[\mathrm{LT}^{-1}\right]$

Option 3:
The dimensions of n/λ is [T]

Option 4:
The dimensions of x is [L]

Correct Answer:
The dimensions of n/λ is [T]

Solution:
Comparing the given equation with standard equation of standing $\frac{2\pi \mathrm{n}}{\lambda}=\omega \& \frac{2\pi}{\lambda}=\mathrm{k}$

$$\left[\frac{\mathrm{n}}{\lambda}\right]=[\omega]=\mathrm{T}^{-1}$$
$$[\mathrm{nt}]=[\lambda]=\mathrm{L} \quad \$$$
$$[\mathrm{n}]=[\lambda\omega]=\mathrm{LT}^{-1}$$
$$[\mathrm{x}]=[\lambda]=\mathrm{L}$$

Hence, the answer is the option 3.

Q. 14 A body travels $102.5\ \mathrm{m}^{\text{in n}^{\text{th}}}$ second and $115.0\ \mathrm{m}$ in $(\mathrm{n}+2)^{\text{th}}$ second. The acceleration is :

Option 1:
$9\ \mathrm{m/s^2}$
(2) $6.25\ \mathrm{m/s^2}$

Option 2:
$6.25\ \mathrm{m/s^2}$

Option 3:
$12.5\ \mathrm{m/s^2}$
(4) $5\ \mathrm{m/s^2}$

Option 4:
$5\ \mathrm{m/s^2}$

Correct Answer:
$6.25\ \mathrm{m/s^2}$

Solution:
Given, $102.5 = \mathrm{u} + \frac{\mathrm{a}}{2}(2\mathrm{n} - 1)$ \&
$115 = u + \frac{a}{2}(2n + 3)$\$

$$\Rightarrow 102.5 = u + an - \frac{a}{2}\&$$

$$115 = u + an + \frac{3a}{2} \quad \$$$

$$12.5 = 2a \Rightarrow a = 6.25\ \mathrm{m/s^2}$$

Hence, the answer is the option 2.

Q. 15 To measure the internal resistance of a battery, potentiometer is used. For $\mathrm{R} = 10\Omega$, the balance point is observed at $\ell = 500\ \mathrm{cm}$ and for $\mathrm{R} = 1\Omega$ the balance point is observed at $\ell = 400\ \mathrm{cm}$. The internal resistance of the battery is approximately :

Option 1:
0.2Ω

Option 2:
0.4Ω

Option 3:
0.1Ω

Option 4:
0.3Ω

Correct Answer:
0.3Ω

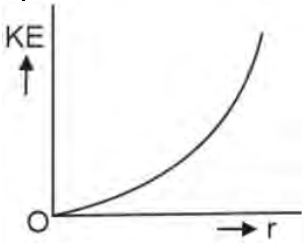
Solution:
Let potential gradient be λ.
$\therefore \mathrm{i} \times 10 = \lambda \times 500 = \varepsilon - \mathrm{ir_s}$\$
$\Rightarrow 500\lambda = \varepsilon - 50\lambda \mathrm{r_s}$

Also,
$\mathrm{i'} \times 1 = \lambda \times 400 = \varepsilon - \mathrm{i'r_s}$
$\Rightarrow 400\lambda = \varepsilon - 400\lambda \mathrm{r_s}$
$\therefore 100\lambda = 350\lambda \mathrm{r_s} \Rightarrow \mathrm{r_s} = \frac{10}{35} \approx 0.3\Omega$ \$

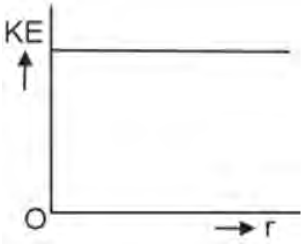
Hence option (4)

Q. 16 An infinitely long positively charged straight thread has a linear charge density $\lambda \mathrm{Cm}^{-1}$. An electron revolves along a circular path having axis along the length of the wire. The graph that correctly represents the variation of the kinetic energy of electron as a function of radius of circular path from the wire is :

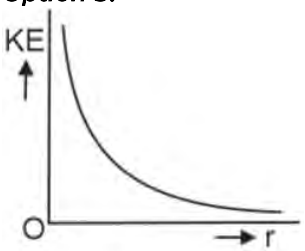
Option 1:



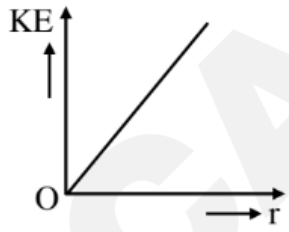
Option 2:



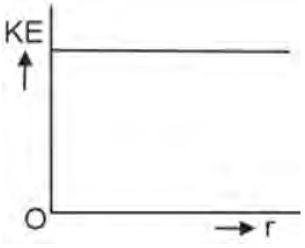
Option 3:



Option 4:



Correct Answer:



Solution:

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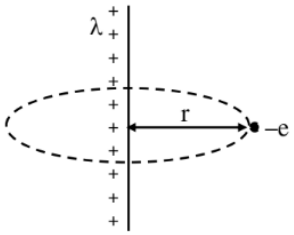


Electric field E at a distance r due to infinite long wire is $E = \frac{2k\lambda}{r}$

Force of electron $\Rightarrow \mathrm{F} = \mathrm{eE}$

$$\mathrm{F} = \mathrm{e}\left(\frac{2k\lambda}{r}\right)$$

$$\mathrm{F} = \frac{2k\lambda e}{r}$$

This force will provide required centripetal force

$$\mathrm{F} = \frac{mv^2}{r} = \frac{2k\lambda e}{r}$$

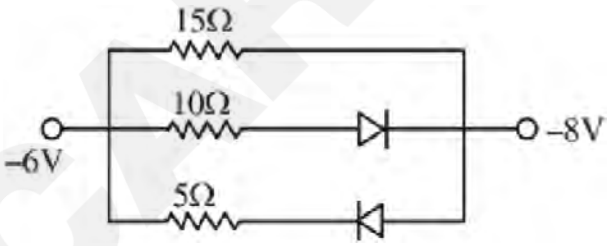
$$v = \sqrt{\frac{2k\lambda e}{m}}$$

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}\,m\left(\frac{2k\lambda e}{m}\right)$$

$$= k\lambda e$$

This is constant so option (2) is correct.

Q. 17 The value of net resistance of the network as shown in the given figure is :



Option 1:
$\left(\frac{5}{2}\right)\Omega$

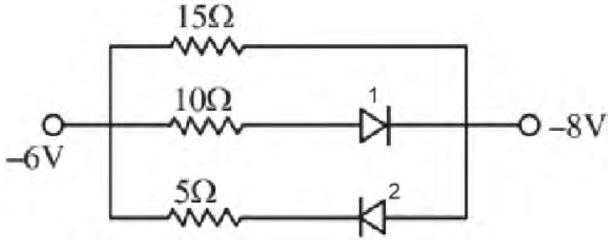
Option 2:
$\left(\frac{15}{4}\right)\Omega$

Option 3:
6Ω

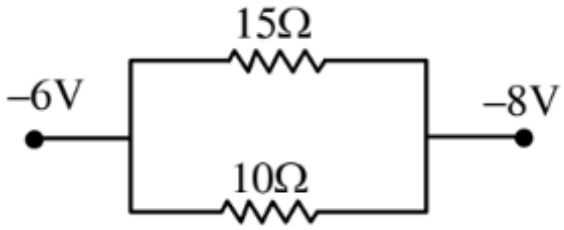
Option 4:
$(\frac{30}{11})\Omega$

Correct Answer:
6Ω

Solution:



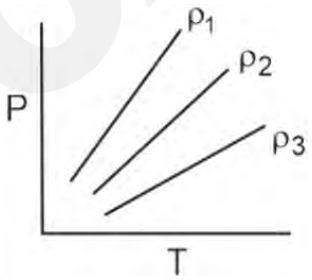
Diode 2 is in reverse bias
So current will not flow in branch of 2^{nd} diode, So we can assume it to be broken wire.
Diode 1 is in forward bias
So it will behave like conducting wire. So new circuit will be



$R_{eq} = \frac{15\times10}{15+10} = \frac{15\times10}{25} = 6\Omega$

Correct answer (3)

Q. 18 P-T diagram of an ideal gas having three different densities ρ_1, ρ_2, ρ_3 (in three different cases) is shown in the figure. Which of the following is correct :



Option 1:
$\rho_2 < \rho_3$

Option 2:
$\rho_1 > \rho_2$

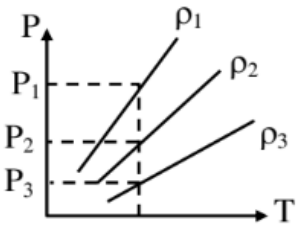
Option 3:
$\rho_1 < \rho_2$

Option 4:
$\rho_1 = \rho_2 = \rho_3$

Correct Answer:
$\rho_1 > \rho_2$

Solution:
For ideal gas
$PV = nRT$
$PV = \frac{m}{M}RT$
$P = \left(\frac{M}{V}\right)\frac{RT}{M}$ $
$P = \frac{\rho RT}{M}$

(Where m is mass of gas and M is molecular mass of gas)



for same temperature $\mathrm{P_1 > P_2 > P_3}$
So $\rho_1 > \rho_2 > \rho_3$
So correct answer is (2)

Q. 19 The coordinates of a particle moving in the x-y plane are given by

x=2+4 t, y=3 t+8 t^2

The motion of the particle is :

Option 1:
non-uniformly accelerated.

Option 2:
uniformly accelerated having motion along a straight line.

Option 3:
uniform motion along a straight line.

Option 4:
uniformly accelerated having motion along a parabolic path.

Correct Answer:
uniformly accelerated having motion along a parabolic path.

Solution:
$x = 2 + 4t$

$\frac{dx}{dt} = v_x = 4$

$\frac{dv_x}{dt} = a_x = 0$

$y = 3t + 8t^2$

$\frac{dy}{dt} = v_y = 3 + 16t$

$\frac{dv_y}{dt} = a_y = 16$

the motion will be uniformly accelerated motion.
For path
$x = 2 + 4t$

$\frac{(x-2)}{4} = t$

Put this value of t is equation of y

$y = 3\left(\frac{x-2}{4}\right) + 8\left(\frac{x-2}{4}\right)^2$

this is a quadratic equation so the path will be a parabola.

Hence, the answer is the option 3.

Q. 20 In an ac circuit, the instantaneous current is zero, when the instantaneous voltage is maximum. In this case, the source may be connected to :

A. pure inductor.
B. pure capacitor.
C. pure resistor.
D. combination of an inductor and capacitor.

Choose the correct answer from the options giver below :

Option 1:
A, B and C only

Option 2:
B, C and D only

Option 3:
A and B only

Option 4:
A, B and D only

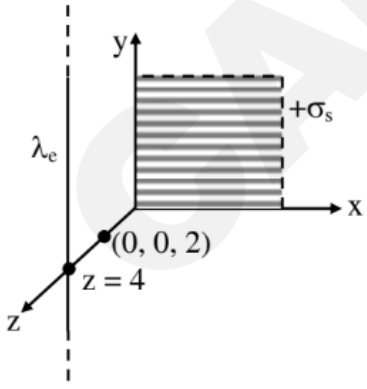
Correct Answer:
A, B and D only

Solution:
This is possible when phase difference is $\frac{\pi}{2}$ between current and voltage so correct answer will be (4)

Q. 21 An infinite plane sheet of charge having uniform surface charge density $+\sigma_s C/m^2$ is placed on $x - y$ plane. Another infinitely long line charge having uniform linear charge density $+\lambda_e C/m$ is placed at $z = 4$ m plane and parallel to y-axis. If the magnitude values $|\sigma_s| = 2|\lambda_e|$ then at point $(0, 0, 2)$, the ratio of magnitudes of electric field values due to sheet charge to that of line charge is $\pi\sqrt{n} : 1$. The value of n is

Correct Answer:
16

Solution:



$$\frac{E_S}{E_\ell} = \frac{\sigma}{2\epsilon_0} \times \frac{2\pi\epsilon_0 r}{\lambda}$$
$$= \frac{\pi \times \sigma r}{\lambda}$$

$= \frac{\pi \times 2\lambda \times 2}{\lambda} = \frac{4\pi}{1}$

$\therefore \mathrm{n} = 16$

Q. 22 A hydrogen atom changes its state from $\mathrm{n} = 3$ to $\mathrm{n} = 2$. Due to recoil, the percentage change in the wave length of emitted light is approximately 1×10^{-n}. The value of n is

[Given Rhc $= 13.6\mathrm{eV}, \mathrm{hc} = 1242\mathrm{eVnm}, \mathrm{h} = 6.6 \times 10^{-34}$ J s, mass of the hydrogen atom $= 1.6 \times 10^{-27}$ kg]

Correct Answer:
7

Solution:
$\Delta \mathrm{E} = 13.6\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = 1.9\mathrm{eV}$

$\Delta \mathrm{E} = \frac{\mathrm{hc}}{\lambda}$

$\lambda = \frac{\mathrm{hc}}{\Delta \mathrm{E}}$

$\mathrm{P_i} = \mathrm{P_f}$

$0 = -mv + \frac{h}{\lambda'}$

$\Rightarrow \mathrm{v} = \frac{\mathrm{h}}{\mathrm{m}\lambda'}$

$\Delta \mathrm{E} = \frac{1}{2}\mathrm{mv}^2 + \frac{\mathrm{hc}}{\lambda'}$

$= \frac{1}{2}\,\mathrm{m}\left(\frac{\mathrm{h}}{\mathrm{m}\lambda'}\right)^2 + \frac{\mathrm{hc}}{\lambda'}$

Now

$\Delta \mathrm{E} = \frac{\mathrm{h}^2}{2\,\mathrm{m}\lambda'^2} + \frac{\mathrm{hc}}{\lambda'}$

$\lambda'^2 \Delta \mathrm{E} - \mathrm{hc}\lambda' - \frac{\mathrm{h}^2}{2\,\mathrm{m}} = 0$

$\lambda' = \frac{\mathrm{hc} \pm \sqrt{\mathrm{h}^2\mathrm{c}^2 + \frac{4\Delta \mathrm{E}\mathrm{h}^2}{2\,\mathrm{m}}}}{2\Delta \mathrm{E}}$

$\lambda' = \frac{hc \pm hc\sqrt{1 + \frac{2\Delta \mathrm{E}}{\mathrm{mc}^2}}}{2\Delta \mathrm{E}}$

$\frac{\lambda'}{\lambda} = \frac{1 + \left(1 + \frac{2\Delta \mathrm{E}}{\mathrm{mc}^2}\right)^{\frac{1}{2}}}{2} = \frac{1 + 1 + \frac{\Delta \mathrm{E}}{\mathrm{mc}^2}}{2}$

$\frac{\lambda'}{\lambda} = 1 + \frac{\Delta E}{2mc^2}$

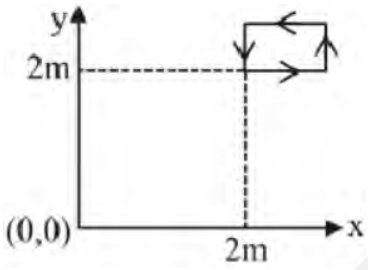
$\frac{\lambda' - \lambda}{\lambda} = \frac{\Delta E}{2mc^2} = \frac{1.9 \times 1.6 \times 10^{-19}}{2 \times 1.67 \times 10^{-27} \times 9 \times 10^{16}} = 10^{-9}$

$\therefore \%\ \text{change} \approx 10^{-7}\$$

Correct answer 7

Q. 23

The magnetic field existing in a region is given by $\vec{B} = 0.2(1 + 2x)\hat{k}T$. A square loop of edge $50\ \text{cm}$ carrying 0.5 A current is placed in $x - y$ plane with its edges parallel to the $x - y$ axes, as shown in figure. The magnitude of the net magnetic force experienced by the loop is $\quad\quad$ mN.



Correct Answer:
50

Solution:
Force on segment parallel to x-axis will cancel each other. Hence F_{net} will be due to portion parallel to y -axis.

$$F = 0.5 \times 0.5 \times 6 \times 0.2 - 0.5 \times 0.5 \times 0.2 \times 5$$
$$= 0.5 \times 0.5 \times 0.2$$
$$= 0.25 \times 0.2$$
$$= 50 \times 10^{-3}\ \text{N}$$
$$= 50\text{mN}$$

Q. 24

A alternating current at any instant is given by $i = \left[6 + \sqrt{56}\sin\left(100\pi t + \frac{\pi}{3}\right)\right]A$. The rms value of the current is _ A.

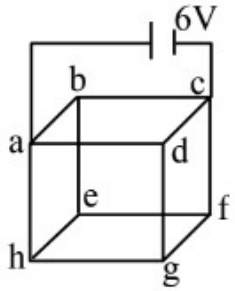
Correct Answer:
8

Solution:

$$I_{\text{rms}} = \sqrt{\frac{\int i^2 dt}{\int dt}}$$

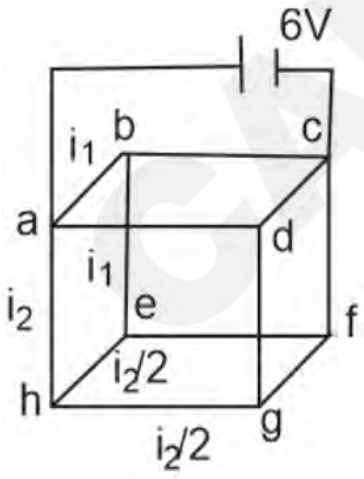
$$I_{rms} = \sqrt{(6)^2 + \frac{(\sqrt{56})^2}{2}}$$
$$= \sqrt{36 + 28}$$
$$= \sqrt{64}$$
$$= 8 \text{ A}$$

Q. 25 Twelve wires each having resistance 2Ω are joined to form a cube. A battery of 6 V emf is joined across point a and c. The voltage difference between e and f is _ V



Correct Answer:
1

Solution:



From symmetry, current through e-b \& g-d =0
$\therefore \text{R}_{\text{eq}} = \frac{3}{4} \times \text{R} = \frac{3}{2}\Omega$\$

$\therefore$ Current through battery $= \dfrac{6 \times 2}{3} = 4\ \text{A}$

$i_2 = \dfrac{4}{8} \times 2 = 1\ \text{A}$

$\therefore \Delta V$ across e-f $= \dfrac{i_2}{2} \times R = \dfrac{1}{2} \times 2 = 1\ \text{V}$

Q. 26 A soap bubble is blown to a diameter of $7\ \text{cm}$. 36960erg of work is done in blowing it further. If surface tension of soap solution is 40dyne/cm then the new radius is ____ cm. Take : $\left(\pi = \frac{22}{7}\right)$.

Correct Answer:
7

Solution:
$\omega = \Delta U = S\Delta A$

$$36960\text{erg} = \frac{40\text{dyne}}{\text{cm}} 8\pi \left[(r)^2 - \left(\frac{7}{2}\right)^2\right]\text{cm}^2$$

$r = 7\ \text{cm}$

Q. 27 Two wavelengths λ_1 and λ_2 are used in Young's double slit experiment $\lambda_1 = 450\ \text{nm}$ and $\lambda_2 = 650\ \text{nm}$. The minimum order of fringe produced by λ_2 which overlaps with the fringe produced by λ_1 is n. The value of n is

Correct Answer:
9

Solution:
$n_2\lambda_2 = n_1\lambda_1$

$$\frac{n_2}{n_1} = \frac{\lambda_1}{\lambda_2} = \frac{450}{650} = \frac{9}{13}$$

$n_2 = 9$

Q. 28 An elastic spring under tension of $3\ \text{N}$ has a lengtha. Its length is b under tension $2\ \text{N}$. For its length $(3a - 2b)$, the value of tension will be _ N.

Correct Answer:
5

Solution:
$3 = K(a - \ell)$

$2 = K(b - \ell)$

$T = K(3a - 2b - \ell)$

$T = K(3(a-\ell) - 2(b-\ell)$
$= K\left[3\left(\frac{3}{K}\right) - 2\left(\frac{2}{K}\right)\right]$
$= 9 - 4$
$= 5\ \mathrm{N}$

Q. 29 Two forces $\vec{F}_1$ and $\vec{F}_2$ are acting on a body. One force has magnitude thrice that of the other force and the resultant of the two forces is equal to the force of larger magnitude. The angle between $\overrightarrow{\mathrm{F}}_1$ and $\vec{F}_2$ is $\cos^{-1}\left(\frac{1}{n}\right)$. The value of $|n|$ is

Correct Answer:
6

Solution:
$\overrightarrow{\mathrm{F}}_1 = \mathrm{F}$

$\overrightarrow{\mathrm{F}}_\mathrm{R} = \overrightarrow{\mathrm{F}}_2 = 3\,\mathrm{F}$

$\mathrm{F}_\mathrm{R}^2 = \mathrm{F}_1^2 + \mathrm{F}_2^2 + 2\,\mathrm{F}_1\,\mathrm{F}_2\cos\theta$
$9\,\mathrm{F}^2 = \mathrm{F}^2 + 9\,\mathrm{F}^2 + 6\,\mathrm{F}^2\cos\theta$

$\cos\theta = -\frac{1}{6}$
$\theta = \cos^{-1}\left(\frac{1}{-6}\right)$
$\mathrm{n} = -6$
$|\mathrm{n}| = 6$

Q. 30 A solid sphere and a hollow cylinder roll up without slipping on same inclined plane with same initial speed v. The sphere and the cylinder reaches upto maximum heights h_1 and h_2 , respectively, above the initial level. The ratio $h_1 : h_2$ is $\frac{n}{10}$. The value of n is _

Correct Answer:
7

Solution:

$$\text{Gain in P.E.} = \text{Loss in K.E.}$$

$$mgh = \frac{1}{2}mv^2\left(1+\frac{K^2}{R^2}\right)$$

$$h \propto 1+\frac{K^2}{R^2}$$

$$\frac{h_1}{h_2} = \frac{1+\frac{2}{5}}{1+1} = \frac{7}{5\times 2} = \frac{7}{10}$$

$$n = 7$$

Chemistry

Q. 1 What pressure (bar) of H_2 would be required to make emf of hydrogen electrode zero in pure water at $25^{\circ}C$?

Option 1:
10^{-14}

Option 2:
10^{-7}

Option 3:
1

Option 4:
0.5

Correct Answer:
10^{-14}

Solution:

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$$2e^- + 2H^+(aq) \rightarrow H_2(g)$$

$$E = E^\circ - \frac{0.059}{n}\log\frac{P_{H_2}}{[H^+]^2}$$

$$0 = 0 - \frac{0.059}{2}\log\frac{P_{H_2}}{(10^{-7})^2}$$

$$\log\frac{P_{H_2}}{(10^{-7})^2} = 0$$

$$\frac{P_{H_2}}{10^{-14}} = 1$$

$$P_{H_2} = 10^{-14}\ \text{bar}$$

Hence, the answer is the option (1).

Q. 2 The correct sequence of ligands in the order of decreasing field strength is :

Option 1:
$CO > H_2O > F^- > S^{2-}$

Option 2:
$^-OH > F^- > NH_3 > CN^-$

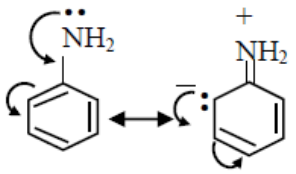
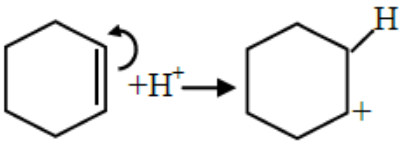
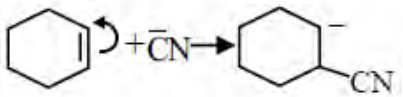
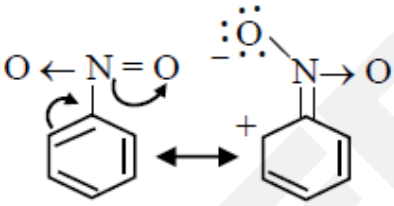
Option 3:
$NCS^- > EDTA^{4-} > CN^- > CO$

Option 4:
$S^{2-} > -OH > EDTA^{4-} > CO$

Correct Answer:
$CO > H_2O > F^- > S^{2-}$

Solution:
According to spectrochemical series ligand field strength is $CO > H_2O > F^- > S^{2-}$

Q. 3 Match List-I with List -II

List - I Mechanism steps		List - II Effect	
(A)		(I)	– E effect
(B)		(II)	– R effect
(C)		(III)	+ E effect
(D)		(IV)	+ R effect

Choose the correct answer from the options given below :

Option 1:
(A) – (IV), (B) – (III), (C) – (I), (D) – (II)

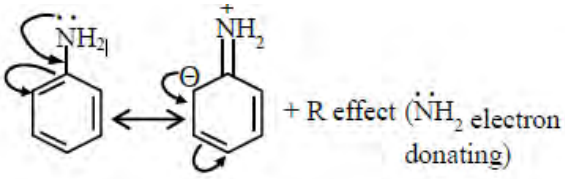
Option 2:
(A) – (III), (B) – (I), (C) – (II), (D) – (IV)

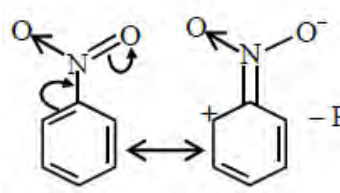
Option 3:
(A) – (II), (B) – (IV), (C) – (III), (D) – (I)

Option 4:
(A) – (I), (B) – (II), (C) – (IV), (D) – (III)

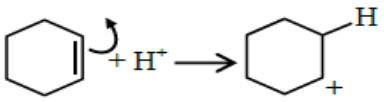
Correct Answer:
(A) – (IV), (B) – (III), (C) – (I), (D) – (II)

Solution:

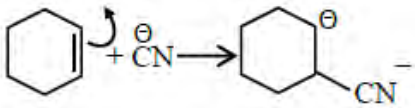




– R effect (NO_2 electron withdrawing)



+E Effect H^+ electron deficient species)



– E Effect ($\bar{C}N$ electron deficient species)

Q. 4 What will be the decreasing order of basic strength of the following conjugate bases ?

$^{-}OH, RO, CH_3CO\bar{O}, C\bar{l}$

Option 1:

$C\bar{l} > OH > R\bar{O} > CH_3CO\bar{O}$

Option 2:

$R\bar{O} > -OH > CH_3CO\bar{O} > C\bar{l}$

Option 3:

$^{-}OH > R\bar{O} > CH_3CO\bar{O} > C\bar{l}$

Option 4:

$C\bar{l} > R\bar{O} >^{-} OH > CH_3CO\bar{O}$

Correct Answer:

$R\bar{O} > -OH > CH_3CO\bar{O} > C\bar{l}$

Solution:

Strong acid have weak conjugate base Acidic strength :

$H - Cl > CH_3COOH > H_2O > R - OH$

Conjugate base strength :

$Cl^- < CH_3COO^- < \bar{O}H < RO^-$

Hence, the answer is the option (2).

Q. 5 In the precipitation of the iron group (III) in qualitative analysis, ammonium chloride is added
before adding ammonium hydroxide to :

Option 1:
prevent interference by phosphate ions

Option 2:
decrease concentration of ^{-}OH ions

Option 3:
increase concentration of Cl^{-} ions

Option 4:
increase concentration of NH_4^+ ions

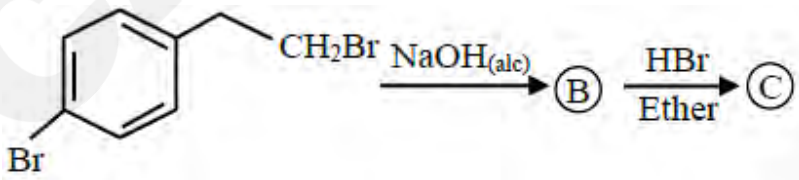
Correct Answer:
decrease concentration of ^{-}OH ions

Solution:
$NH_4OH \rightleftharpoons NH_4^+ + OH^-$
$NH_4Cl \rightarrow NH_4^+ + Cl$

Due to the common ion effect of NH_4^+, $[OH^-]$ decreases in such extent that only group-III cation can be precipitated, due to their very low K_{sp} in the range of 10^{-38}.

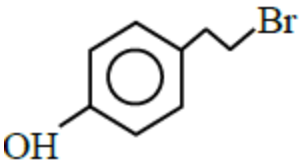
Hence, the answer is the option (2).

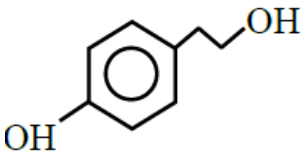
Q. 6



Identify(B) and (C) and how are (A) and (C) related ?

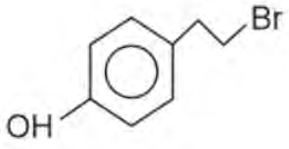
Option 1:

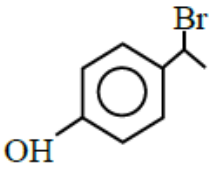




functional group isomers

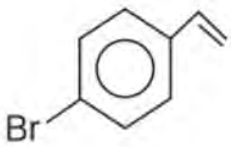
Option 2:

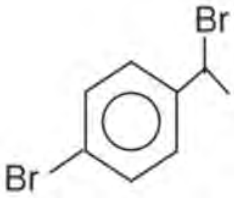




Derivative

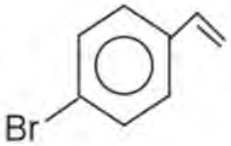
Option 3:

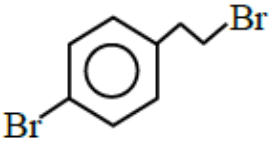




position isomers

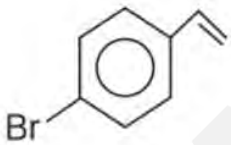
Option 4:

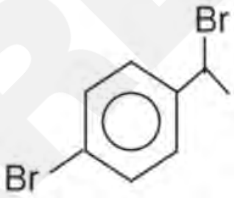




chain isomers

Correct Answer:





position isomers

Solution:

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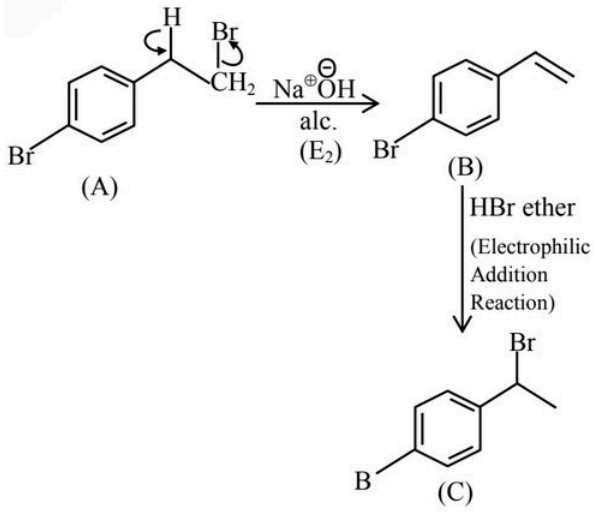

A and C are position isomer.

Q. 7 One of the commonly used electrode is calomel electrode. Under which of the following categories calomel electrode comes ?

Option 1:
Metal – Insoluble Salt – Anion electrodes

Option 2:
Oxidation – Reduction electrodes

Option 3:
Gas – Ion electrodes

Option 4:
Metal ion – Metal electrodes

Correct Answer:
Metal – Insoluble Salt – Anion electrodes

Solution:
Theory based

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Q. 8 Number of complexes from the following with even number of unpaired "d" electrons is ____.

$[V(H_2O)_6]^{3+}$, $[Cr(H_2O)_6]^{2+}$, $[Fe(H_2O)_6]^{3+}$,
$[Ni(H_2O)_6]^{3+}$, $[Cu(H_2O)_6]^{2+}$
[Given atomic numbers : $V = 23, Cr = 24, Fe = 26, Ni = 28, Cu = 29$]

Option 1:
2

Option 2:
4

Option 3:
5

Option 4:
1

Correct Answer:
2

Solution:
$[V(H_2O)_6]^{3+} \rightarrow d^2sp^3$
$_{23}V : -[Ar]3\,d^3 4\,s^2$
$V^{+3} : -[Ar]3\,d^2, n = 2$ (even number of unpaired e)
$[Cr(H_2O)_6]^{2+} \rightarrow sp^3\,d^2$
$_{24}Cr : -[Ar]3\,d^5 4\,s^1$
$Cr^{+2} : -[Ar]3\,d^4, n = 4$ (even number of unpaired e^-)

$[Fe(H_2O)_6]^{3+} \rightarrow sp^3\,d^2$
$Fe^{3+} : -[Ar]3\,d^5 4\,s^0$
$n = 5$ (odd number of unpaired e^-)
$[Ni(H_2O)_6]^{3+} \rightarrow sp^3\,d^2$
$Ni : -[Ar]3\,d^8 4\,s^2$
$Ni^{+3} : -[Ar]3\,d^7, n = 3$ (odd number of unpaired e^-)
$[Cu(H_2O)_6]^{2+} \rightarrow sp^3\,d^2$
$Cu : -[Ar]3\,d^9 4\,s^0$
$n = 1$(odd number of unpaired e)

Hence, the answer is the option (1).

Q. 9 Which one of the following molecules has maximum dipole moment?

Option 1:
NF_3

Option 2:
CH_4

Option 3:
NH_3

Option 4:
PF_5

Correct Answer:
NH_3

Solution:
$CH_4 \& PF_5, \mu_{net} = 0$ (non polar)

$\underset{\text{Vector addition of bond moment \& lone pair moment}}{\mu_{NH_3}} > \underset{\text{Vector subtraction of bond moment \& lone pair moment}}{\mu_{NF_3}}$

Q. 10 Number of molecules/ions from the following in which the central atom is involved in sp^3 hybridization is _____

$NO_3^-, BCl_3, ClO_2^-, ClO_3$

Option 1:
2

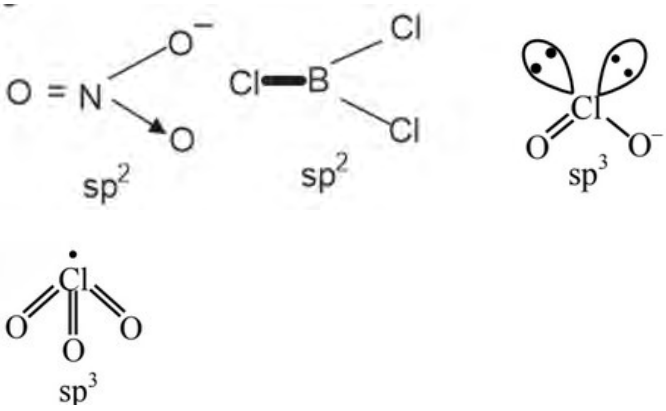
Option 2:
4

Option 3:
3

Option 4:
1

Correct Answer:
2

Solution:



Q. 11 Which among the following is incorrect statement?

Option 1:
Electromeric effect dominates over inductive effect

Option 2:
The electromeric effect is, temporary effect

Option 3:
The organic compound shows electromeric effect in the presence of the reagent only

Option 4:
Hydrogen ion (H^+)shows negative electromeric effect|

Correct Answer:
Hydrogen ion (H^+)shows negative electromeric effect|

Solution:
Hydrogen ion (H^+)shows positive electromeric effect.

Q. 12 Given below are two statements :

Statement I : Acidity of α-hydrogens of aldehydes and ketones is responsible for Aldol reaction.

Statement II : Reaction between benzaldehyde and ethanal will NOT give Cross – Aldol product.

In the light of above statements, choose the most appropriate answer from the options given below.

Option 1:
Both Statement I and Statement II are correct.

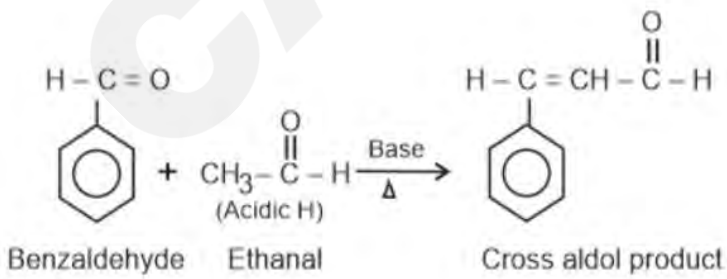
Option 2:
Both Statement I and Statement II are incorrect.

Option 3:
Statement I is incorrect but Statement II is correct.

Option 4:
Statement I is correct but Statement II is incorrect.

Correct Answer:
Statement I is correct but Statement II is incorrect.

Solution:
Aldehyde and ketones having acidic
α-hydrogen show aldol reaction



Q. 13 Which of the following nitrogen containing compound does not give Lassaigne's test ?

Option 1:
Phenyl hydrazine

Option 2:
Glycene

Option 3:
Urea

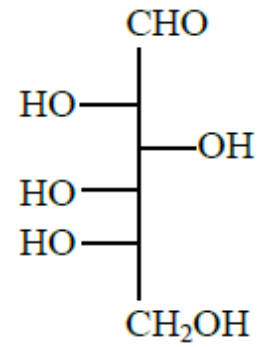
Option 4:
Hydrazine

Correct Answer:
Hydrazine

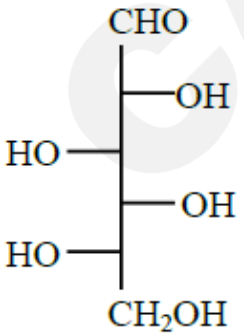
Solution:
Hydrazine $(NH_2 - NH_2)$ have no carbon so does not show Lassaigne's test.

Q. 14 Which of the following is the correct structure of L-Glucose ?

Option 1:



Option 2:



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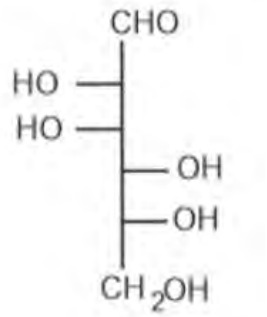
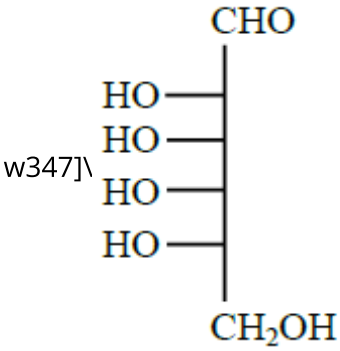
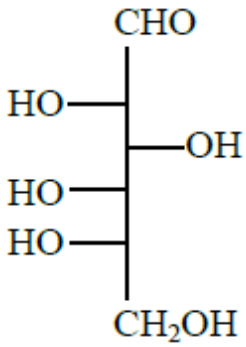
Option 3:



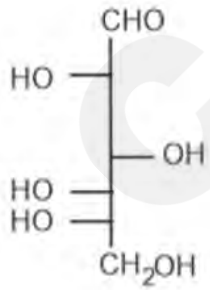
Option 4:



Correct Answer:



Solution:
Structure of L-Glucose is



Q. 15 The element which shows only one oxidation state other than its elemental form is :

Option 1:
Cobalt

Option 2:
Scandium

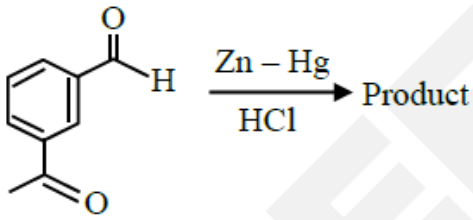
Option 3:
Titanium

Option 4:
Nickel

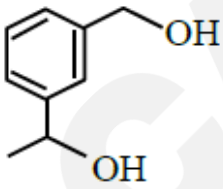
Correct Answer:
Scandium

Solution:
$\mathrm{Co},\mathrm{Ti}, \mathrm{Ni}canshow+2,+3$ and +4 oxidation state, But 'Sc' only shows +3 stable oxidation state.

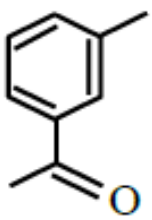
Q. 16 Identify the product in the following reaction :



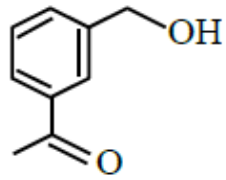
Option 1:



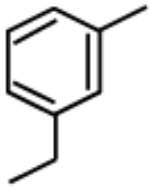
Option 2:



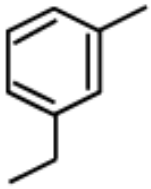
Option 3:



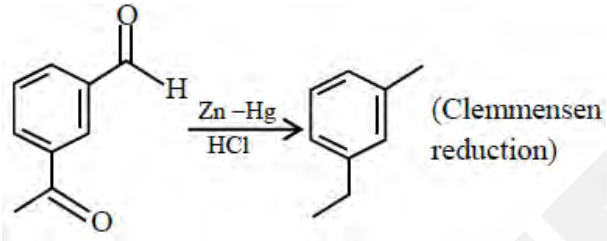
Option 4:



Correct Answer:



Solution:



Q. 17 Number of elements from the following that CANNOT form compounds with valencies which match with their respective group valencies is _____.
B, C, N, S, O, F, P, Al, Si

Option 1:
7

Option 2:
5

Option 3:
6

Option 4:
3

Correct Answer:
3

Solution:

N,O, F can't extend their valencies upto their group number due to the non-availability of vacant 2d like orbital.

Hence, the answer is the option (4).

Q. 18 The Molarity (M) of an aqueous solution containing $5.85\ \mathrm{g}$ of NaCl in $500\ \mathrm{mL}$ water is : (Given : Molar Mass $\mathrm{Na}: 23$ and $\mathrm{Cl}: 35.5 \mathrm{gmol}^{-1}$)

Option 1:
20

Option 2:
0.2

Option 3:
2

Option 4:
4

Correct Answer:
0.2

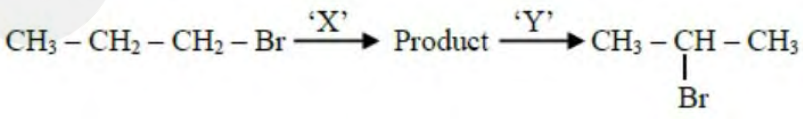
Solution:

$$\mathrm{M}=\frac{\mathrm{n}_{\mathrm{NaCl}}}{\mathrm{V}_{\mathrm{sol}}(\text{ in L})}$$

$$\mathrm{M}=\frac{\frac{5.85}{58.5}}{0.5}=0.2\mathrm{M}$$

Hence, the answer is the option (2).

Q. 19 Identify the correct set of reagents or reaction conditions ' X ' and ' Y ' in the following set of transformation.



Option 1:
$\mathrm{X}=$ conc.alc. $\mathrm{NaOH}, 80^{\circ} \mathrm{C}, \mathrm{Y}=\mathrm{Br}_2 / \mathrm{CHCl}_3$

Option 2:
$\mathrm{X} = \text{dil.aq. } \mathrm{NaOH}, 20^{\circ}\mathrm{C}, \mathrm{Y} = \mathrm{HBr}$ /acetic acid

Option 3:
$\mathrm{X} = \text{conc.alc. } \mathrm{NaOH}, 80^{\circ}\mathrm{C}, \mathrm{Y} = \mathrm{HBr}$ /acetic acid

Option 4:
$\mathrm{X} = \text{dil.aq. } \mathrm{NaOH}, 20^{\circ}\mathrm{C}, \mathrm{Y} = \mathrm{Br_2/CHCl_3}$

Correct Answer:
$\mathrm{X} = \text{conc.alc. } \mathrm{NaOH}, 80^{\circ}\mathrm{C}, \mathrm{Y} = \mathrm{HBr}$ /acetic acid

Solution:

$$\mathrm{CH_3-CH_2-CH_2-Br} \xrightarrow{\mathrm{X=\ conc.alc.\ NaOH}}$$
$$\mathrm{CH_3-CH=CH_2} \xrightarrow{\mathrm{Y\equiv HBr/\ Aceticacid}} \mathrm{CH_3-CHBr-CH_3}$$

Q. 20 The correct order of first ionization enthalpy values of the following elements is :

(A) O

(B) N

(C) Be

(D) F

(E) B

Choose the correct answer from the options given below :

Option 1:
B < D < C < E < A

Option 2:
E < C < A < B < D

Option 3:
C < E < A < B < D

Option 4:
A < B < D < C < E

Correct Answer:
E < C < A < B < D

Solution:

Correct order of $I^{st}IE$

$$\begin{array}{ccccccccccc} Li & < & B & < & Be & < & C & < & O & < & N & < & F & < & Ne \\ & & \downarrow & & \downarrow & & & & \downarrow & & \downarrow & & \downarrow & & \\ & & E & < & C & & < & & A & < & B & < & D & & \end{array}$$

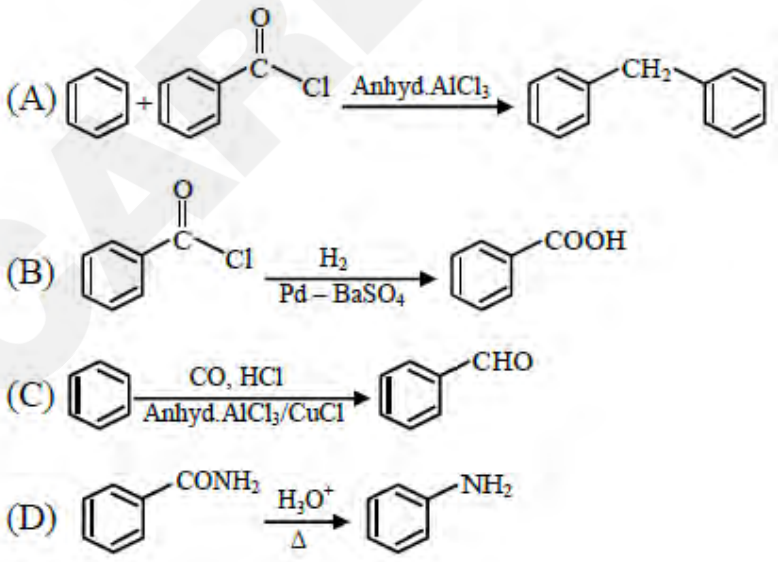
Q. 21 The enthalpy of formation of ethane (C_2H_6) from ethylene by addition of hydrogen where the bondenergies of $C-H, C-C, H-H$ are $414\ kJ, 347\ kJ, 615\ kJ$ and $435\ kJ$ respectively is - _____ kJ.

Correct Answer:
125

Solution:
$C_2H_4(g) + H_2(g) \rightarrow C_2H_6(g)$
$\Delta H = BE(C=C) + 4BE(C-H) + BE(H-H)$
$- BE(C-C) - 6BE(C-H)$
$\Delta H = BE(C=C) + BE(H-H) - BE(C-C)$
$- 2BE(C-H)$
$= 615 + 435 - 347 - 2 \times 414$
$= -125\ kJ$

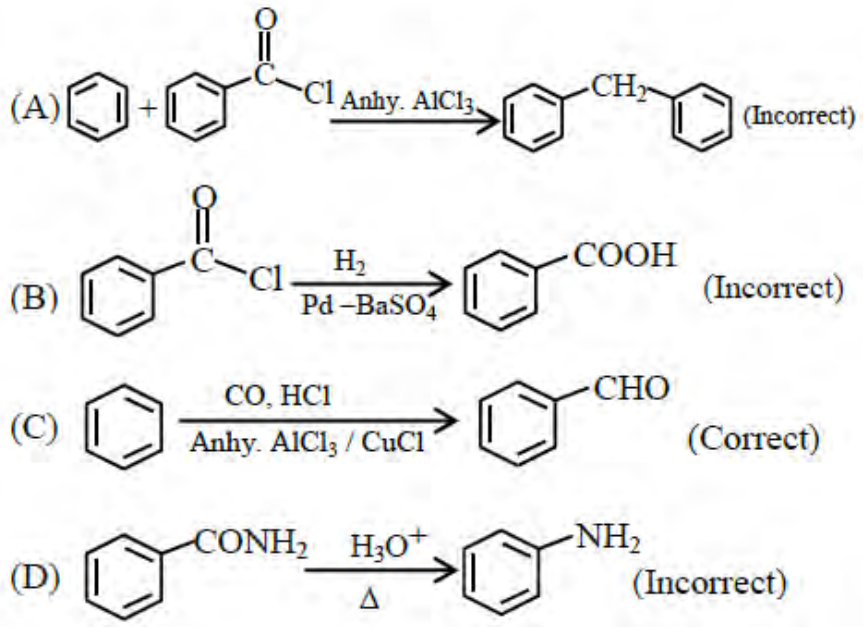
Hence, the answer is the option (1).

Q. 22 The number of correct reaction(s) among the following is _____.



Correct Answer:
1

Solution:



Q. 23 Xg of ethylamine is subjected to reaction with $\mathrm{NaNO_2/HCl}$ followed by water; evolved dinitrogen gas which occupied $2.24\ \mathrm{L}$ volume at STP. X is_____ $\times 10^{-1}$ g.

Correct Answer:
45

Solution:
given : $\mathrm{N_2}$ evolved is $2.24\ \mathrm{L}$ i.e. 0.1 mole.
i.e. $\mathrm{CH_3CH_2NH_2}$ (ethyl amine) will be $4.5\ \mathrm{g}\ (=0.1\ \mathrm{mole})$

Hence the answer $= 45 \times 10^{-1}\ \mathrm{g}$

Hence, the answer is (45).

Q. 24 The de-Broglie's wavelength of an electron in the 4^{th} orbit is _____ πa_0. (a_0 = Bohr's radius)

Correct Answer:
8

Solution:

$2\pi r_n = n\lambda_d$

$2\pi a_0 \frac{n^2}{Z} = n\lambda_d$

$2\pi a_0 \frac{4^2}{1} = 4\lambda_d$

$\lambda_d = 8\pi a_0$

Hence, the answer is (8).

Q. 25 Only $2\ \text{mL}$ of $KMnO_4$ solution of unknown molarity is required to reach the end point of a titration of $20\ \text{mL}$ of oxalic acid ($2M$) in acidic medium. The molarity of $KMnO_4$ solution should be _____ M.

Correct Answer:
8

Solution:
$\text{eq. } (KMnO_4) = \text{ eq. } (H_2C_2O_4)$

$M \times 2 \times 5 = 2 \times 20 \times 2$

$M = 8M$

Q. 26 Consider the following reaction

$MnO_2 + KOH + O_2 \rightarrow A + H_2O.$

Product ' A ' in neutral or acidic medium disproportionate to give products ' B ' and ' C ' along with water. The sum of spin-only magnetic moment values of B and C is _____ BM. (nearest integer)
(Given atomic number of Mn is 25)

Correct Answer:
4

Solution:
$MnO_2 + KOH + O_2 \rightarrow K_2MnO_4 + H_2O$

(A)

$

$$K_2MnO_4 \xrightarrow{\text{Neutral/acidic solution}} KMnO_4 + MnO_2$$

Mn^{+4} :- $[Ar]3\ d^3$

$n = 3, \mu = \sqrt{3(3+2)} = 3.87\ \text{B.M.}$

Hence, the answer is (4).

Q. 27 Consider the following transformation involving first order elementary reaction in each step at constant temperature as shown below.

$$A + B \underset{\text{Step 3}}{\overset{\text{Step 1}}{\rightleftharpoons}} C \xrightarrow{\text{Step 2}} P$$

Some details of the above reaction are listed below.

Step	Rate constant $(sec)^{-1}$	Activation energy (kJ mol^{-1})
1	K_1	300
2	K_2	200
3	K_3	Ea_3

If the overall rate constant of the above transformation (k) is given as $k = \frac{k_1 k_2}{k_3}$ and the overall activation energy (E_a) is $400\ kJ\ mol^{-1}$, then the value of Ea_3 is _____ $kJmol^{-1}$ (nearest integer)

Correct Answer:
100

Solution:

$$K = \frac{K_1\ K_2}{K_3}$$

$$Ae^{\frac{-E_2}{RT}} = \frac{A_1 e^{\frac{-Ea_1}{RT}} A_2 e^{\frac{-Ea_2}{RT}}}{A_3 e^{\frac{-E_{23}}{RT}}}$$

$$Ae^{\frac{-E_2}{RT}} = \frac{A_1\ A_2}{A_3} e^{\frac{-\left(Ea_1 + E_2 - Ea_3\right)}{RT}}$$

$$E_a = E_{a_1} + E_{a_2} - E_{a_3}$$

$$400 = 300 + 200 - E_{a_3}$$

$$E_{a_3} = 100\ kJ/mole$$

Q. 28 2.5 g of a non-volatile, non-electrolyte is dissolved in 100 g of water at $25^\circ C$. The solution showed a boiling point elevation by $2^\circ C$. Assuming the solute concentration in negligible with respect to the solvent concentration, the vapour pressure of the resulting aqueous solution is _____mm of Hg (nearest integer)
[Given : Molal boiling point elevation constant of water $(K_b) = 0.52\ K.\ kgmol^{-1}$, 1 atm pressure $= 760$ mm of Hg, molar mass of water $= 18\ g\ mol^{-1}$]

Correct Answer:
707

Solution:

$2 = 0.52 \times m$

$m = \frac{2}{0.52}$

According to the question, the solution is much diluted

so $\frac{\Delta P}{P^\circ} = \frac{n_{solute}}{n_{solvent}}$

$\frac{\Delta P}{P^\circ} = \frac{m}{1000} \times M_{solvent}$

$\Delta P = P^\circ \times \frac{m}{1000} \times M_{solvent}$

$= 760 \times \frac{\frac{2}{0.52}}{1000} \times 18 = 52.615$

$P_5 = 760 - 52.615 = 707.385 \text{ mm of Hg}$

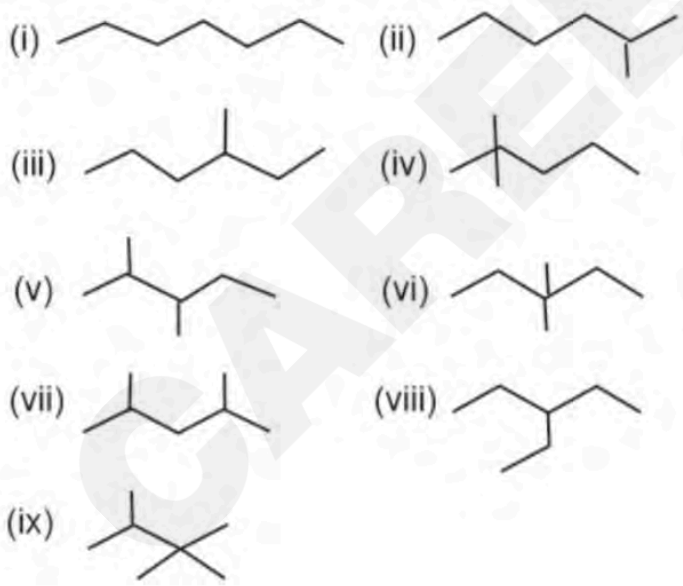
Hence, the answer is (707).

Q. 29 The number of different chain isomers for C_7H_{16} is_____.

Correct Answer:

9

Solution:



Q. 30 Number of molecules/species from the following having one unpaired electron is _____.

$O_2, O_2^{-1}, NO, CN^{-1}, O_2^{2-}$

Correct Answer:

2

Solution:
According to M.O.T.

$O_2 \rightarrow$ no. of unpaired electrons $= 2$

$O_2^- \rightarrow$ no. of unpaired electron $= 1$

$NO \rightarrow$ no. of unpaired electron $= 1$

$CN^- \rightarrow$ no. of unpaired electron $= 0$

$O_2^{2-} \rightarrow$ no. of unpaired electron $= 0$

Hence, the answer is the option (1).

Maths

Q. 1

Let $f : \mathrm{R} \rightarrow \mathrm{R}$ be a function given by $f(x) = \begin{cases} \frac{1-\cos 2x}{x^2} & , x < 0 \\ \alpha & , x = 0, \\ \frac{\beta\sqrt{1-\cos x}}{x} & , x > 0 \end{cases}$ where $\alpha, \beta \in R$. If f is continuous at $\mathrm{x} = 0$, then $\alpha^2 + \beta^2$ is equal to :

Option 1:
48

Option 2:
12

Option 3:
3

Option 4:
6

Correct Answer:
12

Solution:

$$f\left(0^-\right) = \lim_{x \to 0^-} \frac{2\sin^2 x}{x^2} = 2 = \alpha$$

$$f\left(0^+\right) = \lim_{x \to 0^+} \beta \times \sqrt{2}\frac{\sin\frac{x}{2}}{2\frac{x}{2}} = \frac{\beta}{\sqrt{2}} = 2$$

$$\Rightarrow \beta = 2\sqrt{2}$$

$$\alpha^2 + \beta^2 = 4 + 8 = 12$$

Hence, the answer is the option (2).

Q. 2 Three urns A, B and C contain 7 red, 5 black; 5 red, 7 black and 6 red, 6 black balls, respectively. One of the urns is randomly selected and a ball is drawn from it. If the ball drawn is black, then the probability that it is drawn from urn A is:

Option 1:
$\frac{4}{17}$

Option 2:
$\frac{5}{18}$

Option 3:
$\frac{7}{18}$

Option 4:
$\frac{5}{16}$

Correct Answer:
$\frac{5}{18}$

Solution:

A	B	C
7R,5B	5R,7B	6R,6B

$$\mathrm{P(B)} = \frac{1}{3}\cdot\frac{5}{12} + \frac{1}{3}\cdot\frac{7}{12} + \frac{1}{3}\cdot\frac{6}{12}$$

$$\text{Required probability } = \frac{\frac{1}{3}\cdot\frac{5}{12}}{\frac{1}{3}\cdot\left[\frac{5}{12}+\frac{7}{12}+\frac{6}{12}\right]} = \frac{5}{18}$$

Hence, the answer is the option (2).

Q. 3 The vertices of a triangle are A(-1,3), B(-2,2) and C(3,-1). A new triangle is formed by shifting the sides of the triangle by one unit inwards. Then the equation of the side of the new triangle nearest to origin is :

Option 1:
$x - y - (2+\sqrt{2}) = 0$

Option 2:
$-x + y - (2-\sqrt{2}) = 0$

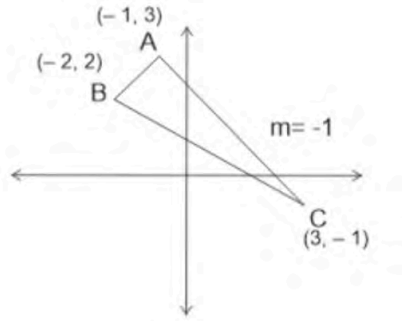
Option 3:
$x + y - (2-\sqrt{2}) = 0$

Option 4:
$x + y + (2-\sqrt{2}) = 0$

Correct Answer:

$x + y - (2 - \sqrt{2}) = 0$

Solution:



equation of $\mathrm{AC} \rightarrow \mathrm{x} + \mathrm{y} = 2$

equation of line parallel to $\mathrm{ACx} + \mathrm{y} = \mathrm{d}$

$$\frac{\mathrm{d} - 2}{\sqrt{2}} = 1$$

$$\mathrm{d} = 2 - \sqrt{2}$$

eq^{n} of new required line

$x + y = 2 - \sqrt{2}$

Q. 4 If the solution $y = y(x)$ of the differential equation $\left(x^4 + 2x^3 + 3x^2 + 2x + 2\right)dy - \left(2x^2 + 2x + 3\right)dx = 0$ satisfies $y(-1) = -\frac{\pi}{4}$, then $y(0)$ is equal to:

Option 1:

$-\frac{\pi}{12}$

Option 2:

0

Option 3:

$\frac{\pi}{4}$

Option 4:

$\frac{\pi}{2}$

Correct Answer:

$\frac{\pi}{4}$

Solution:

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$$\int dy = \int \frac{\left(2x^2+2x+3\right)}{x^4+2x^3+3x^2+2x+2}dx$$

$$y = \int \frac{\left(2x^2+2x+3\right)}{\left(x^2+1\right)\left(x^2+2x+2\right)}dx$$

$$y = \int \frac{dx}{x^2+2x+2} + \int \frac{dx}{x^2+1}$$

$$y = \tan^{-1}(x+1) + \tan^{-1}x + C$$

$$y(-1) = \frac{-\pi}{4}$$

$$\frac{-\pi}{4} = 0 - \frac{\pi}{4} + C \Rightarrow C = 0$$

$$\Rightarrow y = \tan^{-1}(x+1) + \tan^{-1}x$$

$$y(0) = \tan^{-1}1 = \frac{\pi}{4}$$

Hence, the answer is the option (3).

Q. 5 Let the sum of the maximum and the minimum values of the function $f(x) = \frac{2x^2-3x+8}{2x^2+3x+8}$ be $\frac{m}{n}$, where $\gcd(m,n) = 1$. Then m+n is equal to :

Option 1:
182

Option 2:
217

Option 3:
195

Option 4:
201

Correct Answer:
201

Solution:

$$y = \frac{2x^2-3x+8}{2x^2+3x+8}$$

$$x^2(2y-2) + x(3y+3) + 8y - 8 = 0$$

use $D \geq 0$

$$(3y+3)^2 - 4(2y-2)(8y-8) \geq 0$$

$$(11y-5)(5y-11) \leq 0$$

$$\Rightarrow y \in \left[\frac{5}{11}, \frac{11}{5}\right]$$

$y = 1$ is also included

Q. 6 One of the points of intersection of the curves $y = 1 + 3x - 2x^2$ and $y = \frac{1}{x}$ is $\left(\frac{1}{2}, 2\right)$. Let the area of the region enclosed by these curves be $\frac{1}{24}(\ell\sqrt{5} + m) - n\log_e(1 + \sqrt{5})$, where $\ell, m, n \in N$. Then $\ell + m + n$ is equal to

Option 1:
32

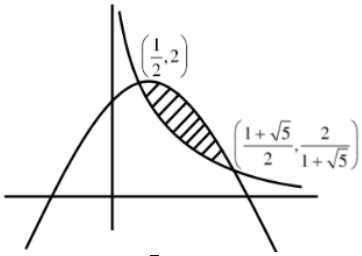
Option 2:
30

Option 3:
29

Option 4:
31

Correct Answer:
30

Solution:



$$A = \int_{\frac{1}{2}}^{\frac{1+\sqrt{5}}{2}} \left(1 + 3x - 2x^2 - \frac{1}{x}\right) dx$$

$$A = \left[x + \frac{3x^2}{2} - \frac{2x^3}{3} - \ln x\right]_{\frac{1}{2}}^{\frac{1+\sqrt{5}}{2}}$$

$$A = \frac{1+\sqrt{5}}{2} + \frac{3}{2}\left(\frac{1+\sqrt{5}}{2}\right)^2 - \frac{2}{3}\left(\frac{1+\sqrt{5}}{2}\right)^3 - \ln\left(\frac{1+\sqrt{5}}{2}\right)$$
$$- \frac{1}{2} - \frac{3}{2}\left(\frac{1}{4}\right) + \frac{2}{3}\left(\frac{1}{8}\right) + \ln\left(\frac{1}{2}\right)$$

$$A = \frac{1}{2} + \frac{\sqrt{5}}{2} + \frac{3}{8} + \frac{3}{4}\sqrt{5} + \frac{15}{8} - \frac{4}{3} - \frac{2}{3}\sqrt{5}$$
$$- \frac{1}{2} - \frac{3}{8} + \frac{1}{12} - \ln(1 + \sqrt{5})$$
$$= \sqrt{5}\left(\frac{1}{2} + \frac{3}{4} - \frac{2}{3}\right) + \frac{15}{8} - \frac{4}{3} + \frac{1}{12} - \ln(1 + \sqrt{5})$$
$$= \frac{14}{24}\sqrt{5} + \frac{15}{24} - \ln(1 + \sqrt{5})$$

Q. 7 If the system of equations
$x + (\sqrt{2}\sin\alpha)y + (\sqrt{2}\cos\alpha)z = 0$
$x + (\cos\alpha)y + (\sin\alpha)z = 0$
$x + (\sin\alpha)y - (\cos\alpha)z = 0$
has a non-trivial solution, then $\alpha \in \left(0, \frac{\pi}{2}\right)$ is equal to :

Option 1:
$\frac{3\pi}{4}$

Option 2:
$\frac{7\pi}{24}$

Option 3:
$\frac{5\pi}{24}$

Option 4:
$\frac{11\pi}{24}$

Correct Answer:
$\frac{5\pi}{24}$

Solution:

$$\begin{vmatrix} 1 & \sqrt{2}\sin\alpha & \sqrt{2}\cos\alpha \\ 1 & \sin\alpha & -\cos\alpha \\ 1 & \cos\alpha & \sin\alpha \end{vmatrix} = 0$$

$$\Rightarrow 1 - \sqrt{2}\sin\alpha(\sin\alpha + \cos\alpha) + \sqrt{2}\cos\alpha(\cos\alpha - \sin\alpha) = 0$$

$$\Rightarrow 1 + \sqrt{2}\cos 2\alpha - \sqrt{2}\sin 2\alpha = 0$$

$$\cos 2\alpha - \sin 2\alpha = -\frac{1}{\sqrt{2}}$$

$$\cos\left(2\alpha + \frac{\pi}{4}\right) = -\frac{1}{2}$$

$$2\alpha + \frac{\pi}{4} = 2n\pi \pm \frac{2\pi}{3}$$

$$\alpha + \frac{\pi}{8} = n\pi \pm \frac{\pi}{3}$$

$$n = 0$$

$$x = \frac{\pi}{3} - \frac{\pi}{8} = \frac{5\pi}{24}$$

Hence, the answer is the option (3).

Q. 8 There are 5 points P_1, P_2, P_3, P_4, P_5 on the side AB, excluding A and B, of a triangle ABC. Similarly there are 6 points $P_6, P_7, \ldots, P_{11}$ on the side BC and 7 points $P_{12}, P_{13}, \ldots, P_{18}$ on the side CA of the triangle. The number of triangles, that can be formed using the points $P_1, P_2, \ldots, P_{18}$ as vertices, is :

Option 1:
776

Option 2:
751

Option 3:
796

Option 4:
771

Correct Answer:
751

Solution:
$^{18}C_3 - {}^5C_3 - {}^6C_3 - {}^7C_3$
$= 751$

Hence, the answer is the option (2).

Q. 9 Let $f(x) = \begin{cases} -2, & -2 \le x \le 0 \\ x-2, & 0 < x \le 2 \end{cases}$ and $h(x) = f(|x|) + |f(x)|$. Then $\int_{-2}^{2} \mathrm{h(x)dx}$ is equal to :

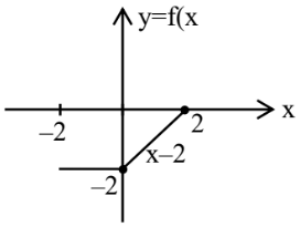
Option 1:
2

Option 2:
4

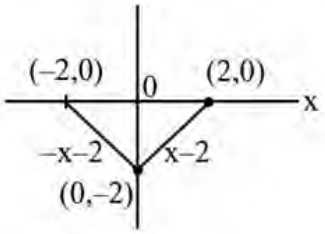
Option 3:
1

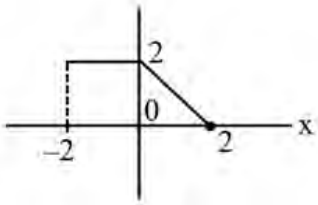
Option 4:
6

Correct Answer:
2

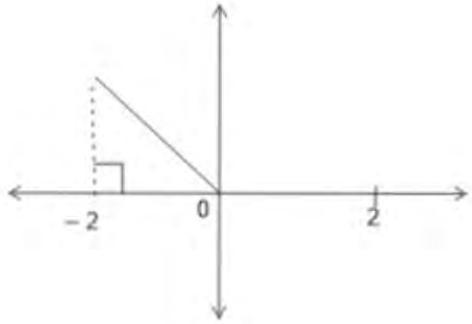
Solution:


$f(|x|) \rightarrow$ $|f(x)|$





$$h(x)=\begin{cases}x-2+2-x=0, & 0\le x\le 2\\ -x-2+2=-x & -2\le x<0\end{cases}$$



$\Rightarrow \int_0^2 h(x)dx=0$ and $\int_{-2}^0 h(x)dx=2$

Q. 10 The sum of all rational terms in the expansion of $\left(2^{\frac{1}{5}}+5^{\frac{1}{3}}\right)^{15}$ is equal to :

Option 1:
3133

Option 2:
633

Option 3:
931

Option 4:
6131

Correct Answer:
3133

Solution:

$T_{r+1}={}^{15}C_r\left(5^{\frac{1}{3}}\right)^r\left(2^{\frac{1}{5}}\right)^{15-r}$

$={}^{15}C_r 5^{\frac{r}{3}}\cdot 2^{\frac{15-r}{5}}$

$R=3\lambda, 15\mu$

$\Rightarrow r=0,15$

2 rational terms

$\Rightarrow {}^{15}C_0 2^3+{}^{15}C_{15}(5)^5$

$=8+3125=3133$

Hence, the answer is the option (1).

Q. 11 Let a unit vector which makes an angle of 60° with $2\hat{i}+2\hat{j}-\hat{k}$ and an angle of 45° with $\hat{i}-\hat{k}$ be $\vec{C}$. Then $\overrightarrow{C}+\left(-\frac{1}{2}\hat{i}+\frac{1}{3\sqrt{2}}\hat{j}-\frac{\sqrt{2}}{3}\hat{k}\right)$ is :

Option 1:
$-\frac{\sqrt{2}}{3}\hat{i}+\frac{\sqrt{2}}{3}\hat{j}+\left(\frac{1}{2}+\frac{2\sqrt{2}}{3}\right)\hat{k}$

Option 2:
$\frac{\sqrt{2}}{3}\hat{i}+\frac{1}{3\sqrt{2}}\hat{j}-\frac{1}{2}\hat{k}$

Option 3:
$\left(\frac{1}{\sqrt{3}}+\frac{1}{2}\right)\hat{i}+\left(\frac{1}{\sqrt{3}}-\frac{1}{3\sqrt{2}}\right)\hat{j}+\left(\frac{1}{\sqrt{3}}+\frac{\sqrt{2}}{3}\right)\hat{k}$

Option 4:
$\frac{\sqrt{2}}{3}\hat{i}-\frac{1}{2}\hat{k}$

Correct Answer:
$\frac{\sqrt{2}}{3}\hat{i}-\frac{1}{2}\hat{k}$

Solution:

$$\overrightarrow{C}=C_1\hat{i}+C_2\hat{j}+C_3\hat{k}$$
$$C_1^2+C_2^2+C_3^2=1$$
$$\overrightarrow{C}\cdot(2\hat{i}+2\hat{j}-\hat{k})=|C|\sqrt{9}\cos 60^\circ$$
$$2C_1+2C_2-C_3=\frac{3}{2}$$
$$C_1-C_3=1$$
$$C_1+2C_2=\frac{1}{2}$$
$$C_1=\frac{\sqrt{2}}{3}+\frac{1}{2}$$
$$C_2=\frac{-1}{3\sqrt{2}}$$
$$C_3=\frac{\sqrt{2}}{3}-\frac{1}{2}$$

Q. 12 Let the first three terms $2, p$ and q, with $q\neq 2$, of a G.P. be respectively the 7^{th}, 8^{th} and 13^{th} terms of an A.P. If the 5^{th} term of the G.P. is the n^{th} term of the A.P., then n is equal to

Option 1:
151

Option 2:
169

Option 3:
177

Option 4:
163

Correct Answer:
163

Solution:
$p^2 = 2q$
$2 = a + 6d \quad \ldots (i)$
$p = a + 7d \quad \ldots (\text{ii})$
$q = a + 12d \quad \ldots (\text{iii})$
$p - 2 = d \quad ((ii) - (i))$
$q - p = 5d \quad ((iii) - (\text{ii}))$
$q - p = 5(p - 2)$
$q = 6p - 10$
$p^2 = 2(6p - 10)$
$p^2 - 12p + 20 = 0$
$p = 10, 2$
$p = 10; q = 50$
$d = 8$
$a = -46$
$2, 10, 50, 250, 1250$
$a^4 = a + (n - 1)d$
$1250 = -46 + (n - 1)8$
$n = 163$
Hence, the answer is the option (4).

Q. 13 Let $a, b \in R$. Let the mean and the variance of 6 observations -3,4,7,-6, a, b be 2 and 23 , respectively. The mean deviation about the mean of these 6 observations is :

Option 1:
$\frac{13}{3}$

Option 2:
$\frac{16}{3}$

Option 3:
$\frac{11}{3}$

Option 4:
$\frac{14}{3}$

Correct Answer:
$\frac{13}{3}$

Solution:
$\frac{\sum x_i}{6} = 2$ and $\frac{\sum x_i^2}{N} - \mu^2 = 23$
$\alpha + \beta = 10$
$\alpha^2 + \beta^2 = 52$
solving we get $\alpha = 4, \beta = 6$
$\frac{\sum|x_i - \bar{x}|}{6} = \frac{5+2+5+8+2+4}{6} = \frac{13}{3}$
Hence, the answer is the option (1).

Q. 14 If 2 and 6 are the roots of the equation $ax^2 + bx + 1 = 0$, then the quadratic equation, whose roots are $\frac{1}{2a+b}$ and $\frac{1}{6a+b}$, is:

Option 1:
$2x^2 + 11x + 12 = 0$

Option 2:
$4x^2 + 14x + 12 = 0$

Option 3:
$x^2 + 10x + 16 = 0$

Option 4:
$x^2 + 8x + 12 = 0$

Correct Answer:
$x^2 + 8x + 12 = 0$

Solution:

$\text{Sum } = 8 = -\frac{b}{a}$

$\text{Product } = 12 = \frac{1}{a} \quad \Rightarrow a = \frac{1}{12}$

$b = -\frac{2}{3}$

$2a + b = \frac{2}{12} - \frac{2}{3} = -\frac{1}{2}$

$6a + b = \frac{6}{12} - \frac{2}{3} = -\frac{1}{6}$

$\text{sum } = -8$

$P = 12$

$x^2 + 8x + 12 = 0$

Hence, the answer is the option (4).

Q. 15 Let α and β be the sum and the product of all the non-zero solutions of the equation $(\bar{z})^2 + |z| = 0, z \in C$. Then $4\left(\alpha^2 + \beta^2\right)$ is equal to:

Option 1:
6

Option 2:
4

Option 3:
8

Option 4:
2

Correct Answer:
4

Solution:

$z = x + iy$

$\bar{z} = x - iy$

$\bar{z}^2 = x^2 - y^2 - 2ixy$

$\Rightarrow x^2 - y^2 - 2ixy + \sqrt{x^2 + y^2} = 0$

$\Rightarrow x = 0 \quad \text{or} \quad y = 0$

$-y^2 + |y| = 0$ | $x^2 + |x| = 0$

$|y| = |y|^2$ | $\Rightarrow x = 0$

$y = 0, \pm 1$ | $\Rightarrow \alpha = i - i = 0$

$\Rightarrow i, -i$ | $\beta = i(-i) = 1$

The roots are | $4(0 + 1) = 4$

Hence, the answer is the option (2).

Q. 16 Let the point, on the line passing through the points P(1,-2,3) and Q(5,-4,7), farther from the origin and at a distance of 9 units from the point P, be (α, β, γ). Then $\alpha^2 + \beta^2 + \gamma^2$ is equal to :

Option 1:
155

Option 2:
150

Option 3:
160

Option 4:
165

Correct Answer:
155

Solution:
PQ line

$$\frac{x-1}{4} = \frac{y+2}{-2} = \frac{z-3}{4}$$

$$\text{pt } (4t+1, -2t-2, 4t+3)$$

$$\text{distance }^2 = 16t^2 + 4t^2 + 16t^2 = 81$$

$$t = \pm\frac{3}{2}$$

$$\text{pt}(7, -5, 9)$$

$$\alpha^2 + \beta^2 + \gamma^2 = 155$$

$$\text{option } (1)$$

Q. 17 A square is inscribed in the circle $x^2 + y^2 - 10x - 6y + 30 = 0$. One side of this square is parallel to $y = x + 3$. If (x_i, y_i) are the vertices of the square, then $\sum \left(x_i^2 + y_i^2\right)$ is equal to:

Option 1:
148

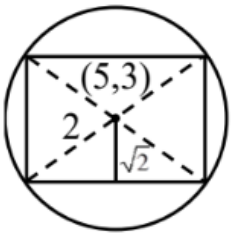
Option 2:
156

Option 3:
160

Option 4:
152

Correct Answer:
152

Solution:



$y = x + c \quad \&$

$\frac{5-3+c}{\sqrt{2}} = \sqrt{2}$

$|c + 2| = 2$

$c = 0, -4$

$x + y + d = 0$

$\frac{8+d}{\sqrt{2}} = \sqrt{2}$

$8 + d = \pm 2$

$d = -10, -6$

$\text{pts}(5, 5), (3, 3), (7, 3), (5, 1)$

$\sum \left(x_i^2 + y_1^2\right) = 25 + 25 + 9 + 9 + 49 + 9 + 25 + 1$

$= 152$

Q. 18 If the domain of the function $\sin^{-1}\left(\frac{3x-22}{2x-19}\right) + \log_e\left(\frac{3x^2-8x+5}{x^2-3x-10}\right)$ is $(\alpha, \beta]$, then $3\alpha + 10\beta$ is equal to :

Option 1:
97

Option 2:
100

Option 3:
95

Option 4:
98

Correct Answer:
97

Solution:

$-1 \leq \frac{3x-22}{2x-19} \leq 1 \quad \frac{3x^2-8x+5}{x^2-3x-10} > 0$

$x \in \left(5, \frac{41}{5}\right]$

$3\alpha + 10\beta = 97$

Q. 19 Let $f(x)=x^5+2e^{x/4}$ for all $x \in R$. Consider a function $g(x)$ such that (gof) $(x)=x$ for all $x \in R$. Then the value of $8g'(2)$ is :

Option 1:
16

Option 2:
4

Option 3:
8

Option 4:
2

Correct Answer:
16

Solution:

$f(x)=2$

when $x=0$

$\because g'(f(x))f'(x)=1$

$g'(2)=\dfrac{1}{f'(0)}$

$\because f'(x)=5x^4+\dfrac{2}{4}e^{x/4}$

$g'(2)=2$

$\text{Ans } =16$

Q. 20 Let $\alpha \in (0,\infty)$ and $A=\begin{bmatrix}1 & 2 & \alpha \\ 1 & 0 & 1 \\ 0 & 1 & 2\end{bmatrix}$. If $\det\left(\text{adj}\left(2A-A^T\right)\cdot \text{adj}\left(A-2A^T\right)\right)=2^8$, then $(\det(A))^2$ is equal to :

Option 1:
1

Option 2:
49

Option 3:
16

Option 4:
36

Correct Answer:
16

Solution:

$$\text{adj}\left(\text{A}-2\,\text{A}^{\text{T}}\right)\left(2\,\text{A}-\text{A}^{\text{T}}\right)=28$$

$$\left(\text{A}-2\,\text{A}^{\text{T}}\right)\left(2\,\text{A}-\text{A}^{\text{T}}\right)=24$$

$$\text{A}-2\,\text{A}^{\text{T}}\;\;2\,\text{A}-\text{A}^{\text{T}}=\pm 16$$

$$\left(A-2A^{T}\right)^{T}=A^{T}-2A$$

$$\text{A}-2\,\text{A}^{\text{T}}=\text{A}^{\text{T}}-2\,\text{A}$$

$$\Rightarrow\;\text{A}-2\,\text{A}^{\text{T}\;2}=16$$

$$\text{A}-2\,\text{A}^{\text{T}}=\pm 4$$

$$\begin{bmatrix}1&2&\alpha\\1&0&1\\0&1&2\end{bmatrix}-\begin{bmatrix}2&2&0\\4&0&2\\2\alpha&2&4\end{bmatrix}$$

$$\begin{matrix}-1&0&\alpha\\-3&0&-1\\-2\alpha&-1&-2\end{matrix}$$

$$1+3\alpha=4$$

$$3\alpha=3$$

$$\alpha=1$$

$$|A|=\begin{matrix}1&2&1\\1&0&1\\0&1&2\end{matrix}=-1-3=-4$$

$$|A|^{2}=16$$

Q. 21 If $\lim_{x\to 1}\frac{(5x+1)^{1/3}-(x+5)^{1/3}}{(2x+3)^{1/2}-(x+4)^{1/2}}=\frac{m\sqrt{5}}{n(2n)^{2/3}}$, where $\gcd(\text{m},\text{n})=1$, then $8\,\text{m}+12\text{n}$ is equal to_____.

Correct Answer:
100

Solution:

$$\lim_{x\to 1}\frac{\frac{1}{3}(5x+1)^{-2/3}5-\frac{1}{3}(x+5)^{-2/3}}{\frac{1}{2}(2x+3)^{-1/2}\cdot 2-\frac{1}{2}(x+4)^{-1/2}}$$

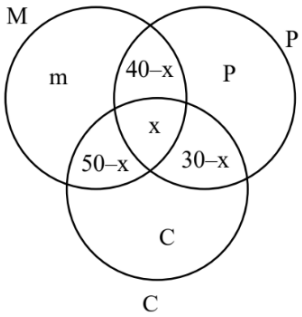
$$=\frac{8}{3}\frac{\sqrt{5}}{6^{2/3}}\begin{matrix}m=8\\n=3\end{matrix}$$

$$8m+12n=100$$

Q. 22 In a survey of 220 students of a higher secondary school, it was found that at least 125 and at most 130 students studied Mathematics; at least 85 and at most 95 studied Physics; at least 75 and at most 90 studied Chemistry; 30 studied both Physics and Chemistry; 50 studied both Chemistry and Mathematics; 40 studied both Mathematics and Physics and 10 studied none of these subjects. Let m and n respectively be the least and the most number of students who studied all the three subjects. Then m+n is equal to_____.

Correct Answer:
45

Solution:



$125 \leq \mathrm{m}+90-\mathrm{x} \leq 130$
$85 \leq \mathrm{P}+70-\mathrm{x} \leq 95$
$75 \leq \mathrm{C}+80-\mathrm{x} \leq 90$
$\mathrm{m}+\mathrm{P}+\mathrm{C}+120-2\mathrm{x}=210$
$\Rightarrow 15 \leq \mathrm{x} \leq 45 \& 30-\mathrm{x} \geq 0$
$\Rightarrow 15 \leq \mathrm{x} \leq 30$
$30+15=45$

Q. 23 Let the solution $y=y(x)$ of the differential equation $\frac{dy}{dx}-y=1+4\sin x$ satisfy $y(\pi)=1$. Then $y\left(\frac{\pi}{2}\right)+10$ is equal to_____.

Correct Answer:
7

Solution:

$$ye^{-x}=\int\left(e^{-x}+4e^{-x}\sin x\right)dx$$
$$ye^{-x}=-e^{-x}-2\left(e^{-x}\sin xe^{-x}\cos x\right)+C$$
$$y=-1-2(\sin x+\cos x)+ce^{x}$$
$$\because y(\pi)=1\Rightarrow c=0$$
$$y(\pi/2)=-1-2=-3$$
$$\text{Ans }=10-3=7$$

Hence, the answer is (7).

Q. 24 If the shortest distance between the lines $\frac{x+2}{2}=\frac{y+3}{3}=\frac{z-5}{4}$ and $\frac{x-3}{1}=\frac{y-2}{-3}=\frac{z+4}{2}$ is $\frac{38}{3\sqrt{5}}k$ and $\int_0^k [x^2]dx=\alpha-\sqrt{\alpha}$, where $[x]$ denotes the greatest integer function, then $6\alpha^3$ is equal to _____.

Correct Answer:
48

Solution:

$$\frac{38}{3\sqrt{5}}\hat{k}=\frac{(5\hat{i}+5\hat{j}-9\hat{k})}{\sqrt{5}}\cdot\begin{matrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 3 & 4\\ 1 & -3 & 2\end{matrix}$$

$$\frac{38}{3\sqrt{5}}\hat{k}=\frac{19}{\sqrt{5}}$$

$$k=\frac{19}{\sqrt{5}}$$

$$k=\frac{3}{2}$$

$$\int_0^{3/2}[x^2]=\int_0^1 0+\int_1^{\sqrt{2}}1+\int_{\sqrt{2}}^{3/2}2$$

$$=\sqrt{2}-1+2\left(\frac{3}{2}-\sqrt{2}\right)$$

$$=2-\sqrt{2}$$

$$\alpha=2$$

$$\Rightarrow 6\alpha^3=48$$

Q. 25 Let A be a square matrix of order 2 such that $|A|=2$ and the sum of its diagonal elements is -3 . If the points (x,y) satisfying $A^2+xA+yI=0$ lie on a hyperbola, whose transverse axis is parallel to the x-axis, eccentricity is e and the length of the latus rectum is ℓ, then $e^4+\ell^4$ is equal to_____.

Correct Answer:
25

Solution:
Given $|A|=2$
trace $A=-3$
and $A^2+xA+yI=0$
$\Rightarrow x=3, y=2$
so, information is incomplete to determine eccentricity of hyperbola (e) and length of latus rectum of hyperbola (ℓ)

Q. 26

$$\text{Let a} = 1 + \frac{{}^2C_2}{3!} + \frac{{}^3C_2}{4!} + \frac{{}^4C_2}{5!} + \ldots$$

$$b = 1 + \frac{{}^1C_0 + {}^1C_1}{1!} + \frac{{}^2C_0 + {}^2C_1 + {}^2C_2}{2!} + \frac{{}^3C_0 + {}^3C_1 + {}^3C_2 + {}^3C_3}{3!} + \ldots$$

Then $\frac{2b}{a^2}$ is equal to_____.

Correct Answer:
8

Solution:

$$f(x) = 1 + \frac{(1+x)}{1!} + \frac{(1+x)^2}{2!} + \frac{(1+x)^3}{3!} + \ldots..$$

$$\frac{e^{(1+x)}}{1+x} = \frac{1}{1+x} + 1 + \frac{(1+x)}{2!} + \frac{(1+x)^2}{3!} + \frac{(1+x)^2}{4!}$$

$$\text{coef } x^2 \text{ in RHS} : 1 + \frac{{}^2C_2}{3} + \frac{{}^3C_2}{4} + \ldots = a$$

$$\text{coeff. } x^2 \text{ in L.H.S.}$$

$$e\left(1 + x + \frac{x^2}{2!}\right) \ldots . \left(1 - x + \frac{x^2}{2!} \ldots..\right)$$

$$\text{is } e - e + \frac{e}{2!} = a$$

$$b = 1 + \frac{2}{1!} + \frac{2^2}{2!} + \frac{2^3}{3!} + \ldots.. = e^2$$

$$\frac{2\,b}{a^2} = 8$$

Hence, the answer is (8).

Q. 27

Let A be a 3×3 matrix of non-negative real elements such that $A \begin{bmatrix}1\\1\\1\end{bmatrix} = 3 \begin{bmatrix}1\\1\\1\end{bmatrix}$. Then the maximum value of $\det(A)$ is_____.

Correct Answer:
27

Solution:

Let $A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$

$A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$\Rightarrow a_1 + a_2 + a_3 = 3$

$\Rightarrow b_1 + b_2 + b_3 = 3$

$\Rightarrow c_1 + a_2 + c_3 = 3$

Now,

$|A| = (a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2) - (a_3b_2c_1 + a_2b_1c_3 + a_1b_3c_2)$

$\therefore$ From above in formation, clearly $|\mathrm{A}|_{\max} = 27$, when $a_1 = 3, b_2 = 3, c_3 = 3$

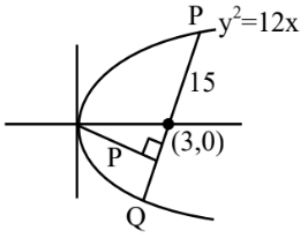
Hence, the answer is (27).

Q. 28 Let the length of the focal chord PQ of the parabola $\mathrm{y}^2 = 12\mathrm{x}$ be 15 units. If the distance of PQ from the origin is p, then $10\mathrm{p}^2$ is equal to_____.

Correct Answer:

72

Solution:



$\text{length of focal chord} = 4a\,\text{cosec}^2\,\theta = 15$

$12\,\text{cosec}^2\,\theta = 15$

$\sin^2\theta = \frac{4}{5}$

$\tan^2\theta = 4$

$\tan\theta = 2$

$\text{equation } \frac{y-0}{x-3} = 2$

$y = 2x - 6$

$2x - y - 6 = 0$

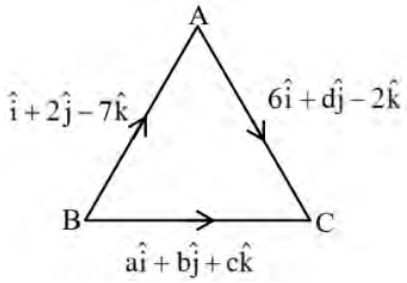
$P = \frac{6}{\sqrt{5}}$

$10p^2 = 10 \cdot \frac{36}{5} = 72$

Q. 29 Let ABC be a triangle of area $15\sqrt{2}$ and the vectors $\overrightarrow{\mathrm{AB}} = \hat{i} + 2\hat{j} - 7\hat{k}, \quad \overrightarrow{BC} = a\hat{i} + b\hat{j} + c\hat{k}$ and $\overrightarrow{\mathrm{AC}} = 6\hat{\mathrm{i}} + \mathrm{d}\hat{\mathrm{j}} - 2\hat{\mathrm{k}}, \mathrm{d} > 0$. Then the square of the length of the largest side of the triangle ABC is

Correct Answer:
54

Solution:



$$\text{Area} = \frac{1}{2}\begin{matrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 2 & -7\\ 6 & \mathrm{d} & -2\end{matrix} = 15\sqrt{2}$$

$(-4+7\,\mathrm{d})\hat{\mathrm{i}} - \hat{\mathrm{j}}(-2+42) + \hat{\mathrm{k}}(\mathrm{d}-12)$

$(7\,\mathrm{d}-4)^2 + (40)^2 + (\mathrm{d}-12)^2 = 1800$

$50\,\mathrm{d}^2 - 80\,\mathrm{d} - 40 = 0$

$5\,\mathrm{d}^2 - 8\,\mathrm{d} - 4 = 0$

$5\,\mathrm{d}^2 - 10\,\mathrm{d} - 2\,\mathrm{d} - 4$

$5\,\mathrm{d}(\,\mathrm{d}-2) + 2(\,\mathrm{d}-2) = 0$

$\mathrm{d} = 2 \text{ or } \mathrm{d} = -\frac{2}{5}$

$\because \mathrm{d} > 0,\ \mathrm{d} = 2$

$(a+1)\hat{i} + (b+2)\hat{j} + (c-7)\hat{k} = 6\hat{i} + 2\hat{j} - 2\hat{k}$

$a+1 = 6 \& b+2 = 2, c-7 = -2$

$a = 5 \quad b = 0 \quad c = 5$

$|AB| = \sqrt{1+4+49} = \sqrt{54}$

$|BC| = \sqrt{25+25} = \sqrt{50}$

$|AC| = \sqrt{86+4+4} = \sqrt{44}$

Q. 30 If $\int_0^{\frac{\pi}{4}} \frac{\sin^2 x}{1+\sin x \cos x}dx = \frac{1}{a}\log_e\left(\frac{a}{3}\right) + \frac{\pi}{b\sqrt{3}}$, where $a, \mathrm{b} \in \mathrm{N}$, then $\mathrm{a}+\mathrm{b}$ is equal to_____.

Correct Answer:
8

Solution:

$$\int_0^{\frac{\pi}{2}} \frac{\sin^2 x}{1+\frac{1}{2}\sin 2x}dx = \int_0^{\frac{\pi}{4}} \frac{1-\cos 2x}{2+\sin 2x}dx$$

$$\int \frac{1}{2+\sin 2x} - \int \frac{\cos 2x}{2+\sin 2x}$$

$(\mathrm{I}_1) - \quad (\mathrm{I}_2)$

$$(\mathrm{I}_1) = \int \frac{\mathrm{dx}}{2+\frac{2\tan \mathrm{x}}{1+\tan^2 \mathrm{x}}}$$

$$\int_0^{\frac{\pi}{4}} \frac{\sec^2 \mathrm{x}\mathrm{dx}}{2\tan^2 \mathrm{x}+2\tan x+2}$$

$$\frac{1}{2}\int_0^1 \frac{dt}{\left(t+\frac{1}{2}\right)^2+\frac{3}{4}} = \frac{\pi}{6\sqrt{3}}$$

$$I_2 = \int_0^{\pi/4} \frac{\cos 2x}{2+\sin 2x}dx = \frac{1}{2}\left(\ln \frac{3}{2}\right)$$

$$I_1 - I_2 = \frac{1}{\sqrt{3}}\frac{\pi}{6} + \frac{1}{2}\ln \frac{2}{3}$$

$\Rightarrow a = 2, b = 6$

The value of a + b = 2 + 6 = 8.

Hence, the answer is (8).

CAREERS360

4 Apr - Evening Shift

Physics

Q. 1 The translational degrees of freedom (f_t) and rotational degrees of freedom ($\mathrm{f_r}$) of $\mathrm{CH_4}$ molecule are :

Option 1:
$\mathrm{f_t} = 2$ and $\mathrm{f_r} = 2$

Option 2:
$\mathrm{f_t} = 3$ and $\mathrm{f_r} = 3$

Option 3:
$\mathrm{f_t} = 3$ and $\mathrm{f_r} = 2$

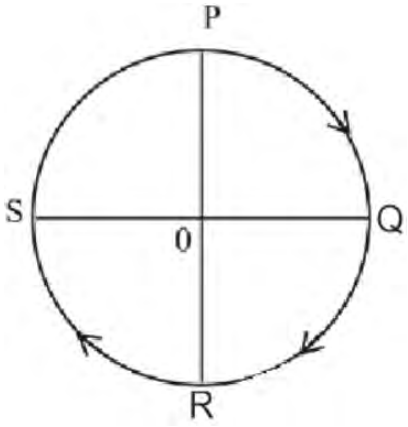
Option 4:
$f_t = 2$ and $f_r = 3$

Correct Answer:
$\mathrm{f_t} = 3$ and $\mathrm{f_r} = 3$

Solution:
Since $\mathrm{CH_4}$ is polyatomic Non-Linear
D.O.F of $\mathrm{CH_4}$
T. $\mathrm{DOF} = 3$
$\mathrm{RDOF} = 3$

CAREERS360

Q. 2 A cyclist starts from the point P of a circular ground of radius $2\ \mathrm{km}$ and travels along its circumference to the point S. The displacement of a cyclist is :



Option 1:
$6\ \mathrm{km}$

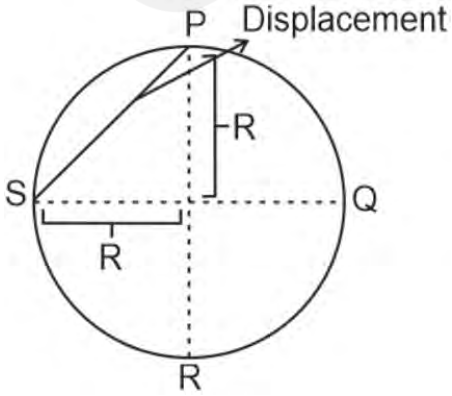
Option 2:
$\sqrt{8}\ \mathrm{km}$

Option 3:
$4\ \mathrm{km}$

Option 4:
$8\ \mathrm{km}$

Correct Answer:
$\sqrt{8}\ \mathrm{km}$

Solution:



$\therefore$ Displacement $= \mathrm{R}\sqrt{2} = 2\sqrt{2} = \sqrt{8}\,\mathrm{km}$

Q. 3 The magnetic moment of a bar magnet is $0.5\mathrm{Am}^2$. It is suspended in a uniform magnetic field of $8 \times 10^{-2}\ \mathrm{T}$. The work done in rotating it from its most stable to most unstable position is :

Option 1:
$16 \times 10^{-2}\ \mathrm{J}$

Option 2:
$8 \times 10^{-2}\ \mathrm{J}$

Option 3:
$4 \times 10^{-2}\ \mathrm{J}$

Option 4:
Zero

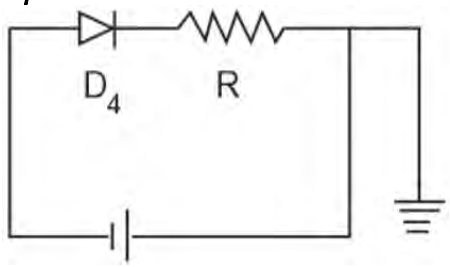
Correct Answer:
$8 \times 10^{-2}\ \mathrm{J}$

Solution:
At stable equilibrium
$\mathrm{U} = -\mathrm{mB}\cos 0^\circ = -\mathrm{mB\$}$

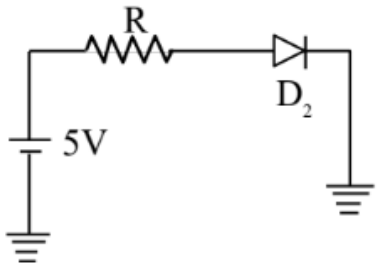
At unstable equilibrium
$\mathrm{U} = -\mathrm{mB}\cos 180^\circ = +\mathrm{mB}$
$\mathrm{W} = \Delta\mathrm{U}$
$\mathrm{W.D.} = 2\mathrm{mB}$
$= 2(0.5)8 \times 10^{-2} = 8 \times 10^{-2}\ \mathrm{J}$

Q. 4 Which of the diode circuit shows correct biasing used for the measurement of dynamic resistance of p-n junction diode :

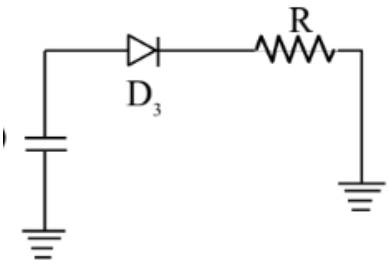
Option 1:



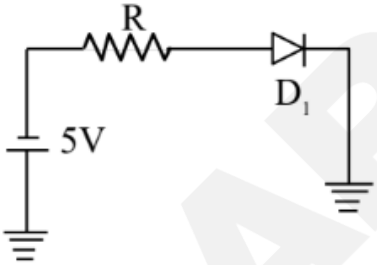
Option 2:



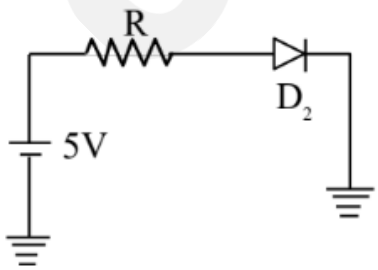
Option 3:



Option 4:



Correct Answer:



Solution:

Diode should be in forward biased to calculate dynamic resistance

Hence correct answer would be 2 .

Q. 5 Arrange the following in the ascending order wavelength :
(A) Gamma rays (λ_1)
(B) x-ray (λ_2)
(C) Infrared waves (λ_3)
(D) Microwaves (λ_4)

Choose the most appropriate answer from th options given below :

Option 1:
$\lambda_4 < \lambda_3 < \lambda_1 < \lambda_2$

Option 2:
$\lambda_4 < \lambda_3 < \lambda_2 < \lambda_1$

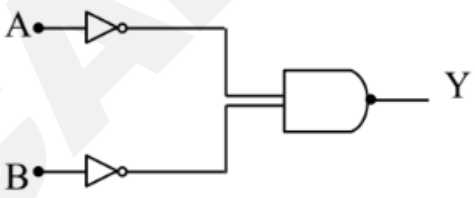
Option 3:
$\lambda_1 < \lambda_2 < \lambda_3 < \lambda_4$

Option 4:
$\lambda_2 < \lambda_1 < \lambda_4 < \lambda_3$

Correct Answer:
$\lambda_1 < \lambda_2 < \lambda_3 < \lambda_4$

Solution:
$\lambda_1 < \lambda_2 < \lambda_3 < \lambda_4$

Q. 6 Identify the logic gate given in the circuit :



Option 1:
NAND - gate

Option 2:
OR - gate

Option 3:
AND gate

Option 4:
NOR gate

Correct Answer:
OR - gate

Solution:
$Y = \overline{\overline{A} \cdot \overline{B}}$$

By De-Morgan Law
$Y = \overline{\overline{A + B}}_{\$}$
$Y = A + B$

Hence OR gate

Q. 7 The width of one of the two slits in a Young's double slit experiment is 4 times that of the other slit. The ratio of the maximum of the minimum intensity in the interference pattern is :

Option 1:
9: 1

Option 2:
16: 1

Option 3:
1: 1

Option 4:
$4 : 1$

Correct Answer:
9: 1

Solution:

Since, Intensity $\propto$ width of slit (ω)

so, $I_1 = I, I_2 = 4I$

$$I_{\min} = \left(\sqrt{I_1} - \sqrt{I_2}\right)^2 = I$$

$$I_{\max} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2 = 9I^{\$}$$

$$\frac{I_{\max}}{I_{\min}} = \frac{9I}{I} = \frac{9}{1}$$

Q. 8 Correct formula for height of a satellite from earths surface is :

Option 1:

$\left(\frac{T^2R^2g}{4\pi}\right)^{1/2} - R$

Option 2:

$\left(\frac{T^2R^2g}{4\pi^2}\right)^{1/3} - R$

Option 3:

$\left(\frac{T^2R^2}{4\pi^2g}\right)^{1/3} - R$

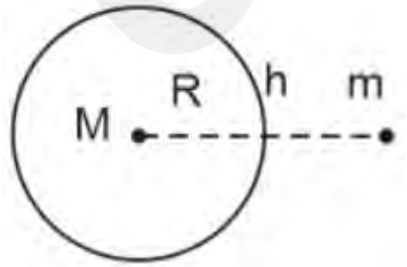
Option 4:

$\left(\frac{T^2R^2}{4\pi^2}\right)^{-1/3} + R$

Correct Answer:

$\left(\frac{T^2R^2g}{4\pi^2}\right)^{1/3} - R$

Solution:



$$\Rightarrow \frac{GMm}{(R+h)^2} = \frac{mv^2}{(R+h)}$$

$$\Rightarrow \frac{GM}{(R+h)} = v^2 \ldots (1)$$

$$\Rightarrow v = (R + h)\omega$$

$$\Rightarrow v = (R + h)\frac{2\pi}{T} \ldots\ldots\ldots (2)$$

$$\Rightarrow \frac{GM}{R^2} = g$$

$$\Rightarrow GM = gR^2 \ldots\ldots\ldots (3)$$

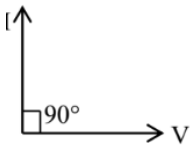
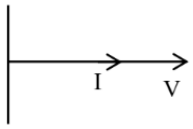
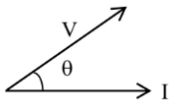
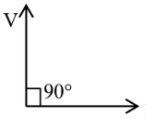
Put value from (2) \& (3) in eq. (1)

$$\Rightarrow \frac{gR^2}{(R+h)} = (R+h)^2\left(\frac{2\pi}{T}\right)^2$$

$$\Rightarrow \frac{T^2R^2g}{(2\pi)^2} = (R+h)^3 \quad \$$$

$$\Rightarrow \left[\frac{T^2R^2g}{(2\pi)^2}\right]^{1/3} - R = h$$

Q. 9 Match List I with List II

	List -I		List -II
A	Purely capacitive circuit	I	
B	Purely inductive circuit	II	
C	LCR series at resonance	III	
D	LCR series circuit	IV	

Choose the correct answer from the options given below:

Option 1:

A-I, B-IV, C-III, D-II

Option 2:
A-IV, B-I, C-III, D-II

Option 3:
A-IV, B-I, C-II, D-III

Option 4:
A-I, B-IV, C-II, D-III

Correct Answer:
A-I, B-IV, C-II, D-III

Solution:
A - V lags by 90° from I hence option (I) is correct.

B - V lead by 90° from I hence option (IV) is correct

C - In LCR resonance $X_L = X_C$. Hence circuit is purely resistive so option (II) is correct

D - In LCR series V is at some angle from I hence
(III) is correct

Hence option (4) is correct.

Q. 10 Given below are two statements :
Statement I : The contact angle between a solid and a liquid is a property of the material of the solid and liquid as well.
Statement II : The rise of a liquid in a capillary tube does not depend on the inner radius of the tube.
In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are false.

Option 2:
Statement I is false but Statement II is true.

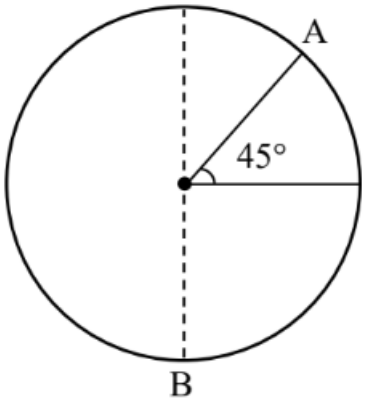
Option 3:
Statement I is true but Statement II is false.

Option 4:
Both Statement I and Statement II are true.

Correct Answer:
Statement I is true but Statement II is false.

Solution:
Statement I is correct as we know contact angle depends on cohesine and adhesive forces.
Statement II is incorrect because height of liquid is given by $h == \frac{2T\cos\theta_C}{\rho g r}$ where r is radius of Tube (assuming length of capillary is sufficient) Hence option (3) is correct.

Q. 11 A body of mkg slides from rest along the curve of vertical circle from point A to B in friction less path. The velocity of the body at B is :



(Vertical Circle)

(given, $\mathrm{R} = 14\ \mathrm{m},\ \mathrm{g} = 10\ \mathrm{m/s^2}$ and $\sqrt{2} = 1.4$)

Option 1:
$19.8\ \mathrm{m/s}$

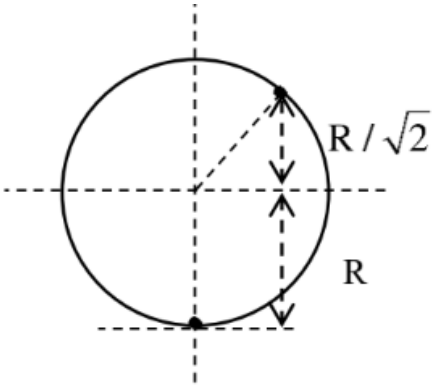
Option 2:
$21.9\ \mathrm{m/s}$

Option 3:
$16.7\ \mathrm{m/s}$

Option 4:
$10.6\ \mathrm{m/s}$

Correct Answer:
$21.9\ \mathrm{m/s}$

Solution:



Apply W.E.T. from A to B

$$\Rightarrow W_{mg} = K_B - K_A$$

$$\Rightarrow mg \times \left(\frac{R}{\sqrt{2}} + R\right) = \frac{1}{2}mv_B^2 - 0 \{v_A = 0 \text{ rest }\}$$

$$\Rightarrow mgR\frac{(\sqrt{2}+1)}{\sqrt{2}} = \frac{1}{2}mv_B^2$$

$$\Rightarrow \sqrt{gR\frac{2(\sqrt{2}+1)}{\sqrt{2}}} = v_B$$

$$\Rightarrow \sqrt{\frac{10 \times 14 \times 2(2.4)}{1.4}} = v_B$$

$$\Rightarrow 21.9 = v_B$$

Hence option (2) is correct

Q. 12 An electric bulb rated $50\ W - 200\ V$ is connected across a $100\ V$ supply. The power dissipation of the bulb is :

Option 1:
$12.5\ W$

Option 2:
$25\ W$

Option 3:
$50\ W$

Option 4:
$100\ W$

Correct Answer:
12.5 W

Solution:
Rated power \& voltage gives resistance

$$R = \frac{V^2}{P} = \frac{(200)^2}{50} = \frac{40000}{50}$$

$$R = 800$$

$$P = \frac{(V_{applied})^2}{R} = \frac{(100)^2}{800}$$

$$P = 12.5 \text{ watt}$$

Hence option 1 is correct.

Q. 13 A $2\ kg$ brick begins to slide over a surface which is inclined at an angle of 45° with respect to horizontal axis. The co-efficient of static friction between their surfaces is :

Option 1:
1

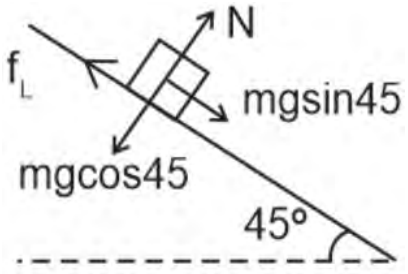
Option 2:
$\frac{1}{\sqrt{3}}$

Option 3:
0.5

Option 4:
1.7

Correct Answer:
1

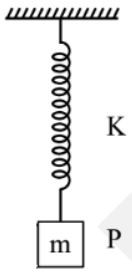
Solution:



$mg\sin 45 = f_L$
$mg\cos 45 = N$
$f_L = \mu_s N$
$\mu_s = \tan 45 = 1$

or
$\tan\theta = \mu_s (\theta \text{ is angle of repose })$
$\tan 45 = \mu_s = 1$ $
correct option (1)

Q. 14 In simple harmonic motion, the total mechanical energy of given system is E. If mass of oscillating particle P is doubled then the new energy of the system for same amplitude is :



Option 1:
$\frac{E}{\sqrt{2}}$

Option 2:
E

Option 3:
$E\sqrt{2}$

Option 4:
$2E$

Correct Answer:
E

Solution:
$T.E. = \frac{1}{2}kA^2\$$
since A is same T.E. will be same

correct option (2)

Q. 15 Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R.**
Assertion A : Number of photons increases with increase in frequency of light.
Reason R : Maximum kinetic energy of emitted electrons increases with the frequency of incident radiation.
In the light of the above statements, choose the most appropriate answer from the options given below :

Option 1:
Both $\mathbf{A}$ and $\mathbf{R}$ are correct and $\mathbf{R}$ is NOT the orrect explanation of $\mathbf{A}$.

Option 2:
$\mathbf{A}$ is correct but $\mathbf{R}$ is not correct.

Option 3:
Both $\mathbf{A}$ and $\mathbf{R}$ are correct and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$.

Option 4:
$\mathbf{A}$ is not correct but $\mathbf{R}$ is correct.

Correct Answer:
$\mathbf{A}$ is not correct but $\mathbf{R}$ is correct.

Solution:
Intensity of light $I = \frac{nh\nu}{A}$
Here n is no. of photons per unit time.
$n = \frac{IA}{h\nu}$ so on increasing frequency ν, n decreases taking intensity constant.

$k_{max} = h\nu - \phi\$$

So on increasing v, kinetic energy increases.

Q. 16 According to Bohr's theory, the moment of momentum of an electron revolving in 4^{th} orbit of hydrogen atom is :

Option 1:
$8\frac{h}{\pi}$

Option 2:
$\frac{h}{\pi}$

Option 3:
$2\frac{h}{\pi}$

Option 4:
$\frac{h}{2\pi}$

Correct Answer:
$2\frac{h}{\pi}$

Solution:

Moment of momentum is $\vec{r} \times \vec{P}$

$\vec{L} = \vec{r} \times m\vec{v}$

$L = mvr = \frac{nh}{2\pi} = \frac{4\,h}{2\pi} = \frac{2\,h}{\pi}$

Q. 17 A sample of gas at temperature T is adiabatically expanded to double its volume. Adiabatic constant for the gas is $\gamma = 3/2$. The work done by the gas in the process is: $(\mu = 1$ mole $)$

Option 1:

$\mathrm{RT}[\sqrt{2}-2]$

$\mathrm{RT}[1-2\sqrt{2}]$

$\mathrm{RT}[2\sqrt{2}-1]$

$\mathrm{RT}[2-\sqrt{2}]$

Option 2:

$\mathrm{RT}[1-2\sqrt{2}]$

$\mathrm{RT}[2\sqrt{2}-1]$

$\mathrm{RT}[2-\sqrt{2}]$

Option 3:

$\mathrm{RT}[2\sqrt{2}-1]$

$\mathrm{RT}[2-\sqrt{2}]$

Option 4:

$\mathrm{RT}[2-\sqrt{2}]$

Correct Answer:

$\mathrm{RT}[2-\sqrt{2}]$

Solution:

$$\mathrm{W}=\frac{\mathrm{nR}\Delta\mathrm{T}}{1-\gamma}$$

$$\mathrm{TV}^{\gamma-1}=\text{ constant }=\mathrm{T_f}(2\ \mathrm{V})^{\gamma-1}$$

$$\mathrm{T_f}=\mathrm{T}\left(\frac{1}{2}\right)^{1/2}=\frac{\mathrm{T}}{\sqrt{2}}$$

$$\mathrm{W}=\frac{\mathrm{R}\left(\frac{\mathrm{T}}{\sqrt{2}}-\mathrm{T}\right)}{1-\frac{3}{2}}=2\mathrm{RT}\frac{(\sqrt{2}-1)}{\sqrt{2}}$$

$$=\mathrm{RT}(2-\sqrt{2})$$

Q. 18 A charge q is placed at the center of one of the surface of a cube. The flux linked with the cube is :-

Option 1:

$\frac{q}{4\epsilon_0}$

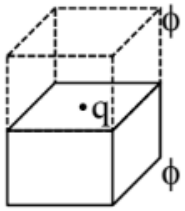
Option 2:
$\frac{q}{2\epsilon_0}$

Option 3:
$\frac{q}{8\epsilon_0}$

Option 4:
Zero

Correct Answer:
$\frac{q}{2\epsilon_0}$

Solution:

From 

$$2\phi = \frac{q}{\epsilon_0}$$
$$\phi = \frac{q}{2\epsilon_0}$$

Q. 19 Applying the principle of homogeneity of dimensions, determine which one is correct. where T is time period, G is gravitational constant,

Option 1:
$T^2 = \frac{4\pi^2 r}{GM^2}$

Option 2:
$T^2 = 4\pi^2 r^3$

Option 3:
$T^2 = \frac{4\pi^2 r^3}{GM}$

Option 4:
$T^2 = \frac{4\pi^2 r^2}{GM}$

Correct Answer:
$T^2 = \frac{4\pi^2 r^3}{GM}$

Solution:
According to principle of homogeneity dimension of LHS should be equal to dimensions of RHS so option (3) is correct.

$$T^2 = \frac{4\pi^2 r^3}{GM}$$

$$[T^2] = \frac{[L^3]}{[M^{-1} L^3 T^{-2}][M]}$$

(Dimension of G is $[M^{-1} L^3 T^{-2}]$)

$$[T^2] = \frac{[L^3]}{[L^3 T^{-2}]} = [T^2]$$

Q. 20 A $90\ kg$ body placed at $2R$ distance from surface of earth experiences gravitational pull of : (R = Radius of earth, $g = 10\ ms^{-2}$)

Option 1:
$300\ N$

Option 2:
$225\ N$

Option 3:
$120\ N$

Option 4:
$100\ N$

Correct Answer:
$100\ N$

Solution:

Value of $g = g_s\left(1+\frac{h}{R}\right)^{-2}$
$= g_s(1+2)^{-2} = \frac{g_s}{9}$$

Here g_s = gravitational acceleration at surface
$\text{Force } = mg = 90 \times \frac{g_s}{0} = 100\ \text{N}$

Q. 21 The displacement of a particle executing SHM is given by $x = 10\sin\left(\omega t + \frac{\pi}{3}\right)\text{m}$. The time period of motion is $3.14\ \text{s}$. The velocity of the particle at $t = 0$ is $\quad\ \text{m/s}$.

Correct Answer:
10

Solution:
Given,
$T = 3.14 = \frac{2\pi}{\omega}$
$\omega = 2\text{rad/s}$
$x = 10\sin\left(\omega t + \frac{\pi}{3}\right)$

$v = \frac{dx}{dt} = 10\omega\cos\left(\omega t + \frac{\pi}{3}\right)$$

at $t = 0$

$v = 10\omega\cos\left(\frac{\pi}{3}\right) = 10 \times 2 \times \frac{1}{2}[\text{ using } \omega = 2\text{rad/s}]$
$v = 10\ \text{m/s}$

Q. 22 A bus moving along a straight highway with speed of $72\ \text{km/h}$ is brought to halt within $4\ \text{s}$ after applying the brakes. The distance travelled by the bus during this time (Assume the retardation is uniform) is m.

Correct Answer:
40

Solution:

$\text{Initial velocity } = u = 72 \text{ km/h} = 20 \text{ m/s}$

$\text{v} = \text{u} + \text{at}$

$\Rightarrow 0 = 20 + \text{a} \times 4$

$\text{a} = -5 \text{ m/s}^2$

$\text{v}^2 - \text{u}^2 = 2 \text{ as}$

$\Rightarrow 0^2 - 20^2 = 2(-5).\text{s}$

$\text{s} = 40 \text{ m}$

Q. 23 A parallel plate capacitor of capacitance 12.5pF is charged by a battery connected between its plates to potential difference of 12.0 V. The battery is now disconnected and a dielectric slab $(\epsilon_\text{r} = 6)$ is inserted between the plates. The change in its potential energy after inserting the dielectric slab is $\times 10^{-12} \text{ J}$.

Correct Answer:
750

Solution:
Before inserting dielectric capacitance is given $\text{C}_0 = 12.5\text{pF}$ and charge on the capacitor $\text{Q} = \text{C}_0 \text{ V}$ After inserting dielectric capacitance will become $\epsilon_\text{r}\text{C}_0$.
Change in potential energy of the capacitor

$$= \text{E}_\text{i} - \text{E}_\text{f}$$

$$= \frac{\text{Q}^2}{2\text{C}_\text{i}} - \frac{\text{Q}^2}{2\text{C}_\text{f}} = \frac{\text{Q}^2}{2\text{C}_0}\left[1 - \frac{1}{\epsilon_\text{r}}\right]$$

$$= \frac{(\text{C}_0 \text{ V})^2}{2\text{C}_0}\left[1 - \frac{1}{\epsilon_\text{r}}\right] = \frac{1}{2}\text{C}_0 \text{ V}^2\left[1 - \frac{1}{\epsilon_\text{r}}\right]$$

$\text{Using } \text{C}_0 = 12.5\text{pF}, \text{V} = 12 \text{ V}, \epsilon_\text{r} = 6$

$$= \frac{1}{2}(12.5) \times 12^2\left[1 - \frac{1}{6}\right] = \frac{1}{2}(12.5) \times 12^2 \times \frac{5}{6}$$

$$= 750\text{pJ} = 750 \times 10^{-12} \text{ J}$$

Q. 24 In a system two particles of masses $\text{m}_1 = 3 \text{ kg}$ and $\text{m}_2 = 2 \text{ kg}$ are placed at certain distance from each other. The particle of mass m_1 is moved towards the center of mass of the system through a distance 2 cm. In order to keep the center of mass of the system at the original position, the particle of mass m_2 should move towards the center of mass by the distance cm.

Correct Answer:
3

Solution:
$m_1 = 3\ kg \quad m_2 = 2\ kg$

$\underset{2\ cm}{\rightarrow} \qquad \underset{x}{\leftarrow}$

$$\Delta X_{C.O.M.} = \frac{m_1\Delta x_1 + m_2\Delta x_2}{m_1 + m_2}$$
$$\Rightarrow 0 = \frac{3 \times 2 + 2(-x)}{3 + 2}$$
$$\Rightarrow x = 3\ cm$$

Q. 25 The disintegration energy Q for the nuclear fission of $^{235}U \rightarrow {}^{140}Ce + {}^{94}Zr + n$ is _____ MeV.

Given atomic masses of
$^{235}U : 235.0439u;\ ^{140}Ce; 139.9054u,$
$^{94}Zr : 93.9063u; n : 1.0086u,$

Value of $c^2 = 931 MeV/u$

Correct Answer:
208

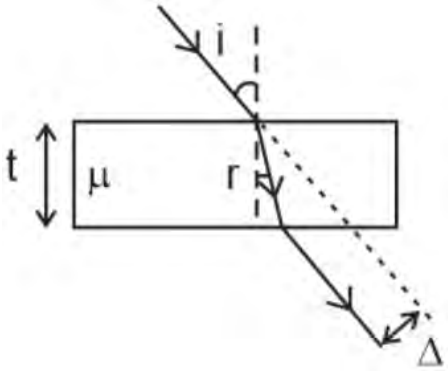
Solution:
$^{235}U \rightarrow {}^{140}Ce + {}^{94}Zr + n$

Disintegration energy
$$Q = (m_R - m_p) \cdot c^2$$
$$m_R = 235.0439u$$
$$m_p = 139.9054u + 93.9063u + 1.0086u$$
$$= 234.8203u$$
$$Q = (235.0439u - 234.8203u)c^2$$
$$= 0.2236c^2$$
$$= 0.2236 \times 931$$
$$Q = 208.1716$$

Q. 26 A light ray is incident on a glass slab of thickness $4\sqrt{3}$ cm and refractive index $\sqrt{2}$. The angle of incidence is equal to the critical angle for the glass slab with air. The lateral displacement of ray after passing through glass slab is cm. (Given $\sin 15^\circ = 0.25$)

Correct Answer:
2

Solution:



$\mathbf{i} = \theta_c$

$\Rightarrow \mathrm{i} = \sin^{-1}\left(\frac{1}{\mu}\right)$ $

$\Rightarrow \mathrm{i} = 45^\circ$

and according to snell's law

$1 \sin 45^\circ = \sqrt{2} \sin \mathrm{r}$ $

$\Rightarrow \mathrm{r} = 30^\circ$

$$\text{Lateral displacement } \Delta = \frac{\mathrm{t}\sin(\mathrm{i}-\mathrm{r})}{\cos \mathrm{r}}$$

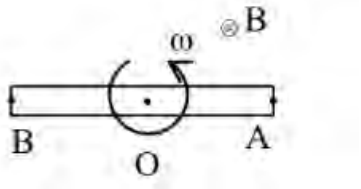
$$\Rightarrow \Delta = \frac{4\sqrt{3} \times \sin 15^\circ}{\cos 30^\circ}$$

$\Rightarrow \Delta = 2$ cm

Q. 27 A rod of length 60 cm rotates with a uniform angular velocity $20 \mathrm{rads}^{-1}$ about its perpendicular bisector, in a uniform magnetic field 0.5 T. The direction of magnetic field is parallel to the axis of rotation. The potential difference between the two ends of the rod is V.

Correct Answer:
0

Solution:



$$\because V_0 - V_A = \frac{B\omega\ell^2}{2}$$

$$V_0 - V_B = \frac{B\omega\ell^2}{2}$$

$$\therefore V_A = V_B \therefore V_A - V_B = 0$$

Q. 28 Two wires A and B are made up of the same material and have the same mass. Wire A has radius of $2.0\ \text{mm}$ and wire B has radius of $4.0\ \text{mm}$. The resistance of wire B is 2Ω. The resistance of wire A is $\quad\quad\ \Omega$.

Correct Answer:
32

Solution:

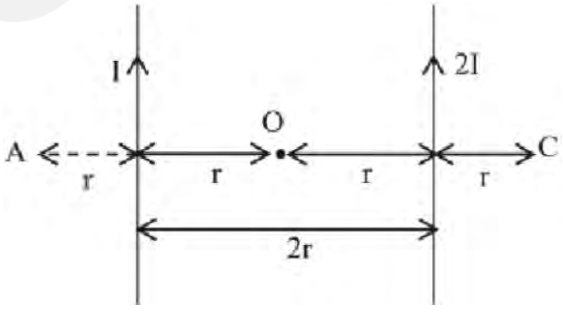
$$\because R = \frac{\rho\ell}{A} = \frac{\rho V}{A^2}$$

$$\therefore \frac{R_A}{R_B} = \frac{A_B^2}{A_A^2} = \frac{r_B^4}{r_A^4}$$

$$\Rightarrow \frac{R_A}{2} = \left[\frac{4\times10^{-3}}{2\times10^{-3}}\right]^4$$

$$\Rightarrow R_A = 32\Omega$$

Q. 29 Two parallel long current carrying wire separated by a distance $2r$ are shown in the figure. The ratio of magnetic field at A to the magnetic field produced at C is $\frac{X}{7}$. The value of X is



Correct Answer:
5

Solution:

$$B_A = \frac{\mu_0 i}{2\pi r} + \frac{\mu_0 (2i)}{2\pi (3r)} = \frac{5\mu_0 i}{6\pi r}$$

$$B_C = \frac{\mu_0 (2i)}{2\pi r} + \frac{\mu_0 i}{2\pi (3r)} = \frac{7\mu_0 i}{6\pi r}$$

$$\therefore \frac{B_A}{B_C} = \frac{5}{7}$$

$$\therefore x = 5$$

Q. 30 Mercury is filled in a tube of radius $2\ \text{cm}$ up to a height of $30\ \text{cm}$. The force exerted by mercury on the bottom of the tube is $\qquad$ N.
(Given, atmospheric pressure $= 10^5 \text{Nm}^{-2}$, density of mercury $= 1.36 \times 10^4\ \text{kg m}^{-3}$, $g = 10\ \text{ms}^{-2}$, $\pi = \frac{22}{7}$)

Correct Answer:
177

Solution:

$$F = P_0\ A + \rho_m\ ghA$$

$$= 10^5 \times \frac{22}{7} \times \left(2 \times 10^{-2}\right)^2$$

$$+ 1.36 \times 10^4 \times 10 \times \left(30 \times 10^{-2}\right) \left(\frac{22}{7} \times \left(2 \times 10^{-2}\right)^2\right)$$

$$F = 51.29 + 125.71 = 177\ \text{N}$$

Chemistry

Q. 1 The equilibrium constant for the reaction

$SO_3(g) \rightleftharpoons SO_2(g) + \frac{1}{2}O_2(g)$
is $K_C = 4.9 \times 10^{-2}$. The value of K_C for the reaction given below is
$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ is

Option 1:
4.9

Option 2:
41.6

Option 3:
49

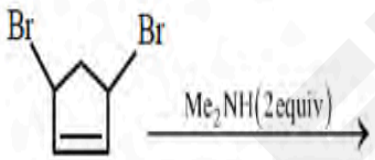
Option 4:
416

Correct Answer:
416

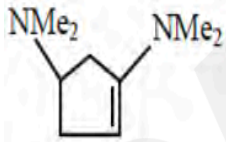
Solution:

$$K'_C = \left(\frac{1}{K_C}\right)^2 = \left(\frac{1}{4.9\times10^{-2}}\right)^2$$

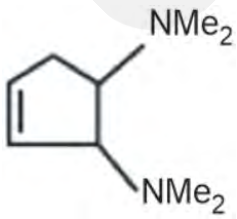
Q. 2 Find out the major product formed from the following reaction. $[\mathrm{Me}:-\mathrm{CH_3}]$



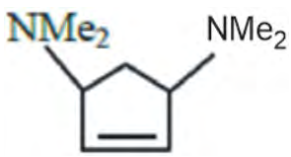
Option 1:



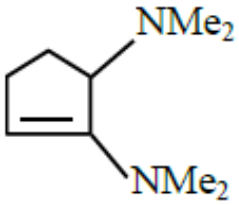
Option 2:



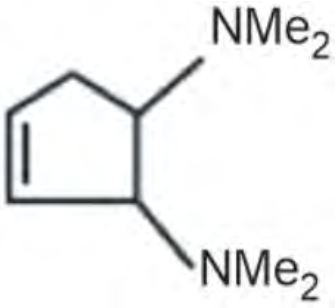
Option 3:



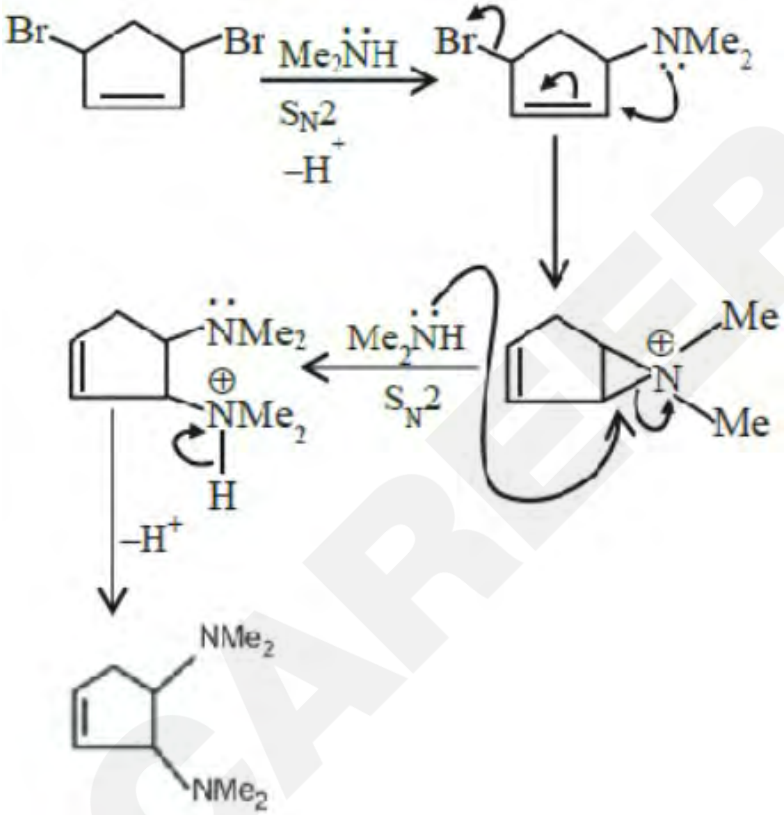
Option 4:



Correct Answer:



Solution:



The above mechanism valid for both cis and trans isomers. So the products are same for both cis and trans isomers.

Q. 3 When MnO_2 and H_2SO_4 is added to a salt (A), the greenish yellow gas liberated as salt (A) is :

Option 1:
$NaBr$

Option 2:
CaI_2

Option 3:
KNO_3

Option 4:
NH_4Cl

Correct Answer:
NH_4Cl

Solution:

$$2NH_4Cl + MnO_2 + 2H_2SO_4 \xrightarrow{\Delta} MnSO_4 + (NH_4)_2SO_4 + 2H_2O + \underset{\substack{\text{greenish}\\\text{yellow}\\\text{solution}}}{Cl_2}\uparrow$$

Q. 4 The correct statement/s about Hydrogen bonding is/are :

A. Hydrogen bonding exists when H is covalently bonded to the highly electro negative atom.

B. Intermolecular H bonding is present in o-nitro phenol

C. Intramolecular H bonding is present in HF.

D. The magnitude of H bonding depends on the physical state of the compound.

E. H-bonding has powerful effect on the structure and properties of compounds.

Choose the correct answer from the options given below :

Option 1:
A only

Option 2:
A, D, E only

Option 3:
A, B, D only

Option 4:
A, B, C only

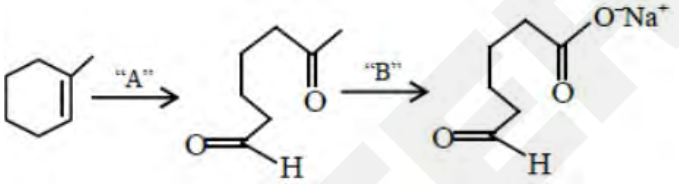
Correct Answer:
A, D, E only

Solution:
(A) Generally hydrogen bonding exists when H is covalently bonded to the highly electronegative atom like F, O, N.
(B) Intramolecular H bonding is present in

(C) Intermolecular Hydrogen bonding is present in HF
(D) The magnitude has Hydrogen bonding in a solid state is greater than liquid state.
(E) Hydrogen bonding has a powerful effect on the structure & properties of compounds like melting point, boiling point, density etc.

Hence, the answer is the option (2).

Q. 5



In the above chemical reaction sequence " A " and " B " respectively are :

Option 1:
$\mathrm{O_3, Zn/H_2O}$ and $\mathrm{NaOH_{(alc.)}\ /I_2}$

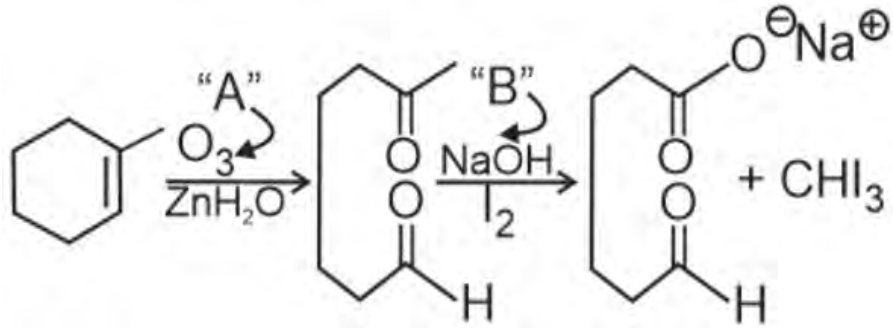
Option 2:
$\mathrm{H_2O, H^+}$ and $\mathrm{NaOH_{(alc.)}\ /I_2}$

Option 3:
$\mathrm{H_2O, H^+}$ and $\mathrm{KMnO_4}$

Option 4:
$\mathrm{O_3, Zn/H_2O}$ and $\mathrm{KMnO_4}$

Correct Answer:
$O_3, Zn/H_2O$ and $NaOH_{(alc.)}$ /I_2

Solution:



Q. 6 Common name of Benzene-1, 2-diol is

Option 1:
quinol

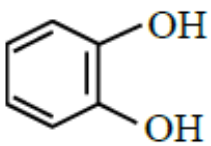
Option 2:
resorcinol

Option 3:
catechol

Option 4:
o-cresol

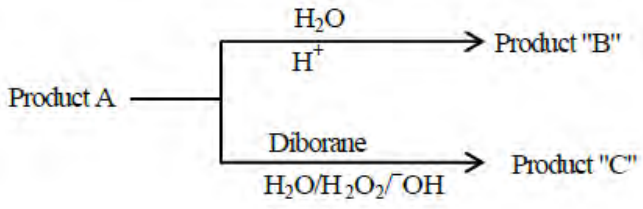
Correct Answer:
catechol

Solution:



IUPAC name : Benzene-1,2-diol
Common name : catechol

Q. 7 $CH_3 - CH_2 - CH_2 - Br + NaOH \xrightarrow{C_2H_5OH}$ Product 'A'



Consider the above reactions, identify product B and product C.

Option 1:
B = C = 2-Propanol

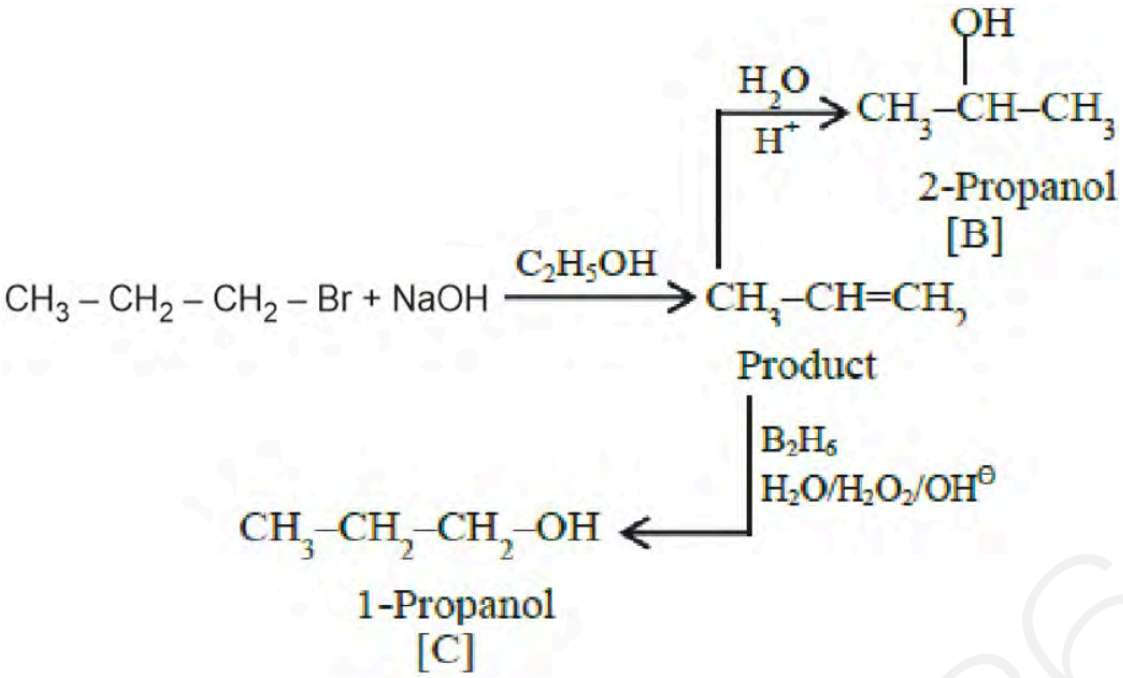
Option 2:
B = 2-Propanol C = 1-Propanol

Option 3:
B = 1-Propanol C = 2-Propanol

Option 4:
B = C = 1-Propanol

Correct Answer:
B = 2-Propanol C = 1-Propanol

Solution:



Q. 8 The adsorbent used in adsorption chromatography is/are

A. silica gel

B. alumina

C. quick lime

D. magnesia

Choose the most appropriate answer from the options given below :

Option 1:
B only

Option 2:
C and D only

Option 3:
A and B only

Option 4:
A only

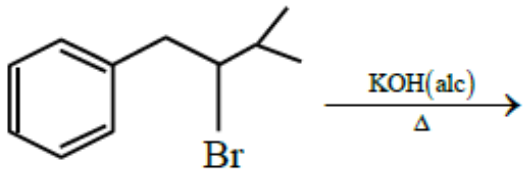
Correct Answer:
A and B only

Solution:

The most common polar and acidic support used in adsorption chromatography is silica. The surface silanol groups on their support to adsorb polar compounds work particularly well for basic substances. Alumina is an example of polar and basic adsorbent that is used in adsorption chromatography.

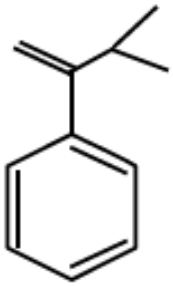
Hence, the answer is the option (3).

Q. 9

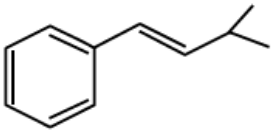


major product "P" Product P is

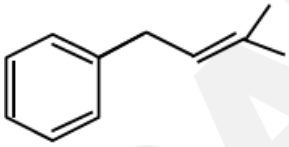
Option 1:



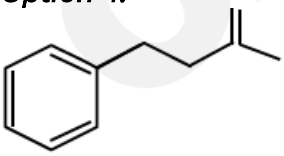
Option 2:



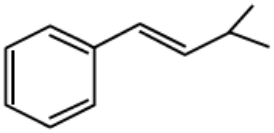
Option 3:



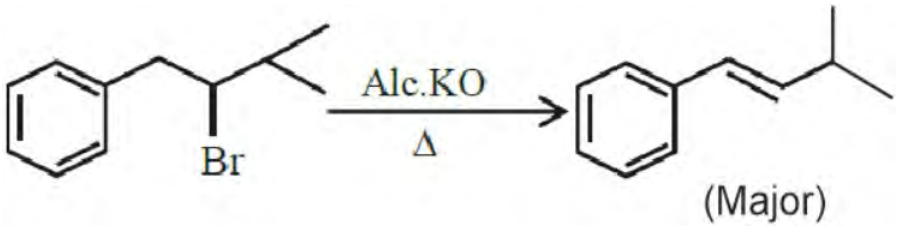
Option 4:



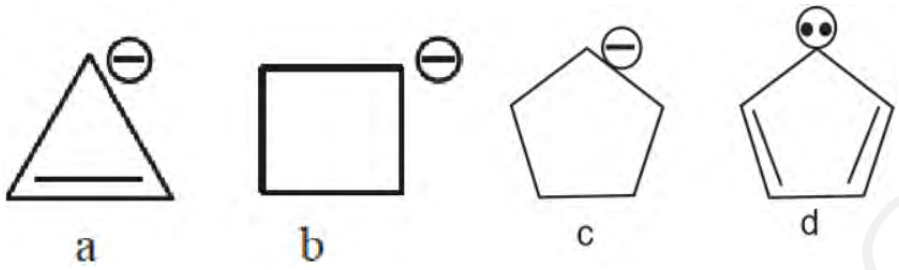
Correct Answer:



Solution:



Q. 10 Correct order of stability of carbanion is



Option 1:
c>b>d>a

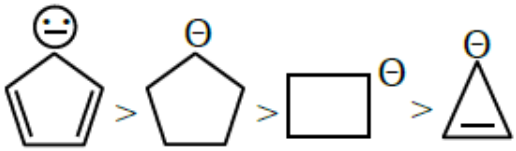
Option 2:
a > b > c >d

Option 3:
d>a>c>b

Option 4:
d>c>b>a

Correct Answer:
d>c>b>a

Solution:
As we know compound (d) is aromatic and the compound (a) is anti-aromatic. Hence compound (d) is most stable and compound (a) is least stable among these in compound (b) and (c) carbon atom of that positive charge is sp^3 hybridised they on the basis of angle strain theory compound (c) is more stable than compound (b).



Q. 11 The correct order of the first ionization enthalpy is

Option 1:
Al > Ga > Tl

Option 2:
Ga > Al > B

Option 3:
B > Al > Ga

Option 4:
Tl > Ga > Al

Correct Answer:
Tl > Ga > Al

Solution:
(i) due to laththanide contraction $\mathrm{T}\ell$ has more I.E. as compared to Ga and $\mathrm{A}\ell$
(ii) due to scandide contraction Ga has more I.E. as compared to $\mathrm{A}\ell$

Hence, the answer is the option (4).

Q. 12 If an iron (III) complex with the formula $[\mathrm{Fe}(\mathrm{NH_3})_{\mathrm{x}}(\mathrm{CN})_{\mathrm{y}}]^{-}$ has no electron in its $\mathrm{e_g}$ orbital, then the value of $x+y$ is

Option 1:
5

Option 2:
6

Option 3:
3

Option 4:
4

Correct Answer:
6

Solution:
Complex is

III

$[Fe(NH_3)_2(CN)_4]^{\Theta}$

$x = 2$

$y = 4$

so $x + y = 6$

Q. 13 Fuel cell, using hydrogen and oxygen as fuels,
A. has been used in spaceship
B. has as efficiency of 40% to produce electricity
C. uses aluminium as catalysts
D. is eco-friendly
E. is actually a type of Galvanic cell only

Option 1:
A,B,C only

Option 2:
A,B,D only

Option 3:
A,B,D,E only

Option 4:
A,D,E only

Correct Answer:
A,D,E only

Solution:
Fuel cell is used in spaceship and it is type of galvanic cell.

Q. 14 Choose the Incorrect Statement about Dalton's Atomic Theory

Option 1:
Compounds are formed when atoms of different elements combine in any ratio

Option 2:
All the atoms of a given element have identical properties including identical mass

Option 3:
Matter consists of indivisible atoms

Option 4:
Chemical reactions involve recorganization of atoms

Correct Answer:
Compounds are formed when atoms of different elements combine in any ratio

Solution:
In compound atoms of different elements combine in fixed ratio by mass.

Q. 15 Match List I with List II

	List I		List II
A.	α-Glucose and α-Galactose	I.	Functional isomers
B.	$\alpha-$-Glucose and $\beta-$ Glucose	II.	Homologous
C.	α-Glucose and α-Fructose	III.	Anomers
D.	α-Glucose and α-Ribose	IV	Epimers

Choose the correct answer from the options given below:

Option 1:
A-III, B-IV, C-II, D-I

Option 2:
A-III, B-IV, C-I, D-II

Option 3:
A-IV, B-III, C-I, D-II

Option 4:
A-IV, B-III, C-II, D-I

Correct Answer:
A-IV, B-III, C-I, D-II

Solution:
Based on biomolecules theory and structure of these named compounds -
(A) α-Glucose and α-Galactose (IV) Epimers.
(B) α-Glucose and β-Glucose (III) Anomers
(C) α-Glucose and α-Fructose (I) Functional isomers
(D) α-Glucose and α-Ribose (II) Homologous

Q. 16 Given below are two statements:
Statement I : The correct order of first ionization enthalpy values of Li, Na, F and Cl is Na < Li < Cl < F.
Statement II : The correct order of negative electron gain enthalpy values of Li, Na, F and Cl is Na < Li < F < Cl
In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is false but Statement II is true

Option 4:
Statement I is true but Statement II is false

Correct Answer:
Both Statement I and Statement II are true

Solution:

(i)

	Na	<	Li	<	Cl	<	F
	↓		↓		↓		↓
$I.E._1$ in kJ/mol	496		520		1256		1681

(ii)

	Na	<	Li	<	F	<	Cl
	↓		↓		↓		↓
$\Delta_{eg}H$ in kJ/mol	−53		−60		−328		−349

Q. 17 For a strong electrolyte, a plot of molar conductivity against (concentration) $^{1/2}$ is a straight line, with a negative slope, the correct unit for the slope is

Option 1:
$\mathrm{Scm^2\ mol^{-3/2}\ L^{1/2}}$

Option 2:
$\mathrm{Scm^2\ mol^{-1}\ L^{1/2}}$

Option 3:
$\mathrm{Scm^2\ mol^{-3/2}\ L}$

Option 4:
$\mathrm{Scm^2\ mol^{-3/2}\ L^{-1/2}}$

Correct Answer:
$\mathrm{Scm^2\ mol^{-3/2}\ L^{1/2}}$

Solution:

$\Lambda_m = \Lambda_m^\circ - A\sqrt{C}$

Units of $A\sqrt{C} = Scm^2\ mole^{-1}$
Units of $A = Scm^2\ mole^{-3/2}\ L^{1/2}$

Hence, the answer is the option (1).

Q. 18 A first row transition metal in its +2 oxidation state has a spin-only magnetic moment value of 3.86 BM.The atomic number of the metal is

Option 1:
25

Option 2:
26

Option 3:
22

Option 4:
23

Correct Answer:
23

Solution:
$_{22}Ti^{+2} \Rightarrow [Ar]3\ d^2$

$_{23}\ V^{+2} \Rightarrow [Ar]3\ d^3$

$_{25}Mn^{+2} \Rightarrow [Ar]3\ d^5$

$_{26}Fe^{+2} \Rightarrow [Ar]3\ d^6$

Hence, the answer is the option (4).

Q. 19 The number of unpaired d-electrons in $[Co(H_2O)_6]^{3+}$ is _____

Option 1:
4

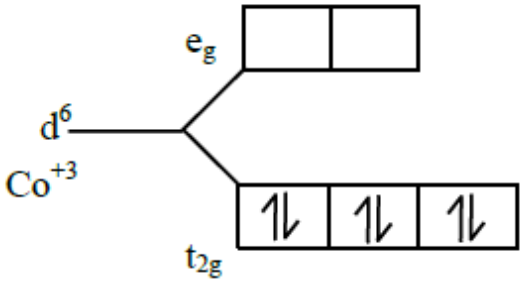
Option 2:
2

Option 3:
0

Option 4:
1

Correct Answer:
0

Solution:
$\Rightarrow [Co(H_2O)_6]^{+3}$



No unpaired electrons

Q. 20 The number of species from the following that have pyramidal geometry around the central atom is_____

$S_2O_3^{2-}, SO_4^{2-}, SO_3^{2-}, S_2O_7^{2-}$

Option 1:
4

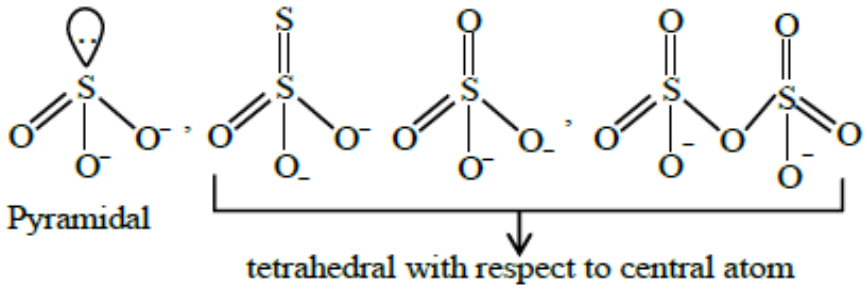
Option 2:
3

Option 3:
1

Option 4:
2

Correct Answer:
1

Solution:

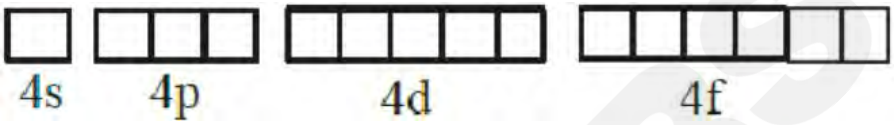


Q. 21 The maximum number of orbitals which can be identified with $n = 4$ and $m_l = 0$ is _____

Correct Answer:
4

Solution:

n = 4 ,



$m_\ell \Rightarrow \quad 1 \quad 1 \quad 1 \quad 1$

So answer is 4.

Q. 22 Number of compounds/species from the following with non-zero dipole moment is $BeCl_2, BCl_3, NF_3, XeF_4, CCl_4, H_2OH_2\ S, HBr, CO_2, H_2, HCl$

Correct Answer:
5

Solution:
Polar molecule: $NF_3, H_2O, H_2\ S, HBr, HCl$

$(\mu \neq 0)$

Non Polar molecule: $BeCl_2, BCl_3, XeF_4, CCl_4, CO_2, H_2$

$(\mu = 0)$

Hence, the answer is (5).

Q. 23 Three moles of an ideal gas are compressed isothermally from 60 L to 20 L using constant
pressure of 5 atm. Heat exchange Q for the compression ____ Lit. atm.

Correct Answer:
200

Solution:
As isothermal $\Delta U = 0$

and the process is irreversible

$$Q = -W = -\left[-P_{ext}\ (V_2 - V_1)\right]$$

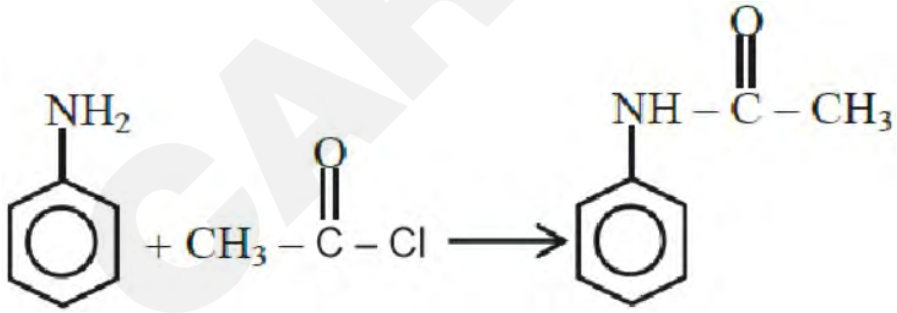
$$Q = 5(20 - 60) = -200\ \text{atm} - \text{L}$$

Hence, the answer is (200).

Q. 24 From 6.55 g of aniline, the maximum amount of acetanilide that can be prepared will be _____ $\times 10^{-1}$ g.

Correct Answer:
95

Solution:



93 g aniline form 135gm acetanlide

so 6.55 g anilne form $\frac{135}{93} \times 6.55 = 9.5$

95×10^{-1}

Q. 25 Consider the following reaction, the rate expression of which is given below

$$A + B \rightarrow C$$
$$\text{rate} = k[A]^{1/2}[B]^{1/2}$$

The reaction is initiated by taking $1M$ concentration A and B each. If the rate constant (k) is $4.6 \times 10^{-2}\ s^{-1}$, then the time taken for A to become $0.1M$ is _____ sec. (nearest integer)

Correct Answer:
50

Solution:
$K = \frac{2.303}{t} \log \frac{1}{0.1}$

$4.6 \times 10^{-2} = \frac{2.303}{t}$

$t = 50 \text{sec.}$

Hence, the answer is (50).

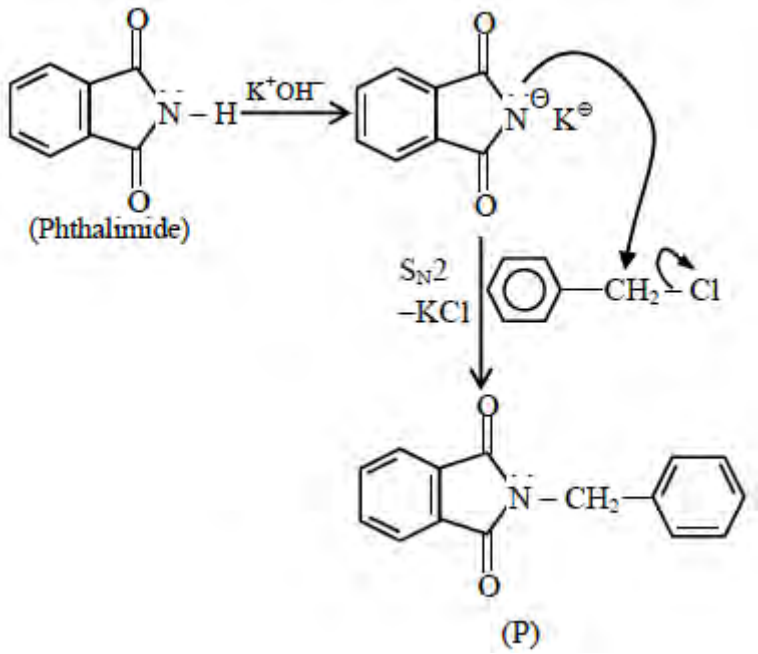
Q. 26 Phthalimide is made to undergo following sequence of reactions.

Phthalimide $\xrightarrow[\text{(ii)Benzylchloride}]{\text{(i)KOH}}$ ' P'

Total number of π bonds present in product 'P' is/are

Correct Answer:
8

Solution:

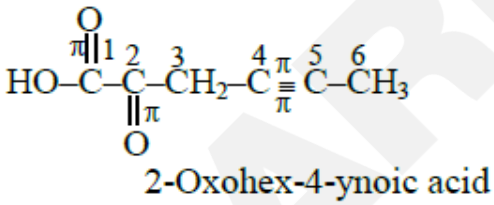


Total number of π-bonds present in product P is 8

Q. 27 The total number of 'sigma' and 'Pi' bonds in 2-oxohex-4-ynoic acid is____.

Correct Answer:
18

Solution:



$$\begin{aligned}\text{Number of } \sigma\text{-bonds} &= 14\\ \text{Number of } \pi\text{-bonds} &= 4\\ &= 18\end{aligned}$$

Hence, the answer is (18).

Q. 28 A first row transition metal with highest enthalpy of atomisation, upon reaction with oxygen at high temperature forms oxides of formula M_2O_n (where $n = 3, 4, 5$). The 'spin-only' magnetic moment value of the amphoteric oxide from the above oxides is _____ BM (near integer)
(Given atomic number: $Sc : 21, Ti : 22, V : 23,$
$Cr : 24, Mn : 25, Fe : 26, Co : 27, Ni : 28, Cu : 29, Zn : 30$)

Correct Answer:
0

Solution:
'V ' has highest enthalpy of atomisation ($515\ kJ/mol$) among first row transition elements.

V_2O_5

Here ' V ' is in +5 oxidation state

$V^{+5} \Rightarrow 1\ s^2 2\ s^2 2p^6 3\ s^2 3p^6$ (no unpaired electrons)

Q. 29 $2.7Kg$ of each of water and acetic acid are mixed, The freezing point of the solution will be $-x^{\circ}C$. Consider the acetic acid does not dimerise in water, nor dissociates in water x = _____ (nearest integer)

[Given : Molar mass of water $= 18\ g\ mol^{-1}$, acetic acid $= 60\ g\ mol^{-1}$]

$^{K_f}H_2O : 1.86\ K\ kg\ mol^{-1}$

K_f acetic acid : $3.90\ K\ kg\ mol^{-1}$

freezing point : $H_2O = 273\ K$, acetic acid $= 290\ K$]

Correct Answer:
31

Solution:
As moles of water $>$ moles of CH_3COOH water is solvent.

$T_F^{\circ} - (T_F)_S = K_F \times M$

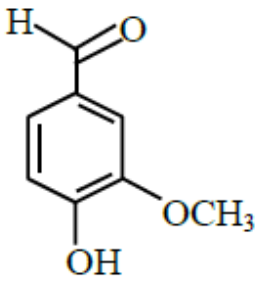
$0 - (T_F)_S = 1.86 \times \frac{2700/60}{2700/1000}$

$(T_F)_S = -31^{\circ}C.$

Q. 30 Vanillin compound obtained from vanilla beans, has total sum of oxygen atoms and π electrons is_____

Correct Answer:
11

Solution:
Vanillin compound is an organic compound molecular formula $C_8H_8O_3$. It is a phenolic aldehyde. Its functional compounds include aldehyde, hydroxyl and ether. It is the primary component of the extract of the vanilla beans.



Total sum of oxygen atoms and π-electrons is $3 + 8 = 11$
Total number of oxygen atoms $= 3$
Total number of π-bonds $= 4$
$\therefore$ Total number of π-electrons $= 8$

Maths

Q. 1 If the function $f(x) = \begin{cases} \frac{72^x - 9^x - 8^x + 1}{\sqrt{2} - \sqrt{1 + \cos x}} & , x \neq 0 \\ a \log_e 2 \log_e 3 & , x = 0 \end{cases}$ is continuous at $x = 0$, then the value of a^2 is equal to

Option 1:
968

Option 2:
1152

Option 3:
746

Option 4:
1250

Correct Answer:
1152

Solution:

$\lim\limits_{x\to 0} f(x) = a\ln 2\ln 3$

$\lim\limits_{n\to 0}\dfrac{72^x - 9^x - 8^x + 1}{\sqrt{2}-\sqrt{1+\cos x}} = \lim\limits_{x\to 0}\dfrac{(8^x-1)(9^x-1)}{\sqrt{2}-\sqrt{1+\cos x}}$

$\lim\limits_{n\to 0}\left(\dfrac{8^x-1}{x}\right)\left(\dfrac{9^x-1}{x}\right)\left(\dfrac{x^2}{1-\cos x}\right)(\sqrt{2}+\sqrt{1+\cos x})$

$\therefore \ln 8\times\ln 9\times 2\times 2\sqrt{2} = 24\sqrt{2}\ln 2\ln 3$

$\therefore a = 24\sqrt{2}, a^2 = 576\times 2 = 1152$

Q. 2 If $\lambda > 0$, let θ be the angle between the vectors $\vec{a} = \hat{i} + \lambda\hat{j} - 3\hat{k}$ and $\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$. If the vectors $\vec{a}+\vec{b}$ and $\vec{a}-\vec{b}$ are mutually perpendicular, then the value of $(14\cos\theta)^2$ is equal to

Option 1:
25

Option 2:
20

Option 3:
50

Option 4:
40

Correct Answer:
25

Solution:

$(\vec{a}+\vec{b})\cdot(\vec{a}-\vec{b})=0, \lambda>0$

$|\vec{a}|^2-|\vec{b}|^2=0 \rightarrow 1+\lambda^2+9=9+1+4$

$\therefore \lambda=2, \cos\theta=\dfrac{\vec{a}-\vec{b}}{|\vec{a}|\cdot|\vec{b}|}=\dfrac{3-\lambda-6}{\sqrt{14}\cdot\sqrt{14}}$

$14\cos\theta=3-8=-5$

$\therefore (14\cos\theta)^2=25$

Hence, the answer is the option (1).

Q. 3 Let C be a circle with radius $\sqrt{10}$ units and centre at the origin. Let the line $x+y=2$ intersects the circle C at the points P and Q. Let MN be a chord of C of length 2 unit and slope -1 . Then, a distance (in units) between the chord PQ and the chord MN is

Option 1:
$2-\sqrt{3}$

Option 2:
$3-\sqrt{2}$

Option 3:
$\sqrt{2}-1$

Option 4:
$\sqrt{2}+1$

Correct Answer:
$3-\sqrt{2}$

Solution:

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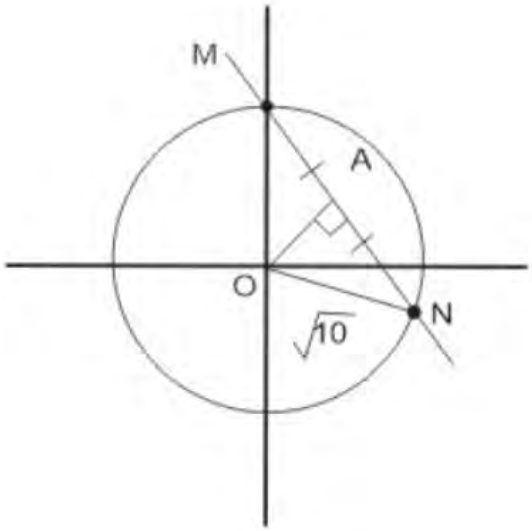


$C : x^2 + y^2 = 10$

$AN = \frac{MN}{2} = 1$

$\therefore In\Delta OAN \rightarrow (ON)^2 = (OA)^2 + (AN)^2$

$10 = (OA)^2 + 1 \rightarrow OA = 3$

Perpendicular distance of center from

$PQ = \frac{|0+0-2|}{\sqrt{2}} = \sqrt{2}$

Perpendicular distance between MN and

$PQ = OA + \sqrt{2}$ or $|OA - \sqrt{2}|$

$= 3 + \sqrt{2}$ or $3 - \sqrt{2}$

Q. 4 Let a relation R on $\mathbb{N} \times \mathbb{N}$ be defined as : $(x_1, y_1)R(x_2, y_2)$ if and only if $x_1 \leq x_2$ or $y_1 \leq y_2$

Consider the two statements :

(I) R is reflexive but not symmetric.

(II) R is transitive

Then which one of the following is true?

Option 1:

Only (II) is correct.

Option 2:

Only (I) is correct.

Option 3:

Both (I) and (II) are correct.

Option 4:
Neither (I) nor (II) is correct.

Correct Answer:
Only (I) is correct.

Solution:
All $((x_1y_1),(x_1,y_1))$ are in R where
$x_1,y_1 \in N \therefore R$ is reflexive
$((1,1),(2,3)) \in R$ but $((2,3),(1,1)) \notin R$
$\therefore R$ is not symmetric
$((2,4),(3,3)) \in R$ and $((3,3),(1,3)) \in R$ but $((2,4),$
$(1,3)) \notin R$
$\therefore R$ is not transitive

Q. 5 Let three real numbers a, b, c be in arithmetic progression and a+1, b, c+3 be in geometric progression. If a>10 and the arithmetic mean of a, b and c is 8 , then the cube of the geometric mean of a, b and c is

Option 1:
120

Option 2:
312

Option 3:
316

Option 4:
128

Correct Answer:
120

Solution:
$2b = a + c, b^2 = (a+1)(c+3)$
$\frac{a+b+c}{3} = 8 \rightarrow b = 8, a + c = 16$
$64 = (a+1)(19-a) = 19 + 18a - a^2$
$a^2 - 18a - 45 = 0 \rightarrow (a-15)(a+3) = 0, (a > 10)$
$a = 15, c = 1, b = 8$
$\left((abc)^{1/3}\right)^3 = abc = 120$

Q. 6 Let $A=\begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}$ and $B=I+\text{adj}(A)+(\text{adj } A)^2+\ldots+(\text{adj } A)^{10}$. Then, the sum of all the elements of the matrix B is :

Option 1:
-110

Option 2:
22

Option 3:
-88

Option 4:
-124

Correct Answer:
-88

Solution:

$$\text{Adj}(A)=\begin{bmatrix}1 & -2\\ 0 & 1\end{bmatrix}$$

$$(\text{AdjA})^2=\begin{bmatrix}1 & -4\\ 0 & 1\end{bmatrix}$$

$$(\text{AdjA})^{10}=\begin{bmatrix}1 & -20\\ 0 & 1\end{bmatrix}$$

$$B=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}+\begin{bmatrix}1 & -2\\ 0 & 1\end{bmatrix}+\begin{bmatrix}1 & -4\\ 0 & 1\end{bmatrix}+\ldots+\begin{bmatrix}1 & -20\\ 0 & 1\end{bmatrix}$$

$$B=\begin{bmatrix}11 & -110\\ 0 & 11\end{bmatrix}\Rightarrow \text{ sum of elements of B}$$

$$=-88$$

Q. 7 The value of $\frac{1\times2^2+2\times3^2+\ldots+100\times(101)^2}{1^2\times2+2^2\times3+\ldots+100^2\times101}$ is

Option 1:
$\frac{306}{305}$

Option 2:
$\frac{305}{301}$

Option 3:
$\frac{32}{11}$

Option 4:
$\frac{31}{30}$

Correct Answer:
$\frac{305}{301}$

Solution:

$$\frac{1\times 2^2+2\times 3^2+\ldots+100\times(101)^2}{1^2\times 2+2^2\times 3+\ldots+100^2\times 101}=\frac{\sum_{r=1}^{100} r(r+1)^2}{\sum_{r=1}^{100} r^2(r+1)}$$

$$=\frac{\sum_{r=1}^{100}\left(r^3+2r^2+r\right)}{\sum_{r=1}^{100}\left(r^3+r^2\right)}=\frac{\left(\frac{n(n+1)^2}{2}\right)+\frac{2\cdot n(n+1)(2n+1)}{6}+\frac{n(n+1)}{2}}{\left(\frac{n(n+1)}{2}\right)^2+\frac{n(n+1)(2n+1)}{6}}$$

$$=\frac{\frac{n(n+1)}{2}\left[\frac{n(n+1)}{2}+\frac{2}{3}\cdot(2n+1)+1\right]}{\frac{n(n+1)}{2}\left[\frac{n(n+1)}{2}+\frac{(2n+1)}{3}\right]};\ \text{Put n}=100$$

$$=\frac{\frac{100(101)}{2}+\frac{2}{3}(201)+1}{\frac{100\times 101}{2}+\frac{201}{3}}=\frac{5185}{5117}=\frac{305}{301}$$

Q. 8 Let $f(x)=\int_0^x\left(t+\sin\left(1-e^t\right)\right)dt, x\in\mathbb{R}$.
Then $\lim_{x\to 0}\frac{f(x)}{x^3}$ is equal to

Option 1:
$\frac{1}{6}$

Option 2:
$-\frac{1}{6}$

Option 3:
$-\frac{2}{3}$

Option 4:
$\frac{2}{3}$

Correct Answer:
$-\frac{1}{6}$

Solution:
$\lim_{x\to 0} \frac{f(x)}{x^3}$

Using L Hopital Rule.
$\lim_{x\to 0} \frac{f'(x)}{3x^2} = \lim_{x\to 0} \frac{x+\sin(1-e^x)}{3x^2}$ (Again L Hopital)
Using L.H. Rule

$$= \lim_{x\to 0} \frac{-\left[\sin\left(1-e^x\right)\left(-e^x\right)\cdot e^x + \cos\left(1-e^x\right)\cdot e^x\right]}{6}$$

$$= -\frac{1}{6}$$

Hence, the answer is the option (2).

Q. 9 The area (in sq. units) of the region described by $\left\{(x, y) : y^2 \leq 2x$, and $y \geq 4x - 1\right\}$ is

Option 1:
$\frac{11}{32}$

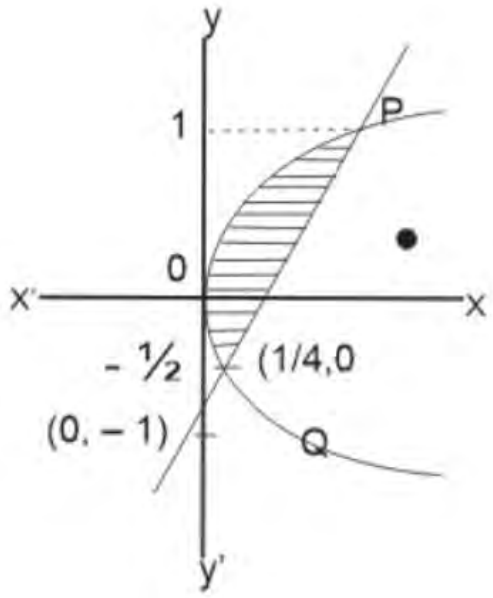
Option 2:
$\frac{8}{9}$

Option 3:
$\frac{11}{12}$

Option 4:
$\frac{9}{32}$

Correct Answer:
$\frac{9}{32}$

Solution:



$$\text{Shaded area } = \int_{-\frac{1}{2}}^{1} (x_{\text{Right}} - x_{\text{Left}})dy$$

$y^2 = 2x$

$y = 4x - 1 \quad$ Solve

$y = 1, y = -\dfrac{1}{2}$

$$\text{Shaded area } = \int_{-\frac{1}{2}}^{1} \left(\frac{y+1}{4} - \frac{y^2}{2}\right)dy$$

$$= \left(\frac{1}{4}\left(\frac{y^2}{2} + y\right) - \frac{y^3}{6}\right)_{-\frac{1}{2}}^{1} = \frac{9}{32}$$

Hence, the answer is the option (4).

Q. 10 The area (in sq. units) of the region
$S = \{z \in \mathbb{C}; |z - 1| \leq 2; (z + \bar{z}) + i(z - \bar{z}) \leq 2, \text{lm}(z) \geq 0\}$ is

Option 1:
$\frac{7\pi}{3}$

Option 2:
$\frac{3\pi}{2}$

Option 3:
$\frac{17\pi}{8}$

Option 4:
$\frac{7\pi}{4}$

Correct Answer:

$\frac{3\pi}{2}$

Solution:

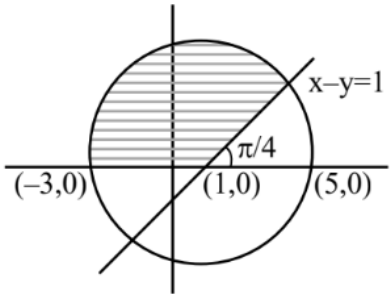
Put $z = x+$ iy

$|z-1| \leq 2 \Rightarrow (x-1)^2 + y^2 \leq 4 \ldots\ldots (1)$

$(z+\bar{z}) + i(z-\bar{z}) \leq 2 \Rightarrow 2x + i(2iy) \leq 2$

$\Rightarrow x - y \leq 1 \ldots\ldots (2)$

$\text{Im}(z) \geq 0 \Rightarrow y \geq 0 \ldots\ldots (3)$



Required area

= Area of semi-circle - area of sector A

$\frac{1}{2}\pi(2)^2 - \frac{\pi}{2}$

$= \frac{3\pi}{2}$

Q. 11 If the value of the integral $\int_{-1}^{1} \frac{\cos \alpha x}{1+3^x} dx$ is $\frac{2}{\pi}$. Then, a value of α is

Option 1:

$\frac{\pi}{6}$

Option 2:

$\frac{\pi}{2}$

Option 3:

$\frac{\pi}{3}$

Option 4:

$\frac{\pi}{4}$

Correct Answer:

$\frac{\pi}{2}$

Solution:

Let $I=\int_{-1}^{+1}\frac{\cos\alpha x}{1+3^x}dx.....(I)$

$$I=\int_{-1}^{+1}\frac{\cos\alpha x}{1+3^{-x}}dx$$

$$\left(u\,\text{sing}\int_a^b f(x)dx=\int_a^b f(a+b-x)dx\right).....(II)$$

Add (I) and (II)

$$2I=\int_{-1}^{+1}\cos(\alpha x)dx=2\int_0^1\cos(\alpha x)dx$$

$$I=\frac{\sin\alpha}{\alpha}=\frac{2}{\pi}(\text{ given })$$

$$\therefore\alpha=\frac{\pi}{2}$$

Q. 12 Let $f(x)=3\sqrt{x-2}+\sqrt{4-x}$ be a real valued function. If α and β are respectively the minimum and the maximum values of f, then $\alpha^2+2\beta^2$ is equal to

Option 1:
44

Option 2:
42

Option 3:
24

Option 4:
38

Correct Answer:
42

Solution:

$f(x) = 3\sqrt{x-2} + \sqrt{4-x}$

$x - 2 \geq 0 \& 4 - x \geq 0$

$\therefore x \in [2, 4]$

Let $x = 2\sin^2\theta + 4\cos^2\theta$

$\therefore f(x) = 3\sqrt{2}|\cos\theta| + \sqrt{2}|\sin\theta|$

$\therefore \sqrt{2} \leq 3\sqrt{2}|\cos\theta| + \sqrt{2}|\sin\theta| \leq \sqrt{9 \times 2 + 2}$

$\sqrt{2} \leq 3\sqrt{2}|\cos\theta| + \sqrt{2}|\sin\theta| \leq \sqrt{20}$

$\therefore \alpha = \sqrt{2} \quad \beta = \sqrt{20}$

$\alpha^2 + 2\beta^2 = 2 + 40 = 42$

Q. 13 If the coefficients of x^4, x^5 and x^6 in the expansion of $(1+x)^n$ are in the arithmetic progression, then the maximum value of n is :

Option 1:
14

Option 2:
21

Option 3:
28

Option 4:
7

Correct Answer:
14

Solution:

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$\text{Coeff. of } x^4 = {}^nC_4$

$\text{Coeff. of } x^5 = {}^nC_5$

$\text{Coeff. of } x^6 = {}^nC_6$

${}^nC_4, {}^nC_5, {}^nC_6 \ldots . AP$

$2.\ {}^nC_5 = {}^nC_4 + {}^nC_6$

$2 = \frac{{}^nC_4}{{}^nC_5} + \frac{{}^nC_6}{{}^nC_5} \quad \left\{ \frac{{}^nC_r}{{}^nC_r - 1} = \frac{n-r+1}{r} \right\}$

$2 = \frac{5}{n-4} + \frac{n-5}{6}$

$12(n-4) = 30 + n^2 - 9n + 20$

$n^2 - 21n + 98 = 0$

$(n-14)(n-7) = 0$

$n_{\max} = 14 \quad n_{\min} = 7$

Q. 14 Consider a hyperbola H having centre at the origin and foci and the x -axis. Let C_1 be the circle touching the hyperbola H and having the centre at the origin. Let C_2 be the circle touching the hyperbola H at its vertex and having the centre at one of its foci. If areas (in sq. units) of C_1 and C_2 are 36π and 4π, respectively, then the length (in units) of latus rectum of H is

Option 1:
$\frac{28}{3}$

Option 2:
$\frac{14}{3}$

Option 3:
$\frac{10}{3}$

Option 4:
$\frac{11}{3}$

Correct Answer:
$\frac{28}{3}$

Solution:

Let $H: \frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

$\left(b^2=a^2\left(e^2-1\right)\right)$

$\therefore \text{eq}^{\text{n}} \text{ of } C_1$

$\text{Ar.} = 36\pi$

$\pi a^2 = 36\pi$

$a = 6$

$\therefore \text{eq}^{\text{n}} \text{ of } C_1 = x^2+y^2=a^2$

Now radius of C_2 can be $a(e-1)$ or $a(e+1)$

for $r=a(e-1)$	for $r=a(e+1)$
$\text{Ar.} = 4\pi$	$\pi r^2 = 4\pi$
$\pi a^2(e-1)^2 = 4\pi$	$a^2(e+1)^2 = 4$
$36\pi(e-1)^2 = 4\pi$	$36(e+1)^2 = 4$
$e-1=\frac{1}{3}$	$e+1=\frac{1}{3}$
$e=\frac{4}{3}$	$-\frac{2}{3}$

Not possible

$\therefore b^2 = 36\left(\frac{16}{9}-1\right) = 28$

$\therefore \text{LR} = \frac{2b^2}{a} = \frac{2\times 28}{6} = \frac{28}{3}$

Q. 15 If the mean of the following probability distribution of a random variable X;

X	0	2	4	6	8
P(X)	a	2a	a+b	2b	3b

is $\frac{46}{9}$, then the variance of the distribution is

Option 1:

$\frac{581}{81}$

Option 2:

$\frac{566}{81}$

Option 3:

$\frac{173}{27}$

Option 4:

$\frac{151}{27}$

Correct Answer:

$\frac{566}{81}$

Solution:

$\sum P_i = 1$

$a + 2a + a + b + 2b + 3b = 1$

$4a + 6b = 1 \ldots . (I)$

$\mathrm{E(x)} = \text{mean} = \frac{46}{9}$

$$\sum \mathrm{P_i X_i} = \frac{46}{9} \Rightarrow 4\mathrm{a} + 4\mathrm{a} + 4\,\mathrm{b} + 12\,\mathrm{b} + 24\,\mathrm{b} = \frac{46}{9}$$

$$8\mathrm{a} + 40\,\mathrm{b} = \frac{46}{9}$$

$$4\mathrm{a} + 20\,\mathrm{b} = \frac{23}{9} \quad \ldots . \ (\mathrm{II})$$

Subtract (I) from (II) we get

$$\mathrm{b} = \frac{1}{9} \& \mathrm{a} = \frac{1}{12}$$

$$\text{Variance } = \mathrm{E}\left(\mathrm{x_i^2}\right) - \mathrm{E(x_i)}^2$$

$$\mathrm{E}\left(\mathrm{x_i^2}\right) = 0^2 \times 9^2 + 2^2 \times 2\mathrm{a} + 4^2(\mathrm{a} + \mathrm{b}) + 6^2(2\,\mathrm{b}) + 8^2(3\,\mathrm{b})$$

$$= 24\mathrm{a} + 280\,\mathrm{b}$$

$$\text{Put a} = \frac{1}{12} \quad \mathrm{b} = \frac{1}{9}$$

$$\mathrm{E}\left(\mathrm{x_i^2}\right) = 2 + \frac{280}{9} = \frac{298}{9}$$

$$\therefore \sigma^2 = \mathrm{E}\left(\mathrm{x_i^2}\right) - \mathrm{E(x_i)}^2$$

$$= \frac{298}{9} - \left(\frac{46}{9}\right)^2$$

$$\sigma^2 = \frac{298}{9} - \frac{2116}{81}$$

$$= \frac{566}{81}$$

Q. 16 Let PQ be a chord of the parabola $y^2 = 12x$ and the midpoint of PQ be at (4,1). Then, which of the following point lies on the line passing through the points P and Q?

Option 1:
$(3, -3)$

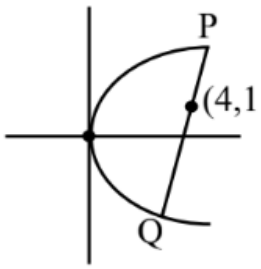
Option 2:
$\left(\frac{3}{2}, -16\right)$

Option 3:
$(2, -9)$

Option 4:
$\left(\frac{1}{2}, -20\right)$

Correct Answer:
$\left(\frac{1}{2}, -20\right)$

Solution:



$T = S_1$
$y - 6(x + 4)$
$= 1 - 48$
$6x - y = 23$

Option 4 $\left(\frac{1}{2}, -20\right)$ will satisfy

Q. 17 Given the inverse trigonometric function assumes principal values only. Let x, y be any two real numbers in [-1,1] such that
$\cos^{-1}\mathrm{x} - \sin^{-1}\mathrm{y} = \alpha, \frac{-\pi}{2} \leq \alpha \leq \pi.$
Then, the minimum value of $x^2 + y^2 + 2xy\sin\alpha$ is

Option 1:
-1

Option 2:
0

Option 3:
$\frac{-1}{2}$

Option 4:
$\frac{1}{2}$

Correct Answer:
0

Solution:
$\cos^{-1} x - \left(\frac{\pi}{2} - \cos^{-1} y\right) = \alpha$

$\cos^{-1} x + \cos^{-1} y = \frac{\pi}{2} + \alpha$

$\alpha \in \left[-\frac{\pi}{2}, \pi\right], \frac{\pi}{2} + \alpha \in \left[0, \frac{3\pi}{2}\right]$

$\cos^{-1}\left(xy - \sqrt{1-x^2}\sqrt{1-y^2}\right) = \frac{\pi}{2} + \alpha$

$xy - \sqrt{1-x^2}\sqrt{1-y^2} = -\sin\alpha$

$(xy + \sin\alpha) = \left(1-x^2\right)\left(1-y^2\right)$

$x^2y^2 + 2xy\sin a + \sin^2 a = 1 - x^2 - y^2 + x^2y^2$

$x^2 + y^2 + 2xy\sin\alpha = 1 - \sin^2\alpha$

$x^2 + y^2 + 2xy\sin\alpha = \cos^2\alpha$

$\text{Min. value of } \cos^2\alpha = 0$

$\text{At } \alpha = \frac{\pi}{2}$

Hence, the answer is the option (2).

Q. 18 Let $y = y(x)$ be the solution of the differential equation $\left(x^2+4\right)^2 dy + \left(2x^3y + 8xy - 2\right)dx = 0$. If $y(0) = 0$, then $y(2)$ is equal to

Option 1:
$\frac{\pi}{8}$

Option 2:
$\frac{\pi}{16}$

Option 3:
2π

Option 4:
$\frac{\pi}{32}$

Correct Answer:

$\frac{\pi}{32}$

Solution:

$$\frac{dy}{dx}+y\left(\frac{2x^3+8x}{\left(x^2+4\right)^2}\right)=\frac{2}{\left(x^2+4\right)^2}$$

$$\frac{dy}{dx}+y\left(\frac{2x}{x^2+4}\right)=\frac{2}{\left(x^2+4\right)^2}$$

$IF=e^{\int\frac{2x}{x^2+4}dx}$

$IF=x^2+4$

$$y\times\left(x^2+4\right)=\int\frac{2}{\left(x^2+4\right)^2}\times\left(x^2+4\right)$$

$$y\left(x^2+4\right)=2\int\frac{dx}{x^2+2^2}$$

$$y\left(x^2+4\right)=\frac{2}{2}\tan^{-1}\left(\frac{x}{2}\right)+c$$

$0=0+c=c=0$

$y\left(x^2+4\right)=\tan^{-1}\left(\frac{x}{2}\right)$

y at x $=2$

$y(4+4)=\tan^{-1}(1)$

$y(2)=\frac{\pi}{32}$

Q. 19 Let $\vec{a}=\hat{i}+\hat{j}+\hat{k},\vec{b}=2\hat{i}+4\hat{j}-5\hat{k}$ and $\vec{c}=x\hat{i}+2\hat{j}+3\hat{k},x\in\mathbb{R}$. If $\vec{d}$ is the unit vector in the direction of $\vec{b}+\vec{c}$ such that $\vec{a}\cdot\vec{d}=1$, then $(\vec{a}\times\vec{b})\cdot\vec{c}$ is equal to

Option 1:
9

Option 2:
6

Option 3:
3

Option 4:
11

Correct Answer:

11

Solution:

$\vec{d} = \lambda(\vec{b} + \vec{c})$

$\vec{a} \cdot \vec{d} = \lambda(\vec{b} \cdot \vec{a} + \vec{c} \cdot \vec{a})$

$1 = \lambda(1 + x + 5)$

$1 = \lambda(x + 6) \ldots\ldots\ldots (1)$

$|\vec{d}| = 1 \quad \frac{1}{\lambda} = x + 6$

$|\lambda(\vec{b} + \vec{c})| = 1$

$|\lambda((x + 2)\hat{i} + 6\hat{j} - 2\hat{k})| = 1$

$\lambda^2 \left((x + 2)^2 + 6^2 + 2^2\right) = 1$

$x^2 + 4x + 4 + 36 + 4 = (x + 6)^2$

$x^2 + 4x + 44 = x^2 + 12x + 36$

$8x = 8, x = 1$

$$\begin{matrix} 1 & 1 & 1 \\ 2 & 4 & -5 \\ x & 2 & 3 \end{matrix} = (\vec{a} \times \vec{b}) \cdot \vec{c}$$

$$\begin{matrix} 0 & 0 & 1 \\ -2 & 9 & -4 \\ x-2 & -1 & 3 \end{matrix} = 2 - 9(x - 2)$$

$= 20 - 9x$

at $x = 1$

$20 - 9 = 11$

Q. 20 Let P the point of intersection of the lines $\frac{x-2}{1} = \frac{y-4}{5} = \frac{z-2}{1}$ and $\frac{x-3}{2} = \frac{y-2}{3} = \frac{z-3}{2}$. Then, the shortest distance of P from the line 4x=2y=z is

Option 1:

$\frac{5\sqrt{14}}{7}$

Option 2:

$\frac{\sqrt{14}}{7}$

Option 3:

$\frac{3\sqrt{14}}{7}$

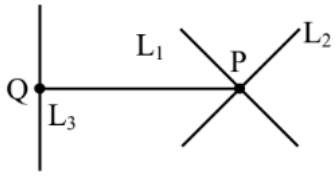
Option 4:

$\frac{6\sqrt{14}}{7}$

Correct Answer:

$\frac{3\sqrt{14}}{7}$

Solution:



$$L_1 \equiv \frac{x-2}{1} = \frac{y-4}{5} = \frac{z-2}{1} = \lambda$$

$$P(\lambda+2, 5\lambda+4, \lambda+2)$$

$$L_2 \equiv \frac{x-3}{2} = \frac{y-2}{3} = \frac{z-3}{2}$$

$$P(2\mu+3, 3\mu+2, 2\mu+3)$$

$$\lambda+2 = 2\mu+3 \quad 3\mu+2 = 5\lambda+4$$

$$\lambda = 2\mu+1 \quad 3\mu = 5\lambda+2$$

$$3\mu = 5(2\mu+1)+2$$

$$3\mu = 10\mu+7$$

$$\mu = -1 \quad \lambda = -1$$

Both satisfies (P)

$$P(1, -1, 1)$$

$$L_3 \equiv \frac{x}{1/4} = \frac{y}{1/2} = \frac{z}{1}$$

$$L_3 = \frac{x}{1} = \frac{y}{2} = \frac{z}{4} = k$$

Coordinates of $Q(k, 2k, 4k)$

DR 's of $PQ = < k-1, 2k+1, 4k-1 >$

$PQ \perp$ to L_3

$(k-1)+2(2k+1)+4(4k-1)=0$

$k-1+4k+2+16k-4=0$

$k=\frac{1}{7}$

$Q\left(\frac{1}{7},\frac{2}{7},\frac{4}{7}\right)$

$PQ=\sqrt{\left(1-\frac{1}{7}\right)^2+\left(-1-\frac{2}{7}\right)^2+\left(1-\frac{4}{7}\right)^2}$

$=\sqrt{\frac{36}{49}+\frac{81}{49}+\frac{9}{49}}=\frac{\sqrt{126}}{7}$

$PQ=\frac{3\sqrt{14}}{7}$

Q. 21 Let $S=\left\{\sin^2 2\theta:\left(\sin^4\theta+\cos^4\theta\right)x^2+(\sin 2\theta)x+\left(\sin^6\theta+\cos^6\theta\right)=0\right.$ has real roots $\}$. If α and β be the smallest and largest elements of the set S, respectively, then $3\left((\alpha-2)^2+(\beta-1)^2\right)$ equals.....

Correct Answer:
4

Solution:

$D=(\sin 2\theta)^2-4\left(1-\frac{\sin^2 2\theta}{2}\right)\left(1-\frac{3}{4}\sin^2 2\theta\right)$

$=(\sin 2\theta)^2-4\left(1-\frac{5}{4}\sin^2 2\theta+\frac{3}{8}\sin^4 2\theta\right)$

$D=-\frac{3}{2}\sin^4 2\theta+6\sin^2 2\theta-4>0$

$3\sin^4 2\theta-12\sin^2 2\theta+8<0$

$\sin^2 2\theta=\frac{12\pm\sqrt{12^2-12.8}}{6}=\frac{12\pm4\sqrt{3}}{6}=\frac{6\pm2\sqrt{3}}{3}$

$\sin^2 2\theta=2\pm\frac{2}{\sqrt{3}},\ \text{but}\ \sin^2 2\theta\in[0,1]$

$\therefore\alpha=2-\frac{2}{\sqrt{3}},\beta=1\rightarrow(\alpha-2)^2=\frac{4}{3},(\beta-1)^2=0$

$3(\alpha-2)^2+(\beta-1)^2=4$

Hence, the answer is (4).

Q. 22 If $\int \text{cosec}^5 x dx = \alpha \cot x \,\text{cosec}\, x \left(\text{cosec}^2 x + \frac{3}{2}\right) + \beta \log_e \tan\frac{x}{2} + C$ where $\alpha, \beta \in \mathbb{R}$ and C is constant of integration , then the value of $8(\alpha + \beta)$ equals ....

Correct Answer:
1

Solution:
$\int \text{cosec}^3 x \cdot \text{cosec}^2 x dx = I$

By applying integration by parts

$I = -\cot x \,\text{cosec}^3 x + \int \cot x \left(-3\,\text{cosec}^2 x \cot x \,\text{cosec}\, x\right) dx$

$I = -\cot x \,\text{cosec}^3 x - 3\int \text{cosec}^3 x \left(\text{cosec}^2 x - 1\right) dx$

$I = -\cot x \,\text{cosec}^3 x - 3I + 3\int \text{cosec}^3 x dx$

let

$I_1 = \int \text{cosec}^3 x dx = -\,\text{cosec}\, x \cot x - \int \cot^2 x \,\text{cosec}\, x dx$

$I_1 = -\,\text{cosec}\, x \cot x - \int \left(\text{cosec}^2 x - 1\right) \text{cosec}\, x dx$

$2I_1 = -\,\text{cosec}\, x \cot x + \ln \tan\frac{x}{2}$

$I_1 = -\frac{1}{2}\text{cosec}\, x \cot x + \frac{1}{2}\ln \tan\frac{x}{2}$

$4I = -\cot x \,\text{cosec}^3 x - \frac{3}{2}\text{cosec}\, x \cot x + \frac{3}{2}\ln \tan\frac{x}{2} + 4c$

$I = -\frac{1}{4}\text{cosec}\, x \cot x \left(\text{cosec}^2 x + \frac{3}{2}\right) + \frac{3}{8}\ln \tan\frac{x}{2} + c$

$\therefore \alpha = \frac{-1}{4}, \beta = \frac{3}{8} \rightarrow 8(\alpha + \beta) = 1$

Hence, the answer is 1.

Q. 23 Let $\text{f} : \mathbb{R} \rightarrow \mathbb{R}$ be a thrice differentiable function such that $f(0) = 0, f(1) = 1, f(2) = -1, f(3) = 2$ and $f(4) = -2$. Then, the minimum number of zeros of $\left(3\text{f}'\text{f}'' + \text{ff}'''\right)(\text{x})$ is .....

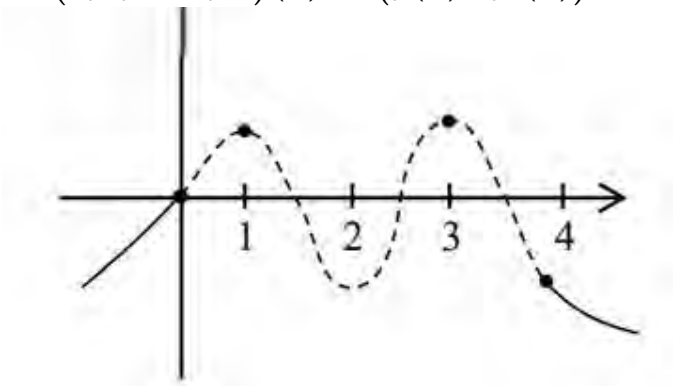
Correct Answer:
5

Solution:

$$(3f'f'' + f''')(x) = \left(\left(ff'' + (f')^2\right)(x)\right)'$$

$$\left(ff'' + (f')^2\right)(x) = ((ff')(x))'$$

$$\therefore (3f'f'' + f''')(x) = (f(x) \cdot f'(x))''$$



$\text{min. roots of } f(x) \to 4$

$\therefore \text{ min. roots of } f'(x) \to 3$

$\therefore \text{ min. roots of } (f(x) \cdot f'(x)) \to 7$

$\therefore \text{min . roots of } (f(x) \cdot f'(x))'^{\to} \to 5$

Q. 24 Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \frac{2x}{\sqrt{1+9x^2}}$. If the composition of f, $\underbrace{(f \text{ o } f \text{ o } f \text{ o} \ldots \text{o })}_{10 \text{ times}}(x) = \frac{2^{10}x}{\sqrt{1+9\alpha x^2}}$, then the value of $\sqrt{3\alpha + 1}$ is equal to .....

Correct Answer:
1024

Solution:

$$f(f(x)) = \frac{2f(x)}{\sqrt{1+9f^2(x)}} = \frac{4x}{\sqrt{1+9x^2+9.2^2x^2}}$$

$$f(f(f(x))) = \frac{2^3x/\sqrt{1+9x^2}}{\sqrt{1+9\left(1+2^2\right)\frac{2^2x^2}{1+9x^2}}} = \frac{2^3x}{\sqrt{1+9x^2\left(1+2^2+2^4\right)}}$$

$\therefore$ By observation

$$\alpha = 1 + 2^2 + 2^4 + \ldots . + 2^{18} = 1\left(\frac{\left(2^2\right)^{10} - 1}{2^2 - 1}\right) = \frac{2^{20} - 1}{3}$$

$$3\alpha + 1 = 2^{20} \to \sqrt{3\alpha + 1} = 2^{10} = 1024$$

Q. 25 Let A be a 2×2 symmetric matrix such that $A\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}3\\7\end{bmatrix}$ and the determinant of A be 1 . If $A^{-1} = \alpha A + \beta I$, where I is an identity matrix of order 2×2, then $\alpha + \beta$ equals . . . . .

Correct Answer:
5

Solution:

$\text{Let } A = \begin{bmatrix}a & b\\ b & d\end{bmatrix}$

$\begin{bmatrix}a & b\\ b & d\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}3\\7\end{bmatrix}, ad - b^2 = 1$

$\mathrm{a} + \mathrm{b} = 3,\ \mathrm{b} + \mathrm{d} = 7, (3 - \mathrm{b})(7 - \mathrm{b}) - \mathrm{b}^2 = 1$

$21 - 10\,\mathrm{b} = 1 \rightarrow \mathrm{b} = 2, \mathrm{a} = 1,\ \mathrm{d} = 5$

$\mathrm{A} = \begin{bmatrix}1 & 2\\ 2 & 5\end{bmatrix}, \mathrm{A}^{-1} = \begin{bmatrix}5 & -2\\ -2 & 1\end{bmatrix}$

$\mathrm{A}^{-1} = \alpha\mathrm{A} + \beta\mathrm{I}$

$\begin{bmatrix}5 & -2\\ -2 & 1\end{bmatrix} = \begin{bmatrix}\alpha+\beta & 2\alpha\\ 2\alpha & 5\alpha+\beta\end{bmatrix}$

$\alpha = -1, \beta = 6 \rightarrow \alpha + \beta = 5$

Q. 26 There are 4 men and 5 women in Group A, and 5 men and 4 women in Group B. If 4 persons are selected from each group, then the number of ways of selecting 4 men and 4 women is

Correct Answer:
5626

Solution:

From Group A	From Group B	Ways of selection
4M	4W	${}^4\mathrm{C}_4\,{}^4\mathrm{C}_4 = 1$
3M 1W	1M 3W	${}^4\mathrm{C}_3\,{}^5\mathrm{C}_1\,{}^5\mathrm{C}_1\,{}^4\mathrm{C}_3 = 400$
2M 2W	2M 2W	${}^4\mathrm{C}_2\,{}^5\mathrm{C}_2\,{}^5\mathrm{C}_2\,{}^4\mathrm{C}_2 = 3600$
1M 3W	3M 1W	${}^4\mathrm{C}_1\,{}^5\mathrm{C}_3\,{}^5\mathrm{C}_3\,{}^4\mathrm{C}_1 = 1600$
4W	4M	${}^5\mathrm{C}_4\,{}^5\mathrm{C}_4 = 25$
Total		5626

Hence, the answer is 5626.

Q. 27 In a tournament, a team plays 10 matches with probabilities of winning and losing each match as $\frac{1}{3}$ and $\frac{2}{3}$ respectively. Let x be the number of matches that the team wins, and y be the number of matches that team loses. If the probability $P(|x-y| \leq 2)$ is p, then $3^9 p$ equals.....

Correct Answer:
8288

Solution:

$P(W) = \frac{1}{3} \quad P(L) = \frac{2}{3}$

$x =$ number of matches that team wins

$y =$ number of matches that team loses

$|x-y| \leq 2$ and $x+y=10$

$|x-y| = 0, 1, 2 \quad x, y \in N$

$\text{Case-I} : |x-y| = 0 \Rightarrow x = y$

$\because x + y = 10 \Rightarrow x = 5 = y$

$$P(|x-y|=0) = {}^{10}C_5 \left(\frac{1}{3}\right)^5 \left(\frac{2}{3}\right)^5$$

Case-II : $|x-y| = 1 \Rightarrow x - y = \pm 1$

$x = y+1$	$x = y-1$
$\because x + y = 10$	$\because x + y = 10$
$2y = 9$	$2y = 11$
Not possible	Not possible

Case-III : $|x-y| = 2 \Rightarrow x - y = \pm 2$

$x - y = 2 \quad \text{OR} \quad x - y = -2$

$\because x + y = 10 \quad \because x + y = 10$

$x = 6, y = 4 \quad x = 4, y = 6$

$$P(|x-y|=2) = {}^{10}C_6 \left(\frac{1}{3}\right)^6 \left(\frac{2}{3}\right)^4 + {}^{10}C_4 \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^6$$

$$p = {}^{10}C_5 \frac{2^5}{3^{10}} + {}^{10}C_6 \frac{2^4}{3^{10}} + {}^{10}C_4 \frac{2^6}{3^{10}}$$

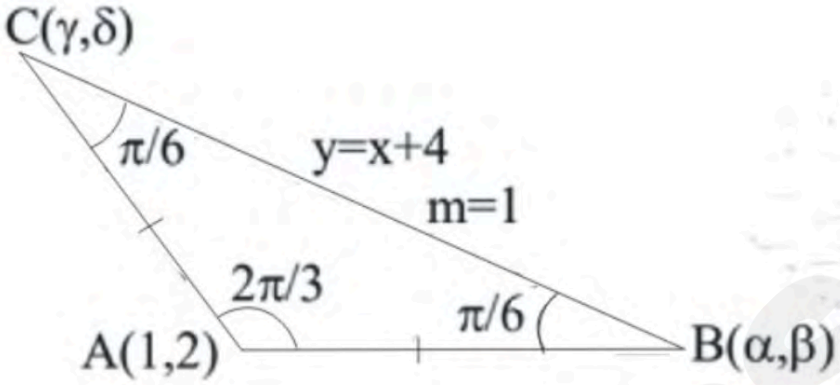
$$3^9 p = \frac{1}{3}\left({}^{10}C_5 2^5 + {}^{10}C_6 2^4 + {}^{10}C_4 2^6\right)$$

$= 8288$

Q. 28 Consider a triangle ABC having the vertices $\mathrm{A}(1,2), \mathrm{B}(\alpha, \beta)$ and $\mathrm{C}(\gamma, \delta)$ and angles $\angle \mathrm{ABC}=\frac{\pi}{6}$ and $\angle \mathrm{BAC}=\frac{2 \pi}{3}$. If the points B and C lie on the line $\mathrm{y}=\mathrm{x}+4$, then $\alpha^2+\gamma^2$ is equal to .....

Correct Answer:
14

Solution:



Equation of line passes through point $\mathrm{A}(1,2)$ which makes angle $\frac{\pi}{6}$ from $y=x+4$ is

$$y-2=\frac{1 \pm \tan \frac{\pi}{6}}{1 \mp \tan \frac{\pi}{6}}(x-1)$$

$$y-2=\frac{\sqrt{3} \pm 1}{\sqrt{3} \mp 1}(x-1)$$

$\oplus$	$\ominus$
$y-2=(2+\sqrt{3})(x-1)$ solve with $y=x+4$ $x+2=(2+\sqrt{3})x-2-\sqrt{3}$ $x=\frac{4+\sqrt{3}}{1+\sqrt{3}}$	$y-2=(2-\sqrt{3})(x-1)$ solve with $y=x+4$ $x+2=(2-\sqrt{3})x-2+\sqrt{3}$ $x=\frac{4-\sqrt{3}}{1-\sqrt{3}}$

$$\alpha^2+\gamma^2=\left(\frac{4+\sqrt{3}}{1+\sqrt{3}}\right)^2+\left(\frac{4-\sqrt{3}}{1-\sqrt{3}}\right)^2$$

$$\alpha^2+\gamma^2=14$$

Q. 29 Consider a line L passing through the points P(1,2,1) and Q(2,1,-1). If the mirror image of the point A(2,2,2) in the line L is (α, β, γ), then $\alpha+\beta+6 \gamma$ is equal to

Correct Answer:
6

Solution:

$\text{DR's of Line L} \equiv -1:1:2$

$\text{DR's of AB} \equiv \alpha-2:\beta-2:\gamma-2$

$\text{AB} \perp_{\text{ar}} \text{L} \Rightarrow 2-\alpha+\beta-2+2\gamma-4=0$

$2\gamma+\beta-\alpha=4$

Let C is mid-point of AB

$\text{C}\left(\frac{\alpha+2}{2}, \frac{\beta+2}{2}, \frac{\gamma+2}{2}\right)$

DR's of PC $= \frac{\alpha}{2}:\frac{\beta-2}{2}:\frac{\gamma}{2}$

line $\text{L}\|\text{PC} \Rightarrow \frac{-\alpha}{2}=\frac{\beta-2}{2}=\frac{\gamma}{4}=\text{K}(\text{ let })$

$\alpha=-2\,\text{K}$

$\beta=2\,\text{K}+2$

$\gamma=4\,\text{K}$

use in (1) $\Rightarrow \text{K}=\frac{1}{6}$

value of $\alpha+\beta+6\gamma=24\,\text{K}+2=6$

Hence, the answer is 6.

Q. 30 Let $y=y(x)$ be the solution of the differential equation $(x+y+2)^2dx=dy, y(0)=-2$. Let the maximum and minimum values of the function $y=y(x)$ in $\left[0,\frac{\pi}{3}\right]$ be α and β, respectively. If $(3\alpha+\pi)^2+\beta^2=\gamma+\delta\sqrt{3}, \gamma, \delta \in \mathbb{Z}$, then $\gamma+\delta$ equals.....

Correct Answer:

31

Solution:

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$\frac{dy}{dx} = (x+y+2)^2 \ldots\ldots (1)$

$y(0) = -2$

Let $x + y + 2 = v$

$1 + \frac{dy}{dx} = \frac{dv}{dx}$

from (1) $\frac{dv}{dx} = 1 + v^2$

$\int \frac{dv}{1+v^2} = \int dx$

$\tan^{-1}(v) = x + C$

$\tan^{-1}(x+y+2) = x + C$

at $x = 0 y = -2 \Rightarrow C = 0$

$\Rightarrow \tan^{-1}(x + y + 2) = x$

$y = \tan x - x - 2$

$f(x) = \tan x - x - 2, x \in \left[0, \frac{\pi}{3}\right]$

$f'(x) = \sec^2 x - 1 > 0 \Rightarrow f(x) \uparrow$

$f_{\min} = f(0) = -2 = \beta$

$f_{\max} = f\left(\frac{\pi}{3}\right) = \sqrt{3} - \frac{\pi}{3} - 2 = \alpha$

now $(3\alpha + \pi)^2 + \beta^2 = \gamma + \delta\sqrt{3}$

$\Rightarrow (3\alpha + \pi)^2 + \beta^2 = (3\sqrt{3} - 6)^2 + 4$

$\gamma + \delta\sqrt{3} = 67 - 36\sqrt{3}$

$\Rightarrow \gamma = 67$ and $\delta = -36 \Rightarrow \gamma + \delta = 31$

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5 Apr - Morning Shift

Physics

Q. 1 Light emerges out of a convex lens when a source of light kept at its focus. The shape of wavefront of the light is:

Option 1:
Both spherical and cylindrical

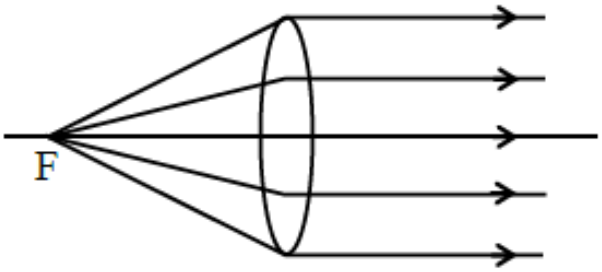
Option 2:
Cylindrical

Option 3:
Spherical

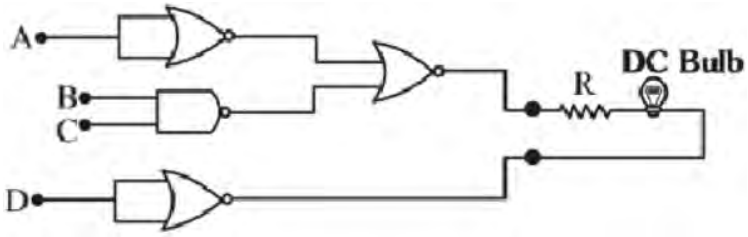
Option 4:
Plane

Correct Answer:
Plane

Solution:
Light emerges parallel
$\therefore$ planor wavefront



Q. 2 Following gates section is connected in a complete suitable circuit.



For which of the following combination, bulb will glow (ON):

Option 1:
$\mathrm{A}=0,\ \mathrm{B}=1, \mathrm{C}=1, \mathrm{D}=1$

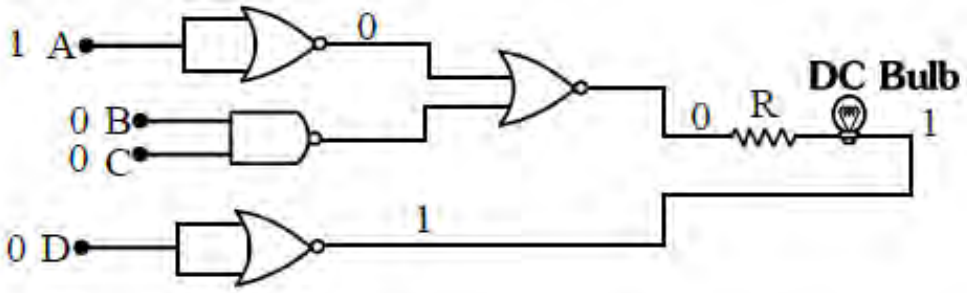
Option 2:
$\mathrm{A}=1,\ \mathrm{B}=0, \mathrm{C}=0, \mathrm{D}=0$

Option 3:
$\mathrm{A}=0,\ \mathrm{B}=0, \mathrm{C}=0, \mathrm{D}=1$

Option 4:
$\mathrm{A}=1,\ \mathrm{B}=1, \mathrm{C}=1, \mathrm{D}=0$

Correct Answer:
$\mathrm{A}=1,\ \mathrm{B}=0, \mathrm{C}=0, \mathrm{D}=0$

Solution:
Bulb will glow if bulb have potential drop on it.
One end of bulb must be at high (1) and other must be at low (0).
Option (2) satisfy this condition



Q. 3 If G be the gravitational constant and u be the energy density then which of the following quantity have the dimension as that the $\sqrt{uG}$:

Option 1:
Pressure gradient per unit mass

Option 2:
Force per unit mass

Option 3:
Gravitational potential

Option 4:
Energy per unit mass

Correct Answer:
Force per unit mass

Solution:

$$[uG] = \left[\left(M^1\ L^{-1}\ T^{-2}\right)\left(M^{-1}\ L^3\ T^{-2}\right)\right]$$

$$[uG] = \left[M^0\ L^2\ T^{-4}\right]$$

$$\left[\sqrt{uG}\right] = \left[L^1\ T^{-2}\right]$$

Option (2) is correct

Q. 4 Given below are two statements :
Statement-I: When a capillary tube is dipped into a liquid, the liquid neither rises nor falls in the capillary. The contact angle may be 0°.
Statement-II: The contact angle between a solid and a liquid is a property of the material of the solid and liquid as well :
In the light of above statement, choose the correct answer from the options given below.

Option 1:
Statement-I is false but Statement-II is true.

Option 2:
Both Statement-I and Statement-II are true.

Option 3:
Both Statement-I and Statement-II are false.

Option 4:
Statement-I is true and Statement-II is false.

Correct Answer:
Statement-I is false but Statement-II is true.

Solution:
Capillary rise

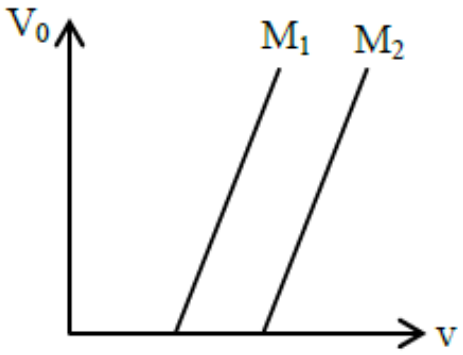
$\mathrm{h}=\frac{2\,\mathrm{T}\cos\theta}{\rho \mathrm{gr}}$;

If $\theta=0^{\circ}$ then rise is non-zero

$\therefore$ Statement-1 is incorrect.

Option(1) is correct

Q. 5 Given below are two statements:



Statement-I: Figure shows the variation of stopping potential with frequency (v) for the two photosensitive materials M_1 and M_2. The slope gives value of $\frac{h}{e}$, where h is Planck's constant, e is the charge of electron.

Statement-II: M_2 will emit photoelectrons of greater kinetic energy for the incident radiation having same frequency.
In the light of the above statements, choose the most appropriate answer from the options given below.

Option 1:
Statement-I is correct and **Statement-II** is incorrect.

Option 2:
Statement-I is incorrect but **Statement-II** is correct.

Option 3:
Both **Statement-I** and **Statement-II** are incorrect.

Option 4:
Both **Statement-I** and **Statement-II** are correct.

Correct Answer:
Statement-I is correct and **Statement-II** is incorrect.

Solution:

$\mathrm{eV_0 = hv - \phi}$

$\mathrm{V_0 = \frac{h}{e}v - \frac{\phi}{e}}$

M_2 material has higher work function, so statement-(II) is incorrect.

Option (1) is correct.

Q. 6 The angle between vector $\vec{Q}$ and the resultant of $(2\vec{Q} + 2\vec{P})$ and $(2\vec{Q} - 2\vec{P})$ is:

Option 1:

0°

Option 2:

$\tan^{-1}\frac{(2\vec{Q}-2\vec{P})}{2\vec{Q}+2\vec{P}}$

Option 3:

$\tan^{-1}\left(\frac{P}{Q}\right)$

Option 4:

$\tan^{-1}\left(\frac{2Q}{P}\right)$

Correct Answer:

0°

Solution:

$\vec{R} = (2\vec{Q} + 2\vec{P}) + (2\vec{Q} - 2\vec{P})$

$\vec{R} = 4\vec{Q}$

Angle between $\vec{Q}$ and $\vec{R}$ is zero

Option (1) is correct

Q. 7 In hydrogen like system the ratio of coulombian force and gravitational force between an electron and a proton is in the order of:

Option 1:
10^{39}

Option 2:
10^{19}

Option 3:
10^{29}

Option 4:
10^{36}

Correct Answer:
10^{39}

Solution:

$$F_e = \frac{kQ_1Q_2}{r^2} = \frac{9\times10^9\times1.6\times10^{-19}\times1.6\times10^{-19}}{r^2}$$

$$F_g = \frac{Gm_1m_2}{r^2} = \frac{6.67\times10^{-11}\times9.1\times10^{-31}\times1.6\times10^{-27}}{r^2}$$

$$\frac{F_e}{F_g} \cong 0.23\times10^{40} \cong 2.3\times10^{39}$$

Option (1)

Q. 8 In a co-axial straight cable, the central conductor and the outer conductor carry equal currents in opposite directions. The magnetic field is zero.

Option 1:
inside the outer conductor

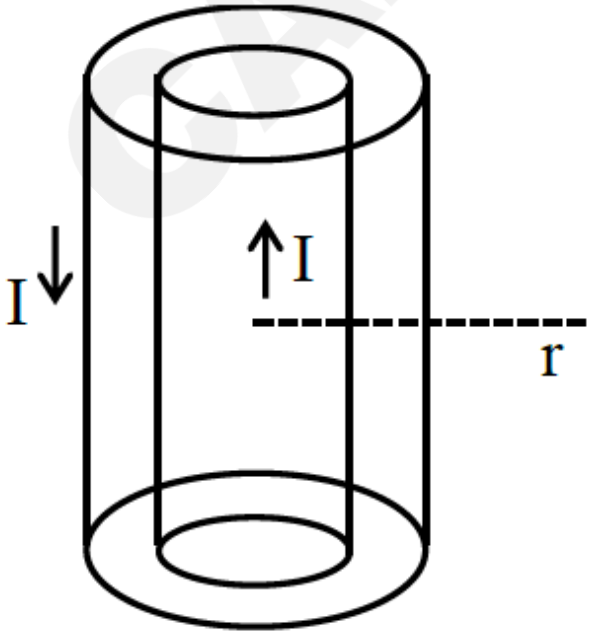
Option 2:
in between the two conductors

Option 3:
outside the cable

Option 4:
inside the inner conductor

Correct Answer:
outside the cable

Solution:



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$$\oint \vec{B} \cdot \vec{d\ell} = \mu_0 i_{enc} = 0$$

$\therefore B = 0$ outside the cable

Q. 9 An electron rotates in a circle around a nucleus having positive charge Ze. Correct relation between total energy (E) of electron to its potential energy (U) is:

Option 1:
E = 2U

Option 2:
2E = 3U

Option 3:
E = U

Option 4:
2E = U

Correct Answer:
2E = U

Solution:

$$F = \frac{k(Ze)(e)}{r^2} = \frac{mv^2}{r}$$

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}\frac{K(Ze)(e)}{r}$$

$$PE = -\frac{K(Ze)(e)}{r}$$

$$TE = \frac{K(Ze)(e)}{2r} - \frac{K(Ze)(e)}{r} = \frac{-K(Ze)(e)}{2r}$$

$$TE = \frac{PE}{2}$$

$$2TE = PE$$

Option (4)

Q. 10 If the collision frequency of hydrogen molecules in a closed chamber at $27^{\circ}\mathrm{C}$ is Z, then the collision frequency of the same system at $127^{\circ}\mathrm{C}$ is :

Option 1:
$\frac{\sqrt{3}}{2}Z$

Option 2:
$\frac{4}{3}\mathrm{Z}$

Option 3:
$\frac{2}{\sqrt{3}}\mathrm{Z}$

Option 4:
$\frac{3}{4}\mathrm{Z}$

Correct Answer:
$\frac{2}{\sqrt{3}}\mathrm{Z}$

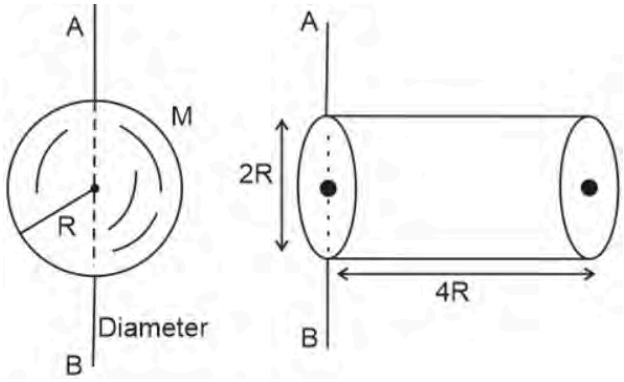
Solution:
Assuming mean free path constant.

$$\mathrm{f} \propto \mathrm{v} \propto \sqrt{\mathrm{T}}$$

$$\frac{\mathrm{f}_1}{\mathrm{f}_2} = \sqrt{\frac{\mathrm{T}_1}{\mathrm{T}_2}} = \sqrt{\frac{300}{400}}$$

$$\mathrm{f}_2 = \sqrt{\frac{4}{3}} = \mathrm{f}_1 = \frac{2}{\sqrt{3}}\mathrm{Z}$$

Q. 11 Ratio of radius of gyration of a hollow sphere to that of a solid cylinder of equal mass, for moment of Inertia about their diameter axis AB as shown in figure is $\sqrt{\frac{8}{x}}$. The value of x is:



Option 1:
34

Option 2:
17

Option 3:
67

Option 4:
51

Correct Answer:
67

Solution:

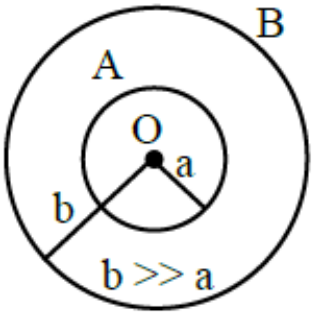
$\mathrm{I_{sphere}} = \frac{2}{3}\mathrm{MR^2} = \mathrm{Mk_1^2}$

$\mathrm{I_{cylinder}} = \frac{1}{12}\mathrm{M}\left(4\mathrm{R}^2\right) + \frac{1}{4}\mathrm{MR^2} + \mathrm{M(2R)^2}$

$= \frac{67}{12}\mathrm{MR^2} = \mathrm{Mk_2^2}$

$\frac{\mathrm{k_1}}{\mathrm{k_2}} = \sqrt{\frac{2}{3} \cdot \frac{12}{67}} = \sqrt{\frac{8}{67}}$

Q. 12 Two conducting circular loops A and B are placed in the same plane with their centres coinciding as shown in figure. The mutual inductance between them is:



Option 1:
$\frac{\mu_0 \pi a^2}{2b}$

Option 2:
$\frac{\mu_0}{2\pi} \cdot \frac{b^2}{a}$

Option 3:
$\frac{\mu_0 \pi b^2}{2a}$

Option 4:
$\frac{\mu_0}{2\pi} \cdot \frac{a^2}{b}$

Correct Answer:
$\frac{\mu_0 \pi a^2}{2b}$

Solution:
$\phi = \mathrm{Mi} = \mathrm{BA}$

$\Rightarrow Mi = \frac{\mu_0 i}{2b}\pi a^2$

$\therefore M = \frac{\mu_0 \pi a^2}{2b}$

Q. 13 Match list-I with list-II:

	List-I		List-II
(A)	Kinetic energy of planet	(I)	$-\frac{GMm}{a}$
(B)	Gravitation Potential energy of Sun-planet system.	(II)	$\frac{GMm}{2a}$
(C)	Total mechanical energy of planet	(III)	$\frac{Gm}{r}$
(D)	Escape energy at the surface of planet for unit mass object	(IV)	$-\frac{GMm}{2a}$

(Where a = radius of planet orbit, r = radius of planet, M = mass of Sun, m = mass of planet) Choose the correct answer from the options given below:

Option 1:
(A) – II, (B) – I, (C) – IV, (D) – III

Option 2:
(A) – III, (B) – IV, (C) – I, (D) – II

Option 3:
(A) – I, (B) – IV, (C) – II, (D) – III

Option 4:
(A) – I, (B) – II, (C) – III, (D) – IV

Correct Answer:
(A) – II, (B) – I, (C) – IV, (D) – III

Solution:
$\mathrm{KE} = \frac{1}{2}\mathrm{mv}^2 = \frac{\mathrm{GMm}}{2\mathrm{a}}$

$\mathrm{PE} = -2\mathrm{KE}$

$\mathrm{TE} = -\mathrm{KE}$

Q. 14 A wooden block of mass 5kg rests on soft horizontal floor. When an iron cylinder of mass 25 kg is placed on the top of the block, the floor yields and the block and the cylinder together go down with an acceleration of $0.1\ \mathrm{ms}^{-2}$. The action force of the system on the floor is equal to:

Option 1:
297N

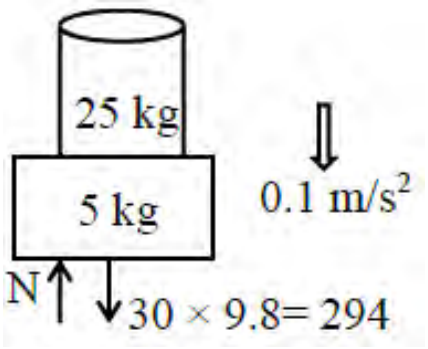
Option 2:
294N

Option 3:
291N

Option 4:
196N

Correct Answer:
291N

Solution:
Taking $\mathrm{g} = 9.8\ \mathrm{m/s}^2$



$294 - \mathrm{N} = 30 \times 0.1$

$\mathrm{N} = 291$

Q. 15 A simple pendulum doing small oscillations at a place R height above earth surface has time period of $T_1 = 4\ \mathrm{s}$. T_2 would be it's time period if it is brought to a point which is at a height $2\mathrm{R}$ from earth surface. Choose the correct relation $[R =$ radius of Earth]:

Option 1:
$T_1 = T_2$

Option 2:
$2\ \mathrm{T}_1 = 3\ \mathrm{T}_2$

Option 3:
$3\ \mathrm{T}_1 = 2\ \mathrm{T}_2$

Option 4:
$2\ \mathrm{T}_1 = \mathrm{T}_2$

Correct Answer:
$3\ \mathrm{T}_1 = 2\ \mathrm{T}_2$

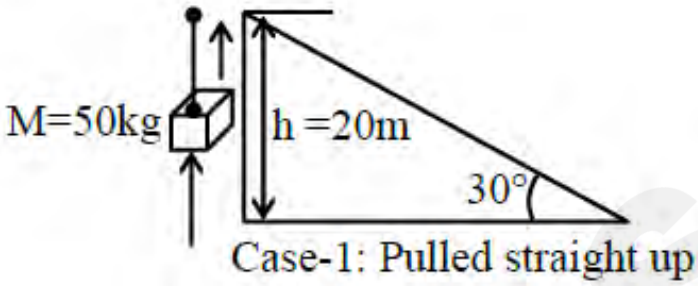
Solution:

$$T_1 = 2\pi\sqrt{\frac{\ell}{GM}(2R)^2}$$

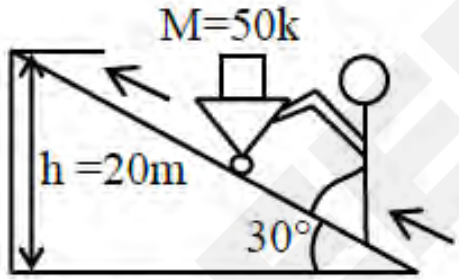
$$T_2 = 2\pi\sqrt{\frac{\ell}{GM}(3R)^2}$$

$$\therefore \frac{T_1}{T_2} = \frac{2}{3}$$

Q. 16 A body of mass 50 kg is lifted to a height of 20 m from the ground in the two different ways as shown in the figures. The ratio of work done against the gravity in both the respective cases, will be:



Case-1: Pulled straight up



Case-2: Along the ramp

Option 1:
$1:1$

Option 2:
$2:1$

Option 3:
$\sqrt{3}:2$

Option 4:
$1:2$

Correct Answer:
$1:1$

Solution:
Work done by gravity is independent of path. It depends only on vertical displacement so work done in both cases will be same.
Option (1) is correct

Q. 17 Time periods of oscillation of the same simple pendulum measured using four different measuring clocks were recorded as 4.62 s, 4.632 s, 4.6 s and 4.64 s. The arithmetic mean of these reading in correct significant figure is.

Option 1:
4.623 s

Option 2:
4.62 s

Option 3:
4.6 s

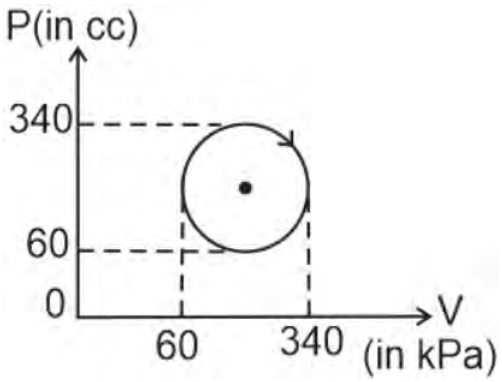
Option 4:
5 s

Correct Answer:
4.6 s

Solution:
Sum of number by considering significant digit sum
$= 4.6 + 4.6 + 4.6 + 4.6 = 18.4$

Arithmetic Mean $= \frac{\text{sum}}{4} = \frac{18.4}{4} = 4.6$

Q. 18 The heat absorbed by a system in going through the given cyclic process is :



Option 1:
61.6 J

Option 2:
431.2 J

Option 3:
616 J

Option 4:
19.6 J

Correct Answer:
61.6 J

Solution:

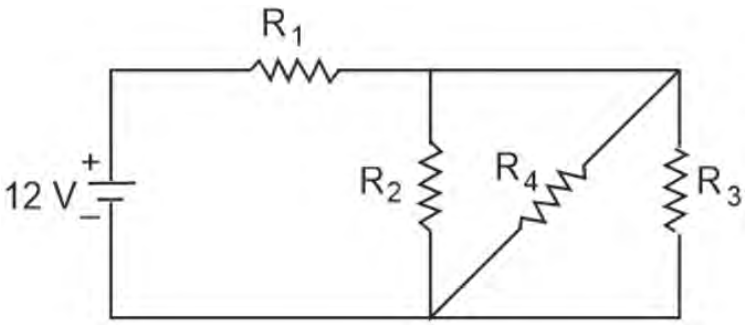
$\Delta \mathrm{U}=0(\text{ Cyclic process })$

$\Delta \mathrm{Q}=\mathrm{W}=\text{ area of } \mathrm{P}-\mathrm{V} \text{ curve}$

$=\pi \times\left(140 \times 10^{3} \mathrm{~Pa}\right) \times\left(140 \times 10^{-6} \mathrm{~m}^{3}\right)$

$\Delta \mathrm{Q}=61.6 \mathrm{~J}$

Q. 19 In the given figure $R_1 = 10\Omega, R_2 = 8\Omega, R_3 = 4\Omega$ and $R_4 = 8\Omega$. Battery is ideal with emf $12\ \text{V}$. Equivalent resistant of the circuit and current supplied by battery are respectively.



Option 1:
12Ω and $11.4\ \text{A}$

Option 2:
10.5Ω and $1.14\ \text{A}$

Option 3:
10.5Ω and $1\ \text{A}$

Option 4:
12Ω and $1\ \text{A}$

Correct Answer:
12Ω and $1\ \text{A}$

Solution:
Here $\text{R}_2, \text{R}_3, \text{R}_4$ are in parallel

$$\frac{1}{\text{R}_{234}} = \frac{1}{\text{R}_2} + \frac{1}{\text{R}_3} + \frac{1}{\text{R}_4}$$

$$\text{R}_{234} = 2\Omega$$

R_{234} is in series with R_1 so

$$\text{R}_{\text{eq}} = \text{R}_{234} + \text{R}_1 = 2 + 10 = 12\Omega$$

$\mathrm{i}=\frac{12}{12}=1\mathrm{Amp}$

Q. 20 An alternating voltage of amplitude $40\ \mathrm{V}$ and frequency $4\mathrm{kHz}$ is applied directly across the capacitor of $12\mu\mathrm{F}$. The maximum displacement current between the plates of the capacitor is nearly:

Option 1:
$13\ \mathrm{A}$

Option 2:
$8\ \mathrm{A}$

Option 3:
$10\ \mathrm{A}$

Option 4:
$12\ \mathrm{A}$

Correct Answer:
$12\ \mathrm{A}$

Solution:
Displacement current is same as conduction current in capacitor.

$X_C=\frac{1}{\omega \mathrm{C}}=\frac{1}{2\pi \mathrm{fC}}$

$=\frac{1}{2\pi\times4\times10^{3}\times12\times10^{-6}}=3.317\Omega$

$\mathrm{I}=\frac{\mathrm{V}}{\mathrm{X_C}}=\frac{40}{3.317}=12\ \mathrm{A}$

Q. 21 In Young's double slit experiment, carried out with light of wavelength 5000 Å, the distance between the slits is $0.3\ \text{mm}$ and the screen is at $200\ \text{cm}$ from the slits. The central maximum is at $x = 0\ \text{cm}$. The value of x for third maxima is _____ mm.

Correct Answer:
10

Solution:
$\beta = \frac{\lambda \text{D}}{\text{d}} = \frac{5\times10^{-7}\times2}{3\times10^{-4}} = \frac{10\times10^{-3}}{3}\ \text{m}$

For 3^{rd} maxima $y_3 = 3\beta = 10 \times 10^{-3}\ \text{m} = 10\ \text{mm}$

Q. 22 A 2A current carrying straight metal wire of resistance 1Ω, resistivity $2 \times 10^{-6}\Omega\text{m}$, area of cross-section $10\ \text{mm}^2$ and mass $500\ \text{g}$ is suspended horizontally in mid air by applying a uniform magnetic field $\vec{B}$. The magnitude of B is _____ $\times10^{-1}\ \text{T}$ (given, $\text{g} = 10\ \text{m/s}^2$)

Correct Answer:
5

Solution:
$\text{R} = \frac{\rho\ell}{\text{A}} \Rightarrow \frac{2\times10^{-6}\times\ell}{10^{-5}} = 1 \Rightarrow \ell = 5$

$\text{mg} = \text{Bi}\ell$

$\text{B} = \frac{\text{mg}}{\text{i}\ell} = \frac{5}{2\times5} = 0.5 = 5 \times 10^{-1}\ \text{Tesla}$

Q. 23 The electric field between the two parallel plates of a capacitor of $1.5\mu\mathrm{F}$ capacitance drops to one third of its initial value in $6.6\mu\mathrm{s}$ when the plates are connected by a thin wire. The resistance of this wire is _____ Ω. (Given, $\log 3 = 1.1$)

Correct Answer:
4

Solution:

$\mathrm{E} = \frac{\mathrm{E}_0}{3} \Rightarrow \mathrm{V} = \frac{\mathrm{V}_0}{3}$

$\frac{\mathrm{V}_0}{3} = \mathrm{V}_0 \mathrm{e}^{\frac{\mathrm{t}}{\tau}}$

$\mathrm{t} = \tau \ell \mathrm{n} 3$

$6.6 \times 10^{-6} = \mathrm{R}\left(1.5 \times 10^{-6}\right)(1.1)$

$\mathrm{R} = \frac{6}{1.5} = 4\Omega$

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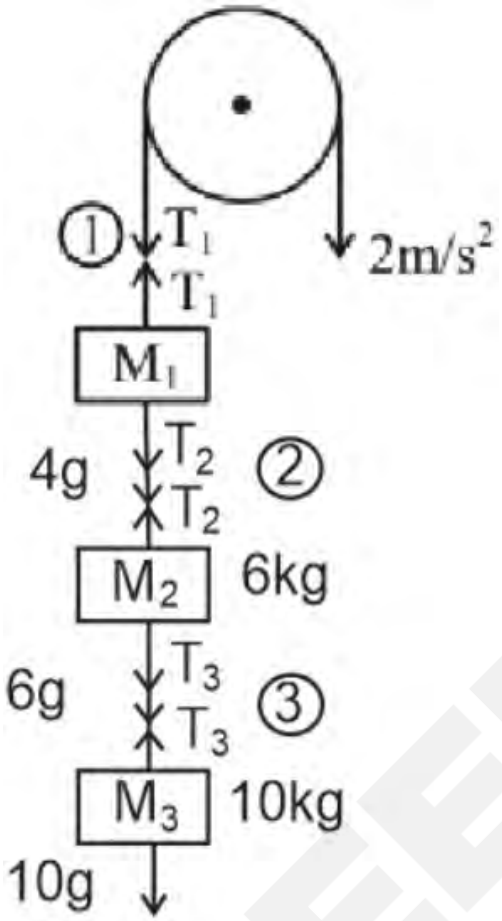
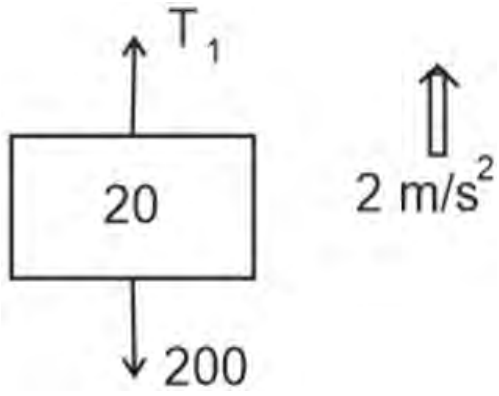
Q. 24 Three blocks M_1, M_2, M_3 having masses $4\ \text{kg}, 6\ \text{kg}$ and $10\ \text{kg}$ respectively are hanging from a smooth pully using rope 1,2 and 3 as shown in figure. The tension in the rope $1,\ \text{T}_1$ when they are moving upward with acceleration of $2\ \text{ms}^{-2}$ is ..... N (if $g = 10\ \text{m/s}^2$)



Correct Answer:
240

Solution:
FBD of M1 :



$T_1 - 200 = (4 + 6 + 10) \times 2$

$\therefore T_1 = 240$

Q. 25 The density and breaking stress of a wire are $6\times 10^4\ \mathrm{kg/m^3}$ and $1.2 \times 10^8\ \mathrm{N/m^2}$ respectively. The wire is suspended from a rigid support on a planet where acceleration due to gravity is $\frac{1^{rd}}{3}$ of the value on the surface of earth. The maximum length of the wire with breaking is _____ m (take, $\mathrm{g} = 10\ \mathrm{m/s^2}$)

Correct Answer:
600

Solution:

$T = mg$

$\sigma = \frac{T}{A} = \frac{mg}{A}$

$\frac{(\sigma A \ell)g}{A}$

$\Rightarrow \ell = \frac{\sigma}{\rho g} = \frac{1.2\times10^8\times3}{6\times10^4\times10} = 600$

Q. 26 A body moves on a frictionless plane starting from rest. If S_n is distance moved between $t = n-1$ and $t = n$ and S_{n-1} is distance moved between $t = n-2$ and $t = n-1$, then the ratio $\frac{S_{n-1}}{S_n}$ is $\left(1-\frac{2}{x}\right)$ for $n = 10$. The value of x is_____

Correct Answer:
19

Solution:
$S_n = \frac{1}{2}a(2n-1) = \frac{19a}{2}$

$S_{n-1} = \frac{1}{2}a(2n-3) = \frac{17a}{2}$

$\frac{S_{n-1}}{S_n} = \frac{17}{19} = 1-\frac{2}{x} \Rightarrow x = 19$

Q. 27 If three helium nuclei combine to form a carbon nucleus then the energy released in this reaction is _____ $\times10^{-2}$MeV. (Given $1u = 931 MeV/c^2$, atomic mass of helium $= 4.002603u$)

Correct Answer:
727

Solution:
Reaction :
$3{}_2^4He \longrightarrow {}_6^{12}C + \gamma$ rays

$\text{Mass defect } = \Delta m = (3\, m_{He} - m_C)$

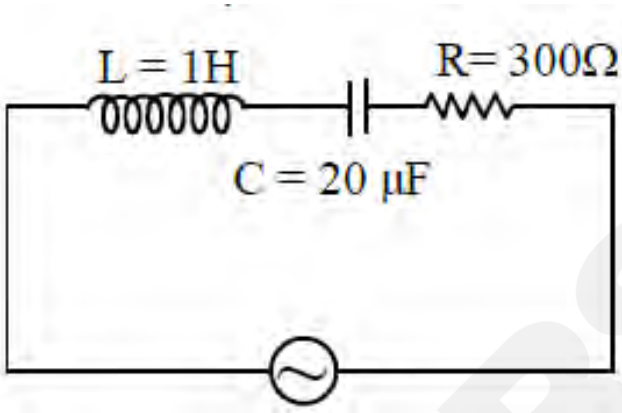
$= (3 \times 4.002603 - 12) = 0.007809\text{u}$

Energy released

$= 931\Delta\text{mMeV}$

$= 7.27\text{MeV} = 727 \times 10^{-2}\text{MeV}$

Q. 28 An ac source is connected in given series LCR circuit. The rms potential difference across the capacitor of $20\mu\text{F}$ is _____ V.



$\text{V} = 50\sqrt{2}\sin 100\text{t}$ volt

Correct Answer:
50

Solution:
$\text{X}_\text{L} = \omega\text{L} = 100 \times 1 = 100\Omega$

$\text{X}_\text{C} = \frac{1}{\omega\text{C}} = \frac{1}{100 \times 20 \times 10^{-6}} = 500\Omega$

$\text{Z} = \sqrt{(\text{X}_\text{L} - \text{X}_\text{C})^2 + \text{R}^2}$

$\sqrt{(100 - 500)^2 + 300^2}$

$\text{Z} = 500\Omega$

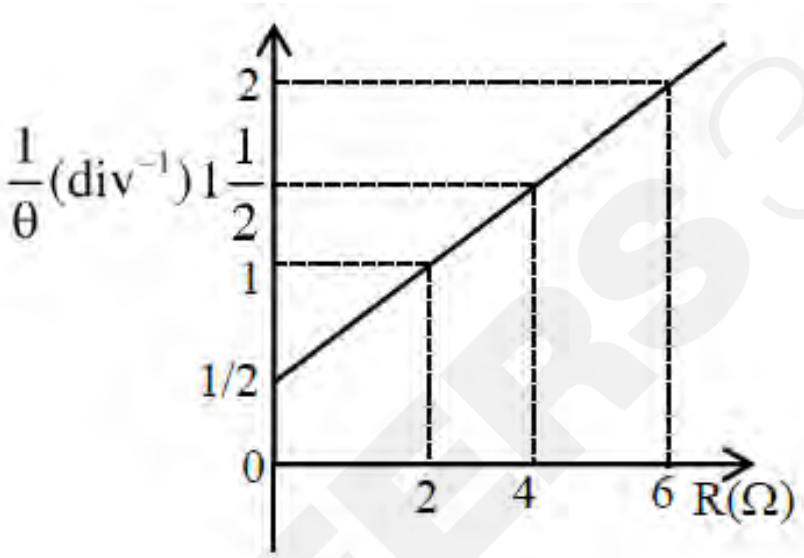
$\text{i}_\text{ms} = \frac{\text{V}_\text{ms}}{\text{Z}} = \frac{50}{500} = 0.1\ \text{A}$

rms voltage across capacitor

$V_{\text{rms}} = X_C i_{\text{rms}}$

$= 500 \times 0.1 = 50 \text{ V}$

Q. 29 In the experiment to determine the galvanometer resistance by half-deflection method, the plot of $\frac{1}{\theta}$ vs the resistance (R) of the resistance box is shown in the figure. The figure of merit of the galvanometer is _____ $\times 10^{-1}$ A /division. [The source has emf 2V]



Correct Answer:
5

Solution:
$i = K\theta$

$\frac{2}{G+R} = K\theta$

$\Rightarrow \frac{1}{\theta} = \frac{(G+R)K}{2} = R\left(\frac{K}{2}\right) + \frac{KG}{2}$

$\text{Slope} = \frac{K}{2} = \frac{1}{4} \Rightarrow K = 0.5 = 5 \times 10^{-1} A$

Q. 30 Three capacitors of capacitances $25\mu\mathrm{F}, 30\mu\mathrm{F}$ and $45\mu\mathrm{F}$ are connected in parallel to a supply of 100 V. Energy stored in the above combination is E. When these capacitors are connected in series to the same supply, the stored energy is $\frac{9}{x}E$. The value of x is _____.

Correct Answer:
86

Solution:
In parallel combination : Potential difference is same across all

$$\text{Energy} = \frac{1}{2}(\mathrm{C}_1 + \mathrm{C}_2 + \mathrm{C}_3)\mathrm{V}^2$$

$$= \frac{1}{2}(25 + 30 + 45) \times (100)^2 \times 10^{-6} = 0.5 = \mathrm{E}$$

In series combination: Charge is same on all.

$$\frac{1}{\mathrm{C}_{\text{equ}}} = \frac{1}{\mathrm{C}_1} + \frac{1}{\mathrm{C}_2} + \frac{1}{\mathrm{C}_3} = \frac{1}{25} + \frac{1}{30} + \frac{1}{45}$$

$$\frac{1}{\mathrm{C}_{\text{equ}}} = \frac{(18+15+10)}{450} = \frac{43}{450} \Rightarrow \mathrm{C}_{\text{equ}} = \frac{450}{43}$$

$$\text{Energy} = \frac{Q^2}{2C_1} + \frac{Q^2}{2C_2} + \frac{Q^2}{2C_3}$$

$$= \frac{Q^2}{2}\left[\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right]$$

$$\frac{(V \times C_{\text{equ}})^2}{2} \times \frac{1}{C_{\text{equ}}} = \frac{V^2 C_{\text{equ}}}{2}$$

$$\frac{(100)^2}{2} \times \frac{450}{43} \times 10^{-6}$$

$$\Rightarrow \frac{4.5}{86} = \frac{9}{x}E = \frac{9}{x} \times 0.5 \Rightarrow x = 86$$

Chemistry

Q. 1 The incorrect postulates of the Dalton's atomic theory are :
(A) Atoms of different elements differ in mass.
(B) Matter consists of divisible atoms.
(C) Compounds are formed when atoms of different element combine in a fixed ratio.
(D) All the atoms of given element have different properties including mass.
(E) Chemical reactions involve reorganisation of atoms.

Choose the correct answer from the options given below :

Option 1:
(B), (D), (E) only

Option 2:
(A), (B), (D) only

Option 3:
(C), (D), (E) only

Option 4:
(B), (D) only

Solution:
from the key concept

All the atoms of a given element have different properties including mass.

Matter consists of divisible atoms.

Hence, the answer is the option (4).

Q. 2 The following reaction occurs in the Blast furnance where iron ore is reduced to iron metal

$$Fe_2O_{3(s)} + 3CO_{(g)} \rightleftharpoons Fe_{(l)} + 3CO_{2(g)}$$

Using the Le-chatelier's principle, predict which one of the following will not disturb the equilibrium.

Option 1:
Addition of Fe_2O_3

Option 2:
Addition of CO_2

Option 3:
Removal of CO

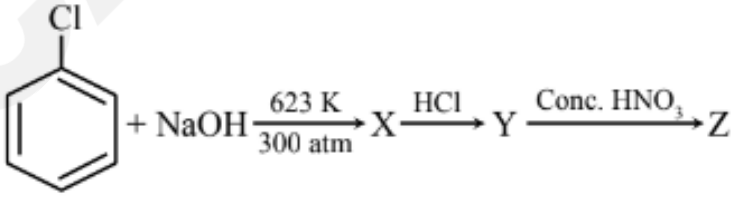
Option 4:
Removal of CO_2

Correct Answer:
Addition of Fe_2O_3

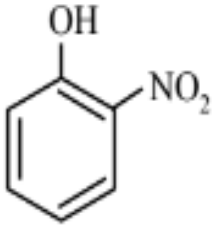
Solution:
When solid is added no effect on equilibrium.

Hence, the answer is the option (1).

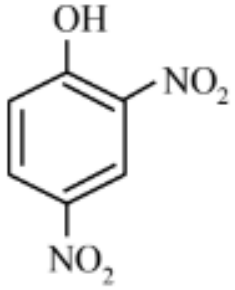
Q. 3 Identify compound (Z) in the following reaction sequence.



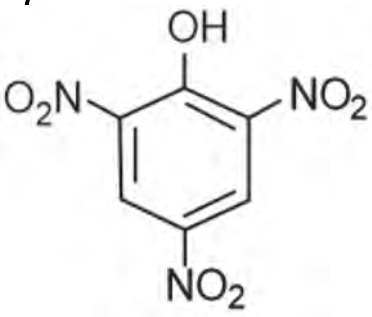
Option 1:



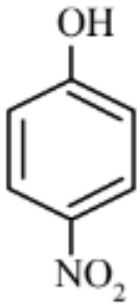
Option 2:



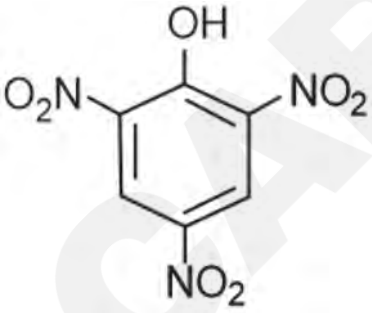
Option 3:



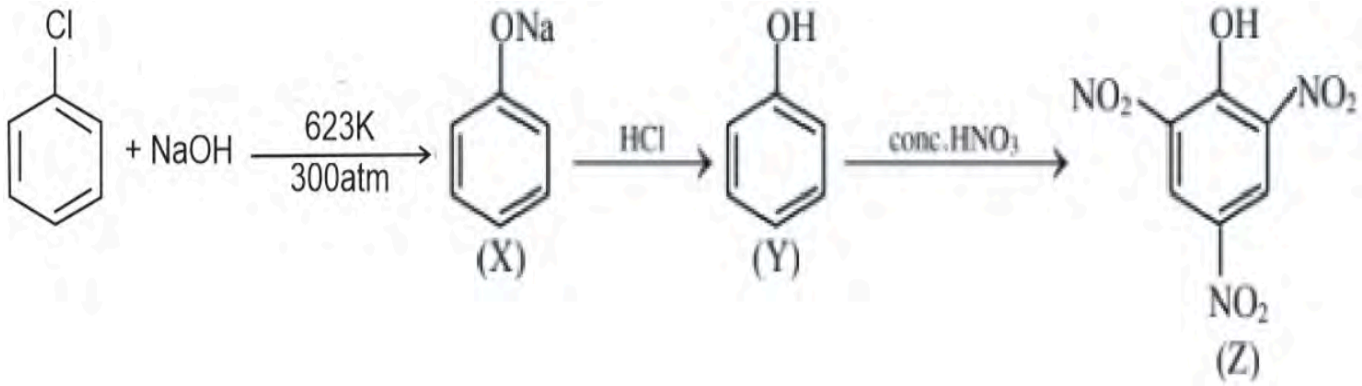
Option 4:



Correct Answer:



Solution:



Q. 4 Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**

Assertion (A): Enthalpy of neutralisation of strong monobasic acid with strong monoacidic base is always $-57\ \mathrm{kJ\ mol^{-1}}$
Reason (R): Enthalpy of neutralisation is the amount of heat liberated when one mole of $\mathrm{H^+}$ ions furnished by acid combine with one mole of $\mathrm{^-OH}$ ions furnished by base to form one mole of water.
In the light of the above statements, choose the **correct** answer from the options given below.

Option 1:
(A) is true but **(R)** is false

Option 2:
Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

Option 3:
(A) is false but **(R)** is true

Option 4:
Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**

Correct Answer:
Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

Solution:
Enthalpy of neutralization of SA & SB is always $-57\ \mathrm{kJ/mol}$ because strong monoacid gives one mole of $\mathrm{H^+}$ and strong mono base gives one mole of $\mathrm{OH^-}$ which form one mole of water.

Hence, the answer is the option (2).

Q. 5 The statement(s) that are **correct** about the species O^{2-}, F^{-}, Na^{+} and Mg^{2+}.

(A) All are isoelectronic

(B) All have the same nuclear charge

(C) O^{2-} has the largest ionic radii

(D) Mg^{2+} has the smallest ionic radii

Choose the **most appropriate** answer from the options given below :

Option 1:
(B), (C) and (D) only

Option 2:
(A), (B), (C) and (D)

Option 3:
(C) and (D) only

Option 4:
(A), (C) and (D) only

Correct Answer:
(A), (C) and (D) only

Solution:

	O^{-2}	F^{-}	Na^{+}	Mg^{+2}
(No. of e^{-})	10	10	10	10
(Ionic radius)	$O^{-2} > F^{-} > Na^{+} > Mg^{+2}$			
Zeff	$O^{-2} < F^{-} < Na^{+} < Mg^{+2}$			

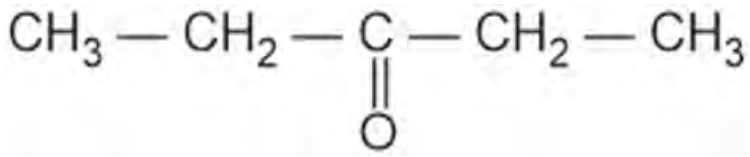
Hence, the answer is the option (4).

Q. 6 For the compounds:

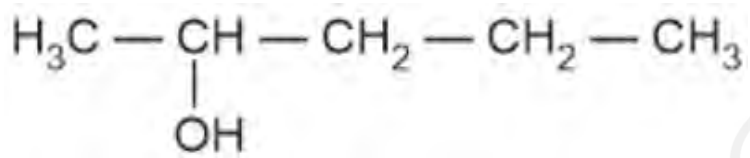
(A) $H_3C-CH_2-O-CH_2-CH_2-CH_3$

(B) $H_3C-CH_2-CH_2-CH_2-CH_3$

(C)



(D)



The increasing order of boiling point is : Choose the **correct** answer from the options given below :

Option 1:

$(A) < (B) < (C) < (D)$

Option 2:

$(B) < (A) < (C) < (D)$

Option 3:

$(D) < (C) < (A) < (B)$

Option 4:

$(B) < (A) < (D) < (C)$

Correct Answer:

$(B) < (A) < (C) < (D)$

Solution:

Compounds having same number of carbon atoms follow the boiling point order as:

$(\text{Boiling point})_{\text{Hydrogen bonding}} > (\text{Boiling point})_{\text{high polanity}} > (\text{Boiling point})_{\text{low polarity}} > (\text{Boiling point})_{\text{non polar}}$

Q. 7 Given below are two statements :
Statement I: In group 13, the stability of +1 oxidation state increases down the group.
Statement II: The atomic size of gallium is greater than that of aluminium.
In the light of the above statements, choose the **most appropriate** answer from the options given below:

Option 1:
Statement I is incorrect but **Statement II** is correct

Option 2:
Both **Statement I** and **Statement II** are correct

Option 3:
Both **Statement I** and **Statement II** are incorrect

Option 4:
Statement I is correct but **Statement II** is incorrect

Correct Answer:
Statement I is correct but **Statement II** is incorrect

Solution:
Statement I : Number of $d\&$ f electrons, increases down the group and due to poor shielding of $d\&f$ e, stability of lower oxidation states increases down the group
Statement II : The atomic size of aluminium is greater than that of gallium.

Q. 8 Number of σ and π bonds present in ethylene molecule is respectively:

Option 1:
3 and 1

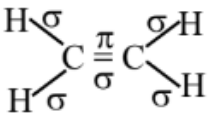
Option 2:
5 and 2

Option 3:
4 and 1

Option 4:
5 and 1

Correct Answer:
5 and 1

Solution:

ethylene is  ,it has 5σ bonds and 1π bond

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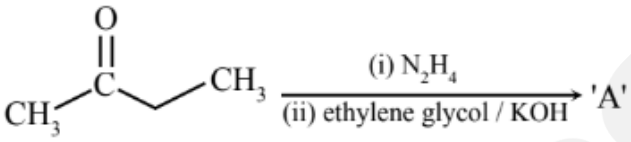
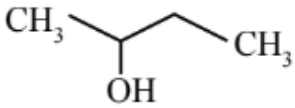
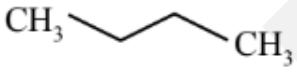
Q. 9 Identify ' A ' in the following reaction :



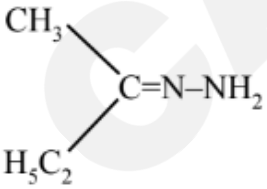
Option 1:



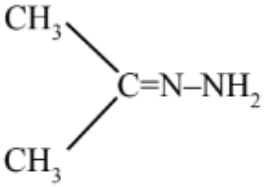
Option 2:



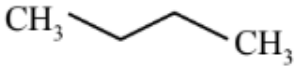
Option 3:



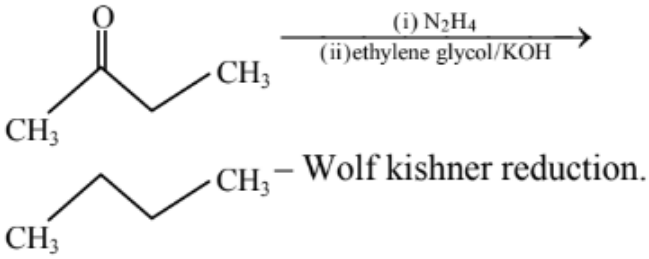
Option 4:



Correct Answer:



Solution:



Q. 10 The reaction at cathode in the cells commonly used in clocks involves.

Option 1:
reduction of Mn from +4 to +3

Option 2:
oxidation of Mn from +3 to +4

Option 3:
reduction of Mn from +7 to +2

Option 4:
oxidation of Mn from +2 to +7

Correct Answer:
reduction of Mn from +4 to +3

Solution:
In the cathode reaction manganese (Mn) is reduced from the +4 oxidation state to the +3 state.

Q. 11 Which one of the following complexes will exhibit the least paramagnetic behaviour?
[Atomic number, $\mathrm{Cr}=24, \mathrm{Mn}=25, \mathrm{Fe}=26, \mathrm{Co}=27$]

Option 1:
$[Co(H_2O)_6]^{2+}$

Option 2:
$[Fe(H_2O)_6]^{2+}$

Option 3:
$[Mn(H_2O)_6]^{2+}$

Option 4:
$[Cr(H_2O)_6]^{2+}$

Correct Answer:
$[Co(H_2O)_6]^{2+}$

Solution:

	Number of unpaired e^-	$\mu = \sqrt{n(n+2)}$ B.M.
$[Co(H_2O)_6]^{2+}$	3	3.87
$[Fe(H_2O)_6]^{2+}$	4	4.89
$[Mn(H_2O)_6]^{2+}$	5	5.92
$[Cr(H_2O)_6]^{2+}$	4	4.89

Least paramagnetic behaviour $= [Co(H_2O)_6]^{2+}$

Hence, the answer is option (1).

Q. 12 Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A): Cis form of alkene is found to be more polar than the trans form
Reason (R): Dipole moment of trans isomer of 2-butene is zero.
In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both **(A)** and **(R)** are true but **(R)** is NOT the correct explanation of **(A)**

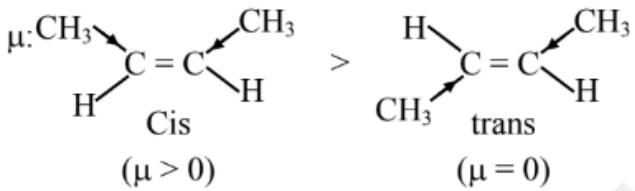
Option 2:
(A) is true but **(R)** is false

Option 3:
Both **(A)** and **(R)** are true and **(R)** is the correct explanation of (A)

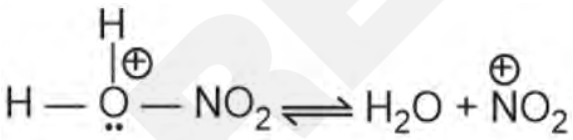
Option 4:
(A) is false but **(R)** is true

Correct Answer:
Both **(A)** and **(R)** are true and **(R)** is the correct explanation of (A)

Solution:
Dipole moment is a vector quantity and for compound net dipole moment is the vector sum of all dipoles hence dipole moment of cis form is greater than trans form.



Q. 13 Given below are two statements :
Statement I: Nitration of benzene involves the following step -



Statement II: Use of Lewis base promotes the electrophilic substitution of benzene.
In the light of the above statements, choose the **most appropriate** answer from the options given below :

Option 1:
Both **Statement I** and **Statement II** are incorrect

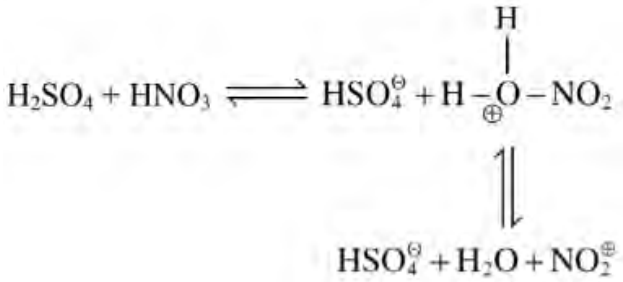
Option 2:
Statement I is correct but **Statement II** is incorrect

Option 3:
Both **Statement I** and **Statement II** are correct

Option 4:
Statement I is incorrect but **Statement II** is correct

Correct Answer:
Statement I is correct but **Statement II** is incorrect

Solution:
In nitration of benzene concentrated H_2SO_4 and HNO_3 is used as reagent which generates electrophile $\overset{\oplus}{O}_2$ in following steps:



Lewis acids can promote the formation of electrophiles not Lewis base

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Q. 14 The correct order of ligands arranged in increasing field strength.

Option 1:
$Cl^- < -OH < Br^- < CN^-$

Option 2:
$F^- < Br^- < I^- < NH_3$

Option 3:
$Br^- < F^- < H_2O < NH_3$

Option 4:
$H_2O < {}^-OH < CN^- < NH_3$

Correct Answer:
$Br^- < F^- < H_2O < NH_3$

Solution:
Experimental order $Br^- < F^- < H_2O < NH_3$

Hence, the answer is the option (3).

Q. 15 Which of the following gives a positive test with ninhydrin?

Option 1:
Cellulose

Option 2:
Starch

Option 3:
Polyvinyl chloride

Option 4:
Egg albumin

Correct Answer:
Egg albumin

Solution:
Ninhydrin test is a test of amino acids. Egg albumin contains protein which is a natural polymer of amino acids which will show positive ninhydrin test

Q. 16 The metal that shows highest and maximum number of oxidation state is:

Option 1:
Fe

Option 2:
Mn

Option 3:
Ti

Option 4:
Co

Correct Answer:
Mn

Solution:
Mn shows highest oxidation state $\left(\mathrm{Mn}^{+7}\right)$ in $3\ \mathrm{d}$ series metals.

Q. 17 Ail organic compound has 42.1% carbon, 6.4% hydrogen and remainder is oxygen. If its molecular weight is 342 , then its molecular formula is :

Option 1:
$\mathrm{C}_{11}\mathrm{H}_{18}\mathrm{O}_{12}$

Option 2:
$\mathrm{C}_{12}\mathrm{H}_{20}\mathrm{O}_{12}$

Option 3:
$\mathrm{C}_{14}\mathrm{H}_{20}\mathrm{O}_{10}$

Option 4:
$\mathrm{C}_{12}\mathrm{H}_{22}\mathrm{O}_{11}$

Correct Answer:
$\mathrm{C}_{12}\mathrm{H}_{22}\mathrm{O}_{11}$

Solution:
only $\mathrm{C}_{12}\mathrm{H}_{22}\mathrm{O}_{11}$ has 42.1% carbon, 6.4% hydrogen & 51.5 percent oxygen.

Hence, the answer is the option (4).

Q. 18 Given below are two statement :

Statement I : Bromination of phenol in solvent with low polarity such as $CHCl_3$ or CS_2 requires Lewis acid catalyst.

Statement II : The lewis acid catalyst polarises the bromine to generate Br^+.

In the light of the above statements, choose the **correct** answer from the options given below :

Option 1:
Statement I is true but Statement II is false.

Option 2:
Both Statement I and Statement II are true

Option 3:
Both Statement I and Statement II are false.

Option 4:
Statement I is false but Statement II is true.

Correct Answer:
Statement I is false but Statement II is true.

Solution:
Phenol is a highly activated compound which can undergo bromination directly with Bromine without any lewis acid.

Q. 19 Molar ionic conductivities of divalent cation and anion are $57\ S\ cm^2\ mol^{-1}$ and $73\ S\ cm^2\ mol^{-1}$ respectively. The molar conductivity of solution of an electrolyte with the above cation and anion will be :

Option 1:
$65\ S\ cm^2\ mol^{-1}$

Option 2:
$130\ \mathrm{S\ cm^2\ mol^{-1}}$

Option 3:
$187\ \mathrm{S\ cm^2\ mol^{-1}}$

Option 4:
$260\ \mathrm{S\ cm^2\ mol^{-1}}$

Correct Answer:
$130\ \mathrm{S\ cm^2\ mol^{-1}}$

Solution:
$\Lambda_{\mathrm{C}}^{+2} = 57 \mathrm{Scm^2\ mol^{-1}}$
$\Lambda_{\mathrm{A}}^{+2} = 73 \mathrm{Scm^2\ mol^{-1}}$
$\Lambda_{\mathrm{Solution}} = \lambda_{\mathrm{C}}^{+2} + \Lambda_{\mathrm{A}}^{-2}$
$= 57 + 73 = 130$

Q. 20 The number of neutrons present in the more abundant isotope of boron is ' x '. Amorphous boron upon heating with air forms a product, in which the oxidation state of boron is ' y '. The value of $x + y$ is ...

Option 1:
4

Option 2:
6

Option 3:
3

Option 4:
9

Correct Answer:
9

Solution:

$\text{More abundant isotope } = \mathrm{B}^{11}$
$[\text{Number of neutrons } = 6]$
$\mathrm{x} = 6$
$\mathrm{B} + \mathrm{O}_2 \rightarrow \mathrm{B}_2\mathrm{O}_3$

Oxidation state of B in $\mathrm{B}_2\mathrm{O}_3 = +3$
So, $y = 3$
Hence $x + y = 9$

Q. 21 The value of Rydberg constant $(\mathrm{R_H})$ is $2.18 \times 10^{-18}\ \mathrm{J}$. The velocity of electron having mass $9.1 \times 10^{-31}\ \mathrm{kg}$ in Bohr's first orbit of hydrogen atom $= \ldots\ldots\ldots \times 10^5\ \mathrm{ms}^{-1}$ (nearest integer)

Correct Answer:
22

Solution:
$$\mathrm{V} = 2.18 \times 10^6 \times \frac{\mathrm{Z}}{\mathrm{n}}$$
$$= 21.8 \times 10^5 \times \frac{1}{1} \approx 22 \times 10^5\ (\text{nearest})$$

Hence, the answer is the option (1).

Q. 22

In a borax bead test under hot condition, a metal salt (one from the given) is heated at point B of the flame, resulted in green colour salt bead. The spin-only magnetic moment value of the salt is BM (Nearest integer)
[Given atomic number of $\mathrm{Cu} = 29, \mathrm{Ni} = 28, \mathrm{Mn} = 25, \mathrm{Fe} = 26$]

Correct Answer:
6

Solution:

Fe^{+3} will give green coloured bead when heated at point B.

Number of unpaired e^- in $Fe^{+3} = 5$

$\mu = 5.92$

Nearest integer $= 6$

Q. 23 The heat of combustion of solid benzoic acid at constant volume is $-321.30\ kJ$ at $27^\circ C$. The heat of combustion at constant pressure is $(-321.30 - xR)\ kJ$, the value of x is

---

Correct Answer:

150

Solution:

$$C_6H_5COOH(S) + \frac{15}{2}O_2(g) \rightarrow 7CO_2(g) + 3H_2O(\ell)$$

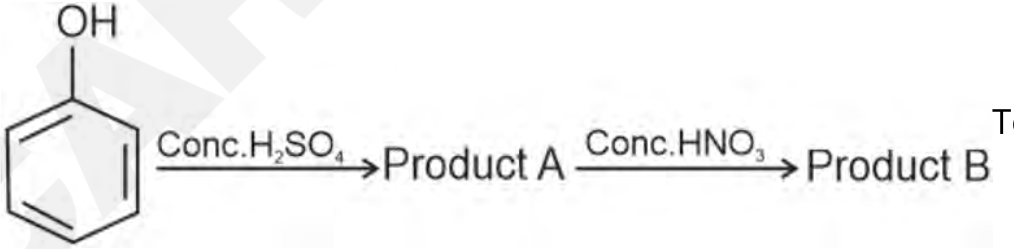
$$\Delta H = \Delta U + \Delta n_g RT$$

$$= -321.30 - \frac{1}{2}\frac{R}{100} \times 300$$

$$= (-321.30 - 150R)kJ$$

Hence, the answer is the option (1).

Q. 24 Consider the given chemical reaction sequence :



Total sum of oxygen atoms in Product A and Product B are.....

Correct Answer:

14

Solution:

Picric acid is prepared by treating phenol first with concentrated sulphuric acid which converts it to phenol-2,4-disulphonic acid and then with concentrated nitric acid to get 2, 4, 6 trinitrophenol.

Q. 25 The spin only magnetic moment value of the ion among Ti^{2+}, V^{2+}, Co^{3+} and Cr^{2+}, that acts as strong oxidising agent in aqueous solution is ..... BM (Near integer).
(Given atomic numbers: $Ti : 22,\ V : 23, Cr : 24$, Co : 27)

Correct Answer:
5

Solution:
Strong oxidising agent $= Co^{+3}$
No. of unpaired e^- in $Co^{+3}\ [3\ d^6] = 4$
Hence $\mu = \sqrt{n(n+2)} = \sqrt{24}BM$
Nearest integer $= 5$

Q. 26 During Kinetic study of reaction $2\ A + B \rightarrow C + D$, the following results were obtained :

	A[M]	B[M]	initial rate of formation of D
I	0.1	0.1	6.0×10^{-3}
II	0.3	0.2	7.2×10^{-2}
III	0.3	0.4	2.88×10^{-1}
IV	0.4	0.1	2.40×10^{-2}

Based on above data, overall order of the reaction is.....

Correct Answer:
3

Solution:

$r = K[A]^x[B]^y$

$(I) 6 \times 10^{-3} = K[0.1]^x[0.1]^y$

$(IV) 2.4 \times 10^{-2} = K[0.4]^x[0.1]^y$

$(IV)/(I)$

$4 = (4)^x$

$x = 1$

$r = K[A]^x[B]^y$

$(III)\ 2.88 \times 10^{-1} = K[0.3]^x[0.4]^y$

$(II)\ 7.2 \times 10^{-2} = K[0.3]^x[0.2]^y$

$(III)/(II)$

$4 = 2^y$

$y = 2$

$\text{Overall order } = x + y = 1 + 2 = 3$

Q. 27 An artificial cell is made by encapsulating 0.2M glucose solution within a semipermeable membrane. The osmotic pressure developed when the artificial cell is placed within a 0.05 M solution of $NaCl$ at $300\ K$ is_____$\times 10^{-1}$ bar. (Nearest Integer)

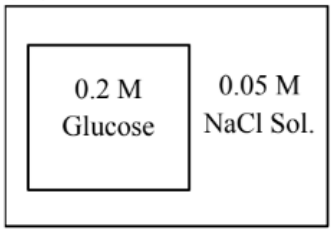
[Given : $R = 0.083\ Lbarmol^{-1}\ K^{-1}$]

Assume complete dissociation of $NaCl$

Correct Answer:

25

Solution:



$NaCl \longrightarrow Na^{+} + Cl^{-}$

$0.05M \quad 0.05M \quad 0.05M$

$\text{Total } C_1 = 0.05 + 0.05 = 0.1M(NaCl)$

$C_2 = 0.2M(\text{ glucose })$

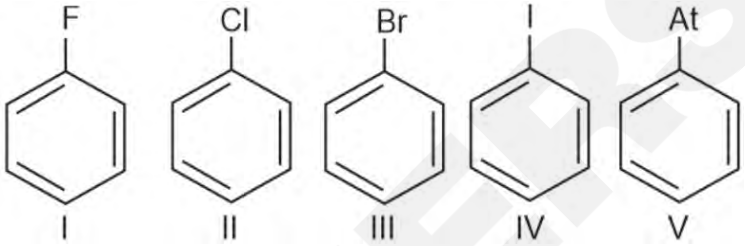
$\pi = (C_2 - C_1)RT$

$= (0.2 - 0.1) \times 0.083 \times 300$

$= 2.49 \text{ bar}$

$= 24.9 \times 10^{-1} \text{bar}$

Q. 28 The number of halobenzenes from the following that can be prepared by Sandmeyer's reaction is.....



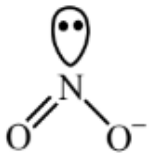
Correct Answer:
2

Solution:
In Sandmayer reaction only bromobenzene \& chlorobenzene are prepared

Q. 29 In the lewis dot structure for NO_2^-, total number of valence electrons around nitrogen is.....

Correct Answer:
8

Solution:

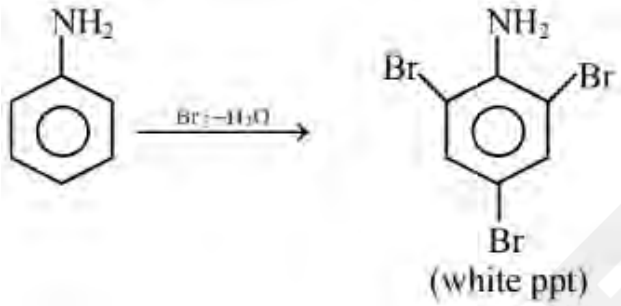


Number of valence e^- around N-atom =8

Q. 30 9.3 g of pure aniline is treated with bromine water at room temperature to give a white precipitate of the product ' P '. The mass of product ' P ' obtained is 26.4 g . The percentage yield is ..... %.

Correct Answer:
80

Solution:



93 g of aniline produces 330 g of 2,4,6- tribromoaniline. Hence 9.3 g of aniline should produce 33 g of 2, 4, 6-tribromoaniline. Hence percentage yield $\frac{26.4\times100}{33} = 80\%$

Maths

Q. 1 Let d be the distance of the point of intersection of the lines $\frac{x+6}{3} = \frac{y}{2} = \frac{z+1}{1}$ and $\frac{x-7}{4} = \frac{y-9}{3} = \frac{z-4}{2}$ from the point (7,8,9). Then $d^2 + 6$ is equal to :

Option 1:
72

Option 2:
69

Option 3:
75

Option 4:
78

Correct Answer:
75

Solution:

$$\frac{x+6}{3}=\frac{y}{2}=\frac{z+1}{1}=\lambda-----(1)$$

$$x=3\lambda-6, y=2\lambda, z=\lambda-1$$

$$\frac{x-7}{4}=\frac{y-9}{3}=\frac{z-4}{2}=\mu------(2)$$

$$x=4\mu+7, y=3\mu+9, z=2\mu+4$$

$$3\lambda-6=4\mu+7\Rightarrow 3\lambda-4\mu=13---(3)\times 2$$

$$2\lambda=3\mu+9\Rightarrow 2\lambda-3\mu=9---(4)\times 3$$

$$6\lambda-8\mu=26$$

$$6\lambda-9\mu=27$$

$$\frac{-+}{\mu=-1}$$

$$\Rightarrow 3\lambda-4(-1)=13$$

$$3\lambda=9$$

$$\lambda=3$$

$$\text{int. point } (3,6,2);(7,8,9)$$

$$d^2=16+4+49=69$$

Ans. $d^2+6=69+6=75$

Q. 2 Let a rectangle ABCD of sides 2 and 4 be inscribed in another rectangle PQRS such that the vertices of the rectangle ABCD lie on the sides of the rectangle PQRS. Let a and b be the sides of the rectangle PQRS when its area is maximum. Then $(a+b)^2$ is equal to :

Option 1:
72

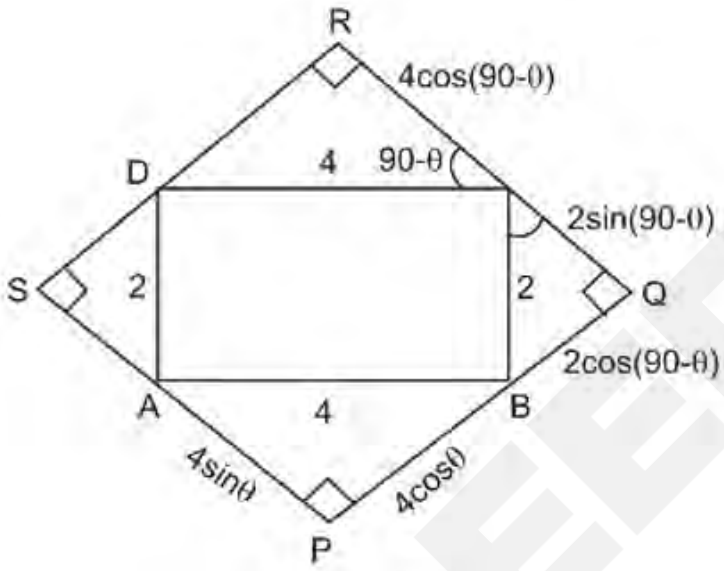
Option 2:
60

Option 3:
80

Option 4:
64

Correct Answer:
72

Solution:



$$\text{Area } = (4\cos\theta + 2\sin\theta)(2\cos\theta + 4\sin\theta)$$
$$= 8\cos^2\theta + 16\sin\theta\cos\theta + 4\sin\theta\cos\theta + 8\sin^2\theta$$
$$= 8 + 20\sin\theta\cos\theta$$
$$= 8 + 10\sin 2\theta$$
$$\text{Max Area } = 8 + 10 = 18(\sin 2\theta = 1)\theta = 45^\circ$$
$$(a+b)^2 = (4\cos\theta + 2\sin\theta + 2\cos\theta + 4\sin\theta)^2$$
$$= (6\cos\theta + 6\sin\theta)^2$$
$$= 36(\sin\theta + \cos\theta)^2$$
$$= 36(\sqrt{2})^2$$
$$= 72$$

Q. 3 Let two straight lines drawn from the origin O intersect the line $3x + 4y = 12$ at the points P and Q such that $\triangle \mathrm{OPQ}$ is an isosceles triangle and $\angle \mathrm{POQ} = 90^{\circ}$. If $l = \mathrm{OP}^2 + \mathrm{PQ}^2 + \mathrm{QO}^2$, then the greatest integer less than or equal to l is :

Option 1:
44

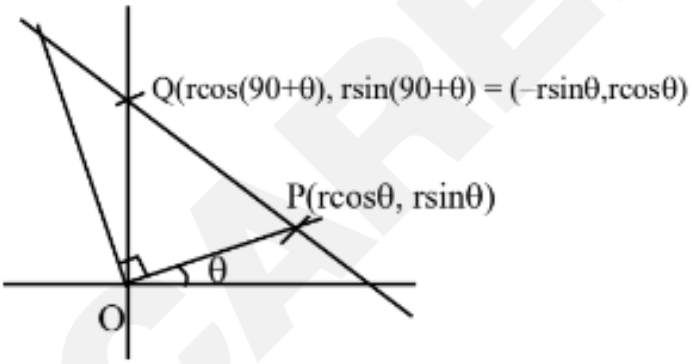
Option 2:
48

Option 3:
46

Option 4:
42

Correct Answer:
46

Solution:



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$3x + 4y = 12$

$3(r\cos\theta) + 4(r\sin\theta) = 12$

$r(3\cos\theta + 4\sin\theta) = 12 \ldots (1)$

$3(-r\sin\theta) + 4(r\cos\theta) = 12$

$r(-3\sin\theta + 4\cos\theta) = 12 \ldots (2)$

$$\left(\frac{12}{r}\right)^2 + \left(\frac{12}{r}\right)^2 = (3\cos\theta + 4\sin\theta)^2 + (-3\sin\theta + 4\cos\theta)^2$$

$$2\left(\frac{12}{r}\right)^2 = 9 + 16$$

$$\frac{2 \times 144}{r^2} = 25 \Rightarrow 288 = 25r^2$$

$$\Rightarrow \frac{288}{25} = \mathrm{r}^2$$

$$\Rightarrow \sqrt{2}\left(\frac{12}{5}\right) = \mathrm{r}$$

$\ell = \mathrm{OP}^2 + \mathrm{PQ}^2 + \mathrm{QO}^2$

$\ell = \mathrm{r}^2 + \mathrm{r}^2 + \mathrm{r}^2(\cos\theta + \sin\theta)^2 + \mathrm{r}^2(\sin\theta + \cos\theta)^2$

$= 2\mathrm{r}^2 + \mathrm{r}^2(1 + \sin 2\theta + 1 - 2\sin 2\theta)$

$= 2\mathrm{r}^2 + 2\mathrm{r}^2$

$= 4\mathrm{r}^2$

$$= 4\left(\frac{288}{25}\right) = \frac{1152}{25} = 46.08$$

$[\ell] = 46$

Q. 4 If $\mathrm{y} = \mathrm{y}(\mathrm{x})$ is the solution of the differential equation $\frac{\mathrm{dy}}{\mathrm{dx}} + 2\mathrm{y} = \sin(2\mathrm{x}), \mathrm{y}(0) = \frac{3}{4}$, then $\mathrm{y}\left(\frac{\pi}{8}\right)$ is equal to :

Option 1:
$\mathrm{e}^{-\pi/8}$

Option 2:
$\mathrm{e}^{-\pi/4}$

Option 3:
$e^{\pi/4}$

Option 4:
$e^{\pi/8}$

Correct Answer:
$e^{-\pi/4}$

Solution:

$$\frac{dy}{dx}+2y=\sin 2x, y(0)=\frac{3}{4}$$

$$I\cdot F=e^{\int 2dx}=e^{2x}$$

$$y\cdot e^{2x}=\int e^{2x}\sin 2xdx$$

$$y\cdot e^{2x}=\frac{e^{2x}(2\sin 2x-2\cos 2x)}{4+4}+C$$

$$x=0, y=\frac{3}{4}\Rightarrow \frac{3}{4}\cdot 1=\frac{1(0-2)}{8}+C$$

$$\frac{3}{4}=-\frac{1}{4}+C$$

$$1=C$$

$$y=\frac{2\sin 2x-2\cos 2x}{8}+1.e^{-2x}$$

$$x=\frac{\pi}{8}, y=\frac{1}{8}\left(2\sin\frac{\pi}{4}-2\cos\frac{\pi}{4}\right)+e^{-2\left(\frac{\pi}{8}\right)}$$

$$y=0+e^{-\frac{\pi}{4}}$$

Hence, the answer is the option (2).

Q. 5 For the function

$f(x)=\sin x+3x-\frac{2}{\pi}\left(x^2+x\right)$, where $x\in\left[0,\frac{\pi}{2}\right]$

Consider the following statements,
(I) f is increasing in $\left(0,\frac{\pi}{2}\right)$.
(II) f' is decreasing in $\left(0,\frac{\pi}{2}\right)$.

Between the above two statements,

Option 1:
only (I) is true.

Option 2:
only (II) is true.

Option 3:
neither (I) nor (II) is true.

Option 4:
both (I) and (II) are true.

Correct Answer:
both (I) and (II) are true.

Solution:

$$f(x) = \sin x + 3x - \frac{2}{\pi}(x^2 + x) \quad x \in \left[0, \frac{\pi}{2}\right]$$

$$f'(x) = \cos x + 3 - \frac{2}{\pi}(2x + 1) > 0 f(x) \uparrow$$

$$f'(x) = -\sin x + 0 - \frac{\pi}{2}(2)$$

$$= -\sin x - \frac{4}{\pi} < 0 \quad f'(x) \downarrow$$

$$0 < x < \frac{\pi}{2}$$

$$\Rightarrow -\tfrac{2}{\pi}(\underset{+1}{0} < \underset{+1}{2x} < \underset{+1}{\pi})$$

$$-\tfrac{2}{\pi} > \tfrac{-2}{\pi}(2x + 1) > -\tfrac{2}{\pi}(\pi + 1)$$

$$\overset{+3}{} \qquad \overset{+3}{} \qquad \overset{+3}{}$$

$$3 - \tfrac{2}{\pi} > 3 - \tfrac{2}{\pi}(2x + 1) > 3 - \tfrac{2}{\pi}(\pi + 1)$$

(+ve) (+ve)

Q. 6 If the system of equations

$$11x + y + \lambda z = -5$$
$$2x + 3y + 5z = 3$$
$$8x - 19y - 39z = \mu$$

has infinitely many solutions, then $\lambda^4 - \mu$ is equal to :

Option 1:
49

Option 2:
45

Option 3:
47

Option 4:
51

Correct Answer:
47

Solution:

$11x + y + \lambda z = -5$
$2x + 3y + 5z = 3$
$8x - 19y - 39z = \mu$

for infinite sol.

$$\mathrm{D} = \begin{vmatrix} 11 & 1 & \lambda \\ 2 & 3 & 5 \\ 8 & -19 & -39 \end{vmatrix} = 0$$

$\Rightarrow 11(-117 + 95) - 1(-78 - 40) + \lambda(-38 - 24)$
$\Rightarrow 11(-22) + 118 - \lambda(62) = 0$
$\Rightarrow 62\lambda = 118 - 242$
$\Rightarrow \lambda = \dfrac{-124}{62} = -2$

$$\mathrm{D}_1 = \begin{vmatrix} -5 & 1 & -2 \\ 3 & 3 & 5 \\ \mu & -19 & -39 \end{vmatrix} = 0$$

$\Rightarrow -5(-117 + 95) - 1(-117 - 5\mu) - 2(-57 - 3\mu) = 0$
$\Rightarrow -5(-22) + 117 + 5\mu + 114 + 6\mu = 0$
$\Rightarrow 11\mu = -110 - 231 = -341$
$\Rightarrow \mu = -31$
$\lambda^4 - \mu = (-2)^4 - (-31) = 16 + 31 = 47$

Q. 7 Let A={1,3,7,9,11} and B={2,4,5,7,8,10,12}. Then the total number of one-one maps $f : A \rightarrow B$, such that f(1)+f(3)=14, is :

Option 1:
180

Option 2:
20

Option 3:
480

Option 4:
240

Correct Answer:
240

Solution:
$\mathrm{A} = \{1, 3, 7, 9, 11\}$
$\mathrm{B} = \{2, 4, 5, 7, 8, 10, 12\}$
$\mathrm{f}(1) + \mathrm{f}(3) = 14$
(i) $2 + 12$
(ii) $4 + 10$
$2 \times (2 \times 5 \times 4 \times 3) = 240$

Hence, the answer is the option (4).

Q. 8 If the function $\mathrm{f}(\mathrm{x}) = \frac{\sin 3\mathrm{x} + \alpha \sin \mathrm{x} - \beta \cos 3\mathrm{x}}{\mathrm{x}^3}$, $\mathrm{x} \in \mathrm{R}$, is continuous at x=0, then f(0) is equal to :

Option 1:
2

Option 2:
-2

Option 3:
4

Option 4:
-4

Correct Answer:
-4

Solution:

$$f(x)=\frac{\sin 3x+\alpha \sin x-\beta \cos 3x}{x^3}$$

is continuous at x=0

$$\lim_{x\to 0}=\frac{3x-\frac{(3x)^3}{3}+\ldots+\alpha\left(x-\frac{x^3}{3}\ldots\right)-\beta\left(1-\frac{(3x)^2}{2}\ldots\right)}{x^3}=f(0)$$

$$\lim_{x\to 0}=\frac{-\beta+x(3+\alpha)+\frac{9\beta x^2}{\lfloor 2}+\left(\frac{-27}{3}-\frac{\alpha}{3}\right)x^3\ldots}{x^3}=f(0)$$

For exist

$$\beta=0, 3+\alpha=0, -\frac{27}{\lfloor 3}-\frac{\alpha}{\lfloor 3}=f(0)$$

$$\alpha=-3, -\frac{27}{6}-\frac{(-3)}{6}=f(0)$$

$$f(0)=\frac{-27+3}{6}=-4$$

Hence, the answer is the option (4).

Q. 9 The integral $\int_0^{\frac{\pi}{4}}\frac{136\sin x}{3\sin x+5\cos x}dx$ is equal to :

Option 1:
$3\pi-50\log_e 2+20\log_e 5$

Option 2:
$3\pi-25\log_e 2+10\log_e 5$

Option 3:

$3\pi - 10\log_e(2\sqrt{2}) + 10\log_e 5$

Option 4:

$3\pi - 30\log_e 2 + 20\log_e 5$

Correct Answer:

$3\pi - 50\log_e 2 + 20\log_e 5$

Solution:

$$I = \int_0^{\pi/4} \frac{136\sin x}{3\sin x + 5\cos x}dx$$

$$136\sin x = A(3\sin x + 5\cos x) + B(3\cos x - 5\sin x)$$

$$136 = 3A - 5B - - - - - -(A)$$

$$0 = 5A + 3B - - - - - -(B)$$

$$3B = -5A \Rightarrow B = -\frac{5}{3}A$$

$$136 = 3A - 5\left(-\frac{5}{3}A\right)$$

$$136 = 3A + \frac{25}{3}A$$

$$136 = \frac{34A}{3}$$

$$\Rightarrow A = \frac{136 \times 3}{34} = 12$$

$$B = \frac{-5}{3}(12) = -20$$

$$I = \int_0^{\pi/4} \frac{A(3\sin x + 5\cos x)}{3\sin x + 5\cos x} + \int_0^{\pi/4} \frac{B(3\cos x - 5\sin x)}{3\sin x + 5\cos x}$$

$$= A(x)_0^{\pi/4} + B[\ln(3\sin x + 5\cos x)]_0^{\pi/4}$$

$$= 12\left(\frac{\pi}{4}\right) - 20\ln\left(\frac{3}{\sqrt{2}} + \frac{5}{\sqrt{2}}\right) - \ln(0+5)$$

$$= 3\pi - 20\ln 4\sqrt{2} + 20\ln 5$$

$$= 3\pi - 20 \times \frac{5}{2}\ln 2 + 20\ln 5$$

$$= 3\pi - 50\ln 2 + 20\ln 5$$

Q. 10 The coefficients a, b, c in the quadratic equation $ax^2+bx+c=0$ are chosen from the set {1,2,3,4,5,6,7,8}. The probability of this equation having repeated roots is :

Option 1:
$\frac{3}{256}$

Option 2:
$\frac{1}{128}$

Option 3:
$\frac{1}{64}$

Option 4:
$\frac{3}{128}$

Correct Answer:
$\frac{1}{64}$

Solution:
$ax^2+bx+c=0$
$a,b,c\in\{1,2,3,4,5,6,7,8\}$

Repeated roots D=0

$\Rightarrow b^2-4ac=0\Rightarrow b^2=4ac$

Prob $=\frac{8}{8\times8\times8}=\frac{1}{64}$

$\Rightarrow (a,b,c)$
$(1,2,1);(2,4,2);(1,4,4);(4,4,1);(3,6,3);$
$(2,8,8);(8,8,2);(4,8,4)$

8 case

Q. 11 Let A and B be two square matrices of order 3 such that $|A| = 3$ and $|B| = 2$.
Then $A^T A(\text{adj}(2A))^{-1}(\text{adj}(4B))(\text{adj}(AB))^{-1}AA^T$ is equal to :

Option 1:
64

Option 2:
81

Option 3:
32

Option 4:
108

Correct Answer:
64

Solution:

$|A| = 3, |B| = 2$

$$A^T A(\text{adj}(2A))^{-1}(\text{adj}(4B))(\text{adj}(AB))^{-1}AA^T$$

$$= 3 \times 3 \times |(\text{adj}(2A))^{-1}| \times |\text{adj}(4B)| \times |(\text{adj}(AB))^{-1}| \times 3 \times 3$$

$$\downarrow \qquad \downarrow \qquad \downarrow$$

$$\frac{1}{|\text{adj}(2A)|} \quad 2^{12} \times 2^2 \quad \frac{1}{|\text{adj}(AB)|}$$

$$= \frac{1}{2^6 |\text{adj}\,A|} \quad = \frac{1}{|\text{adj}\,B \cdot \text{adj}\,A|}$$

$$= \frac{1}{2^6 \cdot 3^2} \quad = \frac{1}{2^2 \cdot 3^2}$$

$$= 3^4 \cdot \frac{1}{2^6 \cdot 3^2} \cdot 2^{12} \cdot 2^2 \cdot \frac{1}{2^2 \cdot 3^2} = 64$$

Hence, the answer is the option (1).

Q. 12 Let a circle C of radius 1 and closer to the origin be such that the lines passing through the point (3, 2) and parallel to the coordinate axes touch it. Then the shortest distance of the circle C from the point (5, 5) is :

Option 1:
$2\sqrt{2}$

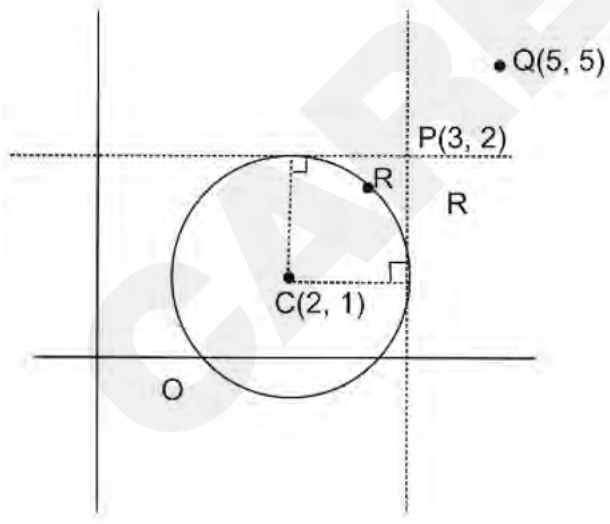
Option 2:
5

Option 3:
$4\sqrt{2}$

Option 4:
4

Correct Answer:
4

Solution:



Coordinates of the centre will be (2, 1)
Equation of circle will be

$(x-2)^2+(y-1)^2=1$

$QC=\sqrt{(5-2)^2+(5-1)^2}$

$QC=5$

shortest distance

$=RQ=CQ-CR$

$=5-1$

$=4$

Q. 13 Let the line $2x+3y-k=0, k>0$, intersect the x-axis and y-axis at the points A and B, respectively. If the equation of the circle having the line segment AB as a diameter is $x^2+y^2-3x-2y=0$ and the length of the latus rectum of the ellipse $x^2+9y^2=k^2$ is $\frac{m}{n}$, where m and n are coprime, then $2m+n$ is equal to

Option 1:
10

Option 2:
11

Option 3:
13

Option 4:
12

Correct Answer:
11

Solution:

$\text{Centre of the circle } =\left(\frac{3}{2},1\right)$

$\text{Equation of diameter } =2x+3y-k=0$

$2\left(\frac{3}{2}\right)+3(1)-k=0$

$\Rightarrow k=6$

Now,Equation of ellipse becomes

$$x^2 + 9y^2 = 36$$
$$\frac{x^2}{6^2} + \frac{y^2}{2^2} = 1$$

$$\text{length of LR} = \frac{2b^2}{a} = \frac{2.2^2}{6} = \frac{8}{6} = \frac{4}{3} = \frac{m}{n}$$
$$\therefore 2m + n = 2(4) + 3 = 11$$

Q. 14 Consider the following two statements :

Statement I : For any two non-zero complex numbers z_1, z_2 $(|z_1| + |z_2|)\ \frac{z_1}{|z_1|} + \frac{z_2}{|z_2|}\ \leq 2\,(|z_1| + |z_2|)$ and

Statement II : If x, y, z are three distinct complex numbers and a, b, c are three positive real numbers such that $\frac{a}{|y-z|} = \frac{b}{|z-x|} = \frac{c}{|x-y|}$, then $\frac{a^2}{y-z} + \frac{b^2}{z-x} + \frac{c^2}{x-y} = 1.$

Between the above two statements,

Option 1:
both Statement I and Statement II are incorrect.

Option 2:
Statement I is incorrect but Statement II is correct.

Option 3:
Statement I is correct but Statement II is incorrect.

Option 4:
both Statement I and Statement II are correct.

Correct Answer:
Statement I is correct but Statement II is incorrect.

Solution:
Statement I

$$(|z_1| + |z_2|)\ \frac{z_1}{|z_1|} + \frac{z_2}{|z_2|}$$

$$\text{Since}\quad \frac{z_1}{|z_1|} + \frac{z_2}{|z_2|} \leq \frac{z_1}{|z_1|} + \frac{z_2}{|z_2|}$$

$$\frac{z_1}{|z_1|} + \frac{z_2}{|z_2|} \leq \frac{|z_1|}{|z_1|} + \frac{|z_2|}{|z_2|}$$

$$\frac{z_1}{|z_1|} + \frac{z_2}{|z_2|} \leq 2$$

$$(|z_1| + |z_2|)\left(\frac{z_1}{|z_1|} + \frac{z_2}{|z_2|}\right) \leq 2\,(|z_1| + |z_2|)$$

$\therefore$ statement I is correct

For **Statement II** :

$$\frac{a}{|y-z|} = \frac{b}{|z-x|} = \frac{c}{|x-y|}$$

$$\frac{a^2}{|y-z|^2} = \frac{b^2}{|z-x|^2} = \frac{c^2}{|x-y|^2} = \lambda$$

$$a^2 = \lambda\left(|y-z|^2\right) = \lambda(y-z)(\bar{y}-\bar{z})$$

$$b^2 = \lambda(z-x)(\bar{z}-\bar{x}) \text{ and } c^2 = \lambda(x-y)(\bar{x}-\bar{y})$$

$$\frac{a^2}{y-z} + \frac{b^2}{z-x} + \frac{c^2}{x-y} = \lambda(\bar{y}-\bar{z}+\bar{z}-\bar{x}+\bar{x}-\bar{y}) = 0$$

Statement II is false

Q. 15 Suppose $\theta \in \left[0, \frac{\pi}{4}\right]$ is a solution of $4\cos\theta - 3\sin\theta = 1$.

Then $\cos\theta$ is equal to :

Option 1:

$\frac{4}{(3\sqrt{6}-2)}$

Option 2:

$\frac{6-\sqrt{6}}{(3\sqrt{6}-2)}$

Option 3:

$\text{frac}6+\sqrt{6}(3\sqrt{6}+2)\frac{4}{(3\sqrt{6}+2)}$

Option 4:

$\frac{4}{(3\sqrt{6}+2)}$

Correct Answer:

$\frac{4}{(3\sqrt{6}-2)}$

Solution:

$4\left(\frac{1-\tan^2\theta/2}{1+\tan^2\theta/2}\right)-3\left(\frac{2\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}\right)=1$

$\text{let } \tan\frac{\theta}{2}=\text{t}$

$$\frac{4-4\text{t}^2-6\text{t}}{1+\text{t}^2}=1$$

$$4-4\text{t}^2-6\text{t}=1+\text{t}^2$$

$$\Rightarrow 5\text{t}^2+6\text{t}-3=0$$

$$\Rightarrow \text{t}=\frac{-6\pm\sqrt{36-4(5)(-3)}}{2(5)}$$

$$=\frac{-6\pm\sqrt{96}}{10}$$

$$=\frac{-6\pm4\sqrt{6}}{10}$$

$$\text{t}=\frac{-3+2\sqrt{6}}{5}$$

$$\cos\theta=\frac{1-\text{t}^2}{1+\text{t}^2}=\frac{1-\left(\frac{2\sqrt{6}-3}{5}\right)^2}{1+\left(\frac{2\sqrt{6}-3}{5}\right)^2}=\frac{1-\left(\frac{24+9-12\sqrt{6}}{25}\right)}{1+\left(\frac{24+9-12\sqrt{6}}{25}\right)}$$

$$=\frac{25-33+12\sqrt{6}}{25+33-12\sqrt{6}}=\frac{12\sqrt{6}-8}{58-12\sqrt{6}}=\frac{6\sqrt{6}-4}{29-6\sqrt{6}}\times\frac{29+6\sqrt{6}}{29+6\sqrt{6}}$$

$$=\frac{100+150\sqrt{6}}{625}=\frac{4+6\sqrt{6}}{25}\times\frac{4-6\sqrt{6}}{4-6\sqrt{6}}$$

$$=\frac{-200}{25(4-6\sqrt{6})}=\frac{-8}{4-6\sqrt{6}}=\frac{4}{3\sqrt{6}-2}$$

Q. 16 If $\frac{1}{\sqrt{1}+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\ldots+\frac{1}{\sqrt{99}+\sqrt{100}}=\mathrm{m}$ and $\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\ldots+\frac{1}{99\cdot 100}=\mathrm{n}$, then the point (m, n) lies on the line

Option 1:
11(x-1)-100(y-2)=0

Option 2:
11(x-2)-100(y-1)=0

Option 3:
11(x-1)-100 y=0

Option 4:
11 x-100 y=0

Correct Answer:
11 x-100 y=0

Solution:

$$\frac{1}{\sqrt{1}+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\ldots+\frac{1}{\sqrt{99}+\sqrt{100}}=\mathrm{m}$$

$$\frac{\sqrt{1}-\sqrt{2}}{-1}+\frac{\sqrt{2}-\sqrt{3}}{-1}\ldots\frac{\sqrt{99}-\sqrt{100}}{-1}=\mathrm{m}$$

$$\sqrt{100}-1=\mathrm{m}\Rightarrow \mathrm{m}=9$$

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\ldots\frac{1}{99\cdot 100}=\mathrm{n}$$

$$\frac{1}{1}-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}\ldots\frac{1}{99}-\frac{1}{100}=\mathrm{n}$$

$$1-\frac{1}{100}=\mathrm{n}$$

$$\frac{99}{100}=\mathrm{n}$$

$$(\mathrm{m},\mathrm{n})=\left(9,\frac{99}{100}\right)$$

$$\Rightarrow 11(9)-100\left(\frac{99}{100}\right)$$

$$=99-99=0$$

Ans. option (4) $11x-100y=0$

Q. 17 Let $f(x)=x^3+2x^3+3x+1, x\in R$, and $g(x)$ be a function such that $g(f(x))=x$ for all $x\in R$. Then $\frac{g(7)}{g'(7)}$ is equal to :

Option 1:
7

Option 2:
42

Option 3:
1

Option 4:
14

Correct Answer:
14

Solution:

$f(x)=x^5+2x^3+3x+1$

$f'(x)=5x^4+6x^2+3$

$f'(1)=5+6+3=14$

$g(f(x))=x$

$g'(f(x))f'(x)=1$

for $f(x)=7$

$\Rightarrow x^5+2x^3+3x+1=7$

$\Rightarrow x=1$

$g'(7)f'(1)=1\Rightarrow g'(7)=\frac{1}{f'(1)}=\frac{1}{14}$

$x=1, f(x)=7\Rightarrow g(7)=1$

$$\frac{g(7)}{g'(7)}=\frac{1}{1/14}=14$$

Hence, the answer is the option (4).

Q. 18 If $\mathrm{A}(1,-1,2), \mathrm{B}(5,7,-6), \mathrm{C}(3,4,-10)$ and $D(-1,-4,-2)$ are the vertices of a quadrilateral ABCD, then its area is :

Option 1:
$12\sqrt{29}$

Option 2:
$24\sqrt{29}$

Option 3:
$24\sqrt{7}$

Option 4:
$48\sqrt{7}$

Correct Answer:
$12\sqrt{29}$

Solution:
$\mathrm{A}(1,-1,2)$
$\mathrm{B}(5,7,-6)$
$\mathrm{C}(3,4,-10)$
$\mathrm{D}(-1,-4,-2)$

$$\text{Area } = \frac{1}{2}|\overrightarrow{\mathrm{AC}} \times \overrightarrow{\mathrm{BD}}| = \frac{1}{2}|(2\hat{\mathrm{i}} + 5\hat{\mathrm{j}} - 12\hat{\mathrm{k}}) \times (6\hat{\mathrm{i}} + 11\hat{\mathrm{j}} - 4\hat{\mathrm{k}})|$$
$$= \frac{1}{2}|112\hat{\mathrm{i}} - 64\hat{\mathrm{j}} - 8\hat{\mathrm{k}}|$$
$$= 4|14\hat{\mathrm{i}} - 8\hat{\mathrm{j}} - \hat{\mathrm{k}}|$$
$$= 4\sqrt{196 + 64 + 1}$$
$$= 4\sqrt{261}$$
$$= 12\sqrt{29}$$

Q. 19 The value of $\int_{-\pi}^{\pi} \frac{2\mathrm{y}(1+\sin \mathrm{y})}{1+\cos^2 \mathrm{y}} \mathrm{dy}$ is :

Option 1:
π^2

Option 2:
$\frac{\pi^2}{2}$

Option 3:
$\frac{\pi}{2}$

Option 4:
$2\pi^2$

Correct Answer:
π^2

Solution:

$\int_{-\pi}^{\pi} \frac{2y(1+\sin y)}{1+\cos^2 y}dy$

$]mathrm{=} \int_{-\pi}^{\pi} \frac{2y}{1+\cos^2 y}dy + \int_{-\pi}^{\pi} \frac{2y\sin y}{1+\cos^2 y}dy$

$= 0 + 2 \cdot 2\int_{0}^{\pi} y\left(\frac{\sin y}{1+\cos^2 y}\right)dy$

$I = 4\int_{0}^{\pi} \frac{y\sin y}{1+\cos^2 y}dy$

$I = 4\int_{0}^{\pi} \frac{(\pi - y)\sin y}{1+\cos^2 y}dy$

$2I = 4\int_{0}^{\pi} \frac{\pi\sin y}{1+\cos^2 y}dy$

$I = 2\pi\int_{0}^{\pi} \frac{\sin y}{1+\cos^2 y}dy$

$= 2\pi\left(-\tan^{-1}(\cos y)\right)_0^{\pi}$

$= -2\pi\left[\left(-\frac{\pi}{4}\right) - \left(\frac{\pi}{4}\right)\right]$

$= -2\pi\left[-\frac{2\pi}{4}\right] = \pi^2$

Q. 20 If the line $\frac{2-x}{3} = \frac{3y-2}{4\lambda+1} = 4 - z$ makes a right angle with the line $\frac{x+3}{3\mu} = \frac{1-2y}{6} = \frac{5-z}{7}$, then $4\lambda + 9\mu$ is equal to :

Option 1:
13

Option 2:
4

Option 3:
5

Option 4:
6

Correct Answer:
4

Solution:

$\frac{2-x}{3} = \frac{3y-2}{4\lambda+1} = 4 - z$ ------(1)

$\frac{x-2}{(-3)} = \frac{y-\frac{2}{3}}{(\frac{4\lambda+1}{3})} = \frac{z-4}{(-1)}$

$\frac{x+3}{3\mu} = \frac{1-2y}{6} = \frac{5-z}{7}$ --------(2)

$\frac{x+3}{3\mu} = \frac{y-\frac{1}{2}}{(-3)} = \frac{z-5}{(-7)}$

$\text{Right angle} \Rightarrow (-3)(3\mu) + \left(\frac{4\lambda+1}{3}\right)(-3) + (-1)(-7) = 0$

$-9\mu - 4\lambda - 1 + 7 = 0$

$4\lambda + 9\mu = 6$

Q. 21 From a lot of 10 items, which include 3 defective items, a sample of 5 items is drawn at random. Let the random variable X denote the number of defective items in the sample. If the variance of X is σ^2, then $96\sigma^2$ is equal to

Correct Answer:
56

Solution:
X = denotes number of defective

x	0	1	2	3
$P(x)$	$\frac{7}{15}$	$\frac{5}{12}$	$\frac{5}{12}$	$\frac{1}{12}$
${x_1}^2$	0	1	4	9
$P_i x_1^2$	0	$\frac{5}{12}$	$\frac{20}{12}$	$\frac{9}{12}$
$P_i x_i$	0	$\frac{5}{12}$	$\frac{10}{12}$	$\frac{3}{12}$

$\mu = \Sigma p_i x_i = \frac{18}{12}$

$\Sigma p_i x_1^2 = \frac{34}{12}$

$\sigma^2 = \Sigma p_i x_1^2 - (\mu)^2$

$= \frac{34}{12} - \left(\frac{18}{12}\right)^2 = \frac{17}{6} - \frac{9}{4}$

$\frac{34-27}{12} = \frac{7}{12}$

$96\sigma^2 = 96 \times \frac{7}{12} = 56$

Q. 22 If the constant term in the expansion of $\left(1+2x-3x^3\right)\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$ is p, then 108 p is equal to

Correct Answer:
54

Solution:

$$(1+2x-3x^3)\left(\frac{3}{2}x^2-\frac{1}{3x}\right)^9$$

$$\text{General term } m\left(\frac{3}{2}x^2-\frac{1}{3x}\right)^9$$

$$= {}^9C_r \cdot \frac{3^{9-2r}}{2^{9-r}}(-1)^r \cdot x^{18-3r}$$

Put $r = 6$ to get coeff. of $x^0 = {}^9C_6 \cdot \frac{1}{6^3} \cdot x^0 = \frac{7}{18}x^0$

Put $r = 7$ to get coeff. of $x^{-3} = {}^9C_r \cdot \frac{3^{-5}}{2^2}(-1)^7 \cdot x^{-3}$

$$= -{}^9C_7 \cdot \frac{1}{3^5 \cdot 2^2} \cdot x^{-3} = \frac{-1}{27}x^{-3}$$

$$(1+2x-3x^3)\left(\frac{7}{18}x^0-\frac{1}{27}x^{-3}\right)$$

$$\frac{7}{18}+\frac{3}{27} = \frac{7}{18}+\frac{1}{9} = \frac{7+2}{18} = \frac{9}{18} = \frac{1}{2}$$

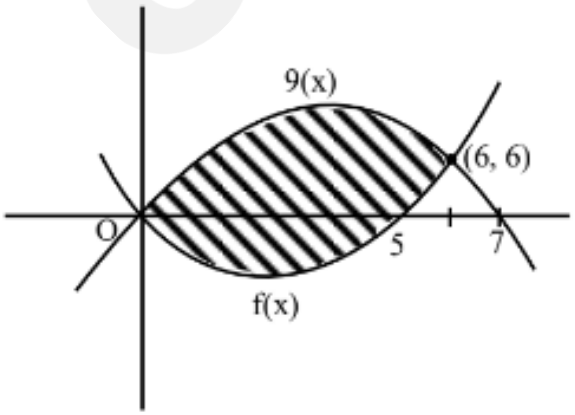
$$\therefore 108 \cdot \frac{1}{2} = 54$$

Hence, the answer is 54.

Q. 23 The area of the region enclosed by the parabolas $y = x^2 - 5x$ and $y = 7x - x^2$ is_____.

Correct Answer:
72

Solution:
$y = x^2 - 5x$ and $y = 7x - x^2$



$$\int_0^6 (g(x) - f(x))dx$$

$$\int_0^6 ((7x - x^2) - (x^2 - 5x))dx$$

$$\int_0^6 (12x - 2x^2)dx = \left[12\frac{x^2}{2} - \frac{2x^3}{3}\right]_0^6$$

$$\Rightarrow 6(6)^2 - \frac{2}{3}(6)^3$$

$$= 216 - 144 = 72 \text{ unit}^2$$

Hence, the answer is (72).

Q. 24 The number of ways of getting a sum 16 on throwing a dice four times is _____.

Correct Answer:
125

Solution:

$$(x^1 + x^2 \ldots + x^6)^4$$

$$x^4 \cdot \left(\frac{1 - x^6}{1 - x}\right)^4$$

$$x^4 \cdot (1 - x^6)^4 \cdot (1 - x)^{-4}$$

$$x^4 \left[1 - 4x^6 + 6x^{12} \ldots.\right] \left[(1 - x)^{-4}\right]$$

$$(x^4 - 4x^{10} + 6x^{16} \ldots)(1 - x)^{-4}$$

$$(x^4 - 4x^{10} + 6x^{16}) \left(1 + {}^{15}C_{12}x^{12} + {}^{9}C_6x^6 \ldots\right)$$

$$({}^{15}C_{12} - 4 \cdot {}^{9}C_6 + 6)x^{16}$$

$$({}^{15}C_3 - 4 \cdot {}^{9}C_6 + 6)$$

$$= 35 \times 13 - 6 \times 8 \times 7 + 6$$

$$= 455 - 336 + 6$$

$$= 125$$

Q. 25 If $S = \{a \in R : |2a - 1| = 3[a] + 2\{a\}\}$, where [t] denotes the greatest integer less than or equal to t and $\{t\}$ represents the fractional part of t, then $72 \sum_{a \in S} a$ is equal to

Correct Answer:
18

Solution:

$|2a - 1| = 3[a] + 2\{a\}$

$|2a - 1| = [a] + 2a$

$\text{Case-1} \ : a > \frac{1}{2}$

$2a - 1 = [a] + 2a$

$[a] = -1 \quad \therefore a \in [-1, 0) \ \text{Reject}$

$\text{Case-2} \ : a < \frac{1}{2}$

$-2a + 1 = [a] + 2a$

$a = I + f$

$-2(I + f) + 1 = I + 2I + 2f$

$I = 0, f = \frac{1}{4} \quad \therefore a = \frac{1}{4}$

$\text{Hence } a = \frac{1}{4}$

$72 \sum_{a \in S} a = 72 \times \frac{1}{4} = 18$

Q. 26 Let f be a differentiable function in the interval $(0, \infty)$ such that $f(1) = 1$ and $\lim_{t \to x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$ for each $x > 0$. Then $2f(2) + 3f(3)$ is equal to _____.

Correct Answer:
24

Solution:

$$\lim_{t\to x}\frac{t^2f(x)-x^2f(t)}{t-x}=1$$

$$\lim_{t\to x}\frac{2t.\,f(x)-x^2f'(x)}{1}=1$$

$$2x\cdot f(x)-x2f'(x)=1$$

$$\frac{dy}{dx}-\frac{2}{x}\cdot y=\frac{-1}{x^2}$$

$$\text{I.f.}=e^{\int-\frac{2}{x}dx}=\frac{1}{x^2}$$

$$\therefore\frac{y}{x^2}=\int-\frac{1}{x^4}dx+C$$

$$\frac{y}{x^2}=\frac{1}{3x^3}+C$$

Put $f(1)=1$

$$C=\frac{2}{3}$$

$$y=\frac{1}{3x}+\frac{2x^2}{3}$$

$$y=\frac{2x^3+1}{3x}$$

$$f(2)=\frac{17}{6}$$

$$f(3)=\frac{55}{9}$$

$$2f(2)+3f(3)=\frac{17}{3}+\frac{55}{3}=\frac{72}{3}=24$$

Q. 27 Let $a_1, a_2, a_3, \ldots$ be in an arithmetic progression of positive terms. Let $A_k = {a_1}^2 - {a_2}^2 + {a_3}^2 - {a_4}^2 + \ldots + {a_{2k-1}}^2 - {a_{2k}}^2$. If $A_3 = -153,\ A_5 = -435$ and ${a_1}^2 + {a_2}^2 + {a_3}^2 = 66$, then $a_{17} - A_7$ is equal to

Correct Answer:
910

Solution:

$d \rightarrow$ common diff.

$A_k = -kd[2a + (2k-1)d]$

$A_3 = -153$

$\Rightarrow 153 = 13\ d[2a + 5\ d]$

$51 = d[2a + 5\ d]$

$A_5 = -435$

$435 = 5\ d[2a + 9\ d]$

$87 = d[2a + 9\ d]$

$(2) - (1)$

$36 = 4\ d^2$

$d = 3, a = 1$

$a_{17} - A_7 = 49 - [-7.3[2 + 39]] = 910$

Hence, the answer is 910.

Q. 28 Let $\vec{a} = \hat{i} - 3\hat{j} + 7\hat{k}, \vec{b} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{c}$ be a vector such that $(\dot{a} + 2\vec{b}) \times \vec{c} = 3(\vec{c} \times \vec{a})$. If $\vec{a} \cdot \vec{c} = 130$, then $\vec{b} \cdot \vec{c}$ is equal to

---

Correct Answer:

30

Solution:

$(\vec{a} + 2\vec{b}) \times \vec{c} = 3(\vec{c} \times \vec{a})$

$(2\vec{b} + 4\vec{a}) \times \vec{c} = 0$

$\vec{c} = \lambda(4\vec{a} + 2\vec{b}) = \lambda(8\hat{i} - 14\hat{j} + 30\hat{k})$

$\vec{a} \cdot \vec{c} = 130$

$8\lambda + 42\lambda + 210\lambda = 130$

$\lambda = \frac{1}{2}$

$\vec{c} = 4\hat{i} - 7\hat{j} + 15\hat{k}$

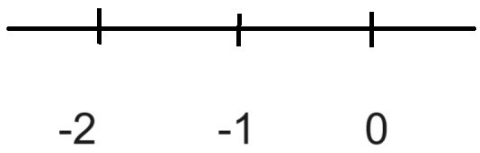
$\vec{b} \cdot \vec{c} = 8 + 7 + 15 = 30$

Hence, the answer is (30).

Q. 29 The number of distinct real roots of the equation $|x||x+2|-5|x+1|-1=0$ is_____.

Correct Answer:
3

Solution:



Case-1

$x \geq 0$

$x^2+2x-5x-5-1=0$

$x^2-3x-6=0$

$x=\frac{3\pm\sqrt{9+24}}{2}=\frac{3\pm\sqrt{33}}{2}$

One positive root

Case-2

$-1 \leq x < 0$

$-x^2-2x-5x-5-1=0$

$x^2+7x+6=0$

$(x+6)(x+1)=0$

$x=-1$

one root in range

Case-3

$-2 \leq x < -1$

$x^2-2x+5x+5-1=0$

$x^2-3x-4=0$

$(x-4)(x+1)=0$

No root in range

Case-4

$x < -2$

$x^2 + 7x + 4 = 0$

$x = \frac{-7 \pm \sqrt{49 - 16}}{2} = \frac{7 \pm \sqrt{33}}{2}$

one root in range

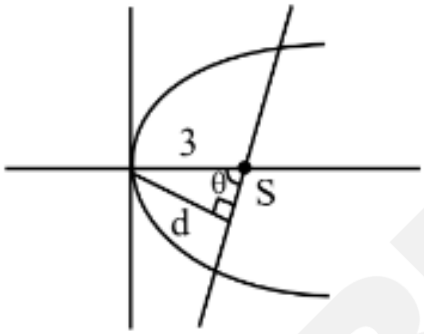
Total number of distinct roots are 3

Q. 30 Suppose A B is a focal chord of the parabola $y^2 = 12x$ of length l and slope $m < \sqrt{3}$. If the distance of the chord A B from the origin is d, then $l\ d^2$ is equal to_____.

Correct Answer:

108

Solution:



$\ell = 4a\ \text{cosec}^2\ \theta$

$\ell = 12 \times \frac{9}{d^2}$

$\ell d^2 = 108$

5 Apr - Evening Shift

Physics

Q. 1 Given below are two statements :

Statement I : When the white light passed through a prism, the red light bends lesser than yellow and violet.

Statement II : The refractive indices are different for different wavelengths in dispersive medium. In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are true.

Option 2:
Statement I is true but Statement II is false.

Option 3:
Both Statement I and Statement II are false.

Option 4:
Statement I is false but Statement II is true.

Correct Answer:
Both Statement I and Statement II are true.

Solution:
As $\lambda_{\text{red}} > \lambda_{\text{yellow}} > \lambda_{\text{violet}}$
Light ray with longer wavelength bends less.

Q. 2 Which of the following statement is not true about stopping potential (V_0) ?

Option 1:
It depends on the nature of emitter material.

Option 2:
It depends upon frequency of the incident light.|

Option 3:
It increases with increase in intensity of the incident light.

Option 4:
It is $1/\mathrm{e}$ times the maximum kinetic energy of electrons emitted.

Correct Answer:
It increases with increase in intensity of the incident light.

Solution:
$\mathrm{KE}_{\max} = \mathrm{h}\nu - \phi_0 = \mathrm{eV}$

Q. 3 The angular momentum of an electron in a hydrogen atom is proportional to: (Where r is the radius of orbit of electron)

Option 1:
$\sqrt{\mathrm{r}}$

Option 2:
$\frac{1}{r}$

Option 3:
r

Option 4:
$\frac{1}{\sqrt{r}}$

Correct Answer:
$\sqrt{r}$

Solution:
$F_C = \frac{mv^2}{r}$

$\frac{Kq_1q_2}{r^2} = \frac{mv^2}{r}$

$mv^2r^2 = Kq_1q_2r$

$\frac{L^2}{m} = Kq_1q_2r$

$L \propto \sqrt{r}$

Q. 4 A galvanometer of resistance 100Ω when connected in series with 400Ω measures a voltage of upto $10\ V$. The value of resistance required to convert the galvanometer into ammeter to read upto $10\ A$ is $x \times 10^{-2}\Omega$. The value of x is :

Option 1:
2

Option 2:
800

Option 3:
20

Option 4:
200

Correct Answer:
20

Solution:
$i_g = \frac{10}{400+100} = 20 \times 10^{-3}\ \mathrm{A}$

For ammeter

Let shunt resistance $= \mathrm{S}$

$i_g R = (i - i_g)S$

$20 \times 10^{-3} \times 100 = 10S$

$S = 20 \times 10^{-2}\Omega$

Q. 5 The vehicles carrying inflammable fluids usually have metallic chains touching the ground :

Option 1:
To conduct excess charge due to air friction to ground and prevent sparking.

Option 2:
To alert other vehicles.

Option 3:
To protect tyres from catching dirt from ground.

Option 4:
It is a custom.

Correct Answer:
To conduct excess charge due to air friction to ground and prevent sparking.

Solution:

Static charge is developed due to air friction. This can result in combustion. So, metallic chains is used to discharge excess charge.

Q. 6 If n is the number density and d is the diameter of the molecule, then the average distance covered by a molecule between two successive collisions (i.e.mean free path) is represented by :

Option 1:
$\frac{1}{\sqrt{2}n\pi d^2}$

Option 2:
$\sqrt{2}n\pi d^2$

Option 3:
$\frac{1}{\sqrt{2}n\pi d^2}$

Option 4:
$\frac{1}{\sqrt{2}n^2\pi^2 d^2}$

Correct Answer:
$\frac{1}{\sqrt{2}n\pi d^2}$

Solution:
$n =$ number of molecule per unit volume

$d =$ diameter of the molecule

$\lambda = \frac{1}{\sqrt{2}\pi d^2 n}$ (By Theory)

CAREERS360

Q. 7 A particle moves in $x-y$ plane under the influence of a force $\vec{F}$ such that its linear momentum is $\vec{P}(t)=\hat{i}\cos(kt)-\hat{j}\sin(kt)$. If k is constant, the angle between $\overrightarrow{\mathrm{F}}$ and $\overrightarrow{\mathrm{P}}$ will be :

Option 1:
$\frac{\pi}{2}$

Option 2:
$\frac{\pi}{6}$

Option 3:
$\frac{\pi}{4}$

Option 4:
$\frac{\pi}{3}$

Correct Answer:
$\frac{\pi}{2}$

Solution:

$\overrightarrow{\mathrm{P}}=\cos(\mathrm{kt})\hat{\mathrm{i}}-\sin(\mathrm{kt})\hat{\mathrm{j}};|\overrightarrow{\mathrm{P}}|=1$

$\because \overrightarrow{\mathrm{P}}=\mathrm{m}\overrightarrow{\mathrm{v}}$

$\therefore \hat{\mathrm{P}}=\hat{\mathrm{v}}$

$\Rightarrow \hat{\mathrm{v}}=\cos(\mathrm{kt})\hat{\mathrm{i}}-\sin(\mathrm{kt})\hat{\mathrm{j}}$

$\hat{\mathrm{a}}=\frac{-\mathrm{k}\sin(\mathrm{kt})\hat{\mathrm{i}}-\mathrm{k}\cos(\mathrm{kt})\hat{\mathrm{j}}}{\mathrm{k}}$

$\Rightarrow \hat{\mathrm{a}}=-\sin kt\hat{\mathrm{i}}-\cos \mathrm{kt}\hat{\mathrm{j}}$

$\because \hat{\mathrm{F}}=\hat{\mathrm{a}}=-\sin \mathrm{kt}\hat{\mathrm{i}}-\cos \mathrm{kt}\hat{\mathrm{j}}$

$$\cos\theta = \frac{\hat{F}\cdot\hat{P}}{|\hat{F}||\hat{P}|} = -\frac{\sin kt\cos t + \sin kt\cos t}{1\times 1} = 0$$

$$\Rightarrow \theta = \frac{\pi}{2}$$

Q. 8 The electrostatic force $\left(\vec{F}_1\right)$ and magnetic force $\left(\vec{F}_2\right)$ acting on a charge q moving with velocity v can be written :

Option 1:
$\vec{F}_1 = q\vec{V}\cdot\vec{E}, \vec{F}_2 = q(\vec{B}\cdot\vec{V})$

Option 2:
$\vec{F}_1 = q\vec{B}, \vec{F}_2 = q(\vec{B}\times\vec{V})$

Option 3:
$\vec{F}_1 = q\vec{E}, \vec{F}_2 = q(\vec{V}\times\vec{B})$

Option 4:
$\vec{F}_1 = q\vec{E}, \vec{F}_2 = q(\vec{B}\times\vec{V})$

Correct Answer:
$\vec{F}_1 = q\vec{E}, \vec{F}_2 = q(\vec{V}\times\vec{B})$

Solution:
$\overrightarrow{F_1} = q\overrightarrow{E}$ (Theory)

$\overrightarrow{F_2} = q(\overrightarrow{V}\times\overrightarrow{B})$

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Q. 9 A man carrying a monkey on his shoulder does cycling smoothly on a circular track of radius $9\ \mathrm{m}$ and completes 120 revolutions in 3 minutes. The magnitude of centripetal acceleration of monkey is (in $\mathrm{m/s^2}$) :

Option 1:
Zero

Option 2:
$16\pi^2\ \mathrm{ms}^{-2}$

Option 3:
$4\pi^2\ \mathrm{ms}^{-2}$

Option 4:
$57600\pi^2\ \mathrm{ms}^{-2}$

Correct Answer:
$16\pi^2\ \mathrm{ms}^{-2}$

Solution:
Given : $\mathrm{R} = 9\ \mathrm{m}$,

120 revolution in $3\ \mathrm{min}$

$\omega = \frac{120\mathrm{Rev.}}{3\ \mathrm{min.}} = \frac{120\times 2\pi\mathrm{rad}}{3\times 60\mathrm{sec}} = \frac{4\pi}{3}\mathrm{rad/s}$

$\mathrm{a_{centripetal}} = \omega^2\mathrm{R} = \left(\frac{4\pi}{3}\right)^2 \times 9 = 16\pi^2\ \mathrm{m/s^2}$

Q. 10 A series LCR circuit is subjected to an AC signal of $200\ \mathrm{V}, 50\ \mathrm{Hz}$. If the voltage across the inductor $(\mathrm{L} = 10\mathrm{mH})$ is $31.4\ \mathrm{V}$, then the current in this circuit is _____:

Option 1:
68A

Option 2:
63A

Option 3:
10A

Option 4:
10mA

Correct Answer:
10A

Solution:
Voltage across inductor $V_{\mathrm{L}} = \mathrm{IX_L}$

$31.4 = \mathrm{I}[\mathrm{L}\omega]$

$31.4 = \mathrm{I}[\mathrm{L}(2\pi\mathrm{f})]$

$31.4 = \mathrm{I}\left[10 \times 10^{-3}(2 \times 3.14) \times 50\right.$

$\Rightarrow \mathrm{I} = 10\ \mathrm{A}$

Q. 11 What is the dimensional formula of ab^{-1} in the equation $\left(\mathrm{P} + \frac{\mathrm{a}}{\mathrm{V}^2}\right)(\mathrm{V} - \mathrm{b}) = \mathrm{RT}$, where letters have their usual meaning.

Option 1:
$\left[\mathrm{M}^{0}\ \mathrm{L}^{3}\ \mathrm{T}^{-2}\right]$

Option 2:
$\left[\mathrm{ML}^{2}\ \mathrm{T}^{-2}\right]$

Option 3:
$\left[\mathrm{M}^{-1}\ \mathrm{L}^{5}\ \mathrm{T}^{3}\right]$

Option 4:
$\left[\mathrm{M}^{6}\ \mathrm{L}^{7}\ \mathrm{T}^{4}\right]$

Correct Answer:
$\left[\mathrm{ML}^{2}\ \mathrm{T}^{-2}\right]$

Solution:
$\because [\mathrm{V}] = [\mathrm{b}]$

$\therefore$ Dimension of $\mathrm{b} = \left[\mathrm{L}^{3}\right]$

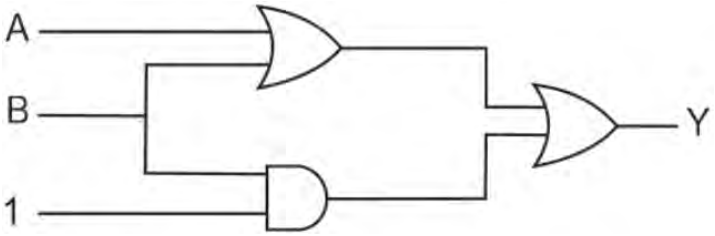
$\&[\mathrm{P}] = \left[\frac{\mathrm{a}}{\mathrm{V}^{2}}\right]$

$[\mathrm{a}] = \left[\mathrm{PV}^{2}\right] = \left[\mathrm{ML}^{-1}\ \mathrm{T}^{-2}\right]\left[\mathrm{L}^{6}\right]$

Dimension of $\mathrm{a} = \left[\mathrm{ML}^{5}\ \mathrm{T}^{-2}\right]$

$\therefore \mathrm{ab}^{-1} = \frac{\left[\mathrm{ML}^{5}\ \mathrm{T}^{-2}\right]}{\left[\mathrm{L}^{3}\right]} = \left[\mathrm{ML}^{2}\ \mathrm{T}^{-2}\right]$

Q. 12 The output (Y) of logic circuit given below is 0 only when :



Option 1:
A = 1, B = 0

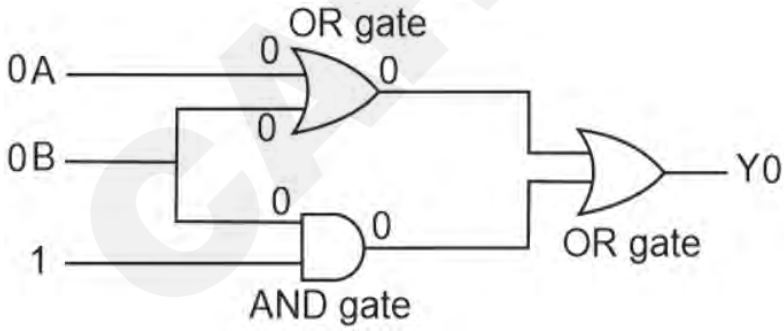
Option 2:
A = 0, B = 0

Option 3:
A = 1, B = 1

Option 4:
A = 0, B = 1

Correct Answer:
A = 0, B = 0

Solution:



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Q. 13 A body is moving unidirectionally under the influence of a constant power source. Its
displacement in time t is proportional to :

Option 1:
t^2

Option 2:
$t^{2/3}$

Option 3:
$t^{3/2}$

Option 4:
t

Correct Answer:
$t^{3/2}$

Solution:
$\mathrm{P} = \text{ costant } \Rightarrow \mathrm{FV} = \text{ constant}$

$\Rightarrow \mathrm{m}\frac{\mathrm{dV}}{\mathrm{dt}}\mathrm{V} = \text{ constant}$

$\int_0^{\mathrm{V}} \mathrm{VdV} = (\mathrm{C}) \int_0^{\mathrm{t}} \mathrm{dt}$

$\left(\frac{\mathrm{V}^2}{2}\right) = \mathrm{Ct}$

$\mathrm{V} \propto \mathrm{t}^{1/2}$

$\frac{\mathrm{ds}}{\mathrm{dt}} \propto \mathrm{t}^{1/2}$

$\int_0^S ds = K \int_0^t t^{1/2} dt$

$S = K \times \frac{2}{3} t^{3/2}$

$S \propto t^{3/2}$

$\therefore$ displacement is proportional to $(t)^{3/2}$

Q. 14 Match **List-I with List-II :-**

	List-I EM-Wave		**List-II Wavelength Range**
(A)	Infra-red	(I)	$< 10^{-3}$ nm
(B)	Ultraviolet	(II)	400 nm to 1 nm
(C)	X-rays	(III)	1 mm to 700 nm
(D)	Gamma rays	(IV)	1 nm to 10^{-3} nm

Choose the correct answer from the options given below :

Option 1:
(A)-(II), (B)-(I), (C)-(IV), (D)-(III)

Option 2:
(A)-(III), (B)-(II), (C)-(IV), (D)-(I)

Option 3:
(A)-(IV), (B)-(III), (C)-(II), (D)-(I)

Option 4:
(A)-(I), (B)-(III), (C)-(II), (D)-(IV)

Correct Answer:
(A)-(III), (B)-(II), (C)-(IV), (D)-(I)

Solution:

Infrared is the least energetic thus having biggest wavelength (λ)& gamma rays are most energetic thus having smallest wavelength (λ).

Q. 15 During an adiabatic process, if the pressure of a gas is found to be proportional to the cube of its absolute temperature, then the ratio of $\frac{C_P}{C_V}$ for the gas is :

Option 1:
$\frac{5}{3}$

Option 2:
$\frac{9}{7}$

Option 3:
$\frac{3}{2}$

Option 4:
$\frac{7}{5}$

Correct Answer:
$\frac{3}{2}$

Solution:
$\mathrm{P} \propto \mathrm{T}^3$

$\mathrm{PT}^{-3} = \text{ constant}$

$\because \frac{\mathrm{PV}}{\mathrm{T}} = \mathrm{nR} = \text{ constant from ideal gas equation}$

$(P)(\mathrm{PV})^{-3} = \text{constant}$

$\mathrm{P}^{-2}\,\mathrm{V}^{-3} = \text{ cosntant }$ .....(1)

$\because$ Process equation for adiabatic process is$

$\mathrm{PV}^{\mathrm{y}} = \text{ constant }$ .....(2)

Comparing equation (1) and (2)

$$\frac{C_P}{C_V} = y = \frac{3}{2}$$

Q. 16 **Match List-I with List-II :**

	List-I		List-II
(A)	A force that restores an elastic body of unit area to its original state	(I)	Bulk modulus
(B)	Two equal and opposite forces parallel to opposite faces	(II)	Young's modulus
(C)	Forces perpendicular everywhere to the surface per unit area same everywhere	(III)	Stress
(D)	Two equal and opposite forces perpendicular to opposite faces	(IV)	Shear modulus

Choose the correct answer from the options given below :

Option 1:
(A)-(II), (B)-(IV), (C)-(I), (D)-(III)

Option 2:
(A)-(IV), (B)-(II), (C)-(III), (D)-(I)

Option 3:
(A)-(III), (B)-(IV), (C)-(I), (D)-(II)

Option 4:
(A)-(III), (B)-(I), (C)-(II), (D)-(IV)

Correct Answer:
(A)-(III), (B)-(IV), (C)-(I), (D)-(II)

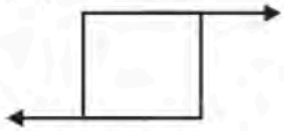
Solution:
(A) stress $= \frac{F_{\text{restoring}}}{A}$

If $A = 1$

Stress $= F_{\text{restoring}}$

(A)-(III)

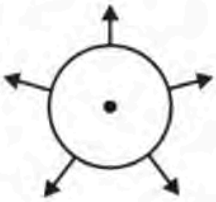
(B)



$\Rightarrow$ Shear stress

(B)-(IV)

(C)



$\Rightarrow$ Volumetric stress

(C)-(I)

(D)

$\Rightarrow$ Longitudinal stress

(D)-(II)

Q. 17 A vernier callipers has 20 divisions on the vernier scale, which coincides with 19^{th} division on the main scale. The least count of the instrument is $0.1\ \text{mm}$. One main scale division is equal to _____mm.

Option 1:
1

Option 2:
0.5

Option 3:
2

Option 4:
5

Correct Answer:
2

Solution:
$20\text{VSD} = 19\text{MSD}$

$1\text{VSD} = \frac{19}{20}\text{MSD}$

$\text{L.C.} = 1\text{MSD} - 1\text{VSD}$

$0.1\ \text{mm} = 1\text{MSD} - \frac{19}{20}\text{MSD}$

$0.1 = \frac{1}{20}\text{MSD}$

$1\text{MSD} = 2\ \text{mm}$

Q. 18 A heavy box of mass 50 kg is moving on a horizontal surface. If co-efficient of kinetic friction between the box and horizontal surface is 0.3 then force of kinetic friction is :

Option 1:
14.7 N

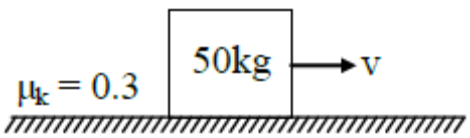
Option 2:
147 N

Option 3:
1.47 N

Option 4:
1470 N

Correct Answer:
147 N

Solution:



$F_k = \mu_k \, \mathrm{N} = 0.3 \times 50 \times 9.8 = 147 \, \mathrm{N}$

Q. 19 A satellite revolving around a planet in stationary orbit has time period 6 hours. The mass of planet is one-fourth the mass of earth. The radius orbit of planet is : (Given = Radius of geo-stationary orbit for earth is 4.2×10^4 km)

Option 1:
1.4×10^4 km

Option 2:
8.4×10^4 km

Option 3:
1.68×10^5 km

Option 4:
1.05×10^4 km

Correct Answer:
1.05×10^4 km

Solution:

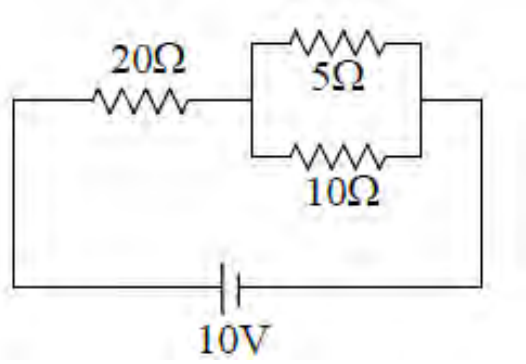
$$\mathrm{T} = \frac{2\pi \mathrm{r}^{3/2}}{\sqrt{\mathrm{GM}}}$$

$$\frac{\mathrm{T_1}}{\mathrm{T_2}} = \left(\frac{\mathrm{r_1}}{\mathrm{r_2}}\right)^{3/2} \left(\frac{\mathrm{M_2}}{\mathrm{M_1}}\right)^{1/2}$$

$$\frac{6}{24} = \frac{(\mathrm{r_1})^{3/2}}{(4.2\times 10^4)^{3/2}} \left(\frac{\mathrm{M}}{\mathrm{M}/4}\right)^{1/2}$$

$$\mathrm{r_1} = 1.05 \times 10^4 \, \mathrm{km}$$

Q. 20 The ratio of heat dissipated per second through the resistance 5Ω and 10Ω in the circuit given below is :



Option 1:
1:2

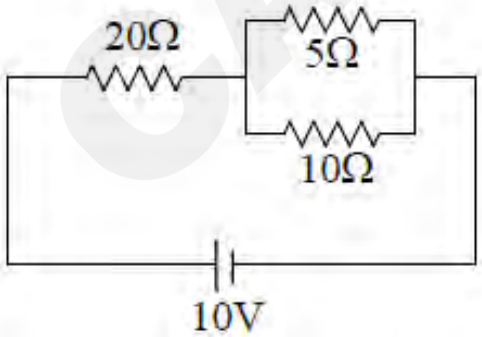
Option 2:
2:1

Option 3:
4:1

Option 4:
1:1

Correct Answer:
2:1

Solution:



$$\frac{i_1}{i_2} = \frac{10}{5} = \frac{2}{1}$$

$$\frac{P_1}{P_2} = \frac{i_1^2 R_1}{i_2^2 R_2} = \left(\frac{2}{1}\right)^2 \times \frac{5}{10} = \frac{2}{1}$$

Q. 21 A solenoid of length $0.5\ \mathrm{m}$ has a radius of $1\ \mathrm{cm}$ and is made up of ' m ' number of turns. It carries a current of $5\ \mathrm{A}$. If the magnitude of the magnetic field inside the solenoid is $6.28 \times 10^{-3}\ \mathrm{T}$, then the value of m is :

Correct Answer:
500

Solution:
$\mu_0 \mathrm{ni} = \mathrm{B} \quad \mathrm{n} =$ number of turns per unit length

$\mu_0 \left(\frac{\mathrm{m}}{\ell}\right)\mathrm{i} = \mathrm{B}$

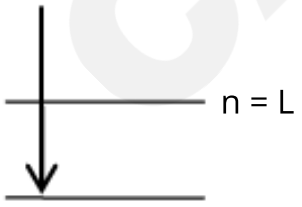
$\mathrm{m} = \frac{\mathrm{B}\ell}{\mu_0 \mathrm{i}} = \frac{6.28\times10^{-3}\times0.5}{12.56\times10^{-7}\times5}$

$\mathrm{m} = 500$

Q. 22 The shortest wavelength of the spectral lines in the Lyman series of hydrogen spectrum is 915Å . The longest wavelength of spectral lines in the Balmer series will be_____ Å.

Correct Answer:
6588

Solution:
Lyman Series



Shortest, $\frac{\mathrm{hc}}{\lambda} = -13.6\left(\frac{1}{\mathrm{n}_1^2} - \frac{1}{\mathrm{n}_2^2}\right)$
$\lambda \downarrow \mathrm{E} \uparrow; \frac{\mathrm{hc}}{\lambda_0} = -13.6(1)$

Balmer Series :

——————— n = 3

——————— n = 2

$\frac{hc}{\lambda_1} = -13.6\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$

$\frac{hc}{\lambda_1} = -13.6\left(\frac{1}{4} - \frac{1}{9}\right)$

$\frac{hc}{\lambda_1} = -13.6 \times \left(\frac{5}{36}\right)$

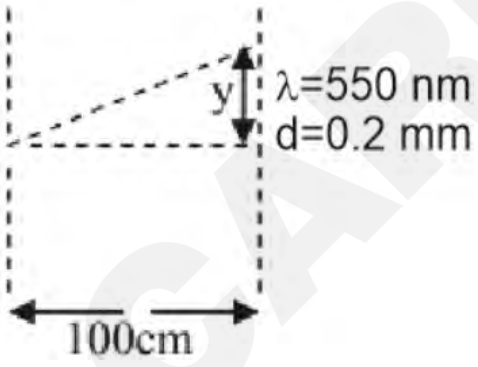
$\Rightarrow \frac{-13.6\lambda_0}{\lambda_1} = -13.6 \times \frac{5}{36}$

$\lambda_1 = \frac{\lambda_0 \times 36}{5} = \frac{915 \times 36}{5} = 6588$

Q. 23 In a single slit experiment, a parallel beam of green light of wavelength $550\ \text{nm}$ passes through a slit of width $0.20\ \text{mm}$. The transmitted light is collected on a screen $100\ \text{cm}$ away. The distance of first order minima from the central maximum will be $\text{x} \times 10^{-5}\ \text{m}$. The value of x is:

Correct Answer:
275

Solution:

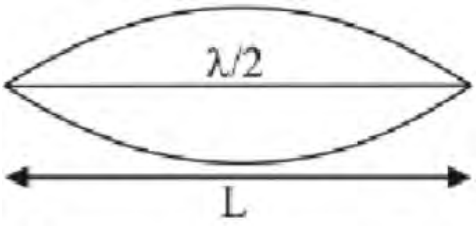


$y = \frac{\lambda D}{d} = \frac{550 \times 10^{-9} \times 100 \times 10^{-2}}{0.2 \times 10^{-3}} = 275$

Q. 24 A sonometer wire of resonating length 90 cm has a fundamental frequency of 400 Hz when kept under some tension. The resonating length of the wire with fundamental frequency of 600 Hz under same tension _____ cm.

Correct Answer:
60

Solution:



$f_0 = 400\ Hz; v = \sqrt{\frac{T}{\mu}} = \text{constant}$

$\frac{\lambda}{2} = L; v = f_0\lambda$

$\frac{v}{2f_0} = L \Rightarrow v = 2Lf_0$

$L' = \frac{v}{2f'} = \frac{2Lf_0}{2f'}$

$= \frac{Lf_0}{f'} = \frac{90\times400}{600} = 60$

Q. 25 A hollow sphere is rolling on a plane surface about its axis of symmetry. The ratio of rotational kinetic energy to its total kinetic energy is $\frac{x}{5}$. The value of x is _____.

Correct Answer:
2

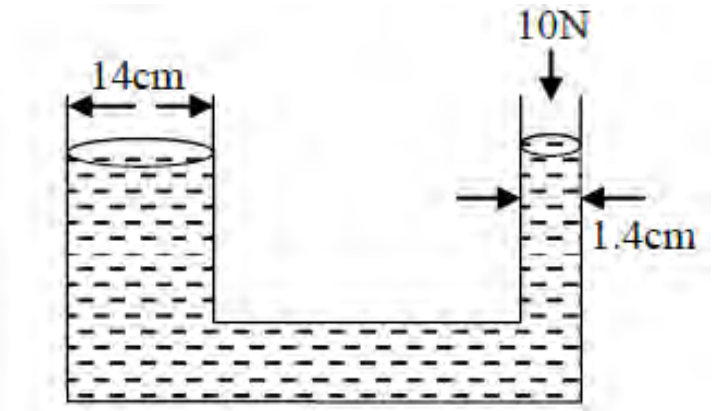
Solution:

$\frac{\frac{1}{2}I\omega^2}{\frac{1}{2}I\omega^2+\frac{1}{2}mv^2} = \frac{(\frac{1}{2})(\frac{2}{3}mR^2)\omega^2}{(\frac{1}{2})(\frac{2}{3}mR^2)\omega^2+\frac{1}{2}m(R\omega)^2}$

$= \frac{\frac{2}{3}}{\frac{2}{3}+1} = \frac{2}{5}$

$x = 2$

Q. 26 A hydraulic press containing water has two arms with diameters as mentioned in the figure. A force of 10 N is applied on the surface of water in the thinner arm. The force required to be applied on the surface of water in the thicker arm to maintain equilibrium of water is _____ N.



Correct Answer:
1000N

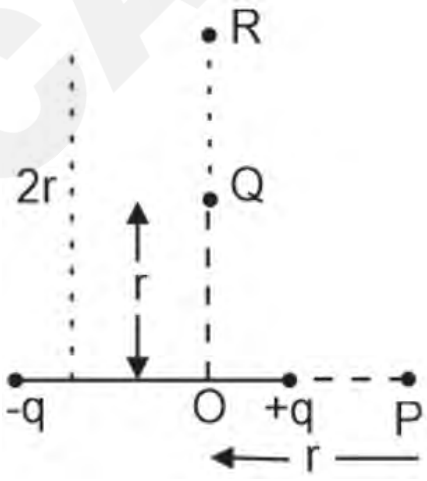
Solution:

$\frac{F_1}{A_1} = \frac{F_2}{A_2}$

$\frac{F_1}{\pi(7)^2} = \frac{10}{\pi \times (0.7)^2}$

$F_1 = 1000 \text{ N}$

Q. 27 The electric field at point p due to an electric dipole is E. The electric field at point R on equitorial line will be $\frac{E}{x}$. The value of x :



Correct Answer:
16

Solution:
$E_P = \frac{2KP}{r^3} = E$

$E_R = \frac{KP}{(2r)^3} = \frac{E}{16}$

$x = 16$

Q. 28 The maximum height reached by a projectile is 64 m. If the initial velocity is halved, the new maximum height of the projectile is _____ m.

Correct Answer:
16

Solution:
$\mathrm{H_{max}} = \frac{\mathrm{u^2 \sin^2 \theta}}{2\,\mathrm{g}}$

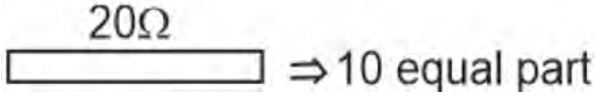
$\frac{\mathrm{H_{1\,max}}}{\mathrm{H_{2\,max}}} = \frac{\mathrm{u_1^2}}{\mathrm{u_2^2}}$

$\frac{64}{\mathrm{H_{2\,max}}} = \frac{\mathrm{u^2}}{(\mathrm{u}/2)^2}$

$\mathrm{H_{2\,max}} = 16\ \mathrm{m}$

Q. 29 A wire of resistance 20Ω is divided into 10 equal parts. A combination of two parts are connected in parallel and so on. Now resulting pairs of parallel combination are connected in series. The equivalent resistance of final combination is _____ Ω.

Correct Answer:
5

Solution:


Each part has resistance $= 2\Omega$

2 parts are connected in parallel so, $\mathrm{R} = 1\Omega$

Now, there will be 5 parts each of resistance 1Ω, they are connected in series.

$\mathrm{R_{eq}} = 5\mathrm{R}, \mathrm{R_{eq}} = 5\Omega$

Q. 30 The current in an inductor is given by $I = (3t + 8)$ where t is in second. The magnitude of induced emf produced in the inductor is $12\mathrm{mV}$. The selfinductance of the inductor _____ mH.

Correct Answer:
4

Solution:
$\mathrm{I} = 3\mathrm{t} + 8$

$\varepsilon = 12\mathrm{mV}$

$|\varepsilon| = \mathrm{L}\ \frac{\mathrm{dI}}{\mathrm{dt}}$

$12 = \mathrm{L} \times 3$

$\mathrm{L} = 4\mathrm{mH}$

Chemistry

Q. 1 Match **List - I** with **List - II.**

	List - I		**List - II.**
(A)	ICl	(I)	T -Shape
(B)	ICl_3	(II)	Square pyramidal
(C)	ClF_5	(III)	Pentagonal bipyramidal
(D)	IF_7	(IV)	Linear

Choose the correct answer from the options given below:

Option 1:
(A)-(I), (B)-(IV), C-(III), D-(II)

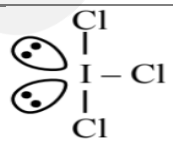
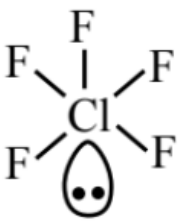
Option 2:
(A)-(I), (B)-(III), C-(II), D-(IV)

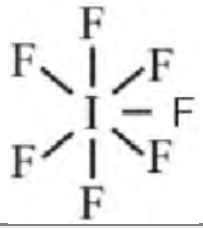
Option 3:
(A)-(IV), (B)-(I), C-(II), D-(III)

Option 4:
(A)-(IV), (B)-(III), C-(II), D-(I)

Correct Answer:
(A)-(IV), (B)-(I), C-(II), D-(III)

Solution:

A.	$I-Cl$	(iv)	linear
B.		(I)	T-shape
C.		(II)	Square pyramidal

D.		(III)	Pentagonal bipyramidal

Q. 2 While preparing crystals of Mohr's salt, dil. H_2SO_4 is added to a mixture of ferrous sulphate and ammonium sulphate, before dissolving this mixture in water, dil. H_2SO_4 is added here to:

Option 1:
prevent the hydrolysis of ferrous sulphate

Option 2:
prevent the hydrolysis of ammonium sulphate

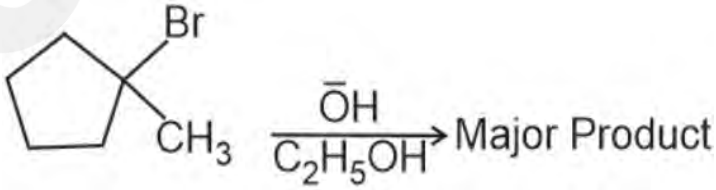
Option 3:
make the medium strongly acidic

Option 4:
increase the rate of formation of crystals

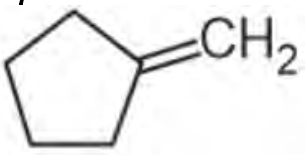
Correct Answer:
prevent the hydrolysis of ferrous sulphate

Solution:
Fe^{+2} ions undergoes hydrolysis, therefore while preparing aqueous solution of ferrous sulphate and ammonium sulphate in water dilute sulphuric acid is added to prevent hydrolysis of ferrous sulphate.

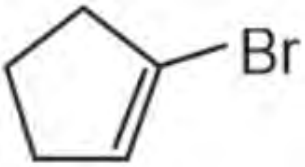
Q. 3 Identify the major product in the following reaction.



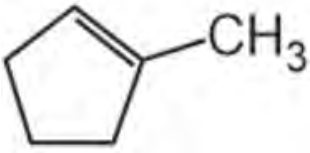
Option 1:



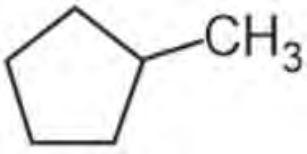
Option 2:



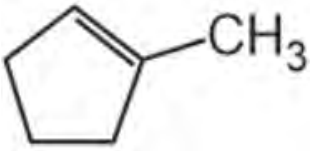
Option 3:



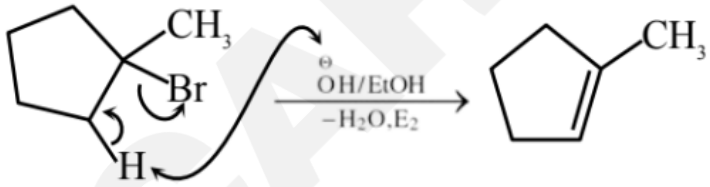
Option 4:



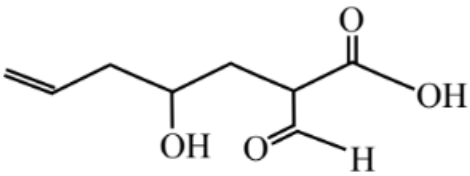
Correct Answer:



Solution:



Q. 4 The correct nomenclature for the following compound is:



Option 1:

2-carboxy-4-hydroxyhept-6-enal

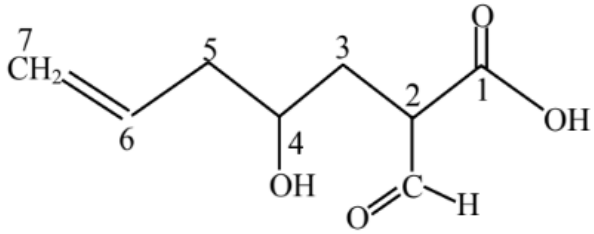
Option 2:
2-carboxy-4-hydroxyhept-7-enal

Option 3:
2-formyl-4-hydroxyhept-6-enoic acid

Option 4:
2-formyl-4-hydroxyhept-7-enoic acid

Correct Answer:
2-formyl-4-hydroxyhept-6-enoic acid

Solution:



2-formly-4-hydroxyhept-6-enoic acid

Q. 5 Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : NH_3 and NF_3 molecule have pyramidal shape with a lone pair of electrons on nitrogen atom. The resultant dipole moment of NH_3 is greater than that of NF_3.

Reason (R): In NH_3, the orbital dipole due to lone pair is in the same direction as the resultant dipole moment of the $N-H$ bonds. F is the most electronegative element.

In the light of the above statements, choose the **correct** answer from the options given below:

Option 1:
Both (A) and (R) are true and (R) is the correct explanation of (A)

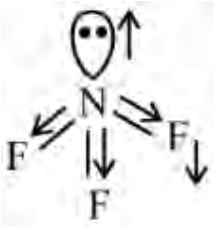
Option 2:
(A) is false but (R) is true

Option 3:
(A) is true but (R) is false

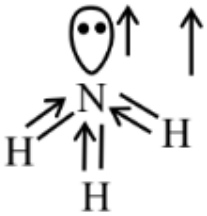
Option 4:
Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

Correct Answer:
Both (A) and (R) are true and (R) is the correct explanation of (A)

Solution:



Resultant dipole moment $= 0.80 \times 10^{-30} \mathrm{Cm}$



Resultant dipole moment $= 4.90 \times 10^{-30}\ \mathrm{cm}$

Q. 6 Given below are two statements:
Statement I : On passing $\mathrm{HCl}_{(g)}$ through a saturated solution of BaCl_2, at room temperature white turbidity appears.
Statement II : When HCl gas is passed through a saturated solution of NaCl, sodium chloride is precipitated due to common ion effect.
In the light of the above statements, choose the **most appropriate** answer from the options given below:

Option 1:
Statement I is correct but Statement II is incorrect

Option 2:
Both Statement I and Statement II are incorrect

Option 3:
Statement I is incorrect but Statement II is correct

Option 4:
Both Statement I and Statement II are correct

Correct Answer:
Statement I is correct but Statement II is incorrect

Solution:
$\mathrm{BaCl_2}, \mathrm{NaCl}$ are soluble but on adding $\mathrm{HCl(g)}$ to $\mathrm{BaCl_2}, \mathrm{NaCl}$ solutions, Sodium or Barium chlorides may precipitate out, as a consequence of the law of mass action.

Q. 7 The metal atom present in the complex MABXL (where A, B, X and L are unidentate ligands and M is metal) involves $\mathrm{sp^3}$ hybridization. The number of geometrical isomers exhibited by the complex is:

Option 1:
4

Option 2:
0

Option 3:
2

Option 4:
3

Correct Answer:
0

Solution:
Tetrahedral complex does not show geometrical isomerism.

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Q. 8 Match **List - I** with **List - II**.

List - I (Pair of Compounds)		**List - II (Isomerism)**	
(A)	n-propanol and Isopropanol	(I)	Metamerism
(B)	Methoxypropane and ethoxyethane	(II)	Chain Isomerism
(C)	Propanone and propanal	(III)	Position Isomerism
(D)	Neopentane and Isopentane	(IV)	Functional Isomerism

Option 1:
(A)-(II), (B)-(I), (C)-(IV), (D)-(III)

Option 2:
(A)-(III),(B)-(I),(C)-(II),(D)-(IV)

Option 3:
(A)-(I), (B)-(III), (C)-(IV), (D)-(II)

Option 4:
(A)-(III), (B)-(I), (C)-(IV), (D)-(II)

Correct Answer:
(A)-(III), (B)-(I), (C)-(IV), (D)-(II)

Solution:

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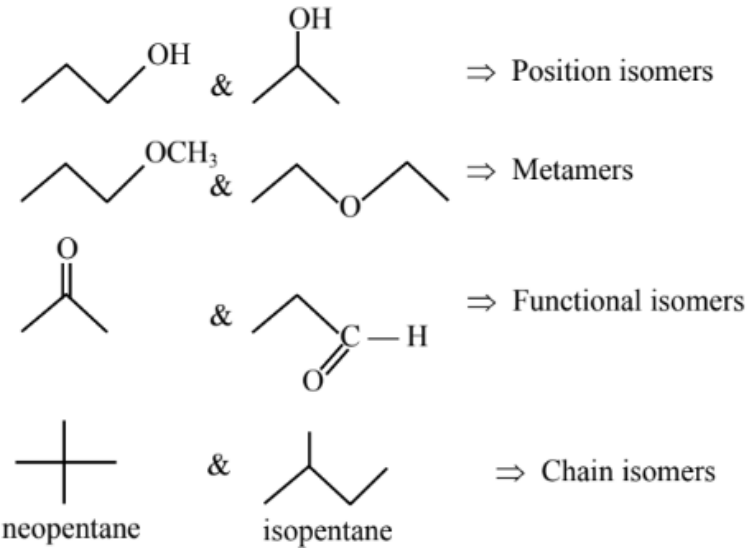


Q. 9 The quantity of silver deposited when one coulomb charge is passed through $\mathrm{AgNO_3}$ solution:

Option 1:
0.1g atom of silver

Option 2:
1 chemical equivalent of silver

Option 3:
1g of silver

Option 4:
1 electrochemical equivalent of silver

Correct Answer:
1 electrochemical equivalent of silver

Solution:

$$\mathrm{W = ZIt}$$

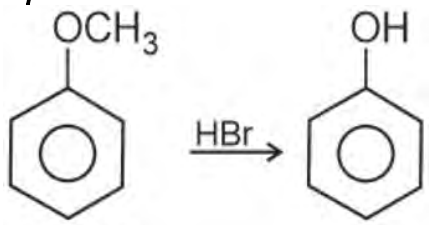
$$\mathrm{W = ZQ}$$

$$\mathrm{Q = \frac{W}{Z}}$$

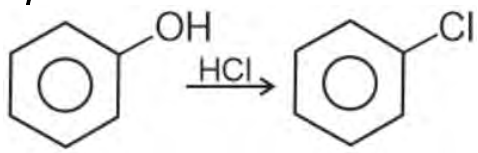
$$\mathrm{W = ZQ = \ (electrochemical\ equivalent)}$$

Q. 10 Which one of the following reactions is NOT possible?

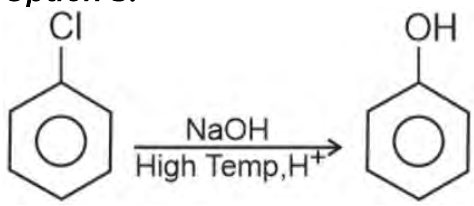
Option 1:



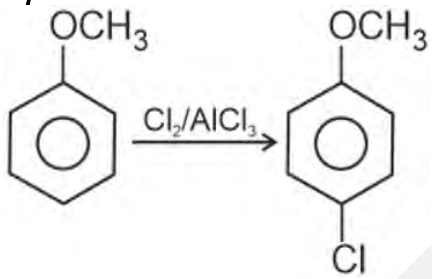
Option 2:



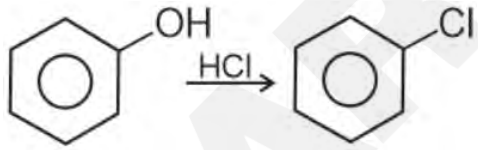
Option 3:



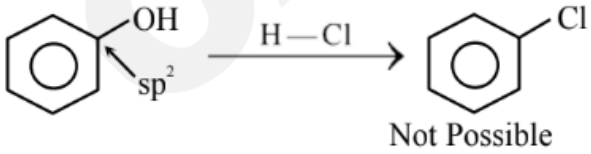
Option 4:



Correct Answer:



Solution:



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Q. 11 Given below are two statements :

Statement I : The metallic radius of Na is $1.86\ \mathrm{A}^{\circ}$ and the ionic radius of $\mathrm{Na}^{+}$is lesser than $1.86\ \mathrm{A}^{\circ}$.

Statement II : Ions are always smaller in size than the corresponding elements.

In the light of the above statements, choose the **correct** answer from the options given below :

Option 1:
Statement I is correct but Statement II is false

Option 2:
Both Statement I and Statement II are true

Option 3:
Both Statement I and Statement II are false

Option 4:
Statement I is incorrect but Statement II is true

Correct Answer:
Statement I is correct but Statement II is false

Solution:

$\mathrm{r_{Na} > r_{Na^{+}}}$

So, Statement (I) is correct but size of anions are greater than size of neutral atoms.
So statement (II) is incorrect.

Q. 12 $\mathrm{CH_3CH_2 - OH \xrightarrow[(ii)\ KMnO_4]{(i)\ \text{Jone's Reagent}} \xrightarrow[(iii)\ NaOH,CaO,\Delta]{}}$

Consider the above reaction sequence and identify the major product P.

Option 1:
Methane

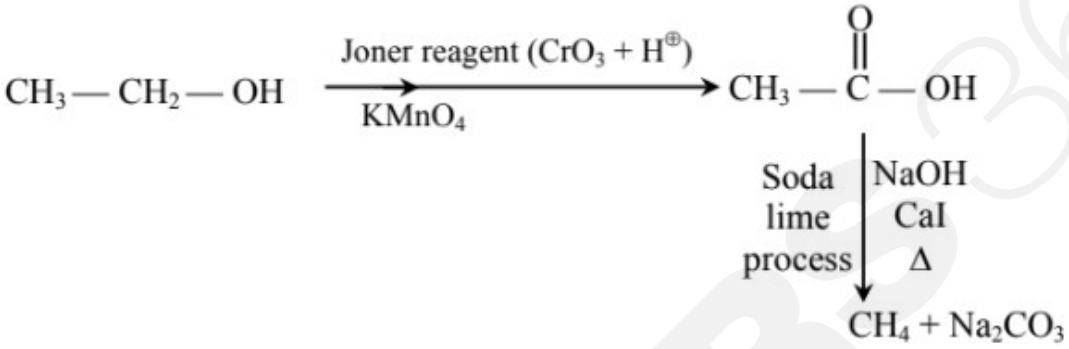
Option 2:
Methanal

Option 3:
Methoxymethane

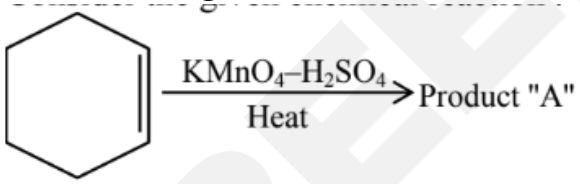
Option 4:
Methanoic acid

Correct Answer:
Methane

Solution:



Q. 13 Consider the given chemical reaction :



Product " A " is :

Option 1:
picric acid

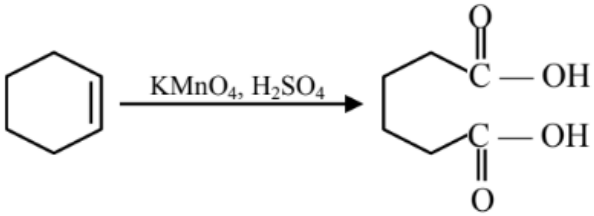
Option 2:
oxalic acid

Option 3:
acetic acid

Option 4:
adipic acid

Correct Answer:
adipic acid

Solution:



Q. 14 For the electro chemical cell
$M\ M^{2+}\ |X|X^{2-}$

If $E^0_{(M^{2+}/M)} = 0.46\ V$ and $E^0_{(X/X^{2-})} = 0.34\ V$.
Which of the following is correct?

Option 1:
$E_{cell} = -0.80\ V$

Option 2:
$M + X \rightarrow M^2 + X^{2-}$ is a spontaneous reaction

Option 3:
$M^{2+} + X^{2-} \rightarrow M + X$ is a spontaneous reaction

Option 4:
$E_{elll} = 0.80\ V$

Correct Answer:
$M^{2+} + X^{2-} \rightarrow M + X$ is a spontaneous reaction

Solution:
$M \mid M^{+2} \| X/X^{2-}$
$E^o_{cell} = E^o_{M/M^{+2}} + E^o_{X/X^{-2}}$
$= -0.46 + 0.34 = -0.12\ V$

As E^0_{cell} is negative so anode becomes cathode and cathode become anode.
Spontaneous reaction will be
$M^{+2} + X^{2-} \longrightarrow M + X$

Q. 15 The number of moles of methane required to produce $11\ \mathrm{g} \mathrm{CO}_2(\mathrm{g})$ after complete combustion is:
(Given molar mass of methane in $\mathrm{gmol}^{-1}: 16$)

Option 1:
0.75

Option 2:
0.25

Option 3:
0.35

Option 4:
0.5

Correct Answer:
0.25

Solution:

$$\mathrm{C_nH_{2n+2}} + \frac{3\mathrm{n}+1}{2}\mathrm{O_2} \longrightarrow \mathrm{nCO_2} + (\mathrm{n}+1)\mathrm{H_2O}$$

$$\mathrm{CH_4} + 2\mathrm{O_2} \longrightarrow \mathrm{CO_2} + 2\mathrm{H_2O}$$

$4\mathrm{gm} \quad 11\mathrm{gm}$

$0.25 \text{ mole} \quad 0.25 \text{ mole}$

$0.25 \text{ mole } \mathrm{CH_4} \text{ gives } 0.25 \text{ mole (or } 11\mathrm{gm} \text{) } \mathrm{CO_2}$

Q. 16 The number of complexes from the following with no electrons in the t_2 orbital is $\mathrm{TiCl_4}, [\mathrm{MnO_4}]^-, [\mathrm{FeO_4}]^{2-}, [\mathrm{FeCl_4}]^-, [\mathrm{CoCl_4}]^{2-}$

Option 1:
3

Option 2:
1

Option 3:
4

Option 4:
2

Correct Answer:
3

Solution:
$\mathrm{TiCl_4 \Rightarrow Ti^{+4}} \quad \mathrm{e^0t_2^0}$
$\mathrm{MnO_4^- \Rightarrow Mn^{+7}} \quad \mathrm{e^0t_2^0}$
$\mathrm{FeO_4^{2-} \Rightarrow Fe^{+6}} \quad \mathrm{e^2t_2^0}$
$\mathrm{FeCl_4^{2-} \Rightarrow Fe^{+2}} \quad \mathrm{e^3t_2^3}$
$\mathrm{CoCl_4^{2-} \Rightarrow Co^{+2}} \quad \mathrm{e^4t_2^3}$

Q. 17 The number of ions from the following that have the ability to liberate hydrogen from a dilute acid is_____ $\mathrm{Ti^{2+}}$, $\mathrm{Cr^{2+}}$ and $\mathrm{V^{2+}}$

Option 1:
0

Option 2:
2

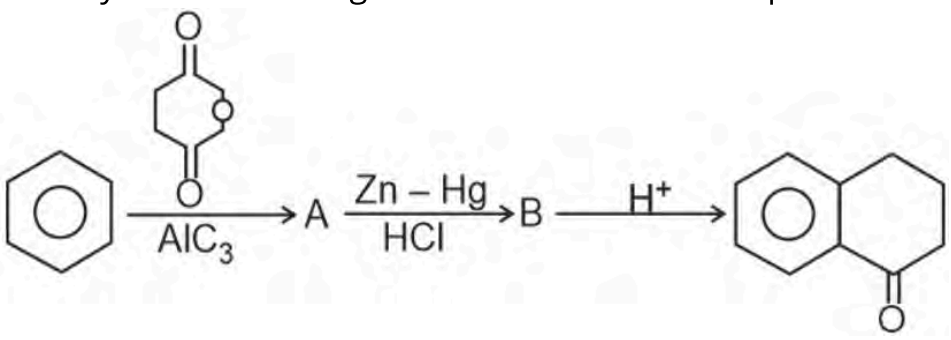
Option 3:
3

Option 4:
1

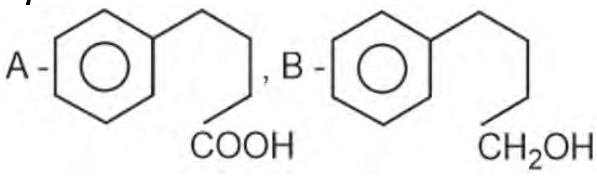
Correct Answer:
3

Solution:
The ions $\mathrm{Ti^{+2}}$, $\mathrm{V^{+2}Cr^{+2}}$ are strong reducing agents and will liberate hydrogen from a dilute acid, eg. $\mathrm{2Cr^{+2}_{(aq.)} + 2H^+_{(aq.)} \longrightarrow 2Cr^{+3}_{(aq.)} + H_2(\,g)}$

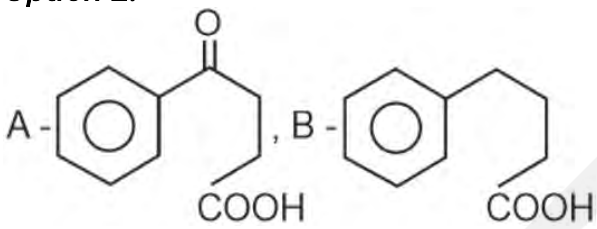
Q. 18 Identify A and B in the given chemical reaction sequence : -



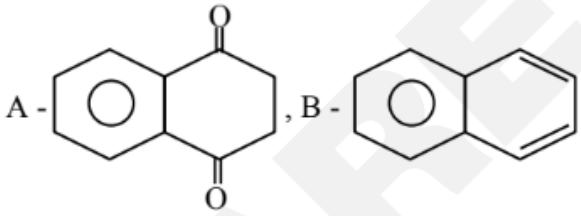
Option 1:



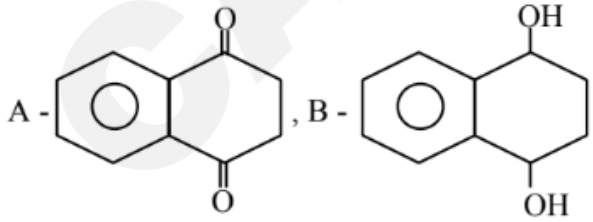
Option 2:



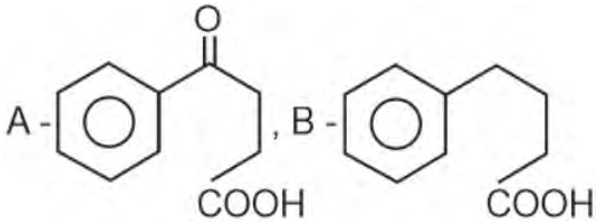
Option 3:



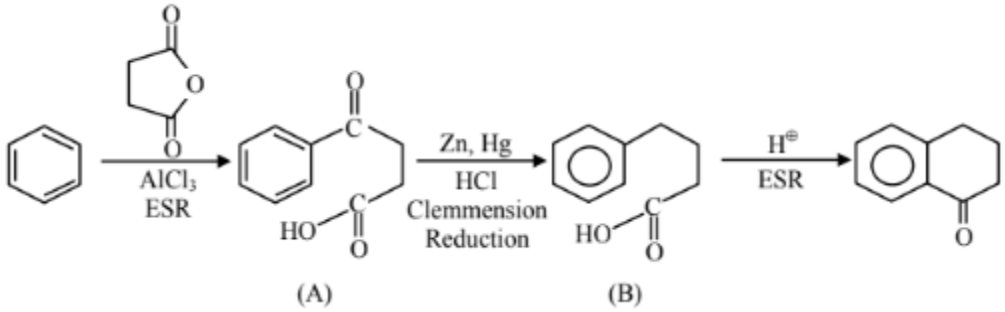
Option 4:



Correct Answer:



Solution:



Q. 19 The correct statements from the following are :

(A) The decreasing order of atomic radii of group 13 elements is $\mathrm{Tl} > \mathrm{In} > \mathrm{Ga} > \mathrm{Al} > \mathrm{B}$.

(B) Down the group 13 electronegativity decreases from top to bottom.

(C) Al dissolves in dil. HCl and liberate $\mathrm{H_2}$ but conc. $\mathrm{HNO_3}$ renders Al passive by forming a protective oxide layer on the surface.

(D) All elements of group 13 exhibits highly stable +1 oxidation state.

(E) Hybridisation of Al in $[\mathrm{Al(H_2O)_6}]^{3+}$ ion is $\mathrm{sp^3\ d^2}$.

Choose the **correct** answer from the options given below :

Option 1:
(C) and (E) only

Option 2:
(A), (C) and (E) only

Option 3:
(A), (B), (C) and (E) only

Option 4:
(A) and (C) only

Correct Answer:
(C) and (E) only

Solution:

A. size order $\mathrm{T\ell} > \mathrm{In} > \mathrm{Al} > \mathrm{Ga} > \mathrm{B}$

B. Electronegativity order $\mathrm{B} > \mathrm{Al} < \mathrm{Ga} < \mathrm{In} < \mathrm{T\ell}$

D. $\mathrm{B, Al}$ are more stable in +3 oxidation state

So, only C, E statements are correct.

Q. 20 Coagulation of egg, on heating is because of :

Option 1:
Denaturation of protein occurs

Option 2:
The secondary structure of protein remains unchanged

Option 3:
Breaking of the peptide linkage in the primary structure of protein occurs

Option 4:
Biological property of protein remains unchanged

Correct Answer:
Denaturation of protein occurs

Solution:
Coagulation of egg give primary structure of protein, which is known as denaturation of protein

Q. 21 Combustion of 1 mole of benzene is expressed at
$C_6H_6(1) + \frac{15}{2}O_2(g) \rightarrow CO_2(g) + 3H_2O(1).$

The standard enthalpy of combustion of $2\ mol$ of benzene is - ' x ' kJ.
$x =$_____

(1) standard Enthalpy of formation of $1\ mol$ of $C_6H_6(1)$, for the reaction $6C\ (graphite) + 3H_2(g) \rightarrow C_6H_6(1)$ is $48.5\ kJ\ mol^{-1}$.

(2) Standard Enthalpy of formation of $1\ mol$ of $CO_2(g)$, for the reaction C (graphite) $+O_{2(g)} \rightarrow CO_2(g)$ is $-393.5\ kJ\ mol^{-1}$.

(3) Standard and Enthalpy of formation of 1 mol of $H_2O(1)$, for the reaction
$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(1)$ is $-286\ kJ\ mol^{-1}$.

Correct Answer:
6535

Solution:
$6C\ (\text{graphite})\ + 3H_2(\ g) \rightarrow C_6H_6(\ell); \Delta H = 48.5\ \text{kJ/mol}$
$C\ (\text{graphite})\ + O_2(\ g) \rightarrow CO_2(\ g); \Delta H = -393.5\ \text{kJ/mol}$
$H_2^{(g)} + \frac{1}{2}(\ g) \longrightarrow H_2O(\ell); \Delta H = -286\ \text{kJ/mol}$
$\text{equation}\ -(1) \times 1 + (2) \times 6 + (3) \times 3$
$-48.5 - 6 \times 393.5 - 3 \times 286$
$= -3267.5\ \text{kJ for 1 mol}$
$= -6535\ \text{kJ for 2 mol}$

Q. 22 The fusion of chromite ore with sodium carbonate in the presence of air leads to the formation of products A and B along with the evolution of CO_2. The sum of spin-only magnetic moment values of A and B is_____B.M. (Nearest integer)
(Given atomic number: C: 6, Na: 11, O: 8, Fe: 26, Cr: 24]

Correct Answer:
6

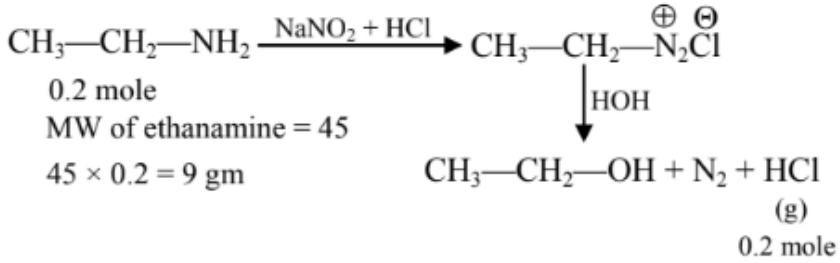
Solution:
$$4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \rightarrow \underset{A}{8Na_2CrO_4} + \underset{B}{2Fe_2O_3} + 8CO_2$$

Spin only magnetic moment
$\text{For } Na_2CrO_4 \quad \mu_B = 0$
$\text{For } Fe_2O_3 \quad \mu_B = 5.9$
$\text{sum} = 5.9$

Q. 23 X of enthanamine was subjected to reaction with $NaNO_2/HCl$ followed by hydrolysis to liberate N_2 and HCl. The HCl generated was completely neutralised by 0.2 moles of $NaOH$. X is_____g.

Correct Answer:
9

Solution:



Q. 24 In an atom, total number of electrons having quantum numbers $n = 4, |m_l| = 1$ and $m_s = -\frac{1}{2}$ is

Correct Answer:
6

Solution:
$n = 4$

ℓ	m_ℓ
0	0
1	$-1, 0, +1$
2	$-2, -1, 0, +1, +2, +3$

So number of orbital associated with
$n = 4, |m_\ell| = 1$ are 6

Now each orbital contain one e^- with $m_s = -\frac{1}{2}$

Q. 25 Using the given figure, the ratio of R_f values of sample A and sample C is $x \times 10^{-2}$. Value of x is_____.

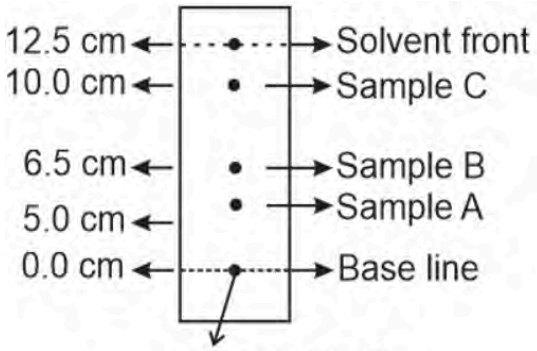


Fig : Paper chromatography of Samples

Correct Answer:
50

Solution:

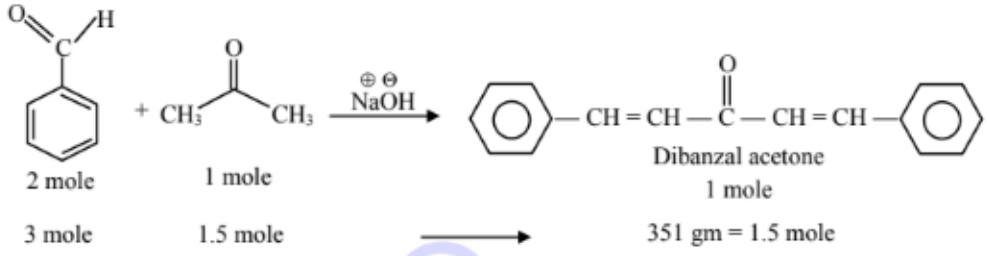
$$R_f \text{ of } A = \frac{5}{12.5} \qquad R_f \text{ of } C = \frac{10}{12.5}$$

$$\text{Ratio } = \frac{R_{f(A)}}{R_{f(C)}} = \frac{1}{2} = 0.5 \text{ or } 50 \times 10^{-2}$$

Q. 26 In the Claisen-Schmidt reaction to prepare 351 g of dibenzalacetone using 87 g of acetone, the amount of benzaldehyde required is_____g. (Nearest integer)

Correct Answer:
318

Solution:
Claisen Schmidt reaction



mw of benzaldehyde $= 106\ 106 \times 3 = 318\text{gm}$. Benzaldehyde is required to give 1.5 mole (or 351gm) product

Q. 27 Consider the following single step reaction in gas phase at constant temperature.

$2\ \mathrm{A}_{(g)} + \mathrm{B}_{(g)} \rightarrow \mathrm{C}_{(g)}$

The initial rate of the reaction is recorded as r_1 when the reaction starts with $1.5\ \mathrm{atm}$ pressure of A and $0.7\ \mathrm{atm}$ pressure of B. After some time, the rate r_2 is recorded when the pressure of C becomes $0.5\ \mathrm{atm}$. The ratio $\mathrm{r}_1 : \mathrm{r}_2$ is_____$\times 10^{-1}$. (Nearest integer)

Correct Answer:
315

Solution:

$2\ \mathrm{A}(\mathrm{g}) + \mathrm{B}(\mathrm{g}) \longrightarrow \mathrm{C}(\mathrm{g})$

$\mathrm{r}_1 \quad 1.5\ \mathrm{atm} \quad 0.7\ \mathrm{atm}$

$\mathrm{r}_2 \quad 0.5\ \mathrm{atm} \quad 0.2\ \mathrm{atm} \quad 0.5\ \mathrm{atm}$

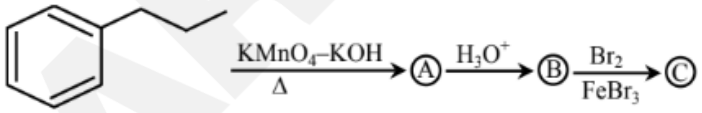
$\because r = K[P_A]^2\,[P_B]$

$r_1 = K[1.5]^2[0.7]$

$r_2 = K[0.5]^2[0.2]$

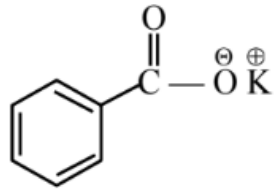
$\frac{r_1}{r_2} = 9 \times \frac{7}{2} = 31.5 = 315 \times 10^{-1}$

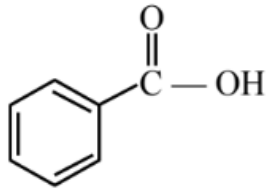
Q. 28 The product (C) in the following sequence of reactions has_____π bonds.

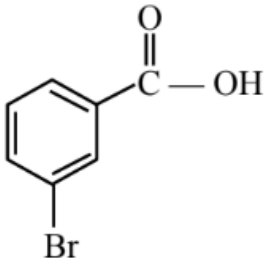


Correct Answer:
4

Solution:

A = 

B = 

C = 

π bonds $= 4$

Q. 29 Considering acetic acid dissociates in water, its dissociation constant is 6.25×10^{-5}. If $5\ \mathrm{mL}$ of acetic acid is dissolved in 1 litre water, the solution will freeze at $-\mathrm{x} \times 10^{-2}{}^{\circ}\mathrm{C}$, provided pure water freezes at $0^{\circ}\mathrm{C}$.

x=_____.(Nearest integer)

Given : $(\mathrm{K_f})_{\text{water}} = 1.86\ \mathrm{K\ kg\ mol^{-1}}$.

density of acetic acid is $1.2\ \mathrm{g\ mol^{-1}}$

molar mass of water $= 18\ \mathrm{g\ mol^{-1}}$.

molar mass of acetic acid $= 60\ \mathrm{g\ mol^{-1}}$.

density of water $= 1\ \mathrm{g\ cm^{-3}}$

Acetic acid dissociates as

$\mathrm{CH_3COOH} \rightleftharpoons \mathrm{CH_3COO^{\Theta}} + \mathrm{H^{\oplus}}$

Correct Answer:

19

Solution:

$\text{Mass of } CH_3COOH = V \times d$
$= 5ml \times 1.2\ g/ml$
$= 6gm$

$$n_{CH_3COOH} = \frac{6}{60} = 0.1 \text{ mol}$$

$$m_{CH_3COOH} \approx M_{CH_3COOH} = \frac{0.1}{1} = 0.1M$$

$$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$$

C

$C - C\alpha \quad C\alpha \quad C\alpha$

$$K_a = \frac{C\alpha^2}{1-\alpha}$$

$$1 - \alpha \approx 1 \Rightarrow K_a = C\alpha^2$$

$$\alpha = \sqrt{\frac{Ka}{C}} = \sqrt{\frac{6.25 \times 10^{-5}}{0.1}} = 25 \times 10^{-3}$$

$\text{V.f. (i)} = 1 + \alpha(n-1) = 1 + \alpha(2-1) = 1 + \alpha$
$= 1 + 25 \times 10^{-3} = 1.025$

$\Delta T_f = iK_t m$
$= (1.025)(1.86)(0.1)$
$= 0.19$
$= 19 \times 10^{-2}$

Q. 30 Number of compounds from the following with zero dipole moment is_____.

$HF, H_2, H_2S, CO_2, NH_3, BF_3, CH_4, CHCl_3, SiF_4, H_2O, BeF_2$

Correct Answer:
6

Solution:
$H_2, CO_2, BF_3, CH_4, SiF_4, BeF_2$
are symm. molecule so dipole moment is zero

Maths

Q. 1 Let $f : [-1, 2] \to R$ be given by $f(x) = 2x^2 + x + [x^2] - [x]$, where $[t]$ denotes the greatest integer less than or equal to t. The number of points, where f is not continuous, is :

Option 1:
6

Option 2:
3

Option 3:
4

Option 4:
5

Correct Answer:
4

Solution:
Doubtful points : $-1,0,1, \sqrt{2}, \sqrt{3}, 2$

at $x = \sqrt{2}, \sqrt{3}$

$$f(x) = \underset{\downarrow}{\left(2x^2 + x - [x]\right)} + \underset{\downarrow}{[x^2]} = \text{Discount}$$

at $x = -1$:

$$\text{RHL} \Rightarrow \left.\begin{matrix} f(x) = (2 - 1 - (-1)) + 0 = 2 \\ f(-1) = 2 - 1 - (-1) + 1 = 3 \end{matrix}\right\} \text{Dis.}$$

at $x = 2$:

$$\text{LHL} \Rightarrow \left.\begin{matrix} f(x) = 8 + 2 - 1 + 3 = 12 \\ f(2) = 8 + 2 - 2 + 4 = 12 \end{matrix}\right\} \text{Cont.}$$

at $x = 0$:

$$\text{LHL} \Rightarrow \left.\begin{matrix} 0 + 0 - (-1) + 0 = 1 \\ f(0) = 0 \end{matrix}\right\} \text{Dis.}$$

at $x = 1$

$$\left.\begin{array}{ll} \text{LHL} \Rightarrow & 2+1-0+0=3 \\ & f(1)=3-1+1=3 \\ \text{RHL} \Rightarrow & 2+1-1+1=3 \end{array}\right\} \text{Cont.}$$

Q. 2 The differential equation of the family of circles passing the origin and having center at the line y=x is :

Option 1:

$(x^2-y^2+2xy)dx=(x^2-y^2+2xy)dy$

Option 2:

$(x^2+y^2+2xy)dx=(x^2+y^2-2xy)dy$

Option 3:

$(x^2-y^2+2xy)dx=(x^2-y^2-2xy)dy$

Option 4:

$(x^2+y^2-2xy)dx=(x^2+y^2+2xy)dy$

Correct Answer:

$(x^2-y^2+2xy)dx=(x^2-y^2-2xy)dy$

Solution:

$C \equiv x^2+y^2+gx+gy=0---(1)$

$2x+2yy'+g+gy'=0$

$g=-\left(\frac{2x+2yy'}{1+y'}\right)$

Put in (1)

$x^2+y^2-\left(\frac{2x+2yy'}{1+y'}\right)(x+y)=0$

$(x^2-y^2-2xy)y'=x^2-y^2+2xy$

Q. 3 Let $S_1 = \{z \in C : |z| \leq 5\}$, $S_2 = \left\{z \in C : Im\left(\frac{z+1-\sqrt{3}i}{1-\sqrt{3}i}\right) \geq 0\right\}$ and $S_3 = \{z \in C : Re(z) \geq 0\}$. Then the area of region $S_1 \cap S_2 \cap S_3$ is

Option 1:
$\frac{125\pi}{6}$

Option 2:
$\frac{125\pi}{24}$

Option 3:
$\frac{125\pi}{4}$

Option 4:
$\frac{125\pi}{12}$

Correct Answer:
$\frac{125\pi}{12}$

Solution:

$S_1 : x^2 + y^2 \leq 25 - - - -(1)$

$S_2 : \text{lm of } \frac{z + (1 - \sqrt{3}i)}{(1 - \sqrt{3}i)} \geq 0$

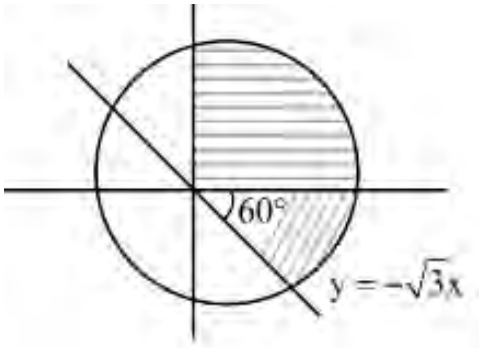
$\text{lm of } \left(\frac{x + iy}{1 - \sqrt{3}i} + 1\right) \geq 0$

$\text{lm of } \left(\frac{(x + iy)(1 + \sqrt{3}i)}{4}\right) \geq 0$

$\Rightarrow \sqrt{3}x + y \geq 0 - - - -(2)$

$S_3 : x \geq 0 - - - - - -(3)$

$\text{Area } = \frac{5}{12}(\pi(5)^2)$



Q. 4 The area enclosed between the curves $y = x|x|$ and $\mathrm{y} = \mathrm{x} - |\mathrm{x}|$ is :

Option 1:
$\frac{8}{3}$

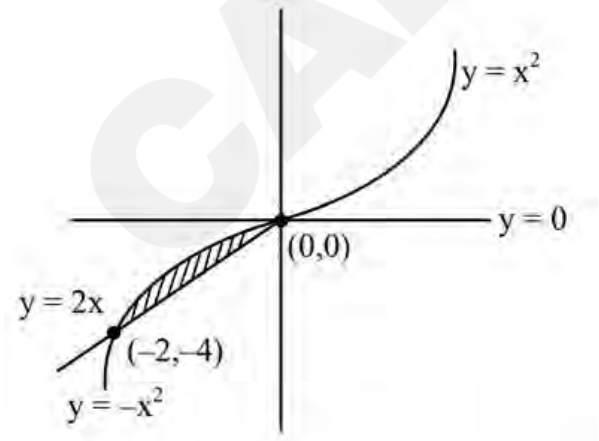
Option 2:
$\frac{2}{3}$

Option 3:
1

Option 4:
$\frac{4}{3}$

Correct Answer:
$\frac{4}{3}$

Solution:



$\mathrm{A} = \int_{-2}^{0} -\mathrm{x}^2 - 2\mathrm{x} = \frac{4}{3}$

Q. 5 60 words can be made using all the letters of the word BHBJO, with or without meaning. If these words are written as in a dictionary, then the 50^{th} word is :

Option 1:
OBBHJ

Option 2:
HBBJO

Option 3:
OBBJH

Option 4:
JBBOH

Correct Answer:
OBBJH

Solution:
B B H J O

$\text{B} - - - 4! = 24$

$\text{H} - - - \frac{4!}{2!} = 12$

$\text{J} - - - \frac{4!}{2!} = 12$

O B B H J
O B B J H $\rightarrow 50^{\text{th}}$ rank

Q. 6 Let $\vec{\text{a}} = 2\hat{\text{i}} + 5\hat{\text{j}} - \hat{\text{k}}, \vec{\text{b}} = 2\hat{\text{i}} - 2\hat{\text{j}} + 2\hat{\text{k}}$ and $\vec{\text{c}}$ be three vectors such that $(\vec{c} + \hat{i}) \times (\vec{a} + \vec{b} + \hat{i}) = \vec{a} \times (\vec{c} + \hat{i}) \cdot \vec{a} \cdot \vec{c} = -29$, then $\vec{\text{c}} \cdot (-2\hat{\text{i}} + \hat{\text{j}} + \hat{\text{k}})$ is equal to :

Option 1:
10

Option 2:
5

Option 3:
15

Option 4:
12

Correct Answer:
5

Solution:

Let's assume $\vec{v} = \vec{a} + \vec{b} + \hat{i}$

$= 5\hat{i} + 3\hat{j} + \hat{k}$

and $\vec{c} + \hat{i} = \vec{p}$

So,

$\vec{p} \times \vec{v} = \vec{a} \times \vec{p}$

$\vec{p} \times \vec{v} + \vec{p} \times \vec{a} = \vec{0}$

$\dot{p} \times (\vec{v} + a) = \vec{0}$

$\Rightarrow \vec{p} = \lambda(\vec{v} + \vec{a})$

$\vec{c} + i = \lambda(7\hat{i} + 8\hat{j})$

$\bar{a} \cdot \bar{c} + \bar{a} \cdot \hat{i} = \lambda \bar{a} \cdot (7\hat{i} + 8\hat{j})$

$-29 + 2 = \lambda(14 + 40)$

$\lambda = -\frac{1}{2}$

$\vec{c} \cdot (-2\hat{i} + \hat{j} + \hat{k}) + \hat{i} \cdot (-2\hat{i} + \hat{j} + \hat{k}) = \lambda(7\hat{i} + 8\hat{j}) \cdot (-2\hat{i} + \hat{j} + \hat{k})$

$= -\frac{1}{2}(-14 + 8) + 2 = 5$

CAREERS360

Q. 7 Consider three vectors $\vec{a}, \vec{b}, \vec{c}$. Let $|\vec{a}| = 2, |\vec{b}| = 3$ and $\vec{a} = \vec{b} \times \vec{c}$. If $\alpha \in \left[0, \frac{\pi}{3}\right]$ is the angle between the vectors $\vec{b}$ and $\vec{c}$, then the minimum value of $27|\vec{c} - \vec{a}|^2$ is equal to :

Option 1:
110

Option 2:
105

Option 3:
124

Option 4:
121

Correct Answer:
124

Solution:

$$|\vec{c} - \vec{a}| = |\vec{c}|^2 + |\vec{a}|^2 - 2\vec{a} \cdot \vec{c}$$

$$= |\vec{c}|^2 + 4 - 0$$

$$\because \vec{a} = \vec{b} \times \vec{c}$$

$$|\vec{a}| = |\vec{b} \times \vec{c}|$$

$$2 = 3|\vec{c}| \sin\alpha$$

$$|\vec{c}| = \frac{2}{3}\operatorname{cosec}\alpha \quad \alpha \in \left[0, \frac{\pi}{3}\right]$$

$$|\vec{c}|_{\min} = \frac{2}{3} \times \frac{2}{\sqrt{3}} \quad \operatorname{cosec}\alpha \in \left[\frac{2}{\sqrt{3}}, \infty\right)$$

$$\Rightarrow 27|\vec{c} - \vec{a}|^2_{\min} = 27\left(\frac{16}{27} + 4\right) = 124$$

Q. 8 Let $A(-1, 1)$ and $B(2, 3)$ be two points and P be a variable point above the line AB such that the area of $\triangle PAB$ is 10 . If the locus of P is $ax + by = 15$, then $5a + 2b$ is :

Option 1:
$-\frac{12}{5}$

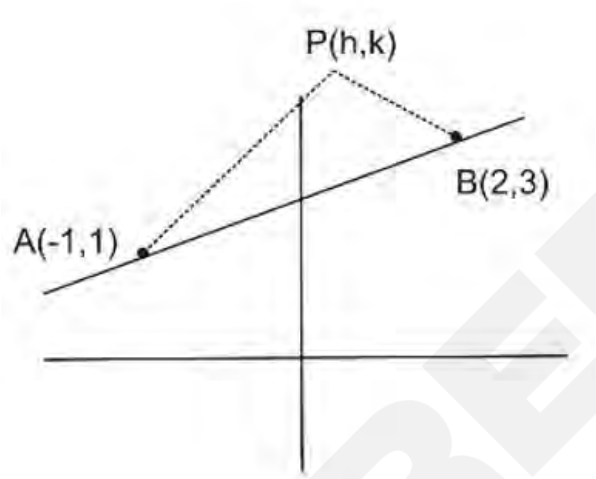
Option 2:
$-\frac{6}{5}$

Option 3:
4

Option 4:
6

Correct Answer:
$-\frac{12}{5}$

Solution:



$$\frac{1}{2}\begin{vmatrix} h & k & 1 \\ -1 & 1 & 1 \\ 2 & 3 & 1 \end{vmatrix} = 10$$

$$-2x + 3y = 25$$

$$-\frac{6}{5}x + \frac{9}{5}y = 15$$

$$a = -\frac{6}{5}, b = \frac{9}{5}$$

$$5a = -6, 2b = \frac{18}{5}$$

Q. 9 Let (α, β, γ) be the image of the point (8,5,7) in the line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-2}{5}$. Then $\alpha + \beta + \gamma$ is equal to

Option 1:
16

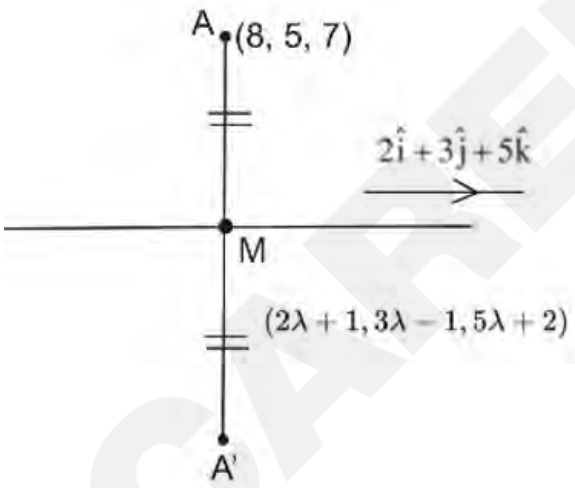
Option 2:
18

Option 3:
14

Option 4:
20

Correct Answer:
14

Solution:



$\mathrm{AM} \cdot (2\hat{\mathrm{i}} + 3\hat{\mathrm{j}} + 5\hat{\mathrm{k}}) = 0$

$(2\lambda - 7)(2) + (3\lambda - 6)(3) + (5\lambda - 5)(5) = 0$

$38\lambda = 57$

$\lambda = \frac{3}{2}$

$\mathrm{M}\left(4, \frac{7}{2}, \frac{19}{2}\right)$

$\mathrm{A}'(0, 2, 12)$

Q. 10 If the constant term in the expansion of $\left(\frac{\sqrt[5]{3}}{x}+\frac{2x}{\sqrt[3]{5}}\right)^{12}, x \neq 0$, is $\alpha \times 2^8 \times \sqrt[5]{3}$, then 25α is equal to :

Option 1:
639

Option 2:
724

Option 3:
693

Option 4:
742

Correct Answer:
693

Solution:

$$T_{r+1} = {}^{12}C_r \left(\frac{3^{1/5}}{x}\right)^{12-r} \left(\frac{2x}{5^{1/3}}\right)^{r}$$

$$T_{r+1} = \frac{{}^{12}C_r (3)^{\frac{12-r}{5}} (2)^r (x)^{2r-12}}{(5)^{r/3}}$$

$$r = 6$$

$$T_7 = \frac{{}^{12}C_6 (3)^{6/5} (2)^6}{5^2} = \left(\frac{9 \times 11 \times 7}{25}\right) 2^8 \cdot 3^{1/5}$$

$$25\alpha = 693$$

Q. 11 Let $f,\ g : R \to R$ be defined as : $f(x) = |x-1|$ and $g(x) = \begin{cases} e^x, & x \geq 0 \\ x+1, & x \leq 0 \end{cases}$. Then the function $f(\ g(x))$ is

Option 1:
neither one-one nor onto.

Option 2:
one-one but not onto.

Option 3:
both one-one and onto.

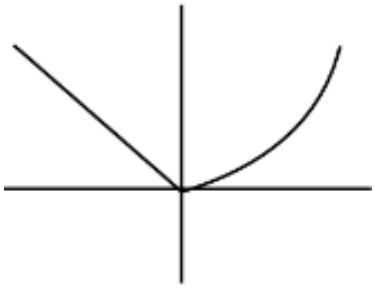
Option 4:
onto but not one-one.

Correct Answer:
neither one-one nor onto.

Solution:
$f(g(x)) = |g(x) - 1|$

$fog \begin{bmatrix} |e^x - 1| & x \geq 0 \\ |x + 1 - 1| & x \leq 0 \end{bmatrix}$

$\text{fog} \begin{bmatrix} e^x - 1 & x \geq 0 \\ -x & x \leq 0 \end{bmatrix}$



Q. 12 Let the circle $C_1 : x^2 + y^2 - 2(x + y) + 1 = 0$ and C_2 be a circle having centre at $(-1, 0)$ and radius 2 . If the line of the common chord of C_1 and C_2 intersects the y-axis at the point P, then the square of the distance of P from the centre of C_1 is :

Option 1:
2

Option 2:
1

Option 3:
6

Option 4:
4

Correct Answer:
2

Solution:
$S_1 : x^2 + y^2 - 2x - 2y + 1 = 0$
$S_2 : x^2 + y^2 + 2x - 3 = 0$
$\text{Common chord } = S_1 - S_2 = 0$
$-4x - 2y + 4 = 0$
$2x + y = 2 \Rightarrow P(0, 2)$
$d^2_{(c,p)} = (1-0)^2 + (2-1)^2 = 2$

Q. 13 Let the set $S = \{2, 4, 8, 16, \ldots..512\}$ be partitioned into 3 sets A, B, C with equal number of elements such that $A \cup B \cup C = S$ and $A \cap B = B \cap C = A \cap C = \phi$. The maximum number of such possible partitions of S is equal to :

Option 1:
1680

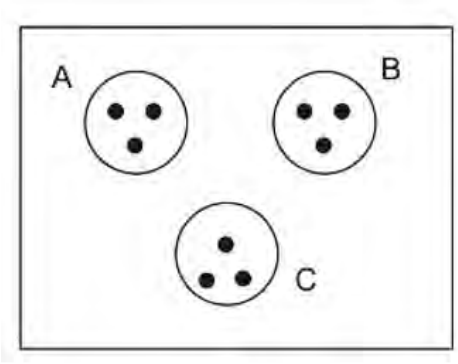
Option 2:
1520

Option 3:
1710

Option 4:
1640

Correct Answer:
1680

Solution:



$\frac{9!}{(3!3!3!)3!} \times 3!$

Q. 14 The values of m, n, for which the system of equations

$x + y + z = 4$

$2x + 5y + 5z = 17$

$x + 2y + mz = n$

has infinitely many solutions, satisfy the equation :

Option 1:

$m^2 + n^2 - m - n = 46$

Option 2:

$m^2 + n^2 + m + n = 64$

Option 3:

$m^2 + n^2 + mn = 68$

Option 4:

$m^2 + n^2 - mn = 39$

Correct Answer:

$m^2 + n^2 - mn = 39$

Solution:

$$D = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 5 & 5 \\ 1 & 2 & m \end{vmatrix} = 0 \Rightarrow m = 2$$

$$D_3 = \begin{vmatrix} 1 & 1 & 4 \\ 2 & 5 & 17 \\ 1 & 2 & n \end{vmatrix} = 0 \Rightarrow n = 7$$

Q. 15 The coefficients a, b, c in the quadratic equation $ax^2 + bx + c = 0$ are from the set {1,2,3,4,5,6}. If the probability of this equation having one real root bigger than the other is P, then 216 p equals :

Option 1:
57

Option 2:
38

Option 3:
19

Option 4:
76

Correct Answer:
38

Solution:

$D > 0$

$b^2 > 4ac$

$b = 3 : (a, c) = (1,1)(1,2)(2,1)$

$b = 4 : (a, c) = (1,1)(1,2)(2,1)(1,3)(3,1)$

$b = 5 : (a, c) = (1,1)(1,2)(2,1)(1,3)(3,1)(1,4)(4,1)$
$(1,5)(5,1)(1,6)(6,1)(2,3)(3,2)(2,2)$

$b = 6 : (a, c) = (1,1)(1,2)(2,1)(1,3)(3,1)(1,4)(4,1)$
$(1,5)(5,1)(1,6)(6,1)(2,3)(3,2)(2,4)(4,2)(2,2)$

$\text{fav. cases } = 38$

$\text{Prob. : } \dfrac{38}{6 \times 6 \times 6}$

Q. 16 Let ABCD and AEFG be squares of side 4 and 2 units, respectively. The point E is on the line segment AB and the point F is on the diagonal AC. Then the radius r of the circle passing through the point F and touching the line segments BC and CD satisfies :

Option 1:
r=1

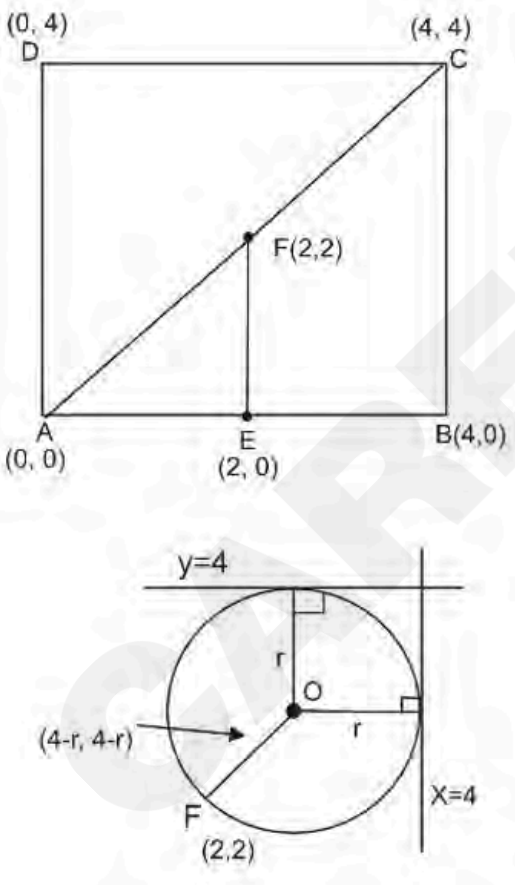
Option 2:
$r^2 - 8r + 8 = 0$

Option 3:
$2r^2 - 4r + 1 = 0$

Option 4:
$2r^2 - 8r + 7 = 0$

Correct Answer:
$r^2 - 8r + 8 = 0$

Solution:



$OF^2 = r^2$

$(2 - r)^2 + (2 - r)^2 = r^2$

$r^2 - 8r + 8 = 0$

Q. 17 Let $\beta(m,n)=\int_0^1 x^{m-1}(1-x)^{n-1}dx, m,n>0$. If $\int_0^1 \left(1-x^{10}\right)^{20}dx = a\times\beta(b,c)$, then $100(a+b+c)$ equals_____.

Option 1:
1021

Option 2:
1120

Option 3:
2012

Option 4:
2120

Correct Answer:
2120

Solution:

$$I=\int_0^1 1\cdot\left(1-x^{10}\right)^{20}dx$$

$$x^{10}=t$$

$$x=t^{1/10}$$

$$dx=\frac{1}{10}(t)^{-9/10}dt$$

$$I=\int_0^1 (1-t)^{20}\frac{1}{10}(t)^{-9/10}dt$$

$$I=\frac{1}{10}\int_0^1 t^{-9/10}(1-t)^{20}dt$$

$$a=\frac{1}{10}\quad b=\frac{1}{10}\quad c=21$$

Q. 18 Let $\alpha\beta \neq 0$ and $A = \begin{bmatrix} \beta & \alpha & 3 \\ \alpha & \alpha & \beta \\ -\beta & \alpha & 2\alpha \end{bmatrix}$.

If $B = \begin{bmatrix} 3\alpha & -9 & 3\alpha \\ -\alpha & 7 & -2\alpha \\ -2\alpha & 5 & -2\beta \end{bmatrix}$ is the matrix of cofactors of the elements of A, then det(A B) is equal to :

Option 1:
343

Option 2:
125

Option 3:
64

Option 4:
216

Correct Answer:
216

Solution:
Equating co-factor fo A_{21}

$(2\alpha^2 - 3\alpha) = \alpha$

$\alpha = 0, 2(\text{ accept })$

$\text{Now, } 2\alpha^2 - \alpha\beta = 3\alpha$

$\alpha = 2 \quad \beta = 1$

$|AB| = |A\,\mathrm{cof}(\mathrm{A})| = |A|^3$

$A = \begin{matrix} 1 & 2 & 3 \\ 2 & 2 & 1 \\ -1 & 2 & 4 \end{matrix} = 6 - 2(9) + 3(6) = 6$

Q. 19 If $\mathrm{y}(\theta) = \frac{2\cos\theta + \cos 2\theta}{\cos 3\theta + 4\cos 2\theta + 5\cos\theta + 2}$, then at $\theta = \frac{\pi}{2}$, $\mathrm{y}'' + \mathrm{y}' + \mathrm{y}$ is equal to:

Option 1:
$\frac{3}{2}$

Option 2:
1

Option 3:
$\frac{1}{2}$

Option 4:
2

Correct Answer:
2

Solution:

$$y = \frac{2\cos\theta + 2\cos^2\theta - 1}{4\cos^3\theta - 3\cos\theta + 8\cos^2\theta - 4 + 5\cos\theta + 2}$$

$$y = \frac{(2\cos^2\theta + 2\cos\theta - 1)}{(2\cos^2\theta + 2\cos\theta - 1)(2\cos\theta + 2)}$$

$$y = \frac{1}{2}\left(\frac{1}{1+\cos\theta}\right)$$

$$\Rightarrow \theta = \frac{\pi}{2} \quad y = \frac{1}{2}$$

$$y' = \frac{1}{2}\left(\frac{-1}{(1+\cos\theta)^2} \times (-\sin\theta)\right)$$

$$\Rightarrow \theta = \frac{\pi}{2} \quad y = \frac{1}{2}$$

$$y'' = \frac{1}{2}\left[\frac{\cos\theta(1+\cos\theta)^2 - \sin\theta(2)(1+\cos\theta)(-\sin\theta)}{(1+\cos\theta)^4}\right]$$

$$\Rightarrow \theta = \frac{\pi}{2} \quad y = 1$$

Q. 20 For $x \geq 0$, the least value of K, for which $4^{1+x} + 4^{1-x}$, $\frac{K}{2}$, $16^x + 16^x$ are three consecutive terms of an A.P. is equal to :

Option 1:
10

Option 2:
4

Option 3:
8

Option 4:
16

Correct Answer:
10

Solution:

$$k = 4\left(4^x + \frac{1}{4^x}\right) + \left(4^{2x} + \frac{1}{4^{2x}}\right)$$
$$\geq 2 \quad \geq 2$$
$$k \geq 10$$

Q. 21 Let the mean and the standard deviation of the probability distribution

X	α	1	0	-3
P(X)	$\frac{1}{3}$	K	$\frac{1}{6}$	$\frac{1}{6}$

be μ and σ, respectively. If $\sigma - \mu = 2$, then $\sigma + \mu$ is equal to_____.

Correct Answer:
5

Solution:

$$\frac{1}{3}+k+\frac{1}{6}+\frac{1}{4}=1 \quad \Rightarrow k=\frac{1}{4}$$

$$\mu=\frac{\alpha}{3}+\frac{1}{4}-\frac{3}{4}$$

$$\mu=\frac{\alpha}{3}-\frac{1}{2}$$

$$\sigma=\sqrt{\left(\alpha^2\frac{1}{3}+\frac{1}{4}+9\frac{1}{4}\right)-\left(\frac{\alpha}{3}-\frac{1}{2}\right)^2}$$

$$\sigma=\sqrt{\frac{2\alpha^2}{9}+\frac{\alpha}{3}+\frac{9}{4}}$$

$$\sigma=\mu+2$$

$$\sigma^2=(\mu+2)^2 \Rightarrow \frac{2\alpha^2}{9}+\frac{\alpha}{3}+\frac{9}{4}=\frac{\alpha^2}{9}+\frac{9}{4}+\alpha$$

$$\frac{\alpha^2}{9}-\frac{2\alpha}{3}=0$$

$\alpha=0,$ (reject) or $\alpha=6$

$(\because x=0$ is already given)

$\Rightarrow \sigma+\mu=2\mu+2$

$=5$

Q. 22 Let $y=y(x)$ be the solution of the differential equation $\frac{dy}{dx}+\frac{2x}{(1+x^2)^2}y=xe^{\frac{1}{(1+x^2)}}$; $y(0)=0$. Then the area enclosed by the curve $f(x)=y(x)e^{\frac{1}{(1+x^2)}}$ and the line $y-x=4$ is_____.

Correct Answer:
18

Solution:

$$\text{IF} = e^{\int \frac{2x}{(1+x^2)^2} dx} = e^{\frac{-1}{1+x^2}}$$

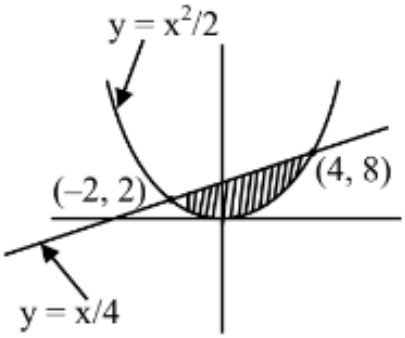
$$y \cdot e^{\frac{-1}{1+x^2}} = \int x \cdot e^{\frac{1}{1+x^2}} \cdot e^{\frac{-1}{1+x^2}} dx$$

$$y \cdot e^{\frac{-1}{1+x^2}} = \frac{x^2}{2} + c$$

$$(0,0) \Rightarrow C = 0$$

$$y(x) = \frac{x^2}{2} e^{\frac{1}{1+x^2}}$$

$$f(x) = \frac{x^2}{2}$$



$$A = \int_{-2}^{4} (x+4) - \frac{x^2}{2} dx = 18$$

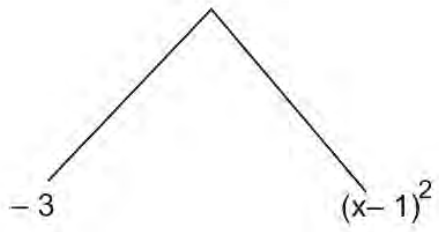
Q. 23 The number of solutions of $\sin^2 x + \left(2 + 2x - x^2\right) \sin x - 3(x-1)^2 = 0$, where $-\pi \leq x \leq \pi$, is

Correct Answer:
2

Solution:

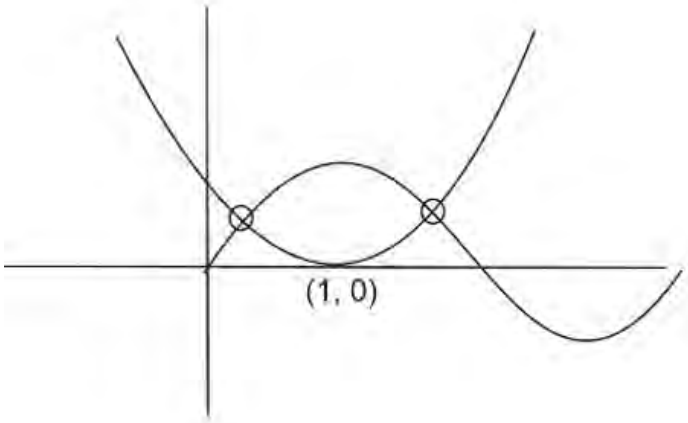
$$\sin^2 x - \left(x^2 - 2x - 2\right) \sin x - 3(x-1)^2 = 0$$

$$\sin^2 x - (x-1)^2) \sin x - 3(x-1)^2 = 0$$



$\sin x = -3(\text{ reject }) \text{ or } (x-1)^2$

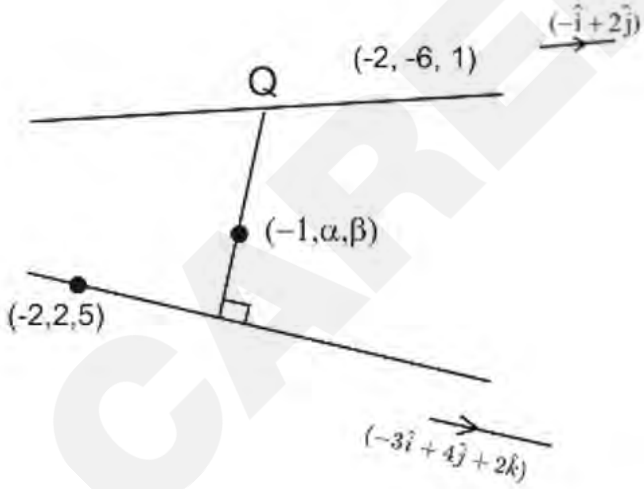
$\sin x = (x-1)^2$



Q. 24 Let the point $(-1, \alpha, \beta)$ lie on the line of the shortest distance between the lines $\frac{x+2}{-3} = \frac{y-2}{4} = \frac{z-5}{2}$ and $\frac{x+2}{-1} = \frac{y+6}{2} = \frac{z-1}{0}$. Then $(\alpha - \beta)^2$ is equal to _____.

Correct Answer:
25

Solution:



$P(-3\lambda - 2, 4\lambda + 2, 2\lambda + 5)$

$Q(-\mu - 2, 2\mu - 6, 1)$

DRS of $PQ = (3\lambda - \mu, 2\mu - 4\lambda - 8, -2\lambda - 4)$

$$\text{DRS of } PQ = \begin{matrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 2 & 0 \\ -3 & 4 & 2 \end{matrix}$$

$= (4\hat{i} + 2\hat{j} + 2\hat{k})$

OR

$(2,1,1)$

$\frac{3\lambda-\mu}{2}=\frac{2\mu-4\lambda-8}{1}=\frac{-2\lambda-4}{1}$

$\Rightarrow \mu=\lambda+2 \& 7\lambda=\mu-8$

$\lambda=-1 \quad \mu=1$

$Q:(-3,-4,1)$

$L_{PQ}=\frac{x+3}{2}=\frac{y+4}{1}=\frac{z-1}{1}$

$(-1,\alpha,\beta)\Rightarrow 1=\frac{\alpha+4}{1}=\frac{\beta-1}{1}$

$\Rightarrow \alpha=-3,\beta=2$

$(\alpha-\beta)^2=25$

Q. 25 If

$1+\frac{\sqrt{3}-\sqrt{2}}{2\sqrt{3}}+\frac{5-2\sqrt{6}}{18}+\frac{9\sqrt{3}-11\sqrt{2}}{36\sqrt{3}}+\frac{49-20\sqrt{6}}{180}+\ldots$

upto $\infty=2\left(\sqrt{\frac{b}{a}}+1\right)\log_e\left(\frac{a}{b}\right)$, where a and b are integers with $\gcd(a,b)=1$, then $11a+18b$ is equal to_____.

Correct Answer:
76

Solution:

$$S = 1 + \frac{x}{2\sqrt{3}} + \frac{x^2}{18} + \frac{x^3}{36\sqrt{3}} + \frac{x^4}{180} + \ldots \infty$$

$\text{Put } \frac{x}{\sqrt{3}} = t, \text{ where } x = \sqrt{3} - \sqrt{2}$

$$S = 1 + \frac{t}{2} + \frac{t^2}{6} + \frac{t^3}{12} + \frac{t^4}{20} + \ldots$$

$$S = 1 + t\left(1 - \frac{1}{2}\right) + t^2\left(\frac{1}{2} - \frac{1}{3}\right) + t^3\left(\frac{1}{3} - \frac{1}{4}\right) + t^4\left(\frac{1}{4} - \frac{1}{5}\right)$$

$$S = \left(1 + t + \frac{t^2}{2} + \frac{t^3}{3} + \frac{t^3}{4} + \ldots\right) - \left(\frac{t}{2} + \frac{t^2}{3} + \frac{t^3}{4} + \frac{t^4}{5} + \ldots\right)$$

$$S = \left(t + \frac{t^2}{2} + \ldots\right) - \frac{1}{t}\left(t + \frac{t^2}{2} + \frac{t^3}{3} + \ldots\right) + 2$$

$$S = 2 + \left(1 - \frac{1}{t}\right)(-\log(1-t)) = \left(\frac{1}{t} - 1\right)\log(1-t) + 2$$

$$\mathrm{S} = 2 + \left(\frac{\sqrt{3}}{\sqrt{3} - \sqrt{2}} - 1\right)\log\left(1 - \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3}}\right)$$

$$\mathrm{S} = 2 + \left(\frac{\sqrt{2}}{\sqrt{3} - \sqrt{2}}\right)\log \mathrm{e}\frac{\sqrt{2}}{\sqrt{3}}$$

$$\mathrm{S} = 2 + \frac{(\sqrt{6} + 2)}{2}\log \mathrm{e}\frac{2}{3} = 2 + \left(\sqrt{\frac{3}{2}} + 1\right)\log \mathrm{e}\frac{2}{3}$$

$\mathrm{a} = 2,\ \mathrm{b} = 3$

$11\mathrm{a} + 18\,\mathrm{b} = 11 \times 2 + 18 \times 3 = 76$

Q. 26 Let $\mathrm{a} > 0$ be a root of the equation $2\mathrm{x}^2 + \mathrm{x} - 2 = 0$.
If $\lim_{x \to \frac{1}{a}} \frac{16(1 - \cos(2 + x - 2x^2))}{(1 - ax^2)} = \alpha + \beta\sqrt{17}$, where $\alpha, \beta \in Z$ then $\alpha + \beta$ is equal to

---

Correct Answer:
170

Solution:

$2x^2+x-2=0 \nearrow a \quad \searrow b$

$2x^2+x-2=0 \nearrow \frac{1}{a} \quad \searrow \frac{1}{b}$

$$\lim_{x\to\frac{1}{a}} 16\cdot\frac{\left(1-\cos 2\left(x-\frac{1}{a}\right)\left(x-\frac{1}{b}\right)\right)}{4\left(x-\frac{1}{b}\right)^2}\times\frac{4\left(x-\frac{1}{b}\right)^2}{a^2\left(x-\frac{1}{a}\right)^2}$$

$$=16\times\frac{2}{a^2}\left(\frac{1}{a}-\frac{1}{b}\right)^2$$

$$=\frac{32}{a^2}\left(\frac{17}{4}\right)=\frac{17.8}{a^2}=\frac{17\times 8\times 16}{(-1+\sqrt{117})^2}$$

$$=\frac{136.16}{18.2\sqrt{7}}\times\frac{18+2\sqrt{7}}{18+2\sqrt{7}}$$

$$=\frac{136}{256}(18+2\sqrt{7})\cdot 16$$

$$=153+17\sqrt{17}=\alpha+\beta\sqrt{17}$$

$$\alpha+\beta=153+17=170$$

Q. 27 If $f(t)=\int_0^{\pi}\frac{2xdx}{1-\cos^2 t\sin^2 x}, 0<t<\pi$, then the value of $\int_0^{\frac{\pi}{2}}\frac{\pi^2 dt}{f(t)}$ equals_____.

Correct Answer:
1

Solution:
$f(t)=\int_0^{\pi}\frac{2x}{1-\cos^2 t\sin^2 x}dx$ ----------(1)

$$= 2\int_0^{\pi} \frac{(\pi - x)dx}{1 - \cos^2 \sin^2 x}$$

$$2f(t) = 2\int_0^{\pi} \frac{\pi}{1 - \cos^2 \sin^2 x}dx$$

$$f(t) = \int_0^{\pi} \frac{\pi}{1 - \cos^2 \sin^2 x}dx$$

divide \& by $\cos^2 x$

$$\mathrm{f(t)} = \pi\int_0^{\pi} \frac{\sec^2 x dx}{\sec^2 x - \cos^2 t \tan^2 x}$$

$$\mathrm{f(t)} = 2\pi\int_0^{\pi/2} \frac{\sec^2 x dx}{\sec^2 x - \cos^2 t \tan^2 x}$$

$\tan x = z$

$\sec^2 x dx = dz$

$$\mathrm{f(t)} = 2\pi\int_0^{\infty} \frac{\mathrm{dz}}{1 + \sin^2 \mathrm{t} \cdot \mathrm{z}^2}$$

$$= \frac{\pi^2}{\sin \mathrm{t}}$$

Then $\int_0^{\pi/2} \frac{\pi^2}{\mathrm{f(t)}}\mathrm{dt}$

$$= \int_0^{\pi/2} \sin \mathrm{t dt}$$

$= 1$

Q. 28 Let the maximum and minimum values of $\left(\sqrt{8\mathrm{x} - \mathrm{x}^2 - 12} - 4\right)^2 + (\mathrm{x} - 7)^2, \mathrm{x} \in R$ be M and m respectively. Then $\mathrm{M}^2 - \mathrm{m}^2$ is equal to

Correct Answer:
1600

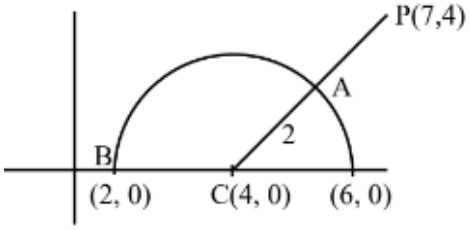
Solution:

$(x-7)^2+(y-4)^2$

$y=\sqrt{8x-x^2-12}$

$y^2=-(x-4)^2+16-12$

$(x-4)^2+y^2=4$



$m=9$

$M=41$

$M^2-m^2=41^2-9^2=1600$

Q. 29 Let a line perpendicular to the line $2x-y=10$ touch the parabola $y^2=4(x-9)$ at the point P. The distance of the point P from the centre of the circle $x^2+y^2-14x-8y+56=0$ is_____.

Correct Answer:
10

Solution:
$y^2=4(x-9)$
slope of tangent $=\frac{-1}{2}$
Point of contact $P\left(9+\frac{1}{\left(-\frac{1}{2}\right)^2},\frac{2\times1}{\frac{-1}{2}}\right)$
$P(13,-4)$
center of circle $C(7,4)$
distance $CP=\sqrt{(13-7)^2+(-4-4)^2}$
$=10$

Q. 30 The number of real solutions of the equation $x|x+5|+2|x+7|-2=0$ is_____.

Correct Answer:
3

Solution:

Case I : $x \geq -5$

$x^2 + 5x + 2x + 12 = 0$

$x^2 + 7x + 12 = 0$

$x = -3, -4$

Case II : $-7 < x < -5$

$-x^2 - 5x + 2x + 14 - 2 = 0$

$-x^2 - 3x + 12 = 0$

$$x = \frac{-3 \pm \sqrt{9 + 48}}{2}$$

$$= \frac{-3 \pm \sqrt{57}}{2}$$

$$x = \frac{-3 - \sqrt{57}}{2}, \frac{-3 + \sqrt{57}}{2} \text{ (rejected)}$$

Case III : $x \leq -7$

$-x^2 - 5x - 2x - 14 - 2 = 0$

$x^2 + 7x + 16 = 0$

$D = 49 - 64 < 0$

No solutions

No. of solutions $= 3$

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6 Apr - Morning Shift

Physics

Q. 1 To find the spring constant (k) of a spring experimentally, a student commits 2% positive error in the measurement of time and 1% negative error in measurement of mass. The percentage error in determining value of k is :

Option 1:
3%

Option 2:
1%

Option 3:
4%

Option 4:
5%

Correct Answer:
5%

Solution:

$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$T^2 \propto \frac{m}{k}$$

$$\frac{2\Delta T}{T}\% = \frac{\Delta m}{m}\% - \frac{\Delta k}{k}\%$$

$$\frac{\Delta k}{k}\% = \frac{\Delta m}{m}\% - \frac{2\Delta T}{T}\%$$

$$\frac{\Delta k}{k}\% = (-1)\% - 2(2)\% = |-5\%| = 5\%$$

Q. 2 A bullet of mass 50 g is fired with a speed 100 m / s on a plywood and emerges with 40 m / s. The percentage loss of kinetic energy is :

Option 1:
32%

Option 2:
44%

Option 3:
16%

Option 4:
84%

Correct Answer:
84%

Solution:

$$K_i = \frac{1}{2} m(100)^2$$

$$K_f = \frac{1}{2} m(40)^2$$

$$\% \text{ loss } = \frac{|K_f - K_i|}{K_i} \times 100$$

$$= \frac{\frac{1}{2} m(40)^2 - \frac{1}{2} m(100)^2}{\frac{1}{2} m(100)^2} \times 100$$

$$= \frac{|1600 - 100 \times 100|}{100} = 84\%$$

Q. 3 The ratio of the shortest wavelength of Balmer series to the shortest wavelength of Lyman series for hydrogen atom is :

Option 1:
4:1

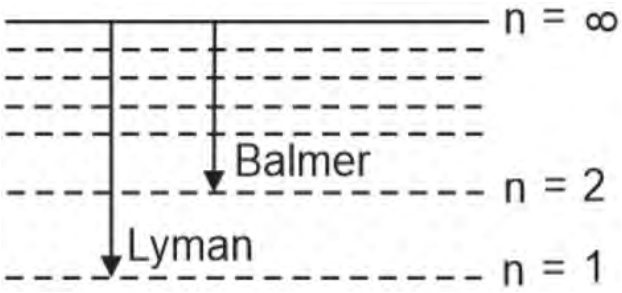
Option 2:
1:2

Option 3:
1:4

Option 4:
2:1

Correct Answer:
4:1

Solution:



$$\frac{1}{\lambda} = \mathrm{Rz}^2\left(\frac{1}{\mathrm{n}_1^2} - \frac{1}{\mathrm{n}_2^2}\right)$$

$$\frac{1}{\lambda_\mathrm{L}} = \mathrm{Rz}^2\left(\frac{1}{1^2}\right)$$

$$\frac{1}{\lambda_\mathrm{B}} = \mathrm{Rz}^2\left(\frac{1}{2^2}\right)$$

$$\frac{\lambda_\mathrm{B}}{\lambda_\mathrm{L}} = 4:1$$

Q. 4 To project a body of mass m from earth's surface to infinity, the required kinetic energy is (assume, the radius of earth is $\mathrm{R_E}, \mathrm{g} =$ acceleration due to gravity on the surface of earth) :

Option 1:
$2\mathrm{mgR_E}$

Option 2:
$\mathrm{mgR_E}$

Option 3:
$\frac{1}{2}\mathrm{mgR_E}$

Option 4:
$4\mathrm{mgR_E}$

Correct Answer:
$\mathrm{mgR_E}$

Solution:

$$\frac{1}{2}\mathrm{mv_e^2} = \frac{\mathrm{GMm}}{\mathrm{R_E}}$$

$$\mathrm{g} = \frac{\mathrm{GM}}{\mathrm{R_E^2}}$$

$$\mathrm{K} = \mathrm{mgR_E}$$

Q. 5 Electromagnetic waves travel in a medium with speed of $1.5 \times 10^8\ \mathrm{ms^{-1}}$. The relative permeability of the medium is 2.0 . The relative permittivity will be :

Option 1:
5

Option 2:
1

Option 3:
4

Option 4:
2

Correct Answer:
2

Solution:

$$\frac{\varepsilon_m \times \mu_m}{\varepsilon_0 \times \mu_0} = \frac{\frac{1}{v^2}}{\frac{1}{c^2}}$$

$$\varepsilon_r \times \mu_r = \frac{c^2}{v^2}$$

$$\varepsilon_r \times 2 = \frac{\left(3 \times 10^8\right)^2}{\left(1.5 \times 10^8\right)^2}$$

$$\varepsilon_r \times 2 = 4$$

$$\varepsilon_r = 2$$

Q. 6 Which of the following phenomena does not explain by wave nature of light.
(A) reflection
(B) diffraction
(C) photoelectric effect
(D) interference
(E) polarization

Choose the most appropriate answer from the options given below :

Option 1:
E only

Option 2:
C only

Option 3:
B, D only

Option 4:
A, C only

Correct Answer:
C only

Solution:
(Theory)
Photoelectric effect prove particle nature of light.

Q. 7 While measuring diameter of wire using screw gauge the following readings were noted. Main scale reading is 1 mm and circular scale reading is equal to 42 divisions. Pitch of screw gauge is 1 mm and it has 100 divisions on circular scale. The diameter of the wire is $\frac{x}{50}$ mm. The value of x is :

Option 1:
142

Option 2:
71

Option 3:
42

Option 4:
21

Correct Answer:
71

Solution:

$$\text{MSR} = 1 \text{ mm}, \text{CSR} = 42, \text{ pitch } = 1 \text{ mm}$$

$$\text{LC} = \frac{\text{pitch}}{\text{No. of CSD}} = \left(\frac{1}{100}\right) = 0.01 \text{ mm}$$

$$\text{Diameter } = \text{MSR} + \text{LC} \times \text{CSD}$$

$$\text{Diameter } = 1 + (0.01) \times 42 \text{ mm}$$

$$\text{Diameter } = 1.42 \text{ mm} = \frac{\text{x}}{50}$$

$$\therefore \text{x} = 71$$

Q. 8 σ is the uniform surface charge density of a thin spherical shell of radius R. The electric field at any point on the surface of the spherical shell is :

Option 1:
$\sigma/\epsilon_0 \mathrm{R}$

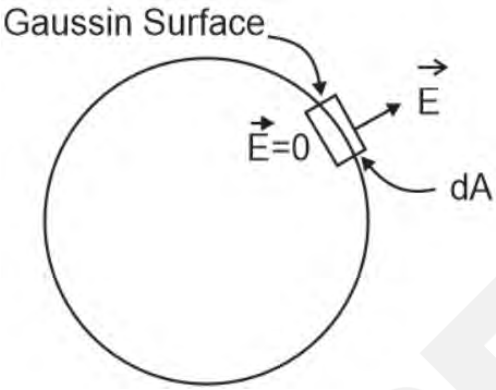
Option 2:
$\sigma/2 \in_0$

Option 3:
$\sigma/ \in_0$

Option 4:
$\sigma/4 \in_0$

Correct Answer:
$\sigma/ \in_0$

Solution:



By Gauss law $\int \vec{E} \cdot d\vec{A} = \frac{q_{\text{in}}}{\varepsilon_0}$

$$\mathrm{EdA} = \frac{\sigma \times \mathrm{dA}}{\varepsilon_0}$$

$$\mathrm{E} = \frac{\sigma}{\varepsilon_0}$$

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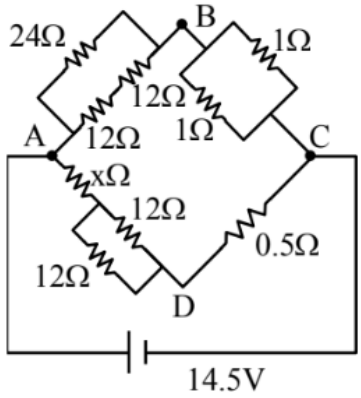
Q. 9 The value of unknown resistance (x) for which the potential difference between B and D will be zero in the arrangement shown, is :



Option 1:
3Ω

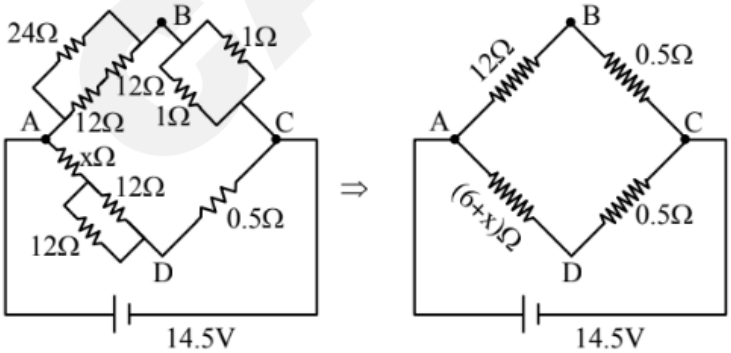
Option 2:
9Ω

Option 3:
6Ω

Option 4:
42Ω

Correct Answer:
6Ω

Solution:



In case of balanced Wheatstone Bridge

$$\frac{V_{AB}}{V_{AD}} = \frac{V_{BC}}{V_{CD}} \Rightarrow \frac{12}{6+x} = \frac{0.5}{0.5}$$

$$x = 6\Omega$$

Q. 10 The specific heat at constant pressure of a real gas obeying $PV^2 = RT$ equation is :

Option 1:
$C_V + R$

Option 2:
$\frac{R}{3} + C_v$

Option 3:
R

Option 4:
$C_V + \frac{R}{2V}$

Correct Answer:
$C_V + \frac{R}{2V}$

Solution:
$dQ = du + dW$

$CdT = C_v dT + PdV \ldots\ldots (1)$

$\therefore \quad PV^2 = RT$

$P = \text{ constant}$

$P(2VdV) = RdT$

$PdV = \frac{RdT}{2V}$

Put in equation (1)

$C = C_V + \frac{R}{2V}$

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Q. 11 Match **List I** with **List II**

	LIST I		**LIST II**
A.	Torque	I.	$\left[M^1 L^1 T^{-2} A^{-2}\right]$
B.	Magnetic field	II.	$\left[L^2 A^1\right]$
C.	Magnetic moment	III.	$\left[M^1 T^{-2} A^{-1}\right]$
D.	Permeability of free space	IV.	$\left[M^1 L^2 T^{-2}\right]$

Choose the correct answer from the options given below:

Option 1:
A-I, B-III, C-II, D-IV

Option 2:
A-IV, B-III, C-II, D-I

Option 3:
A-III, B-I, C-II, D-IV

Option 4:
A-IV, B-II, C-III, D-I

Correct Answer:
A-IV, B-III, C-II, D-I

Solution:

$$[\vec{\tau}] = [\vec{r} \times \vec{F}] = [ML^2\ T^{-2}]$$

$$[F] = [qVB]$$

$$\Rightarrow B = \left(\frac{F}{qV}\right) = \left[\frac{MLT^{-2}}{ATLT^{-1}}\right] = [MA^{-1}\ T^{-2}]$$

$$[M] = [I \times A] = [AL^2]$$

$$B = \frac{\mu_0}{4\pi}\frac{Idl\sin\theta}{r^2}$$

$$\Rightarrow [\mu] = \left[\frac{Br^2}{Idl}\right] = \left[\frac{MT^{-2}\ A^{-1} \times L^2}{AL}\right]$$

$$= [MLT^{-2}\ A^{-2}]$$

Q. 12 Given below are two statements :

Statement I : In an LCR series circuit, current is maximum at resonance.

Statement II : Current in a purely resistive circuit can never be less than that in a series LCR circuit when connected to same voltage source.

In the light of the above statements, choose the **correct** from the options given below :

Option 1:
Statement I is true but Statement II is false

Option 2:
Statement I is false but Statement II is true

Option 3:
Both Statement I and Statement II are true

Option 4:
Both Statement I and Statement II are false

Correct Answer:
Both Statement I and Statement II are true

Solution:

Statement-I

$I_m = \frac{V_m}{\sqrt{R^2+(X_L-X_C)^2}}$ at resonance $X_L = X_C$

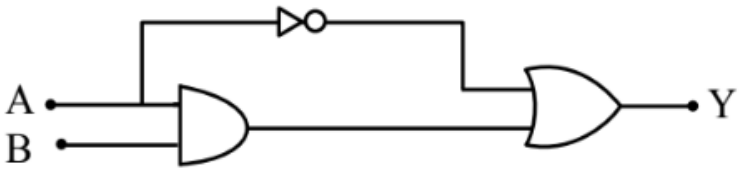
Thus, $I_m = \frac{V_m}{R}$

$\because$ Impendence is minimum therefore I is maximum at resonance.

Statement-II

$\mathrm{I} = \left(\frac{\mathrm{V}}{\mathrm{R}}\right)$ in purely resistive circuit.

Q. 13 The correct truth table for the following logic circuit is :



Option 1:

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

Option 2:

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	1

Option 3:

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

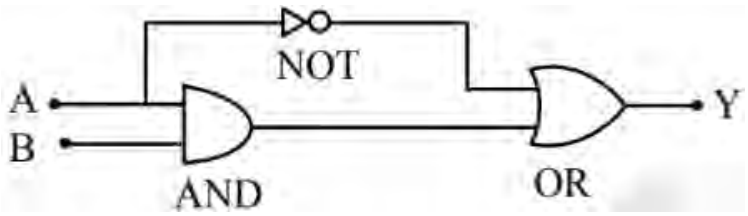
Option 4:

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Correct Answer:

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	1

Solution:



Q. 14 A sample contains mixture of helium and oxygen gas. The ratio of root mean square speed of helium and oxygen in the sample, is :

Option 1:
$\frac{1}{32}$

Option 2:
$\frac{2\sqrt{2}}{1}$

Option 3:
$\frac{1}{4}$

Option 4:
$\frac{1}{2\sqrt{2}}$

Correct Answer:
$\frac{2\sqrt{2}}{1}$

Solution:

$$V_{rms} = \sqrt{\frac{3RT}{M_w}}$$

$$\Rightarrow \frac{V_{O_2}}{V_{He}} = \sqrt{\frac{M_{w,He}}{M_{w,O_2}}}$$

$$= \sqrt{\frac{4}{32}} = \frac{1}{2\sqrt{2}}$$

$$\frac{V_{He}}{V_{O_2}} = \frac{2\sqrt{2}}{1}$$

Q. 15 A light string passing over a smooth light pulley connects two blocks of masses m_1 and m_2 (where $m_2 > m_1$). If the acceleration of the system is $\frac{g}{\sqrt{2}}$, then the ratio of the masses $\frac{m_1}{m_2}$ is :

Option 1:
$\frac{\sqrt{2}-1}{\sqrt{2}+1}$

Option 2:
$\frac{1+\sqrt{5}}{\sqrt{5}-1}$

Option 3:
$\frac{1+\sqrt{5}}{\sqrt{2}-1}$

Option 4:
$\frac{\sqrt{3}+1}{\sqrt{2}-1}$

Correct Answer:
$\frac{\sqrt{2}-1}{\sqrt{2}+1}$

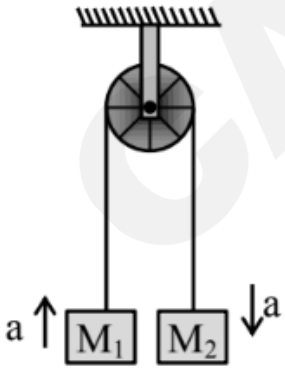
Solution:

$$a = \left(\frac{M_2 - M_1}{M_1 + M_2}\right)g$$

$$\frac{g}{\sqrt{2}} = \left(\frac{M_2 - M_1}{M_1 + M_2}\right)g$$

$$(M_1 + M_2) = \sqrt{2}M_2 - \sqrt{2}M_1$$

$$\frac{M_1}{M_2} = \left(\frac{\sqrt{2}-1}{\sqrt{2}+1}\right)$$



Q. 16 Four particles A, B, C, D of mass $\frac{m}{2}$, m, 2m, 4m, have same momentum, respectively. The particle with maximum kinetic energy is :

Option 1:
D

Option 2:
C

Option 3:
A

Option 4:
B

Correct Answer:
A

Solution:
$\mathrm{KE} = \frac{\mathrm{p}^2}{2\,\mathrm{m}}$

Same momentum, so less mass means more KE.
So $\frac{\mathrm{m}}{2}$ will have max. KE.

Q. 17 A train starting from rest first accelerates uniformly up to a speed of 80 km/h for time t, then it moves with a constant speed for time 3t. The average speed of the train for this duration of journey will be (in km/h) :

Option 1:
80

Option 2:
70

Option 3:
30

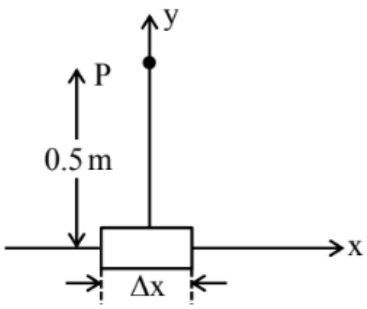
Option 4:
40

Correct Answer:
70

Solution:

$$\text{Average speed} = \frac{\text{total distance}}{\text{time taken}}$$
$$= \frac{\frac{80\times t}{2} + 80 \times 3t}{4t} = 70 \text{ km/hr.}$$

Q. 18 An element $\Delta l = \Delta x\hat{i}$ is placed at the origin and carries a large current $I = 10$ A. The magnetic field on the y-axis at a distance of 0.5 m from the elements Δx of 1 cm length is:



Option 1:
4×10^{-8} T

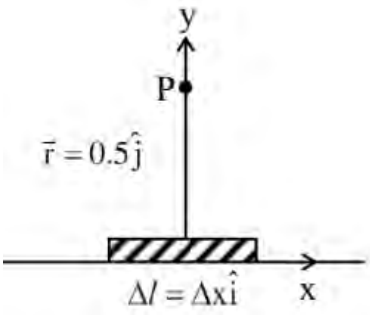
Option 2:
8×10^{-8} T

Option 3:
12×10^{-8} T

Option 4:
10×10^{-8} T

Correct Answer:
4×10^{-8} T

Solution:



$$\overrightarrow{dB}=\frac{\mu_0 I}{4\pi}\frac{(\overrightarrow{dl}\times\overrightarrow{r})}{r^3}(\text{Tesla})$$
$$=\frac{10^{-7}\times 10\times\left(\frac{1}{2}\times\frac{1}{100}\right)(+\hat{k})}{\left(\frac{1}{2}\right)^3}=4\times 10^{-8}\,T(+\hat{k})$$

Q. 19 A small ball of mass m and density ρ is dropped in a viscous liquid of density ρ_0. After sometime, the ball falls with constant velocity. The viscous force on the ball is :

Option 1:
$mg\left(\frac{\rho_0}{\rho}-1\right)$

Option 2:
$mg\left(1+\frac{\rho}{\rho_0}\right)$

Option 3:
$mg\left(1-\rho\rho_0\right)$

Option 4:
$mg\left(1-\frac{\rho_0}{\rho}\right)$

Correct Answer:
$mg\left(1-\frac{\rho_0}{\rho}\right)$

Solution:

$mg - F_B - F_v = ma$

$a = 0$ for constant velocity

$mg - F_B = F_v$

$$F_v = mg - v\rho_0 g = mg - \frac{m}{\rho}\rho_0 g = mg\left(1 - \frac{\rho_0}{\rho}\right)$$

Q. 20 In photoelectric experiment energy of 2.48 eV irradiates a photo sensitive material. The stopping potential was measured to be 0.5 V. Work function of the photo sensitive material is :

Option 1:
0.5 eV

Option 2:
1.68 eV

Option 3:
2.48 eV

Option 4:
1.98 eV

Correct Answer:
1.98 eV

Solution:

$eV_s = h\nu - \phi$

$0.5\ V = 2.48 - \phi$

work function $(\phi) = 2.48\ V - 0.5\ V = 1.98\ V$

Q. 21 If the radius of earth is reduced to three-fourth of its present value without change in its mass then value of duration of the day of earth will be_____hours 30 minutes.

Correct Answer:
13

Solution:

By conservation of angular momentum

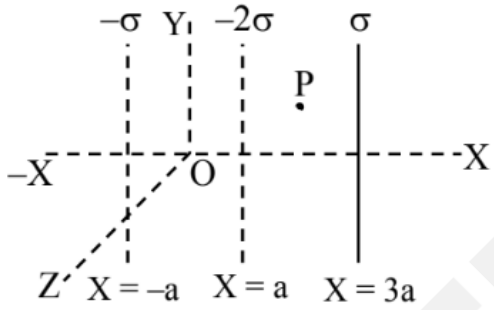
$I_1\omega_1 = I_2\omega_2$

$$\left(\frac{2}{5}MR^2\right)\frac{2\pi}{T_1} = \frac{2}{5}M\left(\frac{3}{4}R\right)^2\frac{2\pi}{T_2}$$

$$\frac{1}{T_1} = \frac{9}{16\,T_2}$$

$$\frac{1}{T_2} = \frac{9}{16} \times T_1 = \frac{9}{16} \times 24\text{hr} = \frac{27}{2}\text{hr} = 13\text{hr}30\text{mins}.$$

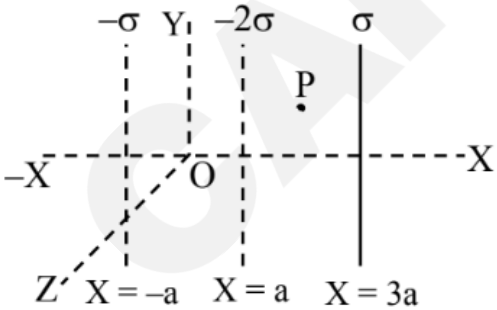
Q. 22 Three infinitely long charged thin sheets are placed as shown in figure. The magnitude of electric field at the point P is $\frac{x\sigma}{\epsilon_0}$. The value of x is_____(all quantities are measured in SI units).



Correct Answer:

2

Solution:



$$\vec{E}_p = \left(\frac{\sigma}{2\varepsilon_0} + \frac{2\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0}\right)(-\hat{i})$$

$$= -\frac{2\sigma}{\varepsilon_0}\hat{i}$$

Q. 23 A big drop is formed by coalescing 1000 small droplets of water. The ratio of surface energy of 1000 droplets to that of energy of big drop is $\frac{10}{x}$. The value of x is_____.

Correct Answer:
1

Solution:

1000 drops Big drop

$$1000\frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3$$

$$10r = R$$

$$R = 10r$$

$$\frac{\text{S.E. of 1000 drops}}{\text{S.E. of Big drop}} = \frac{1000\left(4\pi r^2\right)T}{4\pi R^2 T}$$

$$= \frac{1000 \times r^2}{(10r)^2} = 10 = \frac{10}{x}$$

$$\therefore \quad x = 1$$

Q. 24 When a dc voltage of 100 V is applied to an inductor, a de current of 5 A flows through it. When an ac voltage of 200 V peak value is connected to inductor, its inductive reactance is found to be $20\sqrt{3}\Omega$. The power dissipated in the circuit is_____W.

Correct Answer:
250

Solution:
For DC voltage

$$R = \frac{V}{I} = \frac{100}{5} = 20\Omega$$

for AC voltage

$X_L = 20\sqrt{3}\Omega$

$R = 20\Omega$

$Z = \sqrt{X_L^2 + R^2} = \sqrt{3 \times 400 + 400} = 40\Omega$

$\text{Power } = i_{rms}^2 R$

$= \left(\frac{V_{rms}}{Z}\right)^2 \times R = \left(\frac{\frac{200}{\sqrt{2}}}{40}\right)^2 \times 20 = 250 \text{ W}$

Q. 25 The refractive index of prism is $\mu = \sqrt{3}$ and the ratio of the angle of minimum deviation to the angle of prism is one. The value of angle of prism is_____$^\circ$.

Correct Answer:
60

Solution:

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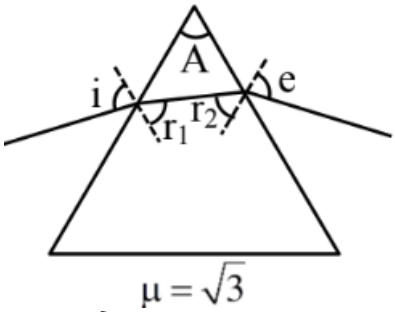


For $\delta_{\min}$

$i = e$

$r_1 = r_2 = \frac{A}{2}$

$\frac{\delta_{\min}}{A} = 1$

$\frac{2i - A}{A} = 1$

$2i = 2A$

$i = A$

Snell's law

$1 \times \sin i = \mu \sin r$

$\sin i = \mu \sin\left(\frac{A}{2}\right)$

$\sin A = \mu \sin\left(\frac{A}{2}\right)$

$2 \sin \frac{A}{2} \cos \frac{A}{2} = \sqrt{3} \sin\left(\frac{A}{2}\right)$

$\cos\left(\frac{A}{2}\right) = \frac{\sqrt{3}}{2}$

$\therefore \frac{A}{2} = 30^\circ$

$\therefore A = 60^\circ$

Q. 26 A wire of resistance R and radius r is stretched till its radius became r / 2. If new resistance of the stretched wire is xR, then value of x is_____.

Correct Answer:

16

Solution:

We know $\mathrm{R}=\frac{\rho l}{\mathrm{A}}, \mathrm{R} \propto \frac{l}{\mathrm{r}^2}$

As we starch the wire, its length will increase but its radius will decrease keeping the volume constant

$V_i = V_f$

$\pi \mathrm{r}^2 l = \pi \frac{\mathrm{r}^2}{4} l_{\mathrm{f}}$

$l_{\mathrm{f}} = 4l$

$\frac{\mathrm{R}_{\mathrm{new}}}{\mathrm{R}_{\mathrm{old}}} = \left(\frac{4l}{\frac{\mathrm{r}^2}{4}}\right)\frac{\mathrm{r}^2}{l} = 16$

$R_{\mathrm{new}} = 16\mathrm{R}$

$\therefore \quad x = 16$

Q. 27 Radius of a certain orbit of hydrogen atom is 8.48 Angstrom. If energy of electron in this orbit is E/x, then x=_____.

Correct Answer:

16

Solution:

We know

$\mathrm{r} = 0.529\frac{\mathrm{n}^2}{\mathrm{Z}} \Rightarrow 8.48 = 0.529\frac{\mathrm{n}^2}{1}$

$\mathrm{n}^2 = 16 \Rightarrow \mathrm{n} = 4$

We know

$\mathrm{E} \propto \frac{1}{\mathrm{n}^2}$

$E_{n^{\mathrm{th}}} = \frac{E}{16}$

$x = 16$

Q. 28 A circular coil having 200 turns, $2.5 \times 10^{-4}\ \mathrm{m}^2$ area and carrying $100\mu\mathrm{A}$ current is placed in a uniform magnetic field of $1\ \mathrm{T}$. Initially the magnetic dipole moment $(\overrightarrow{\mathrm{M}})$ was directed along $\overrightarrow{\mathrm{B}}$. Amount of work, required to rotate the coil through 90° from its initial orientation such that $\overrightarrow{\mathrm{M}}$ becomes perpendicular to $\vec{B}$, is_____$\mu\mathrm{J}$.

Correct Answer:
5

Solution:

We know

$$\begin{aligned}\mathrm{W}_{\text{ext}} &= \Delta\mathrm{U} + \Delta\mathrm{KE} \quad (\mathrm{P.E.} = -\overrightarrow{\mathrm{M}} \cdot \overrightarrow{\mathrm{B}}) \\ &= -\overrightarrow{\mathrm{M}} \cdot \overrightarrow{\mathrm{B}}_{\mathrm{f}} + \overrightarrow{\mathrm{M}} \cdot \overrightarrow{\mathrm{B}}_{\mathrm{i}} + 0 \\ &= -\mathrm{MB}\cos 90 + \mathrm{MB}\cos 0 \\ &= \mathrm{MB} \\ &= \mathrm{NIAB} \\ &= 200 \times 100 \times 10^{-6} \times \frac{5}{2} \times 10^{-4} \times 1 = 5\mu\mathrm{J}\end{aligned}$$

Q. 29 A particle is doing simple harmonic motion of amplitude 0.06 m and time period 3.14 s. The maximum velocity of the particle is_____cm/s.

Correct Answer:
12

Solution:
We know

$$\begin{aligned}\mathrm{V}_{\max} &= \omega\mathrm{A} \quad \text{at mean position} \\ &= \frac{2\pi}{\mathrm{T}}\mathrm{A} = \frac{2\pi}{\pi} \times 0.06 = 0.12\ \mathrm{m/sec} \\ \mathrm{V}_{\max} &= 12\ \mathrm{cm/sec}\end{aligned}$$

Q. 30 For three vectors $\overrightarrow{\mathrm{A}} = (-x\hat{\mathrm{i}} - 6\hat{\mathrm{j}} - 2\hat{\mathrm{k}})$, $\vec{B} = (-\hat{i} + 4\hat{j} + 3\hat{k})$ and $\vec{C} = (-8\hat{i} - \hat{j} + 3\hat{k})$, if $\overrightarrow{\mathrm{A}} \cdot (\overrightarrow{\mathrm{B}} \times \overrightarrow{\mathrm{C}}) = 0$, them value of x is_____.

Correct Answer:
4

Solution:

$$\vec{B} \times \vec{C} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 4 & 3 \\ -8 & -1 & 3 \end{vmatrix} = 15\hat{i} - 21\hat{j} + 33\hat{k}$$

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = (-x\hat{i} - 6\hat{j} - 2\hat{k}) \cdot (15\hat{i} - 21\hat{j} + 33\hat{k})$$
$$0 = -15x + 126 - 66$$
$$15x = 60$$
$$x = 4$$

Chemistry

Q. 1 Functional group present in sulphonic acid is :

Option 1:
$\mathrm{SO_4H}$

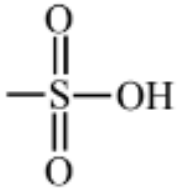
Option 2:
$\mathrm{SO_3H}$

Option 3:
$-\underset{\underset{\mathrm{O}}{\|}}{\mathrm{S}}-\mathrm{OH}$

Option 4:
$-\mathrm{SO_2}$

Correct Answer:
SO_3H

Solution:



Group present in sulphonic acids

Q. 2 Match **List I** with **List II** :

List I (Molecule/Speceis)		**List II (Property/Shape)**	
A.	SO_2Cl_2	I.	Paramagnetic
B.	NO	II.	Diamagnetic
C.	NO_2^-	III.	Tetrahedral
D.	I_3^-	IV.	Linear

Choose the correct answer from the options given below :

Option 1:
A-IV, B-I, C-III, D-II

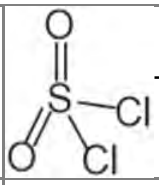
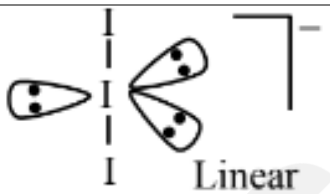
Option 2:
A-III, B-I, C-II, D-IV

Option 3:
A-II, B-III, C-I, D-IV

Option 4:
A-III, B-IV, C-II, D-I

Correct Answer:
A-III, B-I, C-II, D-IV

Solution:

(A)	SO_2Cl_2	sp^3	 Tetrahedral
(B)	NO		Paramagnetic
(C)	NO_2^-		Diamagnetic
(D)	I_3^-	sp^3d	 Linear

Q. 3 Given below are two statements :

Statement I : Picric acid is 2, 4, 6-trinitrotoluene.

Statement II : Phenol-2, 4-disulphuric acid is treated with conc.HNO_3 to get picric acid.In the light of the above statement, choose the **most appropriate** answer from the options given below :

Option 1:
Statement I is incorrect but Statement II is correct.

Option 2:
Both Statement I and Statement II are incorrect.

Option 3:
Statement I is correct but Statement II is incorrect.

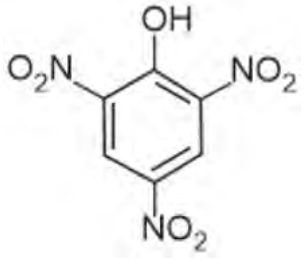
Option 4:

Both Statement I and Statement II are correct.

Correct Answer:

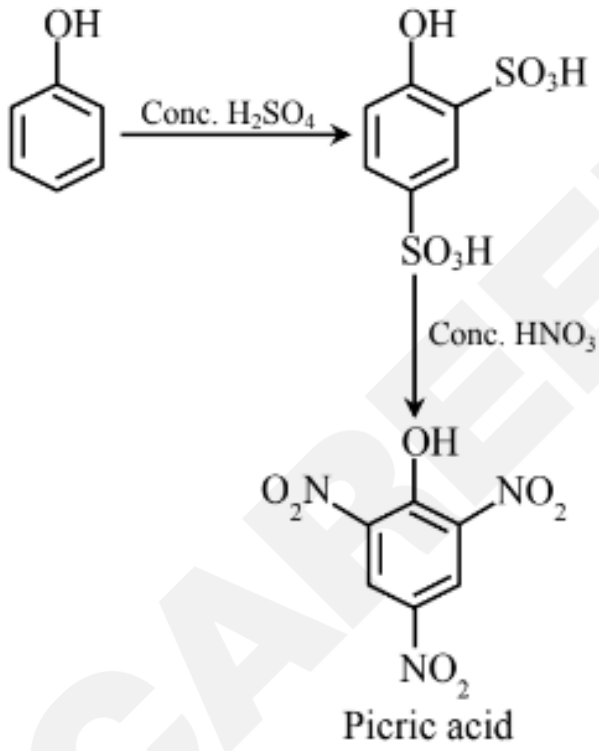
Statement I is incorrect but Statement II is correct.

Solution:



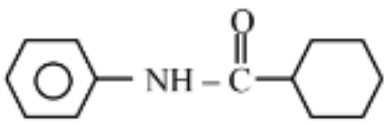
picric acid

(2,4,6 - trinitrophenol)



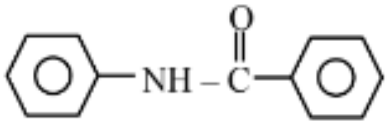
Picric acid

Q. 4 Which of the following is metamer of the given compound (X) ?

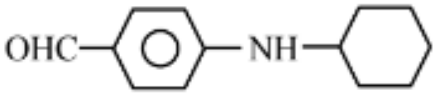


(X)

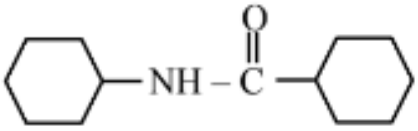
Option 1:



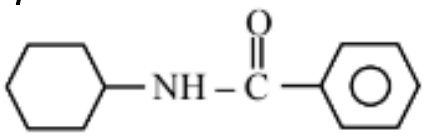
Option 2:



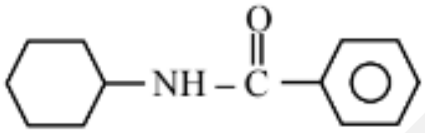
Option 3:



Option 4:



Correct Answer:

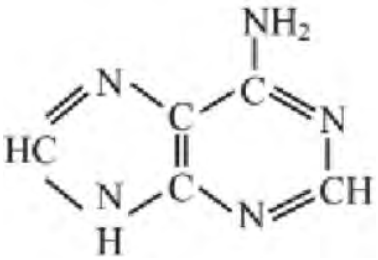


Solution:

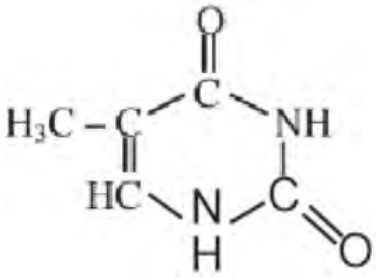
Metamer $\Rightarrow$ Isomer having same molecular formula, same functional group but different alkyl/aryl groups on either side of functional group.

Q. 5 DNA molecule contains 4 bases whoes structure are shown below. One of the structure is not correct, identify the **incorrect** base structure.

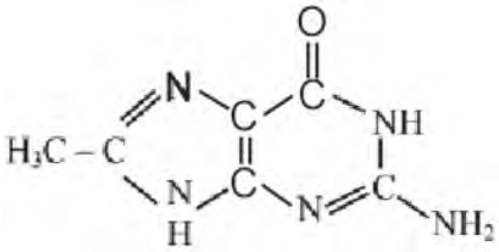
Option 1:



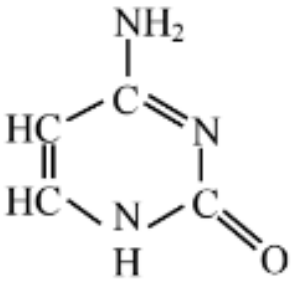
Option 2:



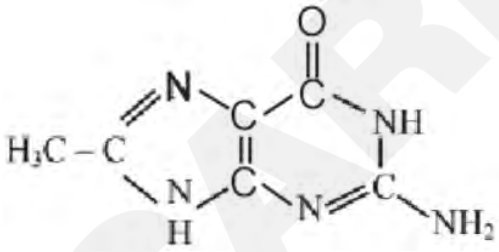
Option 3:



Option 4:

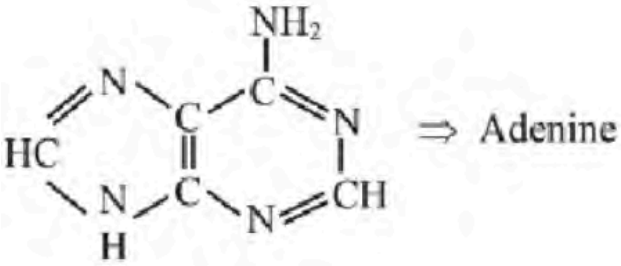


Correct Answer:

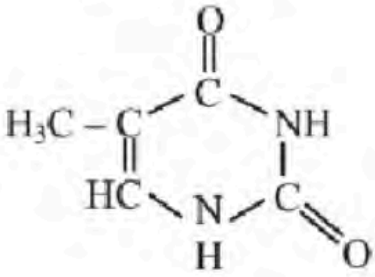


Solution:

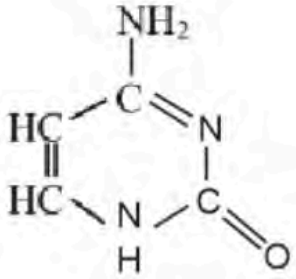
CAREERS360



⇒ Adenine



⇒ Thymine



⇒ Cytosine

Are bases of DNA molecule. As DNA contain four bases, which are adenine, guanine, cytosine and thymine.

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Q. 6 Match **List I** with **List II** :

LIST I (Hybridization)		**LIST II (Orientation in Space)**	
A.	sp^3	I.	Trigonal bipyramidal
B.	dsp^2	II.	Octahedral
C.	$sp^3 d$	III.	Tetrahedral
D.	$sp^3 d^2$	IV.	Square planar

Choose the **correct** answer from the options given below:

Option 1:
A-III, B-I, C-IV, D-II

Option 2:
A-II, B-I, C-IV, D-III

Option 3:
A-IV, B-III, C-I, D-II

Option 4:
A-III, B-IV, C-I, D-II

Correct Answer:
A-III, B-IV, C-I, D-II

Solution:

$sp^3 \rightarrow$ Tetrahedral
$dsp^2 \rightarrow$ Square planar
$sp^3 d \rightarrow$ Trigonal Bipyramidal
$sp^3 d^2 \rightarrow$ Octahedral

Q. 7 Given below are two statements :

Statement I : Gallium is used in the manufacturing of thermometers.

Statement II : A thermometer containing gallium is useful for measuring the freezing point 256 K of brine solution.
In the light of the above statement, choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are false.

Option 2:
Statement I is false but Statement II is true.

Option 3:
Both Statement I and Statement II are true.

Option 4:
Statement I is true but Statement II is false.

Correct Answer:
Statement I is true but Statement II is false.

Solution:
Statement - I $\Rightarrow$ Correct
Statement - II $\Rightarrow$ False

Ga is used to measure high temperature

Q. 8 Which of the following statements are correct?

A. Glycerol is purified by vacuum distillation because it decomposes at its normal boiling point.

B. Aniline can be purified by steam distillation as aniline is miscible in water.

C. Ethanol can be separated from ethanol water mixture by azeotropic distillation because it forms azeotrope.

D. An organic compound is pure, if mixed M.P. is remained same.

Choose the **most appropriate** answer from the options given below :

Option 1:
A, B, C only

Option 2:
A, C, D only

Option 3:
B, C, D only

Option 4:
A, B, D only

Correct Answer:
A, C, D only

Solution:
Option (B) is incorrect because aniline is immisible in water.

CAREERS360

Q. 9 Match **List I** with **List II** :

LIST I (Compound/Species)		**LIST II (Shape/Geometry)**	
A.	SF_4	I.	Tetrahedral
B.	BrF_3	II.	Pyramidal
C.	BrO_3^-	III.	See saw
D.	NH_4^+	IV.	Bent T- Shape

Choose the **correct** answer from the options given below:

Option 1:
A-II, B-III, C-I, D-IV

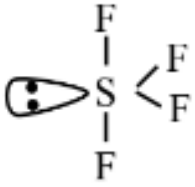
Option 2:
A-III, B-IV, C-II, D-I

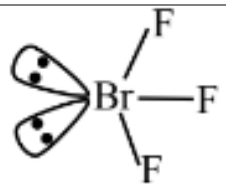
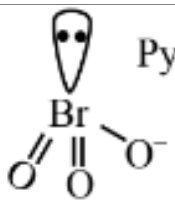
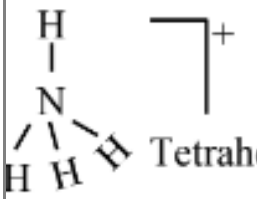
Option 3:
A-II, B-IV, C-III, D-I

Option 4:
A-III, B-II, C-IV, D-I

Correct Answer:
A-III, B-IV, C-II, D-I

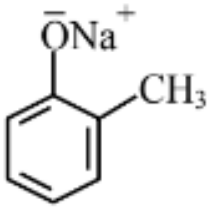
Solution:

(A)	SF_4	sp^3d hybridisation	

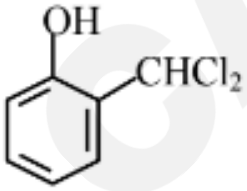
(B)	BrF_3	sp^3d hybridisation	 Bent T-Shape
(C)	BrO_3^-	sp^3 hybridisation	 Pyramidal
(D)	NH_4^+	sp^3 hybridisation	 Tetrahedral

Q. 10 In Reimer - Tiemann reaction, phenol is converted into salicylaldehyde through an intermediate. The structure of intermediate is_____.

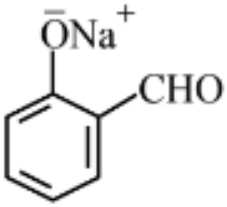
Option 1:



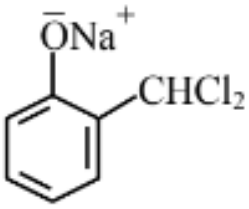
Option 2:



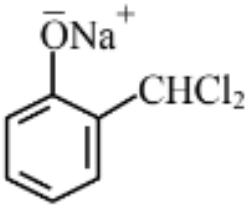
Option 3:



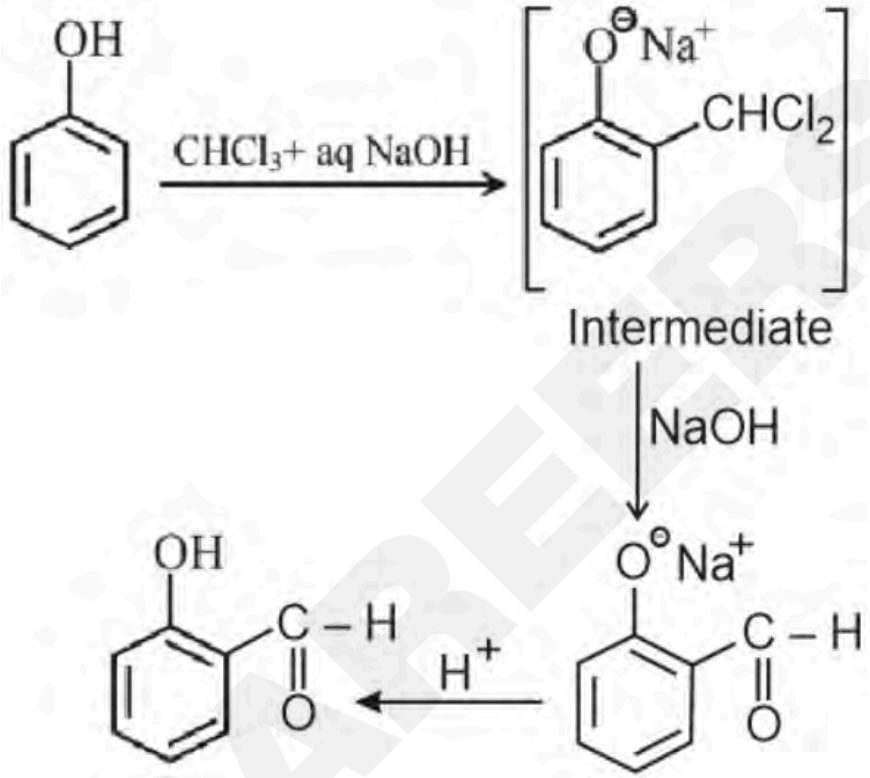
Option 4:



Correct Answer:



Solution:



Q. 11 Which of the following material is not a semiconductor.

Option 1:

Germanium

Option 2:
Graphite

Option 3:
Silicon

Option 4:
Copper oxide

Correct Answer:
Graphite

Solution:
Graphite is conductor

Q. 12 Consider the following complexes.

(A) $[CoCl(NH_3)_5]^{2+}$,

(B) $[Co(CN)_6]^{3-}$,

(C) $[Co(NH_3)_5\,(H_2O)]^{3+}$,

(D) $[Cu(H_2O)_4]^{2+}$

The correct order of A, B, C and D in terms of wavenumber of light absorbed is :

Option 1:
$C < D < A < B$

Option 2:
$D < A < C < B$

Option 3:
$A < C < B < D$

Option 4:
$B < C < A < D$

Correct Answer:
$D < A < C < B$

Solution:
As ligand field increases, light of more energy is absorbed

Energy $\propto$ wave number

$(\bar{v})$

Q. 13 Match **List I** with **List II:**

LIST I (Precipitating reagent and conditions)		**LIST II (Cation)**	
A.	$NH_4Cl + NH_4OH$	I.	Mn^{2+}
B.	$NH_4OH + Na_2CO_3$	II.	Pb^{2+}
C.	$NH_4OH + NH_4Cl + H_2S$ gas	III.	Al^{3+}
D.	dilute HCl	IV.	Sr^{2+}

Choose the correct answer from the options given below:

Option 1:
A-IV, B-III, C-II, D-I

Option 2:
A-IV, B-III, C-I, D-II

Option 3:
A-III, B-IV, C-I, D-II

Option 4:
A-III, B-IV, C-II, D-I

Correct Answer:
A-III, B-IV, C-I, D-II

Solution:
Theory based question

Q. 14 The electron affinity value are negative for :

A. $\mathrm{Be} \rightarrow \mathrm{Be}^{-}$
B. $\mathrm{N} \rightarrow \mathrm{N}^{-}$
C. $\mathrm{O} \rightarrow \mathrm{O}^{2-}$
D. $\mathrm{Na} \rightarrow \mathrm{Na}$
E. $\mathrm{Al} \rightarrow \mathrm{Al}$

Choose the most appropriate answer from the options given below :

Option 1:
D and E only

Option 2:
A, B, D and E only

Option 3:
A and D only

Option 4:
A, B and C only

Correct Answer:
A, B and C only

Solution:
(A) $\mathrm{Be} + \mathrm{e}^{-} \rightarrow \mathrm{Be}^{-}$, E.A =-ive
(B) $\mathrm{N} + \mathrm{e} \rightarrow \mathrm{N}^{-}$ E.A = − ive
(C) $\mathrm{O}^{+}\mathrm{e}^{-} \rightarrow \mathrm{O}^{-}$

$O^- + e^- \rightarrow O^{-2}$ E.A $= -$ ive

(D) $Na + e^- \rightarrow Na^-$ E.A $= +$ ive

(E) $A\ell + e^- \rightarrow A\ell^-$ $E \cdot A = +ive$

A, B and C only

Q. 15 The number of element from the following that do not belong to lanthanoids is :

Eu, Cm, Er, Tb, Yb and Lu

Option 1:

3

Option 2:

4

Option 3:

1

Option 4:

5

Correct Answer:

1

Solution:

cm is Actinide

Q. 16 The density of ' x ' M solution (' x ' molar) of $NaOH$ is $1.12\ g\ mL^{-1}$. while in molality, the concentration of the solution is $3\ m$ (3 molal). Then x is (Given : Molar mass of $NaOH$ is $40\ g/mol$)

Option 1:

3.5

Option 2:
3.0

Option 3:
3.8

Option 4:
2.8

Correct Answer:
3.0

Solution:

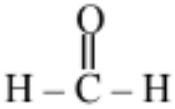
$$\text{Molality} = \frac{1000 \times \text{M}}{1000 \times \text{d} - \text{M} \times (\text{Mw})_{\text{solute}}}$$

$$3 = \frac{1000 \times \text{x}}{1000 \times 1.12 - (\text{x} \times 40)}$$

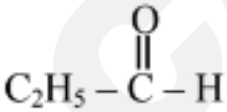
$$\text{x} = 3$$

Q. 17 Which among the following aldehydes is most reactive towards nucleophilic addition reactions?

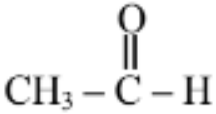
Option 1:



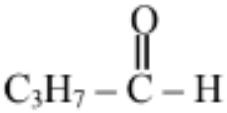
Option 2:



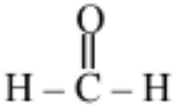
Option 3:



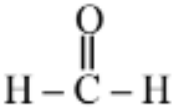
Option 4:



Correct Answer:



Solution:

 has low steric hindrance at carbonyl carbon and high partial positive charge at carbonyl carbon.

Q. 18 At $-20^{\circ}\mathrm{C}$ and $1\ \mathrm{atm}$ pressure, a cylinder is filled with equal number of $\mathrm{H_2}, \mathrm{I_2}$ and HI molecules for the reaction $\mathrm{H_2(\ g) + I_2(\ g) \rightleftharpoons 2HI(g)}$, the $\mathrm{K_P}$ for the process is $\mathrm{x \times 10^{-1} \cdot x =}$ _____.

[Given : $\mathrm{R = 0.082\ L\ atm\ K^{-1}\ mol^{-1}}$]

Option 1:
2

Option 2:
1

Option 3:
10

Option 4:
0.01

Correct Answer:
10

Solution:

$\Delta n_g = 0$ $\quad K_p = \frac{(n_{HI})^2}{n_{H_2} n_{I_2}} \left(\frac{P_T}{n_T}\right)^{\Delta n_g}$

$n_{HI} = n_{H_2} = n_{I_2}$ $\quad$ so $K_P = 1$

$1 = x \times 10^{-1}$ $\quad x = 10$

Q. 19 Match **list I** with **list II** :

LIST I (Compound)		**LIST II (Uses)**	
A.	Iodoform	I.	Fire Extinguisher
B.	Carbon Tetrachloride	II.	Insecticide
C.	CFC	III.	Antiseptic
D.	DDT	IV.	Refrigerants

Choose the correct answer from the options given below :

Option 1:
A-I, B-II, C-III, D-IV

Option 2:
A-III, B-II, C-IV, D-I

Option 3:
A-III, B-I, C-IV, D-II

Option 4:
A-II, B-IV, C-I, D-III

Correct Answer:
A-III, B-I, C-IV, D-II

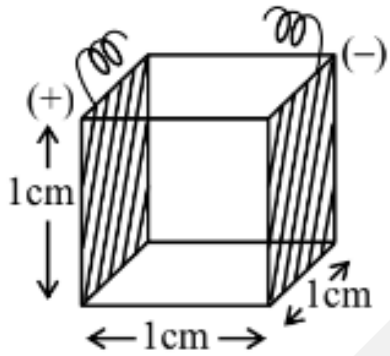
Solution:
Iodoform - Antiseptic

CCl_4 - Fire extinguisher

CFC - Refrigerants

DDT - Insecticide

Q. 20 A conductivity cell with two electrodes (dark side) are half filled with infinitely dilute aqueous solution of a weak electrolyte. If volume is doubled by adding more water at constant temperature, the molar conductivity of the cell will -



Option 1:
increase sharply

Option 2:
remain same or can not be measured accurately

Option 3:
decrease sharply

Option 4:
depend upon type of electrolyte

Correct Answer:
remain same or can not be measured accurately

Solution:

Solution is already infinitely dilute, hence no change in molar conductivity upon addition of water

Q. 21 Consider the dissociation of the weak acid HX as given below
$\mathrm{HX(aq) \rightleftharpoons H^+(aq) + X^-(aq), Ka = 1.2 \times 10^{-5}}$

$[\mathrm{K_a}$: dissociation constant $]$

The osmotic pressure of $0.03\mathrm{M}$ aqueous solution of HX at $300\ \mathrm{K}$ is_____ $\times 10^{-2}$ bar (nearest integer).

[Given : $\mathrm{R = 0.083\ LbarMol^{-1}\ K^{-1}}$]

Correct Answer:
76

Solution:

$$\mathrm{HX \rightleftharpoons H^+ + X^-} \quad \mathrm{K_a = 1.2 \times 10^{-5}}$$
$$0.03\mathrm{M}$$
$$0.03 - \mathrm{x} \quad \mathrm{x} \quad \mathrm{x}$$
$$\mathrm{K_a = 1.2 \times 10^{-5} = \frac{x^2}{0.03 - x}}$$
$$0.03 - \mathrm{x} \approx 0.03\ (\ \mathrm{K_a}\ \text{is very small}\ )$$
$$\mathrm{\frac{x^2}{0.03} = 1.2 \times 10^{-5}}$$
$$\mathrm{x = 6 \times 10^{-4}}$$

$$\text{Final solution} : 0.03 - x + x + x$$
$$= 0.03 + x = 0.03 + 6 \times 10^4$$
$$\Pi = \left(0.03 + \left(6 \times 10^4\right)\right) \times 0.083 \times 300$$
$$= 76.19 \times 10^{-2} \approx 76 \times 10^{-2}$$

Q. 22 The difference in the 'spin-only' magnetic moment values of $KMnO_4$ and the manganese product formed during titration of $KMnO_4$ against oxalic acid in acidic medium is _____ BM. (nearest integer)

Correct Answer:
6

Solution:
Spin only magnetic moment of Mn in $KMnO_4 = 0$ Spin only value of manganese product fromed during titration of $KMnO_4$ aganist oxalic acid in acidic medium is $= 6$

Q. 23 Time required for 99.9% completion of a first order reaction is_____ time the time required for completion of 90% reaction.(nearest integer).

Correct Answer:
3

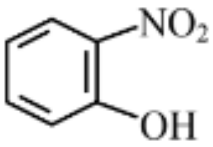
Solution:

$$K = \frac{1}{t_{99.9\%}} \ln\left(\frac{100}{0.1}\right) = \frac{1}{t_{90\%}} \ln\left(\frac{100}{10}\right)$$

$$t_{99.9\%} = t_{90\%} \frac{\ln\left(10^3\right)}{\ln 10}$$

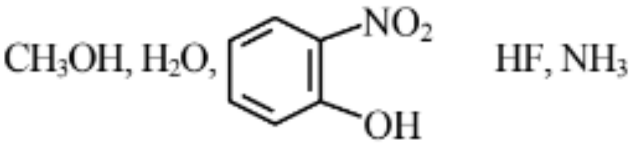
$$t_{99.9\%} = t_{90\%} \times 3$$

Q. 24 Number of molecules from the following which can exhibit hydrogen bonding is . (nearest integer)

$CH_3OH, H_2O, C_2H_6, C_6H_6,$  HF, NH_3

Correct Answer:
5

Solution:

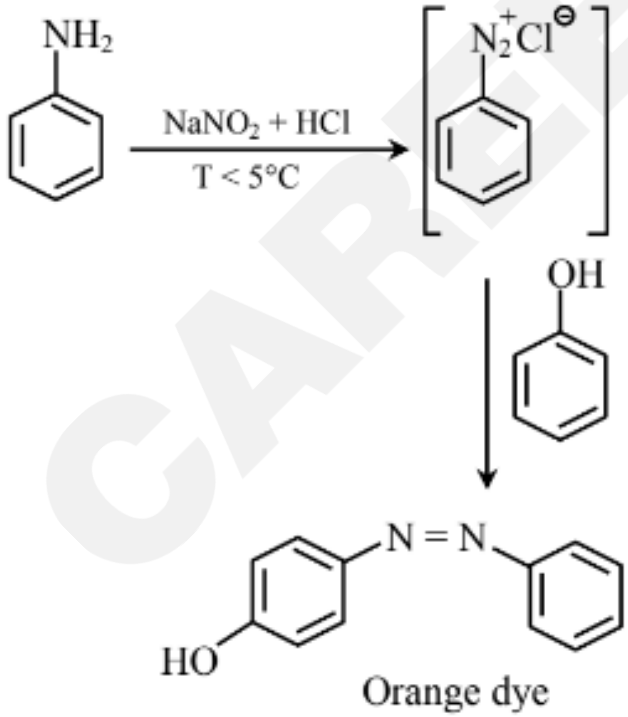


Can show H–bonding.

Q. 25 9.3 g of pure aniline upon diazotisation followed by coupling with phenol gives an orange dye. The mass of orange dye produced (assume 100% yield/ conversion) is_____ g. (nearest integer)

Correct Answer:
20

Solution:



Reaction suggests that 1 mole of aniline give 1 mole of orange dye.

$$\text{so } (\text{mol})_{\text{aniline}} = (\text{ mole })_{\text{orange dye}}$$
$$\frac{9.3 \text{ g}}{93 \text{ g mol}^{-1}} = \frac{\text{mass of orange dye}}{199 \text{ g mol}^{-1}}$$
$$\text{mass of orange dye } = 19.9 \text{ g} \simeq 20 \text{ g}$$

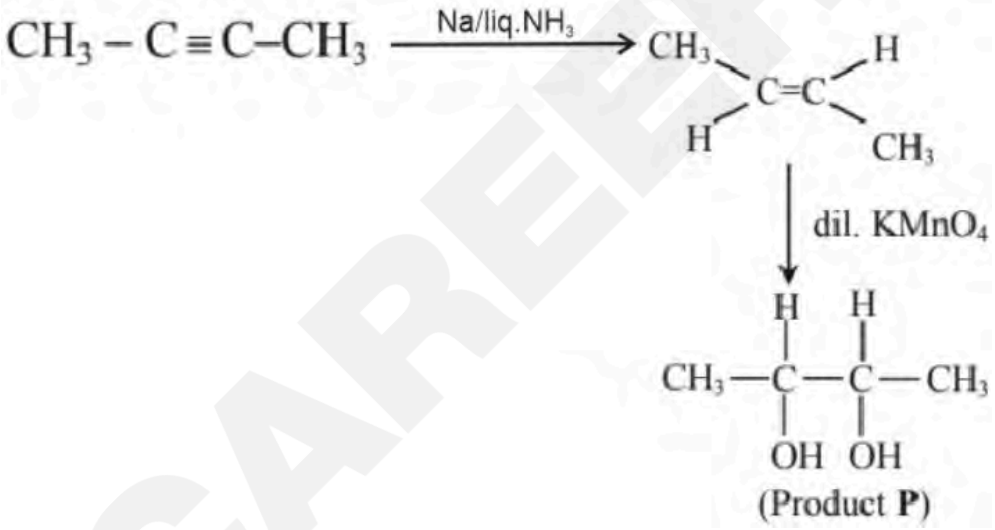
Q. 26 The major product of the following reaction is P.

$$CH_3C \equiv C - CH_3 \xrightarrow[\substack{\text{(ii) dil. KMMnO} \\ 273 \text{ K}}]{\text{(i) Na/K.NH}_3}, ' P'$$

Number of oxygen atoms present in product 'P' is _____ (nearest integer).

Correct Answer:
2

Solution:



Q. 27 Frequency of the de-Broglie wave of election in Bohr's first orbit of hydrogen atom is _____$\times 10^{13}$ Hz (nearest integer).
[Given : R_H (Rydberg constant) $= 2.18 \times 10^{-18}$ J. h(Plank's constant) $= 6.6 \times 10^{-34}$ J.s.]

Correct Answer:
661

Solution:

$$\lambda = \frac{h}{mv}$$

$$\lambda = \frac{hv}{mv^2}$$

$$\frac{mv^2}{h} = \frac{v}{\lambda} = v \text{ (frequency)}$$

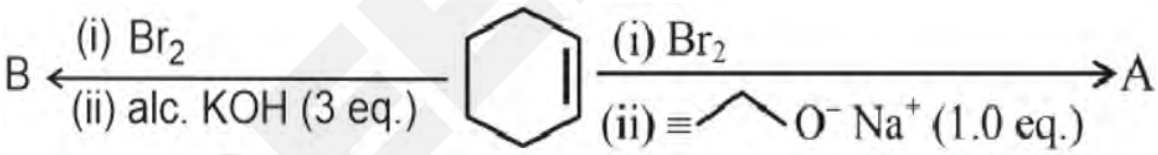
$$\text{Given } \frac{1}{2}mv^2 = 2.18 \times 10^{-18} \text{ J}$$

$$h = 6.6 \times 10^{-34}$$

$$v = \frac{4.36 \times 10^{-18}}{6.6 \times 10^{-34}} = 660.60 \times 10^{13} \text{ Hz}$$

$$\approx 661 \times 10^{13} \text{ Hz}$$

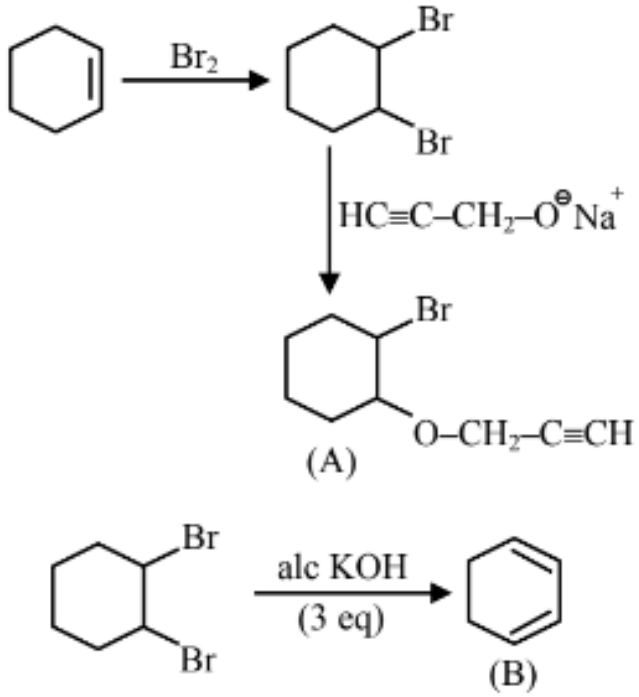
Q. 28 The major products from the following reaction sequence are product A and product B.



The total sum of π electrons in product A and product B are _____ (nearest integer)

Correct Answer:
8

Solution:



Q. 29 Among CrO, Cr_2O_3 and CrO_3, the sum of spin-only magnetic moment values of basic and amphoteric oxides is _____ $10^2 BM$ (nearest integer).
(Given atomic number of Cr is 24)

Correct Answer:
877

Solution:
CrO Basic oxide
Cr_2O_3 Amphoteric oxide
In CrO, Cr exist as Cr^{+2} and have μ only $= 4.90$
In Cr_2O_3, Cr exist as Cr^{+3} and have μ only $= 3.87$
Sum of spin only magnetic moment
$= 4.90 + 3.87 = 8.77$
$\mu_{only} = 877 \times 10^{-2}$
Ans. 877

Q. 30 An ideal gas, $\bar{C}_v = \frac{5}{2}R$, is expanded adiabatically against a constant pressure of $1\ \text{atm}$ untill it doubles in volume. If the initial temperature and pressure is $298\ \text{K}$ and $5\ \text{atm}$, respectively then the final temperature is_____ K (nearest integer). [$\overline{\text{C}}_\text{V}$ is the molar heat capacity at constant volume]

Correct Answer:
274

Solution:

$$\Delta \text{U} = \text{q} + \text{w}(\text{q} = 0)$$

$$\text{nC}_\text{V}\Delta\text{T} = -\text{P}_\text{ext}\left(\text{V}_2 - \text{V}_1\right)$$

$$\text{V}_2 = 2\,\text{V}_1$$

$$\frac{\text{nRT}_2}{\text{P}_2} = \frac{2\text{nRT}_1}{\text{P}_1}$$

$$\text{P}_1 = 5,\ \text{T}_1 = 298$$

$$\text{P}_2 = \frac{5\,\text{T}_2}{2 \times 298}$$

$$\text{n}\frac{5}{2}\text{R}\left(\text{T}_2 - \text{T}_1\right) = -1\left(\frac{\text{nRT}_2}{\text{P}_1} - \frac{\text{nRT}_1}{\text{P}_1}\right)$$

Put $T_1 = 298$
and $\text{P}_2 = \frac{5\,\text{T}_2}{2\times 298}$
Solve and we get $T_2 = 274.16\ \text{K}\ \text{T}_2 \approx 274\ \text{K}$

Maths

Q. 1 If $f(x) = \begin{cases} x^3 \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$, then

Option 1:
$f''(0) = 1$

Option 2:
$f''\left(\frac{2}{\pi}\right)=\frac{24-\pi^2}{2\pi}$

Option 3:
$f''\left(\frac{2}{\pi}\right)=\frac{12-\pi^2}{2\pi}$

Option 4:
$f''(0)=0$

Correct Answer:
$f''\left(\frac{2}{\pi}\right)=\frac{24-\pi^2}{2\pi}$

Solution:

$$f'(x)=3x^2\sin\left(\frac{1}{x}\right)-x\cos\left(\frac{1}{x}\right)$$

$$f''(x)=6x\sin\left(\frac{1}{x}\right)-3\cos\left(\frac{1}{x}\right)-\cos\left(\frac{1}{x}\right)-\frac{\sin\left(\frac{1}{x}\right)}{x}$$

$$f''\left(\frac{2}{\pi}\right)=\frac{12}{\pi}-\frac{\pi}{2}=\frac{24-\pi^2}{2\pi}$$

Q. 2 If $A(3,1,-1), B\left(\frac{5}{3},\frac{7}{3},\frac{1}{3}\right), C(2,2,1)$ and $D\left(\frac{10}{3},\frac{2}{3},\frac{-1}{3}\right)$ are the vertices of a quadrilateral $ABCD$, then its area is

Option 1:
$\frac{4\sqrt{2}}{3}$

Option 2:
$\frac{5\sqrt{2}}{3}$

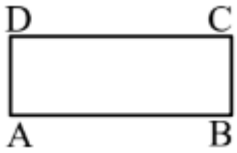
Option 3:
$2\sqrt{2}$

Option 4:
$\frac{2\sqrt{2}}{3}$

Correct Answer:

$\frac{4\sqrt{2}}{3}$

Solution:



Area $= \frac{1}{2}|\overline{BD} \times \overline{AC}|$

$\overline{BD} = \frac{5}{3}\hat{i} - \frac{5}{3}\hat{j} - \frac{2}{3}\hat{k}$

$\overline{AC} = \hat{i} - \hat{j} - 2\hat{k}$

Q. 3 $\int_0^{\pi/4} \frac{\cos^2 x \sin^2 x}{(\cos^3 x + \sin^3 x)^2} dx$ is equal to

Option 1:

$1/12$

Option 2:

$1/9$

Option 3:

$1/6$

Option 4:

$1/3$

Correct Answer:

$1/6$

Solution:

Divide $\mathrm{Nr\&Dr}$ by cosx

$\int_0^{\pi/4} \frac{\tan^2 x \sec^2 x dx}{(1+\tan^3 x)^2} dx$

Let $1+\tan^3 \mathrm{x}=\mathrm{t}$

$\tan^2 x \sec^2 x dx=\frac{dt}{3}$

$\frac{1}{3}\int_1^2 \frac{dt}{t^2}=\frac{1}{6}$

Q. 4 The mean and standard deviation of 20 observations are found to be 10 and 2, respectively. On respectively, it was found that an observation by mistake was taken 8 instead of 12 . The correct standard deviation is

Option 1:
$\sqrt{3.86}$

Option 2:
1.8

Option 3:
$\sqrt{3.96}$

Option 4:
1.94

Correct Answer:
$\sqrt{3.96}$

Solution:

$\text{Mean } (\bar{\mathrm{x}})=10$

$\Rightarrow \frac{\Sigma \mathrm{x_i}}{20}=10$

$\Sigma \mathrm{x_i}=10\times 20=200$

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If 8 is replaced by 12, then $\Sigma x_i = 200 - 8 + 12 = 204$

$\therefore$ Correct mean $(\bar{x}) = \frac{\sum x_i}{20}$

$= \frac{204}{20} = 10.2$

$\because$ Standard deviation $= 2$

$\therefore$ Variance $= (\text{S.D.})^2 = 2^2 = 4$

$$\Rightarrow \frac{\Sigma x_i^2}{20} - \left(\frac{\Sigma x_i}{20}\right)^2 = 4$$

$$\Rightarrow \frac{\Sigma x_i^2}{20} - (10)^2 = 4$$

$$\Rightarrow \frac{\Sigma x_i^2}{20} = 104$$

$$\Rightarrow \Sigma x_i^2 = 2080$$

Now, replaced ' 8 ' observations by ' 12 '

Then, $\Sigma x_i^2 = 2080 - 8^2 + 12^2 = 2160$

$\therefore$ Variance of removing observations

$$\Rightarrow \frac{\Sigma x_i^2}{20} - \left(\frac{\Sigma x_i}{20}\right)^2$$

$$\Rightarrow \frac{2160}{20} - (10.2)^2$$

$$\Rightarrow 108 - 104.04$$

$$\Rightarrow 3.96$$

Correct standard deviation

$= \sqrt{3.96}$

Q. 5 The function $f(x) = \frac{x^2+2x-15}{x^2-4x+9}, x \in R$ is

Option 1:

both one-one and onto.

Option 2:

onto but not one-one.

Option 3:
neither one-one nor onto.

Option 4:
one-one but not onto.

Correct Answer:
neither one-one nor onto.

Solution:
$f(x) = \frac{(x+5)(x-3)}{x^2-4x+9}$
Let $g(x) = x^2 - 4x + 9$
$\mathrm{D} < 0$
$\mathrm{g(x)} > 0 \text{ for } \mathrm{x} \in \mathrm{R}$

$\therefore \quad \begin{bmatrix} \mathrm{f}(-5) = 0 \\ \mathrm{f}(3) = 0 \end{bmatrix}$

So, $\mathrm{f(x)}$ is many-one.
again,

$yx^2 - 4xy + 9y = x^2 + 2x - 15$
$x^2(y-1) - 2x(2y+1) + (9y+15) = 0$
$\text{for } \forall x \in R \Rightarrow D \geq 0$
$D = 4(2y+1)^2 - 4(y-1)(9y+15) \geq 0$
$5y^2 + 2y + 16 \leq 0$
$(5y-8)(y+2) \leq 0$

$\mathrm{y} \in \left[-2, \frac{8}{5}\right]$ range
Note : If function is defined from $f : R \to R$ then only correct answer is option (3)

Q. 6 Let $\mathrm{A} = \{\mathrm{n} \in [100, 700] \cap \mathrm{N} : \mathrm{n}$ is neither a multiple of 3 nor a multiple of $4\}$. Then the number of elements in A is

Option 1:
300

Option 2:
280

Option 3:
310

Option 4:
290

Correct Answer:
300

Solution:

$n(3) \Rightarrow$ multiple of 3

$102, 105, 108, \ldots\ldots, 699$

$T_n = 699 = 102 + (n-1)(3)$

$n = 200$

$n(3) = 200$

$\because n(4) \Rightarrow$ multiple of 4

$100, 104, 108, \ldots., 700$

$T_n = 700 = 100 + (n-1)(4)$

$n = 151$

$n(4) = 151$

$n(3 \cap 4) \Rightarrow$ multiple of 3&4 both

$108, 120, 132, ..., 696$

$T_n = 696 = 108 + (n-1)(12)$

$n = 50$

$n(3 \cap 4) = 50$

$n(3 \cup 4) = n(3) + n(4) - n(3 \cap 4)$

$= 200 + 151 - 50$

$= 301$

$n(\overline{3 \cup 4}) =$ Total $- n(3 \cup 4) =$ neither a multiple of 3 nor a multiple of 4

$= 601 - 301 = 300$

Q. 7 Let C be the circle of minimum area touching the parabola $y = 6 - x^2$ and the lines $y = \sqrt{3}|x|$. Then, which one of the following points lies on the circle C?

Option 1:
(2,4)

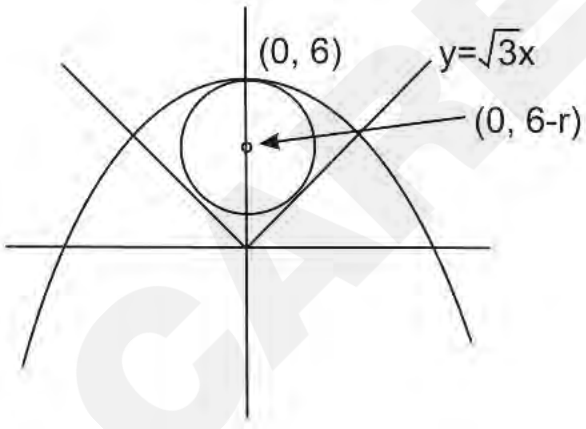
Option 2:
(1,2)

Option 3:
(2,2)

Option 4:
(1,1)

Correct Answer:
(2,4)

Solution:



Equation of circle
$\mathrm{x}^2 + (\mathrm{y} - (6 - \mathrm{r}))^2 = \mathrm{r}^2$
touches $\sqrt{3}x - y = 0$

$p = r$

$\frac{|0-(6-r)|}{2} = r$

$|r-6| = 2r$

$r = 2$

$\therefore$ Circle $x^2 + (y-4)^2 = 4$

$(2, 4)$ Satisfies this equation

Q. 8 For $\alpha, \beta \in R$ and a natural number n, let

$$A_r = \begin{matrix} r & 1 & \frac{n^2}{2} + \alpha \\ 2r & 2 & n^2 - \beta \\ 3r-2 & 3 & \frac{n(3n-1)}{2} \end{matrix}$$

. Then $2A_{10} - A_8$ is

Option 1:
$4\alpha + 2\beta$

Option 2:
$2\alpha + 4\beta$

Option 3:
2n

Option 4:
0

Correct Answer:
$4\alpha + 2\beta$

Solution:

$$A_r = \begin{matrix} r & 1 & \frac{n^2}{2} + \alpha \\ 2r & 2 & n^2 - \beta \\ 3r-2 & 3 & \frac{n(3n-1)}{2} \end{matrix}$$

$$2A_{10} - A_8 = \begin{matrix} 20 & 1 & \frac{n^2}{2} + \alpha \\ 40 & 2 & n^2 - \beta \\ 56 & 3 & \frac{n(3n-1)}{2} \end{matrix} - \begin{matrix} 8 & 1 & \frac{n^2}{2} + \alpha \\ 16 & 2 & n^2 - \beta \\ 22 & 3 & \frac{n(3n-1)}{2} \end{matrix}$$

$$\begin{array}{lll} 12 & 1 & \frac{n^2}{2}+\alpha \\ \Rightarrow\ 24 & 2 & n^2-\beta \\ 34 & 3 & \frac{n(3n-1)}{2} \end{array}$$

$$\begin{array}{lll} 0 & 1 & \frac{n^2}{2}+\alpha \\ \Rightarrow\ 0 & 2 & n^2-\beta \\ -2 & 3 & \frac{n(3n-1)}{2} \end{array}$$

$$\Rightarrow -2\left(\left(n^2-\beta\right)-\left(n^2+2\alpha\right)\right)$$

$$\Rightarrow -2(-\beta-2\alpha) \Rightarrow 4\alpha+2\beta$$

Q. 9 The shortest distance between the lines $\frac{x-3}{2}=\frac{y+15}{-7}=\frac{z-9}{5}$ and $\frac{x+1}{2}=\frac{y-1}{1}=\frac{z-9}{-3}$ is

Option 1:
$6\sqrt{3}$

Option 2:
$4\sqrt{3}$

Option 3:
$5\sqrt{3}$

Option 4:
$8\sqrt{3}$

Correct Answer:
$4\sqrt{3}$

Solution:

$$\frac{x-3}{2}=\frac{y+15}{-7}=\frac{z-9}{5}\&\frac{x+1}{2}=\frac{y-1}{1}=\frac{z-9}{-3}$$

$$\text{S.D}=\frac{(\bar{a}_2\cdot\bar{a}_1)\cdot\left(\bar{b}_1\cdot\bar{b}_2\right)}{\bar{b}_1\times\bar{b}_2}$$

$a_1=3,-15,9$
$b_1=2,-7,5$

$a_2=-1,1,9$

$b_2=2,1,-3$

$a_2-a_1=-4,16,0$

$$\bar{b}_1\times\bar{b}_2=\begin{matrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & -7 & 5\\ 2 & 1 & -3\end{matrix}=\hat{i}(16)-\hat{j}(-16)+\hat{k}(16)$$

$16(\hat{i}+\hat{j}+\hat{k})$

$\bar{b}_1\times\bar{b}_2=16\sqrt{3}$

$\therefore(\bar{a}_2-\bar{a}_1)\cdot\left(\bar{b}_1-\bar{b}_2\right)=16[-4+16]=(16)(12)$

$$\text{S.D.}=\frac{(16)(12)}{16\sqrt{3}}=4\sqrt{3}$$

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Q. 10 A company has two plants A and B to manufacture motorcycles. 60% motorcycles are manufactured at plant A and the remaining are manufactured at plant B. 80% of the motorcycles manufactured at plant A are rated of the standard quality, while 90% of the motorcycles manufactured at plant B are rated of the standard quality. A motorcycle picked up randomly from the total production is found to be of the standard quality. If p is the probability that it was manufactured at plant B, then 126 p is

Option 1:
54

Option 2:
64

Option 3:
66

Option 4:
56

Correct Answer:
54

Solution:

	A	B
Manufactured	60%	40%
Standard quality	80%	90%

$\mathrm{P}(\text{ Manufactured at } \mathrm{B} / \text{ found standard quality }) = ?$

A : Found S.Q

B : Manufacture B

C : Manufacture A

$$P(E_1) = \frac{40}{100}$$

$$P(E_2) = \frac{60}{100}$$

$$P(A/E_1) = \frac{90}{100}$$

$$P(A/E_2) = \frac{80}{100}$$

$$\because P(E_1/A) = \frac{P(A/E_1)P(E_1)}{P(A/E_1)P(E_1) + P(A/E_2)P(E_2)} = \frac{3}{7}$$

$$\therefore 126P = 54$$

Q. 11 Let, α, β be the distinct roots of the equation $x^2 - (t^2 - 5t + 6)x + 1 = 0, t \in R$ and $a_n = \alpha^n + \beta^n$. Then the minimum value of $\frac{a_{2023}+a_{2025}}{a_{2024}}$ is

Option 1:
$1/4$

Option 2:
$-1/2$

Option 3:
$-1/4$

Option 4:
$1/2$

Correct Answer:
$-1/4$

Solution:

by newton's theorem

$a_{n+2} - (t^2 - 5t + 6)a_{n+1} + a_n = 0$

$\therefore a_{2025} + a_{2023} = (t^2 - 5t + 6)a_{2024}$

$\therefore \dfrac{a_{2025} + a_{2023}}{a_{2024}} = t^2 - 5t + 6$

$\because t^2 - 5t + 6 = \left(t - \dfrac{5}{2}\right)^2 - \dfrac{1}{4}$

$\therefore \text{ minimum value } = -\dfrac{1}{4}$

Q. 12 Let the relations R_1 and R_2 on the set $\mathrm{X} = \{1, 2, 3, \ldots, 20\}$ be given by
$\mathrm{R_1} = \{(\mathrm{x}, \mathrm{y}) : 2\mathrm{x} - 3\mathrm{y} = 2\}$ and $R_2 = \{(x, y) : -5x + 4y = 0\}$. If M and N be the minimum number of elements required to be added in $\mathrm{R_1}$ and $\mathrm{R_2}$, respectively, in order to make the relations symmetric, then $\mathrm{M} + \mathrm{N}$ equals

Option 1:
8

Option 2:
16

Option 3:
12

Option 4:
10

Correct Answer:
10

Solution:

$x = \{1, 2, 3, \ldots\ldots 20\}$
$R_1 = \{(x, y) : 2x - 3y = 2\}$
$R_2 = \{(x, y) : -5x + 4y = 0\}$

$R_1 = \{(4, 2), (7, 4), (10, 6), (13, 8), (16, 10), (19, 12)\}$
$R_2 = \{(4, 5), (8, 10), (12, 15), (16, 20)\}$

in $R_1 6$ element needed in $R_2 4$ element needed So, total $6 + 4 = 10$ element

Q. 13 Let a variable line of slope $m > 0$ passing through the point $(4, -9)$ intersect the coordinate axes at the points A and B. the minimum value of the sum of the distances of A and B from the origin is

Option 1:
25

Option 2:
30

Option 3:
15

Option 4:
10

Correct Answer:
25

Solution:

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equation of line is

$y+9=m(x-4)$

$\therefore \quad A=\left(\frac{9+4m}{m},0\right)$

$B=(0,-9-4m)$

$\therefore \quad OA+OB=\frac{9+4m}{m}+9+4m$

$\because \mathrm{m}>0$

$=13+\frac{9}{\mathrm{m}}+4\,\mathrm{m}$

$\because \frac{4\,\mathrm{m}+\frac{9}{\mathrm{m}}}{2} \geq \sqrt{36} \Rightarrow 4\,\mathrm{m}+\frac{9}{\mathrm{m}} \geq 12$

$\therefore \mathrm{OA}+\mathrm{OB} \geq 25$

Q. 14 The interval in which the function $\mathrm{f}(\mathrm{x})=\mathrm{x}^{\mathrm{x}}, \mathrm{x}>0$, is strictly increasing is

Option 1:

$\left(0, \frac{1}{e}\right]$

Option 2:

$\left[\frac{1}{\mathrm{e}^2}, 1\right)$

Option 3:

$(0, \infty)$

Option 4:

$\left[\frac{1}{\mathrm{e}}, \infty\right)$

Correct Answer:

$\left[\frac{1}{\mathrm{e}}, \infty\right)$

Solution:

$f(x) = x^x; x > 0$

$\ell \text{ny} = x \ln x$

$\frac{1}{y}\frac{dy}{dx} = \frac{x}{x} + \ln x$

$\frac{dy}{dx} = x^x(1 + \ell nx)$

for strictly increasing

$\frac{\text{dy}}{\text{dx}} \geq 0 \Rightarrow \text{x}^{\text{x}}(1 + \ell \ln \text{x}) \geq 0$

$\Rightarrow \ln x \geq -1$

$\text{x} \geq \text{e}^{-1}$

$\text{x} \geq \frac{1}{\text{e}}$

$\text{x} \in \left[\frac{1}{\text{e}}, \infty\right)$

Q. 15 A circle in inscribed in an equilateral triangle of side of length 12. If the area and perimeter of any square inscribed in this circle are m and n, respectively, then $m + n^2$ is equal to

Option 1:
396

Option 2:
408

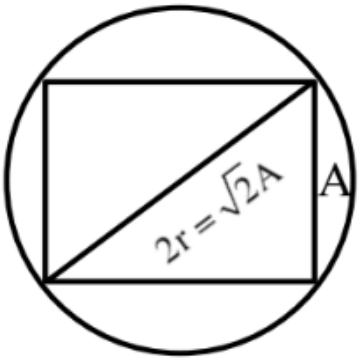
Option 3:
312

Option 4:
414

Correct Answer:
408

Solution:

$\because r = \frac{\Delta}{s} = \frac{\sqrt{3}a^2}{4 \cdot \frac{3a}{2}} = \frac{a}{2\sqrt{3}} = \frac{12}{2\sqrt{3}} = 2\sqrt{3}$



$\therefore A = r\sqrt{2} = 2\sqrt{6}$

$\text{Area } = m = A^2 = 24$

$\text{Perimeter } = n = 4\ A = 8\sqrt{6}$

$\therefore m + n^2 = 24 + 384$

$= 408$

Q. 16 The number of triangles whose vertices are at the vertices of a regular octagon but none of whose sides is a side of the octagon is

Option 1:
24

Option 2:
56

Option 3:
16

Option 4:
48

Correct Answer:
16

Solution:

$\because$ no. of triangles having no side common with a n sided polygon $= \dfrac{{}^{n}C_1 \cdot {}^{n-4}C_2}{3}$

$= \dfrac{{}^{8}C_1 \cdot {}^{4}C_2}{3} = 16$

Q. 17 Let $y = y(x)$ be the solution of the differential equation $(1+x^2)\frac{dy}{dx} + y = e^{\tan^{-1}x}, y(1) = 0$. Then $y(0)$ is

Option 1:
$\frac{1}{4}\left(e^{\pi/2} - 1\right)$

Option 2:
$\frac{1}{2}\left(1 - e^{\pi/2}\right)$

Option 3:
$\frac{1}{4}\left(1 - e^{\pi/2}\right)$

Option 4:
$\frac{1}{2}\left(e^{\pi/2} - 1\right)$

Correct Answer:
$\frac{1}{2}\left(1 - e^{\pi/2}\right)$

Solution:

$$\frac{dy}{dx} + \frac{y}{1+x^2} = \frac{e^{\tan^{-1}x}}{1+x^2}\Bigg)$$

$$\text{I.F.} = e^{\int \frac{1}{1+x^2}dx} = e^{\tan^{-1}x}$$

$$y \cdot e^{\tan^{-1}x} = \int \left(\frac{e^{\tan^{-1}x}}{1+x^2}\right) e^{\tan^{-1}x} \cdot dx$$

Let $\tan^{-1} x = z$

$\therefore \frac{dx}{1+x^2} = dz$

$\therefore y \cdot e^z = \int e^{2z} dz = \frac{e^{2z}}{2} + C$

$y \cdot e^{\tan^{-1} x} = \frac{e^{2\tan^{-1} x}}{2} + C$

$\Rightarrow y = \frac{e^{\tan^{-1} x}}{2} + \frac{C}{e^{\tan^{-1} x}}$

$\because y(1) = 0 \Rightarrow 0 = \frac{e^{\pi/4}}{2} + \frac{C}{e^{\pi/4}} \Rightarrow C = \frac{-e^{\pi/2}}{2}$

$\therefore y = \frac{e^{\tan^{-1} x}}{2} - \frac{e^{\pi/2}}{2e^{\tan^{-1} x}}$

$\therefore y(0) = \frac{1 - e^{\pi/2}}{2}$

Q. 18 Let $y = y(x)$ be the solution of the differential equation $(2x \log_e x)\frac{dy}{dx} + 2y = \frac{3}{x}\log_e x, x > 0$ and $y\left(e^{-1}\right) = 0$. Then, $y(e)$ is equal to

Option 1:
$-\frac{3}{2e}$

Option 2:
$-\frac{2}{3e}$

Option 3:
$-\frac{3}{e}$

Option 4:
$-\frac{2}{e}$

Correct Answer:
$-\frac{3}{e}$

Solution:

$$\frac{dy}{dx}+\frac{y}{x\ln x}=\frac{3}{2x^2}$$

$$\therefore \text{ I.F. } = e^{\int \frac{1}{x\ln x}dx} = e^{\ln(\ln(x))} = \ln x$$

$$\therefore y\ln x = \int \frac{3\ln x}{2x^2}dx$$

$$= \frac{3\ln x}{2}\int x^{-2}dx - \int\left(\frac{3}{2x}\cdot\int x^{-2}dx\right)dx$$

$$= \frac{3\ln x}{2}\left(-\frac{1}{x}\right) - \int \frac{3}{2x}\left(-\frac{1}{x}\right)dx$$

$$y\cdot\ln x = \frac{-3\ln x}{2x} - \frac{3}{2x} + C$$

$$\because y\left(e^{-1}\right) = 0$$

$$\therefore 0(-1) = \frac{3e}{2} - \frac{3e}{2} + C \Rightarrow C = 0$$

$$\therefore y = \frac{-3\ln x}{2x} - \frac{3}{2x}$$

$$\therefore y(e) = \frac{-3}{2e} - \frac{3}{2e} = \frac{-3}{e}$$

Q. 19 Let the area of the region enclosed by the curves $y = 3x, 2y = 27 - 3x$ and $y = 3x - x\sqrt{x}$ be A. Then $10\ \mathrm{A}$ is equal to

Option 1:
184

Option 2:
154

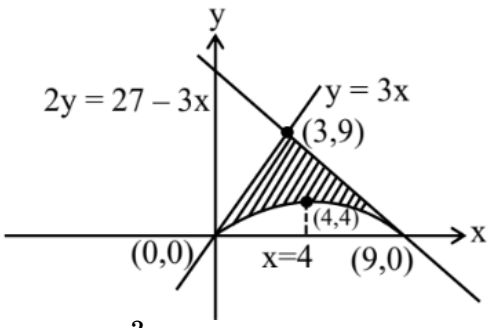
Option 3:
172

Option 4:
162

Correct Answer:
162

Solution:

$y = 3x, 2y = 27 - 3x \& y = 3x - x\sqrt{x}$



$$A = \int_0^3 3x - (3x - x\sqrt{x})dx + \int_3^9 \left(\frac{27 - 3x}{2} - (3x - x\sqrt{x})\right)dx$$

$$A = \int_0^3 x^{3/2}dx + \int_3^9 \frac{27}{2} - \frac{9x}{2} + x^{3/2}dx$$

$$A = \left[\frac{2x^{5/2}}{5}\right]_0^3 + \frac{27}{2}[x]_3^9 - \frac{9}{2}\left[\frac{x^2}{2}\right]_3^9 + \left[\frac{2x^{5/2}}{5}\right]_3^9$$

$$A = \frac{2}{5}\left(3^{5/2}\right) + \frac{27}{2}(6) - \frac{9}{4}(72) + \frac{2}{5}\left(9^{5/2} - 3^{5/2}\right)$$

$$A = \frac{2}{5}\left(3^{5/2}\right) + 81 - 162 + \frac{2}{5} \times 3^5 - \frac{2}{5} \times 3^{5/2}$$

$$A = \frac{486}{5} - 81 = \frac{81}{5}$$

$10\ A = 162$

$\text{Ans.} = 4$

Q. 20 Let $f : (-\infty, \infty) - \{0\} \to R$ be a differentiable function such that $f'(1) = \lim_{a\to\infty} a^2 f\left(\frac{1}{a}\right)$.

Then $\lim_{a\to\infty} \frac{a(a+1)}{2}\tan^{-1}\left(\frac{1}{a}\right) + a^2 - 2\log_e a$ is equal to

Option 1:

$\frac{3}{2} + \frac{\pi}{4}$

Option 2:

$\frac{3}{8} + \frac{\pi}{4}$

Option 3:

$\frac{5}{2} + \frac{\pi}{8}$

Option 4:

$\frac{3}{4}+\frac{\pi}{8}$

Correct Answer:

$\frac{5}{2}+\frac{\pi}{8}$

Solution:

$f:(-\infty,\infty)-\{0\}\to R$

$$f'(1)=\lim_{a\to\infty}a^2f\left(\frac{1}{a}\right)$$

$$\lim_{a\to\infty}\frac{a(a+1)}{2}\tan^{-1}\left(\frac{1}{a}\right)+a^2-2\ln(a)$$

$$\lim_{a\to\infty}a^2\left(\frac{\left(1+\frac{1}{a}\right)}{2}\tan^{-1}\left(\frac{1}{a}\right)+1-\frac{2}{a^2}\ln(a)\right)$$

$$f(x)=\frac{1}{2}(1+x)\tan^{-1}(x)+1-2x^2\ln(x)$$

$$f'(x)=\frac{1}{2}\left(\frac{1+x}{1+x^2}+\tan^{-1}(x)+4x\ln(x)\right)+2x$$

$$f'(1)=\frac{1}{2}\left(1+\frac{\pi}{4}\right)+2$$

$$f'(1)=\frac{5}{2}+\frac{\pi}{8}$$

Q. 21 Let $\alpha\beta\gamma=45;\alpha,\beta,\gamma\in$ R. If $x(\alpha,1,2)+y(1,\beta,2)+z(2,3,\gamma)=(0,0,0)$ for some $x,y,z\in R,xyz\neq 0$, then $6\alpha+4\beta+\gamma$ is equal to_____.

Correct Answer:

55

Solution:

$\alpha\beta\gamma = 45, \alpha\beta\gamma \in R$

$x(\alpha, 1, 2) + y(1, \beta, 2) + z(2, 3, \gamma) = (0, 0, 0)$

$x, y, z \in R, xyz \neq 0$

$\alpha x + y + 2z = 0$

$x + \beta y + 3z = 0$

$2x + 2y + \gamma z = 0$

$xyz \neq 0 \Rightarrow$ non-trivial

$$\begin{matrix} \alpha & 1 & 2 \\ 1 & \beta & 3 \\ 2 & 2 & \gamma \end{matrix} = 0$$

$\Rightarrow \alpha(\beta\gamma - 6) - 1(\gamma - 6) + 2(2 - 2\beta) = 0$

$\Rightarrow \alpha\beta\gamma - 6\alpha - \gamma + 6 + 4 - 4\beta = 0$

$\Rightarrow 6\alpha + 4\beta + \gamma = 55$

Q. 22 Let a conic C pass through the point $(4, -2)$ and $\mathrm{P(x, y)}, \mathrm{x} \geq 3$, be any point on C. Let the slope of the line touching the conic C only at a single point P be half the slope of the line joining the points P and $(3, -5)$. If the focal distance of the point $(7, 1)$ on C is d, then $12\ \mathrm{d}$ equals_____.

Correct Answer:
75

Solution:
$P(x, y) \& x \geq 3$

Slope of line at $\mathrm{P(x, y)}$ will be $\frac{\mathrm{dy}}{\mathrm{dx}} = \frac{1}{2}\left(\frac{\mathrm{y+5}}{\mathrm{x-3}}\right)$

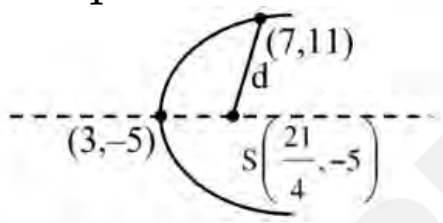
$$\Rightarrow 2\frac{dy}{(y+5)} = \frac{1}{(x-3)}dx$$
$$\Rightarrow 2\ln(y+5) = \ln(x-3) + C$$

Passes through $(4, -2)$

$$\Rightarrow 2\ln(3) = \ln(1) + C$$
$$\Rightarrow C = 2\ln(3)$$
$$\Rightarrow 2\ln(y+5) = \ln(x-3) + 2\ln(3)$$
$$\Rightarrow 2\left(\ln\left(\frac{y+5}{3}\right)\right) = \ln(x-3)$$
$$\Rightarrow \left(\frac{y+5}{3}\right)^2 = (x-3)$$
$$\Rightarrow (y+5)^2 = 9(x-3)$$
$$\downarrow$$

Parabola

$$4a = 9$$
$$a = \frac{9}{4}$$



$$d = \sqrt{\left(\frac{7}{4}\right)^2 + 6^2}$$
$$d = \frac{\sqrt{625}}{4}$$
$$d = \frac{25}{4}$$
$$12d = 75$$

Parabola

Q. 23 Let $r_k = \frac{\int_0^1 (1-x^7)^k dx}{\int_0^1 (1-x^7)^{k+1} dx}, k \in N$. Then the value of $\sum_{\mathrm{k=1}}^{10} \frac{1}{7(\mathrm{r_k}-1)}$ is equal to_____.

Correct Answer:
65

Solution:

$$I_K = \int 1.\left(1-x^7\right)^K dx$$

$$I_K = \left(1-x^7\right)^K x\ {}_0^1 + 7K\int_0^1 \left(1-x^7\right)^{K-1} x^6 \cdot x dx$$

$$I_K = -7K\int_0^1 \left(1-x^7\right)^{K-1}\left(\left(1-x^7\right)-1\right)dx$$

$$I_K = -7KI_K + 7KI_{K-1}$$

$$\Rightarrow \frac{I_K}{I_{K+1}} = \frac{7K+8}{7K+7}$$

$$r_K = \frac{7K}{7K+8}$$

$$r_K - 1 = \frac{1}{7(K+1)}$$

$$\Rightarrow 7\left(r_K - 1\right) = \frac{1}{K+1}$$

$$\sum_{K=1}^{10}(K+1) = 11(6) - 1 = 65$$

Q. 24 Let x_1, x_2, x_3, x_4 be the solution of the equation $4x^4 + 8x^3 - 17x^2 - 12x + 9 = 0$ and $\left(4+x_1^2\right)\left(4+x_2^2\right)\left(4+x_3^2\right)\left(4+x_4^2\right) = \frac{125}{16}m$. Then the value of m is_____.

Correct Answer:
221

Solution:

$$4x^4 + 8x^3 - 17x^2 - 12x + 9$$

$$= 4\left(x-x_1\right)\left(x-x_2\right)\left(x-x_3\right)\left(x-x_4\right)$$

Put $x = 2i \& -2i$

$64-64i+68-24i+9=(2i-x_1)(2i-x_2)(2i-x_3)$

$(2i-x_4)$

$=141-88\mathrm{i}.\ldots\ldots(1)$

$64+64i+68+24i+9=4(-2i-x_1)(-2i-x_2)(-2i$

$-x_3)(-2i-x_4)$

$=141+88i.\ldots\ldots(2)$

$\frac{125}{16}\mathrm{m}=\frac{141^2+88^2}{16}$

$m=221$

Q. 25 Let $\mathrm{L}_1,\ \mathrm{L}_2$ be the lines passing through the point $\mathrm{P}(0,1)$ and touching the parabola $9x^2+12x+18y-14=0$. Let Q and R be the points on the lines L_1 and L_2 such that the $\triangle\mathrm{PQR}$ is an isosceles triangle with base QR. If the slopes of the lines QR are m_1 and m_2. then $16\left(\mathrm{m}_1^2+\mathrm{m}_2^2\right)$ is equal to_____.

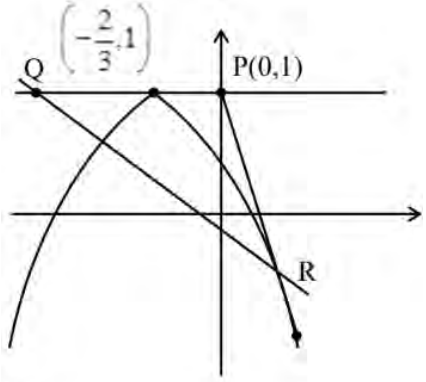
Correct Answer:
68

Solution:

$9x^2 + 12x + 4 = -18(y-1)$

$(3x+2)^2 = -18(y-1)$

$\left(x + \frac{2}{3}\right)^2 = -2(y-1)$



(0, 1)

$y = mx + 1$

$\left(x + \frac{2}{3}\right)^2 = -2(y-1)$

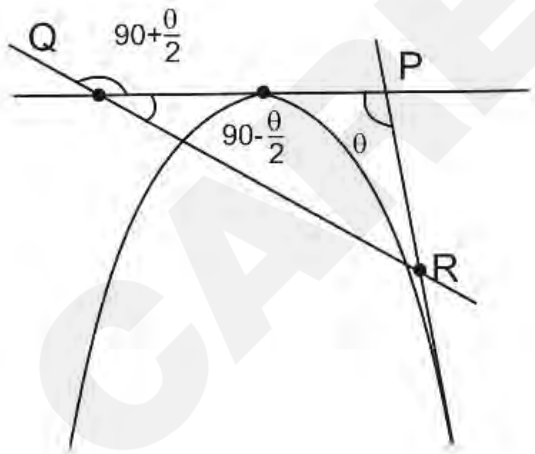
$(3x+2)^2 = -18mx$

$9x^2 + (12 + 18m)x + 4 = 0$

$4(6+9m)^2 = 4(36)$

$6 + 9m = 6, -6$

$m = 0, \frac{-4}{3}$



$\tan\theta = -\frac{4}{3}$

$$\frac{2\tan\frac{\theta}{2}}{1-\tan^2\frac{\theta}{2}} = \frac{-4}{3}$$

$$\left(\tan\frac{\theta}{2}-2\right)\left(2\tan\frac{\theta}{2}+1\right)=0$$

$$\tan\frac{\theta}{2} = 2, \frac{-1}{2}$$

$$m_{QR} = \tan\left(90+\frac{\theta}{2}\right)$$

$$= -\cot\frac{\theta}{2}$$

$$m_1 = \frac{-1}{2} \quad m_2 = \frac{-1}{-1/2} = 2$$

$$16\left(m_1^2+m_2^2\right) = 16\left(\frac{1}{4}+4\right)$$

$$= 4+64 = 68$$

Q. 26 If the second, third and fourth terms in the expansion of $(x+y)^n$ are 135, 30 and $\frac{10}{3}$, respectively, then $6\left(n^3+x^2+y\right)$ is equal to_____.

Correct Answer:
806

Solution:

$${}^{n}C_1 x^{n-1} y = 135 \ldots\ldots (i)$$

$${}^{n}C_2 x^{n-2} y^2 = 30 \ldots\ldots (ii)$$

$${}^{n}C_3 x^{n-3} y^3 = \frac{10}{3} \ldots\ldots (iii)$$

By $\frac{(i)}{(ii)}$

$$\frac{{}^{n}C_1}{{}^{n}C_2}\frac{x}{y} = \frac{9}{2} \ldots\ldots (iv)$$

By $\frac{(ii)}{(iii)}$

$$\frac{{}^{n}C_2}{{}^{n}C_3}\frac{x}{y} = 9 \ldots\ldots (v)$$

By $\frac{(iv)}{(v)}$

$$\frac{{}^{n}C_1 C_3}{{}^{n}C_2 C_2} = \frac{1}{2}$$

$$\frac{2n^2(n-1)(n-2)}{6} = \frac{n(n-1)}{2}\frac{n(n-1)}{2}$$

$$4n - 8 = 3n - 3$$

$$\Rightarrow n = 5$$

put in (v)

$$\frac{x}{y} = 9$$

$$x = 9y$$

put in (i)

$${}^{5}C_1 x^4 \left(\frac{x}{9}\right) = 135$$

$$x^5 = 27 \times 9$$

$$\Rightarrow x = 3, \quad y = \frac{1}{3}$$

$$6\left(n^3 + x^2 + y\right)$$

$$= 6\left(125 + 9 + \frac{1}{3}\right)$$

$$= 806$$

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Q. 27 Let the first term of a series be $T_1 = 6$ and its r^{th} term $T_r = 3T_{r-1} + 6^r, r = 2, 3, \ldots\ldots, n$. If the sum of the first n terms of this series is $\frac{1}{5}\left(n^2 - 12n + 39\right)\left(4.6^n - 5.3^n + 1\right)$. Then n is equal to_____.

Correct Answer:
6

Solution:

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$T_r = 3\,T_{r-1} + 6^r, r = 2, 3, 4, \ldots n$

$T_2 = 3 \cdot T_1 + 6^2$

$T_2 = 3 \cdot 6 + 6^2 \ldots\ldots (1)$

$T_3 = 3\,T_2 + 6^3$

$T_3 = 3\,T_2 + 6^3$

$T_3 = 3\left(3 \cdot 6 + 6^2\right) + 6^3$

$T_3 = 3^2 \cdot 6 + 3 \cdot 6^2 + 6^3 \ldots\ldots\ldots\ldots\ldots (2)$

$T_r = 3^{r-1} \cdot 6 + 3^{r-2} \cdot 6^2 + \ldots + 6^r$

$T_r = 3^{r-1} \cdot 6\left[1 + \frac{6}{3} + \left(\frac{6}{3}\right)^2 + \ldots + \left(\frac{6}{3}\right)^{r-1}\right]$

$T_r = 3^{r-1} \cdot 6\left(1 + 2 + 2^2 + \ldots + 2^{r-1}\right)$

$T_r = 6 \cdot 3^{r-1} 1 \cdot \frac{(1 - 2^r)}{(-1)}$

$T_r = 6 \cdot 3^{r-1} \cdot (2^r - 1)$

$T_r = \frac{6 \cdot 3^r}{3} \cdot (2^r - 1)$

$T_r = 2 \cdot (6^r - 3^r)$

$S_n = 2\Sigma\,(6^r - 3^r)$

$S_n = 2 \cdot \left[\frac{6 \cdot (6^n - 1)}{5} - \frac{3 \cdot (3^n - 1)}{2}\right]$

$S_n = 2\left[\frac{12\,(6^n - 1) - 15\,(3^n - 1)}{10}\right]$

$S_n = \frac{3}{5}\left[4 \cdot 6^4 - 5 \cdot 3^n + 1\right]$

$\therefore n^2 - 12n + 39 = 3$

$n^2 - 12n + 36 = 0$

$n = 6$

Q. 28 For $n \in N$, if $\cot^{-1} 3 + \cot^{-1} 4 + \cot^{-1} 5 + \cot^1 n = \frac{\pi}{4}$, then n is equal to_____.

Correct Answer:
47

Solution:

$$\cot^{-1} 3 + \cot^{-1} 4 + \cot^{-1} 5 + \cot^{1} n = \frac{\pi}{4}$$

$$\tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{4} + \tan^{-1} \frac{1}{5} + \tan^{-1} \frac{1}{n} = \frac{\pi}{4}$$

$$\tan^{-1} \left(\frac{46}{48} \right) + \tan^{-1} \frac{1}{n} = \frac{\pi}{4}$$

$$\tan^{-1} \left(\frac{23}{24} \right) + \tan^{-1} \frac{1}{n} = \frac{\pi}{4}$$

$$\tan^{-1} \frac{1}{n} = \tan^{-1} 1 - \tan^{-1} \frac{23}{24}$$

$$\tan^{-1} \frac{1}{n} = \tan^{-1} \left(\frac{1 - \frac{23}{24}}{1 + \frac{23}{24}} \right)$$

$$\tan^{-1} \frac{1}{n} = \tan^{-1} \left(\frac{1}{\frac{24}{47}} \right.$$

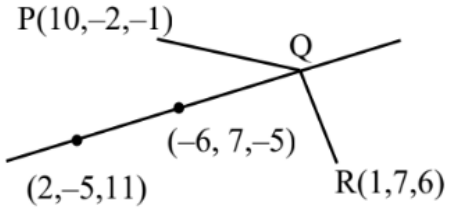
$$\tan^{-1} \frac{1}{n} = \tan^{-1} \frac{1}{47}$$

$$\mathrm{n} = 47$$

Q. 29 Let P be the point $(10, -2, -1)$ and Q be the foot of the perpendicular drawn from the point $\mathrm{R}(1, 7, 6)$ on the line passing through the points $(2, -5, 11)$ and $(-6, 7, -5)$. Then the length of the line segment PQ is equal to_____.

Correct Answer:
13

Solution:



$$\text{Line}: \frac{x+6}{-8} = \frac{y-7}{12} = \frac{z+5}{-16}$$

$$\frac{x+6}{2} = \frac{y-7}{-3} = \frac{z+5}{4} = \lambda$$

$$Q(2\lambda - 6, 7 - 3\lambda, 4\lambda - 5)$$

$$\overline{QR}(2\lambda - 7, -3\lambda, 4\lambda - 11)$$

$$\overline{QR} \cdot \text{ dr's of line } = 0$$

$$4\lambda - 14 + 9\lambda + 16\lambda - 44 = 0$$

$$29\lambda = 58 \Rightarrow \lambda = 2$$

$$Q(-2, 1, 3)$$

$$PQ = \sqrt{144 + 9 + 16} = \sqrt{169} = 13$$

Q. 30 Let $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}, \vec{b} = 3\hat{i} + 4\hat{j} - 5\hat{k}$, and a vector $\vec{c}$ be such that $\vec{a} \times (\vec{b} + \vec{c}) + \vec{b} \times \vec{c} = \hat{i} + 8\hat{j} + 13\hat{k}$. If $\vec{a} \cdot \vec{c} = 13$, then $(24 - \vec{b} \cdot \vec{c})$ is equal to_____.

Correct Answer:
46

Solution:

$\vec{a} \times \vec{b} + \vec{a} \times \vec{c} + \vec{b} \times \vec{c} = (1, 8, 13)$

$\vec{a} \times (\vec{a} \times \vec{b}) + \vec{a} \times (\vec{a} \times \vec{c}) + \vec{a} \times (\vec{b} \times \vec{c})$

$= \vec{a} \times (\hat{i} + 8\hat{j} + 13\hat{k})$

$(\vec{a} \cdot \vec{b})\vec{a} - a^2\vec{b} + (\vec{a} \cdot \vec{c})\vec{a} - a^2\vec{c} + (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} = \vec{a} \times (\hat{i} + 8\hat{j} + 13\hat{k})$

$\Rightarrow -26\vec{a} - 29\vec{b} + 13\vec{a} - 29\vec{c} + 13\vec{b} + 26\vec{c} = \vec{a} \times (\hat{i} + 8\hat{j} + 13\hat{k})$

$\Rightarrow \quad -13\vec{a} - 16\vec{b} - 3\vec{c} = \vec{a} \times (\hat{i} + 8\hat{j} + 13\hat{k})$

$\Rightarrow \quad -13\vec{a} \cdot \vec{b} - 16b^2 - 3\vec{b} \cdot \vec{c} = \{\vec{a} \times (\hat{i} + 8\hat{j} + 13\hat{k})\} \cdot \vec{b}$

$$\Rightarrow \quad (-13)(-26) - 16(50) - 3\vec{b} \cdot \vec{c} = \begin{matrix} 2 & -3 & 4 \\ 1 & 8 & 13 \\ 3 & 4 & -5 \end{matrix}$$

$\Rightarrow \quad -462 - 3\vec{b} \cdot \vec{c} = -396$

$\Rightarrow \quad \vec{b} \cdot \vec{c} = -22$

Hence $24 - \vec{b} \cdot \vec{c} = 46$

CAREERS360

6 Apr - Evening Shift

Physics

Q. 1 The longest wavelength associated with Paschen series is : (Given $R_H = 1.097 \times 10^7 SI$ unit)

Option 1:
$1.094 \times 10^{-6}\ m$

Option 2:
$2.973 \times 10^{-6}\ m$

Option 3:
$3.646 \times 10^{-6}\ m$

Option 4:
$1.876 \times 10^{-6}\ m$

Correct Answer:
$1.876 \times 10^{-6}\ m$

Solution:
For longest wavelength in Paschen's series:

$$\frac{1}{\lambda} = R\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

For longest $n_1 = 3$

$$\frac{1}{2} = R\left[\frac{1}{(3)^2} - \frac{1}{(4)^2}\right]$$

$$\frac{1}{\lambda} = R\left[\frac{1}{9} - \frac{1}{16}\right]$$

$$\frac{1}{\lambda} = R\left[\frac{16-9}{16\times 9}\right]$$

$$\Rightarrow \lambda = \frac{16\times 9}{7R} = \frac{16\times 9}{7\times 1.097\times 10^7}$$

$$\lambda = 1.876 \times 10^{-6}\ m$$

Q. 2 A total of $48\ J$ heat is given to one mole of helium kept in a cylinder. The temperature of helium increases by $2^\circ C$. The work done by the gas is : (Given, $R = 8.3\ J\ K^{-1}\ mol^{-1}$.)

Option 1:
72.9 J

Option 2:
24.9 J

Option 3:
48 J

Option 4:
23.1 J

Correct Answer:
23.1 J

Solution:
1^{st} law of thermodynamics

$$\Delta Q = \Delta U + W$$
$$\Rightarrow +48 = nC_v\Delta T + W$$
$$\Rightarrow 48 = (1)\left(\frac{3R}{2}\right)(2) + W$$
$$\Rightarrow W = 48 - 3 \times R$$
$$\Rightarrow W = 48 - 3 \times (8.3)$$
$$\Rightarrow W = 23.1 \text{ Joule}$$

Q. 3 In finding out refractive index of glass slab the following observations were made through travelling microscope 50 vernier scale division = 49 MSD; 20 divisions on main scale in each cm For mark on paper

$MSR = 8.45\ cm, VC = 26$

For mark on paper seen through slab

$MSR = 7.12\ cm, VC = 41$

For powder particle on the top surface of the glass slab

$MSR = 4.05\ cm, VC = 1$

(MSR = Main Scale Reading, VC = Vernier Coincidence)

Refractive index of the glass slab is:

Option 1:
1.42

Option 2:
1.52

Option 3:
1.24

Option 4:
1.35

Correct Answer:
1.42

Solution:

$1\text{MSD} = \frac{1\text{ cm}}{20} = 0.05\text{ cm}$

$1\text{VSD} = \frac{49}{50}\text{MSD} = \frac{49}{50} \times 0.05\text{ cm} = 0.049\text{ cm}$

$\text{LC} = 1\text{MSD} - 1\text{VSD} = 0.001\text{ cm}$

For mark on paper, $\text{L}_1 = 8.45\text{ cm} + 26 \times 0.001\text{ cm}$
$= 84.76\text{ mm}$

For mark on paper through slab, $\text{L2} = 7.12\text{ cm} +$
$41 \times 0.001\text{ cm} = 71.61\text{ mm}$

For powder particle on top surface, $ZE = 4.05\text{ cm}$
$+1 \times 0.001\text{ cm} = 40.51\text{ mm}$

$\therefore \text{actual } \text{L}_1 = 84.76 - 40.51 = 44.25\text{ mm}$

$\text{actual } \text{L}_2 = 71.61 - 40.51 = 31.10\text{ mm}$

$\text{L}_2 = \frac{\text{L}_1}{\mu}$

$\Rightarrow \mu = \frac{\text{L}_1}{\text{L}_2} = \frac{44.25}{31.10} = 1.42$

Q. 4 In the given electromagnetic wave $E_y = 600\sin(\omega t - kx)\text{Vm}^{-1}$, intensity of the associated light beam is (in W/m^2); (Given $\epsilon_0 = 9 \times 10^{-12}\text{C}^2\ \text{N}^{-1}\ \text{m}^{-2}$)

Option 1:
486

Option 2:
243

Option 3:
729

Option 4:
972

Correct Answer:
486

Solution:

$$\text{Intensity} = \frac{1}{2}\varepsilon_0\text{E}_0^2\text{c}$$
$$= \frac{1}{2} \times 9 \times 10^{-12} \times (600)^2 \times 3 \times 10^8$$
$$= \frac{9}{2} \times 36 \times 3 = 486\text{w}/\text{m}^2$$

Q. 5 Assuming the earth to be a sphere of uniform mass density, a body weighed 300 N on the surface of earth. How much it would weigh at R/4 depth under surface of earth?

Option 1:
75 N

Option 2:
375 N

Option 3:
300 N

Option 4:
225 N

Correct Answer:
225 N

Solution:

$\text{At surface: } mg = 300 \text{ N}$

$m = \frac{300}{g_s}$

$\text{At Depth } \frac{R}{4} : g_d = g_s \left[1 - \frac{d}{R}\right]$

$g_d = g_s \left[1 - \frac{R}{4R}\right]$

$g_d = \frac{3\, g_s}{4}$

weight at depth

$= m \times g_d$

$= m \times \frac{3\, g_s}{4}$

$= \frac{3}{4} \times 300$

$= 225 \text{ N}$

Q. 6 The acceptor level of a p-type semiconductor is 6 eV. The maximum wavelength of light which can create a hole would be : Given hc=1242 eV nm.

Option 1:
407 nm

Option 2:
414 nm

Option 3:
207 nm

Option 4:
103.5 nm

Correct Answer:
207 nm

Solution:

$$\text{Energy} = \frac{hc}{\lambda}$$

$$E = \frac{1240}{\lambda(nm)} eV$$

$$6 = \frac{1240}{\lambda(nm)}$$

$$\lambda = \frac{1240}{6} = 207 \text{ nm}$$

Q. 7 A car of 800 kg is taking turn on a banked road of radius 300 m and angle of banking 30°. If coefficient of static friction is 0.2 then the maximum speed with which car can negotiate the turn safely: $\left(g = 10 \text{ m/s}^2, \sqrt{3} = 1.73\right)$

Option 1:
70.4 m/s

Option 2:
51.4 m/s

Option 3:
264 m/s

Option 4:
102.8 m/s

Correct Answer:
51.4 m/s

Solution:

$m = 800 \text{ kg}$

$r = 300 \text{ m}$

$\theta = 30^\circ$

$\mu_s = 0.2$

$$V_{max} = \sqrt{Rg\left[\frac{\tan\theta + \mu}{1 - \mu\tan\theta}\right]}$$

$$= \sqrt{300 \times g \times \left[\frac{\tan 30^\circ + 0.2}{1 - 0.2 \times \tan 30^\circ}\right]}$$

$$= \sqrt{300 \times 10 \times \left[\frac{0.57 + 0.2}{1 - 0.2 \times 0.57}\right]}$$

$$V_{max} = 51.4 \text{ m/s}$$

Q. 8 Two identical conducting spheres P and S with charge Q on each, repel each other with a force 16 N. A third identical uncharged conducting sphere R is successively brought in contact with the two spheres. The new force of repulsion between P and S is :

Option 1:
4N

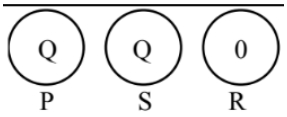
Option 2:
6N

Option 3:
1N

Option 4:
12N

Correct Answer:
6N

Solution:



$F_{PS} \propto Q^2$

$F_{PS} = 16\ \text{N}$

Now If $P\&R$ are brought in contact then

Now If $\text{S}\&\text{R}$ are brought in contact then

New force between $\text{P}\&\ \text{S}$ is :

$$\text{F}_{\text{PS}} \propto \frac{\text{Q}}{2} \times \frac{3\text{Q}}{4}$$

$$\text{F}_{\text{PS}} \propto \frac{3\text{Q}^2}{8} = \frac{3}{8} \times 16 = 6\ \text{N}$$

Q. 9 In a coil, the current changes form $-2\ \text{A}$ to $+2\ \text{A}$ in 0.2 s and induces an emf of $0.1\ \text{V}$. The selfinductance of the coil is :

Option 1:
5 mH

Option 2:
1 mH

Option 3:
2.5 mH

Option 4:
4 mH

Correct Answer:
5 mH

Solution:
$(\mathrm{Emf})_{\text{induced}} = -\mathrm{L}\frac{\mathrm{di}}{\mathrm{dt}}$

In magnitude form,

$|\mathrm{Emf}_{\text{ind}}| = (-)\mathrm{L}\dfrac{\mathrm{di}}{\mathrm{dt}}$

$\Rightarrow 0.1 = \dfrac{(\mathrm{L})[+2-(-2)]}{0.2}$

$\Rightarrow \mathrm{L} = \dfrac{0.1 \times 0.2}{4} = 5\mathrm{mH}$

Q. 10 For the thin convex lens, the radii of curvature are at 15 cm and 30 cm respectively. The focal length the lens is 20 cm. The refractive index of the material is :

Option 1:
1.2

Option 2:
1.4

Option 3:
1.5

Option 4:
1.8

Correct Answer:
1.5

Solution:
$\dfrac{1}{\mathrm{f}} = \left(\dfrac{\mu_{\text{lens}}}{\mu_{\text{air}}} - 1\right)\left(\dfrac{1}{\mathrm{R}_1} - \dfrac{1}{\mathrm{R}_2}\right)$

$\Rightarrow \dfrac{1}{+20} = \left(\dfrac{\mu}{1} - 1\right)\left(\dfrac{1}{+15} - \dfrac{1}{(-30)}\right)$

$\Rightarrow \dfrac{1}{20} = (\mu - 1)\left(\dfrac{3}{30}\right)$

$\Rightarrow \mu - 1 = \dfrac{1}{2}$

$\Rightarrow \mu = 1 + \dfrac{1}{2} = \dfrac{3}{2} = 1 \cdot 5$

Q. 11 Energy of 10 non rigid diatomic molecules at temperature T is:

Option 1:
$\frac{7}{2}\mathrm{RT}$

Option 2:
$70\ \mathrm{K_B T}$

Option 3:
$35\mathrm{RT}$

Option 4:
$35\ \mathrm{K_B T}$

Correct Answer:
$35\ \mathrm{K_B T}$

Solution:

$$\text{Degree of freedom } (f) = 5 + 2(3\,\mathrm{N} - 5)$$

$$f = 5 + 2(3 \times 2 - 1) = 7$$

$$\text{energy of one molecule } = \frac{f^2}{2} K_B T$$

$$\text{energy of 10 molecules}$$

$$= 10\left(\frac{f}{2} K_B T\right) = 10\left(\frac{7}{2} K_B T\right) = 35 K_B T$$

Q. 12 A body of weight 200 N is suspended form a tree branch thought a chain of mass 10 kg. The branch pulls the chain by a force equal to (if $g = 10\ \mathrm{m/s^2}$):

Option 1:
150 N

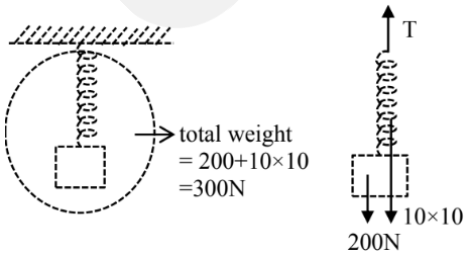
Option 2:
300 N

Option 3:
200 N

Option 4:
100 N

Correct Answer:
300 N

Solution:



Chain block system is in equilibrium so
T=200+100=300 N .

Q. 13 When UV light of wavelength $300\ \mathrm{nm}$ is incident on the metal surface having work function $2.13\mathrm{eV}$, electron emission takes place. The stopping potential is : (Given $\mathrm{hc} = 1240\mathrm{eVnm}$)

Option 1:
4 V

Option 2:
4.1 V

Option 3:
2 V

Option 4:
1.5 V

Correct Answer:
2 V

Solution:

$$\frac{\mathrm{hc}}{\lambda} - \phi = \mathrm{e.\,V_s}$$

$$\Rightarrow \frac{1240}{300}\mathrm{eV} - 2.13\mathrm{eV} = \mathrm{eVs}$$

$$\Rightarrow 4.13\mathrm{eV} - 2.13\mathrm{eV} = \mathrm{eVs}$$

$$\Rightarrow \mathrm{So,} \quad \mathrm{V_s} = 2\mathrm{volt}$$

Q. 14 The number of electrons flowing per second in the filament of a $110\ \mathrm{W}$ bulb operating at $220\ \mathrm{V}$ is : (Given $\mathrm{e} = 1.6 \times 10^{-19}\mathrm{C}$)

Option 1:
31.25×10^{17}

Option 2:
6.25×10^{18}

Option 3:
6.25×10^{17}

Option 4:
1.25×10^{19}

Correct Answer:
31.25×10^{17}

Solution:

$\text{Power } (P) = V.I$
$\Rightarrow 110 = (220)(I)$
$\Rightarrow I = 0.5 \text{ A}$
$\text{Now, } I = \frac{\text{n} \cdot \text{e}}{\text{t}}$
$\Rightarrow 0.5 = \left(\frac{\text{n}}{\text{t}}\right)\left(1.6 \times 10^{-19}\right)$
$\Rightarrow \frac{\text{n}}{\text{t}} = \frac{0.5}{1.6 \times 10^{-19}}$
$\Rightarrow \frac{\text{n}}{\text{t}} = 31.25 \times 10^{17}$

Q. 15 When kinetic energy of a body becomes 36 times of its original value, the percentage increase in the momentum of the body will be :

Option 1:
500%

Option 2:
600%

Option 3:
6%

Option 4:
60%

Correct Answer:
500%

Solution:

$\text{Kinetic energy } (K) = \frac{P^2}{2m}$
$\Rightarrow \text{P} = \sqrt{2\text{mK}}$
$\text{If } \text{K}_\text{f} = 36 \text{ K}_\text{i}$
$\text{So, } \text{P}_\text{f} = 6\text{P}_\text{i}$
$\% \text{ increase in momentum } = \frac{\text{P}_\text{f} - \text{P}_\text{i}}{\text{P}_\text{i}} \times 100\%$
$= \frac{6\text{P}_\text{i} - \text{P}_\text{i}}{\text{P}_\text{i}} \times 100\%$
$= 500\%$

Q. 16 Pressure inside a soap bubble is greater than the pressure outside by an amount :
(given : R= Radius of bubble, S= Surface tension of bubble)

Option 1:
$\frac{4\,\text{S}}{\text{R}}$

Option 2:
$\frac{4R}{S}$

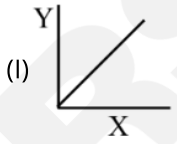
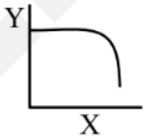
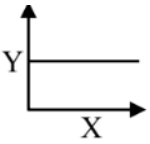
Option 3:
$\frac{S}{R}$

Option 4:
$\frac{2S}{R}$

Correct Answer:
$\frac{4S}{R}$

Solution:
There are two liquid-air surfaces in bubble so
$\Delta P = 2\left(\frac{2S}{R}\right) = \frac{4S}{R}$

Q. 17 Match List-I with List-II

	List-I (Y vs X)	List-II (Shape of Graph)
(A)	$Y =$ magnetic susceptibility $X =$ magnetising field	(I) 
(B)	Y = magnetic field X = distance from centre of a current carrying wire for $x < a$ (where a = radius of wire)	(II) 
(C)	Y = magnetic field X = distance from centre of a current carrying wire for $x > a$ (where a = radius of wire)	(III) 
(D)	Y = magnetic field inside solenoid X = distance from center	(IV)

Choose the correct answer from the options given below :

Option 1:
(A)-(III), (B)-(I), (C)-(IV), (D)-(II)

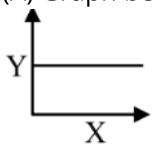
Option 2:
(A)-(I), (B)-(III), (C)-(II), (D)-(IV)

Option 3:
(A)-(IV), (B)-(I), (C)-(III), (D)-(II)

Option 4:
(A)-(III), (B)-(IV), (C)-(I), (D)-(II)

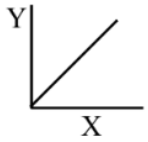
Correct Answer:
(A)-(III), (B)-(IV), (C)-(I), (D)-(II)

Solution:
(A) Graph between Magnetic susceptibility and magnetising field is :



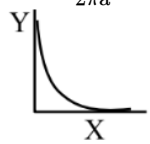
(B) magnetic field due to a current carrying wire for $\mathrm{x} < \mathrm{a}$:

$\mathrm{B} = \frac{\mu_0 \mathrm{ir}}{2\pi \mathrm{a}^2}$

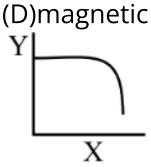


(C) magnetic field due to a current carrying wire for $\mathrm{x} > \mathrm{a}$:

$B = \frac{\mu_0 \mathrm{i}}{2\pi \mathrm{a}}$



(D)magnetic field inside solenoid varies as:



Q. 18 In a vernier calliper, when both jaws touch each other, zero of the vernier scale shifts towards left and its 4^{th} division coincides exactly with a certain division on main scale. If 50 vernier scale divisions equal to 49 main scale divisions and zero error in the instrument is 0.04 mm then how many main scale divisions are there in 1 cm ?

Option 1:
40

Option 2:
5

Option 3:
20

Option 4:
10

Correct Answer:
20

Solution:
4^{th} division coincides with 3^{rd} division then
$0.004\ \text{cm} = 4\text{VSD} - 3\text{MSD}$
$49\text{MSD} = 50\text{VSD}$
$1\text{MSD} = \dfrac{1}{\text{N}}\ \text{cm}$
$0.004 = 4\left\{\dfrac{49}{50}\text{MSD}\right\} - 3\text{MSD}$
$0.004 = \left(\dfrac{196}{50} - 3\right)\left(\dfrac{1}{\text{N}}\right)$
$\text{N} = \dfrac{46}{50} \times \dfrac{1000}{4} = \dfrac{46 \times 1000}{200} = 230$

Q. 19 Given below are two statements :
Statement (I) : Dimensions of specific heat is $\left[\text{L}^2\ \text{T}^{-2}\ \text{K}^{-1}\right]$
Statement (II) : Dimensions of gas constant is $\left[\text{ML}^2\ \text{T}^{-1}\ \text{K}^{-1}\right]$

Option 1:
Statement (I) is incorrect but statement (II) is correct

Option 2:
Both statement (I) and statement (II) are incorrect

Option 3:
Statement (I) is correct but statement (II) is incorrect

Option 4:
Both statement (I) and statement (II) are correct

Correct Answer:
Statement (I) is correct but statement (II) is incorrect

Solution:
$\Delta\text{Q} = \text{mS}\Delta\text{T}$
$\text{s} = \dfrac{\Delta\text{Q}}{\text{m}\Delta\text{T}}$
$[\text{s}] = \left[\dfrac{\text{ML}^2\ \text{T}^{-2}}{\text{MK}}\right]$
$[\text{s}] = \left[\text{L}^2\ \text{T}^{-2}\ \text{K}^{-1}\right]$

Statement-(I) is correct
$\text{PV} = \text{nRT} \Rightarrow \text{R} = \dfrac{\text{PV}}{\text{nT}}$
$[\text{R}] = \dfrac{\left[\text{ML}^{-1}\ \text{T}^{-2}\right]\left[\text{L}^3\right]}{[\text{mol}][\text{K}]}$
$[\text{R}] = \left[\text{ML}^2\ \text{T}^{-2}\ \text{mol}^{-1}\ \text{K}^{-1}\right]$

Statement-II is incorrect

Q. 20 A body projected vertically upwards with a certain speed from the top of a tower reaches the ground in t_1. If it is projected vertically downwards from the same point with the same speed, it reaches the ground in t_2. Time required to reach the ground, if it is dropped from the top of the tower, is :

Option 1:
$\sqrt{t_1 t_2}$

Option 2:
$\sqrt{t_1 - t_2}$

Option 3:
$\sqrt{\frac{t_1}{t_2}}$

Option 4:
$\sqrt{t_1 + t_2}$

Correct Answer:
$\sqrt{t_1 t_2}$

Solution:

$$t_1 = \frac{u + \sqrt{u^2 + 2gh}}{g}$$

$$t_2 = \frac{-u + \sqrt{u^2 + 2gh}}{g}$$

$$t = \frac{\sqrt{2gh}}{g}$$

$$t_1 t_2 = \frac{\left(u^2 + 2gh\right) - u^2}{g^2} = \frac{2gh}{g^2} = t^2$$

$$\Rightarrow t = \sqrt{t_1 t_2}$$

Q. 21 In Franck-Hertz experiment, the first dip in the current-voltage graph for hydrogen is observed at 10.2 V. The wavelength of light emitted by hydrogen atom when excited to the first excitation level is_____nm.
(Given $\mathrm{hc} = 1245\mathrm{eVnm}, \mathrm{e} = 1.6 \times 10^{-19}\mathrm{C}$).

Correct Answer:
122

Solution:

$$10.2\mathrm{eV} = \frac{\mathrm{hc}}{\lambda}$$

$$\lambda = \frac{1245\mathrm{eV} - \mathrm{nm}}{10.2\mathrm{eV}} = 122.06\ \mathrm{nm}$$

Q. 22 For a given series LCR circuit it is found that maximum current is drawn when value of variable capacitance is 2.5 nF . If resistance of 200Ω and 100 mH inductor is being used in the given circuit. The frequency of ac source is_____$\times 10^3$ Hz. (given $\pi^2 = 10$)

Correct Answer:
10

Solution:
for maximum current, circuit must be in resonance.

$$f_0 = \frac{1}{2\pi\sqrt{L \times C}}$$

$$f_0 = \frac{1}{2\pi\sqrt{100 \times 10^{-3} \times 2.5 \times 10^{-9}}}$$

$$= \frac{1}{2\pi\sqrt{25 \times 10^{-11}}}$$

$$= \frac{1}{2\pi \times 5} \times 10^5 \times \sqrt{10}\ \text{Hz}$$

$$= \frac{100}{10} \times 10^3\ \text{Hz}$$

$$f_0 = 10 \times 10^3\ \text{Hz}$$

Q. 23 A particle moves in a straight line so that its displacement x at any time t is given by $x^2 = 1 + t^2$. Its acceleration at any time t is x^{-n} where n=_____.

Correct Answer:
3

Solution:

$$x^2 = 1 + t^2$$

$$2x\frac{dx}{dt} = 2t$$

$$xv = t$$

$$x\frac{dv}{dt} + v\frac{dx}{dt} = 1$$

$$x \cdot a + v^2 = 1$$

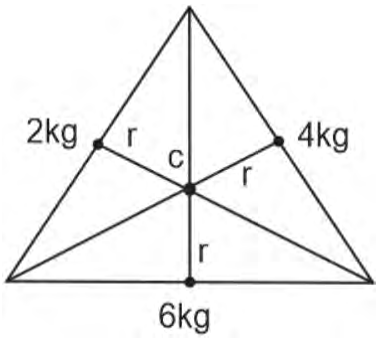
$$a = \frac{1 - v^2}{x} = \frac{1 - t^2/x^2}{x}$$

$$a = \frac{1}{x^3} = x^{-3}$$

Q. 24 Three balls of masses $2\ \text{kg}, 4\ \text{kg}$ and $6\ \text{kg}$ respectively are arranged at centre of the edges of an equilateral triangle of side $2\ \text{m}$. The moment of inertia of the system about an axis through the centroid and perpendicular to the plane of triangle, will be_____kgm^2.

Correct Answer:
4

Solution:



$r = \frac{1}{\sqrt{3}}$

Moment of inertia about C and perpendicular to the plane is :

$\mathrm{I} = \mathrm{r}^2[2+4+6]$

$= \frac{1}{3} \times 12$

$\mathrm{I} = 4\ \mathrm{kg} - \mathrm{m}^2$

Q. 25 A coil having 100 turns, area of $5 \times 10^{-3}\ \mathrm{m}^2$, carrying current of $1\ \mathrm{mA}$ is placed in uniform magnetic field of $0.20\ \mathrm{T}$ such a way that plane of coil is perpendicular to the magnetic field. The work done in turning the coil through 90° is_____$\mu\mathrm{J}$.

Correct Answer:
100

Solution:

$W = \Delta U = U_f - U_i$

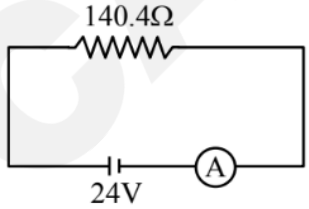
$W = (-\vec{\mu} \cdot \vec{B})_f - (-\vec{\mu}.\vec{B})_i$

$= 0 + (\vec{\mu} \cdot \vec{B})_i$

$= \left(100 \times 5 \times 10^{-3} \times 1 \times 10^{-3}\right) \times 0.2\ \mathrm{J}$

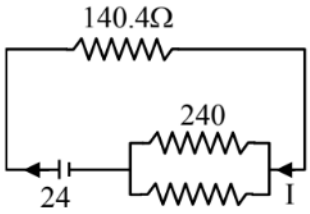
$= 1 \times 10^{-4}\ \mathrm{J} = 100\mu\mathrm{J}$

Q. 26 In the given figure an ammeter A consists of a 240Ω coil connected in parallel to a 10Ω shunt. The reading of the ammeter is_____mA.



Correct Answer:
160

Solution:



Req

$$= 140.4 + \frac{240 \times 10}{240 + 10}$$

$$\text{Req } = 140.4 + \frac{2400}{250}$$

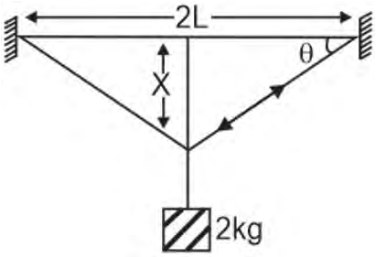
$$\text{Req. } = 150\Omega$$

$$\therefore \text{ Current in ammeter } = \frac{24}{150}$$

$$= 160 \text{ mA}$$

Q. 27 A wire of cross sectional area A, modulus of elasticity $2 \times 10^{11}\text{Nm}^{-2}$ and length 2 m is stretched between two vertical rigid supports. When a mass of 2 kg is suspended at the middle it sags lower from its original position making angle $\theta = \frac{1}{100}$ radian on the points of support. The value of A is_____$\times 10^{-4}$ m^2 (consider $\text{x} << \text{L}$).

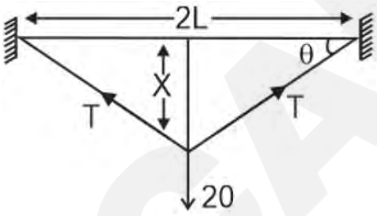
$(\text{given } : \text{g} = 10 \text{ m/s}^2)$



Correct Answer:

1

Solution:



In vertical derection

$2 \text{ T} \sin\theta = 20$

using small angle approximation $\sin\theta = \theta$

$$\theta = \frac{1}{100}$$

$$\therefore \quad \text{T} = \frac{10}{\theta}$$

$$\text{T} = 1000 \text{ N}$$

Change in length

$$\Delta L = 2\sqrt{x^2 + L^2} - 2L$$
$$= 2L\left[1 + \frac{x^2}{2L^2} - 1\right]$$
$$\Delta L = \frac{x^2}{L}$$
$$\therefore \quad \text{Modulus of elasticity} = \frac{\text{stress}}{\text{strain}}$$
$$2 \times 10^{11} = \frac{10^3}{A \times \frac{x^2}{L}} \times 2L$$
$$\therefore \quad A = 1 \times 10^{-4}\ m^2$$

Q. 28 Two coherent monochromatic light beams of intensities I and 4I are superimposed. The difference between maximum and minimum possible intensities in the resulting beam is x. The value of x is_____.

Correct Answer:
8

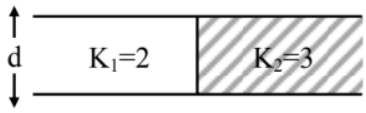
Solution:
$$I_{max} = (\sqrt{I} + \sqrt{4I})^2 = 9I$$
$$I_{min} = (\sqrt{4I} - \sqrt{I})^2 = I$$
$$\therefore I_{max} - I_{min} = 8I$$

Q. 29 Two open organ pipes of length $60\ cm$ and $90\ cm$ resonate at 6^{th} and 5^{th} harmonics respectively. The difference of frequencies for the given modes is_____Hz
(Velocity of sound in air $= 333\ m/s$)

Correct Answer:
740

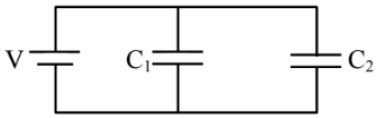
Solution:
The difference in frequency in open organ pipe $=$
$$f = \frac{nv}{2L}$$
$$\Delta f = \frac{6v}{2 \times 0.6} - \frac{5v}{2 \times 0.9}$$
$$v = 333\ m/s$$
$$\Delta f = 740\ Hz$$

Q. 30 A capacitor of $10\mu F$ capacitance whose plates are separated by $10\ mm$ through air and each plate has area $4\ cm^2$ is now filled equally with two dielectric media of $K_1 = 2, K_2 = 3$ respectively as shown in figure. If new force between the plates is $8\ N$. The supply voltage is_____V.



Correct Answer:
80

Solution:



$C_{eq} = C_1 + C_2$

$C_1 = \frac{2\epsilon_0\ A}{2 \times d} = 10\mu F$

$C_2 = \frac{3\epsilon_0\ A}{2\ d} = 15\mu F$

$C_{eq} = 25\mu F$

Now the charge on

$C_1 = 10\ V\mu c$

$C_2 = 1.5\ V\mu C.$

Now force between the plates $\left[F = \frac{Q^2}{2A\epsilon_0}\right]$

$$\frac{100\ V^2 \times 10^{-12}}{2 \times 2 \times 10^{-4}\epsilon_0} + \frac{225\ V^2 \times 10^{-12}}{2 \times 2 \times 10^{-4} \times \epsilon_0} = 8$$

$$325\ V^2 = 8 \times 4 \times 10^{-4} \times 8.85$$

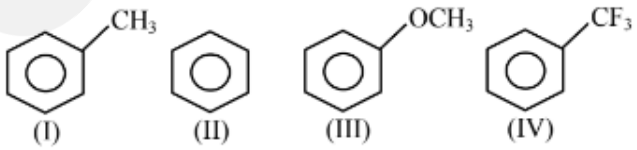
$$V^2 = \frac{32 \times 8.85 \times 10^{-4}}{325}$$

$$\therefore V = \sqrt{\frac{283.2 \times 10^{-4}}{325}}$$

$$V = 0.93 \times 10^{-2}$$

Chemistry

Q. 1



The correct arrangement for decreasing order of electrophilic substitution for above compounds.

Option 1:
(IV) > (I) > (II) > (III)

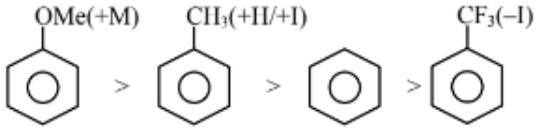
Option 2:
(III) > (I) > (II) > (IV)

Option 3:
(II) > (IV) > (III) > (I)

Option 4:
(III) > (IV) > (II) > (I)

Correct Answer:
(III) > (I) > (II) > (IV)

Solution:



Q. 2 Molality (m) of $3\mathrm{M}$ aqueous solution of NaCl is: (Given : Density of solution $= 1.25\ \mathrm{g\ mL^{-1}}$, Molar mass in $\mathrm{gol^{-1}}$: $\mathrm{Na} - 23, \mathrm{Cl} - 35.5$)

Option 1:
2.90 m

Option 2:
2.79 m

Option 3:
1.90 m

Option 4:
3.85 m

Correct Answer:
2.79 m

Solution:
3 moles are present in 1 litre solution
molality $= \frac{3\times1000}{1.25\times1000-[3\times58.5]} = 2.79\ \mathrm{m}$

Q. 3 The incorrect statements regarding enzymes are:
(A) Enzymes are biocatalysts.
(B) Enzymes are non-specific and can catalyse different kinds of reactions.
(C) Most Enzymes are globular proteins.
(D) Enzyme - oxidase catalyses the hydrolysis of maltose into glucose.

Choose the correct answer from the option given below :

Option 1:
(B) and (C)

Option 2:
(B), (C) and (D)

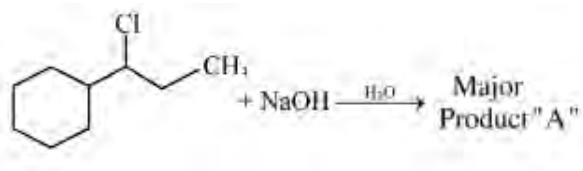
Option 3:
(B) and (D)

Option 4:
(A), (B) and (C)

Correct Answer:
(B) and (D)

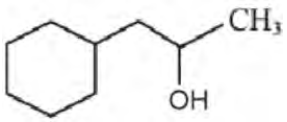
Solution:
Direct NCERT Based

Q. 4

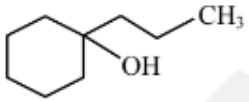


Consider the above chemical reaction. Product "A" is:

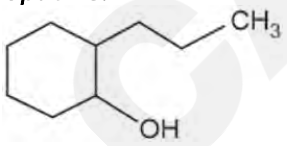
Option 1:



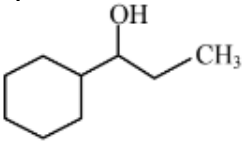
Option 2:



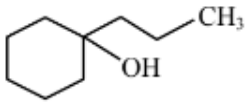
Option 3:



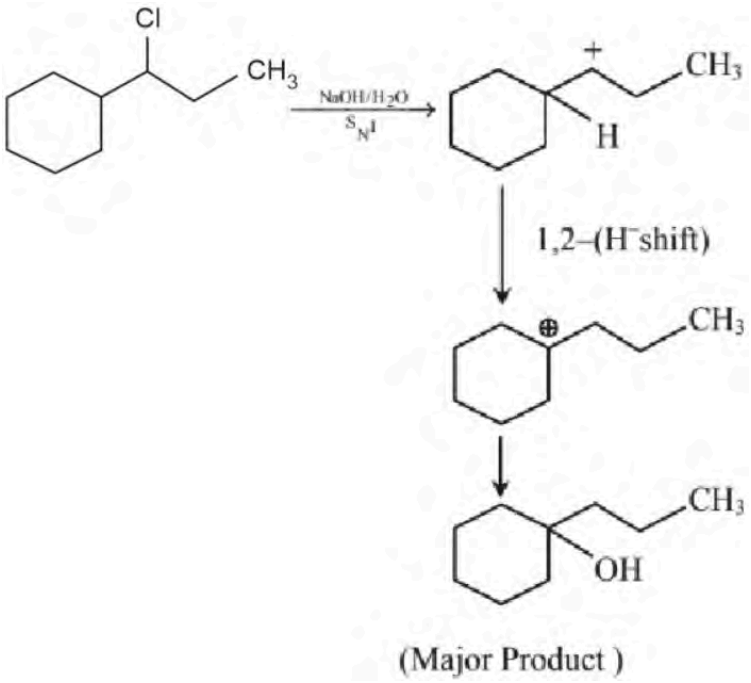
Option 4:



Correct Answer:



Solution:



(Major Product)

Q. 5 During the detection of acidic radical present in a salt, a student gets a pale yellow precipitate soluble with difficulty in $\mathrm{NH_4OH}$ solution when sodium carbonate extract was first acidified with dil. $\mathrm{HNO_3}$ and then $\mathrm{AgNO_3}$ solution was added. This indicates presence of:

Option 1:
$\mathrm{Br^-}$

Option 2:
$\mathrm{CO_3^{2-}}$

Option 3:
$\mathrm{I^-}$

Option 4:
$\mathrm{Cl^-}$

Correct Answer:
$\mathrm{Br^-}$

Solution:
$\mathrm{Ag^+ + I^- \rightarrow AgI}$ Yellow ppt.
$\mathrm{Ag^+ + Cl^- \rightarrow AgCl}$ White ppt
$\mathrm{Ag^+ + Br^- \rightarrow AgBr}$ Pale yellow ppt

Q. 6 How can an electrochemical cell be converted into an electrolytic cell?

Option 1:
Applying an external opposite potential greater than $\mathrm{E^0_{cell}}$

Option 2:
Reversing the flow of ions in salt bridge.

Option 3:
Applying an external opposite potential lower than E^0_{cell} .

Option 4:
Exchanging the electrodes at anode and cathode.

Correct Answer:
Applying an external opposite potential greater than E^0_{cell}

Solution:
Applied external potential should be greater than E^0_{cll} in opposite direction.

Q. 7 Arrange the following elements in the increasing order of number of unpaired electrons in it.
(A) Sc
(B) Cr
(C) V
(D) Ti
(E) Mn
Choose the correct answer from the options given below:

Option 1:
$(C) < (E) < (B) < (A) < (D)$

Option 2:
$(B) < (C) < (D) < (E) < (A)$

Option 3:
$(A) < (D) < (C) < (B) < (E)$

Option 4:
$(A) < (D) < (C) < (E) < (B)$

Correct Answer:
$(A) < (D) < (C) < (E) < (B)$

Solution:
Unpaired electron

$Sc[Ar]4\,s^2 3\,d^1$	1
$Cr[Ar]4\,s^1 3\,d^5$	6
$V[Ar]4\,s^2 3\,d^3$	3
Ti : $[Ar]4\,s^2 3\,d^2$	2
Mn : $[Ar]4\,s^2 3\,d^3$	5

Q. 8 Match List-I with List-II.

LIST I **Alkali Metal**		**LIST II** **Emission Wavelength in nm**	
(A)	Li	I.	589.2
(B)	Na	II.	455.5
(C)	Rb	III.	670.8
(D)	Cs	IV.	780.0

Choose the correct answer from the options given below:

Option 1:
(A)-(I), (B)-(IV), (C)-(III), (D)-(II)

Option 2:
(A)-(III), (B)-(I), (C)-(IV), (D)-(II)

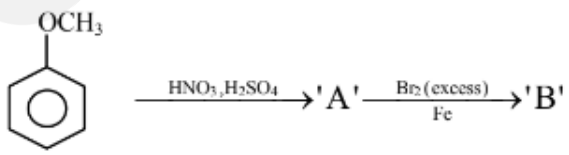
Option 3:
(A)-(IV), (B)-(II), (C)-(I), (D)-(III)

Option 4:
(A)-(II), (B)-(IV), (C)-(III), (D)-(I)

Correct Answer:
(A)-(III), (B)-(I), (C)-(IV), (D)-(II)

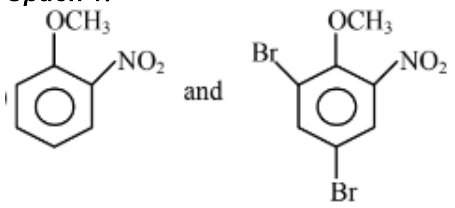
Solution:
Fact Based

Q. 9 The major products formed:

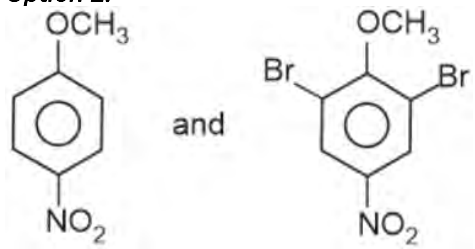


A and B respectively are:

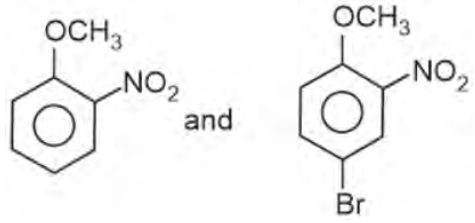
Option 1:



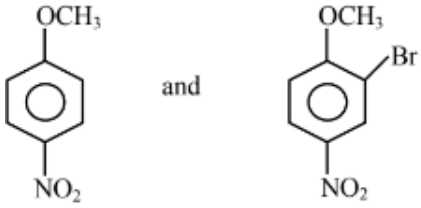
Option 2:



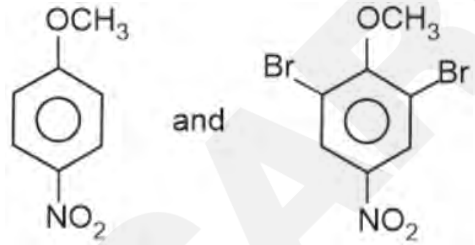
Option 3:



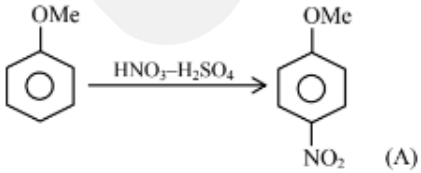
Option 4:



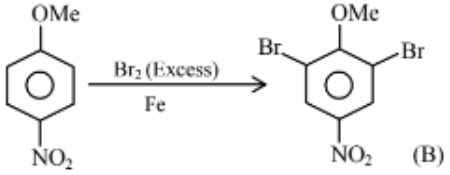
Correct Answer:



Solution:



(A)



(B)

CAREERS360

Q. 10 The incorrect statement regarding the geometrical isomers of 2-butene is:

Option 1:
cis-2-butene and trans-2-butene are not interconvertible at room temperature.

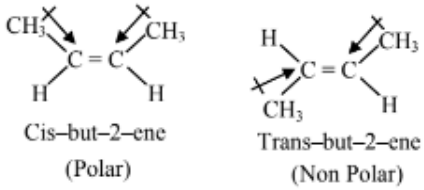
Option 2:
cis-2-butene has less dipole moment than trans-2-butene.

Option 3:
trans-2-butene is more stable than cis-2-butene.

Option 4:
cis-2-butene and trans-2-butene are stereoisomers.

Correct Answer:
cis-2-butene has less dipole moment than trans-2-butene.

Solution:



Cis–but–2–ene has higher Dipole moment than trans–but–2–ene.

Q. 11 Given below are two statements:

Statement I: PF_5 and BrF_5 both exhibit $sp^3\ d$ hybridisation.

Statement II: Both SF_6 and $[Co(NH_3)_6]^{3+}$ exhibit $sp^3\ d^2$ hybridisation.
In the light of the above statements, choose the **correct** answer from the options given below:

Option 1:
Statement I is true but Statement II is false

Option 2:
Both Statement I and Statement II are true

Option 3:
Both Statement I and Statement II are false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Both Statement I and Statement II are false

Solution:

Hybridisation	Hybridisation

PF_5	$sp^3\ d$	SF_6	$sp^3\ d^2$
BrF_5	$sp^3\ d^2$	$[Co(NH_3)_6]^{+3}$	d^2sp^3

Both Statement (1) and (2) are false.

Q. 12 The number of ions from the following that are expected to behave as oxidising agent is:

$$Sn^{4+}, Sn^{2+}, Pb^{2+}, Tl^{3+}, Pb^{4+}, Tl^{+}$$

Option 1:
3

Option 2:
4

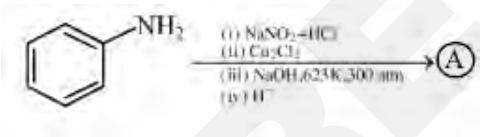
Option 3:
1

Option 4:
2

Correct Answer:
2

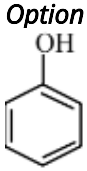
Solution:
Due to inert pair effect; $T\ell^{+3}$ and Pb^{+4} can behave as oxidising agents.

Q. 13 Identify the product A in the following reaction.

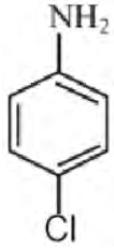


Option 1:

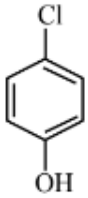
Option 2:



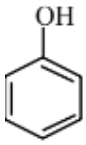
Option 3:



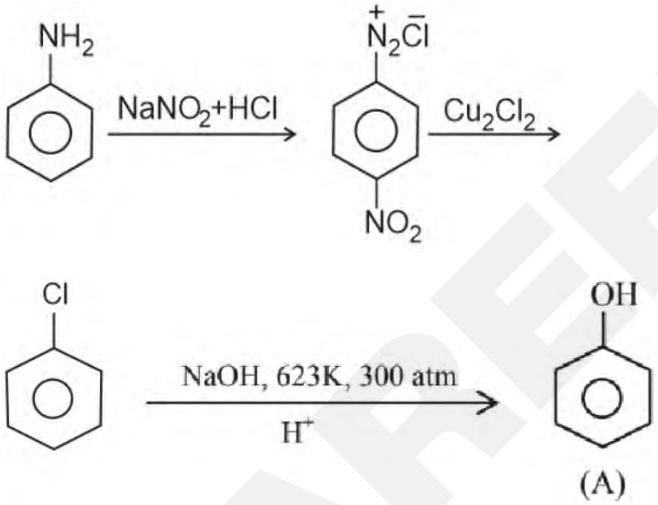
Option 4:



Correct Answer:



Solution:



Q. 14 The correct statements among the following, for a "chromatography" purification method is:

Option 1:

Organic compounds run faster than solvent in the thin layer chromatographic plate.

Option 2:

Non-polar compounds are retained at top and polar compounds come down in column chromatography.

Option 3:

R_f of a polar compound is smaller than that of a non-polar compound.

Option 4:

R_f is an integral value.

Correct Answer:
R_f of a polar compound is smaller than that of a non-polar compound.

Solution:
Non polar compounds are having higher value of R_f than polar compound.

Q. 15 Evaluate the following statements related to group 14 elements for their correctness.
(A)Covalent radius decreases down the group from C to Pb in a regular manner.
(B) Electronegativity decreases from C to Pb down the group gradually.
(C) Maximum covalence of C is 4 whereas other elements can expand their covalence due to presence of d orbitals.
(D) Heavier elements do not form $p\pi - p\pi$ bonds.
(E) Carbon can exhibit negative oxidation states.

Choose the **correct** answer from the options given below:

Option 1:
(C), (D) and (E) Only

Option 2:
(A) and (B) Only

Option 3:
(A), (B) and (C) Only

Option 4:
(C) and (D) Only

Correct Answer:
(C), (D) and (E) Only

Solution:
(A) Down the group; radius increases
(B) EN does not decrease gradually from C to Pb.
(C) Correct.
(D) Correct.
(E) Range of oxidation state of carbon ; -4 to +4

Q. 16 Match **List-I** with the **List-II**

	LIST I **Reaction**		**LIST II** **Type of Redox Reaction**
(A)	$N_{2(g)} + O_{2(g)} \rightarrow 2NO_{(g)}$	(I)	Decomposition
(B)	$2\,Pb(NO_3)_{2(s)} \rightarrow 2PbO_{(s)} + 4NO_{2(g)} + O_{2(g)}$	(II)	Displacement
(C)	$2Na_{(s)} + 2H_2O_{(l)} \rightarrow 2NaOH_{(aq)} + H_{2(g)}$	(III)	Disproportionation
(D)	$2NO_{2(g)} + 2OH_{(aq)} \rightarrow NO^-_{2(aq.)} + NO^-_{3(aq.)} + H_2O_{(l)}$	(IV)	Combintion

Choose the **correct** answer from the options given below:

Option 1:
(A)-(I), (B)-(II), (C)-(III), (D)-(IV)

Option 2:
(A)-(III), (B)-(II), (C)-(I), (D)-(IV)

Option 3:
(A)-(II), (B)-(III), (C)-(IV), (D)-(I)

Option 4:
(A)-(IV), (B)-(I), (C)-(II), (D)-(III)

Correct Answer:
(A)-(IV), (B)-(I), (C)-(II), (D)-(III)

Solution:
$A \rightarrow (IV)$
$B \rightarrow (I)$
$C \rightarrow (II)$
$D \rightarrow (III)$

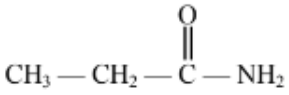
Q. 17 Consider the given reaction, identify the major product P.

$$CH_3 - COOH \xrightarrow[(iv)\ H_2O/\bar{O}H,\Delta]{(i)\ LiAlH_4 \quad (ii)\ PCC \quad (iii)\ HCN/\bar{O}H} \text{"P"}$$

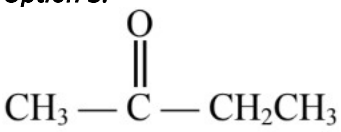
Option 1:

$CH_3 - CH_2 - CH_2 - OH$

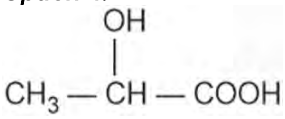
Option 2:



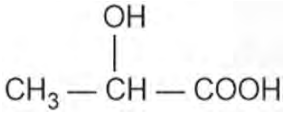
Option 3:



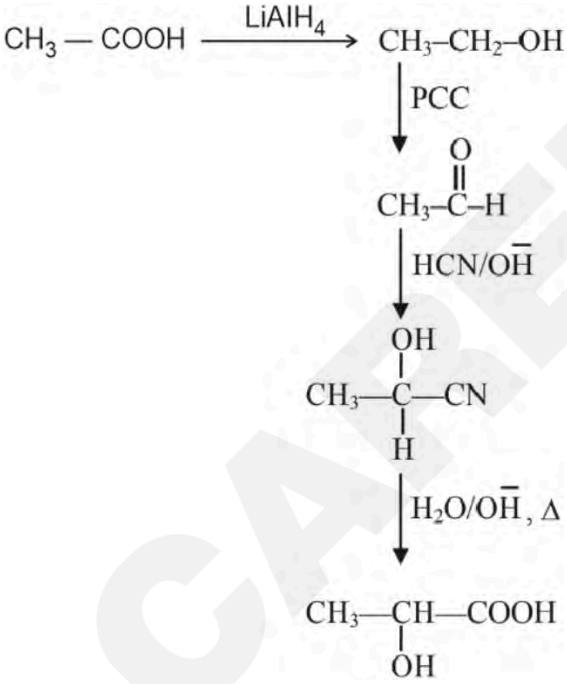
Option 4:



Correct Answer:



Solution:



Q. 18 The correct IUPAC name of $[PtBr_2(PMe_3)_2]$ is:

Option 1:

bis(trimethylphosphine)dibromoplatinum(II)

Option 2:

bis[bromo(trimethylphosphine)]platinum(II)

Option 3:
dibromobis(trimethylphosphine)platinum(II)

Option 4:
dibromodi(trimethylphosphine)platinum(II)

Correct Answer:
dibromobis(trimethylphosphine)platinum(II)

Solution:
Dibromo bis(trimethylphosphine) platinum (II)

Q. 19 Match **List-I** with **List-II**

LIST I **Tetrahedral Complex**		**LIST II** **Electronic configuration**	
(A)	$TiCl_4$	(I)	e^2, t_2^0
(B)	$[FeO_4]^{2-}$	(II)	e^4, t_2^3
(C)	$[FeCl_4]^{-}$	(III)	e^0, t_2^0
(D)	$[CoCl_4]^{2-}$	(IV)	e^2, t_2^3

Choose the **correct** answer from the option given below:

Option 1:
(A)-(I), (B)-(III), (C)-(IV), (D)-(II)

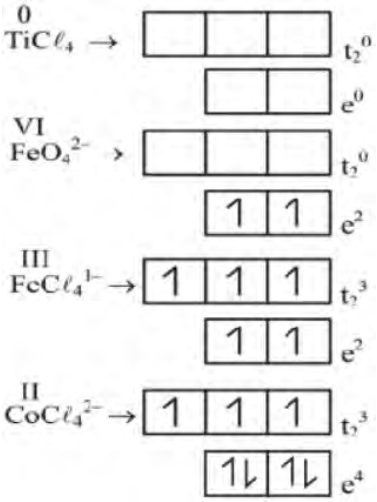
Option 2:
(A)-(IV), (B)-(III), (C)-(I), (D)-(II)

Option 3:
(A)-(III), (B)-(IV), (C)-(II), (D)-(I)

Option 4:
(A)-(III), (B)-(I), (C)-(IV), (D)-(II)

Correct Answer:
(A)-(III), (B)-(I), (C)-(IV), (D)-(II)

Solution:



Q. 20 The ratio $\frac{K_P}{K_C}$ for the reaction:

$CO_{(g)} + \frac{1}{2}O_{2(g)} \rightleftharpoons CO_{2(g)}$ is:

Option 1:
$(RT)^{1/2}$

Option 2:
RT

Option 3:
1

Option 4:
$\frac{1}{\sqrt{RT}}$

Correct Answer:
$\frac{1}{\sqrt{RT}}$

Solution:

$$CO(g) + \frac{1}{2}O_2(g) \rightleftharpoons CO_2(g)$$

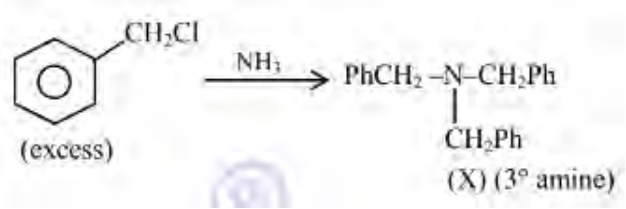
$$\Delta n_g = 1 - \left(1 + \frac{1}{2}\right) = -\frac{1}{2}$$

$$\frac{K_P}{K_C} = (RT)^{\Delta n_g} = \frac{1}{\sqrt{RT}}$$

Q. 21 An amine (X) is prepared by ammonolysis of benzyl chloride. On adding p-toluenesulphonyl chloride to it the solution remains clear. Molar mass of the amine (X) formed is _____$gmol^{-1}$. (Given molar mass in gmol $^{-1}C: 12, H: 1, O: 16,\ N: 14$)

Correct Answer:
287

Solution:



Molar Mass of (X) is $287\ \mathrm{g\ mol^{-1}}$

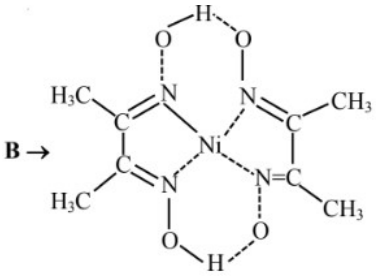
Q. 22 Consider the following reactions

$$\mathrm{NiS + HNO_3 + HCl \rightarrow A + NO + S + H_2O}$$
$$\mathrm{A + NH_4OH + H_3C - C = N - OH \longrightarrow B + NH_4Cl + H_2O}$$

The number of protons that do not involve in hydrogen bonding in the product B is

Correct Answer:
12

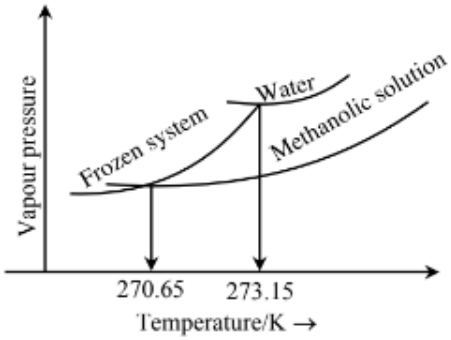
Solution:



$$\mathrm{3NiS + 2HNO_3 + 6HCl \longrightarrow 3NiCl_2 + 2NO + 3\ S + 4H_2O}$$

$$\mathrm{NiCl_2 + 2NH_4OH + \begin{matrix} H_3C - C = N - OH \\ | \\ H_3C - C = N - OH \end{matrix} \rightarrow NH_4Cl + H_2O + (B)}$$

Q. 23 When ' x ' $\times 10^{-2}$ mL methanol (molar mass $= 32$ g; density $= 0.792$ g/cm^3) is added to 100 mL water (density $= 1$ g/cm^3), the following diagram is obtained.



$x =$ ..... (nearest integer)

[Given: Molal freezing point depression constant of water at 273.15 K^2 is 1.86 K kg mol^{-1}]

Correct Answer:

543

Solution:

$\Delta T_f = 273.15 - 270.65 = 2.5$ K

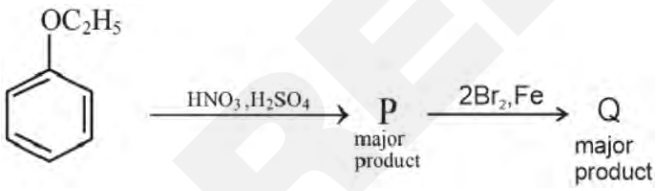
$\Delta T_f = K_f m \Rightarrow 2.5 = 1.86 \times \frac{n}{0.1}$

$\Rightarrow n = 0.1344$ moles

$\Rightarrow w = 0.1344 \times 32 = 4.3$ g

$\text{Volume} = \frac{4.3}{0.792} = 5.43\text{ml} = 543 \times 10^{-2}\text{ml}$

Q. 24

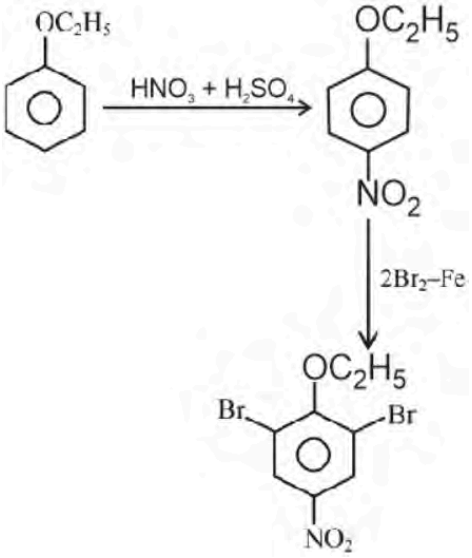


The ratio of number of oxygen atoms to bromine atoms in the product Q is _____ $\times 10^{-1}$.

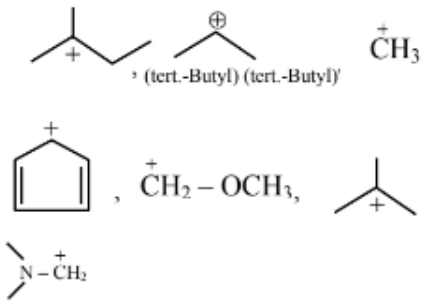
Correct Answer:

15

Solution:

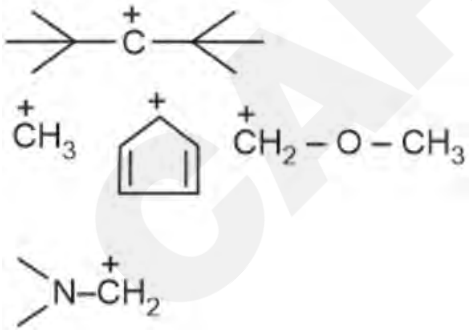


Q. 25 Number of carbocation from the following that are **not** stabilized by hyperconjugation is.....



Correct Answer:
5

Solution:



Q. 26 For the reaction at $298\ \mathrm{K}, 2\ \mathrm{A} + \mathrm{B} \rightarrow \mathrm{C}. \Delta\mathrm{H} = 400\ \mathrm{kJ\ mol^{-1}}$ and $\Delta\mathrm{S} = 0.2\ \mathrm{kJ\ mol^{-1}\ K^{-1}}$. The reaction will become spontaneous above__ K.

Correct Answer:
2000

Solution:

$\Delta G = 0$

$T = \frac{\Delta H}{\Delta S} = \frac{400}{0.2} = 2000\ K$

Q. 27 Total number of species from the following with central atom utilising $2p^2$ hybrid orbitals for bonding is.....

$NH_3, SO_2, SiO_2, BeCl_2, C_2H_2, C_2H_4, BCl_3, HCHO,$
$C_6H_6, BF_3, C_2H_4Cl_2$

Correct Answer:
6

Solution:
Central atom utilising sp^2 hybrid orbitals

$SO_2, C_2H_4, BCl_3, HCHO, C_6H_6, BF_3$

Q. 28 Consider the two different first order reactions given below

$A + B \rightarrow C$(Reaction 1)

$P \rightarrow Q$ (Reaction 2)

The ratio of the half life of Reaction 1 : Reaction 2 is $5:2$. If t_1 and t_2 represent the time taken to complete $2/3^{rd}$ and $4/5$ of Reaction 1 and Reaction 2 , respectively, then the value of the ratio $t_1 : t_2$ is $____ \times 10^{-1}$ (nearest integer).

[Given: $\log_{10}(3) = 0.477$ and $\log_{10}(5) = 0.699$]

Correct Answer:
17

Solution:

$\frac{(t_{1/2})_I}{(t_{1/2})_{II}} = \frac{K_2}{K_1} = \frac{5}{2}$

$\therefore K_1 t_1 = \ln \frac{1}{1-\frac{2}{3}} = \ln 3$

$K_2 t_2 = \ln \frac{1}{1-\frac{4}{5}} = \ln 5$

$\Rightarrow \frac{K_1}{K_2} \times \frac{t_1}{t_2} = \frac{0.477}{0.699}$

$\Rightarrow \frac{t_1}{t_2} = \frac{0.477}{0.699} \times \frac{5}{2} = 1.7 = 17 \times 10^{-1}$

Q. 29 For hydrogen atom, energy of an electron in first excited state is $-3.4eV$, K. E. of the same electron of hydrogen atom is xeV. Value of x is_____ $\times 10^{-1} eV$. (Nearest integer)

Correct Answer:
34

Q. 30 Among VO_2^+, MnO_4^- and $Cr_2O_7^{2-}$, the spin-only magnetic moment value of the species with least oxidising ability is.....BM (Nearest integer).
(Given atomic member $V = 23, Mn = 25, Cr = 24$)

Correct Answer:
0

Solution:
For 3d transition series;

Oxidising power: $V^{+5} < Cr^{+6} < Mn^{+7}$

$V^{+5} : [Ar]4\,s^0 3\,d^0$

Number of unpaired electron =0

$\mu = 0$

Maths

Q. 1 Let ABC be an equilateral triangle. A new triangle is formed by joining the middle points of all sides of the triangle ABC and the same process is repeated infinitely many times. If P is the sum of perimeters and Q is be the sum of areas of all the triangles formed in this process, then:

Option 1:
$P^2 = 36\sqrt{3}Q$

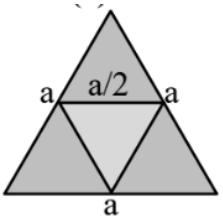
Option 2:
$P^2 = 6\sqrt{3}Q$

Option 3:
$P = 36\sqrt{3}Q^2$

Option 4:
$P^2 = 72\sqrt{3}Q$

Correct Answer:
$P^2 = 36\sqrt{3}Q$

Solution:



Area of first $\Delta = \frac{\sqrt{3}a^2}{4}$

Area of second $\Delta = \frac{\sqrt{3}a^2}{4}\frac{a^2}{4} = \frac{\sqrt{3}a^2}{16}$

Area of third $\Delta = \frac{\sqrt{3}a^2}{64}$

sum of area $= \frac{\sqrt{3}a^2}{4}\left(1+\frac{1}{4}+\frac{1}{16}\ldots\right)$

$Q = \frac{\sqrt{3}a^2}{4}\frac{1}{\frac{3}{4}} = \frac{a^2}{\sqrt{3}}$

perimeter of $1^{st}\ \Delta = 3a$ perimeter of $2^{nd}\ \Delta = \frac{3a}{2}$ perimeter of $3^{rd}\ \Delta = \frac{3a}{4}$

$$P = 3a\left(1+\frac{1}{2}+\frac{1}{4}+\ldots\right)$$

$$P = 3a \cdot 2 = 6a$$

$$a = \frac{P}{6}$$

$$Q = \frac{1}{\sqrt{3}} \cdot \frac{P^2}{36}$$

$$P^2 = 36\sqrt{3}Q$$

Q. 2 Let $A = \{1,2,3,4,5\}$. Let R be a relation on A defined by xRy if and only if $4x \leq 5y$. Let m be the number of elements in R and n be the minimum number of elements from $A \times A$ that are required to be added to R to make it a symmetric relation. Then $m+n$ is equal to:

Option 1:
24

Option 2:
23

Option 3:
25

Option 4:
26

Correct Answer:
25

Solution:
Given : $4x \leq 5y$

then

$$R = \{(1,1),(1,2),(1,3),(1,4),(1,5),(2,2),(2,3),(2,4)$$

$$(2,5),(3,3),(3,4),(3,5),(4,4),(4,5),(5,4),(5,5)\}$$

i.e. 16 elements.

i.e. $m = 16$

Now to make R a symmetric relation add

$$\{(2,1)(3,2)(4,3)(3,1)(4,2)(5,3)(4,1)(5,2)(5,1)\}$$

i.e. $n = 9$

So $m+n=25$

Q. 3 If three letters can be posted to any one of the 5 different addresses, then the probability that the three letters are posted to exactly two addresses is:

Option 1:
$\frac{12}{25}$

Option 2:
$\frac{18}{25}$

Option 3:
$\frac{4}{25}$

Option 4:
$\frac{6}{25}$

Correct Answer:
$\frac{12}{25}$

Solution:

$$\text{Total method } = 5^3$$
$$\text{faverable } = {}^5C_2\left(2^3-2\right)=60$$
$$\text{probability } = \frac{60}{125}=\frac{12}{25}$$

Q. 4 Suppose the solution of the differential equation $\frac{dy}{dx}=\frac{(2+\alpha)x-\beta y+2}{\beta x-2\alpha y-(\beta\gamma-4\alpha)}$ represents a circle passing through origin. Then the radius of this circle is :

Option 1:
$\sqrt{17}$

Option 2:
$\frac{1}{2}$

Option 3:
$\frac{\sqrt{17}}{2}$

Option 4:
2

Correct Answer:
$\frac{\sqrt{17}}{2}$

Solution:

$$\frac{dy}{dx} = \frac{(2+\alpha)x - \beta y + 2}{\beta x - y(2\alpha+\beta) + 4\alpha}$$

$$\beta x dy - (2\alpha+\beta)y dy + 4\alpha dy = (2+\alpha)x dx - \beta y dx + 2dx$$

$$\beta(x dy + y dx) - (2\alpha+\beta)y dy + 4\alpha dy = (2+\alpha)x dx + 2dx$$

$$\beta xy - \frac{(2\alpha+\beta)y^2}{2} + 4\alpha y = \frac{(2+\alpha)x^2}{2}$$

$\Rightarrow \beta = 0$ for this to be circle

$$(2+\alpha)\frac{x^2}{2} + \alpha y^2 + 2x - 4\alpha y = 0$$

$$\text{coeff. of } x^2 = y^2 > 2 + a = 2a$$

$\Rightarrow \alpha = 2$

i.e. $2x^2 + 2y^2 + 2x - 8y = 0$

$x^2 + y^2 + x - 4y = 0$

$$rd = \sqrt{\frac{1}{4} + 4} = \frac{\sqrt{17}}{2}$$

Q. 5 If the locus of the point, whose distances from the point $(2, 1)$ and $(1, 3)$ are in the ratio $5 : 4$, is $ax^2 + by^2 + cxy + dx + ey + 170 = 0$, then the value of $a^2 + 2b + 3c + 4d + e$ is equal to:

Option 1:
5

Option 2:
-27

Option 3:
37

Option 4:
437

Correct Answer:
37

Solution:

let $P(x, y)$

$$\frac{(x-2)^2 + (y-1)^2}{(x-1)^2 + (y-3)^2} = \frac{25}{16}$$

$$9x^2 + 9y^2 + 14x - 118y + 170 = 0$$

$a^2 + 2b + 3c + 4d + e$

$= 81 + 18 + 0 + 56 - 118$

$= 155 - 118$

$= 37$

Q. 6 $\lim_{n\to\infty} \frac{(1^2-1)(n-1)+(2^2-2)(n-2)+\ldots+((n-1)^2-(n-1))\cdot 1}{(1^3+2^3+\ldots+n^3)-(1^2+2^2+\ldots+n^2)}$ is equal to:

Option 1:
$\frac{2}{3}$

Option 2:
$\frac{1}{3}$

Option 3:
$\frac{3}{4}$

Option 4:
$\frac{1}{2}$

Correct Answer:
$\frac{1}{3}$

Solution:

$$\lim_{n\to\infty} \frac{\sum_{r=1}^{n-1}\left(r^2-r\right)(n-r)}{\sum_{r=1}^{n} r^3-\sum_{r=1}^{n} r^2}$$

$$\lim_{n\to\infty} \frac{\sum_{r=1}^{n-1}\left(-r^3+r^2(n+1)-nr\right)}{\left(\frac{n(n+1)}{2}\right)^2-\frac{n(n+1)(2n+1)}{6}}$$

$$\lim_{n\to\infty} \frac{\left(\frac{((n-1)n)}{2}\right)^2+\frac{(n+1)(n-1)n(2n-1)}{6}-\frac{n^2(n-1)}{2}}{\frac{n(n+1)}{2}\left(\frac{n(n+1)}{2}-\frac{2n+1}{3}\right)}$$

$$\lim_{n\to\infty} \frac{\frac{n(n-1)}{2}\left(\frac{-n(n-1)}{2}+\frac{(n+1)(2n-1)}{3}-n\right)}{\frac{n(n+1)}{2}\frac{3n^2+3n-4n-2}{6}}$$

$$\lim_{n\to\infty} \frac{(n-1)\left(-3n^2+3n+2\left(2n^2+n-1\right)-6\right)}{(n+1)\left(3n^2-n-2\right)}$$

$$\lim_{n\to\infty} \frac{(n-1)\left(n^2+5n-8\right)}{(n+1)\left(3n^2-n-2\right)}=\frac{1}{3}$$

Q. 7 Let $0 \le r \le n$. If ${}^{n+1}C_{r+1} : {}^{n}C_{r} : {}^{n-1}C_{r-1} = 55 : 35 : 21$, then $2n + 5r$ is equal to:

Option 1:
60

Option 2:
62

Option 3:
50

Option 4:
55

Correct Answer:
50

Solution:

$$\frac{{}^{n+1}C_r}{{}^{n}C_r} = \frac{55}{35}$$

$$\frac{(n+1)!}{(r+1)!(n-r)!}\,!\frac{r!(n-r)!}{n!} = \frac{11}{7}$$

$$\frac{(n+1)}{r+1} = \frac{11}{7}$$

$$7\mathrm{n} = 4 + 11\mathrm{r}$$

$$\frac{{}^{n}C_r}{{}^{\mathrm{n-1}}C_r - 1} = \frac{35}{21}$$

$$\frac{\mathrm{n!}}{\mathrm{r!(n-r)!}} = \frac{(\mathrm{r}-1)!(\mathrm{n}-\mathrm{r})!}{(\mathrm{n}-1)!} = \frac{5}{3}$$

$$\frac{\mathrm{n}}{\mathrm{r}} = \frac{5}{3}$$

$$3\mathrm{n} = 5\mathrm{r}$$

$$\text{By solving } \mathrm{r} = 6 \quad \mathrm{n} = 10$$

$$2\mathrm{n} + 5\mathrm{r} = 50$$

Q. 8 A software company sets up m number of computer systems to finish an assignment in 17 days. If 4 computer systems crashed on the start of the second day, 4 more computer systems crashed on the start of the third day and so on, then it took 8 more days to finish the assignment. The value of m is equal to :

Option 1:
125

Option 2:
150

Option 3:
180

Option 4:
160

Correct Answer:
150

Solution:
$17\ \mathrm{m} = \mathrm{m} + (\mathrm{m} - 4) + (\mathrm{m} - 4 \times 2) \ldots + \ldots (\mathrm{m} - 4 \times 24)$

$17\ \mathrm{m} = 25\ \mathrm{m} - 4(1 + 2 \ldots 24)$

$8\ \mathrm{m} = \dfrac{4 \cdot 24 \cdot 25}{2} = 150$

Q. 9 If z_1, z_2 are two distinct complex number such that $\frac{z_1 - 2z_2}{\frac{1}{2} - z_1\bar{z}_2} = 2$, then

Option 1:
either z_1 lies on a circle of radius 1 or z_2 lies on a circle of radius $\frac{1}{2}$

Option 2:
either z_1 lies on a circle of radius $\frac{1}{2}$ or z_2 lies on a circle of radius 1 .

Option 3:
z_1 lies on a circle of radius $\frac{1}{2}$ and z_2 lies on a circle of radius 1 .

Option 4:
both z_1 and z_2 lie on the same circle.

Correct Answer:
either z_1 lies on a circle of radius 1 or z_2 lies on a circle of radius $\frac{1}{2}$

Solution:
$$\frac{z_1 - 2z_2}{\frac{1}{2} - z_1\bar{z}_2} \times \frac{\bar{z}_1 - 2\bar{z}_2}{\frac{1}{2} - \bar{z}_1 z_2} = 4$$

$$|z_1|^2 2z_1\bar{z}_2 - 2\bar{z}_1 z_2 + 4|z_2|^2$$

$$= 4\left(\frac{1}{4} - \frac{\bar{z}_1 z_2}{2} - \frac{z_1\bar{z}_2}{2} + |z_1|^2|z_2|^2\right)$$

$$z_1\bar{z}_1 + 2z_2 \cdot 2\bar{z}_2 - z_1\bar{z}_1 2z_2 2\bar{z}_2 - 1 = 0$$

$$(\mathrm{z}, \bar{\mathrm{z}}_1 - 1)\,(1 - 2\mathrm{z}_2 \cdot 2\bar{\mathrm{z}}_2) = 0$$

$$\left(|\mathrm{z}_1|^2 - 1\right)\left(|2\mathrm{z}_2|^2 - 1\right) = 0$$

Q. 10 If the function $f(x)=\left(\frac{1}{x}\right)^{2x}; x>0$ attains the maximum value at $\mathrm{x}=\frac{1}{\mathrm{e}}$ then :

Option 1:
$\mathrm{e}^{\pi}<\pi^{\mathrm{e}}$

Option 2:
$\mathrm{e}^{2\pi}<(2\pi)^{\mathrm{e}}$

Option 3:
$\mathrm{e}^{\pi}>\pi^{\mathrm{e}}$

Option 4:
$(2\mathrm{e})^{\pi}>\pi^{(2e)}$

Correct Answer:
$\mathrm{e}^{\pi}>\pi^{\mathrm{e}}$

Solution:
Let $y=\left(\frac{1}{x}\right)^{2x}$

$$\ell ny=2x\ell n\left(\frac{1}{x}\right)$$

$$\ell ny=-2x\ell nx$$

$$\frac{1}{y}\frac{dy}{dx}=-2(1+\ell nx)$$

for $\mathrm{x}>\frac{1}{\mathrm{e}}\mathrm{f}^{\mathrm{n}}$ is decreasing

$\mathrm{so,\ e}<\pi$

$$\left(\frac{1}{\mathrm{e}}\right)^{2\mathrm{e}}>\left(\frac{1}{\pi}\right)^{2\pi}$$

$\mathrm{e}^{\pi}>\pi^{\mathrm{e}}$

Q. 11 Let $\vec{\mathrm{a}}=6\hat{\mathrm{i}}+\hat{\mathrm{j}}-\hat{\mathrm{k}}$ and $\vec{\mathrm{b}}=\hat{\mathrm{i}}+\hat{\mathrm{j}}$. If $\vec{\mathrm{c}}$ is a is vector such that $|\vec{\mathrm{c}}|\geq 6, \vec{\mathrm{a}}\cdot\vec{\mathrm{c}}=6|\vec{\mathrm{c}}|, |\vec{\mathrm{c}}-\vec{\mathrm{a}}|=2\sqrt{2}$ and the angle between $\vec{\mathrm{a}}\times\vec{\mathrm{b}}$ and $\vec{\mathrm{c}}$ is 60°, then $|(\vec{\mathrm{a}}\times\vec{\mathrm{b}})\times\vec{\mathrm{c}}|$ is equal to:

Option 1:
$\frac{9}{2}(6-\sqrt{6})$

Option 2:
$\frac{3}{2}\sqrt{3}$

Option 3:
$\frac{3}{2}\sqrt{6}$

Option 4:
$\frac{9}{2}(6+\sqrt{6})$

Correct Answer:
$\frac{9}{2}(6+\sqrt{6})$

Solution:

$$|(\vec{a}\times\vec{b}\times\vec{c})| = |\vec{a}\times\vec{b}||\vec{c}|\frac{\sqrt{3}}{2}$$

$$|\vec{c}-\vec{a}| = 2\sqrt{2}$$

$$|c|^2+|a|^2-2\vec{c}\cdot\vec{a} = 8$$

$$|z|^2+38-12|z| = 8$$

$$|z|^2-12|z|+30 = 0$$

$$|z| = \frac{12\pm\sqrt{144-120}}{2}$$
$$= \frac{12\pm 2\sqrt{6}}{2}$$

$$|z| = 6+\sqrt{6}$$

$$\vec{a}\times\vec{b} = \begin{matrix}\hat{\ell} & \hat{j} & \hat{k}\\ 6 & 1 & -1\\ 1 & 1 & 0\end{matrix}$$

$$\hat{\ell}-\hat{j}+5\hat{k}$$

$$|\vec{a}\times\vec{b}| = \sqrt{27}$$

$$|(\vec{a}\times b)\times z| = \sqrt{27}(6+\sqrt{6})\frac{\sqrt{3}}{2}$$

$$\frac{9}{2}(6+\sqrt{6})$$

Q. 12 If all the words with or without meaning made using all the letters of the word "NAGPUR" are arranged as in a dictionary, then the word at 315^{th} position in this arrangement is :

Option 1:
NRAGUP

Option 2:
NRAGPU

Option 3:
NRAPGU

Option 4:
NRAPUG

Correct Answer:
NRAPGU

Solution:
NAGPUR

$\mathrm{A} \rightarrow 5! = 120$

$\mathrm{G}\textcircled{R}5! = 120 \quad 240$

$\mathrm{NA}\textcircled{R}4! = 24 \quad 264$

NG $\textcircled{R}4! = 24 \quad 288$

$\mathrm{NP}(\textcircled{R})4! = 24 \quad 312$

NRAGPU $= 1 \quad 313$

NRAGUP $\quad 314$

NRAPGU $\quad 315$

Q. 13 Suppose for a differentiable function h, $h(0) = 0$, $\mathrm{h}(1) = 1$ and $\mathrm{h}'(0) = \mathrm{h}'(1) = 2$. If $\mathrm{g}(\mathrm{x}) = \mathrm{h}\left(\mathrm{e}^{\mathrm{x}}\right)\mathrm{e}^{\mathrm{h}(\mathrm{x})}$, then $g'(0)$ is equal to:

Option 1:
5

Option 2:
3

Option 3:
8

Option 4:
4

Correct Answer:
4

Solution:

$g(x) = h\left(e^x\right) \cdot e^{h(x)}$
$g'(x) = h\left(e^x\right) \cdot e^{h(x)} \cdot h'(x) + e^{h(x)} h'\left(e^x\right) \cdot e^x$
$g'(0) = h(1)e^{h(0)}h'(0) + e^{h(0)}h'(1)$
$= 2 + 2 = 4$

Q. 14 Let $\mathrm{P}(\alpha, \beta, \gamma)$ be the image of the point $\mathrm{Q}(3, -3, 1)$ in the line $\frac{x-0}{1} = \frac{y-3}{1} = \frac{z-1}{-1}$ and R be the point $(2, 5, -1)$. If the area of the triangle PQR is λ and $\lambda^2 = 14\,\mathrm{K}$, then K is equal to:

Option 1:
36

Option 2:
72

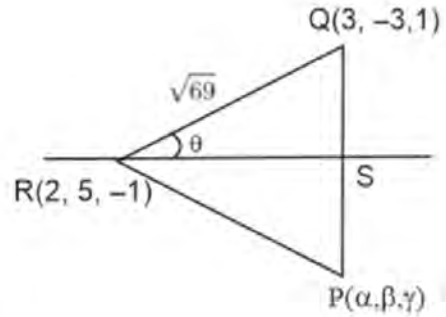
Option 3:
18

Option 4:
81

Correct Answer:
81

Solution:

CAREERS360



$RQ = \sqrt{1+64+4} = \sqrt{69}$

$\overrightarrow{RQ} = \hat{\ell} - 8\hat{j} + 2\hat{k}$

$\overrightarrow{RS} = \hat{\ell} + \hat{j} - \hat{k}$

$$\cos\theta = \frac{\overrightarrow{RQ}\cdot\overrightarrow{RS}}{|\overrightarrow{RQ}||\overrightarrow{RS}|} = \frac{1-8-2}{\sqrt{69}\sqrt{3}} = \frac{9}{3\sqrt{23}}$$

$$\cos\theta = \frac{3}{\sqrt{23}} = \frac{RS}{RQ} = \frac{RS}{\sqrt{69}}$$

$RS = 3\sqrt{3}$

$$\sin\theta = \frac{\sqrt{14}}{\sqrt{23}} = \frac{QS}{\sqrt{69}}$$

$QS = \sqrt{42}$

$$\text{area} = \frac{1}{2}\cdot 2QS\cdot RS = \sqrt{42}\cdot 3\sqrt{3}$$

$\lambda = 9\sqrt{14}$

$\lambda^2 = 81.14 = 14k$

$k = 81$

Q. 15 If $P(6,1)$ be the orthocentre of the triangle whose vertices are $A(5,-2), B(8,3)$ and $C(h,k)$, then the point C lies on the circle.

Option 1:
$x^2 + y^2 - 65 = 0$

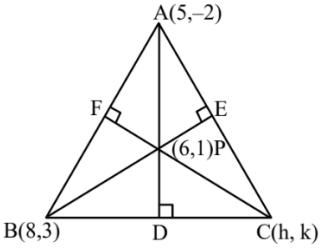
Option 2:
$x^2 + y^2 - 74 = 0$

Option 3:
$x^2+y^2-61=0$

Option 4:
$x^2+y^2-52=0$

Correct Answer:
$x^2+y^2-65=0$

Solution:



Slope of $AD=3$

Slope of $BC=-\frac{1}{3}$

equation of $BC=3y+x-17=0$

slope of $BE=1$

Slope of $AC=-1$

equation of AC is $x+y-3=0$

point C is $(-4,7)$

Q. 16 Let $f(x)=\frac{1}{7-\sin 5x}$ be a function defined on R. Then the range of the function $f(x)$ is equal to:

Option 1:
$\left[\frac{1}{8},\frac{1}{5}\right]$

Option 2:
$\left[\frac{1}{7},\frac{1}{6}\right]$

Option 3:
$\left[\frac{1}{7},\frac{1}{5}\right]$

Option 4:
$\left[\frac{1}{8},\frac{1}{6}\right]$

Correct Answer:
$\left[\frac{1}{8},\frac{1}{6}\right]$

Solution:

$\sin 5x \in [-1, 1]$

$-\sin 5x \in [-1, 1]$

$7 - \sin 5x \in [6, 8]$

$$\frac{1}{7-\sin 5x} \in \left[\frac{1}{8}, \frac{1}{6}\right]$$

Q. 17 Let $\vec{a} = 2\hat{i} + \hat{j} - \hat{k}, \vec{b} = ((\vec{a} \times (\hat{i} + \hat{j})) \times \hat{i}) \times \hat{i}$.
Then the square of the projection of $\vec{a}$ on $\vec{b}$ is :

Option 1:
$\frac{1}{5}$

Option 2:
2

Option 3:
$\frac{1}{3}$

Option 4:
$\frac{2}{3}$

Correct Answer:
2

Solution:

$$\vec{a} \times (\hat{i} + \hat{j}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 1 & 1 & 0 \end{vmatrix}$$

$= \hat{i} - \hat{j} + \hat{k}$

$(\vec{a} \times (\hat{i} \times \hat{j})) \times \hat{i} = \hat{k} + \hat{j}$

$((\vec{a} \times (\hat{i} \times \hat{j})) \times i) \times \hat{i} = \hat{j} - \hat{k}$

projection of $\vec{a}$ on $\hat{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$

$= \frac{1+1}{\sqrt{2}} = \sqrt{2}$

Q. 18 If the area of the region $\left\{(x, y) : \frac{a}{x^2} \leq y \leq \frac{1}{x}, 1 \leq x \leq 2, 0 < a < 1\right\}$ i is $(\log_e 2) - \frac{1}{7}$ then the value of $7a - 3$ is equal to:

Option 1:
2

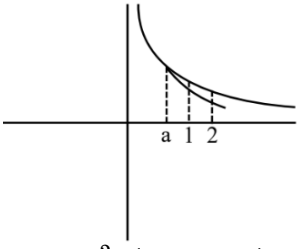
Option 2:
0

Option 3:
-1

Option 4:
1

Correct Answer:
-1

Solution:



$$\text{area} \int_1^2 \left(\frac{1}{x} - \frac{a}{x^2}\right) dx$$

$$\left[\ell \ln x + \frac{a}{x}\right]_1^2$$

$$\ell \operatorname{nn} 2 + \frac{a}{2} - a = \log_e 2 - \frac{1}{7}$$

$$\frac{-a}{2} = -\frac{1}{7}$$

$$a = \frac{2}{7}$$

$$7a = 2$$

$$7a - 3 = -1$$

Q. 19 If $\int \frac{1}{a^2 \sin^2 x + b^2 \cos^2 x} dx = \frac{1}{12} \tan^{-1}(3 \tan x) +$ constant, then the maximum value of $a \sin x + b \cos x$, is :

Option 1:
$\sqrt{40}$

Option 2:
$\sqrt{39}$

Option 3:
$\sqrt{42}$

Option 4:
$\sqrt{41}$

Correct Answer:
$\sqrt{40}$

Solution:
$$\int \frac{\sec^2 x dx}{a^2 \tan^2 x + b^2}$$

$\text{let } \tan x = t$

$\sec^2 dx = dt$

$$\int \frac{dt}{a^2 t^2 + b^2}$$

$$\frac{1}{a^2} \int \frac{dt}{t^2 + \left(\frac{b}{a}\right)^2}$$

$$\frac{1}{a^2} \frac{1}{b} \tan^{-1} \left(\frac{t}{b} a\right) + c$$

$$\frac{1}{ab} \tan^{-1} \left(\frac{\alpha}{b} \tan x\right) + c$$

$\text{on comparing } \frac{a}{b} = 3$

$ab = 12$

$a = 6, b = 2$

maximum value of

$6 \sin + 2 \cos x \text{ is } \sqrt{40}$

maximum value of

Q. 20 If A is a square matrix of order 3 such that $\det(\mathrm{A}) = 3$ and $\det\left(\text{adj}\left(-4 \text{ adj}\left(-3 \text{ adj}\left(3 \text{ adj}\left((2 \mathrm{A})^{-1}\right)\right)\right)\right)\right) = 2^{\mathrm{m}} 3^{\mathrm{n}}$, then $m+ \mid 2n$ is equal to:

Option 1:
3

Option 2:
2

Option 3:
4

Option 4:
6

Correct Answer:
4

Solution:
$|\mathrm{A}| = 3$

$$\operatorname{adj}\left(-4\operatorname{adj}\left(-3\operatorname{adj}\left(3\operatorname{adj}\left((2\ \mathrm{A})^{-1}\right)\right)\right)\right)$$

$$|-4\operatorname{adj}\left(-3\operatorname{adj}\left(3\operatorname{adj}(2\ \mathrm{A})^{-1}\right)\right.^{2}$$

$$4^{6}\ \operatorname{adj}\left(-3\operatorname{adj}\left(3\operatorname{adj}(2\ \mathrm{A})^{-1}\right)\right)^{2}$$

$$2^{12}\cdot 3^{12}\ 3\operatorname{adj}(2\ \mathrm{A})^{-1\ 8}$$

$$2^{12}\cdot 3^{12}\cdot 3^{24}\ \operatorname{adj}(2\ \mathrm{A})^{-1\ 8}$$

$$2^{12}\cdot 3^{36}\ (2\ \mathrm{A})^{-1\ 16}$$

$$2^{12}\cdot 3^{36}\frac{1}{|2\ \mathrm{A}|^{16}}$$

$$2^{12}\cdot 3^{36}\frac{1}{2^{48}|\ \mathrm{A}|^{16}}$$

$$2^{12}\cdot 3^{36}\frac{1}{2^{48}\cdot 3^{16}}$$

$$\frac{3^{20}}{2^{36}} = 2^{-36}\cdot 3^{20}$$

$$\mathrm{m} = -36 \quad \mathrm{n} = 20$$

$$\mathrm{m} + 2\mathrm{n} = 4$$

Q. 21 Let $[t]$ denote the greatest integer less than or equal to t. Let $\mathrm{f}:[0,\infty)\to\mathrm{R}$ be a function defined by $f(x)=\left[\frac{x}{2}+3\right]-[\sqrt{x}]$. Let S be the set of all points in the interval $[0,8]$ at which f is not continuous. Then $\sum_{\mathrm{a}\in\mathrm{S}}\mathrm{a}$ is equal to_____.

Correct Answer:
17

Solution:
$\left[\frac{x}{2}+3\right]$ is discontinuous at $x=2,4,6,8$

$\sqrt{\mathrm{x}}$ is discontinuous at $\mathrm{x}=1,4$

$\mathrm{F}(\mathrm{x})$ is discontinuous at $\mathrm{x}=1,2,6,8$

$\sum a=1+2+6+8=17$

Q. 22 The length of the latus rectum and directrices of a hyperbola with eccentricity e are 9 and $x=\pm\frac{4}{\sqrt{3}}$, respectively. Let the line $y-\sqrt{3}x+\sqrt{3}=0$ touch this hyperbola at $(\mathrm{x}_0,\mathrm{y}_0)$. If m is the product of the focal distances of the point $(\mathrm{x}_0,\mathrm{y}_0)$, then $4\mathrm{e}^2+\mathrm{m}$ is equal to_____.

Correct Answer:
61

Solution:
Given $\frac{2\,\mathrm{b}^2}{\mathrm{a}}=9$ and $\frac{\mathrm{a}}{\mathrm{e}}=\pm\frac{4}{\sqrt{3}}$
equation of tangent $y-\sqrt{3}x+\sqrt{3}=0$
by equation of tangent
Let slope $=\mathrm{S}=\sqrt{3}$
Constant $=-\sqrt{3}$
By condition of tangency
$\Rightarrow 6=6\mathrm{a}^2-9\mathrm{a}$
$\Rightarrow \mathrm{a}=2,\ \mathrm{b}^2=9$

Equation of Hyperbola is
$\frac{x^2}{4}-\frac{y^2}{9}=1$ and for tangent
Point of contact is $(4,3\sqrt{3})=(\mathrm{x}_0,\mathrm{y}_0)$
Now e $=\sqrt{1+\frac{9}{4}}=\frac{\sqrt{13}}{2}$
Again product of focal distances
$\mathrm{m}=(\mathrm{x}_0\mathrm{e}+\mathrm{a})\,(\mathrm{x}_0\mathrm{e}-\mathrm{a})$
$\mathrm{m}+4\mathrm{e}^2=20\mathrm{e}^2-\mathrm{a}^2$
$=20\times\dfrac{13}{4}-4=61$
(There is a printing mistake in the equation of directrix $x=\pm\frac{4}{\sqrt{3}}$.

Corrected equation is $x=\pm\frac{4}{\sqrt{13}}$ for directrix, as eccentricity must be greater than one, so question must be bonus)

Q. 23 If $S(x)=(1+x)+2(1+x)^2+3(1+x)^3+\ldots\ldots\ldots+60(1+x)^{60},x\neq 0,$ and $(60)^2S(60)=a(b$ where $\mathrm{a},\mathrm{b}\in\mathrm{N}$, then $(\mathrm{a}+\mathrm{b})$ equal to_____.

Correct Answer:
3660

Solution:
$S(x) = (1+x) + 2(1+x)^2 + 3(1+x)^3 + .. + 60(1+x)^{60}$

$(1+x)S = (1+x)^2 + ... \quad 59(1+x)^{60} + 60(1+x)^{61}$

$-xS = \dfrac{(1+x)(1+x)^{60}-1}{x} - 60(1+x)^{61}$

Put $\mathrm{x} = 60$

$-60\ \mathrm{S} = \frac{61((61)^{60}-1)}{60} - 60(61)^{61}$

on solving 3660

Q. 24 Let $[t]$ denote the largest integer less than or equal to t. If $\int_0^3 \left([x^2] + \left[\frac{x^2}{2}\right]\right)dx = a + b\sqrt{2} - \sqrt{3} - \sqrt{5} + c\sqrt{6} - \sqrt{7}$, where $\mathrm{a}, \mathrm{b}, \mathrm{c} \in \mathrm{z}$, then $\mathrm{a}+\mathrm{b}+\mathrm{c}$ is equal to_____.

Correct Answer:
23

Solution:
$$\int_0^3 [x^2]dx + \int_0^3 \left[\frac{x^2}{2}\right]dx$$
$$= \int_0^1 0dx + \int_1^{12} 1dx + \int_{\sqrt{2}}^{\sqrt{3}} 2dx$$
$$+ \int_{\sqrt{3}}^2 3\mathrm{dx} + \int_2^{\sqrt{5}} 4\mathrm{dx} + \int_{\sqrt{5}}^{\sqrt{6}} 5\mathrm{dx}$$
$$+ \int_{\sqrt{6}}^{\sqrt{7}} 6\mathrm{dx} + \int_{\sqrt{7}}^{\sqrt{8}} 7\mathrm{dx} + \int_{\sqrt{8}}^3 8\mathrm{dx}$$
$$+ \int_0^{\sqrt{2}} 0\mathrm{dx} + \int_{\sqrt{2}}^2 1\mathrm{dx}$$
$$+ \int_2^{\sqrt{6}} 2\mathrm{dx} + \int_{\sqrt{6}}^{\sqrt{8}} 3\mathrm{dx} + \int_{\sqrt{8}}^3 4\mathrm{dx} = 31 - 6\sqrt{2} - \sqrt{3} - \sqrt{5}$$
$$- 2\sqrt{6} - \sqrt{7}$$
$\mathrm{a} = 31 \quad \mathrm{b} = -6 \quad \mathrm{c} = -2$
$\mathrm{a}+\mathrm{b}+\mathrm{c} = 31 - 6 - 2 = 23$

Q. 25 From a lot of 12 items containing 3 defectives, a sample of 5 items is drawn at random. Let the random variable X denote the number of defective items in the sample. Let items in the sample be drawn one by one without replacement. If variance of X is $\frac{m}{n}$, where $\gcd(m,n) = 1$, then $n - m$ is equal to_____.

Correct Answer:
71

Solution:

$$a = 1 - \frac{^3C_5}{^{12}C_5}$$

$$b = 3 \cdot \frac{^9C_4}{^{12}C_5}$$

$$c = 3 \cdot \frac{^9C_3}{^{12}C_5}$$

$$d = 1 \cdot \frac{^9C_2}{^{12}C_5}$$

$$u = 0 \cdot a + 1 \cdot b + 2 \cdot c + 3 \cdot d = 1.25$$

$$\sigma^2 = 0 \cdot a + 1.b + 4 \cdot c + 9\, d - u^2$$

$$\sigma^2 = \frac{105}{176}$$

Ans. $176 - 105 = 71$

Q. 26 In a triangle $ABC, BC = 7, AC = 8, AB = \alpha \in N$ and $\cos A = \frac{2}{3}$. If $49\cos(3C) + 42 = \frac{m}{n}$, where $\gcd(m, n) = 1$, then $m + n$ is equal to_____.

Correct Answer:
39

Solution:

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\frac{2}{3} = \frac{8^2 + c^2 - 7^2}{2 \times 8 \times c}$$

$$C = 9$$

$$\cos C = \frac{7^2 + 8^2 - 9^2}{2 \times 7 \times 8} = \frac{2}{7}$$

$$49\cos 3C + 42$$

$$49\left(4\cos^3 C - 3\cos C\right) + 42$$

$$49\left(4\left(\frac{2}{7}\right)^3 - 3\left(\frac{2}{7}\right)\right) + 42$$

$$= \frac{32}{7}$$

$$m + n = 32 + 7 = 39$$

Q. 27 If the shortest distance between the lines $\frac{x-\lambda}{3} = \frac{y-2}{-1} = \frac{z-1}{1}$ and $\frac{x+2}{-3} = \frac{y+5}{2} = \frac{z-4}{4}$ is $\frac{44}{\sqrt{30}}$, then the largest possible value of $|\lambda|$ is equal to_____.

Correct Answer:
43

Solution:

$\bar{a}_1 = \lambda\hat{i} + 2\hat{j} + \hat{k}$

$\bar{a}_2 = -2\hat{i} - 5\hat{j} + 4\hat{k}$

$\vec{p} = 3\hat{i} - \hat{j} + \hat{k}$

$\vec{q} = -3\hat{i} + 2\hat{j} + 4\hat{k}$

$(\lambda + 2)\hat{i} + 7\hat{j} - 3\hat{k} = \bar{a}_1 - \bar{a}_2$

$\vec{p} \times \vec{q} = -6\hat{i} - 15\hat{j} + 3\hat{k}$

$$\frac{44}{\sqrt{30}} = \frac{|-6\lambda - 12 - 105 - 9|}{\sqrt{(-6)^2 + (-15)^2 + 3^2}}$$

$$\frac{44}{\sqrt{30}} = \frac{|6\lambda + 126|}{3\sqrt{30}}$$

$132 = |6\lambda + 126|$

$\lambda = 1, \lambda = -43$

$|\lambda| = 43$

Q. 28 Let α, β be roots of $x^2 + \sqrt{2}x - 8 = 0$. If $U_n = \alpha^n + \beta^n$, then $\frac{U_{10} + \sqrt{12}U_9}{2U_8}$ is equal to_____.

Correct Answer:
4

Solution:

$$\frac{\alpha^{10} + \beta^{10} + \sqrt{2}\left(\alpha^9 + \beta^9\right)}{2\left(\alpha^8 + \beta^8\right)}$$

$$\frac{\alpha^8\left(\alpha^2 + \sqrt{2}\alpha\right) + \beta^8\left(\beta^2 + \sqrt{2}\beta\right)}{2\left(\alpha^8 + \beta^8\right)}$$

$$\frac{8\alpha^8 + 8\beta^8}{2\left(\alpha^8 + \beta^8\right)} = 4$$

Q. 29 If the system of equations

$2x + 7y + \lambda z = 3$

$3x + 2y + 5z = 4$

$x + \mu y + 32z = -1$

has infinitely many solutions, then $(\lambda - \mu)$ is equal to

Correct Answer:
38

Solution:
$D = D_1 = D_2 = D_3 = 0$

$$D_3 = \begin{vmatrix} 2 & 7 & 3 \\ 3 & 2 & 4 \\ 1 & \mu & -1 \end{vmatrix} = 0 \Rightarrow \mu = -39$$

$$D = \begin{vmatrix} 2 & 7 & \lambda \\ 3 & 2 & 5 \\ 1 & -39 & 32 \end{vmatrix} = 0 \Rightarrow \lambda = -1$$

$\lambda - \mu = 38$

Q. 30 If the solution $y(x)$ of the given differential equation $(e^y + 1)\cos x dx + e^y \sin x dy = 0$ passes through the point $\left(\frac{\pi}{2}, 0\right)$, then the value of $e^{y\left(\frac{\pi}{6}\right)}$ is equal to_____.

Correct Answer:
3

Solution:
$(e^y + 1)\cos x dx + e^y \sin x dy = 0$

$\Rightarrow d\left((e^y + 1)\sin x\right) = 0$

$(e^y + 1)\sin x = C$

It passes through $\left(\frac{\pi}{2}, 0\right)$

$\Rightarrow c = 2$

Now, $x = \frac{\pi}{6}$

$\Rightarrow e^y = 3$

8 Apr - Morning Shift

Physics

Q. 1 Three bodies A, B and C have equal kinetic energies and their masses are 400 g, 1.2 kg and 1.6 kg respectively. The ratio of their linear momenta is :

Option 1:
$1 : \sqrt{3} : 2$

Option 2:
$1 : \sqrt{3} : \sqrt{2}$

Option 3:
$\sqrt{2} : \sqrt{3} : 1$

Option 4:
$\sqrt{3} : \sqrt{2} : 1$

Correct Answer:
$1 : \sqrt{3} : 2$

Solution:

$$\mathrm{KE} = \frac{\mathrm{P}^2}{2\,\mathrm{m}}$$

$$\mathrm{P} \propto \sqrt{\mathrm{m}}$$

Hence, $\mathrm{P_A} : \mathrm{P_B} : \mathrm{P_C}$

$= \sqrt{400} : \sqrt{1200} : \sqrt{1600} = 1 : \sqrt{3} : 2$

Q. 2 Average force exerted on a non-reflecting surface at normal incidence is $2.4 \times 10^{-4}\ \mathrm{N}$. If $360\ \mathrm{W/cm^2}$ is the light energy flux during span of 1 hour 30 minutes. Then the area of the surface is:

Option 1:
$0.2\ \mathrm{m^2}$

Option 2:
$0.02\ \mathrm{m^2}$

Option 3:
$20\ \mathrm{m}^2$

Option 4:
$0.1\ \mathrm{m}^2$

Correct Answer:
$0.02\ \mathrm{m}^2$

Solution:

$$\text{Pressure} = \frac{I}{C} = \frac{F}{A}$$
$$\Rightarrow \frac{360}{10^{-4} \times 3 \times 10^{8}} = \frac{2.4 \times 10^{-4}}{A}$$
$$\Rightarrow A = 2 \times 10^{-2}\ \mathrm{m}^2 = 0.02\ \mathrm{m}^2$$

Q. 3 A proton and an electron are associated with same de-Broglie wavelength. The ratio of their kinetic energies is :
(Assume $\mathrm{h} = 6.63 \times 10^{-34}\ \mathrm{J\ s},\ \mathrm{m_e} = 9.0 \times 10^{-31}\ \mathrm{kg}$ and $\mathrm{m_p} = 1836$ times $\mathrm{m_e}$)

Option 1:
$1 : 1836$

Option 2:
$1 : \frac{1}{1836}$

Option 3:
$1 : \frac{1}{\sqrt{1836}}$

Option 4:
$1 : \sqrt{1836}$

Correct Answer:
$1 : 1836$

Solution:
λ is same for both
$\mathrm{P} = \frac{\mathrm{h}}{\lambda}$ same for both
$\mathrm{P} = \sqrt{2\mathrm{mK}}$
Hence,

$$\mathrm{K} \propto \frac{1}{\mathrm{m}}$$
$$\Rightarrow \frac{\mathrm{KE_p}}{\mathrm{KE_e}} = \frac{\mathrm{m_e}}{\mathrm{m_p}} = \frac{1}{1836}$$

Q. 4 A mixture of one mole of monoatomic gas and one mole of a diatomic gas (rigid) are kept at room temperature $(27^{\circ}\mathrm{C})$. The ratio of specific heat of gases at constant volume respectively is:

Option 1:
$\frac{7}{5}$

Option 2:
$\frac{3}{2}$

Option 3:
$\frac{3}{5}$

Option 4:
$\frac{5}{3}$

Correct Answer:
$\frac{3}{5}$

Solution:
$\frac{(\mathrm{C_v})_{\mathrm{mono}}}{(\mathrm{C_v})_{\mathrm{dia}}} = \frac{\frac{3}{2}\mathrm{R}}{\frac{5}{2}\mathrm{R}} = \frac{3}{5}$

Q. 5 In an expression $\mathrm{a} \times 10^{\mathrm{b}}$:

Option 1:
a is order of magnitude for $\mathrm{b} \leq 5$

Option 2:
b is order of magnitude for $a \leq 5$

Option 3:
b is order of magnitude for $5 < a \leq 10$

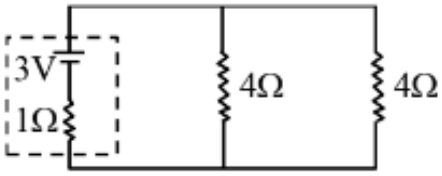
Option 4:
b is order of magnitude for $a \geq 5$

Correct Answer:
b is order of magnitude for $a \leq 5$

Solution:

$a \times 10^b$
if $a \leq 5$ order is b
$a > 5$ order is $b+1$

Q. 6 In the given circuit, the terminal potential difference of the cell is :



Option 1:
2 V

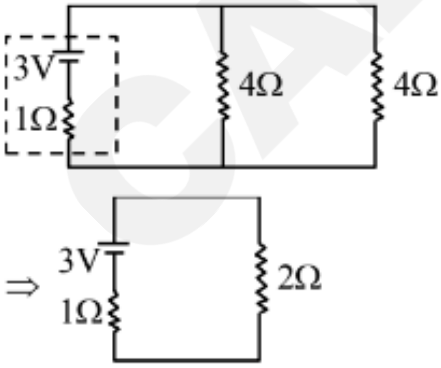
Option 2:
4 V

Option 3:
1.5 V

Option 4:
3 V

Correct Answer:
2 V

Solution:



$$i = \frac{3}{1+2} = 1 \text{ A}$$
$$v = E - ir$$
$$= 3 - 1 \times 1 = 2 \text{ V}$$

Q. 7 Binding energy of a certain nucleus is 18×10^8 J. How much is the difference between total mass of all the nucleons and nuclear mass of the given nucleus:

Option 1:
$0.2\mu g$

Option 2:
$20\mu g$

Option 3:
$2\mu g$

Option 4:
$10\mu g$

Correct Answer:
$20\mu g$

Solution:
$\Delta mc^2 = 18 \times 10^8$
$\Delta m \times 9 \times 10^{16} = 18 \times 10^8$
$\Delta m = 2 \times 10^{-8}\ kg = 20\mu g$

Q. 8 Paramagnetic substances:
A. align themselves along the directions of external magnetic field.
B. attract strongly towards external magnetic field.
C. has susceptibility little more than zero.
D. move from a region of strong magnetic field to weak magnetic field.

Choose the **most appropriate** answer from the options given below:

Option 1:
A, B, C, D

Option 2:
B, D Only

Option 3:
A, B, C Only

Option 4:
A, C Only

Correct Answer:
A, C Only

Solution:
A, C only

Q. 9 A clock has $75\ \text{cm}, 60\ \text{cm}$ long second hand and minute hand respectively. In 30 minutes duration the tip of second hand will travel x distance more than the tip of minute hand. The value of x in meter is nearly (Take $\pi = 3.14$) :

Option 1:
139.4

Option 2:
140.5

Option 3:
220.0

Option 4:
118.9

Correct Answer:
139.4

Solution:

$$\mathrm{x_{min}} = \pi \times \mathrm{r_{min}}$$
$$= \pi \times \frac{60}{100}\ \mathrm{m.}$$
$$\mathrm{x_{second}} = 30 \times 2\pi \times \mathrm{r_{second}}$$
$$= 30 \times 2\pi \times \frac{75}{100}$$
$$\mathrm{x} = \mathrm{x_{sccond}} - \mathrm{x_{min}}$$
$$= 139.4\ \mathrm{m}$$

Q. 10 Young's modulus is determined by the equation given by $Y = 49000 \frac{\mathrm{m}}{\ell} \frac{\text{dyne}}{\text{cm}^2}$ where M is the mass and ℓ is the extension of wire used in the experiment. Now error in Young modules (Y) is estimated by taking data from $\mathrm{M} - \ell$ plot in graph paper. The smallest scale divisions are $5\ \mathrm{g}$ and $0.02\ \text{cm}$ along load axis and extension axis respectively. If the value of M and ℓ are $500\ \mathrm{g}$ and $2\ \text{cm}$ respectively then percentage error of Y is :

Option 1:
0.2%

Option 2:
0.02%

Option 3:
2%

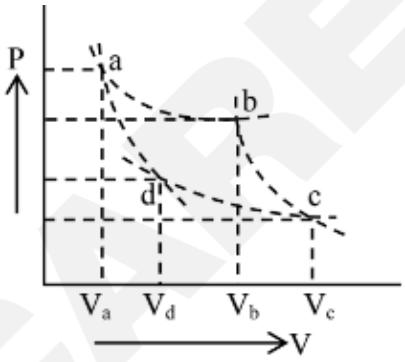
Option 4:
0.5%

Correct Answer:
2%

Solution:

$$\frac{\Delta Y}{Y} = \frac{\Delta m}{m} + \frac{\Delta \ell}{\ell}$$
$$= \frac{5}{500} + \frac{0.02}{2} = 0.01 + 0.01$$
$$\frac{\Delta Y}{Y} = 0.02 \Rightarrow \%\frac{\Delta Y}{Y} = 2\%$$

Q. 11 Two different adiabatic paths for the same gas intersect two isothermal curves as shown in $P - V$ diagram. The relation between the ratio $\frac{V_a}{V_d}$ and the ratio $\frac{V_b}{V_c}$ is:



Option 1:
$\frac{V_u}{V_d} = \left(\frac{V_b}{V_c}\right)^{-1}$

Option 2:
$\frac{V_a}{V_d} \neq \frac{V_b}{V_c}$

Option 3:
$\frac{V_a}{V_d} = \frac{V_b}{V_c}$

Option 4:

$\frac{V_a}{V_d}=\left(\frac{V_b}{V_c}\right)^2$

Correct Answer:

$\frac{V_a}{V_d}=\frac{V_b}{V_c}$

Solution:

For adiabatic process

$$\mathrm{TV}^{\gamma-1}=\text{ constant}$$

$$\mathrm{T_a}\cdot\mathrm{V_a}^{\gamma-1}=\mathrm{T_d}\cdot\mathrm{V_d}^{\gamma-1}$$

$$\left(\frac{V_a}{V_d}\right)^{\gamma-1}=\frac{T_d}{T_a}$$

$$\mathrm{T_b}\cdot\mathrm{V_b}^{\gamma-1}=\mathrm{T_c}\cdot\mathrm{V_c}^{\gamma-1}$$

$$\left(\frac{V_b}{V_c}\right)^{\gamma-1}=\frac{T_c}{T_b}$$

$$\frac{\mathrm{V_a}}{\mathrm{V_d}}=\frac{\mathrm{V_b}}{\mathrm{V_c}}\quad\left(\because\begin{matrix}\mathrm{T_d}=\mathrm{T_c}\\ \mathrm{T_a}=\mathrm{T_b}\end{matrix}\right)$$

Q. 12 Two planets A and B having masses m_1 and m_2 move around the sun in circular orbits of r_1 and r_2 radii respectively. If angular momentum of A is L and that of B is $3L$, the ratio of time period $\left(\frac{T_A}{T_B}\right)$ is:

Option 1:

$\left(\frac{r_2}{r_1}\right)^{\frac{3}{2}}$

Option 2:

$\left(\frac{r_1}{r_2}\right)^3$

Option 3:

$\frac{1}{27}\left(\frac{m_2}{m_1}\right)^3$

Option 4:

$27\left(\frac{m_1}{m_2}\right)^3$

Correct Answer:

$\frac{1}{27}\left(\frac{m_2}{m_1}\right)^3$

Solution:

$$\frac{\pi r_1^2}{T_A} = \frac{L}{2\,m_1} \ldots\ldots\ldots (1)$$

$$\frac{\pi r_2^2}{T_B} = \frac{3\,L}{2\,m_2} \ldots\ldots\ldots (2)$$

$$\Rightarrow \frac{T_A}{T_B} = 3 \cdot \frac{m_1}{m_2} \cdot \left(\frac{r_1}{r_2}\right)^2$$

$$\left(\frac{T_A}{T_B}\right)^2 = \left(\frac{r_1}{r_2}\right)^3 \Rightarrow \left(\frac{r_1}{r_2}\right)^2 = \left(\frac{T_A}{T_B}\right)^{\frac{4}{3}}$$

$$\Rightarrow \frac{1}{27} \cdot \left(\frac{m_2}{m_1}\right)^3 = \left(\frac{T_A}{T_B}\right)^2$$

Q. 13 A LCR circuit is at resonance for a capacitor C, inductance L and resistance R. Now the value of resistance is halved keeping all other parameters same. The current amplitude at resonance will be now:

Option 1:
Zero

Option 2:
double

Option 3:
same

Option 4:
halved

Correct Answer:
double

Solution:
In resonance Z=R

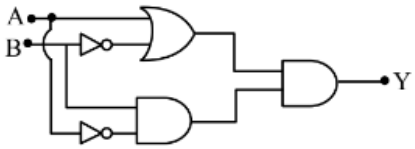
$$I = \frac{V}{R}$$

$$R \rightarrow \text{ halved}$$

$$\Rightarrow I \rightarrow 2I$$

I becomes doubled.

Q. 14 The output Y of following circuit for given inputs is :



Option 1:
$\mathrm{A \cdot B(A + B)}$

Option 2:
$A \cdot B$

Option 3:
0

Option 4:
$\bar{A} \cdot B$

Correct Answer:
0

Solution:
By truth table

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	0

Q. 15 Two charged conducting spheres of radii a and b are connected to each other by a conducting wire. The ratio of charges of the two spheres respectively is :

Option 1:
$\sqrt{\mathrm{ab}}$

Option 2:
ab

Option 3:
$\frac{\mathrm{a}}{\mathrm{b}}$

Option 4:
$\frac{b}{a}$

Correct Answer:
$\frac{a}{b}$

Solution:
Potential at surface will be same

$$\frac{Kq_1}{a} = \frac{Kqq_2}{b}$$
$$\frac{q_1}{q_2} = \frac{a}{b}$$

Q. 16 Correct Bernoulli's equation is (symbols have their usual meaning):

Option 1:
$P + mgh + \frac{1}{2}mv^2 = \text{constant}$

Option 2:
$P + \rho gh + \frac{1}{2}\rho v^2 = \text{constant}$

Option 3:
$P + \rho gh + \rho v^2 = \text{constant}$

Option 4:
$P + \frac{1}{2}\rho gh + \frac{1}{2}\rho v^2 = \text{constant}$

Correct Answer:
$P + \rho gh + \frac{1}{2}\rho v^2 = \text{constant}$

Solution:
$P + \rho gh + \frac{1}{2}\rho v^2 = \text{constant}$

Q. 17 A player caught a cricket ball of mass 150 g moving at a speed of 20 m/s. If the catching process is completed in 0.1 s, the magnitude of force exerted by the ball on the hand of the player is:

Option 1:
150 N

Option 2:
3 N

Option 3:
30 N

Option 4:
300 N

Correct Answer:
30 N

Solution:

$$F = \frac{\Delta P}{\Delta t} = \frac{mv - 0}{0.1}$$
$$= \frac{150 \times 10^{-3} \times 20}{0.1} = 30\ N$$

Q. 18 A stationary particle breaks into two parts of masses m_A and m_B which move with velocities v_A and v_B respectively. The ratio of their kinetic energies $\mathrm{\left(K_B: K_A\right)}$ is :

Option 1:
$v_B : v_A$

Option 2:
$m_B : m_A$

Option 3:
$m_B v_B : m_A v_A$

Option 4:
$1 : 1$

Correct Answer:
$v_B : v_A$

Solution:
Initial momentum is zero.

Hence $|P_A| = |P_B|$
$\Rightarrow m_A v_B = m_B v_B$

$$\frac{(KE)_A}{(KE)_B} = \frac{\frac{1}{2} m_A v_A^2}{\frac{1}{2} m_B v_B^2} = \frac{v_A}{v_B}$$

$\frac{(KE)_B}{(KE)_A} = \frac{v_B}{v_A}$

Q. 19 Critical angle of incidence for a pair of optical media is 45°. The refractive indices of first and second media are in the ratio:

Option 1:
$\sqrt{2} : 1$

Option 2:
$1 : 2$

Option 3:
$1 : \sqrt{2}$

Option 4:
$2 : 1$

Correct Answer:
$\sqrt{2} : 1$

Solution:

$$\sin\theta_c = \frac{\mu_R}{\mu_d} = \frac{\mu_2}{\mu_1}$$

$$\sin 45^\circ = \frac{\mu_2}{\mu_1}$$

$$\Rightarrow \frac{1}{\sqrt{2}} = \frac{\mu_2}{\mu_1}$$

$$\Rightarrow \frac{\mu_1}{\mu_2} = \frac{\sqrt{2}}{1}$$

Q. 20 The diameter of a sphere is measured using a vernier caliper whose 9 divisions of main scale are equal to 10 divisions of vernier scale. The shortest division on the main scale is equal to l mm. The main scale reading is 2 cm and second division of vernier scale coincides with a division on main scale. If mass of the sphere is 8.635 g, the density of the sphere is :

Option 1:
$2.5\ g/cm^3$

Option 2:
$1.7\ g/cm^3$

Option 3:
$2.2\ \mathrm{g/cm^3}$

Option 4:
$2.0\ \mathrm{g/cm^3}$

Correct Answer:
$2.0\ \mathrm{g/cm^3}$

Solution:

$\text{Given } 9\text{MSD} = 10\text{VSD}$

$\text{mass } = 8.635\ \text{g}$

$\text{LC} = 1\text{MSD} - 1\text{VSD}$

$\text{LC} = 1\text{MSD} - \frac{9}{10}\text{MSD}$

$\text{LC} = \frac{1}{10}\text{MSD}$

$\text{LC} = 0.01\ \text{cm}$

$\text{Reading of diameter } = \text{MSR} + \text{LC} \times \text{VSR}$

$= 2\ \text{cm} + (0.01) \times (2)$

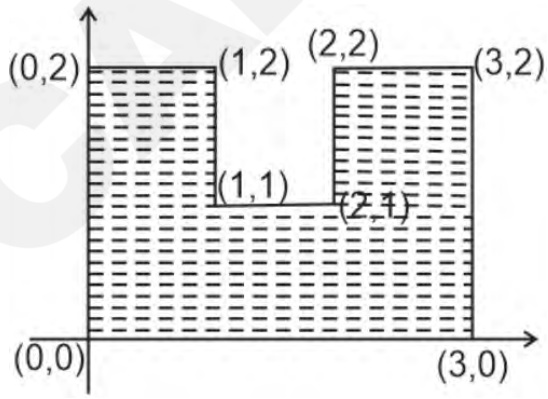
$= 2.02\ \text{cm}$

$\text{Volume of sphere } = \frac{4}{3}\pi\left(\frac{d}{2}\right)^3 = \frac{4}{3}\pi\left(\frac{2.02}{2}\right)^3$

$= 4.32\ \text{cm}^3$

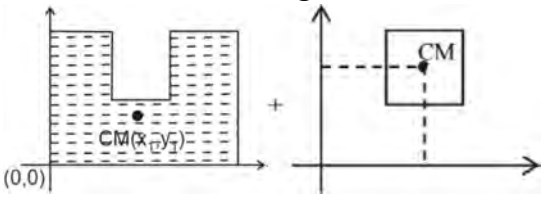
$\text{Density } = \frac{\text{mass}}{\text{volume}} = \frac{8.635}{4.32} = 1.998 \sim 2.00\ \text{g}$

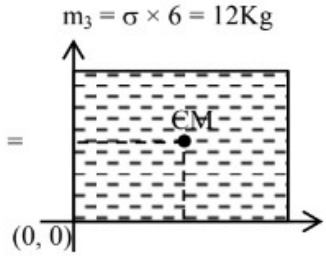
Q. 21 A uniform thin metal plate of mass $10\ \text{kg}$ with dimensions is shown. The ratio of x and y coordinates of center of mass of plate in $\frac{n}{9}$. The value of n is_____.



Correct Answer:
15

Solution:
$m_1 = \sigma \times 5 = 10Kg$
$m_1 = \sigma \times 5 = 10Kg \quad m_2 = \sigma \times 1 = 2Kg$





$\Rightarrow m_1x_1 + m_2x_2 = m_3x_3$
$10x_1 + 2(1.5) = 12(1.5) \Rightarrow x_1 = 1.5\ cm$
$\Rightarrow m_1y_1 + m_2y_2 = m_3y_3$
$10y_1 + 2(1.5) = 12 \times 1 \Rightarrow y_1 = 0.9\ cm$
$\frac{x_1}{y_1} = \frac{1.5}{0.9} = \frac{15}{9}$
$n = 15$

Q. 22 An electron with kinetic energy $5eV$ enters a region of uniform magnetic field of $3\mu T$ perpendicular to its direction. An electric field E is applied perpendicular to the direction of velocity and magnetic field. The value of E, so that electron moves along the same path, is $\quad NC^{-1}$.
(Given, mass of electron $= 9 \times 10^{-31}$ kg, electric charge $= 1.6 \times 10^{-19}C$)

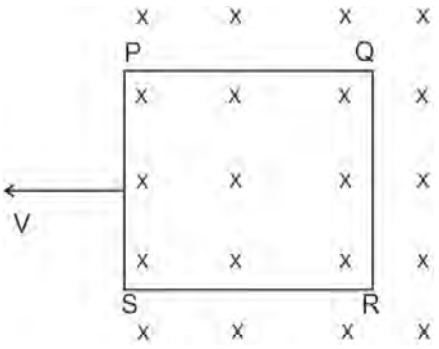
Correct Answer:
4

Solution:

For the given condition of moving undeflected, net force should be zero.

$$qE = qVB$$
$$E = VB$$
$$= \sqrt{\frac{2 \times KE}{m}} \times B$$
$$= \sqrt{\frac{2 \times 5 \times 1.6 \times 10^{-19}}{9 \times 10^{-31}}} \times 3 \times 10^{-6}$$
$$= 4\ N/C$$

Q. 23 A square loop PQRS having 10 turns, area $3.6 \times 10^{-3}\ m^2$ and resistance 100Ω is slowly and uniformly being pulled out of a uniform magnetic field of magnitude $B = 0.5\ T$ as shown. Work done in pulling the loop out of the field in 1.0 s is_____ $\times 10^6$ J



Correct Answer:
3

Solution:

$$\epsilon = NB\ell v$$
$$i = \frac{\epsilon}{R} = \frac{NB\ell v}{R}$$
$$F = N(i\ell B) = \frac{N^2B^2\ell^2 v}{R}$$
$$W = F \times \ell = \frac{N^2\ B^2\ell^3}{R}\left(\frac{\ell}{t}\right)$$
$$A = \ell^2$$
$$W = \frac{(10 \times 10)(0.5)^2 \times \left(3.6 \times 10^{-3}\right)^2}{100 \times 1}$$
$$W = 3.24 \times 10^{-6}\ J$$

Q. 24 Resistance of a wire at $0^\circ C, 100^\circ C$ and $t^\circ C$ is found to be $10\Omega, 10.2\Omega$ and 10.95Ω respectively. The temperature t in Kelvin scale is

Correct Answer:
748

Solution:
$\mathrm{R} = \mathrm{R}_0(1 + \alpha\Delta\mathrm{T})$
$\frac{\Delta\mathrm{R}}{\mathrm{R}_0} = \alpha\Delta\mathrm{T}$

Case - I

$0^\circ\mathrm{C} \rightarrow 100^\circ\mathrm{C}$
$\frac{10.2 - 10}{10} = \alpha(100 - 0)$ --------(1)

Case - II

$0^\circ\mathrm{C} \rightarrow \mathrm{t}^\circ\mathrm{C}$
$\frac{10.95 - 10}{10} = \alpha(\mathrm{t} - 0) - - - - - - (2)$
$\Rightarrow \frac{\mathrm{t}}{100} = \frac{0.95}{0.2} = 475^\circ\mathrm{C}$
$\mathrm{t} = 475 + 273 = 748\ \mathrm{K}$

Q. 25 An electric field, $\vec{\mathrm{E}} = \frac{2\hat{\mathrm{i}}+6\hat{\mathrm{j}}+8\hat{\mathrm{k}}}{\sqrt{6}}$ passes through the surface of $4\ \mathrm{m}^2$ area having unit vector $\hat{\mathrm{n}} = \left(\frac{2\hat{\mathrm{i}}+\hat{\mathrm{j}}+\hat{\mathrm{k}}}{\sqrt{6}}\right)$. The electric flux for that surface is _____ Vm.

Correct Answer:
12

Solution:
$\phi = \vec{\mathrm{E}} \cdot \vec{\mathrm{A}}$
$= \left(\frac{2\hat{\mathrm{i}} + 6\hat{\mathrm{j}} + 8\hat{\mathrm{k}}}{\sqrt{6}}\right) \cdot 4\left(\frac{2\hat{\mathrm{i}} + \hat{\mathrm{j}} + \hat{\mathrm{k}}}{\sqrt{6}}\right)$
$= \frac{4}{6} \times (4 + 6 + 8) = 12\mathrm{Vm}$

Q. 26 A liquid column of height $0.04\ \mathrm{cm}$ balances excess pressure of soap bubble of certain radius. If density of liquid is $8 \times 10^3\ \mathrm{kg\ m^{-3}}$ and surface tension of soap solution is $0.28\mathrm{Nm}^{-1}$, then diameter of the soap bubble is_____ cm.

(if $\mathrm{g} = 10\ \mathrm{ms}^{-2}$)

Correct Answer:
7

Solution:

$$\rho gh = \frac{4\,S}{R}$$

$$\Rightarrow R = \frac{4 \times 0.28}{8 \times 10^3 \times 10 \times 4 \times 10^{-4}}$$

$$\Rightarrow \frac{0.28}{8}\,m = \frac{28}{8}\,cm$$

$$\Rightarrow R = 3.5\,cm$$

$$\text{Diameter } = 7\,cm$$

Q. 27 A closed and an open organ pipe have same lengths. If the ratio of frequencies of their seventh overtones is $\left(\frac{a-1}{a}\right)$ then the value of a is ____.

Correct Answer:
16

Solution:
For closed organ pipe

$$f_c = (2n+1)\frac{v}{4\ell} = \frac{15v}{4\ell}$$

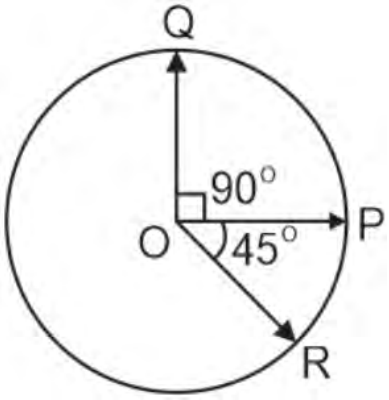
For open organ pipe

$$f_o = (n+1)\frac{v}{2\ell} = \frac{8v}{2\ell}$$

$$\frac{f_c}{f_o} = \frac{15}{16} = \frac{a-1}{a}$$

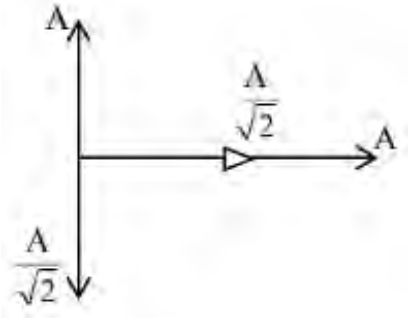
$$\Rightarrow a = 16$$

Q. 28 Three vectors $\overrightarrow{OP}, \overrightarrow{OQ}$ and $\overrightarrow{OR}$ each of magnitude A are acting as shown in figure. The resultant of the three vectors is $A\sqrt{x}$. The value of x is _____.



Correct Answer:
3

Solution:



$$\vec{R} = \left(A + \frac{A}{\sqrt{2}}\right)\hat{i} + \left(A - \frac{A}{\sqrt{2}}\right)\hat{j}$$

$$|\vec{R}| = \sqrt{\left(A + \frac{A}{\sqrt{2}}\right)^2 + \left(A - \frac{A}{\sqrt{2}}\right)^2} = \sqrt{3}A$$

Q. 29 A parallel beam of monochromatic light of wavelength 600 nm passes through single slit of 0.4 mm width. Angular divergence corresponding to second order minima would be_____ \times 10^{-3} \mathrm{rad}$.

Correct Answer:
6

Solution:

$$\sin\theta = \theta = \frac{2\lambda}{b}$$
$$= \frac{2 \times 600 \times 10^{-9}}{4 \times 10^{-4}} = 3 \times 10^{-3}\text{rad}$$

Total divergence $= (3+3) \times 10^{-3} = 6 \times 10^{-3}\text{rad}$

Q. 30 In an alpha particle scattering experiment distance of closest approach for the α particle is 4.5×10^{-14} m. If target nucleus has atomic number 80 , then maximum velocity of α-particle is_____ $\times 10^5$ m/s approximately.

$$\left(\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{SI unit, mass of } \alpha \text{ particle } = 6.72 \times 10^{-27} \text{ kg}\right)$$

Correct Answer:
156

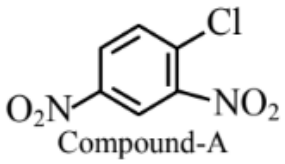
Solution:

$$\text{v} = \sqrt{\frac{4\text{KZe}^2}{\text{mr}_{\text{min}}}}$$
$$= \sqrt{\frac{4 \times 9 \times 10^9 \times 80}{6.72 \times 10^{-27} \times 4.5 \times 10^{-14}}} \times 1.6 \times 10^{-19}$$
$$= 9.759 \times 10^{25} \times 1.6 \times 10^{-19}$$
$$= 156 \times 10^5 \text{ m/s}$$

Chemistry

Q. 1 Given below are two statements:

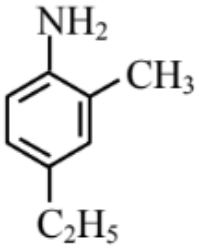
Statement I :



Compound-A

IUPAC name of Compound A is 4-chloro-1, 3-dinitrobenzene:

Statement II:



Compound-B

IUPAC name of Compound B is
4-ethyl-2-methylaniline.
In the light of the above statements, choose the most appropriate answer from the options given below:

Option 1:
Both Statement I and Statement II are correct

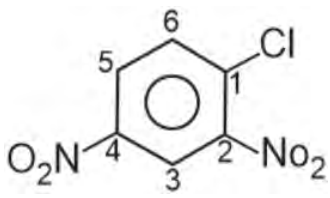
Option 2:
Statement I is incorrect but Statement II is correct

Option 3:
Statement I is correct but Statement II is incorrect

Option 4:
Both Statement I and Statement II are incorrect

Correct Answer:
Statement I is incorrect but Statement II is correct

Solution:
Statement I :

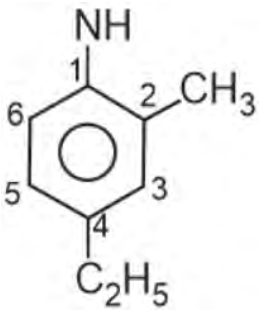


IUPAC name

$\Rightarrow$ 1-chloro-2, 4-dinitrobenzene

$\Rightarrow$ statement-I is incorrect

Statement II :

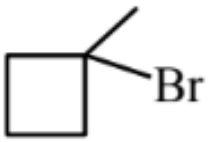


$\Rightarrow$ 4-ethyl-2-methylaniline

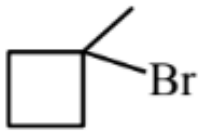
$\Rightarrow$ statement-II is correct

Q. 2 Which among the following compounds will undergo fastest S_N2 reaction.

Option 1:

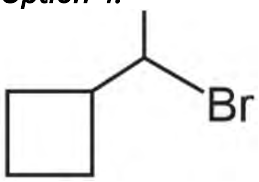


Option 2:



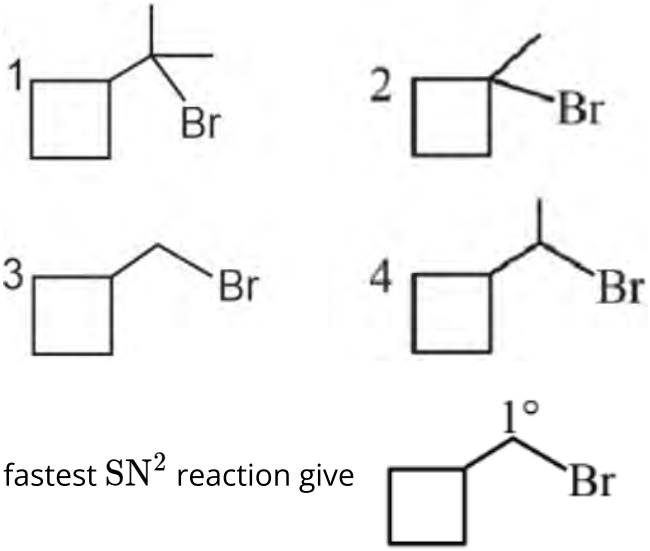
Option 3:

Option 4:



Correct Answer:

Solution:



fastest SN^2 reaction give

Rate of SN^2 is $Me - x > 1^\circ - x > 2^\circ - x > 3^\circ - x$

Q. 3 Combustion of glucose $(C_6H_{12}O_6)$ produces CO_2 and water. The amount of oxygen (in g) required for the complete combustion of 900 g of glucose is: [Molar mass of glucose in $gmol^{-1} = 180$]

Option 1:
480

Option 2:
960

Option 3:
800

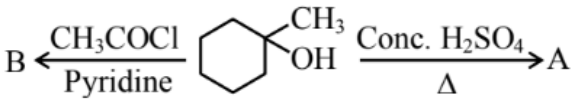
Option 4:
32

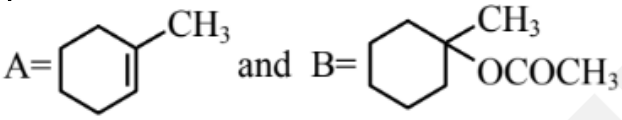
Correct Answer:
960

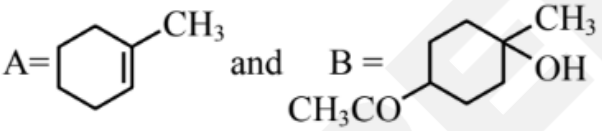
Solution:
$C_6H_{12}O_{6(s)} + 6O_{2(g)} \longrightarrow 6CO_{2(g)} + 6H_2O_{(\ell)}$

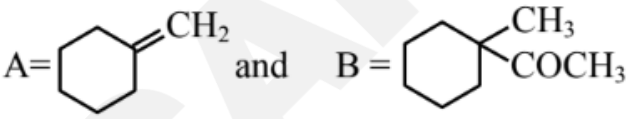
$\frac{900}{180}$
$= 5 \text{ mol} \quad 30 \text{ mol}$
$\text{Mass of } O_2 \text{ required } = 30 \times 32 = 960\text{gm}$

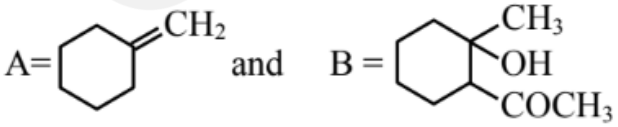
Q. 4 Identify the major products A and B respectively in the following set of reactions.

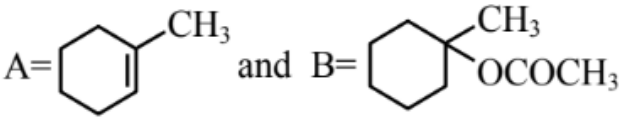


Option 1:


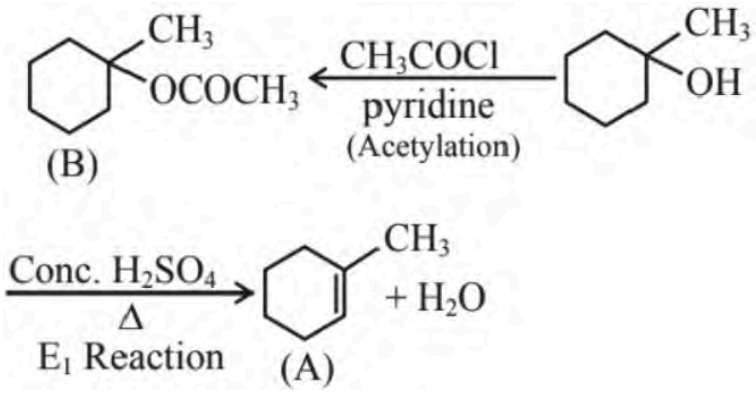
Option 2:


Option 3:


Option 4:


Correct Answer:


Solution:



Q. 5 Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R**: **Assertion A :** The stability order of +1 oxidation state of Ga, In and Tl is $\mathrm{Ga} < \mathrm{In} < \mathrm{Tl}$.

Reason R : The inert pair effect stabilizes the lower oxidation state down the group.

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$.

Option 2:
$\mathbf{A}$ is true but $\mathbf{R}$ is false.

Option 3:
Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$.

Option 4:
$\mathbf{A}$ is false but $\mathbf{R}$ is true.

Correct Answer:
Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$.

Solution:
The relative stability of +1 oxidation state progressively increases for heavier elements due to inert pair effect.
$\therefore$ Stability of $\mathrm{A}\ell^{+1} < \mathrm{Ga}^{+1} < \mathrm{In}^{+1} < \mathrm{T}\ell^{+1}$

Q. 6 Match **List I** with **List-II**

List-I (Name of the test)	**List-II (Reaction sequence involved) [M is metal]**
A Borax bead test	I. $MCO_3 \rightarrow MO \xrightarrow[+\Delta]{Co(NO_3)_2} CoO.MO$
B Charcoal cavity test	II. $MCO_3 \rightarrow MCl_2 \rightarrow M^{2+}$
C Cobalt nitrate | test	III $MSO_4 \xrightarrow[\Delta]{Na_2B_4O_7}$ $M(BO_2)_2 \rightarrow MBO_2 \rightarrow M$
D Flame test	IV $MSO_4 \xrightarrow[\Delta]{Na_2CO_3} MCO_3 \rightarrow$ $MO \rightarrow M$

Choose the correct answer from the option below :

Option 1:
A-III, B-I, C-IV, D-II

Option 2:
A-III, B-II, C-IV, D-I

Option 3:
A-III, B-I, C-II, D-IV

Option 4:
A-III, B-IV, C-I, D-II

Correct Answer:
A-III, B-IV, C-I, D-II

Solution:
Cobalt nitrate test

$$MCO_3 \rightarrow MO \xrightarrow[+\Delta]{Co(NO_3)_2} CoO.MO$$

Flame test

$$MCO_3 \rightarrow MCl_2 \rightarrow M^{2+}$$

Borax Bead test

$$MSO_4Na_2 \underset{\Delta}{B_4O_7M(BO_2)_2} \rightarrow MBO_2 \rightarrow M$$

Charcoal cavity test

$$MSO_4 \xrightarrow[\Delta]{Na_2CO_3} MCO_3 \rightarrow MO \rightarrow M$$

Q. 7 Match **List I** and with **List II**

List-I (Molecule)	**List-II(Shape)**
A NH_3	I. Square pyramid
B BrF_5	II.Tetrahedral
C PCl_5	III Trigonal pyramidal
D CH_4	IV Trigonal bipyramidal

Choose the correct answer from the option below :

Option 1:
A-IV, B-III, C-I, D-II

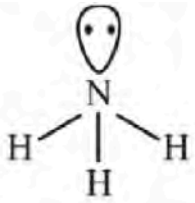
Option 2:
A-II, B-IV, C-I, D-III

Option 3:
A-III, B-I, C-IV, D-II

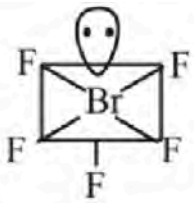
Option 4:
A-III, B-IV, C-I, D-II

Correct Answer:
A-III, B-I, C-IV, D-II

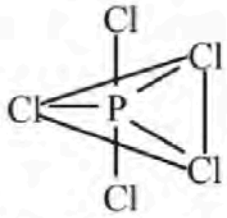
Solution:



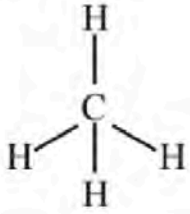
Trigonal pyramidal



Square pyramidal



Trigonal bipyramidal



Tetrahedral

Q. 8 For the given hypothetical reactions, the equilibrium constants are as follows:

$X \rightleftharpoons Y; K_1 = 1.0$
$Y \rightleftharpoons Z; K_2 = 2.0$

$Z \rightleftharpoons W; K_3 = 4.0$

The equilibrium constant for the reaction $X \rightleftharpoons W$ is

Option 1:
6.0

Option 2:
12.0

Option 3:
8.0

Option 4:
7.0

Correct Answer:
8.0

Solution:
$X \rightleftharpoons Y \quad k_1 = 1$
$Y \rightleftharpoons Z \quad k_2 = 2$

$Z \rightleftharpoons \omega \quad k_3 = 4$
$X \rightleftharpoons \omega \quad k_1 \cdot k_2 \cdot k_3$

$k = 1 \times 2 \times 4$
$k = 8$

Q. 9 Thiosulphate reacts differently with iodine and bromine in the reaction given below :

$2\ S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + 2I^-$

$S_2O_3^{2-} + 5Br_2 + 5H_2O \rightarrow 2SO_4^{2-} + 4Br^- + 10H^+$ $

Which of the following statement justifies the above dual behaviour of thiosulphate?

Option 1:
Bromine undergoes oxidation and iodine undergoes reduction by iodine in these reactions

Option 2:
Thiosulphate undergoes oxidation by bromine and reduction by iodine in these reaction

Option 3:
Bromine is a stronger oxidant than iodine

Option 4:
Bromine is a weaker oxidant than iodine

Correct Answer:
Bromine is a stronger oxidant than iodine

Solution:
In the reaction of $S_2O_3{}^{2-}$ with I_2, oxidation state of sulphur changes to +2 to +2.5

In the reaction of $S_2O_3{}^{2-}$ with Br_2, oxidation state of sulphur changes from +2 to +6 .
$\therefore$ Both I_2 and Br_2 are oxidant (oxidising agent) and Br_2 is stronger oxidant than I_2.

Q. 10 An octahedral complex with the formula $CoCl_3nNH_3$ upon reaction with excess of $AgNO_3$ solution given 2 moles of $AgCl$. Consider the oxidation state of Co in the complex is ' x '. The value of " $x + n$ " is

Option 1:
3

Option 2:
6

Option 3:
8

Option 4:
5

Correct Answer:
8

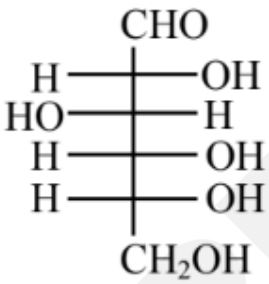
Solution:

$$\left[\overset{+3}{\mathrm{C}}\ (\mathrm{NH_3})_5\mathrm{Cl}\right]\mathrm{Cl_2} + \text{ excess } \mathrm{AgNO_3} \longrightarrow$$

$$(2\text{moles})$$

$x + 0 - 1 - 2 = 0$
$x = +3$
$n = 5$
$\therefore x + n = 8$

Q. 11



The incorrect statement regarding the given structure is

Option 1:
Can be oxidized to a dicarboxylic acid with $\mathrm{Br_2}$ water

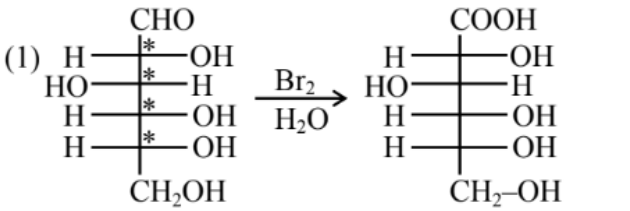
Option 2:
despite the presence of - CHO does not giveSchiff's test

Option 3:
has 4-asymmetric carbon atom

Option 4:
will coexist in equilibrium with 2 other cyclic structure

Correct Answer:
Can be oxidized to a dicarboxylic acid with Br_2 water

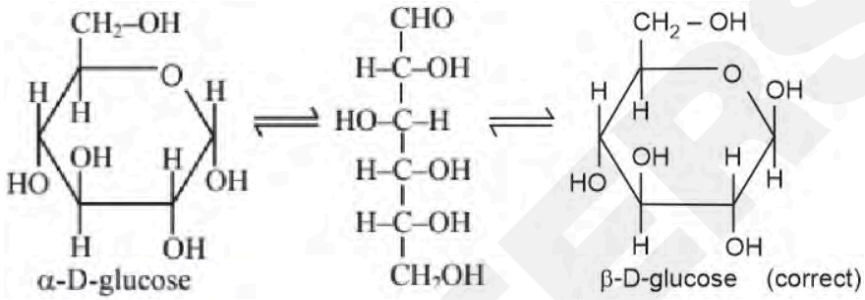
Solution:

(1) 

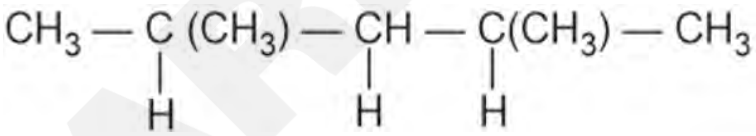
statement 1 is incorrect (monocarboxylic acid)

(2) correct

(3) c.c is 4 (correct)

(4) 

Q. 12 In the given compound, the number of 2° carbon atom /s is



Option 1:
Three

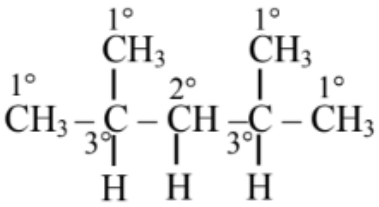
Option 2:
One

Option 3:
Two

Option 4:
Four

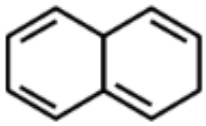
Correct Answer:
One

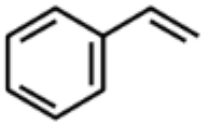
Solution:

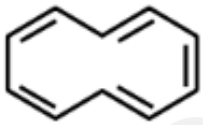


only one 2° carbon is present in this compound.

Q. 13 Which of the following are aromatic?

A. 

B. 

C. 

D.

Option 1:
B and D only

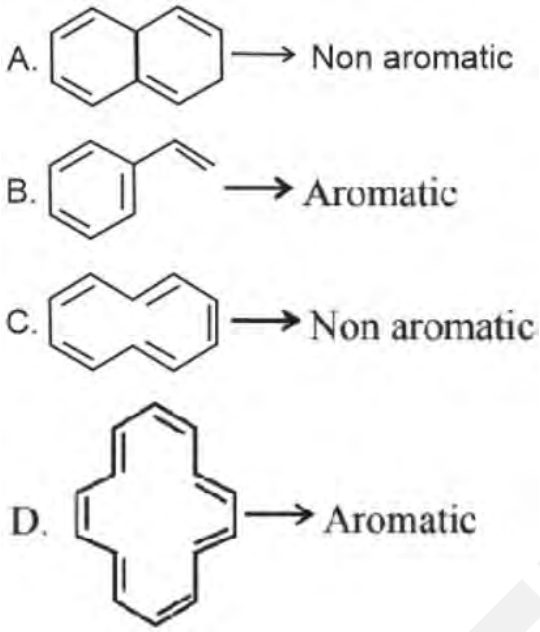
Option 2:
A and C only

Option 3:
A and B only

Option 4:
C and D only

Correct Answer:
B and D only

Solution:



Q. 14 Among the following halogens
F_2, Cl_2, Br_2 and I_2
Which can undergo disproportionation reaction?

Option 1:
Only I_2

Option 2:
Cl_2, Br_2 and I_2

Option 3:
F_2, Cl_2 and Br_2

Option 4:
F_2 and Cl_2

Correct Answer:
Cl_2, Br_2 and I_2

Solution:
F_2 do not disproportionate because fluorine do not exist in positive oxidation state however $Cl_2, Br_2 \& I_2$ undergoes disproportionation.

Q. 15 Given below are two statements:
Statement I : $N(CH_3)_3$ and $P(CH_3)_3$ can act as ligands to form transition metal complexes.
Statement II: As N and P are from same group, the nature of bonding of $N(CH_3)_3$ and $P(CH_3)_3$ is always same with transition metals.
In the light of the above statements, choose the most appropriate answer from the options given below:

Option 1:
Statement I is incorrect but Statement II is correct

Option 2:
Both Statement I and Statement II are correct

Option 3:
Statement I is correct but Statement II is incorrect

Option 4:
Both Statement I and Statement II are incorrect

Correct Answer:
Statement I is correct but Statement II is incorrect

Solution:
$N(CH_3)_3$ and $P(CH_3)_3$ both are Lewis base and acts as ligand, However, $P(CH_3)_3$ has a π-acceptor character.

Q. 16 Match **List I** with **List II**

List-I (Elements)	**List-II(Properties in their respective groups)**
A. Cl,S	I. Elements with highest electronegativity
B. Ge,As	II. Elements with largest atomic size
C. Fr, Ra	III. Elements which show properties of both metals and non metal
D. F,O	IV. Elements with highest negative electron gain enthalpy

Choose the correct answer from the options given below :

Option 1:
A-II, B-III, C-IV, D-I

Option 2:
A-III, B-II, C-I, D-IV

Option 3:
A-IV, B-III, C-II, D-I

Option 4:
A-II, B-I, C-IV, D-III

Correct Answer:
A-IV, B-III, C-II, D-I

Solution:
Elements with highest electronegativity $\rightarrow \mathrm{F}, \mathrm{O}$
Elements with largest atomic size $\rightarrow \mathrm{Fr}, \mathrm{Ra}$
Elements which shows properties of both metal and non-metals i.e. metalloids $\rightarrow \mathrm{Ge}$, As
Elements with highest negative electron gain enthalpy $\rightarrow \mathrm{Cl}, \mathrm{S}$

Q. 17 Iron (III) catalyses the reaction between iodide and persulphate ions, in which
A. Fe^{3+} oxidises the iodide ion
B. Fe^{3+} oxidises the persulphate ion
C. Fe^{2+} reduces the iodide ion
D. Fe^{2+} reduces the persulphate ion

Choose the most appropriate answer from the options given below:

Option 1:
B and C only

Option 2:
B only

Option 3:
A only

Option 4:
A and D only

Correct Answer:
A and D only

Solution:
$2Fe^{3+} + 2I^- \longrightarrow 2Fe^{2+} + I_2$
$2Fe^{2+} + {S_2O_8}^{2-} \longrightarrow 2Fe^{3+} + 2{SO_4}^{2-}$

Fe^{+3} oxidises I^- to I_2 and convert itself into Fe^{+2}. This Fe^{+2} reduces ${S_2O_8}^{2-}$ to ${SO_4}^{2-}$ and converts itself into Fe^{+3}

Q. 18 Match **List I** with **List II**

List-I (Compound)	**List-II (Colour)**
A. $Fe_4[Fe(CN)_6]_3 \cdot xH_2O$	I. Violet
B. $[Fe(CN)_5NOS]^4$	II.Blood Red
C. $[Fe(SCN)]^{2+}$	III.Prussian Blue
D.$(NH_4)_3PO_4.12MoO_3$	IV. Yellow

Choose the correct answer from the options given below:

Option 1:
A-III, B-I, C-II, D-IV

Option 2:
A-IV, B-I, C-II, D-III

Option 3:
A-II, B-III, C-IV, D-I

Option 4:
A-I, B-II, C-III, D-IV

Correct Answer:
A-III, B-I, C-II, D-IV

Solution:
$Fe_4[Fe(CN)_6]_3 \cdot xH_2O \rightarrow$ Prussian Blue
$[Fe(CN)_5NOS]^{4-} \rightarrow$ Violet

$[Fe(SCN)]^{2+} \rightarrow$ Blood Red
$(NH_4)_3PO_4 \cdot 12MoO_3 \rightarrow$ Yellow

Q. 19 Number of complexes with even number of electrons in t_{2g} orbitals is -

$[Fe(H_2O)_6]^{2+}, [Co(H_2O)_6]^{2+}, [Co(H_2O)_6]^{3+},$
$[Cu(H_2O)_6]^{2+}, [Cr(H_2O)_6]^{2+}$

Option 1:
1

Option 2:
3

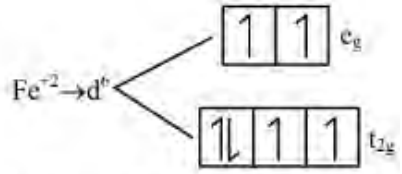
Option 3:
2

Option 4:
5

Correct Answer:
3

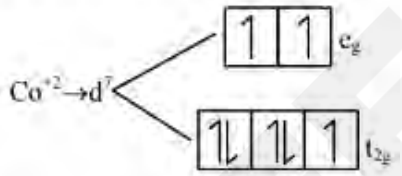
Solution:

$[Fe(H_2O)_6]^{2+}$ $Fe^{+2} \rightarrow d^6$



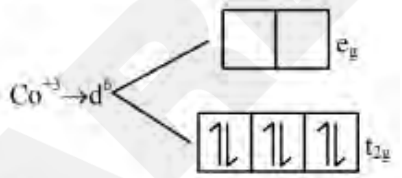
Electron in t_{2g} = 4(even)

$[Co(H_2O)_6]^{2+}$ $Co^{+2} \rightarrow d^7$



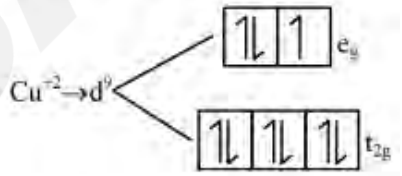
Electron in t_{2g} = 5(odd)

$[Co(H_2O)_6]^{3+}$ $Co^{+3} \rightarrow d^6$



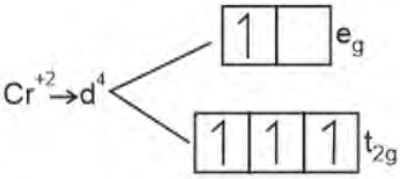
Electron in t_{2g} = 6(even)

$[Cu(H_2O)_6]^{2+}$ $Cu^{+2} \rightarrow d^9$



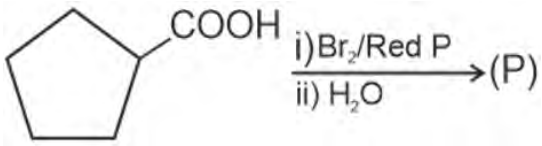
Electron in t_{2g} = 6(even)

$[Cr(H_2O)_6]^{2+}$ $Cr^{+2} \rightarrow d^4$

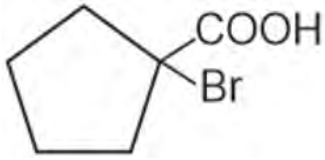


Electron in t_{2g} = 3 (odd)

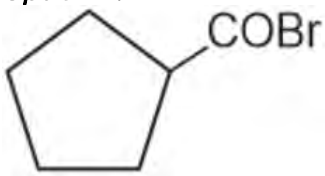
Q. 20 Identify the product (P) in the following reaction:



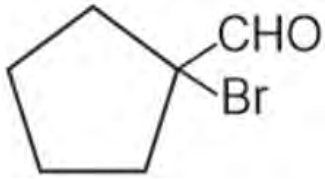
Option 1:



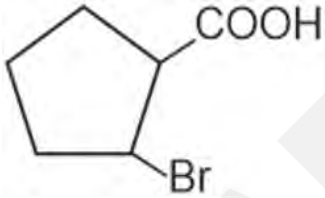
Option 2:



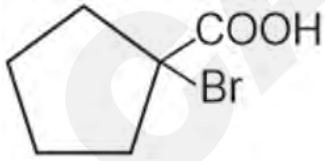
Option 3:



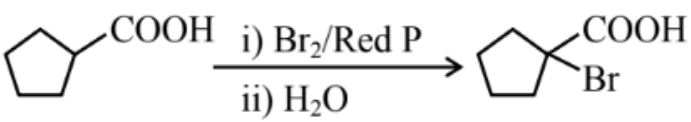
Option 4:



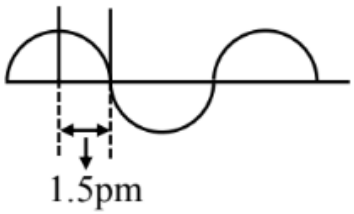
Correct Answer:



Solution:
HVZ Reaction



Q. 21 A hypothetical electromagnetic wave is show below.

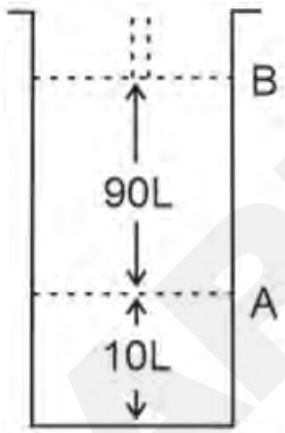


The frequency of the wave is $\mathrm{x} \times 10^{19}$ Hz. $x =$ (nearest integer)

Correct Answer:
5

Solution:
$\lambda = 1.5 \times 4\text{pm}$
$= 6 \times 10^{-12}$ meter
$\lambda v = \mathrm{C}$
$6 \times 10^{-12} \times v = 3 \times 10^{8}$
$v = 5 \times 10^{19}$ Hz

Q. 22



Consider the figure provided.

1 mol of an ideal gas is kept in a cylinder, fitted with a piston, at the position A, at $18^{\circ}\mathrm{C}$. If the piston is moved to position B, keeping the temperature unchanged, then ' x ' L atm work is done in this reversible process.

x = L atm. (nearest integer)
[Given : Absolute temperature $= {}^{\circ}\mathrm{C} + 273.15$,
$\mathrm{R} = 0.08206$ L atm mol^{-1} K^{-1}]

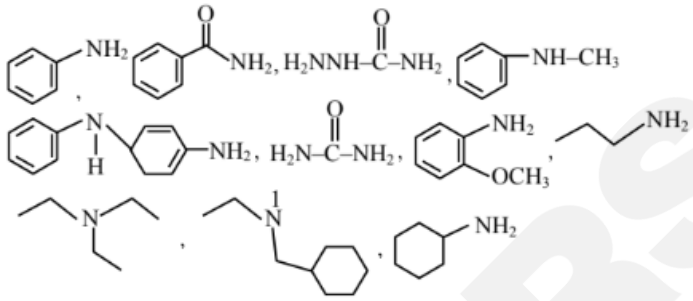
Correct Answer:
55

Solution:
$\omega = -\mathrm{nRT}\ln\left(\frac{V_2}{V_1}\right)$

$= -1 \times .08206 \times 291.15 \ln\left(\frac{100}{10}\right)\$\$$
$= -55.0128$

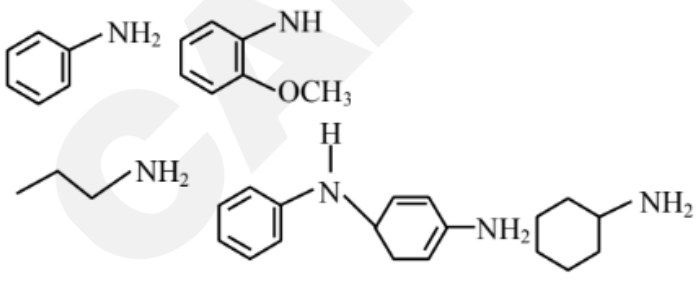
Work done by system $\approx 55\ \mathrm{atm}$ lit.

Q. 23 Number of amine compounds from the following giving solids which are soluble in NaOH upon reaction with Hinsberg's reagent is

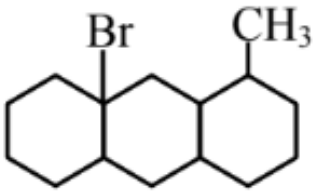


Correct Answer:
5

Solution:
Primary amine give an ionic solid upon reaction with Hinsberg reagent which is soluble in NaOH.

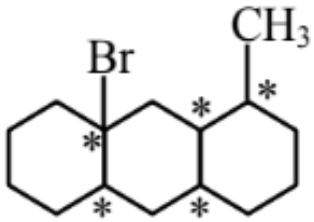


Q. 24 The number of optical isomers in following compound is 32



Correct Answer:
32

Solution:



Total chiral centre $= 5$
No. of optical isomers $= 2^5 = 32$

Q. 25 The 'spin only' magnetic moment value of $MO_4{}^{2-}$ is _ BM. (Where M is a metal having least metallic radii. among Sc, Ti, V, Cr, Mn and Zn). (Given atomic number: $Sc = 21, Ti = 22,\ V = 23, Cr = 24, Mn = 25$ and $Zn = 30$)

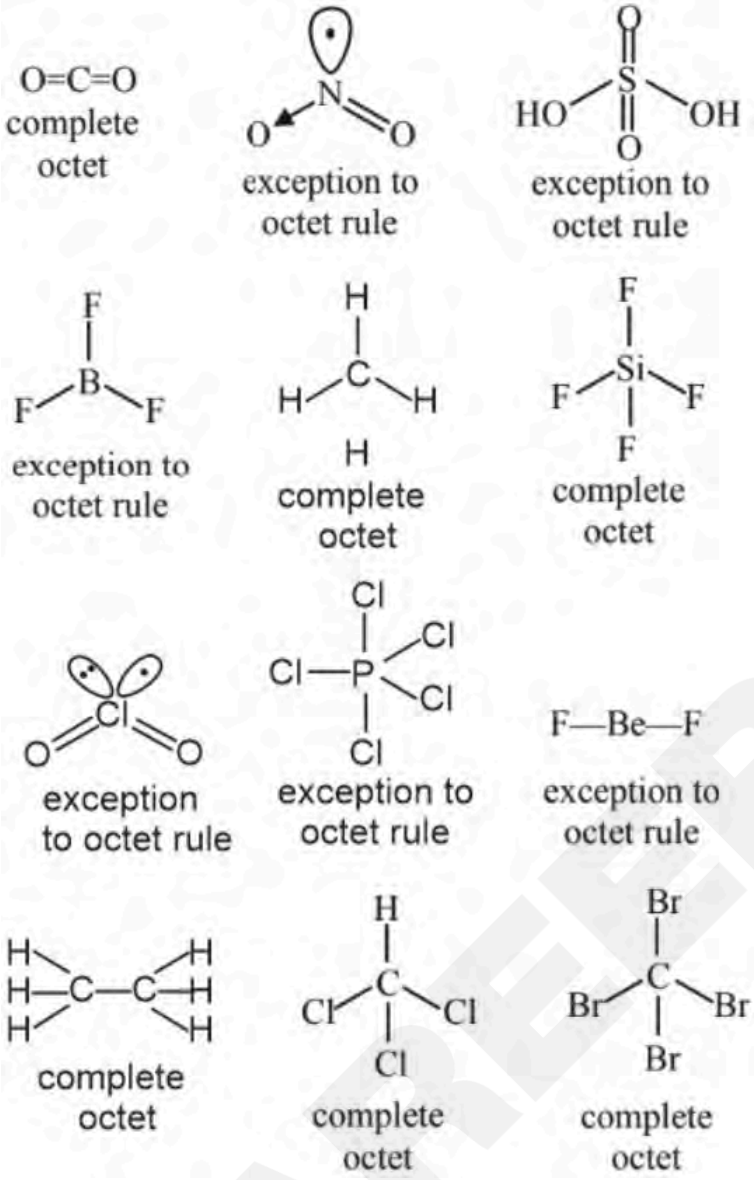
Correct Answer:
0

Solution:
Metal having least metallic radii among $Sc, Ti, V, Cr, Mn\&Zn$ is Cr.
Spin only magnetic moment of $CrO_4{}^{2-}$.
Here Cr^{+6} is in d^0 configuration (diamagnetic).

Q. 26 Number of molecules from the following which are exceptions to octet rule is_
$CO_2, NO_2, H_2SO_4, BF_3, CH_4, SiF_4, ClO_2, PCl_5, BeF_2, C_2H_6, CHCl_3, CBr_4$

Correct Answer:
6

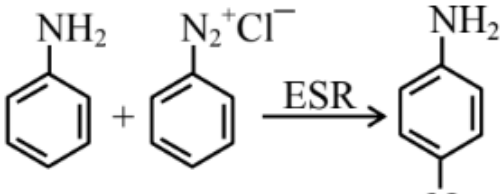
Solution:



Q. 27 If 279 g of aniline is reacted with one equivalent of benzenediazonium chloride, the mximum amount of aniline yellow formed will be _g. (nearest integer)
(consider complete conversion)

Correct Answer:
591

Solution:



m.wt.=93
given wt.=279gm

moles = $\frac{279}{93}=3$

yellow dye

moles formed =3
m.wt. = 197
amount formed
=197 × 3 = 591 gm

Q. 28 Consider the following reaction

$$A + B \rightarrow C$$

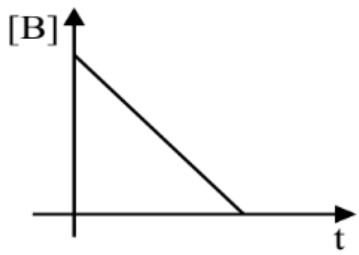
The time taken for A to become $1/4^{th}$ of its initial concentration is twice the time taken to become $1/2$ of the same. Also, when the change of concentration of B is plotted against time, the resulting graph gives a straight line with a negative slope and a positive intercept on the concentration axis.

The overall order of the reaction is

Correct Answer:
1

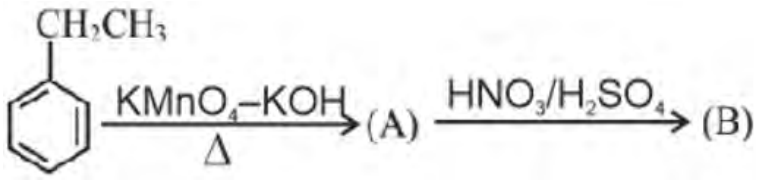
Solution:
For 1^{st} order reaction
75% life $= 2 \times 50\%$ life

So order with respect to A will be first order.



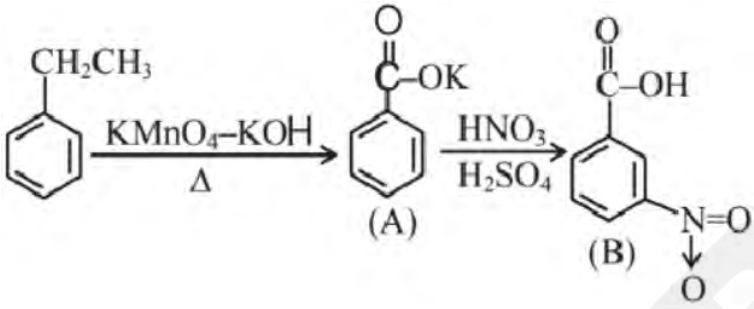
So order with respect to B will be zero. Overall order of reaction $= 1 + 0 = 1$

Q. 29 Major product B of the following reaction has _ π-bond.



Correct Answer:
5

Solution:
Major product B is $\rightarrow$



Total number of π bonds in B are 5

Q. 30 A solution containing $10\ \mathrm{g}$ of an electrolyte $\mathrm{AB_2}$ in $100\ \mathrm{g}$ of water boils at $100.52^{\circ}\mathrm{C}$. The degree of ionization of the electrolyte (α) is $\times 10^{-1}$. (nearest integer)

[Given : Molar mass of $\mathrm{AB_2} = 200\ \mathrm{g\ mol^{-1}} \cdot \mathrm{K_b}$ (molal boiling point elevation const. of water) $= 0.52\ \mathrm{K\ kg\ mol^{-1}}$, boiling point of water $= 100^{\circ}\mathrm{C}$; $\mathrm{AB_2}$ ionises as $\mathrm{AB_2} \rightarrow \mathrm{A^{2+}} + 2\ \mathrm{B^{-}}$]

Correct Answer:
5

Solution:
$\mathrm{AB_2} \rightarrow \mathrm{A^{+2}} + 2\ \mathrm{B^{\ominus}}$

$\mathrm{i} = 1 + (3-1)\alpha$

$\mathrm{i} = 1 + 2\alpha$

$\Delta \mathrm{T_b} = \mathrm{k_b i m}$

$$0.52 = 0.52(1+2\alpha)\frac{\frac{10}{200}}{\frac{100}{1000}}$$

$$1 = (1+2\alpha)\frac{10}{20}$$

$2 = 1 + 2\alpha$
$\alpha = 0.5$

Ans. $\alpha = 5 \times 10^{-1}$

Maths

Q. 1 The value of $k \in \mathbb{N}$ for which the integral $I_n = \int_0^1 \left(1 - x^k\right)^n dx, n \in \mathbb{N},$ satisfies $147I_{20} = 148I_{21}$

Option 1:
10

Option 2:
8

Option 3:
14

Option 4:
7

Correct Answer:
7

Solution:
$I_n = \int_0^1 \left(1 - x^k\right)^n \cdot 1dx$

$I_n = \left(1 - x^k\right)^n \cdot x - nk \int_0^1 \left(1 - x^k\right)^{n-1} \cdot x^{k-1} \cdot dx$

$I_n = nk \int_0^1 \left[\left(1 - x^k\right)^n - \left(1 - x^k\right)^{n-1}\right] dx$

$I_n = nkI_n - nkI_n$

$\frac{I_n}{I_{n-1}} = \frac{nk}{nk+1}$

$\frac{I_{21}}{I_{20}} = \frac{21k}{1+21k}$

$= \frac{147}{148} \Rightarrow k = 7$

Q. 2 The sum of all the solutions of the equation $(8)^{2x} - 16 \cdot (8)^{x} + 48 = 0$ is :

Option 1:
$1 + \log_6(8)$

Option 2:
$\log_8(6)$

Option 3:
$1 + \log_8(6)$

Option 4:
$\log_8(4)$

Correct Answer:
$1 + \log_8(6)$

Solution:
$(8)^{2x} - 16 \cdot (8)^{x} + 48 = 0$

$\Rightarrow t = 4 \text{ or } t = 12$

$\Rightarrow 8^x = 4 \quad 8^x = 12 \quad x = \log_8 12$

$\Rightarrow x = \log_8 x \quad \log_8 12$

$\text{sum of solution } = \log_8 4 + \log_8$

$= \log_8 48 = \log_8(6.8)$

$= 1 + \log_8 6$

Q. 3 Let the circles $C_1 : (x-\alpha)^2 + (y-\beta)^2 = r_1^2$ and $C_2 : (x-8)^2 + \left(y - \frac{15}{2}\right)^2 = r_2^2$ touch each other externally at the point $(6, 6)$. If the point $(6, 6)$ divides the line segment joining the centres of the circles C_1 and C_2 internally in the ratio $2 : 1$, then $(\alpha + \beta) + 4\left(r_1^2 + r_2^2\right)$ equals

Option 1:
110

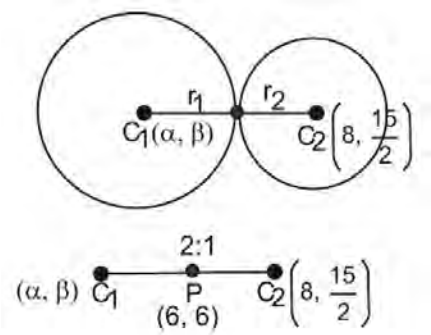
Option 2:
130

Option 3:
125

Option 4:
145

Correct Answer:
130

Solution:



$\because \frac{16+\alpha}{3} = 6 \text{ and } \frac{15+\beta}{3} = 6$

$\Rightarrow (\alpha, \beta) \equiv (2, 3)$

Also, $C_1C_2 = r_1 + r_2$

$\Rightarrow \sqrt{(2-8)^2 + \left(3 - \frac{15}{2}\right)^2} = 2r_2 + r_2$

$\Rightarrow r_2 = \frac{5}{2} \Rightarrow r_1 = 2r_2 = 5$

$\therefore (\alpha + \beta) + 4\left(r_1^2 + r_2^2\right)$

$= 5 + 4\left(\frac{25}{4} + 25\right) = 130$

Q. 4 Let $P(x, y, z)$ be a point in the first octant, whose projection in the xy-plane is the point Q. Let $OP = \gamma$; the angle between OQ and the positive x-axis be θ; and the angle between OP and the positive z-axis be ϕ, where O is the origin. Then the distance of P from the x-axis is :

Option 1:

$\gamma\sqrt{1-\sin^2\phi\cos^2\theta}$

Option 2:

$\gamma\sqrt{1+\cos^2\theta\sin^2\phi}$

Option 3:

$\gamma\sqrt{1-\sin^2\theta\cos^2\phi}$

Option 4:

$\gamma\sqrt{1+\cos^2\phi\sin^2\theta}$

Correct Answer:

$\gamma\sqrt{1-\sin^2\phi\cos^2\theta}$

Solution:

$P(x,y,z), Q(x,y,O); x^2+y^2+z^2=\gamma^2$

$\overline{OQ}=x\hat{i}+y\hat{j}$

$\cos\theta=\frac{x}{\sqrt{x^2+y^2}}$

$\cos\phi=\frac{z}{\sqrt{x^2+y^2+z^2}}$

$\Rightarrow \sin^2\phi=\frac{x^2+y^2}{x^2+y^2+z^2}$

distance of P from x-axis $\sqrt{\mathrm{y}^2+\mathrm{z}^2}$

$\Rightarrow \sqrt{\gamma^2-x^2} \Rightarrow \gamma\sqrt{1-\frac{x^2}{\gamma^2}}$

$=\gamma\sqrt{1-\cos^2\theta\sin^2\phi}$

Q. 5 The number of critical points of the function $f(x)=(x-2)^{2/3}(2x+1)$ is :

Option 1:
2

Option 2:
0

Option 3:
1

Option 4:
3

Correct Answer:
2

Solution:

$f(x) = (x-2)^{2/3}(2x+1)$

$f'(x) = \frac{2}{3}(x-2)^{-1/3}(2x+1) + (x-2)^{2/3}(2)$

$f'(x) = 2 \times \frac{(2x+1)+(x-2)}{3(x-2)^{1/3}}$

$\frac{3x-1}{(x-2)^{1/3}} = 0$

Critical points $\mathbf{x} = \frac{1}{3}$ and $\mathbf{x} = 2$

Q. 6 Let $f(x)$ be a positive function such that the area bounded by $y = f(x), y = 0$ from $x = 0$ to $x = a > 0$ is $e^{-a} + 4a^2 + a - 1$. Then the differential equation, whose general solution is $y = c_1 f(x) + c_2$, where c_1 and c_2 are arbitrary constants, is :

Option 1:
$(8e^x - 1)\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

Option 2:
$(8e^x + 1)\frac{d^2y}{dx^2} - \frac{dy}{dx} = 0$

Option 3:
$(8e^x + 1)\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

Option 4:

$(8e^x - 1)\frac{d^2y}{dx^2} - \frac{dy}{dx} = 0$

Correct Answer:

$(8e^x + 1)\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

Solution:

$\int_0^a f(x)dx = e^{-a} + 4a^2 + a - 1$

$f(a) = -e^{-a} + 8a + 1$

$f(x) = -e^{-x} + 8x + 1$

Now $y = C_1 f(x) + C_2$

$\frac{dy}{dx} = C_1 f'(x) = C_1 (e^{-x} + 8)$ .....(1)

$\frac{d^2y}{dx^2} = -C_1 e^{-x} \Rightarrow -e^x \frac{d^2y}{dx^2}$

Put in equation (1)

$\frac{dy}{dx} = -e^x \frac{d^2y}{dx^2}(e^{-x} + 8)$

$(8e^x + 1)\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

Q. 7 Let $f(x) = 4\cos^3 x + 3\sqrt{3}\cos^2 x - 10$. The number of points of local maxima of f in interval $(0, 2\pi)$ is:

Option 1:

1

Option 2:

2

Option 3:

3

Option 4:

4

Correct Answer:

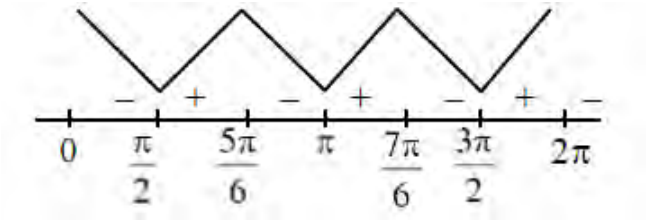
2

Solution:

$f(x) = 4\cos^3(x) + 3\sqrt{3}\cos^2(x) - 10; x \in (0, 2\pi)$

$\Rightarrow f'(x) = 12\cos^2 x[-\sin(x)] + 3\sqrt{3}(2\cos(x))[-\sin(x)]$

$\Rightarrow f'(x) = -6\sin(x)\cos(x)[2\cos(x) + \sqrt{3}]$



local maxima at $\mathrm{x} = \frac{5\pi}{6}, \frac{7\pi}{6}$

Q. 8 Let $A = \begin{bmatrix} 2 & a & 0 \\ 1 & 3 & 1 \\ 0 & 5 & b \end{bmatrix}$. If $A^3 = 4A^2 - A - 21I$, where I is the identity matrix of order 3×3, then $2a + 3b$ is equal to:

Option 1:
-10

Option 2:
-13

Option 3:
-9

Option 4:
-12

Correct Answer:
-13

Solution:
$A^3 - 4A^2 + A + 21I = 0$

$\mathrm{tr}(A) = 4 = 5 + 6 \Rightarrow b = -1$

$|A| = -21$

$-16 + a = -21 \Rightarrow a = -5$

$2a + 3b = -13$

Q. 9 If the shortest distance between the lines

$$L_1 : \vec{r} = (2+\lambda)\hat{i} + (1-3\lambda)\hat{j} + (3+4\lambda)\hat{k}, \lambda \in \mathbb{R}$$

$$L_2 : \vec{r} = 2(1+\mu)\hat{i} + 3(1+\mu)\hat{j} + (5+\mu)\hat{k}, \mu \in \mathbb{R}$$

is $\frac{m}{\sqrt{n}}$, where $\gcd(m, n) = 1$, then the value of $m + n$ equals.

Option 1:
384

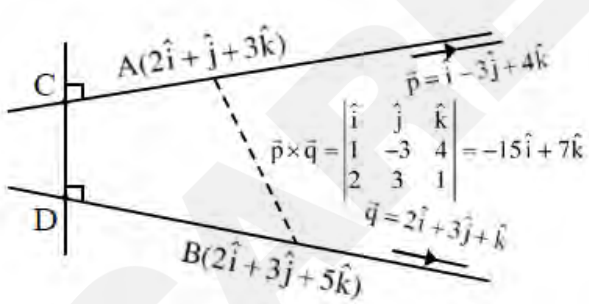
Option 2:
387

Option 3:
377

Option 4:
390

Correct Answer:
387

Solution:



$$\text{Shortes distance (CD)} = \frac{\overrightarrow{AB}\cdot\vec{p}\times\vec{q}}{|\vec{p}\times\vec{q}|}$$

$$= \frac{(0\hat{i}+2\hat{j}+2\hat{k})\cdot(-15\hat{i}+7\hat{j}+9\hat{k})}{\sqrt{355}}$$

$$= \frac{0+14+18}{\sqrt{355}} = \frac{32}{\sqrt{355}}$$

$$\therefore m + n = 32 + 355 = 387$$

Q. 10 Let the sum of two positive integers be 24 . If the probability, that their product is not less than $\frac{3}{4}$ times their greatest positive product, is $\frac{\mathrm{m}}{\mathrm{n}}$, where $\gcd(\mathrm{m}, \mathrm{n}) = 1$, then $\mathrm{n} - \mathrm{m}$ equals :

Option 1:
9

Option 2:
11

Option 3:
8

Option 4:
10

Correct Answer:
10

Solution:
$x + y = 24, x, y \in N$

$AM > GM \Rightarrow xy \leq 144$

$xy \geq 108$

Favorable pairs of (x, y) are

$(13, 11), (12, 12), (14, 10), (15, 9), (16, 8),$

$(17, 7), (18, 6), (6, 18), (7, 17), (8, 16), (9, 15),$

$(10, 14), (11, 13)$

i.e. 13 cases

Total choices for $\mathrm{x} + \mathrm{y} = 24$ is 23

Probability $= \frac{13}{23} = \frac{\mathrm{m}}{\mathrm{n}}$

$\mathrm{n} - \mathrm{m} = 10$

Q. 11 If $\sin x = -\frac{3}{5}$, where $\pi < x < \frac{3\pi}{2}$ then $80\left(\tan^2 x - \cos x\right)$ is equal to :

Option 1:
109

Option 2:
108

Option 3:
18

Option 4:
19

Correct Answer:
109

Solution:
$\sin x = \frac{-3}{5}, \pi < x < \frac{3\pi}{2}$

$\tan x = \frac{3}{4} \cos x = -\frac{4}{5}$

$80\left(\tan^2 x - \cos x\right)$

$= 80\left(\frac{9}{16} + \frac{4}{5}\right) = 45 + 64 = 109$

Q. 12 Let $I(x) = \int \frac{6}{\sin^2 x(1-\cot x)^2} dx$. If $I(0) = 3$, then $\mathrm{I}\left(\frac{\pi}{12}\right)$ is equal to :

Option 1:
$\sqrt{3}$

Option 2:
$3\sqrt{3}$

Option 3:
$6\sqrt{3}$

Option 4:
$2\sqrt{3}$

Correct Answer:
$3\sqrt{3}$

Solution:
$I(x) = \int \frac{6dx}{\sin^2 x(1-\cot x)^2} = \int \frac{6\,\text{cosec}^2\, xdx}{(1-\cot x)^2}$

$\text{Put } 1-\cot x = t$

$\text{cosec}^2\, xdx = dt$

$I = \int \frac{6dt}{t^2} = \frac{-6}{t} + c$

$I(x) = \frac{-6}{1-\cot x} c, c = 3$

$I(x) = 3 - \frac{6}{1-\cot x}, I\left(\frac{\pi}{12}\right) = 3 - \frac{6}{1-(2+\sqrt{3})}$

$I\left(\frac{\pi}{12}\right) = 3 + \frac{6}{\sqrt{3}+1} = 3 + \frac{6(\sqrt{3}-1)}{2} = 3\sqrt{3}\sqrt{2}$

Q. 13 The equations of two sides AB and AC of a triangle ABC are $4x + y = 14$ and $3x - 2y = 5$, respectively. The point $\left(2, -\frac{4}{3}\right)$ divides the third side BC internally in the ratio $2:1$. The equation of the side BC is :

Option 1:
$x - 6y - 10 = 0$

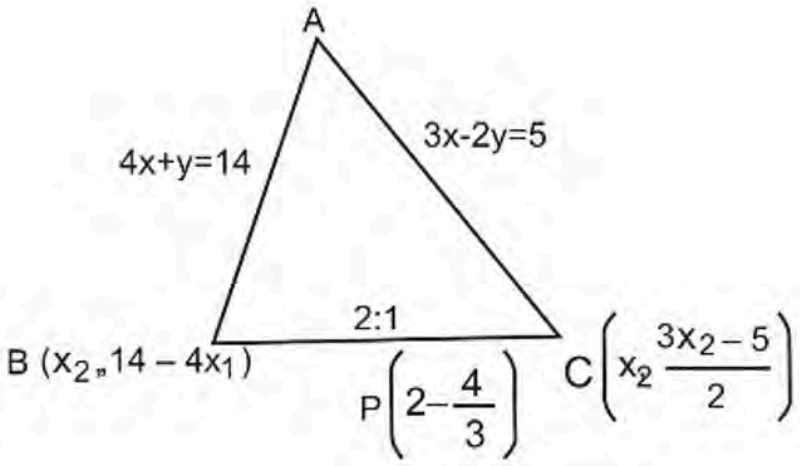
Option 2:
$x - 3y - 6 = 0$

Option 3:
$x + 3y + 2 = 0$

Option 4:
$x + 6y + 6 = 0$

Correct Answer:
$x + 3y + 2 = 0$

Solution:



$\frac{2x_2+x_1}{3} = 2, \frac{2\left(\frac{3x_2-5}{2}\right)+(14-4x_1)}{3} = \frac{-4}{3}$

$2x_2 + x_1 = 6, 3x_2 - 4x_1 = -13$

$x_2 = 1, x_1 = 4$

So, $\mathrm{C}(1,-1), \mathrm{B}(4,-2)$

$\mathrm{m} = \frac{-1}{3}$

Equation of $BC : y + 1 = \frac{-1}{3}(x-1)$

$3y + 3 = -x + 1$

$x + 3y + 2 = 0$

Q. 14 Let $[t]$ be the greatest integer less than or equal to t. Let A be the set of al prime factors of 2310 and $f : A \to \mathbb{Z}$ be the function $f(x) = \left[\log_2\left(x^2 + \left[\frac{x^3}{5}\right]\right)\right]$ The number of one-to-one functions from A to the range of f is :

Option 1:
20

Option 2:
120

Option 3:
25

Option 4:
24

Correct Answer:
120

Solution:
$N = 2310 = 231 \times 10$

$= 3 \times 11 \times 7 \times 2 \times 5$

$A = \{2, 3, 5, 7, 11\}$

$f(x) = \left[\log_2 \left(x^2 + \left[\frac{x^3}{5}\right]\right)\right]$

$f(2) = [\log_2(5)] = 2$

$f(3) = [\log_2(14)] = 3$

$f(5) = [\log_2(25 + 25)] = 5$

$f(7) = [\log_2(117)] = 6$

$f(11) = [\log_2 387] = 8$

Range of $f : B = \{2, 3, 5, 6, 8\}$

No. of one-one functions $= 5! = 120$

Q. 15 Let z be a complex number such that $|z + 2| = 1$ and $\text{lm}\left(\frac{z+1}{z+2}\right) = \frac{1}{5}$. Then the value of $|\text{Re}(\overline{z + 2})|$ is:

Option 1:
$\frac{\sqrt{6}}{5}$

Option 2:
$\frac{1+\sqrt{6}}{5}$

Option 3:
$\frac{24}{5}$

Option 4:
$\frac{2\sqrt{6}}{5}$

Correct Answer:
$\frac{2\sqrt{6}}{5}$

Solution:
$|z+2|=1, \operatorname{Im}\left(\frac{z+1}{z+2}\right)=\frac{1}{5}$

$\text{Let } z+2=\cos\theta+i\sin\theta$

$\frac{1}{z+2}=\cos\theta-i\sin\theta$

$\Rightarrow \frac{z+1}{z+2}=1-\frac{1}{z+2}=1-(\cos\theta-i\sin\theta)$

$=(1-\cos\theta)+i\sin\theta$

$\operatorname{Im}\left(\frac{z+1}{z+2}\right)=\sin\theta, \sin\theta=\frac{1}{5}$

$\cos\theta=\pm\sqrt{1-\frac{1}{25}}=\pm\frac{2\sqrt{6}}{5}$

$|\operatorname{Re}(\overline{z+2})|=\frac{2\sqrt{6}}{5}$

Q. 16 If the set $R=\{(a,b); a+5b=42, a,b\in N\}$ has m elements and $\sum_{n=1}^{m}\left(1+i^{n!}\right)=x+iy$, where $I=\sqrt{-1}$, then the value of $m+x+y$ is :

Option 1:
8

Option 2:
12

Option 3:
4

Option 4:
5

Correct Answer:
12

Solution:
$a+5b=42, a,b\in N$
$a=42-5b, b=1, a=37$
$b=2, a=32$
$b=3, a=27$

$\vdots$

$b = 8, a = 2$

R has " 8 " elements $\Rightarrow$ m $= 8$

$$\sum_{n=1}^{8}\left(1 - i^{n!}\right) = x + iy$$

for $n \geq 4, i^{n!} = 1$

$\Rightarrow (1 - i) + \left(1 - i^{2!}\right) + \left(1 - i^{3!}\right)$

$= 1 - I + 2 + 1 + 1$

$= 5 - I = x + iy$

$m + x + y = 8 + 5 - 1 = 12$

Q. 17 For the function $f(x) = (\cos x) - x + 1, x \in \mathbb{R}$, between the following two statements (S1) $f(x) = 0$ for only one value of x is $[0, \pi]$.
(S2) $f(x)$ is decreasing in $\left[0, \frac{\pi}{2}\right]$ and increasing in $\left[\frac{\pi}{2}, \pi\right]$.

Option 1:
Both (S1) and (S2) are correct

Option 2:
Only (S1) is correct

Option 3:
Both (S1) and (S2) are incorrect

Option 4:
Only (S2) is correct

Correct Answer:
Only (S1) is correct

Solution:
$f(x) = \cos x - x + 1$
$f'(x) = -\sin x - 1$

f is decreasing $\forall$x $\in$ R

$f(x) = 0$
$f(0) = 2, f(\pi) = -\pi$

f is strictly decreasing in $[0, \pi]$ and $\mathrm{f}(0) \cdot \mathrm{f}(\pi) < 0$

$\Rightarrow$ only one solution of $\mathrm{f}(\mathrm{x}) = 0$

S1 is correct and S2 is incorrect.

Q. 18 The set of all α, for which the vector $\vec{a} = \alpha t\hat{i} + 6\hat{j} - 3\hat{k}$ and $\vec{b} = t\hat{i} - 2\hat{j} - 2\alpha t\hat{k}$ are inclined at an obtuse angle for all $t \in \mathbb{R}$ is :

Option 1:
$[0, 1)$

Option 2:
$(-2, 0]$

Option 3:
$\left(-\frac{4}{3}, 0\right]$

Option 4:
$\left(-\frac{4}{3}, 1\right)$

Correct Answer:
$\left(-\frac{4}{3}, 0\right]$

Solution:
$\vec{a} = \alpha t\hat{i} + 6\hat{j} - 3\hat{k}$

$\vec{b} = t\hat{i} - 2\hat{j} - 2\alpha t\hat{k}$

so $\vec{a} \cdot \vec{b} < 0, \forall t \in R$

$\alpha t^2 - 12 + 6\alpha t < 0$

$\alpha t^2 + 6\alpha t - 12 < 0, \forall t \in R$

$\alpha < 0$, and $D < 0$

$36\alpha^2 + 48\alpha < 0$

$12\alpha(3\alpha + 4) < 0$

$\frac{-4}{3} < \alpha < 0$

$\text{also for a } = 0, \vec{a} \cdot \vec{b} < 0$

$\text{hence a } \alpha \in \left(\frac{-4}{3}, 0\right]$

Q. 19 Let $y = y(x)$ be the solution of the differential equation $\left(1 + y^2\right)e^{\tan x}dx + \cos^2 x\left(1 + e^{2\tan x}\right)dy = 0, y(0) = 1$. Then $y\left(\frac{\pi}{4}\right)$ is equal to :

Option 1:
$\frac{2}{e}$

Option 2:
$\frac{1}{e^2}$

Option 3:
$\frac{1}{e}$

Option 4:
$\frac{2}{e^2}$

Correct Answer:
$\frac{1}{e}$

Solution:
$\left(1 + y^2\right)e^{\tan x}dx + \cos^2 x\left(1 + e^{2\tan x}\right)dy = 0$

$\int \frac{\sec^2 x e^{\tan x}}{1+e^{2\tan x}}dx + \int \frac{dy}{1+y^2} = C$

$\Rightarrow \tan^{-1}\left(e^{\tan x}\right) + \tan^{-1} y = C$

$\text{for } x = 0, y = 1, \tan^{-1}(1) + \tan^{-1} 1 = C$

$C = \frac{\pi}{2}$

$\tan^{-1}\left(e^{\tan x}\right) + \tan^{-1} y = \frac{\pi}{2}$

Put $\mathrm{x}=\pi, \tan^{-1} \mathrm{e}+\tan^{-1} \mathrm{y}=\frac{\pi}{2}$

$\tan^{-1} y=\cot^{-1} e$

$y=\frac{1}{e}$

Q. 20 Let $\mathrm{H}: \frac{-\mathrm{x}^2}{\mathrm{a}^2}+\frac{\mathrm{y}^2}{\mathrm{b}^2}=1$ be the hyperbola, whose eccentricity is $\sqrt{3}$ and the length of the latus rectum is $4\sqrt{3}$. Suppose the point $(\alpha, 6), \alpha>0$ lies on H. If β is the product of the focal distances of the point $(\alpha, 6)$, then $\alpha^2+\beta$ is equal to :

Option 1:
170

Option 2:
171

Option 3:
169

Option 4:
172

Correct Answer:
171

Solution:
$\mathrm{H}: \frac{\mathrm{y}^2}{\mathrm{b}^2}-\frac{\mathrm{x}^2}{\mathrm{a}^2}=1, \mathrm{e}=\sqrt{3}$

$\mathrm{e}=\sqrt{1+\frac{\mathrm{a}^2}{\mathrm{b}^2}}=\sqrt{3} \Rightarrow \frac{\mathrm{a}^2}{\mathrm{b}^2}=2$

$a^2=2b^2$

$\text{length of L.R.}=\frac{2a^2}{b}=4\sqrt{3}$

$\mathrm{a}=\sqrt{6}$

$P(\alpha, 6)$ lie on $\frac{y^2}{3}-\frac{x^2}{6}=1$

$12-\frac{\alpha^2}{6}=1 \Rightarrow \alpha^2=66$

$$\text{Foci}=(0, \pm \text{ be})=(0,3) \& (0,-3)$$

Let $d_1 \& d_2$ be focal distances of $P(\alpha, 6)$

$d_1 = \sqrt{\alpha^2 + (6+be)^2}, d_2 = \sqrt{\alpha^2 + (6-be)^2}$

$d_1 = \sqrt{66+81}, d_2 = \sqrt{66+9}$

$\beta = d_1 d_2 = \sqrt{147 \times 75} = 105$

$\alpha^2 + \beta = 66 + 105 = 171$

Q. 21 Let $A = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$. If the sum of the diagonal elements of A^{13} is 3^{n}, then n is equal to _____

Correct Answer:
7

Solution:

$A = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$

$A^2 = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -3 \\ 3 & 0 \end{bmatrix}$

$A^3 = \begin{bmatrix} 3 & -3 \\ 3 & 0 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -6 \\ 6 & -3 \end{bmatrix}$

$A^4 = \begin{bmatrix} 3 & -6 \\ 6 & -3 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -9 \\ 9 & -9 \end{bmatrix}$

$A^5 = \begin{bmatrix} 0 & -9 \\ 9 & -9 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} -9 & -9 \\ 9 & -18 \end{bmatrix}$

$A^6 = \begin{bmatrix} -9 & -9 \\ 9 & -18 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} -27 & 0 \\ 0 & -27 \end{bmatrix}$

$A^7 = \begin{bmatrix} -27 & -0 \\ 0 & -27 \end{bmatrix}\begin{bmatrix} -54 & 27 \\ -27 & -27 \end{bmatrix} = \begin{bmatrix} 3^6 \times 2 & -27^2 \\ 27^2 & 3^6 \end{bmatrix}$

$3^7 = 3^n \Rightarrow \mathrm{n} = 7$

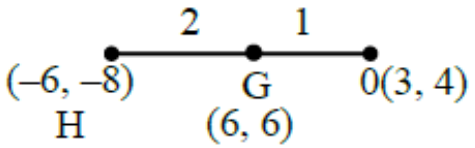
Q. 22 If the orthocentre of the triangle formed by the lines $2x + 3y - 1 = 0, x + 2y - 1 = 0$ and $ax + by - 1 = 0$, is the centroid of another triangle, whose circumecentre and orthocentre respectively are $(3, 4)$ and $(-6, -8)$, then the value of $|a - b|$ is_____.

Correct Answer:
16

Solution:
$2x + 3y - 1 = 0$

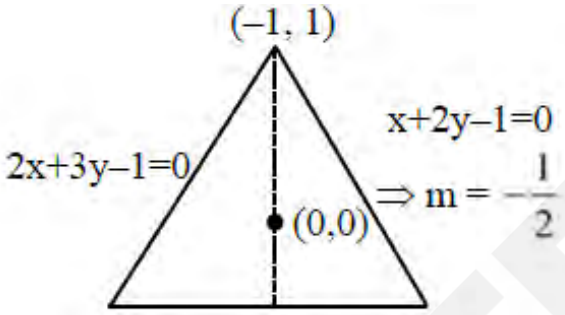
$x + 2y - 1 = 0$

$ax + by - 1 = 0$



$\left(\frac{6-6}{3}, \frac{8-8}{3}\right)$

$= (0, 0)$



$ax + by - 1 = 0$

$\left(\frac{1-0}{-1-0}\right)\left(\frac{-a}{b}\right) = -1$

$\Rightarrow -a = b$

$\Rightarrow \quad ax - ay - 1 = 0$

$ax - a\left(1 - \frac{2x}{3}\right) - 1$

$x\left(a + \frac{2a}{3}\right) = \frac{a}{3}$

$x = \frac{a+3}{5a}$

$2\left(\frac{a+3}{5a}\right) + 3y - 1 = 0$

$y = \frac{1-\frac{2a+6}{5a}}{3} = \frac{3a-6}{3\times 5a}$

$y = \frac{a-2}{5a}$

$\frac{\left(\frac{a-2}{5a}\right)}{\left(\frac{a+3}{5a}\right)} = 2 \Rightarrow a - 2 = 2a + 6$

$a = -8$

$b = 8$

$-8x + 8y - 1 = 0$

$|a - b| = 16$

Q. 23 Three balls are drawn at random from a bag containing 5 blue and 4 yellow balls. Let the random variables X and Y respectively denote the number of blue and Yellow balls. If $\bar{X}$ and $\bar{Y}$ are the means of X and Y respectively, then $7\bar{X} + 4\bar{Y}$ is equal to _____.

Correct Answer:
17

Solution:

Blue balls	0	1	2	3	4	5
Pr ob	$\frac{{}^5C_0 \cdot {}^4C_1}{{}^9C_3}$	$\frac{{}^5C_1 \cdot {}^4C_2}{{}^9C_3}$	$\frac{{}^5C_2 {}^4C_1}{{}^9C_3}$	$\frac{{}^5C_3 \cdot {}^4C_0}{{}^9C_3}$	0	0

$$7\overline{x} = \frac{{}^5C_1 {}^4C_2 + {}^5C_2 \cdot {}^4C_1 \times 2 + {}^5C_3 \cdot {}^4C_0 \times 3}{{}^9C_3} \times 7$$

$$\frac{30 + 80 + 30}{84} \times 7$$

$$= \frac{140}{12} = \frac{70}{6} = \frac{35}{3}$$

yellow	0	1	2	3	4
		${}^5C_2 {}^4C_1$	${}^5C_1^4C_2$	${}^5C_0 {}^4C_3$	0

$4\bar{y} = \frac{40+60+12}{84} \times 4 = \frac{112}{21} = \frac{16}{3}$

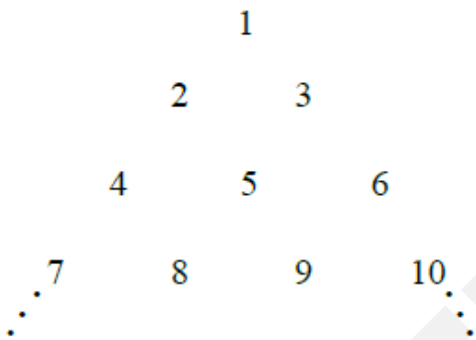
Q. 24 The number of 3-digit numbers, formed using the digits 2, 3, 4, 5 and 7, when the repetition of digits is not allowed, and which are not divisible by 3, is equal to _____.

Correct Answer:
36

Solution:
2, 3, 4, 5, 7

total number of three digit numbers not divisible
by 3 will be formed by using the digits
(4, 5, 7)
(3, 4, 7)
(2, 5, 7)
(2, 4, 7)
(2, 4, 5)
(2, 3, 5)
number of ways$= 6 \times 3! = 36$

Q. 25 Let the positive integers be written in the form :



If the k^{th} row contains exactly k numbers for every natural number k, then the row in which the number 5310 will be, is_____.

Correct Answer:
103

Solution:

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$S = 1 + 2 + 4 + 7 + \ldots\ldots + T_n$

$S = 1 + 2 + 4 + \ldots\ldots$

$Tn = 1 + 1 + 2 + 3 + \ldots\ldots + (T_n - T_{n-1})$

$T_n = 1 + \left(\frac{n-1}{2}\right)[2 + (n-2) \times 1]$

$T_n = 1 + 1 + \frac{n(n-1)}{2}$

$n = 100 \quad T_n = 1 + \frac{100 \times 99}{2} = 4950 + 1$

$n = 101 \quad T_n = 1 + \frac{101 \times 100}{2} = 5050 + 1 = 5051$

$n = 102 \quad T_n = 1 + \frac{102 \times 101}{2} = 5151 + 1 = 5152$

$n = 103 \quad T_n = 1 + \frac{103 \times 102}{2} = 5254$

$n = 104 \quad T_n = 1 + \frac{104 \times 103}{2} = 5357$

Q. 26 If the range of $f(\theta) = \frac{\sin^4\theta + 3\cos^2\theta}{\sin^4\theta + \cos^2\theta}, \theta \in \mathbb{R}\ [\alpha, \beta]$, then the sum of the infinite G.P., whose firs term is 64 and the common ratio is $\frac{\alpha}{\beta}$, is equal to _____.

Correct Answer:
96

Solution:

$$f(\theta) = \frac{\sin^4\theta + 3\cos^2\theta}{\sin^4\theta + \cos^2\theta}$$

$$f(\theta) = 1 + \frac{2\cos^2\theta}{\sin^4\theta + \cos^2\theta}$$

$$f(\theta) = \frac{2\cos^2\theta}{\cos^4\theta - \cos^2\theta + 1} + 1$$

$$f(\theta) = \frac{2}{\cos^2\theta + \sec^2\theta - 1} + 1$$

$$f(\theta)|_{\text{min.}} = 1$$

$$f(\theta)_{\text{max.}} = 3$$

$$S = \frac{64}{1 - 1/3} = 96$$

Q. 27 Let $\alpha = \sum_{r=0}^{n} \left(4r^2 + 2r + 1\right) {}^nC_r$ and $\beta = \left(\sum_{r=0}^{n} \frac{{}^nC_r}{r+1}\right) + \frac{1}{n+1}$. If $140 < \frac{2\alpha}{\beta} < 281$,then the value of n is_____.

Correct Answer:
5

Solution:

$$\alpha = \sum_{r=0}^{n} \left(4r^2 + 2r + 1\right) \cdot {}^{n}C_r$$

$$\alpha = 4\sum_{r=0}^{n} r^2 \cdot \frac{n}{r} \cdot {}^{n-1}C_{r-1} + 2\sum_{r=0}^{n} r \cdot \frac{n}{r} \cdot {}^{n-1}C_{r-} + \sum_{r=0}^{n} {}^{n}C_r$$

$$+ 4n\sum_{r=0}^{n} {}^{n-1}C_{r-1} + 2n\sum_{r=0}^{n} {}^{n-1}C_{r-1} + \sum_{r=0}^{n} {}^{n}C_r$$

$$\alpha = 4n(n-1) \cdot 2^{n-2} + 4n \cdot 2^{n-1} + 2n \cdot 2^{n-1} + 2^n$$

$$\alpha = 2^{n-2}[4n(n-1) + 8n + 4n + 4]$$

$$\alpha = 2^{n-2}\left[4n^2 + 8n + 4\right]$$

$$\alpha = 2n(n+1)^2$$

$$\beta = \sum_{r=0}^{n} \frac{{}^{n}C_r}{r+1} + \frac{1}{n+1}$$

$$= \sum_{r=0}^{n} \frac{{}^{n+1}C_{r+1}}{n+1} + \frac{1}{n+1}$$

$$= \frac{1}{n+1}\left(1 + {}^{n+1}C_1 + \ldots. + {}^{n+1}C_{n+1}\right)$$

$$= \frac{2^{n+1}}{n+1}$$

$$\frac{2\alpha}{\beta} = \frac{2^{n+1}(n+1)^2}{2^{n+1}} \cdot (n+1) = (n+1)^3$$

$$140 < (n+1)^3 < 281$$

$$n = 4 \Rightarrow (n+1)^3 = 125$$

$$n = 5 \Rightarrow (n+1)^3 = 216$$

$$n = 6 \Rightarrow (n+1)^3 = 343$$

$$\therefore n = 5$$

Q. 28 Let $\vec{a} = 9\hat{i} - 13\hat{j} + 25\hat{k}, \vec{b} = 3\hat{i} + 7\hat{j} - 13\hat{k}$ and $\vec{c} = 17\hat{i} - 2\hat{j} + \hat{k}$ be three given vectros. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{a} = (\vec{b} + \vec{c}) \times \vec{a}$ and $\vec{r} \cdot (\vec{b} - \vec{c}) = 0$, then $\frac{|593\vec{r} + 67\vec{a}|^2}{(593)^2}$ is equal to _____.

Correct Answer:
569

Solution:

$\vec{a} = 9\hat{i} - 13\hat{j} + 25\hat{k}$

$\vec{b} = 3\hat{i} + 7\hat{j} - 13\hat{k}$

$\vec{c} = 17\hat{i} - 2\hat{j} + \hat{k}$

$\vec{b} + \vec{c} = 20\hat{i} + 5\hat{j} - 12\hat{k}$

$\vec{b} - \vec{c} = -14\hat{i} + 9\hat{j} - 14\hat{k}$

$(\vec{r} - (\vec{b} + \vec{c})) \times \vec{a} = 0$

$r - (\vec{b} + \vec{c}) = \lambda\vec{a}$

$\vec{r} = \lambda\vec{a} + \vec{b} + \vec{c}$

$\text{But } \vec{r} \cdot (\vec{b} - \vec{c}) = 0$

$\Rightarrow (\lambda\vec{a} + \vec{b} + \vec{c}) \cdot (\vec{b} - \vec{c}) = 0$

$\Rightarrow \lambda\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{b} + \vec{c} \cdot \vec{b} - \lambda\vec{a} \cdot \vec{c} - \vec{b} \cdot \vec{c} - \vec{c} \cdot \vec{c} = 0$

$\lambda = \dfrac{\vec{c} \cdot \vec{c} - \vec{b} \cdot \vec{b}}{\vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{c}} = \dfrac{294 - 227}{-389 = 204} = \dfrac{-67}{593}$

$\therefore \vec{r} = \vec{b} + \vec{c} - \dfrac{67}{593}\vec{a}$

$\Rightarrow 593\vec{r} + 67\vec{a} = 593(\vec{b} + \vec{c})$

$\Rightarrow |\vec{b} + \vec{c}|^2 = 569$

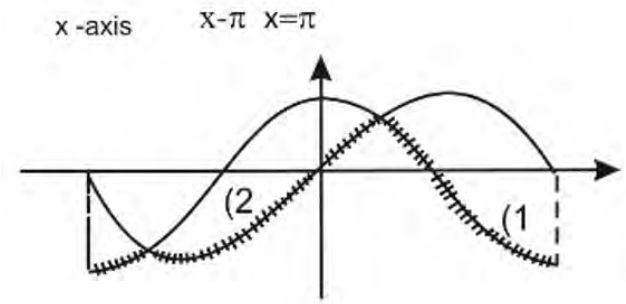
Q. 29 Let the area of the region enclosed by the curve $y = \min\{\sin x, \cos x\}$ and the x-axis between $x = -\pi$ to $x = \pi$ be A. Then A^2 is equal to _____.

Correct Answer:

16

Solution:

$y = \min\{\sin x, \cos x\}$



$$\int_0^{\pi/4} \sin x = (\cos x)_{\pi/4}^{0} = 1 - \frac{1}{\sqrt{2}}$$

$$\int_{-\pi}^{-3\pi/4} (\sin x - \cos x) = (-\cos x - \sin x)_{-\pi}^{-3\pi/4}$$

$$= (\cos x + \sin x)_{-3\pi/4}^{-\pi}$$

$$= (-1 + 0) - \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right)$$

$$= -1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}$$

$$\int_{\pi/4}^{\pi/2} \cos x dx = (\sin x)_{\pi/4}^{\pi/2} = 1 - \frac{1}{\sqrt{2}}$$

$$A_{=4}^{2}$$

$$A^2 = 16$$

Q. 30 The value of $\lim_{x\to 0} 2\left(\frac{1-\cos x\sqrt{\cos 2x}\sqrt[3]{\cos 3x}.....\sqrt[10]{\cos 10x}}{x^2}\right)$ is _____.

Correct Answer:
55

Solution:

$$\lim_{x\to 0} 2\left(\frac{1-\left(1-\frac{x^2}{2!}\right)\left(1-\frac{4x^2}{2!}\right)\left(1-\frac{9x^2}{2!}\right).....\left(1-\frac{100x^2}{2!}\right)}{x^2}\right)$$

By expansion

$$\lim_{x\to 0} \frac{2\left(1-\left(1-\frac{x^2}{2}\right)\right)\left(1-\frac{1}{2}\cdot\frac{4x^2}{2}\right)\left(1-\frac{1}{3}\cdot\frac{9x^2}{2}\right)....\left(1-\frac{1}{10}\cdot\frac{100x^2}{2}\right)}{x^2}\Bigg)$$

$$\lim_{x\to 0} 2\left(\frac{1-\left(1-\frac{x^2}{2}\right)\left(1-\frac{2x^2}{2}\right)\left(1-\frac{3x^2}{2}\right).....\left(1-\frac{10x^2}{2}\right)\right)}{x^2}\Bigg)$$

$$\lim_{x\to 0} \frac{2\left(1-1+x^2\left(\frac{1}{2}+\frac{2}{2}+\frac{3}{2}+....+\frac{10}{2}\right)\right)}{x^2}$$

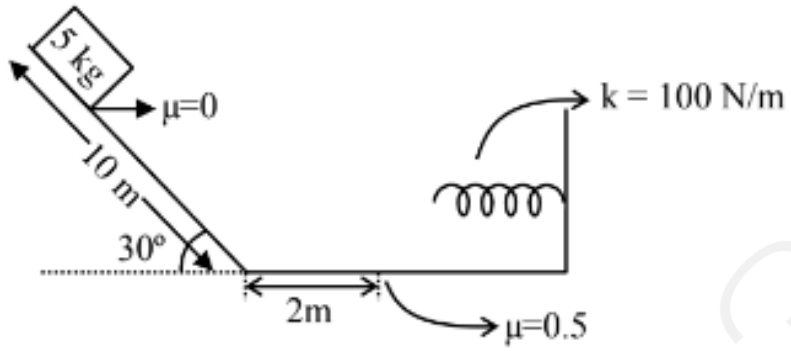
$$2\left(\frac{1}{2}+\frac{2}{2}+\frac{3}{2}+....+\frac{10}{2}\right)$$

$$1+2+.....+10 = \frac{10\times 11}{2} = 55$$

8 Apr - Evening Shift

Physics

Q. 1



A block is simply released from the top of an inclined plane as shown in the figure above. The maximum compression in the spring when the block hits the spring is:

Option 1:
$\sqrt{6}$ m

Option 2:
2 m

Option 3:
1 m

Option 4:
$\sqrt{5}$ m

Correct Answer:
2 m

Solution:

CAREERS360

$$w_g + w_{Fr} + w_s = \Delta KE$$
$$5 \times 10 \times 5 - 0.5 \times 5 \times 10 \times x - \frac{1}{2}Kx^2 = 0 - 0$$
$$250 = 25x + 50x^2$$
$$2x^2 + x - 10 = 0$$
$$x = 2$$

Q. 2 In a hypothetical fission reaction
$_{92}X^{236} \rightarrow {}_{56}Y^{141} + {}_{36}Z^{92} + 3R$
The identity of emitted particles (R) is :

Option 1:
Proton

Option 2:
Electron

Option 3:
Neutron

Option 4:
γ-radiations

Correct Answer:
Neutron

Solution:
Z in LHS = 92
Z in RHS = 56 + 36 = 92
A in LHS = 236
A in RHS = 141 + 92 = 233
So 3 neutrons are released.

Q. 3 If ϵ_0 is the permittivity of free space and E is the electric field, then $\in_0 E^2$ has the dimensions :

Option 1:
$\left[\mathrm{M}^{0}\ \mathrm{L}^{-2}\ \mathrm{T}\ \mathrm{A}\right]$

Option 2:
$\left[\mathrm{ML}^{-1}\ \mathrm{T}^{-2}\right]$

Option 3:
$\left[\mathrm{M}^{-1}\ \mathrm{L}^{-3}\ \mathrm{T}^{4}\ \mathrm{A}^{2}\right]$

Option 4:
$\left[\mathrm{ML}^{2}\ \mathrm{T}^{2}\right]$

Correct Answer:
$\left[\mathrm{ML}^{-1}\ \mathrm{T}^{-2}\right]$

Solution:

$$\mathrm{E}=\frac{\mathrm{KQ}}{\mathrm{R}^{2}}$$

$$\mathrm{E}=\frac{\mathrm{Q}}{4\pi\varepsilon_{0}\mathrm{R}^{2}}$$

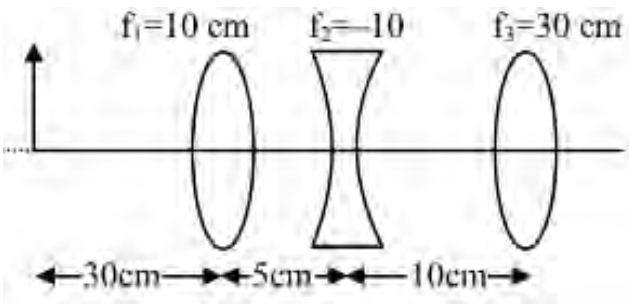
$$\varepsilon_{0}=\frac{\mathrm{Q}}{4\pi\mathrm{R}^{2}\mathrm{E}}$$

$$\text{Now, } \varepsilon_{0}\mathrm{E}^{2}=\frac{\mathrm{Q}}{4\pi\mathrm{R}^{2}\mathrm{E}}\cdot\mathrm{E}^{2}=\frac{\mathrm{Q}}{4\pi\mathrm{R}^{2}}\cdot\mathrm{E}$$

$$\left[\varepsilon_{0}\mathrm{E}^{2}\right]=\left[\frac{\mathrm{QE}}{\mathrm{R}^{2}}\right]=\frac{[\mathrm{Q}][\mathrm{E}]}{[\mathrm{R}^{2}]}=\frac{[\mathrm{Q}]}{[\mathrm{R}]}\frac{[\mathrm{W}]}{[\mathrm{Q}][\mathrm{R}]}$$

$$=\frac{[\mathrm{W}]}{[\mathrm{R}^{3}]}=\frac{\mathrm{ML}^{2}\ \mathrm{T}^{-2}}{\mathrm{L}^{3}}=\mathrm{ML}^{-1}\ \mathrm{T}^{-2}$$

Q. 4 The position of the image formed by the combination of lenses is :



Option 1:
30 cm (right of third lens)

Option 2:
15 cm (left of second lens)

Option 3:
30 cm (left of third lens)

Option 4:
15 cm (right of second lens)

Correct Answer:
30 cm (right of third lens)

Solution:
For lens 1: $\mathrm{f_l} = 10, \mathrm{u} = -30, \mathrm{v} = ?$

$$\mathrm{v} = \frac{\mathrm{uf}}{\mathrm{u+f}} = \frac{-30\times10}{-30+10} = 15$$

For lens 2:

$$f_1 = -10, \mathrm{u} = 10, \mathrm{v} = ?$$

$$\mathrm{v} = \frac{\mathrm{uf}}{\mathrm{u+f}} = \frac{10\times-10}{10-10} = \infty$$

For lense 3 : $f = 30, u = -\infty, v = ?$

So v will be 30 .

Q. 5 A plane progressive wave is given by $\mathrm{y} = 2\cos 2\pi(330\mathrm{t} - \mathrm{x})\mathrm{m}$. The frequency of the wave is :

Option 1:
165 Hz

Option 2:
330 Hz

Option 3:
660 Hz

Option 4:
340 Hz

Correct Answer:
330 Hz

Solution:
$y = 2\cos 2\pi(330t - x)m$
$y = A\cos(\omega t - kx)$
by comparing $\omega = 2\pi \times 330$
$2\pi f = 2\pi \times 330$
$f = 330$

Q. 6 A thin circular disc of mass M and radius R is rotating in a horizontal plane about an axis passing through its centre and perpendicular to its plane with angular velocity ω. If another disc of same dimensions but of mass $\mathrm{M}/2$ is placed gently on the first disc co-axially, then the new angular velocity of the system is :

Option 1:
$\frac{4}{5}\omega$

Option 2:
$\frac{5}{4}\omega$

Option 3:
$\frac{2}{3}\omega$

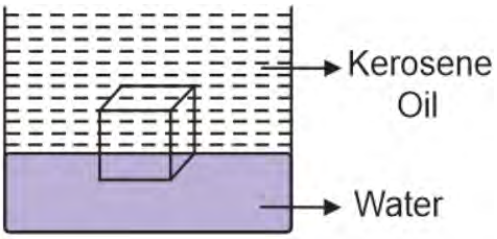
Option 4:
$\frac{3}{2}\omega$

Correct Answer:
$\frac{2}{3}\omega$

Solution:

$$I_1\omega = I_2\omega_2$$
$$\frac{MR^2}{2}\omega = \frac{3}{2}\left(\frac{MR^2}{2}\right)\omega_2$$
$$\omega_2 = \frac{2}{3}\omega$$

Q. 7 A cube of ice floats partly in water and partly in kerosene oil. The radio of volume of ice immersed in water to that in kerosene oil (specific gravity of Kerosene oil $= 0.8$, specific gravity of ice $= 0.9$)



Option 1:
8:9

Option 2:
5:4

Option 3:
9:10

Option 4:
1:1

Correct Answer:
1:1

Solution:

CAREERS360

$v_1 =$ volume immersed in water.

$v_2 =$ volume immersed in oil.

$$v_1\rho_w g + v_2\rho_o g = (v_1 + v_2)\rho_c g$$

$$v_1 + \frac{v_2\rho_v}{\rho_w} = (v_1 + v_2)\frac{\rho_c}{\rho_w}$$

$$= v_1 + 0.8v_2 = 0.9v_1 + 0.9v_2$$

$$= 0.1v_1 = 0.1v_2$$

$$v_1 : v_2 = 1 : 1$$

Q. 8 Given below are two statements :

Statement (I) : The mean free path of gas molecules is inversely proportional to square of molecular diameter.

Statement (II) : Average kinetic energy of gas molecules is directly proportional to absolute temperature of gas.
In the light of the above statements, choose the correct answer from the option given below:

Option 1:
Statement I is false but Statement II is true.

Option 2:
Statement I is true but Statement II is false.

Option 3:
Both Statement I and Statement II are false.

Option 4:
Both Statement I and Statement II are true.

Correct Answer:
Both Statement I and Statement II are true.

Solution:

$$\lambda = \frac{RT}{\sqrt{2}\pi d^2 N_A P}$$

$$KE = \frac{f}{2}nRT$$

Q. 9 Two satellite A and B go round a planet in circular orbits having radii 4 R and R respectively. If the speed of A is $3v$, the speed of B will be :

Option 1:
$3v$

Option 2:
$\frac{4}{3}v$

Option 3:
$6v$

Option 4:
$12v$

Correct Answer:
$6v$

Solution:

$$v = \sqrt{\frac{GM}{R}}$$

$$\frac{v_A}{v_B} = \sqrt{\frac{R_B}{R_A}} = \sqrt{\frac{R}{4R}} = \frac{1}{2}$$

$$v_B = 2v_A = 6v$$

Q. 10 A long straight wire of radius a carries a steady current I. The current is uniformly distributed across its cross section. The ratio of the magnetic field at $\frac{a}{2}$ and 2 a from axis of the wire is :

Option 1:
1:4

Option 2:
4:1

Option 3:
1:1

Option 4:
3:4

Correct Answer:
1:1

Solution:

$$B_1 2\pi \frac{a}{2} = \mu_0 \frac{1}{4}$$

$$B_1 = \frac{\mu_0 I}{4\pi a}$$

$$B_2 2\pi 2a = \mu_0 I$$

$$B_2 = \frac{\mu_0 I}{4\pi a}$$

Q. 11 The angle of projection for a projectile to have same horizontal range and maximum height is :

Option 1:
$\tan^{-1}(2)$

Option 2:
$\tan^{-1}(4)$

Option 3:
$\tan^{-1}\left(\frac{1}{4}\right)$

Option 4:
$\tan^{-1}\left(\frac{1}{2}\right)$

Correct Answer:
$\tan^{-1}(4)$

Solution:

$$\frac{u^2 \sin 2\theta}{g} = \frac{u^2 \sin^2 \theta}{2g}$$

$$4 \sin\theta \cos\theta = \sin^2\theta$$

$$4 = \tan\theta$$

Q. 12 Water boils in an electric kettle in 20 minutes after being switched on. Using the same main supply, the length of the heating element should be ..... to ..... times of its initial length if the water is to be boiled in 15 minutes.

Option 1:
increased, $\frac{3}{4}$

Option 2:
increased, $\frac{4}{3}$

Option 3:
decreased, $\frac{3}{4}$

Option 4:
decreased, $\frac{4}{3}$

Correct Answer:
decreased, $\frac{3}{4}$

Solution:

$$\mathrm{P} = \frac{\mathrm{v}^2}{\mathrm{R}}, \mathrm{R} = \frac{\rho\ell}{\mathrm{A}}$$

$$\mathrm{P} \propto \frac{1}{\ell}$$

$$\frac{\mathrm{P}_1}{\mathrm{P}_2} = \frac{\mathrm{t}_2}{\mathrm{t}_1} = \frac{15}{20} = \frac{\ell_2}{\ell_1}$$

$$\ell_2 = \frac{3}{4}\ell_1$$

Q. 13 A capacitor has air as dielectric medium and two conducting plates of area $12\ \mathrm{cm}^2$ and they are $0.6\ \mathrm{cm}$ apart. When a slab of dielectric having area $12\ \mathrm{cm}^2$ and $0.6\ \mathrm{cm}$ thickness is inserted between the plates, one of the conducting plates has to be moved by $0.2\ \mathrm{cm}$ to keep the capacitance same as in previous case. The dielectric constant of the slab is : $\left(\text{ Given } \epsilon_0 = 8.834 \times 10^{-12}\ \mathrm{F/m}\right)$

Option 1:
1.50

Option 2:
1.33

Option 3:
0.66

Option 4:
1

Correct Answer:
1.50

Solution:

$$\frac{\mathrm{A}\varepsilon_\mathrm{o}}{\mathrm{d}} = \frac{\mathrm{A}\varepsilon_\mathrm{o}}{\left(0.2 + \frac{\mathrm{d}}{\mathrm{k}}\right)}$$

$$0.6 = 0.2 + \frac{0.6}{\mathrm{k}}$$

$$\mathrm{k} = \frac{3}{2}$$

Q. 14 A given object takes n times the time to slide down 45° rough inclined plane as it takes the time to slide down an identical perfectly smooth 45° inclined plane. The coefficient of kinetic friction between the object and the surface of inclined plane is:

Option 1:
$1-\frac{1}{n^2}$

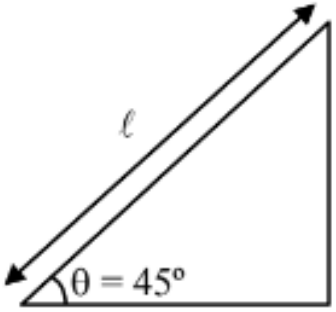
Option 2:
$1-n^2$

Option 3:
$\sqrt{1-\frac{1}{n^2}}$

Option 4:
$\sqrt{1-n^2}$

Correct Answer:
$1-\frac{1}{n^2}$

Solution:



Case - 1 No friction

$$a = g\sin\theta$$

$$\ell = \frac{1}{2}(g\sin\theta)t_1^2$$

$$t_1 = \sqrt{\frac{2\ell}{g\sin\theta}}$$

Case - 2 With friction

$$a = g\sin\theta - \mu g\cos\theta$$

$$\ell = \frac{1}{2}(g\sin\theta - \mu g\cos\theta)t_2^2$$

$$\sqrt{\frac{2\ell}{g\sin\theta - \mu g\cos\theta}} = n\sqrt{\frac{2\ell}{g\sin\theta}}$$

$$\mu = 1 - \frac{1}{n^2}$$

Q. 15 A coil of negligible resistance is connected in series with 90Ω resistor across $120\ V, 60\ Hz$ supply. A voltmeter reads $36\ V$ across resistance. Inductance of the coil is :

Option 1:
0.76 H

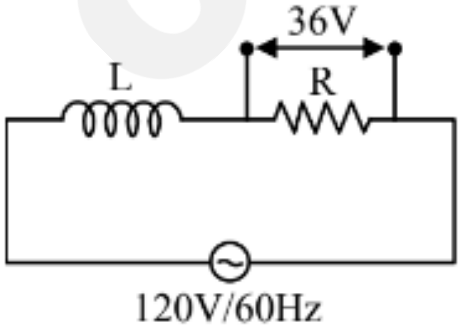
Option 2:
2.86 H

Option 3:
2.286 H

Option 4:
0.91 H

Correct Answer:
0.76 H

Solution:



$$36 = I_{\text{rms}}\, R$$

$$36 = \frac{120}{\sqrt{X_L^2 + R^2}} \times R$$

$$R = 90\Omega \Rightarrow 36 = \frac{120 \times 90}{\sqrt{X_L^2 + 90^2}}$$

$$\sqrt{X_L^2 + 90^2} = 300$$

$$X_L^2 = 81900$$

$$X_L = 286.18$$

$$\omega L = 286.18$$

$$L = \frac{286.18}{376.8}$$

$$L = 0.76\text{H}$$

Q. 16 There are 100 divisions on the circular scale of a screw gauge of pitch 1 mm. With no measuring quantity in between the jaws, the zero of the circular scale lies 5 divisions below the reference line. The diameter of a wire is then measured using this screw gauge. It is found the 4 linear scale divisions are clearly visible while 60 divisions on circular scale coincide with the reference line. The diameter of the wire is :

Option 1:
4.65 mm

Option 2:
4.55 mm

Option 3:
4.60 mm

Option 4:
3.35 mm

Correct Answer:
4.55 mm

Q. 17 A proton and an electron have the same de Broglie wavelength. If K_p and K_e be the kinetic energies of proton and electron respectively. Then choose the correct relation :

Option 1:
$K_p > {K_e}^2$

Option 2:
$K_p = K_e$

Option 3:
$K_p = {K_e}^2$

Option 4:
$K_p < K_e$

Correct Answer:
$K_p < K_e$

Solution:
De Broglie wavelength of proton \& electron $= \lambda$

$$\because \lambda = \frac{h}{p}$$

$$\therefore p_{proton} = p_{elcectron}$$

$$\because KE = \frac{p^2}{2m}$$

$$\therefore KE_{proton} < KE_{electron}$$

$$[K_p < K_e]$$

Q. 18 Least count of a vernier caliper is $\frac{1}{20\ N}\ cm$. The value of one division on the main scale is 1 mm. Then the number of divisions of main scale that coincide with N divisions of vernier scale is :

Option 1:
$\left(\frac{2\,N-1}{20\,N}\right)$

Option 2:
$\left(\frac{2\,N-1}{2}\right)$

Option 3:
$(2\,N-1)$

Option 4:
$\left(\frac{2\,N-1}{2\,N}\right)$

Correct Answer:
$\left(\frac{2\,N-1}{2}\right)$

Solution:
Least count of vernier calipers $=\frac{1}{20\,N}\ \text{cm}$
$\because$ Least count $=1$ MSD -1 VSD
let x no. of divisions of main scale coincides with N division of vernier scale, then

$$1\text{VSD}=\frac{x\times 1\ \text{mm}}{N}$$
$$\therefore\ \frac{1}{20\,N}\ \text{cm}=1\ \text{mm}-\frac{x\times 1\ \text{mm}}{N}$$
$$\frac{1}{2\,N}\ \text{mm}=1\ \text{mm}-\frac{x}{N}\text{mm}$$
$$x=\left(1-\frac{1}{2\,N}\right)N$$
$$x=\frac{2\,N-1}{2}$$

Q. 19 If M_0 is the mass of isotope $^{12}_{5}B$, M_p and M_n are the masses of proton and neutron, then nuclear binding energy of isotope is :

Option 1:
$(5M_p+7M_n-M_o)C^2$

Option 2:
$(\mathrm{M_0} - 5\mathrm{M_p})\mathrm{C}^2$

Option 3:
$(\mathrm{M_0} - 12\mathrm{M_n})\mathrm{C}^2$

Option 4:
$(\mathrm{M_0} - 5\mathrm{M_p} - 7\mathrm{M_n})\mathrm{C}^2$

Correct Answer:
$(5\mathrm{M_p} + 7\mathrm{M_n} - \mathrm{M_o})\mathrm{C}^2$

Solution:
$\mathrm{B.E.} = \Delta mC^2$

$(5M_p + 7M_n - M_0)C^2$

Q. 20 A diatomic gas $(\gamma = 1.4)$ does 100 J of work in an isobaric expansion. The heat given to the gas is :

Option 1:
350 J

Option 2:
490 J

Option 3:
150 J

Option 4:
250 J

Correct Answer:
350 J

Solution:

For Isobaric process

$$w = P\Delta v = nR\Delta T = 100\ J$$

$$Q = \Delta u + w$$

$$\Delta Q = \frac{F}{2} nR \wedge T + nR\Lambda T$$

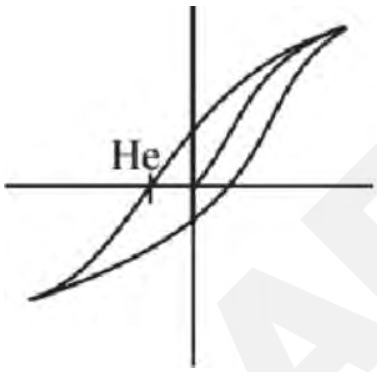
$$\left(\frac{f}{2} + 1\right) nR\Delta T$$

$$\left(\frac{5}{2} + 1\right) 100 = 350\ J$$

Q. 21 The coercivity of a magnet is $5 \times 10^3\ A/m$. The amount of current required to be passed in a solenoid of length 30 cm and the number of turns 150 , so that the magnet gets demagnetised when inside the solenoid is _____ A.

Correct Answer:
10

Solution:



$$H_c = \frac{\mu_o ni}{\mu_o}$$

$$5 \times 10^3 = \frac{150}{30} \times 100 \times i$$

$$\frac{50}{5} = i$$

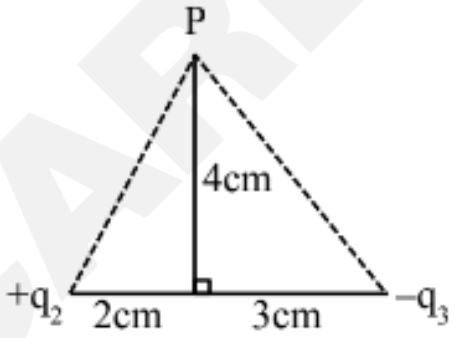
$$I = 10$$

Q. 22 Small water droplets of radius $0.01\ \mathrm{mm}$ are formed in the upper atmosphere and falling with a terminal velocity of $10\ \mathrm{cm/s}$. Due to condensation, if $8\mathrm{such}$ droplets are coalesced and formed a larger drop, the new terminal velocity will be_____cm/s.

Correct Answer:
40

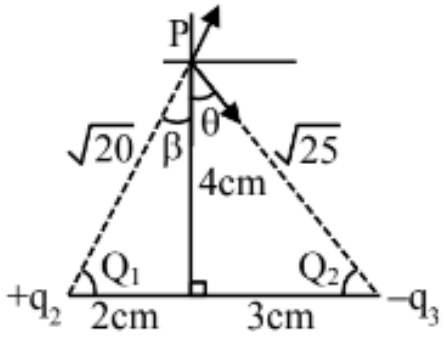
Solution:
$\mathrm{m} =$ mass of small drop
$\mathrm{M} =$ mass of bigger drop
$$\mathrm{V_t} = \frac{2}{9}\frac{\mathrm{R}^2(\rho-\sigma)\mathrm{g}}{\eta}$$
$8 \propto \mathrm{m} = \mathrm{M}$
$8\mathrm{r}^3 = \mathrm{R}^3 \Rightarrow \mathrm{R} = 2\mathrm{R}$
as $\mathrm{V_t} \times \mathrm{R}^2$ $\because$ Radius double so $\mathrm{V_t}$ becomes 4 time
$\therefore 4 \times 10 = 40\ \mathrm{cm/s}$

Q. 23 If the net electric field at point P along Y axis is zero, then the ratio of $\frac{q_2}{q_3}$ is $\frac{8}{5\sqrt{x}}$, where $\mathrm{x}=$_____.



Correct Answer:
5

Solution:



$$\frac{Kq_2}{20}\cos\beta = \frac{Kq_3}{25}\cos\theta$$

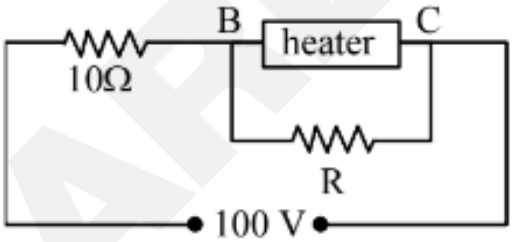
$$\frac{Kq_2}{20}\frac{4}{\sqrt{20}} = \frac{Kq_3}{25}\frac{4}{\sqrt{25}}$$

$$\frac{q_2}{q_3} = \frac{20}{25}\sqrt{\frac{20}{25}} = \frac{8}{5\sqrt{x}}$$

$$\Rightarrow \sqrt{x} = \frac{8 \times 25\sqrt{25}}{5 \times 20\sqrt{20}}$$

$$x = 5$$

Q. 24 A heater is designed to operate with a power of 1000 W in a 100 V line. It is connected in combination with a resistance of 10Ω and a resistance R, to a 100 V mains as shown in figure. For the heater to operate at 62.5 W, the value of R should be_____Ω.



Correct Answer:
5

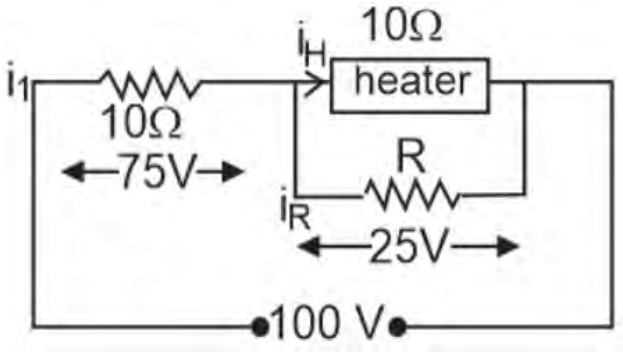
Solution:

$$R_{\text{heater}} = \frac{V^2}{P} = \frac{(100)^2}{1000} = 10\Omega$$

$$\text{For heater } P = \frac{V^2}{R} \Rightarrow V = \sqrt{PR}$$

$$V = \sqrt{62.5 \times 10}$$

$$V = 25v$$



$$i_1 = \frac{75}{10} = 7.5A, i_H = \frac{25}{10} = 2.5 \text{ A},$$

$$i_R = i_1 - i_H = 5$$

$$V = IR$$

$$R = \frac{25}{5} = 5\Omega$$

Q. 25 An alternating emf $\mathrm{E} = 110\sqrt{2}\sin 100\mathrm{t}$ volt is applied to a capacitor of $2\mu\mathrm{F}$, the rms value of current in the circuit is_____$mA.

Correct Answer:
22

Solution:

$$C = 2\mu f; \quad E = 110\sqrt{2}\sin(100t)$$

$$X_C = \frac{1}{\omega c} = \frac{1}{100 \times 2 \times 10^6}$$

$$= \frac{10000}{2} = 5000\Omega$$

$$i_o = \frac{110\sqrt{2}}{5000}$$

$$i_{ms} = \frac{110\sqrt{2}}{5000\sqrt{2}}$$

$$= \frac{110}{5}\,mA$$

$$= 22\,mA$$

Q. 26 Two slits are 1 mm apart and the screen is located 1 m away from the slits. A light wavelength 500 nm is used. The width of each slit to obtain 10 maxima of the double slit pattern within the central maximum of the single slit pattern is _____\times 10^{-4} \mathrm{~m}$.

Correct Answer:
2

Solution:

$$d = 1\,mm, D = 1\,m, \lambda = 500\,nm$$

$$10\left(\frac{\lambda D}{d}\right) = \frac{2\lambda D}{a}$$

$$a = \frac{d}{5}$$

$$= \frac{10 \times 10^{-4}\,m}{5}$$

$$= 2 \times 10^4$$

Q. 27 An object of mass 0.2 kg executes simple harmonic motion along x axis with frequency of $\left(\frac{25}{\pi}\right)$ Hz. At the position $x = 0.04$ m the object has kinetic energy 0.5 J and potential energy 0.4 J. The amplitude of oscillation is _____cm.

Correct Answer:
6

Solution:

$$\text{Total energy} = \text{K.E.} + \text{P.E.}$$

$$\text{at } x = 0.04 \text{ m}, \text{ T.E.} = 0.5 + 0.4 = 0.9 \text{ J}$$

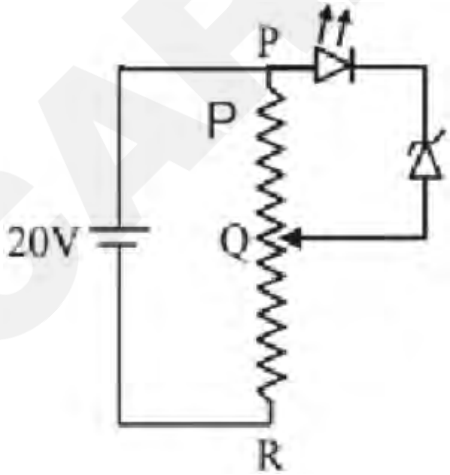
$$\text{T.E} = 1 \text{ m}^2 \text{ A}^2 = 0.9$$

$$= \frac{1}{2} \times 0.2 \left(2\pi \times \frac{25}{\pi}\right)^2 \times \text{A}^2 = 0.9$$

$$\Rightarrow \text{A} = 0.06 \text{ m}$$

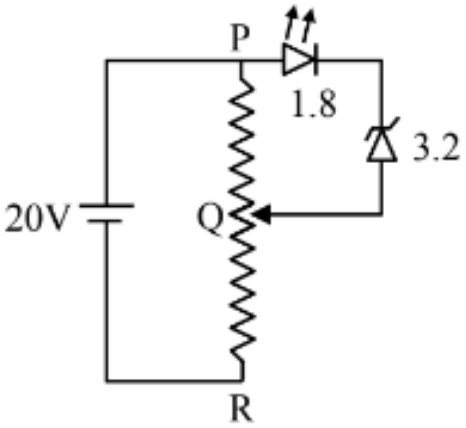
$$\text{A} = 6 \text{ cm}$$

Q. 28 A potential divider circuit is connected with a dc source of 20 V, a light emitting diode of glow in voltage 1.8 V and a zener diode of breakdown voltage of 3.2 V. The length (PR) of the resistive wire is 20 cm. The minimum length of PQ to just glow the LED is ..... cm.



Correct Answer:
5

Solution:



$$PR = 20 \text{ cm}$$

$$V_{PQ} = \frac{1}{4} \times R_{PR}$$

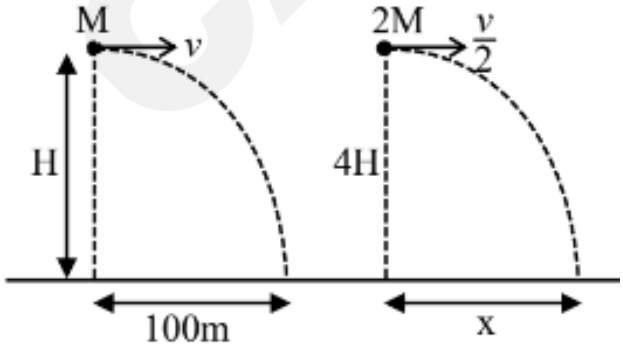
$$\ell_{min}(PQ) = \frac{1}{4} \times 20$$

$$= 5 \text{ cm}$$

Q. 29 A body of mass M thrown horizontally with velocity v from the top of the tower of height H touches the ground at a distance of 100 M from the foot of the tower. A body of mass 2 M thrown at a velocity $\frac{v}{2}$ from the top of the tower of height 4 H will touch the ground at a distance of .....m.

Correct Answer:
100

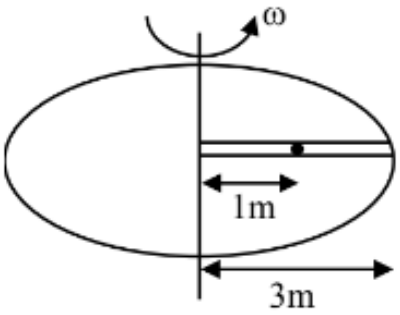
Solution:



$$100 = v\sqrt{\frac{2H}{g}}; \quad x = \frac{v}{2}\sqrt{\frac{2(4H)}{g}} = v\sqrt{\frac{2H}{g}}$$

$$\Rightarrow x = 100$$

Q. 30 A circular table is rotating with an angular velocity of $\omega \text{rad/s}$ about its axis (see figure). There is a smooth groove along a radial direction on the table. A steel ball is gently placed at a distance of $1\ \text{m}$ on the groove. All the surface are smooth. If the radius of the table is $3\ \text{m}$, the radial velocity of the ball w.r.t. the table at the time ball leaves the table is $x\sqrt{2}\omega \text{m/s}$, where the value of x is _____



Correct Answer:
2

Solution:

$a_c = \omega^2 x$

$\frac{vdv}{dx} = \omega^2 x$

$\int_0^v vdv = \int_1^3 \omega^2 x dx$

$\frac{v^2}{2} = \omega^2 \left[\frac{x^2}{2}\right]$

$\frac{v^2}{2} = \frac{\omega^2}{2}\left[3^2 - 1^2\right]$

$v = 2\sqrt{2}\omega$

$x = 2$

Chemistry

Q. 1 In qualitative test for identification of presence of phosphorous, the compound is heated with an oxidising agent. Which is further treated with nitric acid and ammonium molybdate respectively. The yellow coloured precipitate obtained is :

Option 1:
$Na_3PO_4 \cdot 12MoO_3$

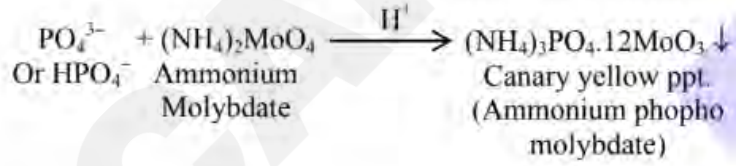
Option 2:
$(NH_4)_3PO_4 \cdot 12(NH_4)_2MoO_4$

Option 3:
$(NH_4)_3PO_4 \cdot 12MoO_3$

Option 4:
$MoPO_4 \cdot 21NH_4NO_3$

Correct Answer:
$(NH_4)_3PO_4 \cdot 12MoO_3$

Solution:



Q. 2 For a reaction $A \xrightarrow{K_1} B \xrightarrow{K_2} C$ If the rate of formation of B is set to be zero then the concentration of B is given by:

Option 1:
$K_1\ K_2[A]$

Option 2:
$(\mathrm{K_1 - K_2})[\mathrm{A}]$

Option 3:
$(\mathrm{K_1 + K_2})[\mathrm{A}]$

Option 4:
$(\mathrm{K_1/K_2})[\mathrm{A}]$

Correct Answer:
$(\mathrm{K_1/K_2})[\mathrm{A}]$

Solution:
Rate of formation of B is

$$\frac{\mathrm{d[B]}}{\mathrm{dt}} = \mathrm{k_1[\,A] - k_2[\,B]}$$

$$0 = \mathrm{k_1[\,A] - k_2[\,B]}$$

$$\left(\frac{\mathrm{k_1}}{\mathrm{k_2}}\right)[\mathrm{A}] = [\mathrm{B}]$$

Q. 3 When ψ_A and Ψ_B are the wave functions of atomic orbitals, then σ^* is represented by:

Option 1:
$\psi_\mathrm{A} - 2\psi_\mathrm{B}$

Option 2:
$\psi_\mathrm{A} - \psi_\mathrm{B}$

Option 3:
$\psi_\mathrm{A} + 2\psi_\mathrm{B}$

Option 4:
$\psi_\mathrm{A} + \psi_\mathrm{B}$

Correct Answer:
$\psi_\mathrm{A} - \psi_\mathrm{B}$

Solution:
Antibonding molecular orbitals are formed by destructive interference of wave functions.

$$(\mathrm{ABMO})\sigma^* = \psi_\mathrm{A} - \psi_\mathrm{B}$$

Q. 4 Which one the following compounds will readily react with dilute NaOH ?

Option 1:
$\mathrm{C_6H_5CH_2OH}$

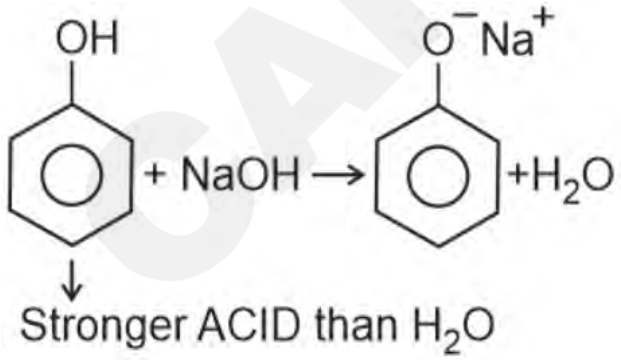
Option 2:
$\mathrm{C_2H_5OH}$

Option 3:
$\mathrm{(CH_3)_3COH}$

Option 4:
$\mathrm{C_6H_5OH}$

Correct Answer:
$\mathrm{C_6H_5OH}$

Solution:



Stronger ACID than $\mathrm{H_2O}$

Q. 5 The shape of carbocation is:

Option 1:
trigonal planar

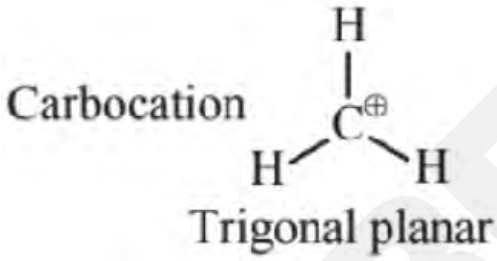
Option 2:
diagonal pyramidal

Option 3:
tetrahedral

Option 4:
diagonal

Correct Answer:
trigonal planar

Solution:



Q. 6 Given below are two statements :

Statement (I) : S_N2 reactions are 'stereospecific', indicating that they result in the formation only one stereo-isomers as the product.

Statement (II) : S_N1 reactions generally result in formation of product as racemic mixtures. In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Statement I is true but Statement II is false

Option 2:
Statement I is false but Statement II is true

Option 3:
Both Statement I and Statement II is true

Option 4:
Both Statement I and Statement II is false

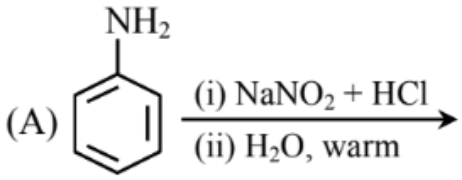
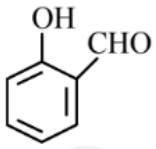
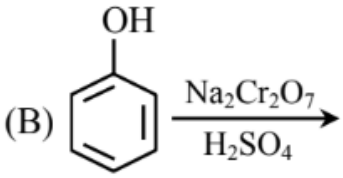
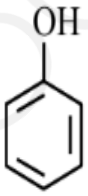
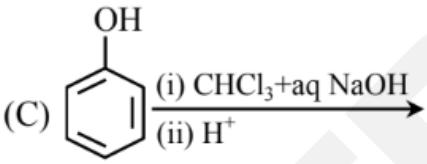
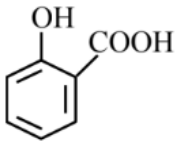
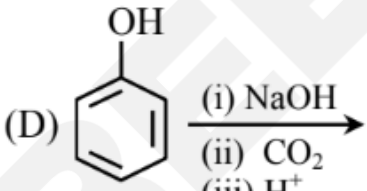
Correct Answer:
Both Statement I and Statement II is true

Solution:
$SN^2 \rightarrow$ Inversion

$SN^1 \rightarrow$ Racemisation

CAREERS360

Q. 7 Match **List-I** with **List-II.**

List-I (Reactions)	**List-II (Products)**
(A) 	(I) 
(B) 	(II) 
(C) 	(III) 
(D) 	(IV)

Choose the correct answer from the options given below:

Option 1:

(A)-(III), (B)-(II), (C)-(I), (D)-(IV)

Option 2:

(A)-(IV), (B)-(II), (C)-(III), (D)-(I)

Option 3:

(A)-(I), (B)-(IV), (C)-(II), (D)-(III)

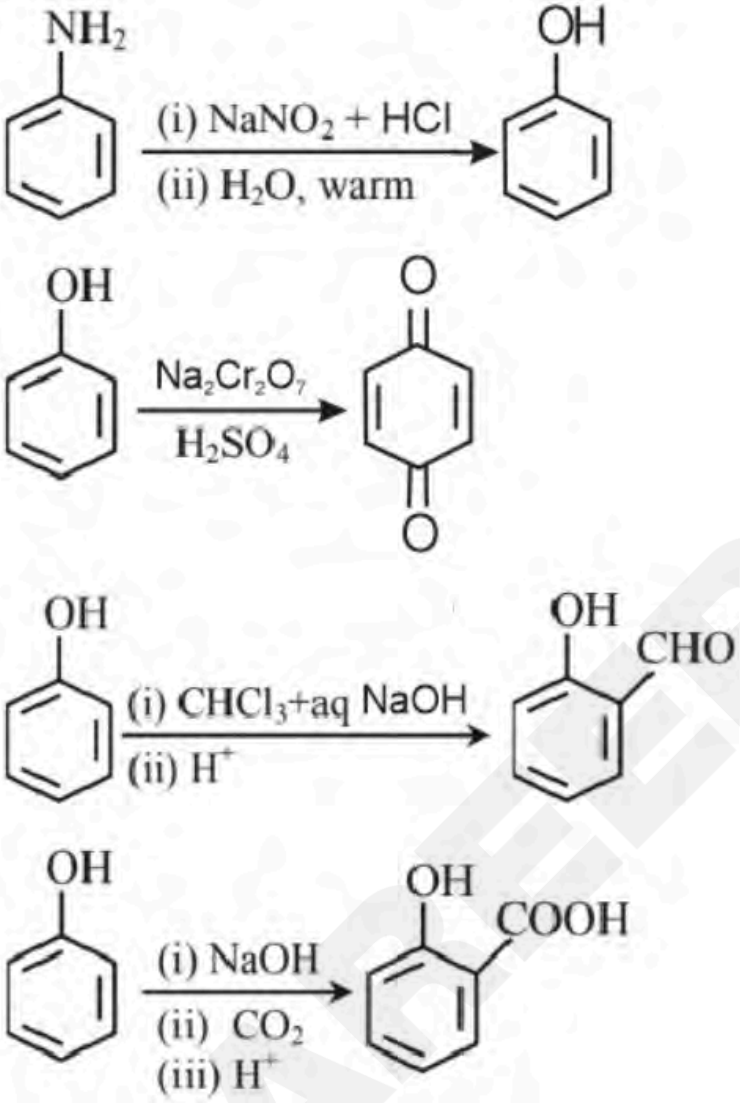
Option 4:

(A)-(II), (B)-(IV), (C)-(I), (D)-(III)

Correct Answer:

(A)-(II), (B)-(IV), (C)-(I), (D)-(III)

Solution:



Q. 8 Match **List-I** with **List-II.**

List-I (Test)	**List-II (Identification)**
(A) Bayer's test	(I) Phenol
(B) Ceric ammonium nitrate test	(II) Aldehyde
(C) Phthalein dye test	(III) Alcoholic-OH group
(D) Schiff's test	(IV) Unsaturation

Choose the correct answer from the options given below :

Option 1:
(A)-(III), (B)-(I), (C)-(IV), (D)-(II)

Option 2:
(A)-(II), (B)-(III), (C)-(IV), (D)-(I)

Option 3:
(A)-(IV), (B)-(I), (C)-(II), (D)-(III)

Option 4:
(A)-(IV), (B)-(III), (C)-(I), (D)-(II)

Correct Answer:
(A)-(IV), (B)-(III), (C)-(I), (D)-(II)

Solution:
(A) Bayer's test $\rightarrow$ Unsaturation
(B) Ceric ammonium nitrate test $\rightarrow$ Alcoholic-OH group
(C) Phthalein dye test $\rightarrow$ Phenol
(D) Schiff's test $\rightarrow$ Aldehyde

Q. 9 Identify the incorrect statements about group 15 elements :

(A) Dinitrogen is a diatomic gas which acts like an inert gas at room temperature.
(B) The common oxidation states of these elements are $-3, +3$ and +5 .
(C) Nitrogen has unique ability to form $\mathrm{p}\pi - \mathrm{p}\pi$ multiple bonds.
(D) The stability of +5 oxidation states increases down the group.
(E) Nitrogen shows a maximum covalency of 6 .

Option 1:
(A), (B), (D) only

Option 2:
(A), (C), (E) only

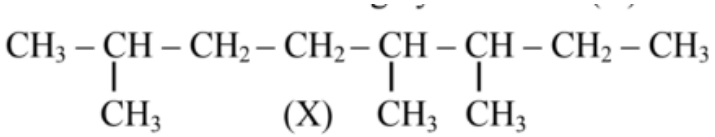
Option 3:
(B), (D), (E) only

Option 4:
(D) and (E) only

Correct Answer:
(D) and (E) only

Solution:
(D) Due to inert pair effect lower oxidation state is more stable.
(E) Nitrogen belongs to 2^{nd} period and cannot expand its octet.

Q. 10 IUPAC name of following hydrocarbon (X) is :



Option 1:
2-Ethyl-3,6-dimethylheptane

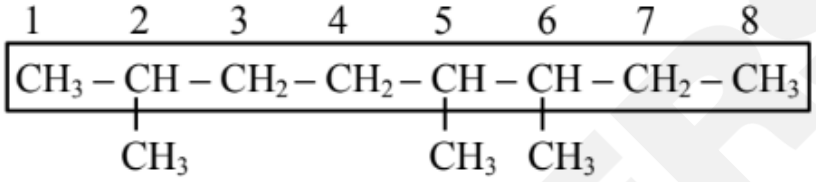
Option 2:
2-Ethyl-2,6-diethylheptane

Option 3:
2,5,6-Trimethyloctane

Option 4:
3,4,7-Trimethyloctane

Correct Answer:
2,5,6-Trimethyloctane

Solution:



2,5,6-Trimethyloctane

Q. 11 The equilibrium $Cr_2O_7^{2-} \rightleftharpoons 2CrO_4^{2-}$ is shifted to the right in :

Option 1:
an acidic medium

Option 2:
a basic medium

Option 3:
a weakly acidic medium

Option 4:
a neutral mediu

Correct Answer:
a basic medium

Solution:
\mathrm{Cr}_2 \mathrm{O}_7^{2-} \stackrel{\mathrm{OH}^{-}}{\rightleftharpoons \mathrm{H}^{+}} 2 \mathrm{CrO}_4^{2-}

Q. 12 Given below are two statements :

Statement (I) : A Buffer solution is the mixture of a salt and an acid or a base mixed in any particular quantities.

Statement (II) : Blood is naturally occurring buffer solution whose pH is maintained by $\mathrm{H_2CO_3/HCO_3^{\ominus}}$ concentrations.

In the light of the above statements, choose the correct answer from the options given below.

Option 1:
Statement I is false but Statement II is true

Option 2:
Both Statement I and Statement II is true

Option 3:
Both Statement I and Statement II is false

Option 4:
Statement I is true but Statement II is false

Correct Answer:
Statement I is false but Statement II is true

Solution:
Buffer solution is a mixture of either weak acid weak base and its respective conjugate.

Blood is a buffer solution of carbonic acid H_2CO_3 and bicarbonate HCO_3^-

Statement 1 is false but Statement II is true.

Q. 13 The correct sequence of acidic strength of the following aliphatic acids in their decreasing order is :

$CH_3CH_2COOH, CH_3COOH, CH_3CH_2CH_2COOH, HCOOH$

Option 1:
$HCOOH > CH_3COOH > CH_3CH_2COOH > CH_3CH_2CH_2COOH$

Option 2:
$HCOOH > CH_3CH_2CH_2COOH > CH_3CHCOOH > CH_3COOH$

Option 3:
$CH_3CH_2CH_2COOH > CH_3CH_2COOH > CH_3COOH > HCOOH$

Option 4:
$CH_3COOH > CH_3CH_2COOH > CH_3CH_2CH_2COOH > HCOOH$

Correct Answer:
$HCOOH > CH_3COOH > CH_3CH_2COOH > CH_3CH_2CH_2COOH$

Solution:

$CH_3CH_2COOH, CH_3COOH, CH_3CH_2CH_2COOH, HCOOH$

The correct order is :

$HCOOH > CH_3COOH > CH_3CH_2COOH > CH_3CH_2CH_2COOH$

Q. 14 Given below are two statements :

Statement (I) : All the following compounds react with n-toluenesulfonvl chloride

$C_6H_5NH_2 \quad (C_6H_5)_2NH \quad (C_6H_5)_3\ N$

Statement (II) : Their products in the above reaction are soluble in aqueous $NaOH$.

In the light of the above statements, choose the correct answer from the options given below.

Option 1:

Both Statement I and Statement II is false

Option 2:

Statement I is true but Statement II is false

Option 3:

Statement I is false but Statement II is true

Option 4:

Both Statement I and Statement II is true

Correct Answer:

Both Statement I and Statement II is false

Solution:

Hinsberg test given by 1° amine only.

Q. 15 The emf of cell $T1 \underset{(0.001M)}{Tl^+} \underset{(0.01\ m)}{Cu^{2+}} Cu$ is $0.83\ V$ at 298 K. It could be increased by :

Option 1:
increasing concentration of T^{+}ions

Option 2:
increasing concentration of both $Tl^{+}$and Cu^{2+} ions

Option 3:
decreasing concentration of both $T1^{+}$and Cu^{2+} ions

Option 4:
increasing concentration of Cu^{2+} ions

Correct Answer:
increasing concentration of Cu^{2+} ions

Solution:

Anodic Reaction	$\left[T\ell_{(s)} \rightarrow T^{+}{}_{(aq)} + e^{-}\right]2$
Cathodic Reaction	$Cu^{+2}{}_{(aq)} + 2e^{-} \rightarrow Cu_{(s)}$
Overall Redox Reaction	$2\,T_{(s)} + Cu^{+2}{}_{(aq)} \rightarrow 2\,T\ell^{+}_{(aq)} + Cu_{(s)}$

$$E_{cell} = E^{o}_{cell} - \frac{0.0591}{2}\log\frac{[T^{+}]^{2}}{[Cu^{+2}]}$$

E_{cell} increases by increasing concentration of $[Cu^{+2}]$ ions.

Q. 16 Identify the correct statements about p-block elements and their compounds.

(A) Non metals have higher electronegativity than metals.
(B) Non metals have lower ionisation enthalpy than metals.
(C) Compounds formed between highly reactive nonmetals and highly reactive metals are generally ionic.

(D) The non-metal oxides are generally basic in nature.
(E) The metal oxides are generally acidic or neutral in nature.

Option 1:
(D) and (E) only

Option 2:
(A) and (C) only

Option 3:
(B) and (E) only

Option 4:
(B) and (D) only

Correct Answer:
(A) and (C) only

Solution:
As electronegativity increases non-metallic nature increases.

Along the period ionisation energy increases.
High electronegativity difference results in ionic bond formation.

Oxides of metals are generally basic and that of non-metals are acidic in nature.

Q. 17 Given below are two statements :
Statement (I) : Kjeldahl method is applicable to estimate nitrogen in pyridine.

Statement (II) : The nitrogen present in pyridine can easily be converted into ammonium sulphate in Kjeldahl method.

In the light of the above statements, choose the correct answer from the options given below.

Option 1:
Both Statement I and Statement II is false

Option 2:
Statement I is false but Statement II is true

Option 3:
Both Statement I and Statement II is true

Option 4:
Statement I is true but Statement II is false

Correct Answer:
Both Statement I and Statement II is false

Solution:
Nitrogen present in pyridine can not be estimated by Kjeldahl method as the nitrogen present in pyridine can not be easily converted into ammonium sulphate.

Q. 18 The reaction;

$$\frac{1}{2}\mathrm{H_{2(\,g)}} + \mathrm{AgCl_{(s)}} \rightarrow \mathrm{H^{+}_{(aq)}} + \mathrm{Cl^{-}_{(aq)}} + \mathrm{Ag_{(s)}}$$

occurs in which of the following galvanic cell:

Option 1:
$\mathrm{Pt\ \ H_{2(\,g)}\ \ HCl_{(soln.)}\ \ \ AgCl_{(s)}\ \ Ag}$

Option 2:
$\mathrm{Pt\ \ H_{2(\,g)}\ \ HCl_{(soln.)}\ \ \ AgNO_{3(aq)}\ \ Ag}$

Option 3:
$\mathrm{Pt\ \ H_{2(\,g)}\ \ KCl_{(soln.)}\ \ \ AgCl_{(s)}\ \ Ag}$

Option 4:

$Ag\ \ AgCl_{(s)}\ \ KCl_{(soln.)}\ \ AgNO_{3\,(aq.)}\ \ Ag$

Correct Answer:

$Pt\ \ H_{2(\,g)}\ \ KCl_{(soln.)}\ \ AgCl_{(s)}\ \ Ag$

Solution:

Anodic half cell

Gas - gas ion electrode

$\frac{1}{2}H_{2(\,g)} \rightarrow H^{+}_{(aq)} + e^{-}$

Cathodic Reaction

Metal-metal insoluble salt anion electrode

$Ag^{+}_{(aq)} + e^{-} \rightarrow Ag_{(s)}$

$AgCl_{(s)} \rightleftharpoons Ag^{+}_{(aq)} + Cl^{-}_{(aq)}$

$AgCl_{(s)} + e^{-} \rightarrow Ag_{(s)} + Cl^{-}{}_{(aq)}$

Overall redox reaction

$\frac{1}{2}H_{2(\,g)} + AgCl_{(s)} \rightarrow H^{+}_{(aq)} + Cl^{-}_{(aq)} + Ag_{(s)}$

Cell Representation

$Pt\ \ H_{2(\,g)}\ \ kCl_{(sol)}\ \ AgCl_{(s)}\ \ Ag$

Q. 19 Given below are two statements :

Statement (I) : Fusion of MnO_2 with KOH and an oxidising agent gives dark green K_2MnO_4.

Statement (II) : Manganate ion on electrolytic oxidation in alkaline medium gives permanganate ion.

In the light of the above statements, choose the correct answer from the options given below.

Option 1:
Both Statement I and Statement II is true

Option 2:
Both Statement I and Statement II is false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Both Statement I and Statement II is true

Solution:

$$\mathrm{MnO_2 + 4KOH + O_2 \xrightarrow{fused} 2\,K_2MnO_4 + 2H_2O}$$

Dark green

Electrolytic oxidation in alkaline medium :

At anode :

$$\mathrm{MnO_4^{2-} \rightarrow MnO_4^- + e^-}$$

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Q. 20 Match **List-I** with **List-II.**

List-I (Complex ion)	**List-II (Spin only magnetic moment in B.M.)**
(A) $[Cr(NH_3)_6]^{3+}$	(I) 4.90
(B) $[NiCl_4]^{2-}$	(II) 3.87
(C) $[CoF_6]^{3-}$	(III) 0.0
(D) $[Ni(CN)_4]^{2-}$	(IV) 2.83

Choose the correct answer from the options given below :

Option 1:
(A)-(I), (B)-(IV), (C)-(II), (D)-(III)

Option 2:
(A)-(IV), (B)-(III), (C)-(I), (D)-(II)

Option 3:
(A)-(II), (B)-(IV), (C)-(I), (D)-(III)

Option 4:
(A)-(II), (B)-(III), (C)-(I), (D)-(IV)

Correct Answer:
(A)-(II), (B)-(IV), (C)-(I), (D)-(III)

Solution:

$(A)\ [Cr(NH_3)_6]^{3+}$

$Cr^{3+} : 3\,d^3$

$n = 3\ (\text{unpaired electrons})$

$\mu \simeq 3.87\ \text{B.M.}\ (II)$

(B) $[NiCl_4]^{2-}$

Ni^{2+} : 3 d^8

$n = 2$

$\mu \simeq 2.83$ B.M. (IV)

(C) $[CoF_6]^{3-}$

Co^{3+} : 3 d^6

$n = 4$

$\mu \simeq 4.90$ B.M. (I)

(D) $[Ni(CN)_4]^{2-}$

Ni^{2+} : 3 d^8

$n = 0$

$\mu = 0$ B.M. (III)

Q. 21 $\Delta_{vap} H^{\ominus}$ for water is $+40.49$ kJ mol^{-1} at 1 bar and $100^{\circ}C$. Change in internal energy for this vapourisation under same condition is kJ mol^{-1}. (Integer answer)
(Given $R = 8.3 JK^{-1}$ mol^{-1})

Correct Answer:
38

Solution:

$H_2O(\ell) \rightleftharpoons H_2O(g) \quad \Delta H^0_{vap} = 40.79$ kJ/ mole

$\Delta H^0_{vap} = \Delta U^0_{vap} + \Delta n_g RT$

$40.79 = \Delta U^0_{vap} + \frac{1 \times 8.3 \times 373.15}{1000}$

$\Delta U^0_{vap} = 40.79 - 3.0971$

$= 37.6929$

$\Delta U^0_{vap} \simeq 38$

Q. 22 Number of molecules having bond order 2 from the following molecule is

$C_2, O_2, Be_2, Li_2, Ne_2, N_2, He_2$

Correct Answer:
2

Solution:

C_2

$(12e^-) : \sigma 1\,s^2, \sigma^* 1\,s^2, \sigma 2\,s^2, \sigma^* 2\,s^2\ [\pi 2p_x^2 = \pi 2p_y^2]$

B.O. $= \frac{8-4}{2} = 2$

O_2

$(16e^-) : \sigma 1\,s^2, \sigma^* 1\,s^2, \sigma 2\,s^2, \sigma^* 2\,s^2, \sigma 2pz^2$

$[\pi 2p_x^2 = \pi 2p_y^2]\ [\pi * 2p_x^1 = \pi * 2p_y^1]$

B.O. $= \frac{10-6}{2} = 2$

Be_2

$(8e^-) : \sigma 1\,s^2, \sigma^* 1\,s^2, \sigma 2\,s^2, \sigma^* 2\,s^2$

$$\text{B.O.} = \frac{4-4}{2} = 0$$

Li_2

$(6e^-) : \sigma 1\,s^2, \sigma^* 1\,s^2, \sigma 2\,s^2$

B.O. $= \frac{4-2}{2} = 1$

Ne_2

$(20e^-) : \sigma 1\,s^2, \sigma^* 1\,s^2, \sigma 2\,s^2, \sigma^* 2\,s^2, \sigma 2pz^2$

$[\pi 2p_x^2 = \pi 2p_y^2]\ [\pi * 2p_x^2 = \pi * 2p_y^2] \sigma * 2p_z^2$

B.O. $= \frac{10-10}{2} = 0$

N_2

$(14e^-) : \sigma 1\,s^2, \sigma^* 1\,s^2, \sigma 2\,s^2, \sigma^* 2\,s^2 \left[\pi 2p_x^2 = \pi 2p_y^2\right] \sigma 2p_z^2$

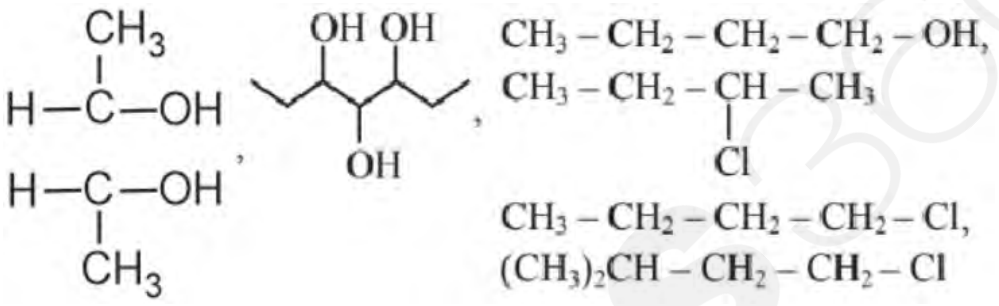
B.O. $= \frac{10-4}{2} = 6$

He_2

$(4e^-) : \sigma 1\,s^2, \sigma^* 1\,s^2$

B.O. $= \frac{2-2}{2} = 0$

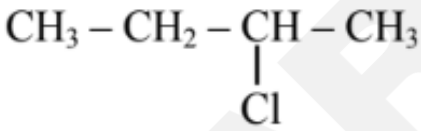
Q. 23 Total number of optically active compounds from the following is



Correct Answer:

1

Solution:



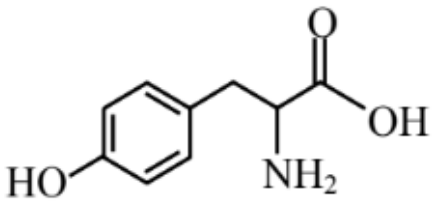
Q. 24 The total number of carbon atoms present in tyrosine, an amino acid, is

Correct Answer:

9

Solution:

Tyrosine

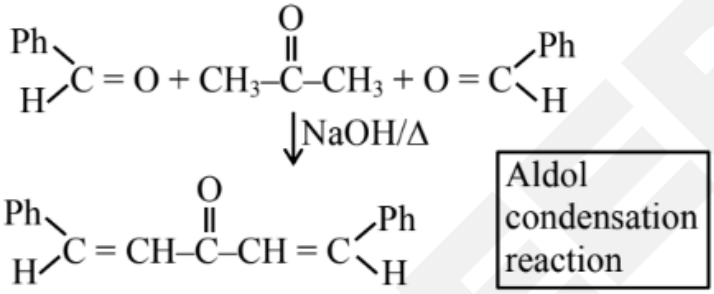


Number of carbon atoms $= 9$

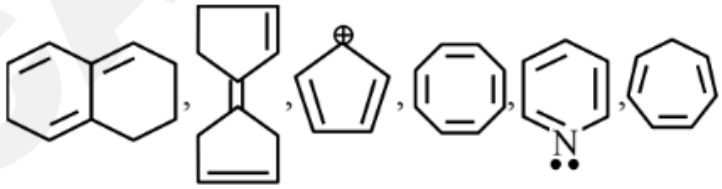
Q. 25 Two moles of benzaldehyde and one mole of acetone under alkaline conditions using aqueous NaOH after heating gives x as the major product. The number of π bonds in the product x is

Correct Answer:
9

Solution:

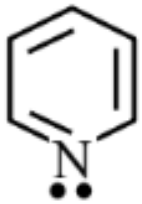


Q. 26 Total number of aromatic compounds among the following compounds is .



Correct Answer:
1

Solution:



Q. 27 Molality of an aqueous solution of urea is $4.44\ \mathrm{m}$.
Mole fraction of urea in solution is $x \times 10^{-3}$.
Value of x is (integer answer)

Correct Answer:
74

Solution:
Molality of urea is $4.44\ \mathrm{m}$, that means 4.44 moles of urea present in $1000\mathrm{gm}$ of water

$$\therefore \mathrm{X_{urea}} = \frac{4.44}{4.44 + \frac{1000}{18}}.$$

$$= 0.0740$$

OR

74×10^{-3}
$X = 74$

Q. 28 Total number of unpaired electrons in the complex ion $[\mathrm{Co(NH_3)_6}]^{3+}$ and $[\mathrm{NiCl_4}]^{2-}$ is

Correct Answer:
2

Solution:

$Co^{+3}: 3\,d^6 \quad t_{2\,g}^{2,2,2} e_g^{0,0}$

$\text{Unpaired e} = 0$

$Ni^{+2}: 3\,d^8 \quad e^{2,2} t_2^{2,1,1}$

$\text{Unpaired e} = 2$

Q. 29 Wavenumber for a radiation having 5800 angstorm wavelength is $x \times 10\ \text{cm}^{-1}$. The value of x is

Option 1:
1724

Option 2:

Option 3:

Option 4:

Correct Answer:
1724

Solution:
$\bar{v}(\text{ wave no. }) = \frac{1}{\lambda} = \frac{1}{5800 \times 10^{-8}\ \text{cm}} = 1724$

OR

$1724 \times 10\ \text{cm}^{-1} \Rightarrow x = 1724$

Q. 30 A solution is prepared by adding 1 mole ethyl alcohol in 9 mole water. The mass percent of solute in the solution is (Integer Answer)
(Given : Molar mass in gmol^{-1} Ethyl alcohol : 46, water : 18)

Correct Answer:
22

Solution:
Mass percent of Alcohol

$$= \frac{\text{Mass of ethyl alcohol}}{\text{Total mass of solution}} \times 100$$

$$= \frac{1 \times 46}{1 \times 46 + 9 \times 18} \times 100 = \frac{4600}{208}$$
$$= 22.11 \text{ Or } 22$$

Maths

Q. 1 If the image of the point $(-4, 5)$ in the line $x + 2y = 2$ lies on the circle $(x + 4)^2 + (y - 3)^2 = r^2$, then r is equal lo:

Option 1:
1

Option 2:
2

Option 3:
75

Option 4:
3

Correct Answer:
2

Solution:

Image of point $(-4, 5)$

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = -2\left(\frac{ax_1 + by_1 + c}{a^2 + b^2}\right)$$

Line: $x + 2y - 2 = 0$

$$\frac{x+4}{1} = \frac{y-5}{2} = -2\left(\frac{-4+10-2}{1^2+2^2}\right)$$

$$= \frac{-8}{5}$$

$$x = -4 - \frac{8}{5} = -\frac{28}{5}$$

$$y = -\frac{16}{5} + 5 = \frac{9}{5}$$

Point lies on circle $(x+4)^2 + (y-3)^2 = r^2$

$$\frac{64}{25} + \left(\frac{9}{5} - 3\right)^2 = r^2$$

$$\frac{100}{25} = r^2, r = 2$$

Q. 2

Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = 2\hat{i} + 3\hat{j} - 5\hat{k}$ and $\vec{c} = 3\hat{i} - \hat{j} + \lambda\hat{k}$ be three vectors. Let $\vec{r}$ be a unit vector along $\vec{b} + \vec{c}$. If $\vec{r}.\vec{a} = 3$, then 3λ is equal to :

Option 1:
27

Option 2:
25

Option 3:
25

Option 4:
21

Correct Answer:
25

Solution:

$\vec{r} = k(\vec{b} + \vec{c})$

$\vec{r} \cdot \vec{a} = 3$

$\vec{r} \cdot \vec{a} = k(\vec{b} \cdot \vec{a} + \vec{c} \cdot \vec{a})$

$3 = k(2 + 6 - 15 + 3 - 2 + 3\lambda)$

$3 = k(-6 + 3\lambda) \quad \ldots(1)$

$\vec{r} = k(5\hat{i} + 2\hat{j} - (5 - \lambda)\hat{k})$

$|\vec{r}| = k\sqrt{25 + 4 + 25 + \lambda^2 - 10\lambda} = 1$...(2)

$k = \frac{3}{-6+3\lambda} = \frac{1}{-2+\lambda}$ put in (2)

$4 + \lambda^2 - 4\lambda = 54 + \lambda^2 - 10\lambda$

$6\lambda = 50$

$3\lambda = 25.$

Q. 3 If $\alpha \neq a, \beta \neq b, \gamma \neq c$ and $\begin{vmatrix} \alpha & b & c \\ a & \beta & c \\ a & b & \gamma \end{vmatrix} = 0$, then $\frac{a}{\alpha - a} + \frac{b}{\beta - b} + \frac{\gamma}{\gamma - c}$ is equal to :

Option 1:
2

Option 2:
3

Option 3:
0

Option 4:
1

Correct Answer:
0

Solution:

$R_1 \to R_1 - R_2, R_2 \to R_2 - R_3$

$$\begin{vmatrix} \alpha - a & b - \beta & 0 \\ 0 & \beta - b & c - \gamma \\ a & b & \gamma \end{vmatrix} = 0$$

$$(\alpha - a)(\gamma(\beta - b) - b(c - \gamma)) - (b - \beta)(-a(c - \gamma)) = 0$$

$$\gamma(\alpha - a)(\beta - b) - b(\alpha - a)(c - \gamma) + a(b - \beta)(c - \gamma)$$

$$\frac{\gamma}{\gamma - c} + \frac{b}{\beta - b} + \frac{a}{\alpha - a} = 0$$

Q. 4 In an increasing geometric progression ol positive terms, the sum of the second and sixth terms is $\frac{70}{3}$ and the product of the third and fifth terms is 49. Then the sum of the 4^{th} , 6^{th} and 8^{th} terms is :-

Option 1:
96

Option 2:
78

Option 3:
91

Option 4:
84

Correct Answer:
91

Solution:

$T_2 + T_6 = \frac{70}{3}$

$ar + ar^5 = \frac{70}{3}$

$T_3 \cdot T_5 = 49$

$ar^2 \cdot ar^4 = 49$

$a^2r^6 = 49$

$ar^3 = +7, a = \frac{7}{r^3}$

$ar\left(1 + r^4\right) = \frac{70}{3}$

$\frac{7}{r^2}\left(1 + r^4\right) = \frac{70}{3}, r^2 = t$

$\frac{1}{t}\left(1 + t^2\right) = \frac{10}{3}$

$3t^2 - 10t + 3 = 0$

$t = 3, \frac{1}{3}$

Increasing G.P. $r^2 = 3, r = \sqrt{3}$

$T_4 + T_6 + T_8$

$= ar^3 + ar^5 + ar^7$

$= ar^3\left(1 + r^2 + r^4\right)$

$= 7(1 + 3 + 9) = 91$

Q. 5 The number of ways five alphabets can be chosen from the alphabets of the word MATHEMATICS, where the chosen alphabets are not necessarily distinct, is equal to :

Option 1:
175

Option 2:
181

Option 3:
177

Option 4:
179

Correct Answer:
179

Solution:
AA, MM, TT, H, I, C, S, E

(1) All distinct

$^{8}\mathrm{C}_{5} \rightarrow 56$

(2) 2 same, 3 different

$^{3}\mathrm{C}_{1} \times {}^{7}\mathrm{C}_{3} \rightarrow 105$

(3) 2 same I^{st} kind, 2 same 2^{nd} kind, 1 different

$^{3}\mathrm{C}_{2} \times {}^{6}\mathrm{C}_{1} \rightarrow 18$

Total $\rightarrow 179$

Q. 6 The sum of all possible values of $\theta \in [-\pi, 2\pi]$, for which $\frac{1+i\cos\theta}{1-2i\cos\theta}$ is purely imaginary, is equal to

Option 1:
2π

Option 2:
3π

Option 3:
5π

Option 4:
4π

Correct Answer:
3π

Solution:

$Z = \frac{1+i\cos\theta}{1-2i\cos\theta}$

$Z = -\bar{Z} \Rightarrow \frac{1+i\cos\theta}{1-2i\cos\theta} = -\left(\frac{\overline{1+i\cos\theta}}{1-2i\cos\theta}\right)$

$(1+i\cos\theta)(\overline{1-2i\cos\theta}) = -(1-2i\cos\theta)(\overline{1+i\cos\theta})$
$(1+i\cos\theta)(1+2i\cos\theta) = -(1-2i\cos\theta)(1-i\cos\theta)$

$1+3i\cos\theta-2\cos^2\theta = -\left(1-3i\cos\theta-2\cos^2\theta\right)$

$2-4\cos^2\theta = 0$

$\Rightarrow \cos^2\theta = \frac{1}{2} \Rightarrow \theta = -\frac{\pi}{4}, -\frac{3\pi}{4}, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

$\text{sum} = 3\pi$

Q. 7 If the system of equations $x+4y-z=\lambda$, $7x+9y+\mu z=-3, 5x+y+2z=-1$ has infinitely many solutions, then $(2\mu.+3\lambda)$ is equal to :

Option 1:
2

Option 2:
-3

Option 3:
3

Option 4:
-2

Correct Answer:
-3

Solution:

$$\Delta = \begin{vmatrix} 1 & 4 & -1 \\ 7 & 9 & \mu \\ 5 & 1 & 2 \end{vmatrix} = 0$$

$$\Rightarrow (18-\mu) - 4(14-5\mu) - (7-45) = 0 \Rightarrow \mu = 0$$

$$\Delta = \Delta_x = \Delta_y = \Delta_z = 0 \text{ (For infinite solution)}$$

$$\Delta_x = \begin{vmatrix} \lambda & 4 & -1 \\ -3 & 9 & \mu \\ -1 & 1 & 2 \end{vmatrix} = 0$$

$$\lambda(18-\mu) - 4(-6+\mu) - 1(-3+9) = 0$$

$$18\lambda + 24 - 6 = 0 \Rightarrow \lambda = -1$$

Q. 8 If the shortest distance between the lines $\frac{x-\lambda}{2} = \frac{y-4}{3} = \frac{z-3}{4}$ and $\frac{x-2}{4} = \frac{y-4}{6} = \frac{z-7}{8}$ is $\frac{13}{\sqrt{29}}$, then a value of λ is :

Option 1:
$-\frac{13}{25}$

Option 2:
$\frac{13}{25}$

Option 3:
1

Option 4:
-1

Correct Answer:
1

Solution:

$$\overline{r}_1 = (\lambda\hat{i} + 4\hat{j} + 3\hat{k}) + \alpha(2\hat{i} + 3\hat{j} + 4\hat{k})\overline{b}_1 = 2\hat{i} + 3\hat{j} + 4\hat{j} + 3\hat{k}$$

$$\overline{r_2} = (2\hat{i} + 4\hat{j} + 7\hat{k}) + \beta(2\hat{i} + 3\hat{j} + 4\hat{k})\Big\}\begin{matrix} a_1 + \lambda 1 + 4\hat{j} + 3k \\ a_2 = 2\hat{i} + 4\hat{j} + 7\hat{k} \end{matrix}$$

$$\text{Shortest dist.} = \frac{\overline{b} \times (\overline{a}_2 - \overline{a}_1)}{|b|} = \frac{13}{\sqrt{29}}$$

$$\frac{|(2\hat{i} + 3\hat{j} + 4\hat{k}) \times ((2-\lambda)\hat{i} + 4\hat{k})|}{\sqrt{29}} = \frac{13}{\sqrt{29}}$$

$$|-8\hat{j} - 3(2-\lambda)\hat{k} + 12\hat{i} + 4(2-\lambda)\hat{j}| = 13$$

$$|12\hat{i} - 4\lambda\hat{j} + (3\lambda - 6)\hat{k}| = 13$$

$$144 + 16\lambda^2 + (3\lambda - 6)^2 = 169$$

$$16\lambda^2 + (3\lambda - 6)^2 = 25 = \lambda \Rightarrow = 1$$

Q. 9 If the value of $\frac{3\cos 36^\circ + 5\sin 18^\circ}{5\cos 36^\circ - 3\sin 18^\circ}$ is $\frac{a\sqrt{5}-b}{c}$ where a, b, c are natural numbers and $\gcd(a, c) = 1$, then $a + b + c$ is equal to :

Option 1:
50

Option 2:
40

Option 3:
52

Option 4:
54

Correct Answer:
52

Solution:

$$\frac{\frac{3(\sqrt{5}+1)}{4}+5\left(\frac{\sqrt{5}-1}{4}\right)}{5\left(\frac{\sqrt{5}+1}{4}\right)-3\left(\frac{\sqrt{5}-1}{4}\right)}=\frac{8\sqrt{5}-2}{2\sqrt{5}+8}$$

$$=\frac{4\sqrt{5}-1}{\sqrt{5}+4}\times\frac{\sqrt{5}-4}{\sqrt{5}-4}$$

$$=\frac{20-16\sqrt{5}-\sqrt{5}+4}{-11}$$

$$=\frac{17\sqrt{5}-24}{11}\Rightarrow a=17,b=27,c=11$$

$$a+b+c=52$$

Q. 10 Let $y=y(x)$ be the solution curve of the differential equation secy $\frac{dy}{dx}+2x\sin y=x^3\cos y$, $y(1)=0$. Then $y(\sqrt{3})$ is equal to :

Option 1:
$\frac{\pi}{3}$

Option 2:
$\frac{\pi}{6}$

Option 3:
$\frac{\pi}{4}$

Option 4:
$\frac{\pi}{12}$

Correct Answer:

$\frac{\pi}{4}$

Solution:

$$\sec^2 y\frac{dy}{dx} + 2x\sin y\,\text{secy} = x^3\cos y\,\text{secy}$$

$$\sec^2 y\frac{dy}{dx} + 2x\tan y = x^3$$

$$\tan y = t \Rightarrow \sec^2 y\frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} + 2xt = x^3,\ \text{If } = e^{\int 2xdx} = e^{x^2}$$

$$t^{x^2} = \int x^3 \cdot e^{x^2}dx + c$$

$$x^2 = Z \Rightarrow t\cdot e^z = \frac{1}{2}\int e^z \cdot ZdZ = \frac{1}{2}[e^z\cdot Z - e^z] + c$$

$$2\tan y = \left(x^2 - 1\right) + 2ce^{-x^2}$$

$$y(1) = 0 \Rightarrow c = 0 \Rightarrow y(\sqrt{3}) = \frac{\pi}{4}$$

Q. 11 The area of the region in the first quadrant inside the circle $x^2 + y^2 = 8$ and outside the pnrabola $y^2 = 2x$ is equal to :

Option 1:

$\frac{\pi}{2} - \frac{1}{3}$

Option 2:

$\pi - \frac{2}{3}$

Option 3:

$\frac{\pi}{2} - \frac{2}{3}$

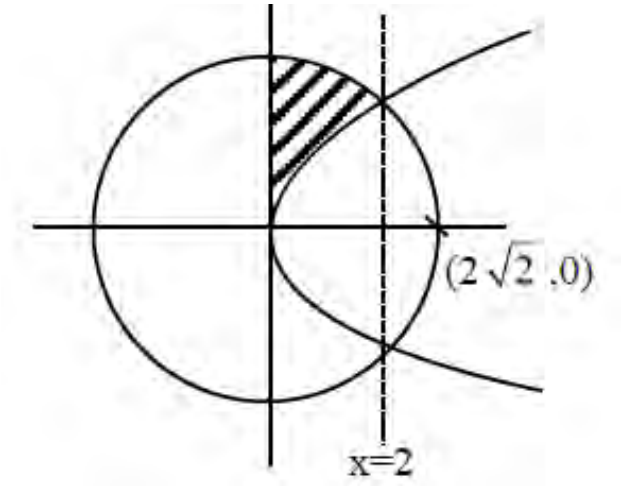
Option 4:

$\pi - \frac{1}{3}$

Correct Answer:

$\pi - \frac{2}{3}$

Solution:



$$\text{Required area } = \text{Ar(circle from 0 to 2)} - \text{ar(para from 0 to 2)}$$

$$= \int_0^2 \sqrt{8-x^2}dx - \int_0^2 \sqrt{2x}dx$$

$$= \left[\frac{x}{2}\sqrt{8-x^2} + \frac{8}{2}\sin^{-1}\frac{x}{2\sqrt{2}}\right]_0^2 - \sqrt{2}\left[\frac{x\sqrt{x}}{3/2}\right]_0^2$$

$$= \frac{2}{2}\sqrt{8-4} + \frac{8}{2}\sin^{-1}\frac{2}{2\sqrt{2}} - \frac{2\sqrt{2}}{3}(2\sqrt{2}-0)$$

$$\Rightarrow 2 + 4\cdot\frac{\pi}{4} - \frac{8}{3} = \pi - \frac{2}{3}$$

Q. 12 If the line segment joining the points $(5, 2)$ and (2, a) subtends an angle $\frac{\pi}{4}$ at the origin, then the absolute value of the product of all possible values of a is:

Option 1:

6

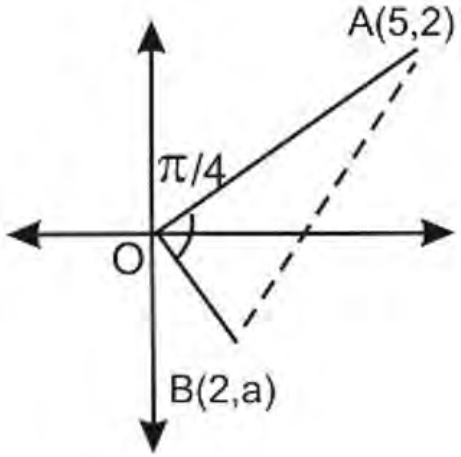
Option 2:
8

Option 3:
2

Option 4:
4

Correct Answer:
4

Solution:



$m_{OA} = \frac{2}{5}$

$m_{OB} = \frac{a}{2}$

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$$\tan\frac{\pi}{4} = \frac{2}{5} - \frac{a}{2}$$

$$1 = \frac{4-5a}{10+2a}$$

$$4-5a = \pm(10+2a)$$

$$4-5\mathrm{a} = 10+2\mathrm{a}$$

$$4-5\mathrm{a} = -10-2\mathrm{a}$$

$$\Rightarrow 7\mathrm{a}+6=0$$

$$3a = 14$$

$$\Rightarrow \mathrm{a} = -\frac{6}{7}$$

$$\mathrm{a} = +\frac{14}{3}$$

$$-\frac{6}{7}\times\frac{14}{3} = -4$$

Q. 13

Let $\vec{\mathrm{a}} = 4\hat{\mathrm{i}} - \hat{\mathrm{j}} + \hat{\mathrm{k}}, \vec{\mathrm{b}} = 11\hat{\mathrm{i}} - \hat{\mathrm{j}} + \hat{\mathrm{k}}$ and $\vec{\mathrm{c}}$ be a vector such that $(\vec{a}+\vec{b})\times\vec{c} = \vec{c}\times(-2\vec{a}+3\vec{b})$.

If $(2\vec{a}+3\vec{b})\cdot\vec{c} = 1670$, then $|\vec{c}|^2$ is equal to :

Option 1:
1627

Option 2:
1618

Option 3:
1600

Option 4:
1609

Correct Answer:
1618

Solution:

$(\vec{a}+\vec{b})\times\vec{c}-\vec{c}\times(-2\vec{a}+3\vec{b})=0$

$(\vec{a}+\vec{b})\times\vec{c}+(-2\vec{a}+3\vec{b})\times\vec{c}=0$

$\Rightarrow(\vec{a}+\vec{b})-2\vec{a}+3\vec{b})\times\vec{c}=0$

$\Rightarrow\vec{c}=\lambda(4\vec{b}-\vec{a})$

$\Rightarrow=\lambda(44\hat{i}-4\hat{j}+4\hat{k}-4\hat{i}+\hat{j}-\hat{k})$

$=\lambda(40\hat{i}-3\hat{j}+3\hat{k})$

Now

$(8\hat{i}-2\hat{j}+2\hat{k}+33\hat{i}-3\hat{j}+3\hat{k})\cdot\lambda(40\hat{i}-3\hat{j}+3\hat{k})=1670$

$\Rightarrow(41\hat{i}-5\hat{j}+5\hat{k})\cdot(40\hat{i}-3\hat{j}+3\hat{k})\times\lambda=1670)$

$\Rightarrow(1640+15+15)\lambda=1670\Rightarrow\lambda=1$

so $\overrightarrow{c}=40\hat{i}-3\hat{j}-3\hat{k}$

$\Rightarrow|\overrightarrow{c}|^2=1600+9+9=1618$

Q. 14 If the function $f(x)=2x^3-9ax^2+12a^2x+1, a>0$ has a local maximum at $\mathrm{x}=\alpha$ and a local minimum $\mathrm{x}=\alpha^2$, then α and α^2 are the roots of the equation :

Option 1:

$x^2-6x+8=0$

Option 2:

$8x^2+6x-8=0$

Option 3:

$8x^2-6x+1=0$

Option 4:

$x^2+6x+8=0$

Correct Answer:
$x^2 - 6x + 8 = 0$

Solution:

$f'(x) = 6x^2 - 18ax + 12a^2 = 0$ $\begin{matrix}\nearrow \alpha \\ \searrow \alpha^2\end{matrix}$

$\alpha + \alpha^2 = 3\mathrm{a} \& \alpha \times \alpha^2 = 2\mathrm{a}^2$

$\downarrow$

$\left(\alpha + \alpha^2\right)^3 = 27\mathrm{a}^3$

$\Rightarrow 2a^2 + 4a^4 + 3(3a)\left(2a^2\right) = 27a^3$
$\Rightarrow 2 + 4a^2 + 18a = 27a$
$\Rightarrow 4a^2 - 9a + 2 = 0$
$\Rightarrow 4a^2 - 8a - a + 2 = 0$
$\Rightarrow (4a - 1)(a - 2) = 0 \Rightarrow a = 2$
so $6x^2 - 36x + 48 = 0$
$\Rightarrow x^2 - 6x + 8 = 0$

If we take $\mathrm{a} = \frac{1}{4}$ then $\alpha = \frac{1}{2}$ which is not possible

---

Q. 15 There are three bags X, Y and Z. Bag X contains 5 one-rupee coins and 4 five-rupee coins; Bag Y contains 4 one-rupee coins and 5 five-rupee coins and Bag Z contains 3 one-rupee coins and 6 five-rupee coins. A bag is selected at random and a coin drawn from it at random is found to be a one-rupee coin. Then the probability, that it came from bag Y, is :

Option 1:
$\frac{1}{3}$

Option 2:
$\frac{1}{2}$

Option 3:
$\frac{1}{4}$

Option 4:
$\frac{5}{12}$

Correct Answer:
$\frac{1}{3}$

Solution:

X	Y	Z
5 one & 4 five	4 one & 5 five	3 one & 6 five

$$P = \frac{4/9}{5/9+4/9+3/9} = \frac{4}{12} = \frac{1}{3}$$

Q. 16 Let $\int_{\alpha}^{\log_e 4} \frac{dx}{\sqrt{e^x-1}} = \frac{\pi}{6}$. Then e^{α} and $e^{-\alpha}$ are the roots of the equation :

Option 1:
$2x^2 - 5x + 2 = 0$

Option 2:
$x^2 - 2x - 8 = 0$

Option 3:
$2x^2 - 5x - 2 = 0$

Option 4:

$x^2 + 2x - 8 = 0$

Correct Answer:

$2x^2 - 5x + 2 = 0$

Solution:

$$\int_{\alpha}^{\log_e 4} \frac{dx}{\sqrt{e^x - 1}} = \frac{\pi}{6}$$

Let $e^x - 1 = t^2$

$e^x dx = 2tdt$

$$= \int \frac{2dt}{t^2 + 1}$$

$$= 2\tan^{-1} t$$

$$= 2\tan^{-1}\left(\sqrt{e^x - 1}\right)_{\alpha}^{\log_e^4}$$

$$= 2\left[\tan^{-1}\sqrt{3} - \tan^{-1}\sqrt{e^{\alpha} - 1}\right] = \frac{\pi}{6}$$

$$= \frac{\pi}{3} - \tan^{-1}\sqrt{e^{\alpha} - 1} = \frac{\pi}{12}$$

$$\Rightarrow \tan^{-1}\sqrt{e^{\alpha} - 1} = \frac{\pi}{4}$$

$e^{\alpha} = 2 \quad e^{-\alpha} = \frac{1}{2}$

$$x^2 - \left(2 + \frac{1}{2}\right)x + 1 = 0$$

$2x^2 - 5x + 2 = 0$

Q. 17 Let $f(x) = \begin{cases} -a & \text{if } -a \le x \le 0 \\ x + a & \text{if } 0 < x \le a \end{cases}$ where $a > 0$ and $g(x) = (f|x|) - |f(x)|)/2.$

Then the function $g : [-a, a] \to [-a, a]$ is

Option 1:
neither one-one nor onto.

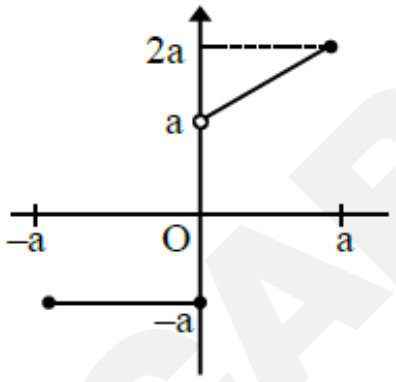
Option 2:
both one-one and onto.

Option 3:
one-one.

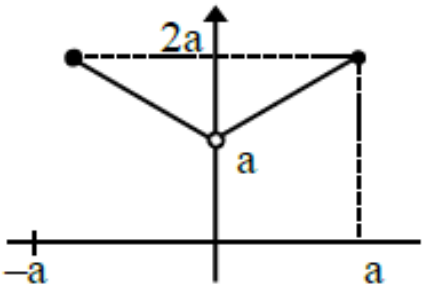
Option 4:
onto

Correct Answer:
neither one-one nor onto.

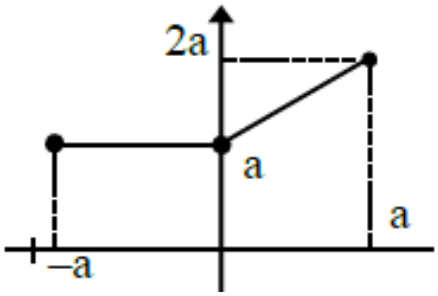
Solution:
$y = f(x)$



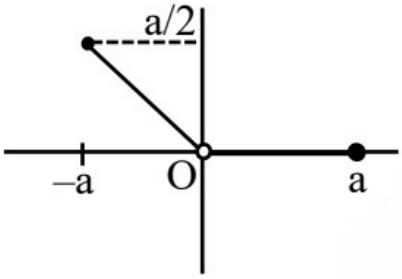
$y = f|x|$



$y = |f(x)|$



$g(x)=\frac{f(|x|)-|f(x)|}{2}$



Q. 18 Let $A=\{2,3,6,8,9,11\}$ and $B=\{1,4,5,10,15\}$ Let R be a relation on $\mathrm{A}\times\mathrm{B}$ define by $(\mathrm{a},\mathrm{b})\mathrm{R}(\mathrm{c},\mathrm{d})$ if and only if $3ad-7bc$ is an even integer. Then the relation R is

Option 1:
reflexive but not symmetric.

Option 2:
transitive but not symmetric.

Option 3:
reflexive and symmetric but not transitive.

Option 4:
an equivalence relation.

Correct Answer:
reflexive and symmetric but not transitive.

Solution:
$A = \{2, 3, 6, 8, 9, 11\} \quad (a, b)R(c, d)$
$B = \{1, 4, 5, 10, 15\} \quad 3ad - 7bc$
$\text{Reflexive} : (a, b)\ R(a, b)$

$\Rightarrow 3ab - 7ba = -4ab$ always even so it is reflexive.
Symmetric : If $3ad - 7bc =$ Even
Case-I : odd odd
Case-II: even even
(c, d) $R(a, b) \Rightarrow 3bc - 3ab$
Case-I : odd odd
Case-II : even even
so symmetric relation
Transitive :
Set $(3, 4)R(6, 4)$ Satisfy relation
Set $(6, 4)R(3, 1)$ Satisfy relation
but $(3, 4)R(3, 1)$ does not satisfy relation so not transitive.

Q. 19 For $a, b > 0$, let

be a continous function at $x = 0$. Then $\frac{b}{a}$ is equal to

Option 1:
5

Option 2:
4

Option 3:
8

Option 4:
6

Correct Answer:
6

Solution:

$$\lim_{x\to 0} f(x) = f(0) = 3$$

$$\lim_{x\to 0^+} \frac{\sqrt{ax+b^2x^2}-\sqrt{ax}}{b\sqrt{a}x\sqrt{x}} = 3$$

$$\lim_{x\to 0^+} \frac{ax+b^2x^2-ax}{b\sqrt{a}x^{3/2}\left(\sqrt{ax+b^2x^2}+\sqrt{ax}\right)}$$

$$\lim_{x\to 0^+} \frac{b^2}{b\sqrt{a}\left(\sqrt{a+b^2x}+\sqrt{a}\right)}$$

$$\frac{b}{\sqrt{a}\cdot 2\sqrt{a}} \Rightarrow \frac{b}{2a} = 3 \Rightarrow \frac{b}{a} = 6$$

Q. 20 If the term independent of x in the expansion of $\left(\sqrt{ax^2}+\frac{1}{2x^3}\right)^{10}$ is 105 , then a^2 is equal to :

Option 1:
4

Option 2:
9

Option 3:
6

Option 4:
2

Correct Answer:
4

Solution:

$$\left(\sqrt{ax}^2+\frac{1}{2x^3}\right)^{10}$$

$$\text{General term } = {}^{10}C_r(\sqrt{ax})^2)^{10-r}\left(\frac{1}{2x^3}\right)^r$$

$$20-2r-3r=0$$

$$r=4$$

$${}^{10}C_4a^3\cdot\frac{1}{16}=105$$

$$a^3=8$$

$$a^2=4$$

Q. 21 Let A be the region enclosed by the parabola $y^2=2x$ and the line $x=24$. Then the maximum area of the rectangle inscribed in the region A is_____.

Correct Answer:
128

Solution:

$$A=2\left(24-\frac{b^2}{2}\right)\cdot b$$

$$\frac{dA}{db}=0 \quad \Rightarrow \quad b=4$$

$$A=2(24-8)4$$

$$=128$$

Q. 22 If $\alpha=\lim_{x\to0^+}\left(\frac{e^{\sqrt{\tan x}}-e^{\sqrt{x}}}{\sqrt{\tan x}-\sqrt{x}}\right)$ and $\beta=\lim_{x\to0}(1+\sin x)^{\frac{1}{2}\cot x}$ are the roots of the quadratic equation $ax^2+bx-\sqrt{e}=0$, then $12\log_e(a+b)$ is equal to _____.

Correct Answer:
6

Solution:

$$\alpha = \lim_{x \to 0^+} e^{\sqrt{x}} \frac{\left(e^{\sqrt{\tan x} - \sqrt{x}} - 1\right)}{\sqrt{\tan x} - \sqrt{x}}$$

$$= 1$$

$$\beta = \lim_{x \to 0} (1 + \sin x)^{\frac{1}{2}\cot x}$$

$$= e^{1/2}$$

$$x^2 - (1 + \sqrt{e}) + \sqrt{e} = 0$$

$$ax^2 + bx - \sqrt{e} = 0$$

On comparing

$$a = -1, b = \sqrt{e} + 1$$

$$12 \ln(a + b) = 12 \times \frac{1}{2} = 6$$

Q. 23 Let S be the focus of the hyperbola $\frac{\mathrm{x}^2}{3} - \frac{\mathrm{y}^2}{5} = 1$, on the positive x-axis. Let C be the circle with its centre at $\mathrm{A}(\sqrt{6}, \sqrt{5})$ and passing through the point S. if O is the origin and SAB is a diameter of C then the square of the area of the triangle OSB is equal to -

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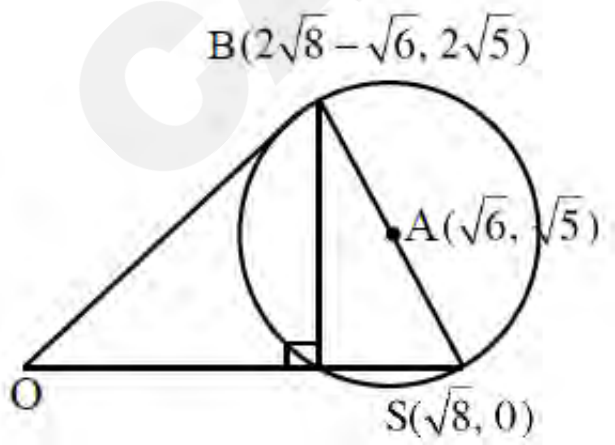
Correct Answer:
40

Solution:

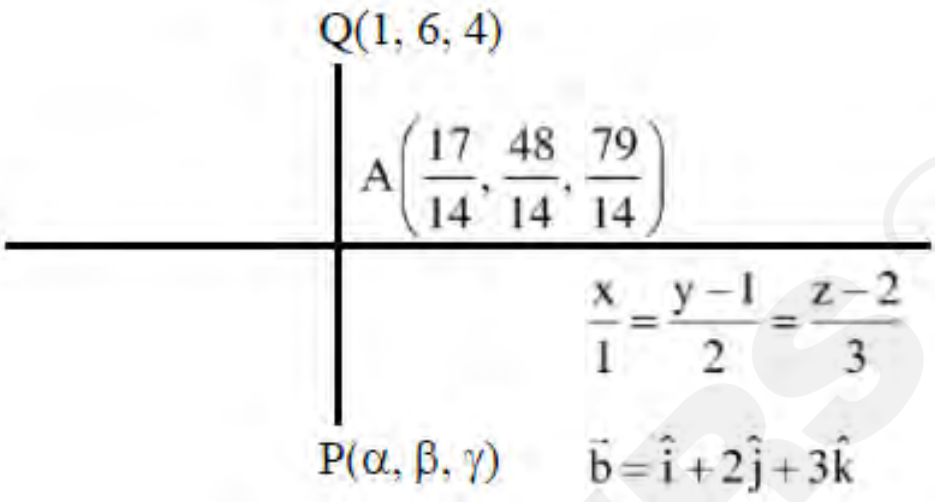


Area $= \frac{1}{2}(\mathrm{OS})\mathrm{h} = \frac{1}{2}\sqrt{8}2\sqrt{5} = \sqrt{40}$

Q. 24 Let $P(\alpha, \beta, \gamma)$ be the image of the point $Q(1, 6, 4)$ in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$. Then $2\alpha + \beta + \gamma$ is equal to _____.

Correct Answer:
11

Solution:



$A(t, 2t+1, 3t+2)$

$\overrightarrow{QA} = (t-1)\hat{i} + (2t-5)\hat{j} + (3t-2)\hat{k}$

$\overrightarrow{QA} \cdot \vec{b} = 0$

$(t-1) + 2(2t-5) + 3(3t-2) = 0$

$14t = 17$

$\alpha = \frac{20}{14} \quad \beta = \frac{12}{14} \quad \gamma = \frac{102}{14}$

$2\alpha + \beta + \gamma = \frac{154}{14} = 11$

Q. 25 An arithmetic progression is written in the following way

The sum of all the terms of the 10^{th} row is

Correct Answer:
1505

Solution:

$2, 5, 11, 20, \ldots\ldots$

$\text{General term } = \frac{3n^2-3n+4}{2}$

$T_{10} = \frac{3(100)-3(10)+4}{2}$

$= 137$

$10 \text{ terms with c.d. } = 3$

$\text{sum } = \frac{10}{2}(2(137) + 9(3))$

$= 1505$

Q. 26 The number of distinct real roots of the equation $|x+1||x+3| - 4|x+2| + 5 = 0$, is_____.

Correct Answer:

2

Solution:

$|x+1||x+3| - 4|x+2| + 5 = 0$

case-1

$x \leq -3$

$(x+1)(x+3) + 4(x+2) + 5 = 0$

$x^2 + 4x + 3 + 4x + 8 + 5 = 0$

$x^2 + 8x + 16 = 0$

$(x+4)^2 = 0$

$x = -4$

case- 2

$-3 \leq x \leq -2$

$-x^2 - 4x - 3 + 4x + 8 + 5 = 0$

$-x^2+10=0$

$x=\pm\sqrt{10}$

case- 3

$-2\leq x\leq -1$

$-x^2-4x-3-4x-8+5=0$

$-x^2-8x-6=0$

$x^2+8x+6=0$

$x=\frac{-8\pm 2\sqrt{10}}{2}=-4\pm\sqrt{10}$

case-4

$x\geq -1$

$x^2+4x+3-4x-8+5=0$

$x^2=0$

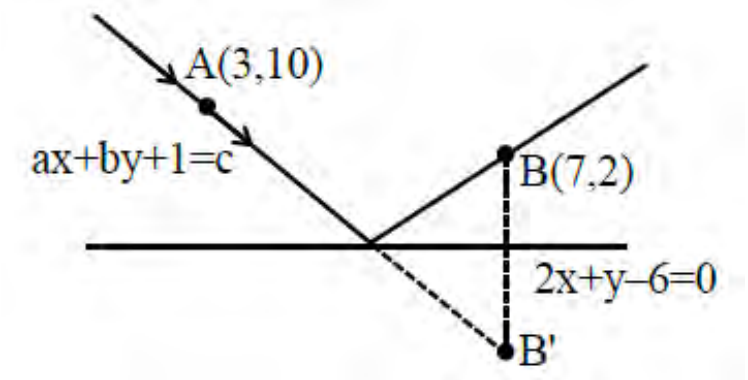
$x=0$

$\text{No. of solution } =2$

Q. 27 Let a ray of light passing through the point $(3,10)$ reflects on the line $2x+y=6$ and the reflected ray passes through the point $(7,2)$. If the equation of the incident ray is $ax+$ by $+1=0$, then a^2+b^2+3ab is equal to_____.

Correct Answer:
1

Solution:



For B' $\quad \frac{x-7}{2} = \frac{y-2}{1} = -2\left(\frac{14+2-6}{5}\right)$

$\frac{x-7}{2} = \frac{y-2}{1} = -4$

$x = -1 \quad y = -2 \quad B'(-1,-2)$

incident ray AB'

$M_{AB'} = 3$

$y + 2 = 3(x + 1)$

$3x - y + 1 = 0$

$a = 3 \quad b = -1$

$a^2 + b^2 + 3ab = 9 + 1 - 9 = 1$

Q. 28 Let $a, b, c \in N$ and $a < b < c$. Let the mean, the mean deviation about the mean and the variance of the 5 observations $9, 25, a, b, c$ be 18,4 and $\frac{136}{5}$, respectively. Then $2a + b - c$ is equal to _____.

Correct Answer:
33

Solution:

$a, b, c \in N \quad a < b < c$

$\bar{x} = \text{ mean } = \frac{9+25+a+b+c}{5} = 18$

$a + b + c = 56$

$\text{Mean deviation } = \frac{\sum|x_i - \bar{x}|}{n} = 4$

$= 9 + 7 + |18 - a| + |18 - b| + |18 - c| = 20$

$= |18-a| + |18-b| + |18-c| = 4$

$\text{Variance} = \frac{\sum |x_i - \bar{x}|^2}{n} = \frac{136}{5}$

$= 81 + 49 + |18-a|^2 + |18-b|^2 + |18-c|^2 = 136$

$= (18-a)^2 + (18-b)^2 + (18-c)^2 = 6$

Possible values $(18-a)^2 = 1, \quad (18-b)^2 = 1 \quad (18-c)^2 = 4$

$a < b < c \quad 18-a = 1 \quad 18-b = -1 \quad 18-c = -2$

so $\quad c = 17 \quad b = 19 \quad 20$

$a + b + c = 56$

$2a + b - c \quad 34 = 19 - 20 = 33$

Q. 29 Lei $\alpha|x| = |y|e^{xy-\beta}, \alpha, \beta \in N$ be the solution of the differential equation $xdy - ydx + xy(xdy + ydx) = 0, y(1) = 2$. Then $\alpha + \beta$ is equal to___

Correct Answer:
4

Solution:
$a|x| = |y|e^{yx-\beta}, a, b \in N$

$xdy - ydx + xy(xdy + ydx) = 0$

$\frac{dy}{y} - \frac{dx}{x} + (xdy + ydx) = 0$

$\ln|y| - \ln|x| + xy = c$

$y(1) = 2$

$\ell n|2| - 0 + 2 = c$

$c = 2 + \ln 2$

$\ln|y| - \ln|x| + xy = 2 + \ln 2$

$\ln|x| = \ln \frac{y}{2} - 2 + xy$

$|x| = \frac{y}{2} e^{xy-2}$

$2|x| = |y|e^{xy-2}$

$\alpha = 2 \quad \beta = 2 \quad \alpha + \beta = 4$

Q. 30 If $\int \frac{1}{\sqrt[5]{(x-1)^4(x+3)^6}}dx = A\left(\frac{\alpha x-1}{\beta x+3}\right)^B + C$ where C is the constant of integration, then the value of $\alpha + \beta + 20\text{AB}$ is_____.

Correct Answer:
7

Solution:

$$\int \frac{1}{\sqrt[5]{(x-1)^4(x+3)^6}}dx = A\left(\frac{\alpha x-1}{\beta x+3}\right)^B + C$$

$$I = \int \frac{1}{(x-1)^{4/5}(x+3)^{6/5}}dx$$

$$I = \int \frac{1}{\left(\frac{x-1}{x+3}\right)^{4/5}(x+3)^2}dx$$

$$\left(\frac{x-1}{x+3}\right) = t \quad \Rightarrow \frac{4}{(x+3)^2}dx = dt \quad t^{-4/5+1}$$

$$I = \frac{1}{4}\int \frac{1}{t^{4/5}}dt = \frac{1}{4}\frac{t^{1/5}}{1/5} + c$$

$$I = \frac{5}{4}\left(\frac{x-1}{x+3}\right)^{1/5} + C$$

$$A = \frac{5}{4} \quad \alpha = \beta = 1 \quad B = \frac{1}{5}$$

$$\alpha + \beta + 20AB = 2 + 20 \times \frac{5}{4} \times \frac{1}{5} = 7$$

9 Apr - Morning Shift

Physics

Q. 1 A proton, an electron and an alpha particle have the same energies. Their de-Broglie wavelengths will be compared as:

Option 1:
$\lambda_e > \lambda_\alpha > \lambda_p$

Option 2:
$\lambda_\alpha < \lambda_\mathrm{p} < \lambda_\mathrm{e}$

Option 3:
$\lambda_p < \lambda_e < \lambda_\alpha$

Option 4:
$\lambda_p > \lambda_e > \lambda_\alpha$

Correct Answer:
$\lambda_\alpha < \lambda_\mathrm{p} < \lambda_\mathrm{e}$

Solution:
$\lambda_{\mathrm{DB}} = \frac{\mathrm{h}}{\mathrm{p}} = \frac{\mathrm{h}}{\sqrt{2\mathrm{mk}}}$

$\Rightarrow \lambda_{\mathrm{DB}} \alpha \frac{1}{\sqrt{\mathrm{m}}}$

$\Rightarrow \lambda_\mathrm{a} < \lambda_\mathrm{p} < \lambda_\mathrm{e}$

Q. 2 A particle moving in a straight line covers half the distance with speed 6 m/s. The other half is covered in two equal time intervals with speeds 9 m/s and 15 m/s respectively. The average speed of the particle during the motion is :

Option 1:
8.8 m/s

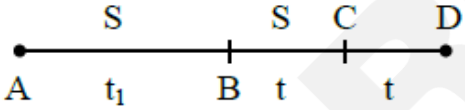
Option 2:
10 m/s

Option 3:
9.2 m/s

Option 4:
8 m/s

Correct Answer:
8 m/s

Solution:



$\mathrm{BD} \Rightarrow \mathrm{S} = 9\mathrm{t} + 15\mathrm{t} = 24\mathrm{t}$

$\mathrm{AB} \Rightarrow \mathrm{S} = 6\mathrm{t}_1 = 24\mathrm{t} \Rightarrow \mathrm{t}_1 = 4\mathrm{t}$

$< \text{ speed } >= \frac{\text{dist.}}{\text{time}} = \frac{48\mathrm{t}}{2\mathrm{t}+\mathrm{t}_1}$

$= \frac{48\mathrm{t}}{2\mathrm{t}+4\mathrm{t}} \Rightarrow \frac{48\mathrm{t}}{6\mathrm{t}} \Rightarrow 8\ \mathrm{m/s}$

Q. 3 A plane EM wave is propagating along $\mathbf{x}$ direction. It has a wavelength of $4\ \mathrm{mm}$. If electric field is in y direction with the maximum magnitude of $60\mathrm{Vm}^{-1}$, the equation for magnetic field is:

Option 1:

$B_z = 60 \sin\left[\frac{\pi}{2}\left(x - 3 \times 10^8 t\right)\right] k\hat{T}$

Option 2:

$B_z = 2 \times 10^{-7} \sin\left[\frac{\pi}{2} \times 10^3 \left(x - 3 \times 10^8 t\right)\right] k\hat{T}$

Option 3:

$B_x = 60 \sin\left[\frac{\pi}{2}\left(x - 3 \times 10^8 t\right)\right] i\hat{T}$

Option 4:

$B_z = 2 \times 10^{-7} \sin\left[\frac{\pi}{2}\left(x - 3 \times 10^8 t\right)\right] k\hat{T}$

Correct Answer:

$B_z = 2 \times 10^{-7} \sin\left[\frac{\pi}{2} \times 10^3 \left(x - 3 \times 10^8 t\right)\right] k\hat{T}$

Solution:

$E = BC \Rightarrow 60 = B \times 3 \times 10^8$

$\Rightarrow B = 2 \times 10^{-7}$

Also $C = f\lambda$

$\Rightarrow 3 \times 10^8 = f \times 4 \times 10^{-3}$

$\Rightarrow f = \frac{3}{4} \times 10^{11}$

$\Rightarrow \omega = 2\pi f = \frac{3}{4} \times 2\pi \times 10^{11}$

$\Rightarrow \omega = \frac{\pi}{2} \times 10^3 C$

$\Rightarrow$ Electric field $\Rightarrow y$ direction

Propagation $\Rightarrow x$ direction

Magnetic field $\Rightarrow z$-direction

Q. 4 Given below are two statements:

Statement (I): When an object is placed at the centre of curvature of a concave lens, image is formed at the centre of curvature of the lens on the other side.

Statement (II): Concave lens always forms a virtual and erect image.In the light of the above statements,

choose the correct answer from the options given below:

Option 1:
Statement I is false but Statement II is true.

Option 2:
Both Statement I and Statement II are false.

Option 3:
Statement I is true but Statement II is false.

Option 4:
Both Statement I and Statement II are true.

Correct Answer:
Both Statement I and Statement II are false.

Solution:

$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$

$\frac{1}{v}-\frac{1}{-2f}=\frac{1}{-f}$

$\Rightarrow \frac{1}{v}=\frac{-1}{2f} \Rightarrow v=-2f$

$\frac{1}{v}=\frac{1}{u}+\frac{1}{f} \Rightarrow$ Virtual image of Real object.

In statement II, it is not mentioned that object is real or virtual hence Statement II is false.

Q. 5 A light emitting diode (LED) is fabricated using GaAs semiconducting material whose band gap is 1.42 eV. The wavelength of light emitted from the LED is:

Option 1:
650 nm

Option 2:
1243 nm

Option 3:
875 nm

Option 4:
1400 nm

Correct Answer:
875 nm

Solution:
$\lambda = \frac{1240}{1.42} = 875 \text{ nm}$ (Approx)

Q. 6 A sphere of relative density σ and diameter D has concentric cavity of diameter d. The ratio of $\frac{D}{d}$, if it just floats on water in a tank is:

Option 1:
$\left(\frac{\sigma}{\sigma-1}\right)^{\frac{1}{3}}$

Option 2:
$\left(\frac{\sigma+1}{\sigma-1}\right)^{\frac{1}{3}}$

Option 3:

$\left(\frac{\sigma-1}{\sigma}\right)^{\frac{1}{3}}$

Option 4:

$\left(\frac{\sigma-2}{\sigma+2}\right)^{\frac{1}{3}}$

Correct Answer:

$\left(\frac{\sigma}{\sigma-1}\right)^{\frac{1}{3}}$

Solution:

$\text{weight } (\text{w}) = \frac{4}{3}\pi\left(\frac{\text{D}^3-\text{d}^3}{8}\right)\sigma\text{g}$

$\text{Buoyant force } (\text{F}_\text{b}) = 1 \times \frac{4}{3}\pi\left(\frac{\text{D}^3}{8}\right)\cdot\text{g}$

$\text{For Just Float } \Rightarrow \text{w} = \text{F}_\text{b}$

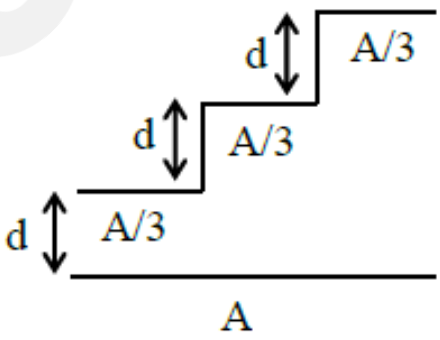
$\Rightarrow (\text{D}^3 - \text{d}^3)\sigma = \text{D}^3$

$\Rightarrow 1 - \frac{\text{d}^3}{\text{D}^3} = \frac{1}{\sigma}$

$\Rightarrow 1 - \frac{1}{\sigma} = \left(\frac{\text{d}}{\text{D}}\right)^3$

$\Rightarrow \left(\frac{\sigma}{\sigma-1}\right)^{\frac{1}{3}} = \left(\frac{\text{D}}{\text{d}}\right)$

Q. 7 A capacitor is made of a flat plate of area A and a second plate having a stair-like structure as shown in figure. If the area of each stair is $\frac{A}{3}$ and the height is d, the capacitance of the arrangement is:



Option 1:
$\frac{11\varepsilon_0 A}{18d}$

Option 2:
$\frac{13\varepsilon_0 A}{17\ \mathrm{d}}$

Option 3:
$\frac{11\varepsilon_0 A}{20d}$

Option 4:
$\frac{18\varepsilon_0 A}{11\ \mathrm{d}}$

Correct Answer:
$\frac{11\varepsilon_0 A}{18d}$

Solution:
All capacitor are in parallel combination.

Also effective area is common area only

$\Rightarrow C_{eq} = C_1 + C_2 + C_3$

$\Rightarrow C_{eq} = \frac{A\varepsilon_0}{3d} + \frac{A\varepsilon_0}{3(2d)} + \frac{A\varepsilon_0}{3(3d)}$

$\Rightarrow C_{eq} = \frac{A\varepsilon_0}{3}\left(\frac{11}{6d}\right)$

$\Rightarrow C_{eq} = \frac{11A\varepsilon_0}{18d}$

Q. 8 A light unstretchable string passing over a smooth light pulley connects two blocks of masses m_1 and $\mathrm{m_2}$. If the acceleration of the system is $\frac{\mathrm{g}}{8}$, then the ratio of the masses $\frac{\mathrm{m_2}}{\mathrm{m_1}}$ is:

Option 1:
9:7

Option 2:
4:3

Option 3:
5:3

Option 4:
8:1

Correct Answer:
9:7

Solution:

$a_{\text{sys}} = \left(\frac{m_2 - m_1}{m_1 + m_2}\right) g = \frac{g}{8}$

$\Rightarrow \frac{m_2}{m_1} = \frac{9}{7}$

Q. 9 The dimensional formula of latent heat is:

Option 1:
$\left[\mathrm{M^0LT^{-2}}\right]$

Option 2:
$\left[\mathrm{MLT^{-2}}\right]$

Option 3:
$\left[\mathrm{M^0\ L^2\ T^{-2}}\right]$

Option 4:
$\left[\mathrm{ML^2\ T^{-2}}\right]$

Correct Answer:
$\left[\mathrm{M^0\ L^2\ T^{-2}}\right]$

Solution:
Latent heat is specific heat

$\Rightarrow \frac{\mathrm{ML^2\ T^{-2}}}{\mathrm{M}} = \mathrm{M^0\ L^2\ T^{-2}}$

Q. 10 The volume of an ideal gas $(\gamma = 1.5)$ is changed adiabatically from 5 litres to 4 litres. The ratio of initial pressure to final pressure is:

Option 1:
$\frac{4}{5}$

Option 2:
$\frac{16}{25}$

Option 3:
$\frac{8}{5\sqrt{5}}$

Option 4:
$\frac{2}{\sqrt{5}}$

Correct Answer:
$\frac{8}{5\sqrt{5}}$

Solution:
For Adiabatic process

$$P_iV_i = P_fV_f^{\gamma}$$

$$P_i(5)^{1.5} = P_f(4)^{1.5}$$

$$\frac{P_i}{P_f} = \left(\frac{4}{5}\right)^{\frac{3}{2}} = \frac{4}{5}\cdot\left(\frac{4}{5}\right)^{\frac{1}{2}} \Rightarrow \frac{8}{5\sqrt{5}}$$

Q. 11 The energy equivalent of 1 g of substance is:

Option 1:
$11.2 \times 10^{24}\text{MeV}$

Option 2:
$5.6 \times 10^{12}\text{MeV}$

Option 3:
5.6eV

Option 4:
$5.6 \times 10^{26}\text{MeV}$

Correct Answer:
$5.6 \times 10^{26}\text{MeV}$

Solution:
$$\text{E} = \text{mC}^2$$

$$\Rightarrow \text{E} = \left(1 \times 10^{-3}\right) \times \left(3 \times 10^{8}\right)^2 \text{J}$$

$$\Rightarrow \text{E} = \left(10^{-3}\right)\left(9 \times 10^{16}\right)\left(6.241 \times 10^{18}\right)\text{eV}$$

$\mathrm{E} = 56.169 \times 10^{31} \mathrm{eV}$

$\mathrm{E} \approx 5.6 \times 10^{26} \mathrm{MeV}$

Q. 12 An astronaut takes a ball of mass m from earth to space. He throws the ball into a circular orbit about earth at an altitude of $318.5\ \mathrm{km}$. From earth's surface to the orbit, the change in total mechanical energy of the ball is $x\frac{\mathrm{GM}_e\ \mathrm{m}}{21\mathrm{R}_e}$. The value of x is (take $R_e = 6370\ \mathrm{km}$) :

Option 1:
11

Option 2:
9

Option 3:
12

Option 4:
10

Correct Answer:
11

Solution:

$\mathrm{h} = 318.5 \approx \left(\frac{\mathrm{R_e}}{20}\right)$

$\mathrm{T} \cdot \mathrm{E_i} = \frac{-\mathrm{GM_e m}}{\mathrm{R_e}}$

$\mathrm{T} \cdot \mathrm{E_f} = \frac{-\mathrm{GM_e m}}{2(\mathrm{R_e}+\mathrm{h})} = \frac{-\mathrm{GM_e m}}{2\left(\mathrm{R_e}+\frac{\mathrm{R_e}}{20}\right)}$

$\Rightarrow \mathrm{T} \cdot \mathrm{E_f} = \frac{-10\mathrm{GM_e m}}{21\mathrm{R_e}}$

Change in total mechanical energy

$= \mathrm{TEE_f} - \mathrm{TEE_i} = \frac{\mathrm{GM_e m}}{\mathrm{Re}}\left[1 - \frac{10}{21}\right] = \frac{11\mathrm{GM_e m}}{21\mathrm{Re}^2}$

Q. 13 Given below are two statements:

Statement (I) : When currents vary with time, Newton's third law is valid only if momentum carried by the electromagnetic field is taken into account.

Statement (II) : Ampere's circuital law does not depend on Biot-Savart's law. In the light of the above statements,

choose the correct answer from the options given below:

Option 1:
Both Statement I and Statement II are false.

Option 2:
Statement I is true but Statement II is false.

Option 3:
Statement I is false but Statement II is true.

Option 4:
Both Statement I and Statement II are true.

Correct Answer:
Statement I is true but Statement II is false.

Solution:
Conceptual.

Q. 14 A particle of mass m moves on a straight line with its velocity increasing with distance according to the equation $\mathrm{v} = \alpha\sqrt{\mathrm{x}}$, where α is a constant. The total work done by all the forces applied on the particle during its displacement from $\mathrm{x} = 0$ to $\mathrm{x} = \mathrm{d}$, will be:

Option 1:
$\frac{\mathrm{m}}{2\alpha^2\,\mathrm{d}}$

Option 2:
$\frac{md}{2\alpha^2}$

Option 3:
$\frac{m\alpha^2 d}{2}$

Option 4:
$2m\alpha^2 d$

Correct Answer:
$\frac{m\alpha^2 d}{2}$

Solution:
$\mathrm{v} = \alpha\sqrt{\mathrm{x}}$

$\text{at } \mathrm{x} = 0 : \mathrm{v} = 0$

$\text{at } \mathrm{x} = \mathrm{d}; \mathrm{v} = \alpha\sqrt{\mathrm{d}}$

$\mathrm{W.D} = \mathrm{K_f} - \mathrm{K_i}$

$\mathrm{W.D} = \frac{1}{2}\,\mathrm{m}(\alpha\sqrt{\mathrm{d}})^2 - \frac{1}{2}\,\mathrm{m}(0)^2$

$\Rightarrow \mathrm{W.D} = \frac{\mathrm{m}\alpha^2\,\mathrm{d}}{2}$

Q. 15 A galvanmeter has a coil of resistance 200Ω with a full scale deflection at $20\mu\mathrm{A}$. The value of resistance to be added to use it as an ammeter of range $(0 - 20)\mathrm{mA}$ is:

Option 1:
0.40Ω

Option 2:
0.20Ω

Option 3:
0.50Ω

Option 4:
0.10Ω

Correct Answer:
0.20Ω

Solution:
$\mathrm{G} = 200\Omega$

$\mathrm{i_g} = 20\mu\mathrm{A}$

$\mathrm{i} = \mathrm{i_g}\left(\frac{\mathrm{G}}{\mathrm{S}} + 1\right)$

$\Rightarrow 20 \times 10^{-3} = 20 \times 10^{-6}\left(\frac{200}{\mathrm{S}} + 1\right)$

$\Rightarrow \frac{200}{\mathrm{S}} = 999$

$\Rightarrow \mathrm{S} \approx 0.2\Omega$

Q. 16 A heavy iron bar, of weight W is having its one end on the ground and the other on the shoulder of a person. The bar makes an angle θ with the horizontal. The weight experienced by the person is:

Option 1:
$\frac{W}{2}$

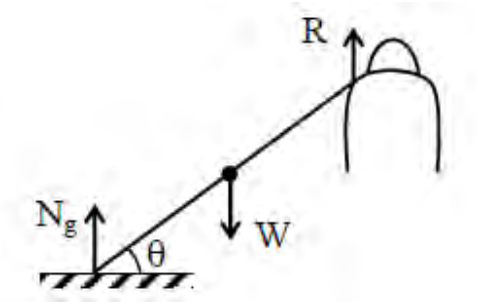
Option 2:
W

Option 3:
$\mathrm{W}\cos\theta$

Option 4:
$\mathrm{W}\sin\theta$

Correct Answer:
$\frac{W}{2}$

Solution:



$R =$ net reaction force by shoulder

Balancing torque about pt of contact on ground:

$\mathrm{W}\left(\frac{\mathrm{L}}{2}\cos\theta\right) = \mathrm{R}(\mathrm{L}\cos\theta)$

$\Rightarrow \mathrm{R} = \frac{\mathrm{W}}{2}$

Q. 17 One main scale division of a vernier caliper is equal to m units. If n^{th} division of main scale coincides with $(n+1)^{\mathrm{th}}$ division of vernier scale, the least count of the vernier caliper is:

Option 1:
$\frac{\mathrm{n}}{(\mathrm{n}+1)}$

Option 2:
$\frac{\mathrm{m}}{(\mathrm{n}+1)}$

Option 3:
$\frac{1}{(n+1)}$

Option 4:
$\frac{\mathrm{m}}{\mathrm{n}(\mathrm{n}+1)}$

Correct Answer:
$\frac{\mathrm{m}}{(\mathrm{n}+1)}$

Solution:
$n\mathrm{MSD} = (\mathrm{n}+1)\mathrm{VSD}$

$\Rightarrow 1\mathrm{VSD} = \frac{\mathrm{n}}{\mathrm{n}+1}\mathrm{MSD}$

$L \cdot C = 1\mathrm{MSD} - 1\mathrm{VSD}$

$L \cdot C = m - m\left(\frac{\mathrm{n}}{\mathrm{n}+1}\right)$

$L \cdot C = m\left(\frac{\mathrm{n}+1-\mathrm{n}}{\mathrm{n}+1}\right)$

$\Rightarrow L \cdot C = \left(\frac{\mathrm{m}}{\mathrm{n}+1}\right)$

Q. 18 A bulb and a capacitor are connected in series across an ac supply. A dielectric is then placed between the plates of the capacitor. The glow of the bulb:

Option 1:
increases

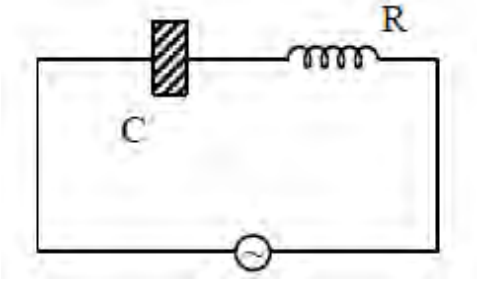
Option 2:
remains same

Option 3:
becomes zero

Option 4:
decreases

Correct Answer:
increases

Solution:



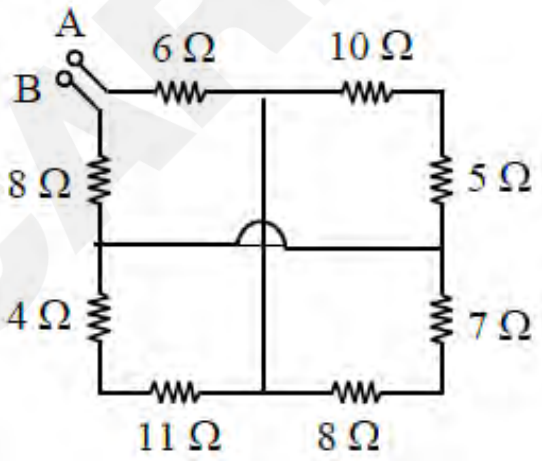
$$Z = \sqrt{R^2 + X_C^2} \& X_C = \frac{1}{WC}$$

due to dielectric

$$C \uparrow \Rightarrow X_c \downarrow \Rightarrow Z \downarrow$$

So, current increases \& thus bulb will glow more brighter.

Q. 19 The equivalent resistance between A and B is:



Option 1:
18Ω

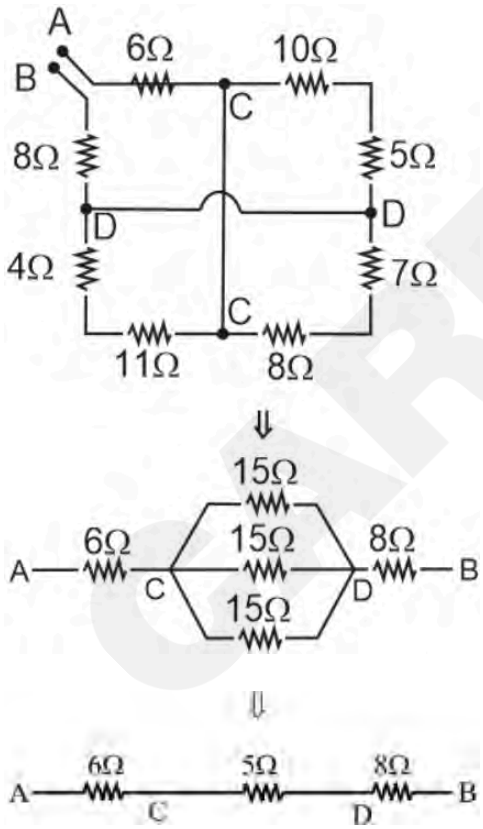
Option 2:
25Ω

Option 3:
27Ω

Option 4:
19Ω

Correct Answer:
19Ω

Solution:



$\Rightarrow R_{eq} = 6\Omega + 5\Omega + 8\Omega = 19\Omega$

Q. 20 A sample of 1 mole gas at temperature T is adiabatically expanded to double its volume. If adiabatic constant for the gas is $\gamma = \frac{3}{2}$, then the work done by the gas in the process is:

Option 1:
$\mathrm{RT}[2-\sqrt{2}]$

Option 2:
$\frac{\mathrm{R}}{\mathrm{T}}[2-\sqrt{2}]$

Option 3:
$\mathrm{RT}[2+\sqrt{2}]$

Option 4:
$\frac{T}{R}[2+\sqrt{2}]$

Correct Answer:
$\mathrm{RT}[2-\sqrt{2}]$

Solution:

$\mathrm{TV}^{\gamma-1} = \text{ constant}$

$\Rightarrow \mathrm{T}(\mathrm{V})^{\frac{3}{2}-1} = \mathrm{T_f}(2\ \mathrm{V})^{\frac{3}{2}-1}$

$\Rightarrow \mathrm{TV}^{\frac{1}{2}} = \mathrm{T_f}(2)^{\frac{1}{2}}(\ \mathrm{V})^{\frac{1}{2}}$

$\Rightarrow \mathrm{T_f} = \left(\frac{\mathrm{T}}{\sqrt{2}}\right)$

$\text{Now, W.D.} = \frac{\mathrm{nR\Delta T}}{1-\gamma} = \frac{1\cdot\mathrm{R}\left[\frac{\mathrm{T}}{\sqrt{2}}-\mathrm{T}\right]}{1-\frac{3}{2}}$

$\Rightarrow \text{ W.D.} = 2\mathrm{RT}\left[1-\frac{1}{\sqrt{2}}\right]$

$\Rightarrow \text{ W.D.} = \mathrm{RT}[2-\sqrt{2}]$

Q. 21 If $\vec{a}$ and $\vec{b}$ makes an angle $\cos^{-1}\left(\frac{5}{9}\right)$ with each other, then $|\vec{a}+\vec{b}| = \sqrt{2}|\vec{a}-\vec{b}|$ for $|\vec{a}| = n|\vec{b}|$ The integer value of n is _____.

Correct Answer:
3

Solution:
$\cos\theta = \frac{5}{9}$

$\frac{\vec{a}\cdot\vec{b}}{ab} = \frac{5}{9}$

$|\vec{a}+\vec{b}| = \sqrt{2}|\vec{a}-\vec{b}|$

$a^2+b^2+2\vec{a}\cdot\vec{b} = 2a^2+2b^2-4\vec{a}\cdot\vec{b}$

$6\vec{a}\cdot\vec{b} = a^2+b^2$

$6\times\frac{5}{9}ab = a^2+b^2$

$\frac{10}{3}ab = a^2+b^2 \quad \& \quad a = nb$

$\frac{10}{3}n^2 = n^2b^2+b^2$

$3n^2-10n+3=0$

$n = \frac{1}{3}$ and $n = 3$

integer value $n = 3$

Q. 22 At the centre of a half ring of radius $\mathrm{R} = 10\ \mathrm{cm}$ and linear charge density $4\mathrm{nCm}^{-1}$, the potential is $x\pi V$. The value of x is _____.

Correct Answer:
36

Solution:

Potential at centre of half ring

$$V = \frac{KQ}{R}$$

$$V = \frac{K\lambda\pi R}{R}$$

$$V = K\lambda\pi \Rightarrow V = 9 \times 10^{9} \times 4 \times 10^{-9}\pi$$

$$V = 36\pi$$

Q. 23 A star has 100% helium composition. It starts to convert three ^{4}He into one ^{12}C via triple alpha process as $^{4}He + {}^{4}He + {}^{4}He \rightarrow {}^{12}C + Q$. The mass of the star is 2.0×10^{32} kg and it generates energy at the rate of 5.808×10^{30} W. The rate of converting these ^{4}He to ^{12}C is $n \times 10^{42}$ s^{-1}, where n is _____ [Take, mass of $^{4}He = 4.0026u$, mass of $^{12}C = 12u$]

Correct Answer:
15

Solution:
$^{4}He + {}^{4}He + {}^{4}He \rightarrow {}^{12}C + Q$

$$\text{power generated } = \frac{N}{t}Q$$

where, $N \rightarrow$ No. of reaction /sec.

$$Q = (3\, m_{He} - m_{C})C^{2}$$

$$Q = (3 \times 4.0026 - 12)\left(3 \times 10^{8}\right)^{2}$$

$$Q = 7.266MeV$$

$$\frac{N}{t} = \frac{\text{power}}{Q} = \frac{5.808 \times 10^{30}}{7.266 \times 10^{6} \times 1.6 \times 10^{-19}}$$

$$\frac{N}{t} = 5 \times 10^{42}$$

rate of conversion of ^{4}He into $^{12}C = 15 \times 10^{42}$ Hence, $n = 15$

Q. 24 In a Young's double slit experiment, the intensity at a point is $\left(\frac{1}{4}\right)^{\text{th}}$ of the maximum intensity, the minimum distance of the point from the central maximum is _____ μm.
(Given : $\lambda = 600\ \text{nm},\ \text{d} = 1.0\ \text{mm}, \text{D} = 1.0\ \text{m}$)

Correct Answer:
200

Solution:
$\text{I} = \text{I}_0 \cos^2\left(\frac{\Delta\phi}{2}\right)$

$\frac{\text{I}_0}{4} = \cos^2\left(\frac{\Delta\phi}{2}\right)$

$\Delta\phi = \frac{2\pi}{3}$

$\frac{2\pi}{\lambda}\left(\frac{\text{yd}}{\text{D}}\right) = \frac{2\pi}{3}$

$\text{y} = \frac{\lambda\text{D}}{3\,\text{d}} = \frac{600\times10^{-9}\times1}{3\times10^{-3}} = 2\times10^{-4}\ \text{m}$

Q. 25 A string is wrapped around the rim of a wheel of moment of inertia 0.40kgm^2 and radius $10\ \text{cm}$. The wheel is free to rotate about its axis. Initially the wheel is at rest. The string is now pulled by a force of $40\ \text{N}$. The angular velocity of the wheel after 10 s is $\text{x}\text{rad}/\text{s}$, where x is_____

Correct Answer:
100

Solution:
$\tau = \text{FR} = \text{I}\alpha \Rightarrow 40\times0.1 = 0.4\alpha$

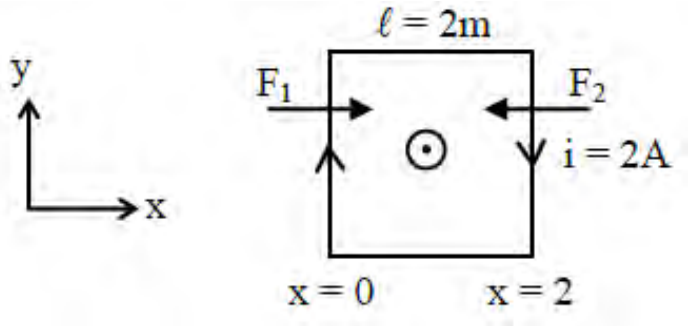
$\alpha = 10\text{rad}/\text{s}^2$

$\text{W}_\text{f} = 10\times10 = 100\text{rad}/\text{s}$

Q. 26 A square loop of edge length $2\ \mathrm{m}$ carrying current of $2\ \mathrm{A}$ is placed with its edges parallel to the $x-y$ axis. A magnetic field is passing through the $x-y$ plane and expressed as $\vec{B}=B_0(1+4x)\hat{k}$, where $\mathrm{B_0}=5\ \mathrm{T}$. The net magnetic force experienced by the loop is _____ N

Correct Answer:
160

Solution:



$\mathrm{B(x=0)=B_0},\quad \mathrm{B(x=2)=9\,B_0}$

$\mathrm{Also,\ F=i\ell B}$

$\Rightarrow \mathrm{F_1=i\ell B_0}$

$\mathrm{F_2=9i\ell B_0}$

$\mathrm{F=F_2-F_1=8i\ell B_0=8\times 2\times 2\times 5}$

$\mathrm{F=160\ N}$

Q. 27 Two persons pull a wire towards themselves. Each person exerts a force of $200\ \mathrm{N}$ on the wire. Young's modulus of the material of wire is $1\times 10^{11}\ \mathrm{N\ m^{-2}}$. Original length of the wire is $2\ \mathrm{m}$ and the area of cross section is $2\ \mathrm{cm^2}$. The wire will extend in length by _____$\mu\mathrm{m}$.

Correct Answer:
20

Solution:

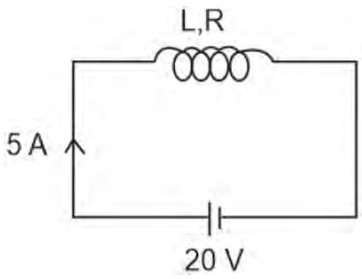
$\frac{F}{A} = Y\frac{\Delta\ell}{\ell} \Rightarrow \Delta\ell = \frac{F\ell}{AY}$

$\Delta\ell = \frac{200\times 2}{2\times 10^{-4}\times 10^{11}} = 2\times 10^{-5} = 20\mu m$

Q. 28 When a coil is connected across a $20\ V$ dc supply, it draws a current of $5\ A$. When it is connected across $20\ V, 50\ Hz$ ac supply, it draws a current of $4\ A$. The self inductance of the coil is _____ mH. (Take $\pi = 3$)

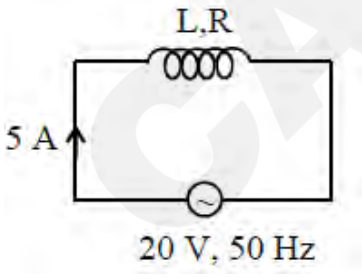
Correct Answer:
10

Solution:
Case-I:



$i = \frac{20}{R} \Rightarrow R = 4\Omega$

Case-II:



$i = \frac{20}{Z}$

$4 = \frac{20}{\sqrt{R^2+X_L^2}} \Rightarrow \sqrt{R^2+X_L^2} = 5$

$R^2 + X_L^2 = 25 \Rightarrow X_L = 3\Omega$

$L = \frac{3}{2\pi f} = \frac{1}{2\times 50} = \frac{1000}{100}\text{mH}$

$L = 10\text{mH}$

Q. 29 The position, velocity and acceleration of a particle executing simple harmonic motion are found to have magnitudes of $4\ \text{m}, 2\ \text{ms}^{-1}$ and $16\ \text{ms}^{-2}$ at a certain instant. The amplitude of the motion is $\sqrt{\text{x}}\text{m}$ where x is_____.

Correct Answer:
17

Solution:
$x = 4\ \text{m}, V = 2\ \text{m/s}, \text{a} = 16\ \text{m/s}^2$

$|\text{a}| = \omega^2\text{x}$

$\Rightarrow 16 = \omega^2(4)$

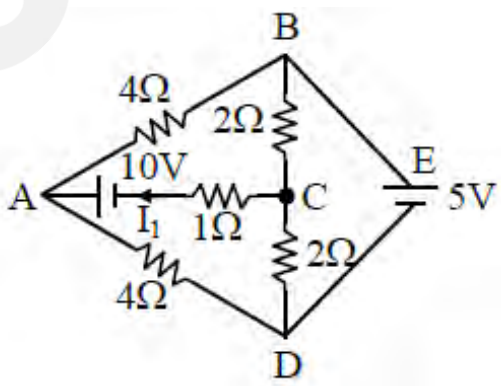
$\omega = 2\text{rad/s}$

$\text{v} = \omega\sqrt{\text{A}^2 - \text{x}^2}$

$A = \sqrt{\frac{\text{v}^2}{\omega^2} + \text{x}^2} \Rightarrow \text{A} = \sqrt{\frac{4}{4} + 16}$

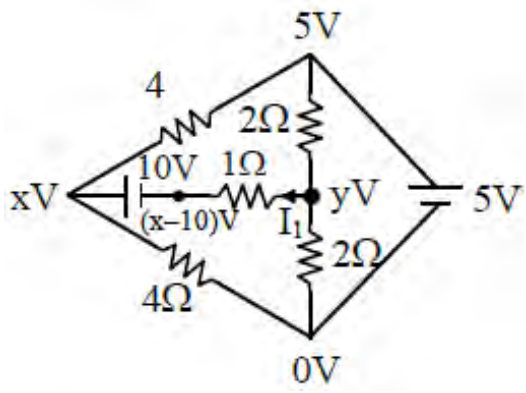
$A = \sqrt{17}\ \text{m}$

Q. 30 The current flowing through the 1Ω resistor is $\frac{\text{n}}{10}$ A. The value of n is _____.



Correct Answer:
25

Solution:



$\frac{y-5}{2}+\frac{y-0}{2}+\frac{y-x+10}{1}=0$

$y-5+y+2y-2x+20=0$

$4y-2x+15=0 \quad \ldots.(i)$

$\frac{x-5}{4}+\frac{x-0}{4}+\frac{x-10-y}{1}=0$

$x-5+x+4x-40-4y=0$

$6x-4y-45=0 \quad ..(i)$

$\frac{-2x+4y+15=0 \quad ..(ii)}{4x-30=0}$

$x=\frac{15}{2} 4y-15+15=0$

$y=0$

$i=\frac{y-x+10}{1}$

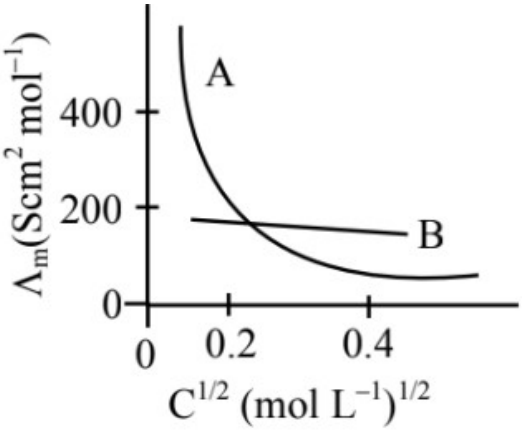
$i=\frac{0-7.5+10}{1}$

$i=2.5A=\frac{n}{10}A$

$n=25$

Chemistry

Q. 1 The molar conductivity for electrolytes A and B are plotted against $\mathrm{C}^{1/2}$ as shown below. Electrolytes A and B respectively are :



Option 1:

A	B
Weak electrolyte	weak electrolyte

Option 2:

A	B
Strong electrolyte	Strong electrolyte

Option 3:

A	B
Weak electrolyte	Strong electrolyte

Option 4:

A	B
Strong electrolyte	weak electrolyte

Correct Answer:

A	B
Weak electrolyte	Strong electrolyte

Solution:
A $\rightarrow$ Weak electrolyte
B $\rightarrow$ Strong electrolyte

Q. 2 Methods used for purification of organic compounds are based on :

Option 1:
neither on nature of compound nor on the impurity present.

Option 2:
nature of compound only.

Option 3:
nature of compound and presence of impurity.

Option 4:
presence of impurity only.

Correct Answer:
nature of compound and presence of impurity.

Solution:
Organic compounds are purified based on their nature and impruity present in it.

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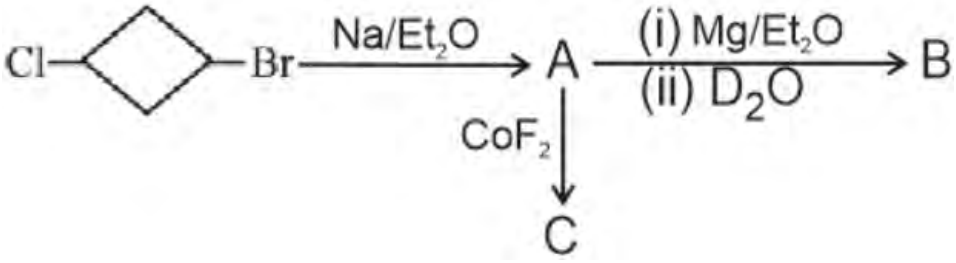
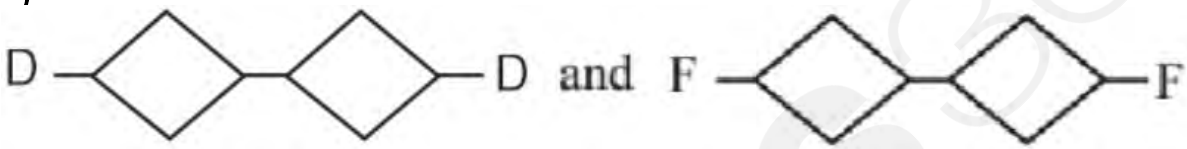
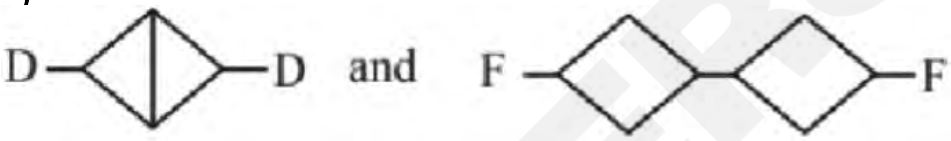
Q. 3 In the following sequence of reaction, the major products B and C respectively are :



Option 1:

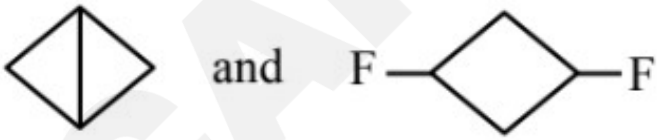


Option 2:

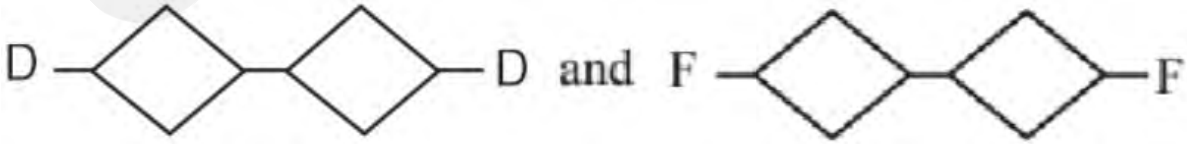


Option 3:

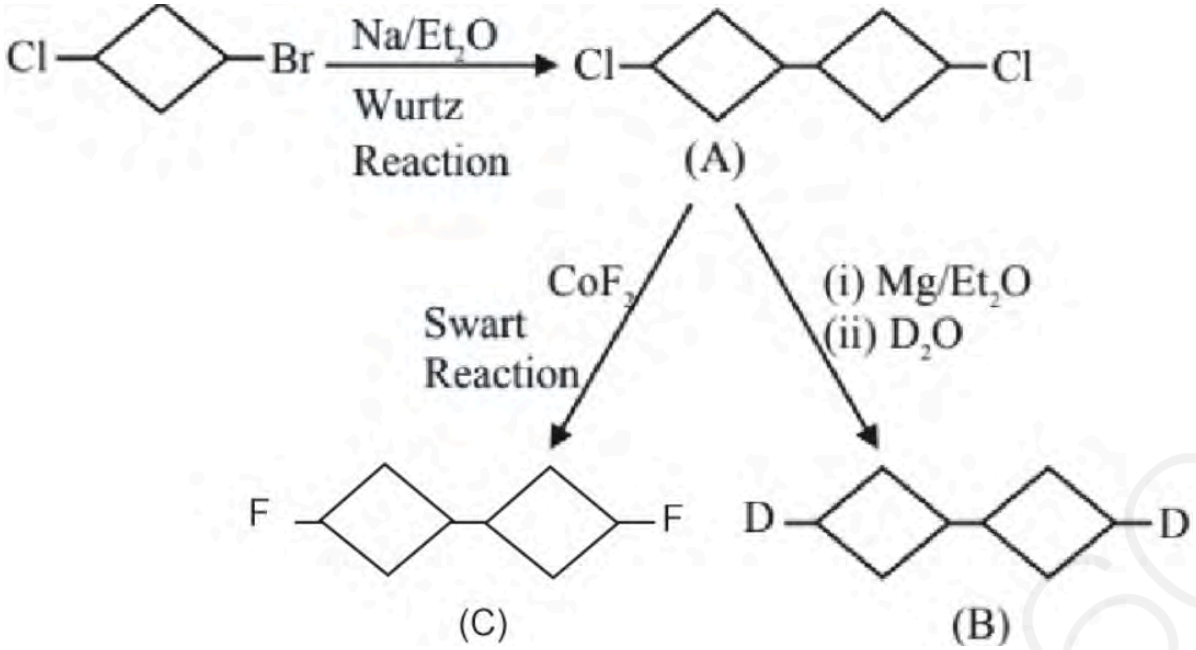
Option 4:



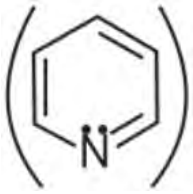
Correct Answer:



Solution:



Q. 4

Correct order of basic strength of Pyrrole , Pyridine 

and Piperidine is :

Option 1:
Piperidine > Pyridine > Pyrrole

Option 2:
Pyrrole > Pyridine > Piperidine

Option 3:
Pyridine > Piperidine > Pyrrole

Option 4:
Pyrrole > Piperidine > Pyridine

Correct Answer:
Piperidine > Pyridine > Pyrrole

Solution:

Order of basic strength is
$\mathrm{N}\left(\mathrm{sp}^3\right.$, localized lone pair $)>\mathrm{N}\left(\mathrm{sp}^2\right.$, localized lone pair $)\mathrm{N}\left(\mathrm{sp}^2\right.$, delocalized lone pair, aromatic)
$\therefore$ Piperidine $>$ Pyridine $>$ Pyrrole

Q. 5 In which one of the following pairs the central atoms exhibit sp^2 hybridization?

Option 1:
BF_3 and NO_2^-

Option 2:
NH_2^- and $\mathrm{H}_2\mathrm{O}$

Option 3:
$\mathrm{H}_2\mathrm{O}$ and NO_2

Option 4:
NH_2^- and BF_3

Correct Answer:
BF_3 and NO_2^-

Solution:
$\mathrm{BF}_3 \rightarrow \mathrm{sp}^2$
$\mathrm{NO}_2^- \rightarrow \mathrm{sp}^2$
$\mathrm{H}_2\mathrm{O} \rightarrow \mathrm{sp}^3$
$\mathrm{NO}_2 \rightarrow \mathrm{sp}^2$
$\mathrm{NH}_2{}^- \rightarrow \mathrm{sp}^3$

Q. 6 The F^- ions make the enamel on teeth much harder by converting hydroxyapatite (the enamel on the surface of teeth) into much harder fluoroapatite having the formula.

Option 1:
$\left[3\left(\mathrm{Ca}_3\left(\mathrm{PO}_4\right)_2\right) \cdot \mathrm{CaF}_2\right]$

Option 2:
$[3\,(Ca_2(PO_4)_2)\cdot Ca(OH)_2]$

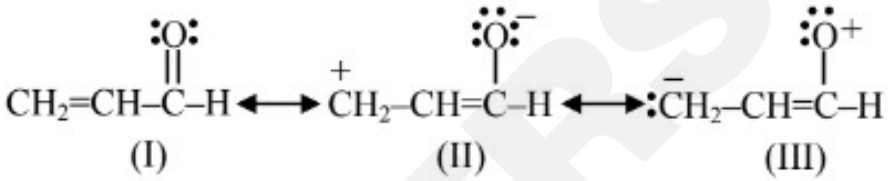
Option 3:
$[3\,(Ca_3(PO_4)_3)\cdot CaF_2]$

Option 4:
$[3\,(Ca_3(PO_4)_2)\cdot Ca(OH)_2]$

Correct Answer:
$[3\,(Ca_3(PO_4)_2)\cdot CaF_2]$

Solution:
Fluoroapatite $\Rightarrow [3Ca_3(PO_4)_2\cdot CaF_2]$

Q. 7 Relative stability of the contributing structures is :



Option 1:
(I) > (III) > (II)

Option 2:
(I) > (II) > (III)

Option 3:
(II) > (I) > (III)

Option 4:
(III) > (II) > (I)

Correct Answer:
(I) > (II) > (III)

Solution:
(1) Neutral structures are more stable than charged ones. Therefore I is more stable than II and III.
(2) +ve charge on less electronegative atom is more stable i.e., $C^{\oplus}$ is more stable than

$O^{\oplus}$

$\therefore$ Order is $I > II > III$

Q. 8 Given below are two statements :

Statement (I) : The oxidation state of an element in a particular compound is the charge acquired by its atom on the basis of electron gain enthalpy consideration from other atoms in the molecule.

Statement (II) : $p\pi - p\pi$ bond formation is more prevalent in second period elements over other periods.

In the light of the above statements, choose the most appropriate answer from the options given below :

Option 1:
Both Statement I and Statement II are incorrect

Option 2:
Statement I is correct but Statement II is incorrect

Option 3:
Both Statement I and Statement II are correct

Option 4:
Statement I is incorrect but Statement II is correct

Correct Answer:
Statement I is incorrect but Statement II is correct

Solution:
Oxidation state of an element in a particular compound is defined by the charge acquired by its atom on the basis of electronegativity consideration from other atoms in molecule.

Q. 9 Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R) :

Assertion (A) : S_N2 reaction of $C_6H_5CH_2Br$ occurs more readily than the S_N2 reaction of CH_3CH_2Br.

Reason (R) : The partially bonded unhybridized p-orbital that develops in the trigonal bipyramidal transition state is stabilized by conjugation with the phenyl ring.

In the light of the above statements, choose the most appropriate answer from the options given below :

Option 1:
(A) is not correct but (R) is correct

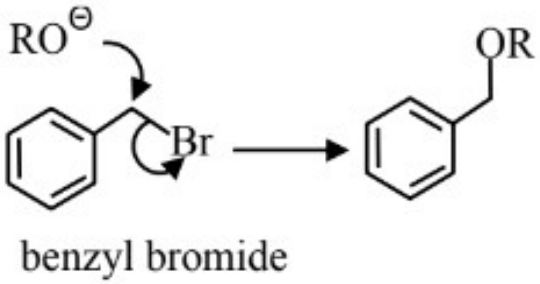
Option 2:
Both (A) and (R) are correct but (R) is not the correct explanation of (A)

Option 3:
Both (A) and (R) are correct and (R) is the correct explanation of (A)

Option 4:
(A) is correct but (R) is not correct

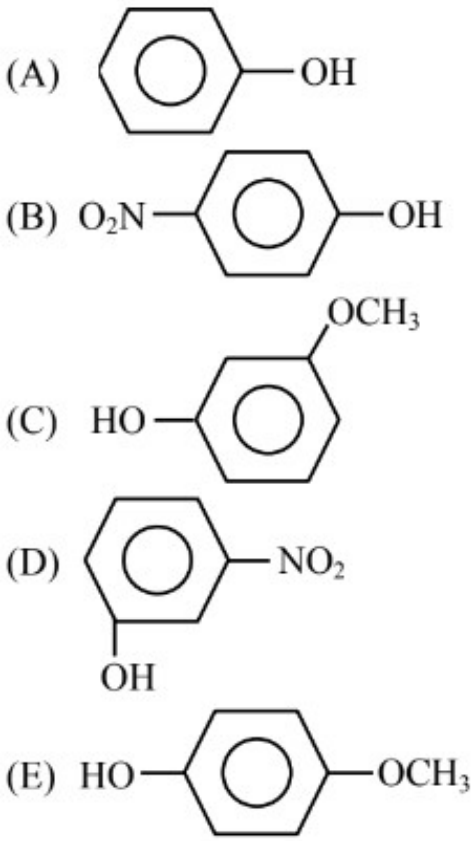
Correct Answer:
Both (A) and (R) are correct and (R) is the correct explanation of (A)

Solution:
The benzyl group acts in much the same way using the π-system of the benzene ring for conjugation with the p-orbital in the transition state.



benzyl bromide

Q. 10 For the given compounds, the correct order of increasing pK_a value :



Option 1:
(E) < (D) < (C) < (B) < (A)

Option 2:
(D) < (E) < (C) < (B) < (A)

Option 3:
(E) < (D) < (B) < (A) < (C)

Option 4:
(B) < (D) < (A) < (C) < (E)

Correct Answer:
(B) < (D) < (A) < (C) < (E)

Solution:
Acidic strength order :-

$$B > D > C > A > E$$

Correct pKa Order :

$$\mathrm{B} < \mathrm{D} < \mathrm{C} < \mathrm{A} < \mathrm{E}$$

All options are incorrect

Q. 11 Given below are two statements : one is labelled as **Assertion (A)**: and the other is labelled as **Reason (R)**.

Assertion (A) : Both rhombic and monoclinic sulphur exist as S_8 while oxygen exists as O_2.

Reason (R) : Oxygen forms $\mathrm{p}\pi$-p π multiple bonds with itself and other elements having small size and high electronegativity like C, N, which is not possible for sulphur.

In the light of the above statements, choose the most appropriate answer from the options given below :

Option 1:
Both (A) and (R) are correct and (R) is the correct explanation of $(\mathbf{A})$.

Option 2:
Both (A) and (R) are correct but (R) is not the correct explanation of (A).

Option 3:
(A) is correct but (R) is not correct.

Option 4:
(A) is not correct but (R) is correct.

Correct Answer:
(A) is correct but (R) is not correct.

Solution:
Oxygen can form $2p\pi - 2p\pi$ multiple bond with itself due to its small size while sulphur cannot form multiple bond with itself as $3p\pi - 3p\pi$ bond will be unstable due to large size of sulphur, but sulphur can form multiple bond with small size atom like C and N.
eg. $\mathrm{S} = \mathrm{C} = \mathrm{S}$

$$\mathrm{S} = \mathrm{C} = \mathrm{N}^- \leftrightarrow \mathrm{S}^{\ominus} - \mathrm{C} \equiv \mathrm{N}$$

Q. 12 Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): The total number of geometrical isomers shown by $[Co(en)_2Cl_2]^+$ complex ion is three

Reason (R): $[Co(en)_2Cl_2]^+$ complex ion has an octahedral geometry.

In the light of the above statements, choose the most appropriate answer from the options given below :

Option 1:
Both (A) and (R) are correct and (R) is the correct explanation of $(\mathbf{A})$.

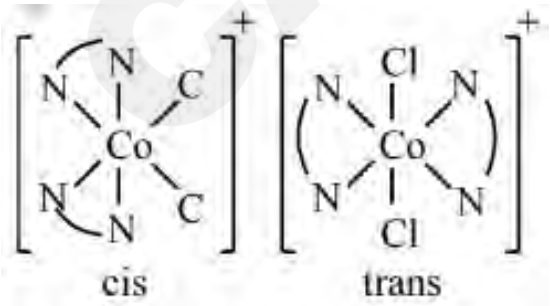
Option 2:
(A) is correct but (R) is not correct.

Option 3:
(A) is not correct but ($\mathbf{R}$) is correct.

Option 4:
Both (A) and (R) are correct but (R) is not the correct explanation of $(\mathbf{A})$.

Correct Answer:
(A) is not correct but ($\mathbf{R}$) is correct.

Solution:
$[Co(en)_2Cl_2]^+$ has octahedral geometry with two geometrical isomers.



Q. 13 The electronic configuration of $Cu(II)$ is $3\ d^9$ whereas that of $Cu(I)$ is $3\ d^{10}$. Which of the following is correct?

Option 1:
$Cu(II)$ is less stable

Option 2:
Stability of $Cu(I)$ and Cu (II) depends on nature of copper salts

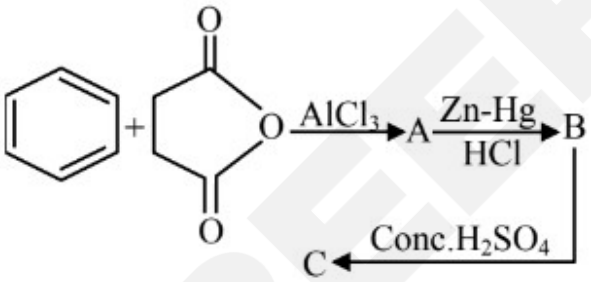
Option 3:
$Cu($ II) is more stable

Option 4:
$Cu(I)$ and Cu (II) are equally stable

Correct Answer:
$Cu($ II) is more stable

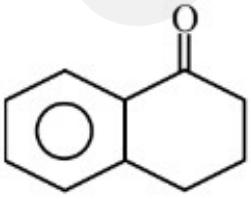
Solution:
$Cu(II)$ is more stable than $Cu(I)$ because hydration energy of Cu^{+2} ion compensate IE_2 of Cu.

Q. 14

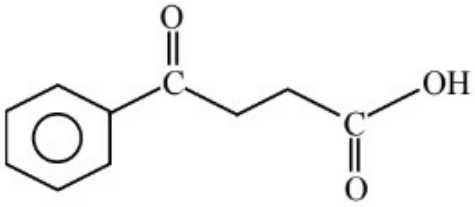


What is the structure of C ?

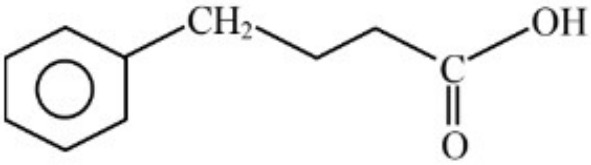
Option 1:



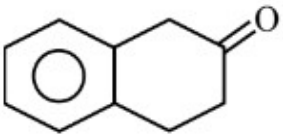
Option 2:



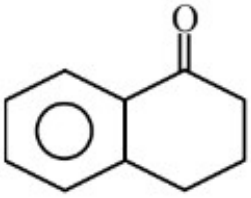
Option 3:



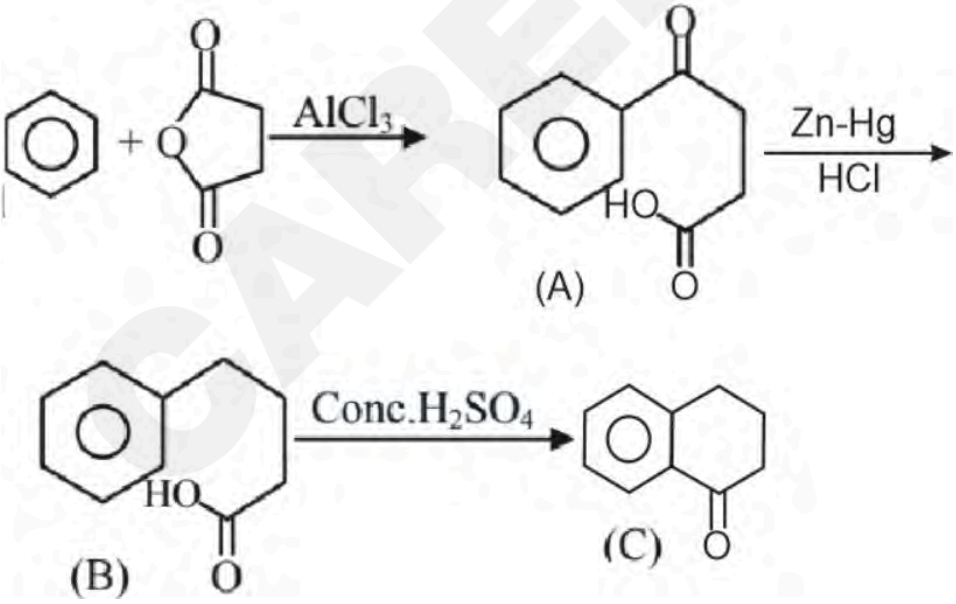
Option 4:



Correct Answer:



Solution:



Q. 15 Compare the energies of following sets of quantum numbers for multielectron system.

(A) $\mathrm{n}=4,1=1$

(B) $\mathrm{n}=4,1=2$

(C) $\mathrm{n}=3,1=1$

(D) $\mathrm{n}=3,1=2$

(E) $n=4,1=0

Choose the correct answer from the options given below :

Option 1:
(B) > (A) > (C) > (E) > (D)

Option 2:
(E) > (C) < (D) < (A) < (B)

Option 3:
(E) > (C) > (A) > (D) > (B)

Option 4:
(C) < (E) < (D) < (A) < (B)

Correct Answer:
(C) < (E) < (D) < (A) < (B)

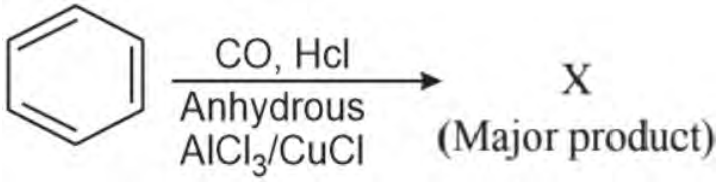
Solution:
Energy level can be determined by comparing $(\mathrm{n}+\ell)$ values

(A) $\mathrm{n}=4, \ell=1 \Rightarrow (\mathrm{n}+\ell)=5$
(B) $\mathrm{n}=4, \ell=2 \Rightarrow (\mathrm{n}+\ell)=6$
(C) $\mathrm{n}=3, \ell=1 \Rightarrow (\mathrm{n}+\ell)=4$
(D) $\mathrm{n}=3, \ell=2 \Rightarrow (\mathrm{n}+\ell)=5$
(E) $\mathrm{n}=4, \ell=0 \Rightarrow (\mathrm{n}+\ell)=4$

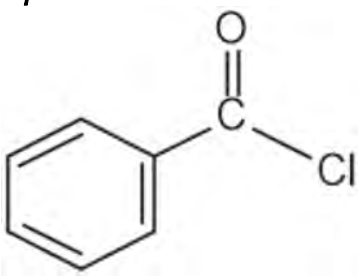
For same value of $(n+\ell)$, orbital having higher value of n, will have more energy.

$(\mathrm{B}) > (\mathrm{A}) > (\mathrm{D}) > (\mathrm{E}) > (\mathrm{C})$

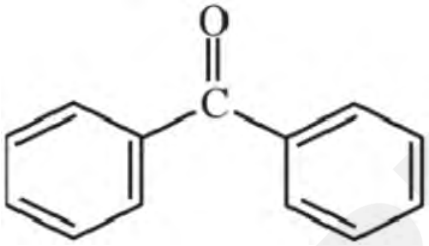
Q. 16 Identify major product " X " formed in the following reaction:



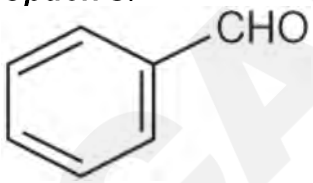
Option 1:



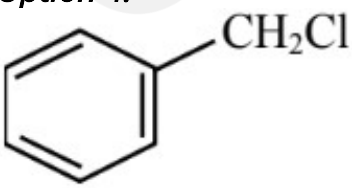
Option 2:



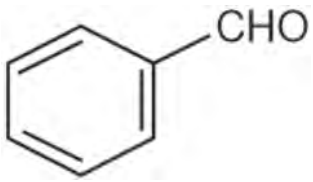
Option 3:



Option 4:

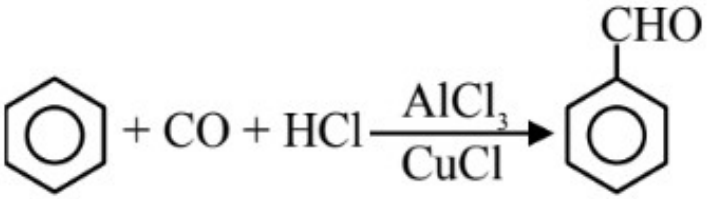


Correct Answer:

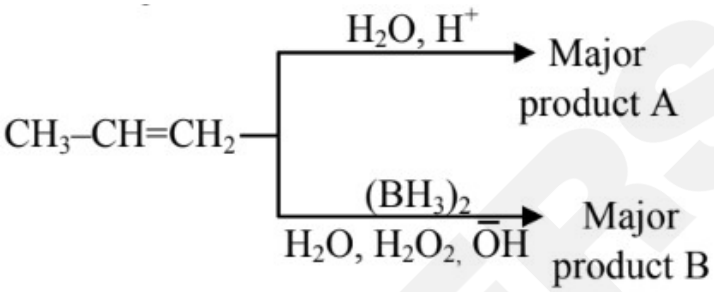


Solution:

This Gattermann-Koch reaction



Q. 17 Identify the product A and product B in the following set of reactions.



Option 1:

$A - CH_3CH_2CH_2 - OH, B - CH_3CH_2CH_2 - OH$

Option 2:

A-$CH_3CH_2CH_2$–OH, B-$CH_3\underset{\underset{OH}{|}}{CH}–CH_3$

Option 3:

A- $CH_3\underset{\underset{OH}{|}}{CH}–CH_3$, B-$CH_3CH_2CH_2$–OH

Option 4:

$A - CH_3CH_2CH_3,\ B - CH_3CH_2CH_3$

Correct Answer:

A- $CH_3-\underset{\underset{OH}{|}}{CH}-CH_3$, B-$CH_3CH_2CH_2-OH$

Solution:

(1) Hydration Reaction :

$$CH_3-CH=CH_2+H^+ \longrightarrow \underset{\text{(More stable)}}{CH_3-\overset{+}{C}H-CH_3}$$

$$CH_3-\overset{+}{C}H-CH_3 + H_2O \longrightarrow (CH_3-\underset{\underset{OH}{|}}{CH}-CH_3) + H^+$$

(A)

(2) Hydroboration Oxidation Reaction :

$$3CH_3-CH=CH_2 + B_2H_6 \xrightarrow{THF} 2(CH_3CH_2CH_2)_3\,B$$

$$(CH_3CH_2CH_2)_3\,B + 3H_2O_2 \xrightarrow{OH^-} 3CH_3CH_2CH_2OH + H_3BO_3$$

(B)

Q. 18 On reaction of Lead Sulphide with dilute nitric acid which of the following is not formed?

Option 1:
Lead nitrate

Option 2:
Sulphur

Option 3:
Nitric oxide

Option 4:
Nitrous oxide

Correct Answer:
Nitrous oxide

Solution:
$\mathrm{PbS} + \mathrm{HNO_3} \rightarrow \mathrm{Pb(NO_3)_2} + \mathrm{NO} + \mathrm{S} + \mathrm{H_2O}$

Nitrous oxide $(\mathrm{N_2O})$ is not formed during the reaction.

Q. 19 Identify the incorrect statements regarding primary standard of titrimetric analysis

(A) It should be purely available in dry form.

(B) It should not undergo chemical change in air.

(C) It should be hygroscopic and should react with another chemical instantaneously and stoichiometrically.

(D) It should be readily soluble in water.

(E) $\mathrm{KMnO_4 \& NaOH}$ can be used as primary standard.

Choose the correct answer from the options given below :

Option 1:
(C) and (D) only

Option 2:
(B) and (E) only

Option 3:
(A) and (B) only

Option 4:
(C) and (E) only

Correct Answer:
(C) and (E) only

Solution:
$\mathrm{KMnO_4 \& NaOH} \rightarrow$ Secondary standard.
Primary standard should not be Hygroscopic.

Q. 20 $0.05\mathrm{M}\mathrm{CuSO_4}$ when treated with $0.01\mathrm{M}\mathrm{K_2Cr_2O_7}$ gives green colour solution of $\mathrm{Cu_2Cr_2O_7}$. The [SPM : Semi Permeable Membrane]

$\mathrm{K_2Cr_2O_7}$		$\mathrm{CuSO_4}$
Side X	SPM	Side Y

Due to Osmosis :

Option 1:
Green colour formation observed on side Y.

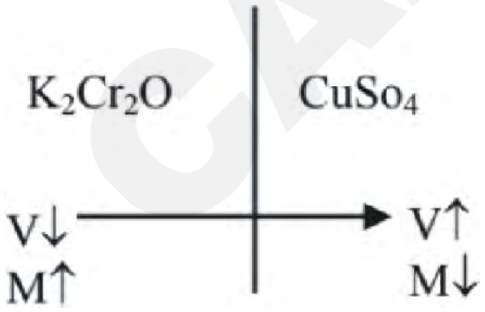
Option 2:
Green colour formation observed on side X.

Option 3:
Molarity of $\mathrm{K_2Cr_2O_7}$ solution is lowered.

Option 4:
Molarity of $\mathrm{CuSO_4}$ solution is lowered.

Correct Answer:
Molarity of $\mathrm{CuSO_4}$ solution is lowered.

Solution:
Only solvent Molecules are allowed to pass through the SPM.



Q. 21 The heat of solution of anhydrous $CuSO_4$ and $CuSO_4 \cdot 5H_2O$ are $-70\ kJ\ mol^{-1}$ and $+12\ kJ\ mol^{-1}$ respectively.

The heat of hydration of $CuSO_4$ to $CuSO_4 \cdot 5H_2O$ is $-xkJ$. The value of x is _____.

Correct Answer:
82

Solution:

$$(1)\ CuSO_4(s) + 5H_2O \xrightarrow{x} CuSO_4 \cdot 5H_2O$$

$$(2)\ CuSO_4 \cdot 5H_2O + H_2O \xrightarrow{12} CuSO_4(aq)$$

$$CuSO_4 + H_2O \xrightarrow{-70} CuSO_4(aq)$$

$$\text{from } (1) \text{ \& } (2)$$

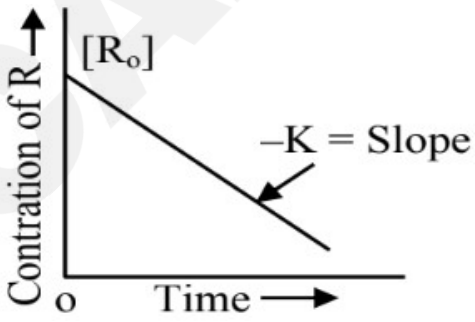
$$-70 = x + 12$$

$$x = -82$$

Q. 22 Given below are two statements :

Statement I : The rate law for the reaction $A + B \rightarrow C$ is rate $(r) = k[A]^2[B]$. When the concentration of both A and B is doubled, the reaction rate is increased " x " times.

Statement II :



The figure is showing "the variation in concentration against time plot" for a "y" order reaction.

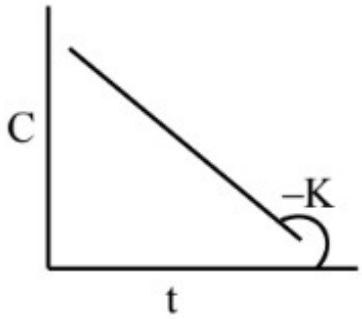
The value of x + y is _____.

Correct Answer:
8

Solution:
$\mathrm{r} = \mathrm{K}[\mathrm{A}]^2|\ \mathrm{B}|$

if conc. are doubled

$\mathrm{r}' = \mathrm{K}[2\ \mathrm{A}]^2[2\ \mathrm{B}]^1$
$\mathrm{r}' = 8\mathrm{r} \Rightarrow \mathrm{x} = 8$



$\Rightarrow$ Zero order, $y = 0$
$x + y = 8$

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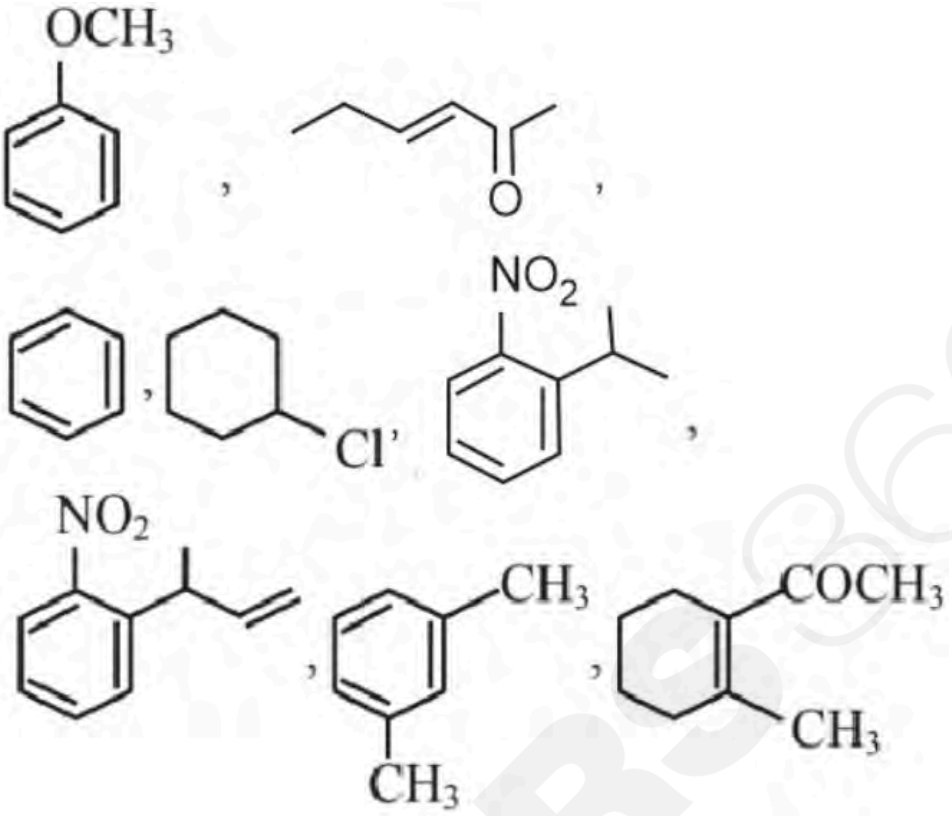
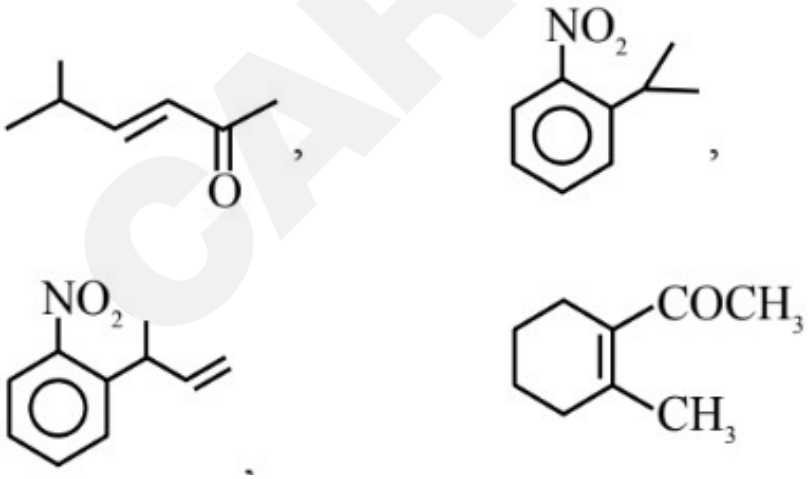
Q. 23 How many compounds among the following compounds show inductive, mesomeric as well as hyperconjugation effects?



Correct Answer:
4

Solution:



Q. 24 The standard reduction potentials at $298\ \mathrm{K}$ for the following half cells are given below :

$\mathrm{Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O, E^\circ = 1.33\ V}$

$\mathrm{Fe^{3+}(aq) + 3e^- \rightarrow Fe}$ $\quad E^\circ = -0.04\ V$

$\mathrm{Ni^{2+}(aq) + 2e^- \rightarrow Ni}$ $\quad E^\circ = -0.25\ V$

$\mathrm{Ag^{+}(aq) + e^- \rightarrow Ag}$ $\quad E^\circ = 0.80\ V$

$\mathrm{Au^{3+}(aq) + 3e^- \rightarrow Au}$ $\quad E^\circ = 1.40\ V$

Consider the given electrochemical reactions, The number of metal(s) which will be oxidized be $\mathrm{Cr_2O_7^{2-}}$, in aqueous solution is _____.

Correct Answer:
3

Solution:
Fe, Ni, Ag will be oxidized due to lower S.R.P.

Q. 25 When equal volume of $1\mathrm{MHCl}$ and $1\mathrm{MH_2SO_4}$ are separately neutralised by excess volume of $1\mathrm{M\ NaOH}$ solution. X and yJ of heat is liberated respectively. The value of y/x is _____.

Correct Answer:
2

Solution:
$\mathrm{H^+ + OH \rightarrow H_2O \Rightarrow x}$
$\mathrm{2H^+ + 2OH^- \rightarrow 2H_2O \Rightarrow 2x = y}$
$\mathrm{y/x = 2}$

Q. 26 Molarity (M) of an aqueous solution containing $\mathrm{x}g$ of anhyd. $\mathrm{CuSO_4}$ in $500\ \mathrm{mL}$ solution at $32^\circ\mathrm{C}$ is $2 \times 10^{-1}\mathrm{M}$. Its molality will be _____ $\times 10^{-3}\ \mathrm{m}$ (nearest integer).
[Given density of the solution $= 1.25\ \mathrm{g/mL}$.]

Correct Answer:
81

Solution:

$$\begin{aligned}\mathrm{M_{soln}}^{\mathrm{n}} &= \mathrm{v_{sol}}^{1} \times \mathrm{d_{sol}}^{\mathrm{n}} \\ &= 500 \times 1.25 = 625\ \mathrm{g}\end{aligned}$$

$$\begin{aligned}\text{Mass of solute (x)} &= 0.2 \times 0.5 \times 159.5 \\ &= 15.95\end{aligned}$$

$$\mathrm{n_{solute}} = 0.1,$$

$$\text{Mass of solvent} = \text{Mass of solution} - \text{Mass of solute}$$

$$= 625 - 15.95$$

$$= 609.05$$

$$\mathrm{m} = \frac{0.1}{\frac{609.05}{1000}}$$

$$\mathrm{m} = 0.164 = 164 \times 10^{-3}$$

Q. 27 The total number of species from the following in which one unpaired electron is present, is _____.

$$\mathrm{N_2, O_2, C_2^-, O_2^-, O_2^{2-}, H_2^+, CN^-, He_2^+}$$

Correct Answer:
4

Solution:
One unpaired $\mathrm{e^-}$ is present in : $\mathrm{C_2^-; O_2^-; H_2^+; He_2^+}$

Q. 28 Number of ambidentate ligands among the following is_____.

$$\mathrm{NO_2^-, SCN^-, C_2O_4^{2-}, NH_3, CN^-, SO_4^{2-}, H_2O.}$$

Correct Answer:
3

Solution:

Ligands which have two different donor sites but at a time connects with only one donor site to central metal are ambidentate ligands.

Ambidentate ligands are $NO_2^-; SCN^-; CN^-$

Q. 29 Total number of essential amino acid among the given list of amino acids is_____.

Arginine, Phenylalanine, Aspartic acid, Cysteine, Histidine, Valine, Proline

Correct Answer:
4

Solution:
Essential Amino acids are :-
Arginine, Phenylalanine, Histidine, Valine

Q. 30 Number of colourless lanthanoid ions among the following is _____
$Eu^{3+}, Lu^{3+}, Nd^{3+}, La^{3+}, Sm^{3+}$

Correct Answer:
2

Solution:
$La^{+3} - [Xe]4f^0$
$Nd^{+3} - [Xe]4f^3$
$Sm^{+3} - [Xe]4f^5$
$Eu^{+3} - [Xe]4f^6$
$Lu^{+3} - [Xe]4f^{14}$

La^{+3} and Lu^{+3} do not show any colour because no unpaired electron is present.

Maths

Q. 1 Let the line L intersect the lines

$x-2=-y=z-1, 2(x+1)=2(y-1)=z+1$ and be parallel to the line $\frac{x-2}{3}=\frac{y-1}{1}=\frac{z-2}{2}$. $

Then which of the following points lies on L ?

Option 1:
$\left(-\frac{1}{3}, 1, 1\right)$

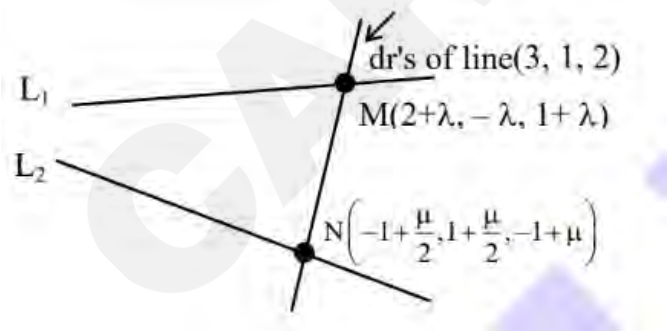
Option 2:
$\left(-\frac{1}{3}, 1, -1\right)$

Option 3:
$\left(-\frac{1}{3}, -1, -1\right)$

Option 4:
$\left(-\frac{1}{3}, -1, 1\right)$

Correct Answer:
$\left(-\frac{1}{3}, 1, -1\right)$

Solution:



$$L_1: \frac{x-2}{1}=\frac{y}{-1}=\frac{z-1}{1}=\lambda$$
$$L_2: \frac{x+1}{\frac{1}{2}}=\frac{y-1}{\frac{1}{2}}=\frac{z+1}{1}=\mu$$

dr of line MN will be
$<3+\lambda-\frac{\mu}{2}, -1-\lambda-\frac{\mu}{2}, 2+\lambda-\mu>$ & it will be proportional to $<3, 1, 2>$

$\therefore \frac{3+\lambda-\frac{\mu}{2}}{3}=\frac{-1-\lambda-\frac{\mu}{2}}{1}=\frac{2+\lambda-\mu}{2}$

$4\lambda+\mu=-6 \qquad 4+3\lambda=0$

$\Rightarrow \lambda=-\frac{4}{3} \& \mu=-\frac{2}{3}$

$\therefore$ Coordinate of M will be $<\left(\frac{2}{3}, \frac{4}{3}, -\frac{1}{3}\right)$

and equation of required line will be.

$\frac{x-\frac{2}{3}}{3}=\frac{y-\frac{4}{3}}{1}=\frac{z+\frac{1}{3}}{2}=k$

So any point on this line will be
$\left(\frac{2}{3}+3\mathrm{k}, \frac{4}{3}+\mathrm{k}, -\frac{1}{3}+2\mathrm{k}\right)$

$\because \frac{2}{3}+3\mathrm{k}=-\frac{1}{3} \Rightarrow \mathrm{k}=-\frac{1}{3}$
$\therefore$ Point lie on the line for $\mathrm{k}=-\frac{1}{3}$ is $\left(-\frac{1}{3}, 1, -1\right)$

Q. 2 The parabola $y^2=4x$ divides the area of the circle $x^2+y^2=5$ in two parts. The area of the smaller part is equal to :

Option 1:
$\frac{2}{3}+5\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)\$$

Option 2:
$\frac{1}{3}+5\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$

Option 3:
$\frac{1}{3}+\sqrt{5}\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$

Option 4:
$\frac{2}{3}+\sqrt{5}\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$

Correct Answer:

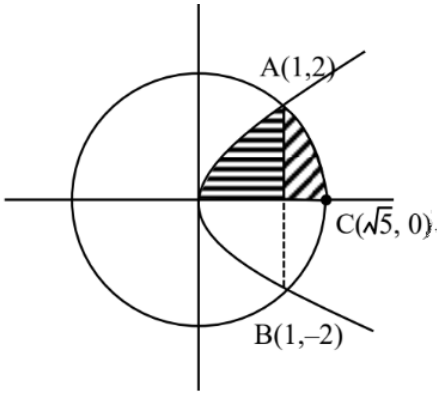
$\frac{2}{3}+5\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$$

Solution:

$y^2=4x$

$x^2+y^2=5$

$\therefore$ Area of shaded region as shown in the figure will be



$$A_1=\int_0^1\sqrt{4x}dx+\int_1^{\sqrt{5}}\sqrt{5-x^2}dx$$

$$=\frac{4}{3}\cdot\left[x^{\frac{3}{2}}\right]_0^1+\left[\frac{x}{2}\sqrt{5-x^2}+\frac{5}{2}\sin^{-1}\frac{x}{\sqrt{5}}\right]_1^{\sqrt{x}}$$

$$=\frac{1}{3}+\frac{5\pi}{4}-\frac{5}{2}\sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$$

$\therefore$ Required Area $=2\,A_1$

$$=\frac{2}{3}+\frac{5\pi}{2}-5\sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$$

$$=\frac{2}{3}+5\left(\frac{\pi}{2}-\sin^{-1}\frac{1}{\sqrt{5}}\right)$$

$$=\frac{2}{3}+5\cos^{-1}\frac{1}{\sqrt{5}}$$

$$=\frac{2}{3}+5\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$$

Q. 3 The solution curve, of the differential equation $2y\frac{dy}{dx}+3=5\frac{dy}{dx}$, passing through the point $(0,1)$ is a conic, whose vertex lies on the line :

Option 1:
$2x + 3y = 9$

Option 2:
$2x + 3y = -9$

Option 3:
$2x + 3y = -6$

Option 4:
$2x + 3y = 6$

Correct Answer:
$2x + 3y = 9$

Solution:

$(2y - 5)\frac{dy}{dx} = -3$

$(2y - 5)dy = -3dx$

$2 \cdot \frac{y^2}{2} - 5y = -3x + \lambda$

$\because$ Curve passes through $(0, 1)$

$\Rightarrow \lambda = -4$

$\because$ Curve will be

$$\left(y - \frac{5}{2}\right)^2 = -3\left(x - \frac{3}{4}\right)$$

$\therefore$ Vertex of parabola will be $\left(\frac{3}{4}, \frac{5}{2}\right)$

$\because 2x + 3y = 9$

Q. 4 A ray of light coming from the point $\mathrm{P}(1, 2)$ gets reflected from the point Q on the x-axis and then passes through the point $R(4, 3)$. If the point $S(h, \mathrm{k})$ is such that PQRS is a parallelogram, then hk^2 is equal to :

Option 1:
80

Option 2:
90

Option 3:
60

Option 4:
70

Correct Answer:
70

Solution:

Image of P wrt x-axis will be $\mathrm{P}'(1,-2)$ equation of line joining $\mathrm{P}'\mathrm{R}$ will be
$\mathrm{y}-3=\frac{5}{3}(\mathrm{x}-4)$

Above line will meet X-axis at Q where
$\mathrm{y}=0 \Rightarrow \mathrm{x}=\dfrac{11}{5}$
$\therefore \mathrm{Q}\left(\dfrac{11}{5}, 0\right)$
$\because \mathrm{PQRS}$ is parallelogram so their diagonals will bisects each other

$$\Rightarrow \frac{4+1}{2}=\frac{\frac{11}{5}+h}{2} \& \frac{2+3}{2}=\frac{\mathrm{k}+0}{2}$$
$$\Rightarrow \mathrm{h}=\frac{14}{5} \& \mathrm{k}=5$$
$$\therefore \mathrm{hk}^2=\frac{14}{5} \times 5^2=70$$

Q. 5 Let $\lambda, \mu \in \mathrm{R}$. If the system of equations
$3x+5y+\lambda z=3$
$7x+11y-9z=2$
$97x+155y-189z=\mu$
has infinitely many solutions, then $\mu+2\lambda$ is equal to :

Option 1:
25

Option 2:
24

Option 3:
27

Option 4:
22

Correct Answer:
25

Solution:

$3x+5y+\lambda z=3$

$7x+11y-9z=2$

$97x+155y-189z=\mu$

$93x+155y+31\lambda z=93$

$97x+155y-189z=\mu$

$-\quad+\quad+$

$-4x+(31\lambda+189)z=93-\mu$

$1085x+1705y-1395z=310$

$1067x+1705y-2079z=11\mu$

$-\quad+\quad-$

$18x+684z=310-11\mu$

$-36x+9(31\lambda+189)z=9(93-\mu)$

$36x+1368z=2(310-11\mu)$

$(279\lambda+3069)z=1457-31\mu$

for infinite solutions -

$\lambda=\frac{-3069}{279}=\frac{-341}{31}$

$\mu=\frac{1457}{31}$

$\mu+2\lambda=\frac{1457-682}{31}=\frac{775}{31}=25$

Q. 6 The coefficient of x^{70} in $x^2(1+x)^{98}+x^3(1+x)^{97}+$ $x^4(1+x)^{96}+\ldots\ldots\ldots+x^{54}(1+x)^{46}$ is ${}^{99}C_p-{}^{46}C_q$. Then a possible value to $p+q$ is :

Option 1:
55

Option 2:
61

Option 3:
68

Option 4:
83

Correct Answer:
83

Solution:
$x^2(1+x)^{98}+x^3\left(1+x^{97}\right)+x^4(1+x)^{96}+\ldots\ldots$
$x^{54}(1+x)^{46}$

Coeff. of x^{70} : ${}^{98}C_{68}+{}^{97}C_{67}+{}^{96}C_{66}+\ldots\ldots\ldots$
${}^{47}C_{17}+{}^{46}C_{16}$
$={}^{46}C_{30}+{}^{47}C_{30}+\ldots\ldots\ldots{}^{98}C_{30}$

$=\left({}^{46}C_{31}+{}^{46}C_{30}\right)+{}^{47}C_{30}+\ldots\ldots\ldots\ldots{}^{98}C_{30}-{}^{46}C_{31}$
$={}^{47}C_{31}+{}^{47}C_{30}+\ldots\ldots\ldots\ldots{}^{98}C_{30}-{}^{46}C_{31}$
$\ldots\ldots\ldots$
$={}^{99}C_{31}-{}^{46}C_{31}={}^{99}C_p-{}^{46}C_q$

Possible values of $(p+q)$ are $62, 83, 99, 46$
$\Rightarrow p+q=83$

Q. 7 Let $\int \frac{2-\tan x}{3+\tan x}dx = \frac{1}{2}(\alpha x + \log_e|\beta \sin x + \gamma \cos x|) + C$, where C is the constant of integration. Then $\alpha + \frac{\gamma}{\beta}$ is equal to :

Option 1:
3

Option 2:
1

Option 3:
4

Option 4:
7

Correct Answer:
4

Solution:

$$\int \frac{2-\tan x}{3+\tan x}dx = \int \frac{2\cos x - \sin x}{3\cos x + \sin x}dx$$

$$2\cos x - \sin x = A(3\cos x + \sin x) + B(\cos x - 3\sin x)$$

$$3A + B = 2$$

$$A - 3B = -1$$

$$\Rightarrow \mathrm{A} = \frac{1}{2}, \mathrm{B} = \frac{1}{2}$$

$$\therefore \int \frac{2\cos \mathrm{x} - \sin \mathrm{x}}{3\cos \mathrm{x} + \sin \mathrm{x}}\mathrm{x}$$

$$= \frac{\mathrm{x}}{2} + \frac{1}{2}\ln|3\cos \mathrm{x} + \sin \mathrm{x}| + \mathrm{C}$$

$$= \frac{1}{2}(\mathrm{x} + \ln|3\cos \mathrm{x} + \sin \mathrm{x}|) + \mathrm{C}$$

$$= \frac{1}{2}(\alpha \mathrm{x} + \ln|\beta \sin \mathrm{x} + \gamma \cos \mathrm{x}|) + \mathrm{C}$$

$$\alpha = 1, \beta = 1, \gamma = 3$$

$$\therefore \alpha + \frac{\gamma}{\beta} = 1 + \frac{3}{1} = 4$$

Q. 8 A variable line L passes through the point $(3,5)$ and intersects the positive coordinate axes at the points A and B. The minimum area of the triangle OAB, where O is the origin, is :

Option 1:
30

Option 2:
25

Option 3:
40

Option 4:
35

Correct Answer:
30

Solution:

$$\frac{x}{a}+\frac{y}{b}=1$$

$$\frac{3}{a}+\frac{5}{b}=1 \Rightarrow b=\frac{5a}{a-3}, a>3$$

$$\mathrm{A}=\frac{1}{2}\mathrm{ab}=\frac{1}{2}\mathrm{a}\frac{5\mathrm{a}}{(\mathrm{a}-3)}=\frac{5}{2}\cdot\frac{\mathrm{a}^2}{\mathrm{a}-3}$$

$$=\frac{5}{2}\left(\frac{\mathrm{a}^2-9+9}{\mathrm{a}-3}\right)$$

$$=\frac{5}{2}\left(\mathrm{a}+3+\frac{9}{\mathrm{a}-3}\right)$$

$$=\frac{5}{2}\left(\mathrm{a}-3+\frac{9}{\mathrm{a}-3}+6\right)\geq 30$$

Q. 9 Let
$|\cos\theta\cos(60-\theta)\cos(60-\theta)| \leq \frac{1}{8}, \theta \in [0, 2\pi]$

Then, the sum of all $\theta \in [0, 2\pi]$, where $\cos 3\theta$ attains its maximum value, is :

Option 1:
9π

Option 2:
18π

Option 3:
6π

Option 4:
15π

Correct Answer:
6π

Solution:
We know that
$(\cos\theta)\left(\cos(60^\circ-\theta)\right)\left(\cos(60^\circ+\theta)\right) = \frac{1}{4}\cos 3\theta$

So equation reduces to $\frac{1}{4}\cos 3\theta \leq \frac{1}{8}$

$$\Rightarrow |\cos 3\theta| \leq \frac{1}{2}$$

$$\Rightarrow -\frac{1}{2} \leq \cos 3\theta \leq \frac{1}{2}$$

$\Rightarrow$ maximum value of $\cos 3\theta = \frac{1}{2}$, here

$$\Rightarrow 3\theta = 2n\pi \pm \frac{\pi}{3}$$

$$\theta = \frac{2n\pi}{3} \pm \frac{\pi}{9}$$

As $\theta \in [0, 2\pi]$ possible values are
$\theta = \left\{\frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}, \frac{11\pi}{9}, \frac{13\pi}{9}, \frac{17\pi}{9}\right\}$

Whose sum is
$\frac{\pi}{9} + \frac{5\pi}{9} + \frac{7\pi}{9} + \frac{11\pi}{9} + \frac{13\pi}{9} + \frac{17\pi}{9} = \frac{54\pi}{9} = 6\pi$

Q. 10 Let $\overrightarrow{OA} = 2\vec{a}, \overrightarrow{OB} = 6\vec{a} + 5\vec{b}$ and $\overrightarrow{OC} = 3\vec{b}$, where O is the origin. If the area of the parallelogram with adjacent sides $\overrightarrow{OA}$ and $\overrightarrow{OC}$ is 15 sq. units, then the area (in sq. units) of the quadrilateral $OABC$ is equal to :

Option 1:
38

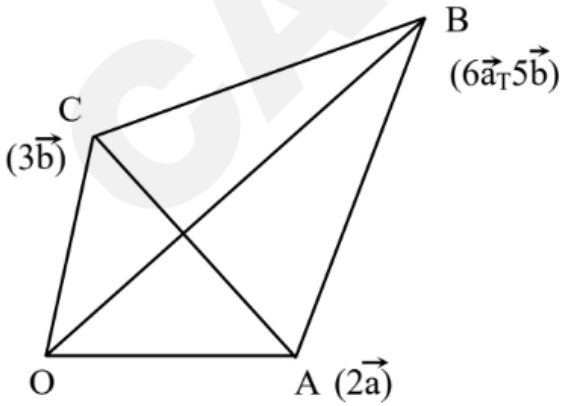
Option 2:
40

Option 3:
32

Option 4:
35

Correct Answer:
35

Solution:



Area of parallelogram having sides

$\overrightarrow{OA}\&\overrightarrow{OC} = |\overrightarrow{OA} \times \overrightarrow{OC}| = |2\vec{a} \times 3\vec{b}| = 15$

$6|\vec{a} \times \vec{b}| = 15$

$\Rightarrow |\vec{a} \times \vec{b}| = \frac{5}{2} \ldots\ldots\ldots (1)$

Area of quadrilateral

$OABC = \frac{1}{2} \vec{d}_1 \times \vec{d}_2$

$= \frac{1}{2}|\overrightarrow{AC} \times \overrightarrow{OB}| = \frac{1}{2}|(3\vec{b} - 2\vec{a}) \times (6\vec{a} + 5\vec{b})|$

$= \frac{1}{2}|18\vec{b} \times \vec{a} - 10\vec{a} \times \vec{b}| = 14|\vec{a} \times \vec{b}|$

$= 14 \times \frac{5}{2} = 35$

Q. 11 If the domain of the function $f(x) = \sin^{-1}\left(\frac{x-1}{2x+3}\right)$ is $R - (\alpha, \beta)$ then $12\alpha\beta$ is equal to :

Option 1:
36

Option 2:
24

Option 3:
40

Option 4:
32

Correct Answer:
32

Solution:

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Domain of $f(x)=\sin^{-1}\left(\frac{x-1}{2x+3}\right)$ is

$2x+3\neq 0 \& x\neq \frac{-3}{2}$ and $\frac{(x-1)}{2x+3}\leq 1$

$|x-1|\leq|2x+3|$

For $|2x+3|\geq|x-1|$

$x\in(-\infty,-4]\cup\left(-\frac{2}{3},\infty\right)$

$\alpha=-4 \& \beta=-\frac{2}{3}: 12\alpha\beta=32$

Q. 12 If the sum of series $\frac{1}{1\cdot(1+d)}+\frac{1}{(1+d)(1+2d)}+\ldots\ldots+\frac{1}{(1+9d)(1+10d)}$ is equal to 5 , then $50\ \mathrm{d}$ is equal to :

Option 1:
20

Option 2:
5

Option 3:
15

Option 4:
10

Correct Answer:
5

Solution:

$$\frac{1}{1\cdot(1+\mathrm{d})}+\frac{1}{(1+\mathrm{d})(1+2\,\mathrm{d})}+\ldots\ldots\ldots\ldots$$

$$\frac{1}{(1+9\,\mathrm{d})(1+10\,\mathrm{d})}=5$$

$$\frac{1}{d}\left[\frac{(1+d)-1}{1\cdot(1+d)}+\frac{(1+2d)-(1-d)}{(1+d)(1+2d)}\right]+\ldots\ldots\ldots\ldots$$

$$\frac{(1+10d)-(1+9d)}{(1+9d)(1+10d)}=5$$

$$\frac{1}{d}\left[\left(1-\frac{1}{1+d}\right)+\left(\frac{1}{1+d}-\frac{1}{1+2d}\right)+\ldots\ldots\ldots\ldots\right.$$

$$\left.\left(\frac{1}{1+9d}-\frac{1}{1+10d}\right)\right]=5$$

$$\frac{1}{d}\left[1-\frac{1}{(1+10d)}\right]=5$$

$$\frac{10d}{1+10d}=5\ \mathrm{d}$$

$$50\ \mathrm{d}=5$$

Q. 13 Let $f(x)=ax^3+bx^2+ex+41$ be such that $f(1)=40, f'(1)=2$ and $f'(1)=4$.
Then $a^2+b^2+c^2$ is equal to :

Option 1:
62

Option 2:
73

Option 3:
54

Option 4:
51

Correct Answer:
51

Solution:

$f(x)=ax^3+bx^2+cx+41$

$f'(x)=3ax^2+2bx+cx$

$\Rightarrow f'(1)=3a+2b+c=2\ldots\ldots\ldots(1)$

$f''(n) = 6ax + 2b$

$\Rightarrow f''(1) = 6a + 2b = 4$

$3a + b = 2 \ldots\ldots\ldots (2)$

$(1) - (2)$

$b + c = 0 \ldots\ldots\ldots (3)$

$f(1) = 40$

$a + b + c + 41 = 40$

use (3)

$a + 41 = 40$

by (2)

$-3 + b = 2 \Rightarrow b = 5 \& c = -5$

$a^2 + b^2 + c^2 = 1 + 25 + 25 = 51$

Q. 14 Let a circle passing through $(2, 0)$ have its centre at the point (h, k). Let $(\mathrm{x_c}, \mathrm{y_c})$ be the point of intersection of the lines $3x + 5y = 1$ and $(2 + c)x + 5c^2y = 1$. If $h = \lim_{c \to 1} x_c$ and $k = \lim_{c \to 1} y_c$, then the equation of the circle is :

Option 1:

$25x^2 + 25y^2 - 20x + 2y - 60 = 0$

Option 2:

$5x^2 + 5y^2 - 4x - 2y - 12 = 0$

Option 3:

$25x^2 + 25y^2 - 2x + 2y - 60 = 0$

Option 4:

$5x^2 + 5y^2 - 4x + 2y - 12 = 0$

Correct Answer:

$25x^2 + 25y^2 - 20x + 2y - 60 = 0$

Solution:

$$(2+c)x+5c^2\left(\frac{1-3x}{5}\right)=1$$

$$x=\frac{1-c^2}{2+c-3c^2}, y=\frac{1-3x}{5}=\frac{c-1}{5\left(2+c-3c^2\right)}$$

$$h=\lim_{c\to 1}\frac{(1-c)(1+c)}{(1-c)(2+3c)}=\frac{2}{5}$$

$$K=\lim_{c\to 1}\frac{c-1}{-5(c-1)(3c+2)}=-\frac{1}{25}$$

$$\text{Centre}\left(\frac{2}{25},-\frac{1}{25}\right)$$

$$r=\sqrt{\left(2-\frac{2}{5}\right)^2+\left(0-\frac{1}{25}\right)^2}=\sqrt{\frac{64}{25}+\frac{1}{625}}$$

$$r=\frac{\sqrt{161}}{25}$$

$$\left(x-\frac{2}{5}\right)^2+\left(y+\frac{1}{25}\right)^2=\frac{161}{125}$$

$$\Rightarrow 25x^2+25y^2-20x+2y-60=0$$

Q. 15 The shortest distance between the line $\frac{x-3}{4}=\frac{y+7}{-11}=\frac{z-1}{5}$ and $\frac{x-5}{3}=\frac{y-9}{-6}=\frac{z+2}{1}$ is :

Option 1:
$\frac{187}{\sqrt{563}}$

Option 2:
$\frac{178}{\sqrt{563}}$

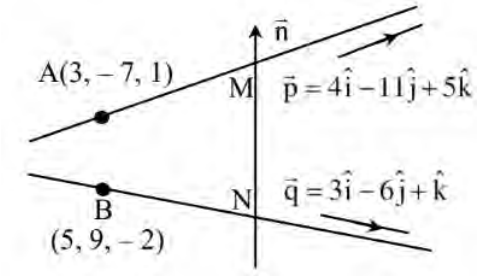
Option 3:
$\frac{185}{\sqrt{563}}$

Option 4:
$\frac{179}{\sqrt{563}}$

Correct Answer:

$\frac{187}{\sqrt{563}}$

Solution:



$$\vec{n}=\vec{p}\times\vec{q}$$

$$\vec{n}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 4 & -11 & 5\\ 3 & -6 & 1\end{vmatrix}=19\hat{i}+11\hat{j}+9\hat{k}$$

$$\text{S.d.} = \text{projection of } \overrightarrow{AB} \text{ on } \vec{n}$$

$$=\frac{\overrightarrow{AB}\cdot\vec{n}}{|\vec{n}|}=\frac{(2\hat{i}+16\hat{j}-3\hat{k})\cdot(19\hat{i}+11\hat{j}+9\hat{k})\ |}{\sqrt{361+121+81}}$$

$$=\frac{38+176-27}{\sqrt{563}}$$

$$\text{S.d.} = \frac{187}{\sqrt{563}}$$

Q. 16 The frequency distribution of the age of students in a class of 40 students is given below.

Age	15	16	17	18	19	20
No. of students	5	8	5	12	x	y

If the mean deviation about the median is 1.25 , then $4x+5y$ is equal to :

Option 1:

43

Option 2:
44

Option 3:
47

Option 4:
46

Correct Answer:
44

Solution:

$$x+y=10 \ldots\ldots\ldots(1)$$

$$\text{Median } = 18 = M$$

$$\text{M.D. } = \frac{\sum f_i\,|x_i - M|}{\sum f_i}$$

$$1.25 = \frac{36+x+2y}{40}$$

$$x+2y=14 \ldots\ldots\ldots(1)$$

$$\text{by } (1)\&(2)$$

$$x=6, y=4$$

$$\Rightarrow 4x+5y=24+20=44$$

Age (x_i)	f	$\lvert x_i - M\rvert$	$f_i\,\lvert x_i - M\rvert$
15	5	3	15
16	8	2	16
17	5	1	5
18	12	0	0
19	x	1	x
20	y	2	2y

Q. 17 The solution of the differential equation $\left(x^2+y^2\right)dx - 5xydy = 0, y(1)=0$, is :

Option 1:
$x^2 - 4y^{2\ 5} = x^2$

Option 2:

$x^2 - 2y^2\ ^6 = x$

Option 3:

$x^2 - 4y^2\ ^6 = x$

Option 4:

$x^2 - 4y^2\ ^6 = x$

Correct Answer:

$x^2 - 4y^2\ ^5 = x^2$

Solution:

$\left(x^2 + y^2\right)dx = 5xydy$

$\Rightarrow \frac{dy}{dx} = \frac{x^2 + y^2}{5xy}$

Put $y = Vx$

$\Rightarrow \mathrm{V} + \mathrm{x}\frac{\mathrm{dv}}{\mathrm{dx}} = \frac{1 + \mathrm{V}^2}{5\ \mathrm{V}}$

$\Rightarrow \frac{\mathrm{xdv}}{\mathrm{dx}} = \frac{1 - 4\ \mathrm{V}^2}{5\ \mathrm{V}}$

$\Rightarrow \int \frac{\mathrm{V}}{1 - 4\ \mathrm{V}^2}\mathrm{dV} = \int \frac{\mathrm{dx}}{5\mathrm{x}}$

Let $1 - 4\ \mathrm{V}^2 = \mathrm{t}$

$\Rightarrow -8\mathrm{VdV} = \mathrm{dt}$

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$$\Rightarrow \int \frac{dt}{(-8)(t)} = \int \frac{dx}{5x}$$

$$\Rightarrow \frac{-1}{8}\ln|t| = \frac{1}{5}\ln|x| + \ln C$$

$$\Rightarrow -5\ln|t| = 8\ln|x| + \ln K$$

$$\Rightarrow \ln x^8 + \ln\ t^5\ + \ln K = 0$$

$$\Rightarrow x^8\ t^5\ = C$$

$$\Rightarrow x^8\ 1 - 4V^2\ ^5 = C$$

$$\Rightarrow x^8\ \frac{x^2 - 4y^2}{x^2}\ ^5 = C$$

$$\Rightarrow\ x^2 - 4y^2\ ^5 = Cx^2$$

given $y(1) = 0$

$$\Rightarrow |1|^5 = C \Rightarrow C = 1$$

$$\Rightarrow\ x^2 - 4y^2\ ^5 = x^2$$

Q. 18 Let three vectors $\vec{a} = \alpha\hat{i} + 4\hat{j} + 2\hat{k}, \vec{b} = 5\hat{i} + 3\hat{j} + 4\hat{k}, \vec{c} = x\hat{i} + y\hat{j} + z\hat{k}$ from a triangle such that $\vec{c} = \vec{a} - \vec{b}$ and the area of the triangle is $5\sqrt{6}$. if α is a positive real number, then $|\vec{c}|^2$ is :

Option 1:
16

Option 2:
14

Option 3:
12

Option 4:
10

Correct Answer:
14

Solution:

$\overrightarrow{c}=\overrightarrow{a}-\overrightarrow{b}$

$\Rightarrow (x,y,z)=(\alpha-5,1,-2)$

$\Rightarrow x=\alpha-5, y=1, z=-2$

Area of $\Delta=5\sqrt{6}$ (given)

$$\frac{1}{2}|\vec{a}\times\vec{c}|=5\sqrt{6}$$

$$\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ \alpha & 4 & 2\\ x & 1 & -2\end{vmatrix}=10\sqrt{6}$$

$\Rightarrow |-10\hat{i}-\hat{j}(-2\alpha-2x)+\hat{k}(\alpha-4x)|=10\sqrt{6}$

$\Rightarrow (2\alpha+2\alpha-10)^2+(\alpha-4\alpha+20)^2=500$

$\Rightarrow (4\alpha-10)^2+(20-3\alpha)^2=500$

$\Rightarrow 25\alpha^2-80\alpha-120\alpha=0$

$\Rightarrow \alpha(25\alpha-200)=0$

$\Rightarrow \alpha=8$ (given α is +ve number)

$\Rightarrow x=\alpha-5=3$

$|\vec{c}|^2=x^2+y^2+z^2$

$=9+1+4$

$=14$

Q. 19 Let α, β be the roots of the equation $x^2+2\sqrt{2}x-1=0$. The quadratic equation, whose roots are $\alpha^4+\beta^4$ and $\frac{1}{10}\left(\alpha^6+\beta^6\right)$, is :

Option 1:
$x^2-190x+9466=0$

Option 2:
$x^2-195x+9466=0$

Option 3:
$x^2-195x+9506=0$

Option 4:
$x^2-180x+9506=0$

Correct Answer:
$x^2 - 195x + 9506 = 0$

Solution:
$x^2 + 2\sqrt{2}x - 1 = 0$
$\alpha + \beta = -2\sqrt{2}$
$\alpha\beta = -1$
$\alpha^4 + \beta^4 = \left(\alpha^2 + \beta^2\right)^2 - 2\alpha^2\beta^2$
$= \left((\alpha + \beta)^2 - 2\alpha\beta\right)^2 - 2(\alpha\beta)^2$
$= (8 + 2)^2 - 2(-1)^2$
$= 100 - 2 = 98$

$\alpha^6 + \beta^6 = \left(\alpha^3 + \beta^3\right)^2 - 2\alpha^3\beta^3$
$= \left((\alpha + \beta)\left((\alpha + \beta)^2 - 3\alpha\beta\right)^2 - 2(\alpha\beta)^3\right.$

$= (-2\sqrt{2}(8 + 3))^2 + 2$
$= (8)(121) + 2 = 970$
$\frac{1}{10}\left(\alpha^6 + \beta^6\right) = 97$
$x^2 - (98 + 97)x + (98)(97) = 0$
$\Rightarrow x^2 - 195x + 9506 = 0$

Q. 20 Let $f(x) = x^2 + 9, g(x) = \frac{x}{x-9}$ and $a = \text{fog}(10), b = \text{gof}(3)$. If e and 1 denote the eccentricity and the length of the latus rectum of the ellipse $\frac{x^2}{a} + \frac{y^2}{b} = 1$, then $8e^2 + 1^2$ is equal to.

Option 1:
16

Option 2:
8

Option 3:
6

Option 4:
12

Correct Answer:
8

Solution:

$f(x) = x^2 + 9 \quad g(x) = \dfrac{x}{x-9}$

$a = f(g(10)) = f\left(\dfrac{10}{10-9}\right)$

$= f(10) = 109$

$b = g(f(3)) = g(9+9)$

$= g(18) = \dfrac{18}{9} = 2$

$E : \dfrac{x^2}{109} + \dfrac{y^2}{2} = 1$

$e^2 = 1 - \dfrac{2}{109} = \dfrac{107}{109}$

$\ell = \dfrac{2(2)}{\sqrt{109}} = \dfrac{4}{\sqrt{109}}$

$8e^2 + \ell^2 = \dfrac{8(107)}{109} + \dfrac{16}{109}$

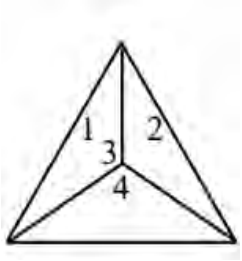
$= 8$

Q. 21 Let a, b and c denote the outcome of three independent rolls of a fair tetrahedral die, whose four faces are marked $1, 2, 3, 4$. If the probability that $ax^2 + bx + c = 0$ has all real roots is $\frac{m}{n}$, $\gcd(m, n) = 1$, then $m + n$ is equal to

Correct Answer:
19

Solution:

$a, b, c \in \{1, 2, 3, 4\}$



Tetrahedral dice

$ax^2 + bx + c = 0$

has all real roots

$\Rightarrow D \geq 0$

$\Rightarrow b^2 - 4ac \geq 0$

$\Rightarrow \mathrm{D} \geq 0$

$\Rightarrow \mathrm{b}^2 - 4\mathrm{ac} \geq 0$

Let $b = 1 \Rightarrow 1 - 4ac \geq 0$ (Not feasible)

$b = 2 \Rightarrow 4 - 4ac \geq 0$

$1 \geq ac \Rightarrow a = 1, c = 1,$

$b = 3 \Rightarrow 9 - 4ac \geq 0$

$\frac{9}{4} \geq ac$

$\Rightarrow a = 1, c = 1$

$\Rightarrow a = 1, c = 2$

$\Rightarrow a = 2, c = 1$

$b = 4 \Rightarrow 16 - 4ac \geq 0$

$4 \geq ac$

$\Rightarrow a = 1, c = 1$

$\Rightarrow a = 1, c = 2 \quad \Rightarrow a = 2, c = 1$

$\Rightarrow a = 1, c = 3 \quad \Rightarrow a = 3, c = 1$

$\Rightarrow a = 1, c = 4 \quad \Rightarrow a = 4, c = 1$

$\Rightarrow a = 2, c = 2$

$$\text{Probability} = \frac{12}{(4)(4)(4)} = \frac{3}{16} = \frac{\mathrm{m}}{\mathrm{m}}$$

$m + n = 19$

Q. 22 The sum of the square of the modulus of the elements in the set $\{z = a + ib : a, b \in Z, z \in C, |z - 1| \leq 1, |z - 5| \leq |z - 5i|\}$ is

Correct Answer:
9

Solution:
$|z-1| \leq 1$
$\Rightarrow |(x-1)+iy| \leq 1$
$\Rightarrow \sqrt{(x-1)^2+y^2} \leq 1$
$\Rightarrow (x-1)^2+y^2 \leq 1 \ldots\ldots\ldots\ldots (1)$
Also $|z-5| \leq |z-5i|$
$(x-5)^2+y^2 \leq x^2+(y-5)^2$
$-10x \leq -10y$
$\Rightarrow x \geq y \ldots\ldots\ldots\ldots\ldots$

Solving (1) and (2)
$\Rightarrow (x-1)^2+x^2=1$
$\Rightarrow 2x^2-2x=0$
$\Rightarrow x(x-1)=0$
$\Rightarrow x=0$ or $x=1$
$y=0$ or $y=1$

Given $x, y \in I$

Points $(0,0),(1,0),(2,0),(1,1),(1,-1)$ to find
$|z_1|^2+|z_2|^2+|z_3|^2+|z_4|^2+|z_5|^2$
$=0+1+4+1+1+1+1=9$

Q. 23 Let the set of all positive values of λ, for which the point of local minimum of the function $\left(1+x\left(\lambda^2-x^2\right)\right)$ satisfies $\frac{x^2+x+2}{x^2+5x+6}<0$, be (α, β).

Then $\alpha^2+\beta^2$ is equal to

Correct Answer:
39

Solution:

$$\frac{x^2+x+2}{x^2+5x+6}<0$$
$$\Rightarrow \frac{1}{(x+2)(x+3)}<0$$

$x \in (-3,-2) \ldots\ldots\ldots\ldots (1)$
$f(x)=1+x\left(\lambda^2-x^2\right)$

Finding local minima
$f'(x)=\left(\lambda^2-x^2\right)+(-2x)\cdot x$

Put $f'(x)=0$
$\Rightarrow \lambda^2=3x^2$
$\Rightarrow x=\pm\frac{\lambda}{\sqrt{3}}$

$\underset{\frac{-\lambda}{\sqrt{3}}\ \frac{\lambda}{\sqrt{3}}}{\overset{-\ +}{\ }}$

Local min Local max

We want local min
$\Rightarrow x=\frac{-\lambda}{\sqrt{3}}$
from (1)
$x \in (-3,-2)$

$$-3<\frac{-\lambda}{\sqrt{3}}<-2$$
$3\sqrt{3}>\lambda>2\sqrt{3}$
$\alpha=2\sqrt{3}, \beta=3\sqrt{3}$
$\alpha^2+\beta^2=12+27=39$

Q. 24 Let

$$\lim_{n\to\infty}\left(\frac{n}{\sqrt{n^4+1}}-\frac{2n}{(n^2+1)\sqrt{n^4+1}}+\frac{n}{\sqrt{n^4+16}}-\frac{8n}{(n^2+4)\sqrt{n^4+16}}+\ldots\ldots\ldots+\frac{n}{\sqrt{n^4+n^4}}-\frac{2n\cdot n^2}{(n^2+n^2)\sqrt{n^4+n^4}}\right) \text{ be } \frac{\pi}{k},$$

using only the principal values of the inverse trigonometric functions. Then k^2 is equal to

Correct Answer:
32

Solution:

$$\sum_{r=1}^{\infty}\frac{n}{\sqrt{n^4+r^4}}-\frac{2nr^2}{(n^2+r^2)\sqrt{n^4+r^4}}$$

$$\sum_{r=1}^{\infty}\frac{\frac{1}{n}}{\sqrt{1+\left(\frac{r}{n}\right)^4}}-\frac{2\left(\frac{1}{n}\right)\left(\frac{r}{n}\right)^2}{\left(1+\left(\frac{r}{n}\right)^2\right)\sqrt{1+\left(\frac{r}{n}\right)^4}}$$

$$\Rightarrow\int_0^1\frac{dx}{\sqrt{1+x^4}}-\frac{2x^2dx}{(1+x^2)\sqrt{1+x^4}}$$

$$\Rightarrow\int_0^1\frac{1-x^2}{(1+x^2)\sqrt{1+x^4}}dx$$

$$\Rightarrow\int_0^1\frac{\frac{1}{x^2}-1}{\left(x+\frac{1}{x}\right)\sqrt{x^2+\frac{1}{x^2}}}dx$$

$$\Rightarrow-\int_0^1\frac{1-\frac{1}{x^2}}{\left(x+\frac{1}{x}\right)\sqrt{\left(x+\frac{1}{x}\right)^2-2}}dx$$

$$x+\frac{1}{x}=t\Rightarrow1-\frac{1}{x^2}dx=dt$$

$$\Rightarrow-\int_{\infty}^{2}\frac{dt}{t\sqrt{t^2-2}}$$

$$\Rightarrow-\int_{\infty}^{2}\frac{tdt}{t^2\sqrt{t^2-2}}$$

take $t^2-2=\alpha^2$

$tdt=\alpha d\alpha$

$$\Rightarrow-\int_{\infty}^{\sqrt{2}}\frac{\alpha d\alpha}{(\alpha^2+2)\alpha}$$

$$\Rightarrow-\int_{\infty}^{\sqrt{2}}\frac{d\alpha}{\alpha^2+2}$$

$$\Rightarrow -\int_{\infty}^{\sqrt{2}} \frac{d\alpha}{\alpha^2+2}$$

$$\Rightarrow \frac{-1}{\sqrt{2}}\tan^{-1}\frac{\alpha}{\sqrt{2}}\Big]_{\infty}^{\sqrt{2}}$$

$$\Rightarrow \frac{-1}{\sqrt{2}}\{\tan^{-1}1\}+\frac{1}{\sqrt{2}}\tan^{-1}\infty$$

$$\Rightarrow \frac{1}{\sqrt{2}}\left\{\frac{\pi}{2}-\frac{\pi}{4}\right\}$$

$$\Rightarrow \frac{\pi}{4\sqrt{2}}=\frac{\pi}{K}$$

So $K=4\sqrt{2}$

$K^2=32$

Q. 25 The remainder when 428^{2024} is divided by 21 is

Correct Answer:

1

Solution:

$(428)^{2024}=(420+8)^{2024}$

$=(21\times 20+8)^{2024}$

$=21\,\mathrm{m}+8^{2024}$

Now $8^{2024}=\left(8^2\right)^{1012}$

$=(64)^{1012}$

$=(63+1)^{1012}$

$=(21\times 3+1)^{1012}$

$=2\,\mathrm{ln}+1$

$\Rightarrow$ Remainder is 1.

Q. 26 Lef $\mathrm{f}:(0,\pi)\to\mathrm{R}$ be a function given by

$$f(x)=\begin{cases}\left(\frac{8}{7}\right)^{\frac{\tan 8x}{\tan 7x}}, & 0<x<\frac{\pi}{2}\\ a-8, & x=\frac{\pi}{2}\\ (1+|\cot x)^{\frac{b}{a}\tan x}, & \frac{\pi}{2}<x<\pi\end{cases}$$

Where $a,b\in Z$. If f is continuous at $x=\frac{\pi}{2}$, then $\mathrm{a}^2+\mathrm{b}^2$ is equal to

Correct Answer:
81

Solution:

$$\mathrm{LHL\ at}\ x=\frac{\pi}{2}$$

$$\lim_{x\to\frac{\pi}{2}}\left(\frac{8}{7}\right)^{\frac{\tan 8x}{\tan 7x}}=\left(\frac{8}{7}\right)^{0}=1$$

$$\mathrm{RHL\ at}\ x=\frac{\pi}{2}$$

$$\lim_{x\to\frac{\pi}{2}}(1+|\cot x|)^{\frac{b}{a}|\tan x|}$$

$$=e^{\lim_{x\to\frac{\pi}{2}}|\cot x|\frac{b}{a}|\tan x|}=e^{\frac{b}{a}}$$

$$\Rightarrow 1=a-8=e^{\frac{b}{a}}$$
$$\Rightarrow a=9,b=0$$
$$\Rightarrow a^2+b^2=81$$

Q. 27 Let A be a non-singular matrix of order 3 . If $\det(3\,\mathrm{adj}(2\,\mathrm{adj}((\det A)A)))=3^{-13}\cdot 2^{-10}$ and $\det(\mathrm{adj}(2A))=2^m\cdot 3^n$, then $|3m+2n|$ is equal to

Correct Answer:
14

Solution:

$$|3\,\mathrm{adj}(2\,\mathrm{adj}(|A|A))| = |\,3\,\mathrm{adj}\,(2|\,A|^2\,\mathrm{adj}(A)\,|$$
$$= \;3.2^2\;A\;^4\,\mathrm{adj}\,\left(\mathrm{adj}(A)\;= 2^6 3^3\;A\;^{12}|\,A|^4\right.$$
$$= 2^6 3^3|\,A|^{16} = 2^{-10}3^{-13}$$
$$\Rightarrow |A|^{16} = 2^{-16}3^{-16} \Rightarrow |A| = 2^{-1}3^{-1}$$

$$\text{Now } |3\,\mathrm{adj}(2\,A)| = \;3.2^2\,\mathrm{adj}(A)$$
$$= 2^6 3^3|\,A|^2 = 2^{-m}3^{-n}$$
$$\Rightarrow 2^6 3^3 2^{-2}3^{-2} = 2^{-m}3^{-n}$$
$$\Rightarrow 2^{-m}3^{-n} = 2^4 3^1$$
$$\Rightarrow m = -4, n = -1$$
$$\Rightarrow |3\,m + 2n| = |-12-2| = 14$$

Q. 28 Let the centre of a circle, passing through the point $(0,0)$, $(1,0)$ and touching the circle $x^2+y^2=9$, be (h,k). Then for all possible values of the coordinates of the centre (h,k), $4\left(h^2+k^2\right)$ is equal to

Correct Answer:
9

Solution:

$$(x-h)^2+(y-k)^2 = h^2+k^2$$
$$x^2+y^2-2hx-2ky=0$$
$$\because \text{ passes through } (1,0)$$
$$\Rightarrow 1+0-2h=0$$
$$\Rightarrow h = 1/2$$
$$\because OC = \frac{OP}{2}$$
$$\sqrt{\left(\frac{1}{2}\right)^2+k^2} = \frac{3}{2}$$

$$\frac{1}{4}+k^2=\frac{9}{4}$$
$$k^2=2$$
$$k=\pm\sqrt{2}$$
$\therefore$ Possible coordinate of
$$c(h,k)\left(\frac{1}{2},\sqrt{2}\right)\left(\frac{1}{2},-\sqrt{2}\right)$$
$$4\left(h^2+k^2\right)=4\left(\frac{1}{4}+2\right)=4\left(\frac{9}{4}\right)=9$$

Q. 29 If a function f satisfies $f(m+n)=f(m)+f(n)$ for all $m,n\in N$ and $f(1)=1$, then the largest natural number λ such that $\sum_{k=1}^{2022} f(\lambda+k)\leq(2022)^2$ is equal to

Correct Answer:
1010

Solution:
$$f(m+n)=f(m)+f(n)$$
$$\Rightarrow f(x)=kx$$
$$\Rightarrow f(1)=1$$
$$\Rightarrow k=1$$
$$f(x)=x$$
Now
$$\sum_{k=1}^{2022} f(\lambda+k)\leq(2022)^2$$
$$\Rightarrow \sum_{k=1}^{2022}(\lambda+k)\leq(2022)^2$$
$$\Rightarrow 2022\lambda+\frac{2022\times2023}{2}\leq(2022)^2$$
$$\Rightarrow \lambda\leq 2022-\frac{2023}{2}$$
$$\Rightarrow \lambda\leq 1010.5$$
$\therefore$ largest natural no. λ is 1010 .

Q. 30 Let $A = \{2, 3, 6, 7\}$ and $B = \{4, 5, 6, 8\}$. Let R be a relation defined on $A \times B$ by $(a_1, b_1)R(a_2, b_2)$ is and only if $a_1 + a_2 = b_1 + b_2$. Then the number of elements in R is

Correct Answer:
25

Solution:
$A = \{2, 3, 6, 7\}$
$B = \{2, 5, 6, 8\}$
$(a_1, b_1)R(a_2, b_2)$
$a_1 + a_2 = b_1 + b_2$

1. $(2, 4)\mathrm{R}(6, 4)$ 2. $(2, 4)\mathrm{R}(7, 5)$
3. $(2, 5)\mathrm{R}(7, 4)$ 4. $(3, 4)\mathrm{R}(6, 5)$
5. $(3, 5)\mathrm{R}(6, 4)$ 6. $(3, 5)\mathrm{R}(7, 5)$
7. $(3, 6)\mathrm{R}(7, 4)$ 8. $(3, 4)\mathrm{R}(7, 6)$
$\times 2$
9. $(6, 5)\mathrm{R}(7, 8)$ 10. $(6, 8)\mathrm{R}(7, 5)$
11. $(7, 8)\mathrm{R}(7, 6)$ 12. $(6, 8)\mathrm{R}(6, 4)$
13. $(6, 6)\mathrm{R}(6, 6)$

Total $24 + 1 = 25$

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9 Apr - Evening Shift

Physics

Q. 1 A nucleus at rest disintegrates into two smaller nuclei with their masses in the ratio of 2:1. After disintegration they will move :-

Option 1:
In opposite directions with speed in the ratio of 1:2 respectively

Option 2:
In opposite directions with speed in the ratio of 2:1 respectively

Option 3:
In the same direction with same speed.

Option 4:
In opposite directions with the same speed.

Correct Answer:
In opposite directions with speed in the ratio of 1:2 respectively

Solution:
By conservation of momentum

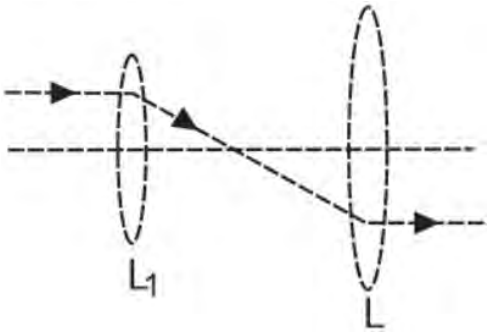
$$p_i = p_f$$

$$O = m_1u_1 + m_2u_2$$

$$\frac{u_1}{u_2} = -\left[\frac{1}{2}\right] \text{ as } \frac{m_1}{m_2} = \frac{2}{1}$$

move in opposite direction with speed ratio $1:2$

Q. 2 The following figure represents two biconvex lenses L_1 and L_2 having focal length $10\ \text{cm}$ and $15\ \text{cm}$ respectively. The distance between $\text{L}_1 \& \text{L}_2$ is :



Option 1:
10 cm

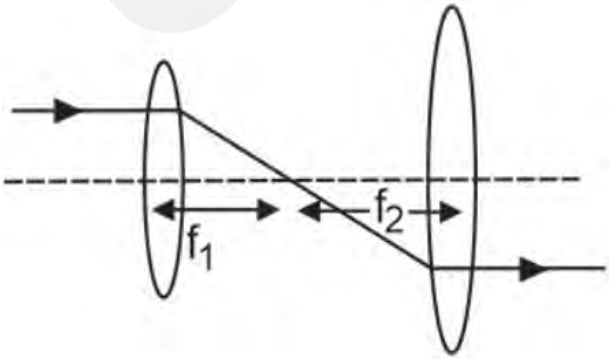
Option 2:
15 cm

Option 3:
25 cm

Option 4:
35 cm

Correct Answer:
25 cm

Solution:



$\text{D} = \text{f}_1 + \text{f}_2 = 25\ \text{cm}$

Paraxial parallel rays pass through focus and ray from focus of convex lens will become parallel

Q. 3 The temperature of a gas is $-78^{\circ}\mathrm{C}$ and the average translational kinetic energy of its molecules is K. The temperature at which the average translational kinetic energy of the molecules of the same gas becomes $2\ \mathrm{K}$ is :

Option 1:
$-39^{\circ}\mathrm{C}$

Option 2:
$117^{\circ}\mathrm{C}$

Option 3:
$127^{\circ}\mathrm{C}$

Option 4:
$-78^{\circ}\mathrm{C}$

Correct Answer:
$117^{\circ}\mathrm{C}$

Solution:
$\mathrm{K.E} = \frac{\mathrm{nf_1RT}}{2}$

$\mathrm{T_i} = -78^{\circ}\mathrm{C} \rightarrow 273 + [-78^{\circ}\mathrm{C}] = 195\ \mathrm{K}$

$\mathrm{K.E} \alpha \mathrm{T}$

K.E $\alpha \mathrm{T}$

To double the K.E energy temp also become double

Q. 4 A hydrogen atom in ground state is given an energy of 10.2 eV. How many spectral lines will be emitted due to transition of electrons ?

Option 1:
6

Option 2:
3

Option 3:
10

Option 4:
1

Correct Answer:
1

Solution:
Hydrogen will be in first excited state therefore itwill emit one spectral line corresponding to transition b/w energy level 2 to 1

Q. 5 The magnetic field in a plane electromagnetic wave is $B_y = \left(3.5 \times 10^{-7}\right) \sin\left(1.5 \times 10^{3} x + 0.5 \times 10^{11} \mathrm{t}\right) \mathrm{T}$. The corresponding electric field will be

Option 1:
$\mathrm{E_y} = 1.17 \sin\left(1.5 \times 10^{3} \mathrm{x} + 0.5 \times 10^{11} \mathrm{t}\right) \mathrm{Vm}^{-1}$

Option 2:
$\mathrm{E_z} = 105 \sin\left(1.5 \times 10^{3} \mathrm{x} + 0.5 \times 10^{11} \mathrm{t}\right) \mathrm{Vm}^{-1}$

Option 3:

$E_z = 1.17 \sin\left(1.5 \times 10^3 x + 0.5 \times 10^{11} t\right) V m^{-1}$

Option 4:

$E_y = 10.5 \sin\left(1.5 \times 10^3 x + 0.5 \times 10^{11} t\right) V m^{-1}$

Correct Answer:

$E_z = 105 \sin\left(1.5 \times 10^3 x + 0.5 \times 10^{11} t\right) V m^{-1}$

Solution:

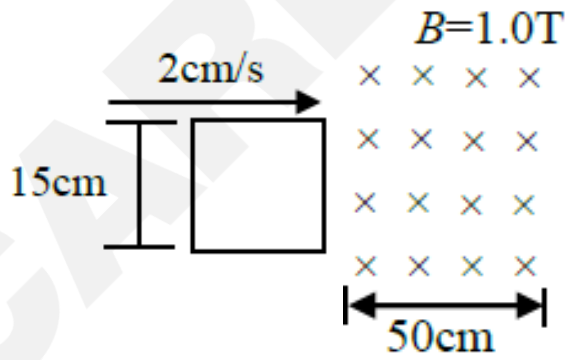
$E_0 = B_0 C$

$E_0 = 3 \times 10^8 \times \left(3.5 \times 10^{-7}\right) \sin\left(1.5 \times 10^3 x + 0.5 \times 10^{11} t\right)$

$E_0 = 105 \sin\left(1.5 \times 10^3 x + 0.5 \times 10^{11} t\right) V m^{-1}$

Data inconsistent while calculating speed of wave. You can challenge for data.

Q. 6 A square loop of side $15\ cm$ being moved towards right at a constant speed of $2\ cm/s$ as shown in figure. The front edge enters the $50\ cm$ wide magnetic field at $t = 0$. The value of induced emf in the loop at $t = 10\ s$ will be :



Option 1:

$0.3 mV$

Option 2:
$4.5\mathrm{mV}$

Option 3:
zero

Option 4:
$3\mathrm{mV}$

Correct Answer:
zero

Solution:
At t = 10 sec complete loop is in magnetic field therefore no change in flux

$\mathrm{e}=\frac{\mathrm{d}\phi}{\mathrm{dt}}=0$

$\mathrm{e}=0$ for complete loop

Q. 7 Two cars are travelling towards each other at speed of $20\ \mathrm{m\ s^{-1}}$ each. When the cars are $300\ \mathrm{m}$ apart, both the drivers apply brakes and the cars retard at the rate of $2\ \mathrm{m\ s^{-2}}$. The distance between them when they come to rest is :

Option 1:
200 m

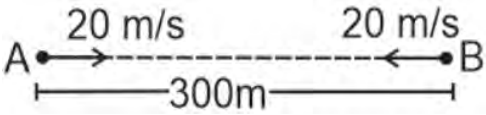
Option 2:
50 m

Option 3:
100 m

Option 4:
25 m

Correct Answer:
100 m

Solution:



$\overrightarrow{u}_{BA} = 40\ m/s$

$\overrightarrow{a}_{BA} = 4\ m/s$

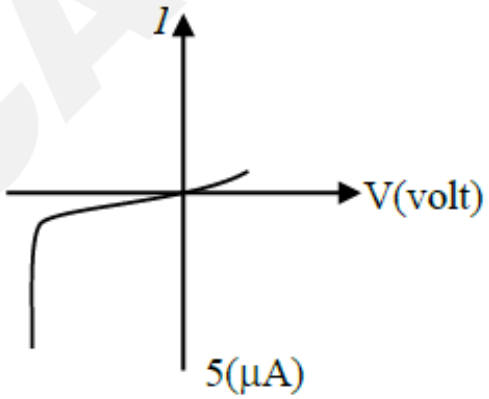
$\text{Apply } (v^2 = u^2 + 2as)_{\text{relative}}$

$O = (40)^2 + 2(-4)(S)$

$S = 200\ m$

$\text{Remaining distance } = 300 - 200 = 100\ m$

Q. 8 The $I - V$ characteristics of an electronic device shown in the figure. The device is:



Option 1:
a solar cell

Option 2:
a transistor which can be used as an amplifier

Option 3:
a zener diode which can be used as voltage regulator

Option 4:
a diode which can be used as a rectifier

Correct Answer:
a zener diode which can be used as voltage regulator

Solution:
Theory
Zener diode used as voltage regulator

Q. 9 The excess pressure inside a soap bubble is thrice the excess pressure inside a second soap bubble. The ratio between the volume of the first and the second bubble is :

Option 1:
1:9

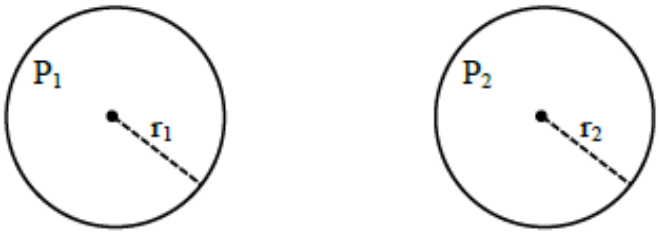
Option 2:
1:3

Option 3:
1:81

Option 4:
1:27

Correct Answer:
1:27

Solution:



$$P_1 - P_0 = \frac{4T}{r_1} \quad P_2 - P_0 = \frac{4T}{r_2}$$

$$P_1 - P_0 = 3(P_2 - P_0)$$

$$\frac{4T}{r_1} = 3\frac{4T}{r_2}$$

$$r_2 = 3r_1$$

$$\frac{V_1}{V_2} = \frac{\frac{4}{3}\pi r_1^3}{\frac{4}{3}\pi r_2^3} = \frac{1}{27}$$

Q. 10 The de-Broglie wavelength associated with a particle of mass m and energy E is $h/\sqrt{2mE}$. The dimensional formula for Planck's constant is :

Option 1:
$[ML^{-1}T^{-2}]$

Option 2:
$[ML^{2}T^{-1}]$

Option 3:
$[MLT^{-2}]$

Option 4:
$[M^{2}L^{2}T^{-2}]$

Correct Answer:
$\left[\mathrm{ML}^2\ \mathrm{T}^{-1}\right]$

Solution:
$\lambda = \frac{\mathrm{h}}{\sqrt{2\mathrm{mE}}}$ or $\mathrm{E} = \mathrm{h}\nu$

$\left[\mathrm{ML}^2\ \mathrm{T}^{-2}\right] = \mathrm{h}\left[\mathrm{T}^{-1}\right]$

$\mathrm{h} = \left[\mathrm{ML}^2\ \mathrm{T}^{-1}\right]$

Q. 11 A satellite of $10^3\ \mathrm{kg}$ mass is revolving in circular orbit of radius $2\mathrm{R}$. If $\frac{10^4\mathrm{R}}{6}\ \mathrm{J}$ energy is supplied to the satellite, it would revolve in a new circular orbit of radius :
(use $\mathrm{g} = 10\ \mathrm{m/s^2}, \mathrm{R} =$ radius of earth)

Option 1:
2.5 R

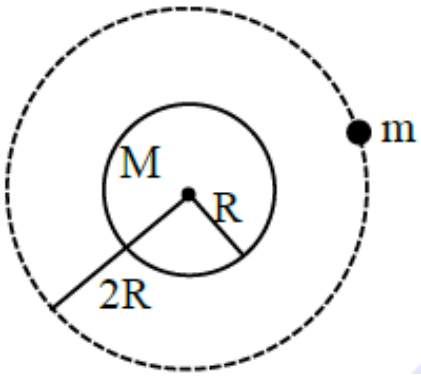
Option 2:
3 R

Option 3:
4 R

Option 4:
6 R

Correct Answer:
6 R

Solution:



$$\text{Total energy } = \frac{-GMm}{2(2R)}$$

$$\text{if energy } = \frac{10^4 R}{6} \text{ is added then}$$

$$\frac{-GMm}{4R} + \frac{10^4 R}{6} = \frac{-GMm}{2r}$$

$$\text{where r is new radius of revolving and } g = \frac{GM}{R^2}$$

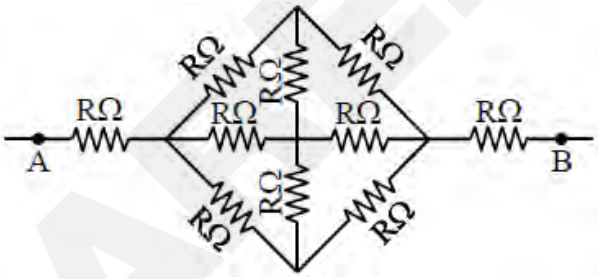
$$-\frac{mgR}{4} + \frac{10^4 R}{6} = -\frac{mgR^2}{2r} \left(m = 10^3 \text{ kg}\right)$$

$$-\frac{10^3 \times 10 \times R}{4} + \frac{10^4 R}{6} = -\frac{10^3 \times 10 \times R^2}{2r}$$

$$-\frac{1}{4} + \frac{1}{6} = -\frac{R}{2r}$$

$$r = 6R$$

Q. 12 The effective resistance between A and B, if resistance of each resistor is R, will be



Option 1:
$\frac{2}{3}R$

Option 2:
$\frac{8R}{3}$

Option 3:

$\frac{5R}{3}$

Option 4:

$\frac{4R}{3}$

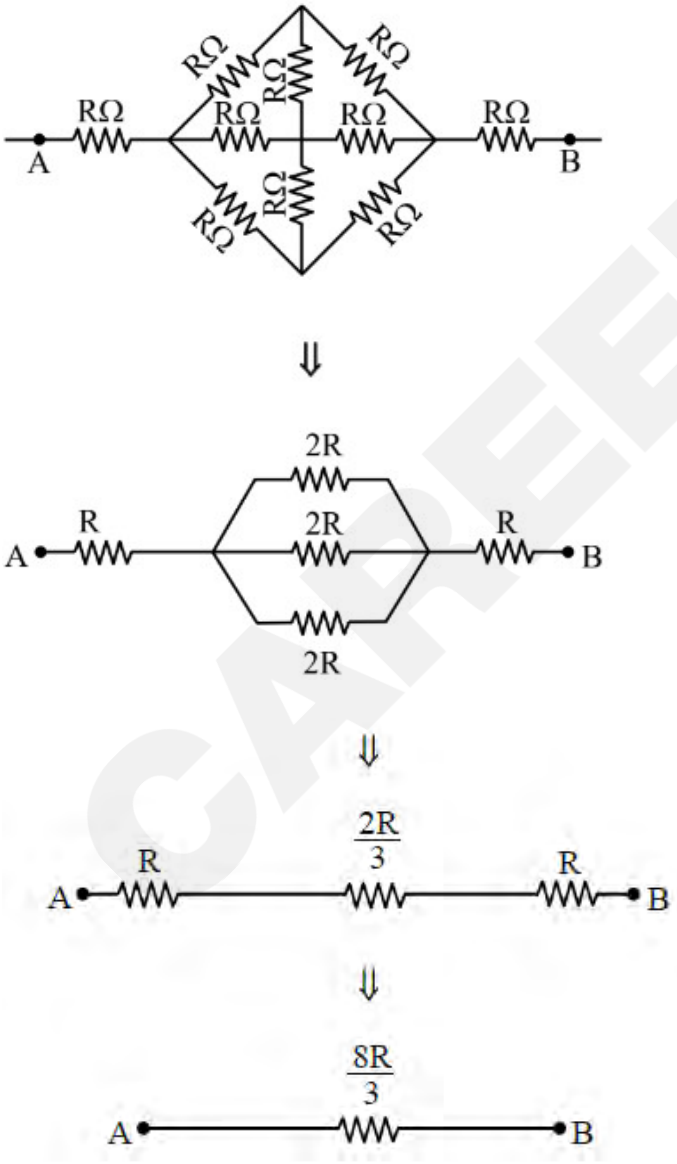
Correct Answer:

$\frac{8R}{3}$

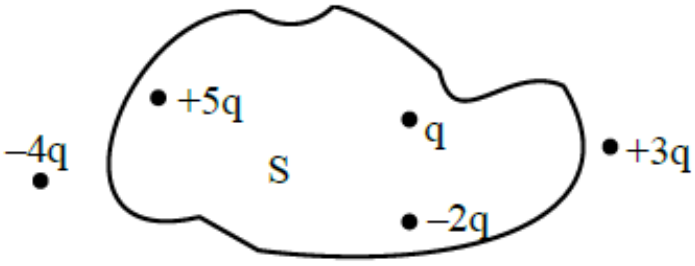
Solution:

From symmetry we can remove two middle resistance.

New circuit is



Q. 13 Five charges +q, +5q, –2q, +3q and –4q are situated as shown in the figure. The electric flux due to this configuration through the surface S is :



Option 1:
$\frac{5q}{\epsilon_0}$

Option 2:
$\frac{4q}{\epsilon_0}$

Option 3:
$\frac{3q}{\epsilon_0}$

Option 4:
$\frac{q}{\epsilon_0}$

Correct Answer:
$\frac{4q}{\epsilon_0}$

Solution:
As per gauss theorem,

$$\phi = \frac{q_{in}}{\epsilon_0} = \frac{q + (-2q) + 5q}{\epsilon_0}$$

$$\frac{4q}{\epsilon_0}$$

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Q. 14 A proton and a deutron ($\mathrm{q}=+\mathrm{e}, m=2.0 \mathrm{u}$) having same kinetic energies enter a region of uniform magnetic field $\vec{B}$, moving perpendicular to $\vec{B}$. The ratio of the radius r_d of deutron path to the radius r_p of the proton path is :

Option 1:
$1:1$

Option 2:
$1:\sqrt{2}$

Option 3:
$\sqrt{2}:1$

Option 4:
$1:2$

Correct Answer:
$\sqrt{2}:1$

Solution:
In uniform magnetic field,

$$\mathrm{R}=\frac{\mathrm{m}v}{\mathrm{qB}}=\frac{\sqrt{2\,\mathrm{m}(\mathrm{K.E})}}{\mathrm{qB}}$$

Since same K.E

$$\mathrm{R} \propto \frac{\sqrt{\mathrm{m}}}{\mathrm{q}}$$

$$\therefore \frac{\mathrm{R_{deutron}}}{\mathrm{R_{proton}}}=\sqrt{\frac{\mathrm{m_d}}{\mathrm{m_p}}} \times \frac{\mathrm{q_p}}{\mathrm{q_d}}$$

$$=\sqrt{2} \times 1$$

$\therefore \gamma_d : \gamma_p = \sqrt{2} : 1$

Q. 15 UV light of 4.13 eV is incident on a photosensitive metal surface having work function 3.13 eV. The maximum kinetic energy of ejected photoelectrons will be :

Option 1:
4.13 eV

Option 2:
1 eV

Option 3:
3.13 eV

Option 4:
7.26 eV

Correct Answer:
1 eV

Solution:

$E_{photon} = (\text{work function}) + K \cdot E_{max}$

$\therefore 4.13 = 3.13 + K \cdot E_{max}$

$\therefore K \cdot E_{max} = 1eV$

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Q. 16 The energy released in the fusion of $2\ \mathrm{kg}$ of hydrogen deep in the sun is $\mathrm{E_H}$ and the energy released in the fission of $2\ \mathrm{kg}$ of $^{235}\mathrm{U}$ is E_U. The ratio $\frac{E_H}{E_U}$ is approximately :

(Consider the fusion reaction as $4_1^1\mathrm{H} + 2\mathrm{e}^- \rightarrow {}_2^4\mathrm{He} + 2\mathrm{v} + 6\gamma + 26.7\mathrm{MeV}$, energy released in the fission reaction of $^{235}\mathrm{U}$ is $200\mathrm{MeV}$ per fission nucleus and $\mathrm{N_A} = 6.023 \times 10^{23}$)

Option 1:
9.13

Option 2:
15.04

Option 3:
7.62

Option 4:
25.6

Correct Answer:
7.62

Solution:
In each fusion reaction, $4_1^1\mathrm{H}$ nucleus are used.

Energy released per Nuclei of $_1^1\mathrm{H} = \frac{26.7}{4}\mathrm{MeV}$

$\therefore$ Energy released by $2\ \mathrm{kg}$ hydrogen $(\mathrm{E_H})$

$= \frac{2000}{1} \times \mathrm{N_A} \times \frac{26.7}{4}\mathrm{MeV}$

$\therefore$ Energy released by $2\ \mathrm{kg}$ Vranium $(\mathrm{E_v})$

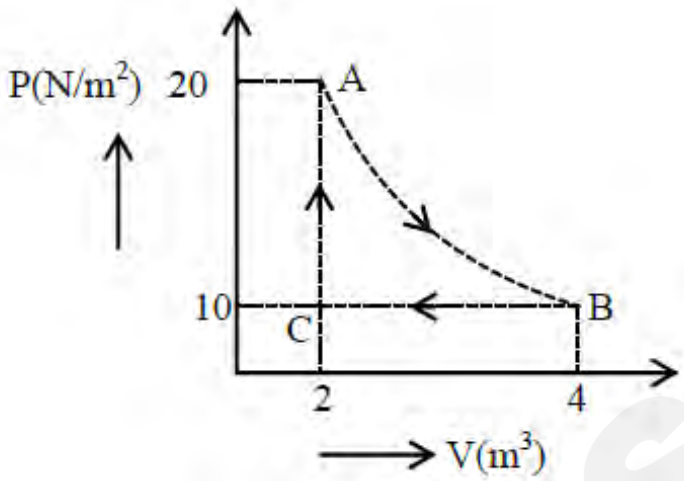
$= \frac{2000}{235} \times \mathrm{N_A} \times 200\mathrm{MeV}$

So,

$\frac{E_H}{E_V} = 235 \times \frac{26.7}{4\times200} = 7.84$

$\therefore$ Approximately close to 7.62

Q. 17 A real gas within a closed chamber at $27^{\circ}\mathrm{C}$ undergoes the cyclic process as shown in figure. The gas obeys $PV^3 = \mathrm{RT}$ equation for the path A to B. The net work done in the complete cycle is (assuming $R = 8\ \mathrm{J/molK}$):



Option 1:
225 J

Option 2:
205 J

Option 3:
20 J

Option 4:
-20 J

Correct Answer:
205 J

Solution:

$\mathrm{W_{AB}} = \int \mathrm{PdV} \quad (\text{Assuming T to be constant})$

$= \int \frac{\mathrm{RTdV}}{\mathrm{V^3}}$

$= \mathrm{RT} \int_2^4 \mathrm{V^{-3}dV}$

$= 8 \times 300 \times \left(-\frac{1}{2}\left[\frac{1}{4^2} - \frac{1}{2^2}\right]\right)$

$= 225\ \text{J}$

$\text{W}_{\text{BC}} = \text{P}\int_4^2 \text{dV} = 10(2-4) = -20\ \text{J}$

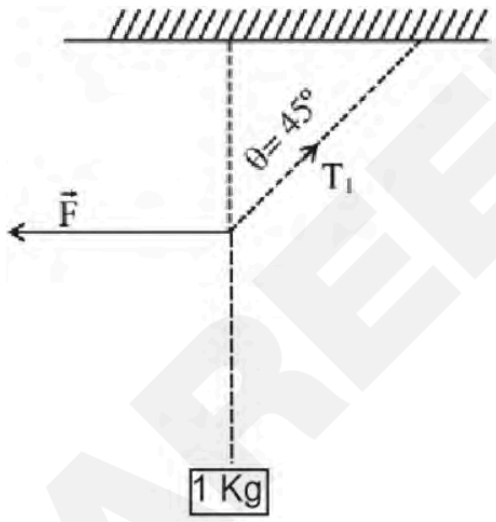
$\text{W}_{\text{CA}} = 0$

$\therefore \text{W}_{\text{cycle}} = 205\ \text{J}$

Note : Data is inconsistent in process AB.

So needs to be challenged.

Q. 18 A $1\ \text{kg}$ mass is suspended from the ceiling by a rope of length $4\ \text{m}$. A horizontal force ' F ' is applied at the mid point of the rope so that the rope makes an angle of 45° with respect to the vertical axis as shown in figure. The magnitude of F is :



Option 1:

$\frac{10}{\sqrt{2}}\ \text{N}$

Option 2:
$1\ \mathrm{N}$

Option 3:
$\frac{1}{10\times\sqrt{2}}\ \mathrm{N}$

Option 4:
$10\ \mathrm{N}$

Correct Answer:
$10\ \mathrm{N}$

Solution:
$\mathrm{T}_1 \sin 45^\circ = \mathrm{F}$

$\mathrm{T}_1 \cos 45^\circ = \mathrm{T}_2 = 1 \times \mathrm{g}$

$\therefore \tan 45^\circ = \frac{\mathrm{F}}{\mathrm{g}}$

$\therefore \mathrm{F} = 10\ \mathrm{N}$

Q. 19 A spherical ball of radius $1 \times 10^{-4}\ \mathrm{m}$ and density $10^5\ \mathrm{kg/m^3}$ falls freely under gravity through a distance h before entering a tank of water, If after entering in water the velocity of the ball does not change, then the value of h is approximately :
(The coefficient of viscosity of water is $9.8 \times 10^{-6}\ \mathrm{Ns/m^2}$)

Option 1:
2296 m

Option 2:
2249 m

Option 3:
2518 m

Option 4:
2396 m

Correct Answer:
2518 m

Solution:

$$V_T = \frac{2\,g}{9}\frac{R^2[\rho_B - \rho_L]}{\eta}$$

$$\Rightarrow V_T = \frac{2}{9} \times \frac{10\times(10^{-4})^2}{9.8\times10^{-6}}\left[10^5 - 10^3\right]$$

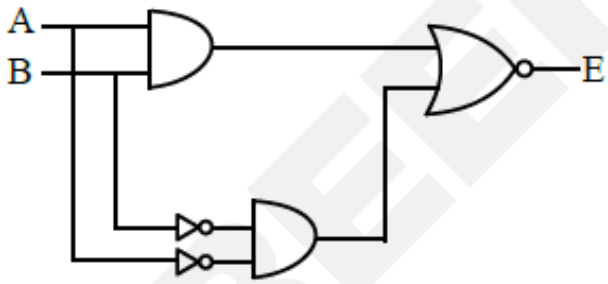
$$\Rightarrow V_T = 224.5$$

when ball fall from height (h)

$$V = \sqrt{2gh}$$

$$h = \left(\frac{V^2}{2\,g}\right) = 2518\ m$$

Q. 20



A	B	E
0	0	0
0	1	X
1	0	Y
1	1	0

In the truth table of the above circuit the value of X and Y are :

Option 1:
1,1

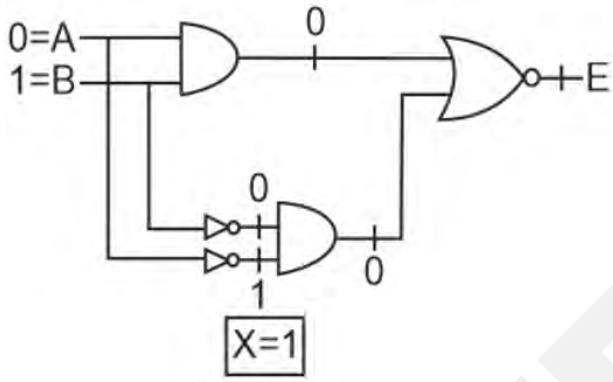
Option 2:
1,0

Option 3:
0,1

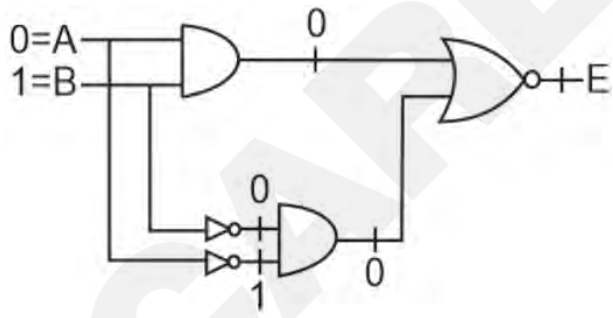
Option 4:
0,0

Correct Answer:
1,1

Solution:
For x



For y

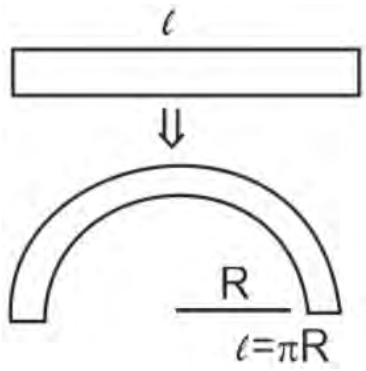


Q. 21 A straight magnetic strip has a magnetic moment of $44\mathrm{Am}^2$. If the strip is bent in a semicircular shape, its magnetic moment will be _____ Am^2 (Given $\pi = \frac{22}{7}$)

Correct Answer:
28

Solution:

Magnetic moment of straight wire $= \mathrm{mx}\ell = 44$



Magnetic moment of arc

$= \mathrm{m} \times 2\mathrm{r}$

$= \mathrm{m} \times \frac{2\ell}{\pi}$

$= \frac{44 \times 2}{\pi} = \frac{88}{\pi} = 28$

Q. 22 A particle of mass $0.50\ \mathrm{kg}$ executes simple harmonic motion under force $\mathrm{F} = -50\left(\mathrm{Nm}^{-1}\right)\mathrm{x}$. The time period of oscillation is $\frac{x}{35}\ s$. The value of x is_____ (Given $\pi = \frac{22}{7}$)

Correct Answer:

22

Solution:

$\mathrm{m} = 0.5\ \mathrm{kg}$

$\mathrm{F} = -50(\mathrm{x})$

$\mathrm{ma} = (-50\mathrm{x})$

$0.5\mathrm{a} = -50\mathrm{x}$

$\mathrm{a} = (-100\mathrm{x})$

$\mathrm{W}^2 = 100 \Rightarrow (\mathrm{w} = 10)$

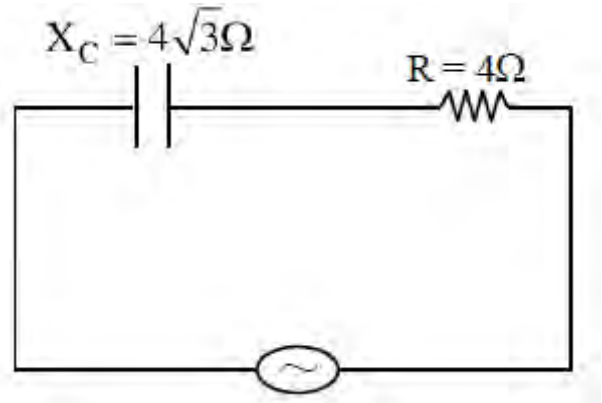
$\mathrm{T} = \frac{2\pi}{10} = \left(\frac{\pi}{5}\right) = \frac{22}{7 \times 15} = \left(\frac{22}{35}\right)$

$\frac{\pi}{35} = \frac{22}{35} \Rightarrow \mathrm{x} = 22$

Q. 23 A capacitor of reactance $4\sqrt{3}\Omega$ and a resistor of resistance 4Ω are connected in series with an ac source of peak value $8\sqrt{2}\ \mathrm{V}$. The power dissipation in the circuit is .....W.

Correct Answer:
4

Solution:



$$Z = \sqrt{R^2 + X^2\ L}$$

$$Z = \sqrt{4^2 + (4\sqrt{3})^2} = 8\Omega$$

$$V_{rms} = \frac{V}{\sqrt{2}} = \frac{8\sqrt{2}}{\sqrt{2}} = (8\ V)$$

$$I_{rms} = \frac{V_{rms}}{Z} = \frac{8}{8} = 1\ A$$

Power dissipated $= I_{rms}^2 \times R = 1 \times 4 = (4\ W)$

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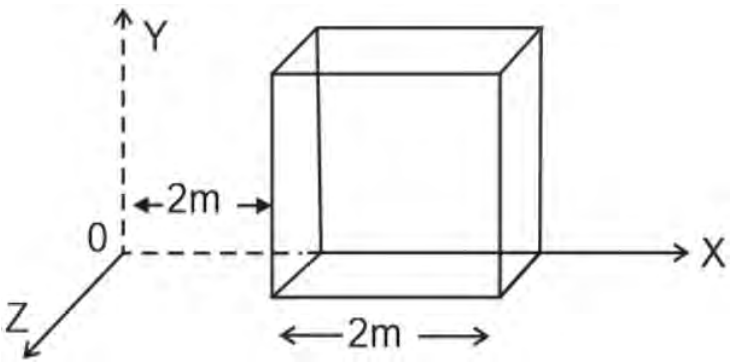
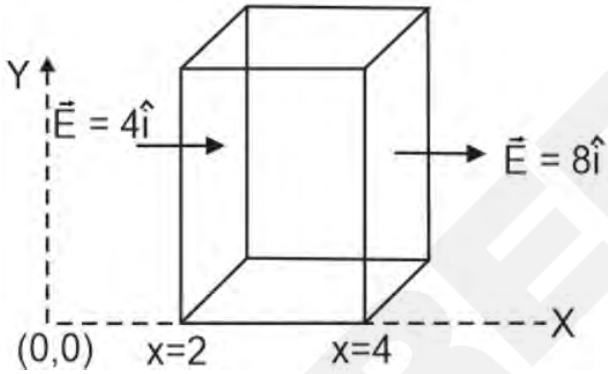
Q. 24

An electric field $\overrightarrow{\mathrm{E}} = (2x\mathrm{i})\mathrm{NC}^{-1}$ exists in space. A cube of side $2\ \mathrm{m}$ is placed in the space as per figure given below. The electric flux through the cube is ..... Nm^2/C.



Correct Answer:
16

Solution:



$\overrightarrow{\mathrm{E}} = 2 \times \hat{\mathrm{i}}$

$\phi = \overrightarrow{\mathrm{E}} \cdot \overrightarrow{\mathrm{A}}$

$\phi_{\text{in}} = -4 \times 4 = -16\mathrm{Nm}^2/\mathrm{c}$

$\phi_{\text{out}} = 8 \times 4 = 32\mathrm{Nm}^2/\mathrm{c}$

$d_{\text{net}} = \phi_{\text{in}} + \phi_{\text{out}} = -16 + 32 = 16\mathrm{Nm}^2/\mathrm{c}$

Q. 25 A circular disc reaches from top to bottom of an inclined plane of length l . When it slips down the plane, if takes t s. When it rolls down the plane then it takes $\left(\frac{\alpha}{2}\right)^{1/2}$ts, where α is .....

Correct Answer:
3

Solution:
For slipping

$a = g\sin\theta$

$\ell = \frac{1}{2}at^2 \Rightarrow t = \sqrt{\frac{2\ell}{g\sin\theta}}$

For rolling

$\mathrm{a'} = \frac{\mathrm{g}\sin\theta}{1+\frac{\mathrm{k}^2}{\mathrm{R}^2}}\left[\mathrm{k} = \frac{\mathrm{R}}{\sqrt{2}}\right]$

$\Rightarrow \mathrm{a'} = \frac{2\,\mathrm{g}\sin\theta}{3}$

$\ell = \frac{1}{2}\mathrm{a'}(\mathrm{t'})^2$

$\Rightarrow \mathrm{t'} = \sqrt{\frac{6\ell}{2\,\mathrm{g}\sin\theta}} = \sqrt{\frac{\alpha}{2}}\sqrt{\frac{2\ell}{\mathrm{s}\sin\theta}}$

$\Rightarrow \alpha = 3$

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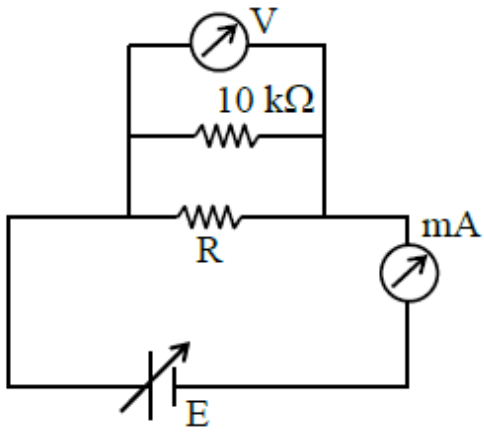
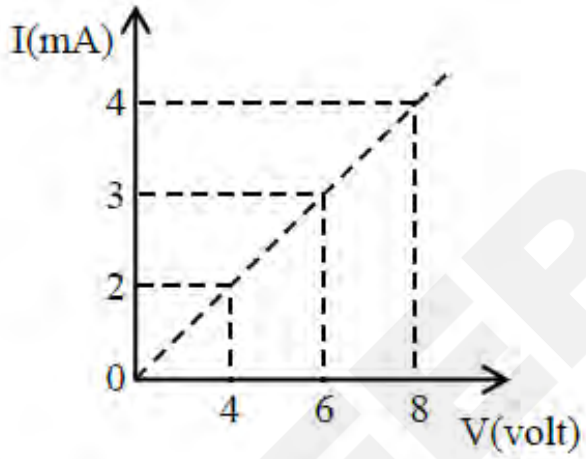
Q. 26 To determine the resistance (R) of a wire, a circuit is designed below, The V-I characteristic curve for this circuit is plotted for the voltmeter and the ammeter readings as shown in figure. The value of R is _____ Ω.





Correct Answer:
2500

Solution:

$\text{Req} = \frac{10^4 R}{10^4 + R}$

$E = 4V, I = 2mA$

$I = \frac{E}{\text{Req}} \Rightarrow 2 \times 10^{-3} = \frac{4(10^4+R)}{10^4 R}$

$\Rightarrow 20R = 40000 + 4R$

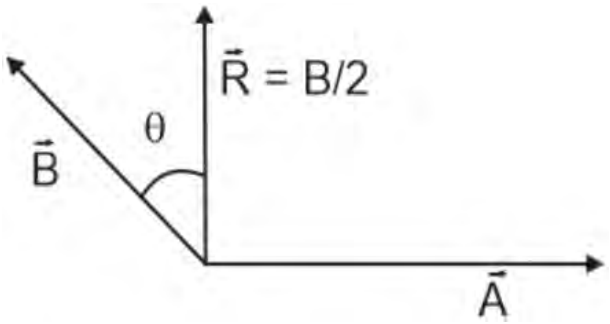
$16R = 40000$

$R = 2500\Omega$

Q. 27 The resultant of two vectors $\vec{A}$ and $\vec{B}$ is perpendicular to $\overrightarrow{\mathrm{A}}$ and its magnitude is half that of $\vec{B}$. The angle between vectors $\vec{A}$ and $\vec{B}$ is_____

Correct Answer:
150

Solution:



$B\cos\theta = \frac{B}{2}$

$\Rightarrow \theta = 60^\circ$

So, angle between $\overrightarrow{\mathrm{A}}\&\overrightarrow{\mathrm{B}}$ is $90^\circ + 60^\circ = 150^\circ$

Q. 28 Monochromatic light of wavelength $500\ \mathrm{nm}$ is used in Young's double slit experiment. An interference pattern is obtained on a screen When one of the slits is covered with a very thin glass plate (refractive index $= 1.5$), the central maximum is shifted to a position previously occupied by the 4^{th} bright fringe. The thickness of the glass-plate is .....$\mu\mathrm{m}$

Correct Answer:
4

Solution:
$(\mu - 1)\mathrm{t} = \mathrm{n}\lambda$

$(1.5 - 1)\mathrm{t} = 4 \times 500 \times 10^{-9}\ \mathrm{m}$

$\mathrm{t} = 4000 \times 10^{-9}\ \mathrm{m}$

$t = 4\mu m$

Q. 29 A force $(3x^2 + 2x - 5)N$ displaces a body from $x = 2$ m to $x = 4$ m. Work done by this force is_____J.

Correct Answer:
58

Solution:
$W = \int_{x_1}^{x_2} F dx$

$W = \int_2^4 (3x^2 + 2x - 5) dx$

$W = [x^3 + x^2 - 5x]_2^4$

$W = [60 - 2]J = 58$ J

Q. 30 At room temperature $(27^\circ C)$, the resistance of a heating element is 50Ω . The temperature coefficient of the material is $2.4 \times 10^{-4} {}^\circ C^{-1}$. The temperature of the element, when its resistance is 62Ω, is_____ $^\circ C$.

Correct Answer:
1027

Solution:
$R = R_0(1 + \alpha \Delta T)$

$62 = 50 \left[1 + 2.4 \times 10^{-4} \Delta T\right]$

$\Delta T = 1000^\circ C$

$\Rightarrow T - 27^\circ = 1000^\circ C$

$T = 1027^\circ C$

Chemistry

Q. 1 The candela is the luminous intensity, in a given direction, of a source that emits monochromatic radiation of frequency ' A ' $\times 10^{12}$ hertz and that has a radiant intensity in that direction of $\frac{1}{'B'}$ watt per steradian. 'A' and 'B' are respectively

Option 1:
540 and $\frac{1}{683}$

Option 2:
540 and 683

Option 3:
450 and $\frac{1}{683}$

Option 4:
450 and 683

Correct Answer:
540 and 683

Solution:
The candela is the luminous intensity of a source that emits monochromatic radiation of frequency radiation of frequency $540 \times 10^{12}\ \mathrm{Hz}$ and has a radiant intensity in that direction of $\frac{1}{683}\mathrm{w/sr}$. It is unit of Candela.

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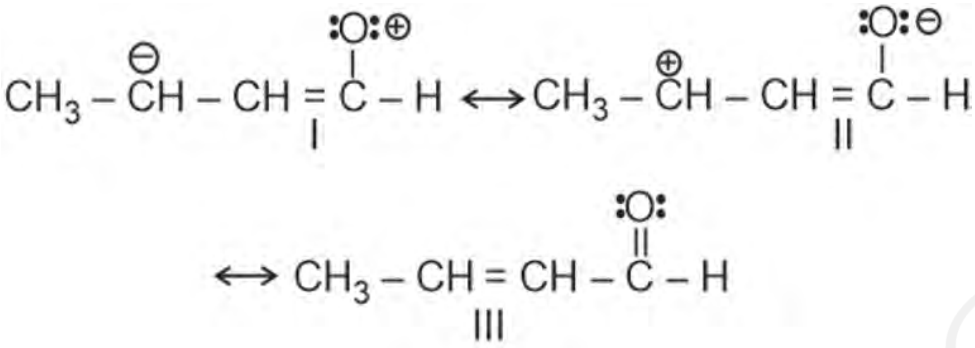
Q. 2 The correct stability order of the following resonance structures of $CH_3 - CH = CH - CHO$ is



Option 1:
II > III > I

Option 2:
III > II > I

Option 3:
I > II > III

Option 4:
II > I > III

Correct Answer:
III > II > I

Solution:

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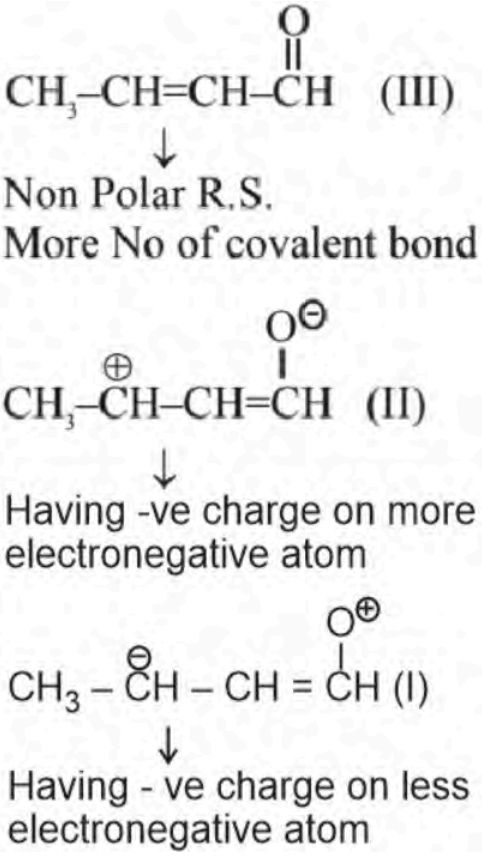
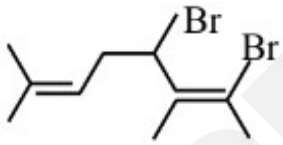


Stability order III > II > I

Q. 3 Total number of stereo isomers possible for the given structure:



Option 1:
8

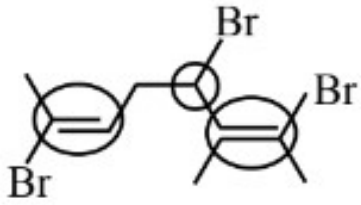
Option 2:
2

Option 3:
4

Option 4:
3

Correct Answer:
8

Solution:



There are three stereo center

So No of stereoisomer $= 2^3 = 8$

Q. 4 The correct increasing order for bond angles among BF_3, PF_3 and $C\ell F_3$ is :

Option 1:
$PF_3 < BF_3 < C / F_3$

Option 2:
$BF_3 < PF_3 < C$ /F 3

Option 3:
$C\ell F_3 < PF_3 < BF_3$

Option 4:
$BF_3 = PF_3 < C/F_3$

Correct Answer:
$C\ell F_3 < PF_3 < BF_3$

Solution:

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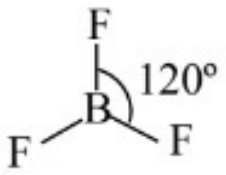
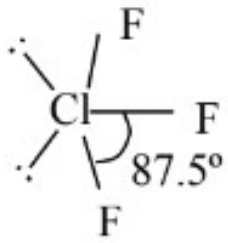
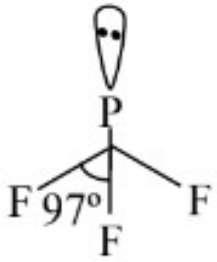






Order of bond angle is

$\mathrm{ClF_3} < \mathrm{PF_3} < \mathrm{BF_3}$

Q. 5 Match List I with List II

List - I(Test)		List - II (Observation)	
A.	$\mathrm{Br_2}$ water test	I.	Yellow orange or orange red precipitate formed
B.	Ceric ammonium nitrate test	II.	Reddish orange colour disappears
C.	Ferric Chloride test	III.	Red colour appears
D.	2,4-DNP test	IV.	Blue, Green, Violet or Red colour appear

Choose the correct answer from the options given below:

Option 1:
A-I, B-II, C-III, D-IV

Option 2:
A-II, B-III, C-IV, D-I

Option 3:
A-III, B-IV, C-I, D-II

Option 4:
A-IV, B-I, C-II, D-III

Correct Answer:
A-II, B-III, C-IV, D-I

Solution:
(A) Br_2 water test is test of unsaturation in which reddish orange colour of bromine water disappears.
(B) Alcohols given Red colour with ceric ammonium nitrate.
(C) Phenol gives Violet colour with natural ferric chloride.
(D) Aldehyde \& Ketone give Yellow/Orange/Red Colour compounds with 2, 4-DNP i.e., 2, 4-Dinitrophenyl hydrazine.

Q. 6 Match List I with List II

List - I (Cell)		**List - II (Use/Property/Reaction)**	
A.	Leclanche cell	I.	Converts energy of combustion into electrical energy
B.	Ni-Cd cell	II.	Does not involve any ion in solution and is used in hearing aids
C.	Fuel Cell	III.	Rechargeable
D.	Mercury Cell	IV.	Reaction at anode $Zn \rightarrow Zn^{2+} + 2e^-$

Choose the correct answer from the options given below:

Option 1:
A-I, B-II, C-III, D-IV

Option 2:
A-III, B-I, C-IV, D-II

Option 3:
A-IV, B-III, C-I, D-II

Option 4:
A-II, B-III, C-IV, D-I

Correct Answer:
A-II, B-III, C-IV, D-I

Solution:
A-IV, B-III, C-I, D-II

Q. 7 Match List I with List II

List-I		List - II	
A.	$K_2[Ni(CN)_4]$	I.	sp^3
B.	$[Ni(CO)_4]$	II.	$sp^3\ d^2$
C.	$[Co(NH_3)_6]Cl_3$	III.	dsp^2
D.	$Na_3[CoF_6]$	IV.	d^2sp^3

Choose the correct answer from the options given below:

Option 1:
A-III, B-I, C-II, D-IV

Option 2:
A-III, B-II, C-IV, D-I

Option 3:
A-I, B-III, C-II, D-IV

Option 4:
A-III, B-I, C-IV, D-II

Correct Answer:
A-III, B-I, C-IV, D-II

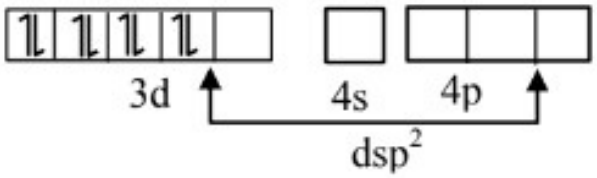
Solution:

+2

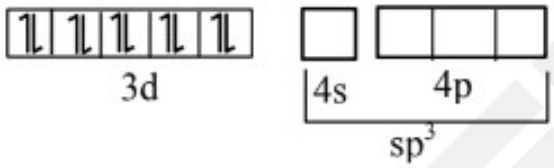
(A) \text { (A) } \mathrm{K}_2\left[\mathrm{Ni (\mathrm{CN})}_4\right\rceil

Ni^{2+} : $[Ar]3\,d^8 4\,s^\circ$, (CN is S.F.L)

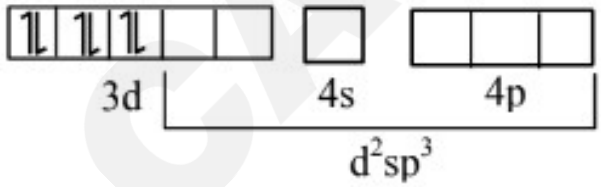
Pre hybridization state of Ni^{+2}



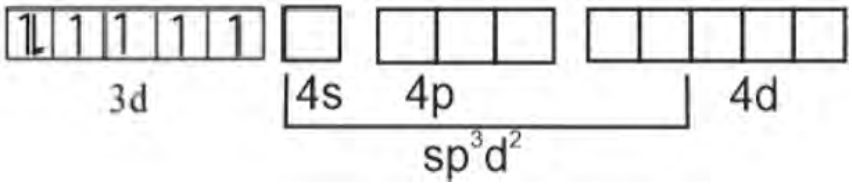
(B) $[Ni(CO)_4]$
$Ni : [Ar]3\,d^8 4\,s^2$
CO is S.F.L , so pairing occur
Pre hybridization state of Ni



(C) $[Co(NH_3)_6]Cl_3$
$Co^{+3} : [Ar]3\,d^6 4\,s^0$
With Co^{3+}, NH_3 act as S.F.L



(d) $Na_3\,[CoF_6]$
$Co^{3+} : [Ar]3\,d^6$ ($F^{\oplus}$: W.F.L)



Q. 8 The coordination environment of Ca^{2+} ion in its complex with EDTA^{4-} is :

Option 1:
tetrahedral

Option 2:
octahedral

Option 3:
square planar

Option 4:
trigonal prismatic

Correct Answer:
octahedral

Solution:
$\mathrm{EDTA}^{4-} \rightarrow$ Hexadentate ligand
$[\mathrm{Ca}(\mathrm{EDTA})]^{2-}$

So Coordination environment is octahedral

Q. 9 The incorrect statement about Glucose is :

Option 1:
Glucose is soluble in water because of having aldehyde functional group

Option 2:
Glucose remains in multiple isomeric form in its aqueous solution

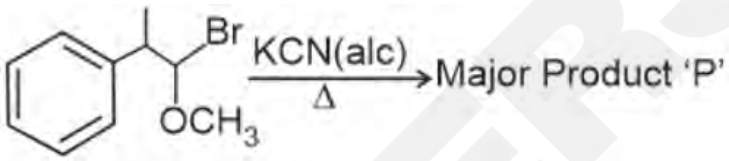
Option 3:
Glucose is an aldohexose

Option 4:
Glucose is one of the monomer unit in sucrose

Correct Answer:
Glucose is soluble in water because of having aldehyde functional group

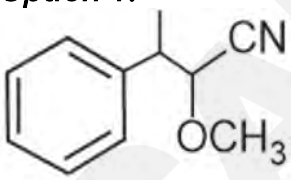
Solution:
Glucose is soluble in water due to presence of alcohol functional group and extensive hydrogen bonding.Glucose exist is open chain as well as cyclic forms in its aqueous solution.Glucose having 6C atoms so it is hexose and having aldehyde functional group so it is aldose.Thus, aldohexose.Glucose is monomer unit in sucrose with fructose.

Q. 10

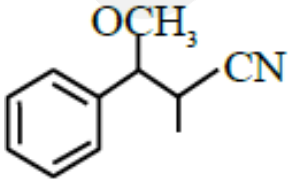


In the above reaction product 'P' is

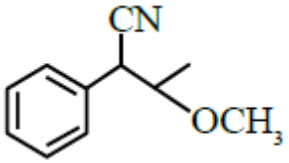
Option 1:



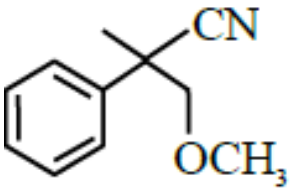
Option 2:



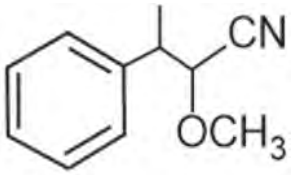
Option 3:



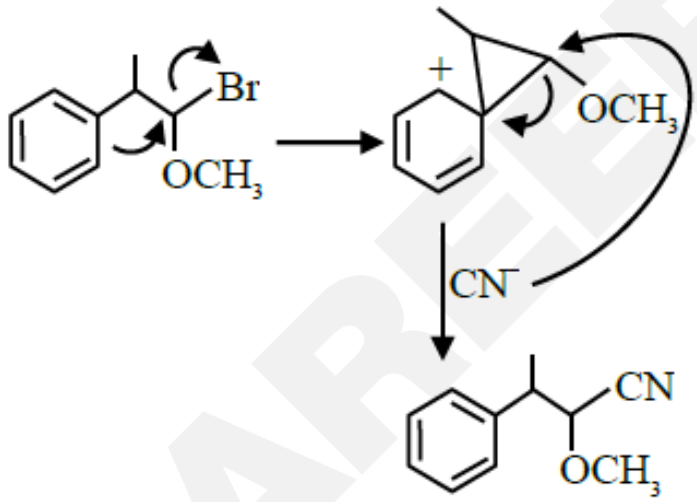
Option 4:



Correct Answer:



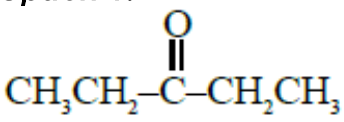
Solution:



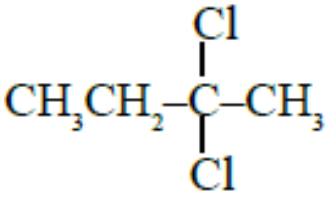
Due to NGP effect of phenyl ring Nucleophilic substitution of Br will occurs.

Q. 11 Which of the following compound can give positive iodoform test when treated with aqueous
KOH solution followed by potassium hypoiodite.

Option 1:



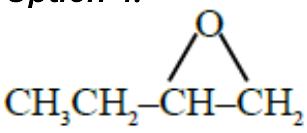
Option 2:



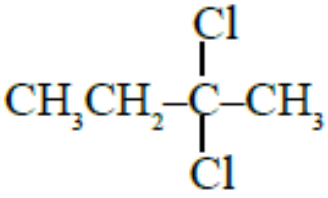
Option 3:

$CH_3CH_2CH_2CHO$

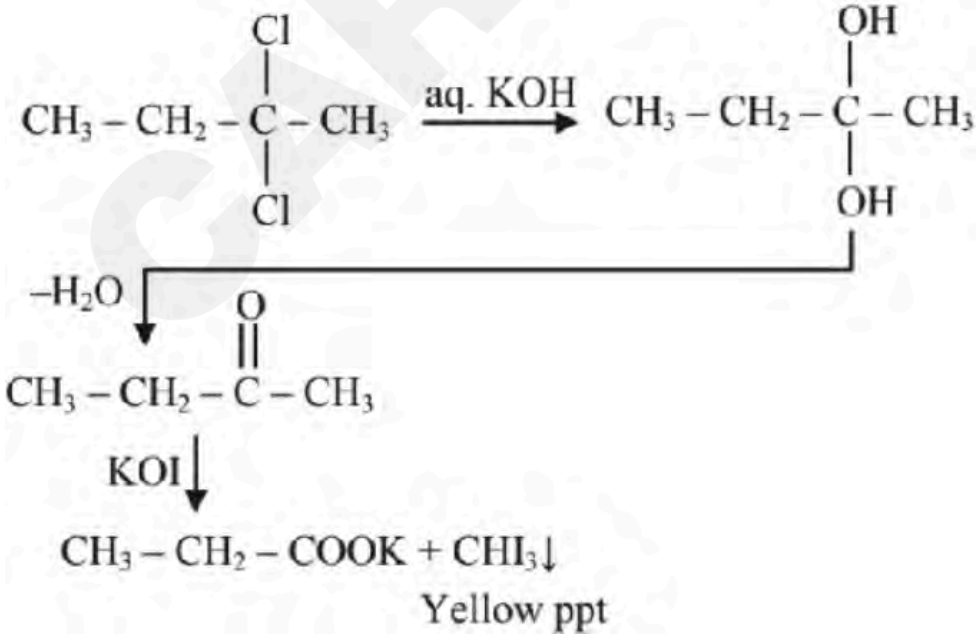
Option 4:



Correct Answer:



Solution:



Q. 12 For a sparingly soluble salt AB_2, the equilibrium concentrations of A^{2+} ions and B^- ions are $1.2 \times 10^{-4}M$ and $0.24 \times 10^{-3}M$, respectively. The solubility product of AB_2 is :

Option 1:
0.069×10^{-12}

Option 2:
6.91×10^{-12}

Option 3:
0.276×10^{-12}

Option 4:
27.65×10^{-12}

Correct Answer:
6.91×10^{-12}

Solution:

$AB_{2(s)} \rightleftharpoons A^{+2}_{(aq)} + 2\,B^{-}_{(aq)}$

$K_{sp} = [A^{+2}][B^-]^2$

$= 1.2 \times 10^{-4} \times (2.4 \times 10^{-4})^2$

$= 6.91 \times 10^{-12}M^3$

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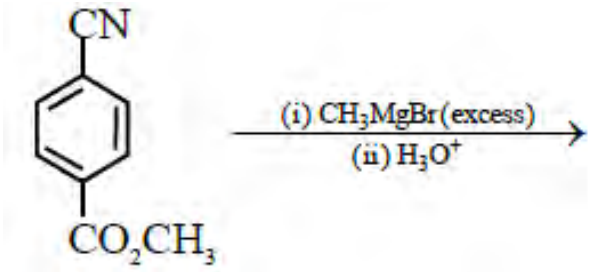
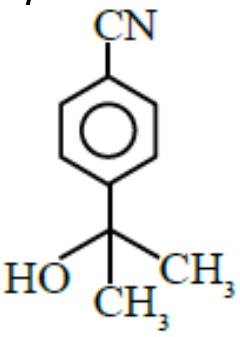
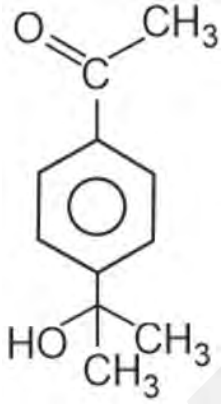
Q. 13 Major product of the following reaction is



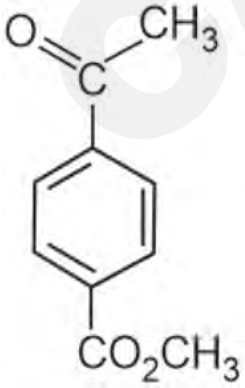
Option 1:



Option 2:



Option 3:



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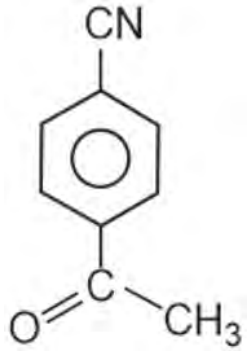
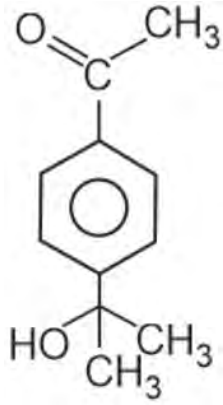
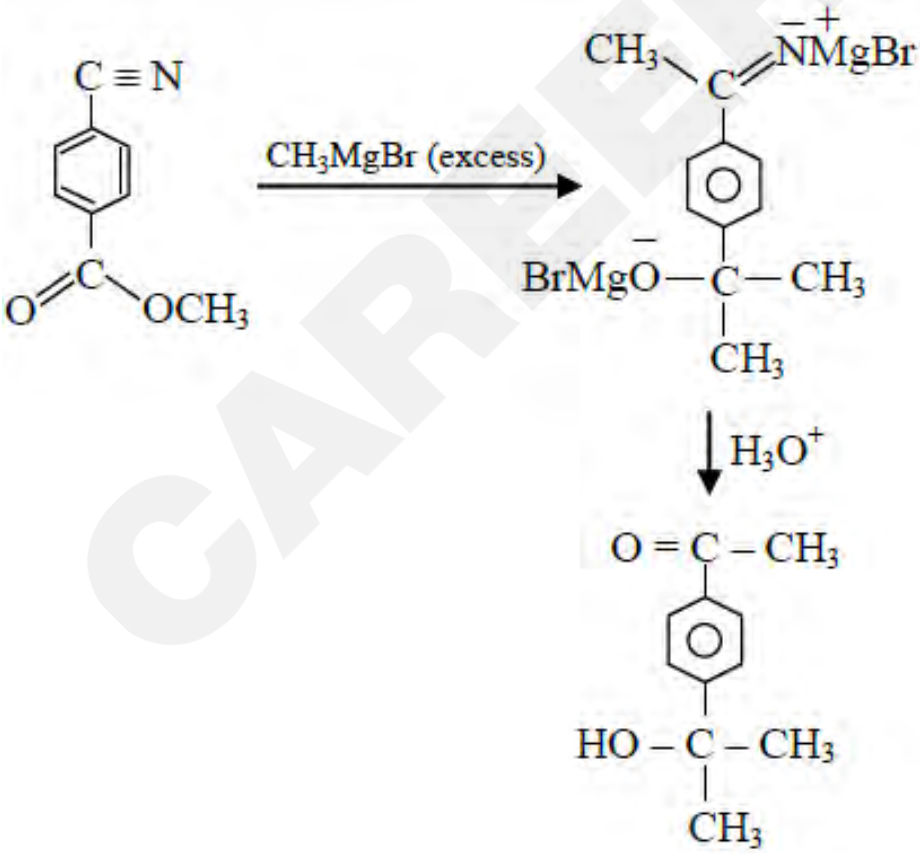
Option 4:



Correct Answer:



Solution:



Q. 14 Given below are two statements :
Statement I : The higher oxidation states are more stable down the group among transition elements unlike p-block elements.
Statement II : Copper can not liberate hydrogen from weak acids.
In the light of the above statements,

choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are false

Option 2:
Statement I is false but Statement II is true

Option 3:
Both Statement I and Statement II are true

Option 4:
Statement I is true but Statement II is false

Correct Answer:
Both Statement I and Statement II are true

Solution:
On moving down the group in transition elements, stability of higher oxidation state increases, due to increase in effective nuclear charge.

$\Rightarrow \mathrm{E}^{\circ}_{\mathrm{Cu}^{+2}/\mathrm{Cu}} = 0.34\ \mathrm{V}$

$\Rightarrow \mathrm{E}^{0}_{\mathrm{H}^{+}/\mathrm{H}_2} = 0$

$\mathrm{SRP}\ : \mathrm{Cu}^{2+} > \mathrm{H}^{+}$

Cu can't liberate hydrogen gas from weak acid.

Q. 15 The incorrect statement regarding ethyne is

Option 1:
The $\mathrm{C}-\mathrm{C}$ bonds in ethyne is shorter than that in ethene

Option 2:
Both carbons are sp hybridised

Option 3:
Ethyne is linear

Option 4:
The carbon-carbon bonds in ethyne is weaker than that in ethene

Correct Answer:
The carbon-carbon bonds in ethyne is weaker than that in ethene

Solution:
The carbon-carbon bonds in ethyne is stronger than that in ethene.
$(\mathrm{H}-\mathrm{C}\equiv\mathrm{C}-\mathrm{H})$ Ethyne is linear and carbon atoms are SP hybridised.

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Q. 16 Match List I with List II

List-I (Element)		List-II (Electronic Configuration)	
A	N	I.	$[Ar]3\,d^{10}4\,s^2 4p^5$
B	S	II.	$[Ne]3\,s^2 3p^4$
C	Br.	III.	$[He]2\,s^2 2p^3$
D	Kr.	IV.	$[Ar]3\,d^{10}4\,s^2 4p^6$

Choose the correct answer from the options given below :

Option 1:
A-IV, B-III, C-II, D-I

Option 2:
A-III, B-II, C-I, D-IV

Option 3:
A-I, B-IV, C-III, D-II

Option 4:
A-II, B-I, C-IV, D-III

Correct Answer:
A-III, B-II, C-I, D-IV

Solution:
(A) ${}_7N : [He]2\,s^2 2p^3$
(B) ${}_{16}S : [Ne]2\,s^2 3p^4$
(C) ${}_{35}Br : [Ar]3\,d^{10}4\,s^2 4p^5$
(D) ${}_{36}Kr : [Ar]3\,d^{10}4\,s^2 4p^6$

Q. 17 Match List I with List II

List-I		List-II	
A.	Melting point [K]	I.	$Tl > In > Ga > Al > B$
B.	Ionic Radius $\left[M^{+3}/pm\right]$	II.	$B > Tl > Al \approx Ga > In$
C.	$\Delta_i H_1$ $\left[kJ\ mol^{-1}\right]$	III.	$Tl > In > Al > Ga > B$
D.	Atomic Radius [pm]	IV.	$B > Al > Tl > In > Ga$

Choose the correct answer from the options given below :

Option 1:
A-III, B-IV, C-I, D-II

Option 2:
A-II, B-III, C-IV, D-I

Option 3:
A-IV, B-I, C-II, D-III

Option 4:
A-I, B-II, C-III, D-IV

Correct Answer:
A-IV, B-I, C-II, D-III

Solution:
Melting point : $B > A\ell > T\ell > In > Ga$

Ionic radius $\left(M^{+3}/pm\right)$: $T\ell > In > Ga > A\ell > B$

$(\Delta_{IE}H)_1\left[\frac{kJ}{mol}\right]: B > T\ell > A\ell \approx Ga > In$

Atomic radius (in pm) : $T\ell > In > A\ell > Ga > B$

Q. 18 Which of the following compounds will give silver mirror with ammoniacal silver nitrate?
(A) Formic acid
(B) Formaldehyde
(C) Benzaldehyde
(D) Acetone
Choose the correct answer from the options given below :

Option 1:
C and D only

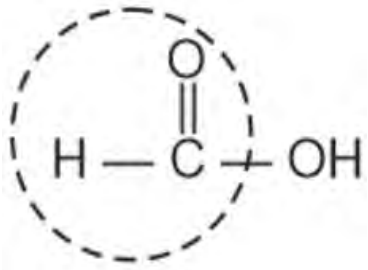
Option 2:
A, B and C only

Option 3:
A only

Option 4:
B and C only

Correct Answer:
A, B and C only

Solution:

Apart from aldehyde, Formic acid 

also gives silver mirror test with ammonical silver nitrate.

Q. 19 Which out of the following is a correct equation to show change in molar conductivity with respect to concentration for a weak electrolyte, if the symbols carry their usual meaning :

Option 1:
$\Lambda^2{}_{\mathrm{m}}\mathrm{C} - \mathrm{K_a}\Lambda_{\mathrm{m}}^{\circ 2} + \mathrm{K_a}\Lambda_{\mathrm{m}}\Lambda_{\mathrm{m}}^{\circ} = 0$

Option 2:
$\Lambda_m - \Lambda_m^{\circ} + AC^{\frac{1}{2}} = 0$

Option 3:
$\Lambda_{\mathrm{m}} - \Lambda_{\mathrm{m}}^{\circ} - \mathrm{AC}^{\frac{1}{2}} = 0$

Option 4:
$\Lambda_{\mathrm{m}}^2\mathrm{C} + \mathrm{K_a}\Lambda_{\mathrm{m}}^{\circ 2} - \mathrm{K_a}\Lambda_{\mathrm{m}}\Lambda_{\mathrm{m}}^{\circ} = 0$

Correct Answer:
$\Lambda^2{}_{\mathrm{m}}\mathrm{C} - \mathrm{K_a}\Lambda_{\mathrm{m}}^{\circ 2} + \mathrm{K_a}\Lambda_{\mathrm{m}}\Lambda_{\mathrm{m}}^{\circ} = 0$

Solution:
$\mathrm{HA(aq)} \rightleftharpoons \mathrm{H^+(aq)} + \mathrm{A^-(aq)}$

$\mathrm{K_a} = \frac{\alpha^2\mathrm{C}}{1-\alpha}$

$\alpha^2\mathrm{C} + \mathrm{K_a}\alpha - \mathrm{K_a} = 0$

$\left(\frac{\lambda_{\mathrm{m}}}{\lambda_{\mathrm{m}}^{\infty}}\right)^2\mathrm{C} + \mathrm{K_a}\frac{\lambda_{\mathrm{m}}}{\lambda_{\mathrm{m}}^{\infty}} - \mathrm{K_a} = 0$

$\lambda_{\mathrm{m}}^2\mathrm{C} + \mathrm{K_a}\lambda_{\mathrm{m}}\lambda_{\mathrm{m}}^{\infty} - \mathrm{K_a}(\lambda_{\mathrm{m}}^{\infty})^2 = 0$

Q. 20 The electronic configuration of Einsteinium is:
(Given atomic number of Einsteinium $= 99$)

Option 1:
$[\mathrm{Rn}]5\mathrm{f}^{12}6\ \mathrm{d}^{0}7\ \mathrm{s}^{2}$

Option 2:
$[\mathrm{Rn}]5\mathrm{f}^{11}6\ \mathrm{d}^{0}7\ \mathrm{s}^{2}$

Option 3:
$[\mathrm{Rn}]5\mathrm{f}^{13}6\ \mathrm{d}^{0}7\ \mathrm{s}^{2}$

Option 4:
$[\mathrm{Rn}]5\mathrm{f}^{10}6\ \mathrm{d}^{0}7\ \mathrm{s}^{2}$

Correct Answer:
$[\mathrm{Rn}]5\mathrm{f}^{11}6\ \mathrm{d}^{0}7\ \mathrm{s}^{2}$

Solution:
Einsteinium (atomic No $= 99$) : $[Rn]5\mathrm{f}^{11}6\ \mathrm{d}^{0}7\ \mathrm{s}^{2}$

Q. 21 Number of oxygen atoms present in chemical formula of fuming sulphuric acid is _____.

Correct Answer:
7

Solution:
Fuming sulphuric acid is a mixture of conc. $\mathrm{H_2SO_4} + \mathrm{SO_3}$ Or $\mathrm{H_2\ S_2O_7}$

So, Number of Oxygen atoms $= 7$

Q. 22 A transition metal ' M ' among $\mathrm{Sc}, \mathrm{Ti}, \mathrm{V}, \mathrm{Cr}, \mathrm{Mn}$ and Fe has the highest second ionisation enthalpy. The spin only magnetic moment value of M^{+} ion is _____ BM (Near integer)
(Given atomic number $\mathrm{Sc}: 21, \mathrm{Ti}: 22, \ \mathrm{V}: 23, \mathrm{Cr}: 24, \mathrm{Mn}: 25, \mathrm{Fe}: 26$)

Correct Answer:
6

Solution:
Among given metals, Cr has maximum IE_2
because Second electron is removed from stable configuration $3\,\mathrm{d}^5$
$\mathrm{Cr}^{+}: [\mathrm{Ar}] 3\,\mathrm{d}^5 4\,\mathrm{s}^0$
$\therefore$ No of unpaired e^{-} in Cr^{+} is $5, \mathrm{n}=5$

$$\text{So, Magnetic moment} = \sqrt{\mathrm{n}(\mathrm{n}+2)}\ \mathrm{B.M}$$
$$= \sqrt{5(5+2)} = 5.92\mathrm{BM} \approx 6$$

Q. 23 The vapour pressure of pure benzene and methyl benzene at $27^{\circ}\mathrm{C}$ is given as $80\mathrm{Torr}$ and $24\mathrm{Torr}$, respectively. The mole fraction of methyl benzene in vapour phase, in equilibrium with an equimolar mixture of those two liquids (ideal solution) at the same temperature is _____ $\times 10^{-2}$ (nearest integer)

Correct Answer:
23

Solution:
$$\mathrm{X}_{\text{methylbenzene}} = 0.5\mathrm{Y}_{\text{methylenzene}} = \frac{\mathrm{P}_{\text{methylbenzene}}}{\mathrm{P}_{\text{total}}}$$

$$\mathrm{Y}_{\text{methylbenzene}} = \frac{0.5\times 24}{0.5\times 80 + 0.5\times 24}$$

$$= \frac{12}{40+12} = 0.23 = 23 \times 10^{-2}$$

Q. 24 Consider the following test for a group-IV cation.

$M^{2+} + H_2S \rightarrow A$ (Black precipitate) + byproduct

$A+$ aqua regia $\rightarrow B + NOCl + S + H_2O$

$B + KNO_2 + CH_3COOH \rightarrow C+$ byproduct

The spin only magnetic moment value of the metal complex C is _____ BM.

(Nearest integer)

Correct Answer:

0

Solution:

$Co^{2+} + H_2S \rightarrow CoS\downarrow$ (Black)

$CoS +$ Aqua-regia $\rightarrow Co^{2+}(aq) + NOCl + S + H_2O$

(A) (B)

$Co^{2+}(aq) + KNO_2 + CH_3COOH$

$\downarrow$

$K_3[Co(NO_2)_6] + NO + S + H_2O$

$InK_3[Co(NO_2)_6], Co^{+3} : 3d^6 4s^0$

$Co^{3+} : d^2sp^3$ Hybridisation

Number of unpaired $e^- = 0$

Magnetic moment $= \sqrt{n(n+2)} = 0$ B.M

Q. 25 Consider the following first order gas phase reaction at constant temperature

$$A(g) \rightarrow 2B(g) + C(g)$$

If the total pressure of the gases is found to be 200 torr after 23sec. and 300 torr upon the complete decomposition of A after a very long time, then the rate constant of the given reaction is _____ $\times 10^{-2}$ s^{-1}

(nearest integer)

[Given : $\log_{10}(2) = 0.301$]

Correct Answer:
3

Solution:
$A(g) \rightarrow 2\,B(\,g) + C(g)$

$P_{23} = P_0 + 2x = 200$

$P_\infty = 3P_0 = 300$

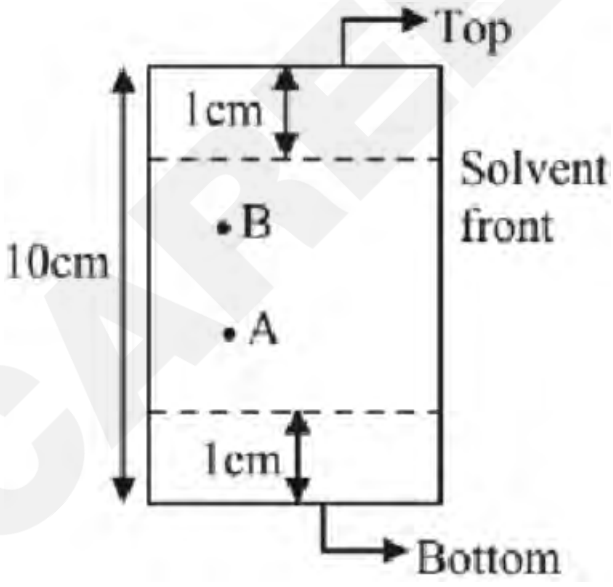
$P_0 = 100$

$K = \frac{1}{t} \ln \frac{P_\infty - P_0}{P_\infty - P_t}$

$K = \frac{2.3}{23} \log \frac{300-100}{300-200}$

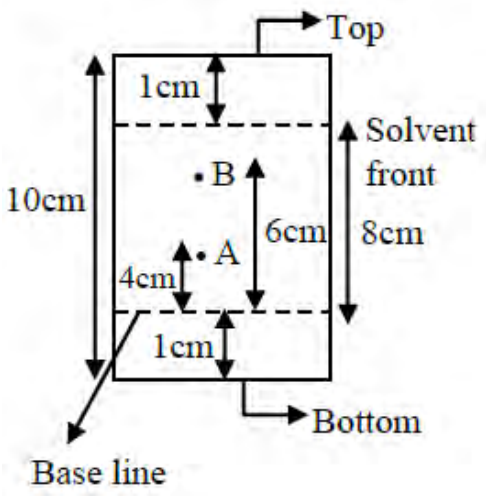
$= \frac{2.3 \times 0.301}{23} = 0.0301 = 3.01 \times 10^{-2} \text{sec}^{-1}$

Q. 26 In the given TLC, the distance of spot A \& B are $5\ \text{cm} \& 7\ \text{cm}$, from the bottom of TLC plate, respectively.
R_f value of B is $x \times 10^{-1}$ times more than A. The value of x is _____.



Correct Answer:
15

Solution:
$R_f = \frac{\text{Distance moved by substance from base line}}{\text{Distance moved by solvent from base line}}$



$(R_f)_A = \frac{4}{8} \quad (R_f)_B = \frac{6}{8}$

$\frac{(R_f)_B}{(R_f)_A} = \frac{6}{8} \times \frac{8}{4}$

$(R_f)_B = 1.5(R_f)_A$

$x = 15$

Q. 27 Based on Heisenberg's uncertainty principle, the uncertainty in the velocity of the electron to be found within an atomic nucleus of diameter 10^{-15} m is _____ $\times 10^9$ ms^{-1} (nearest integer)
[Given : mass of electron $= 9.1 \times 10^{-31}$ kg, Plank's constant $(h) = 6.626 \times 10^{-34} Js$]
(Value of $\pi = 3.14$)

Correct Answer:
58

Solution:
$m\Delta V.\ \Delta x = \frac{h}{4\pi}$

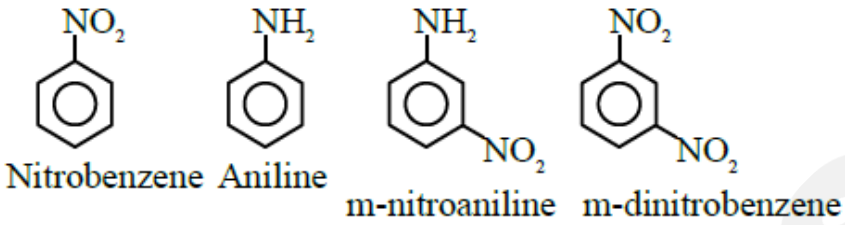
$\Delta V = \frac{6.626 \times 10^{-34}}{9.1 \times 10^{-31} \times 10^{-15} \times 4 \times 3.14}$

$= 57.97 \times 10^{+9}$ m/sec

Q. 28 Number of compounds from the following which cannot undergo Friedel-Crafts reactions is :____ toluene, nitrobenzene, xylene, cumene, aniline, chlorobenzene, m-nitroaniline, m-dinitrobenzene

Correct Answer:
4

Solution:
Compounds which can not undergo Friedel Crafts reaction are



Nitrobenzene Aniline m-nitroaniline m-dinitrobenzene

Q. 29 Total number of electron present in (π^*) molecular orbitals of O_2, O_2^+ and O_2^- is _____

Correct Answer:
6

Solution:
$O_2(16e) : (\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2\,s})^2(\sigma_{2\,s}^*)^2$

$(\sigma_{2p})^2\left[(\pi_{2p})^2 = (\pi_{2p})^2\right], \left[(\pi_{2p}^*)^1 = (\pi_{2p}^*)^1\right]$

Number of e^- present in (π^*) of $O_2 = 2$

Number of e^- present in (π^*) of $O_2^+ = 1$

Number of e^- present in (π^*) of $O_2^- = 3$

So total e^- in $(\pi^*) = 2 + 1 + 3 = 6$

Q. 30 When $\Delta H_{vap} = 30\ kJ/mol$ and $\Delta S_{vap} = 75\ J\ mol^{-1}\ K^{-1}$, then the temperature of vapour, at one atmosphere is _____ K.

Correct Answer:
400

Solution:
At equilibrium $\Delta G_{PT} = 0$

$\Delta H_{vap} = T\Delta S_{vap}$

$30 \times 1000 = T \times 75$

$T = 400\ K$

Maths

Q. 1 $Lim_{x\to 0} \frac{e-(1+2x)^{\frac{1}{2x}}}{x}$ is equal to :

Option 1:
e

Option 2:
$\frac{-2}{e}$
$e - e^2$

Option 3:
0

Option 4:
$e - e^2$

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Correct Answer:
e

Solution:

$$\text{Lim}_{x\to 0}\ \frac{e - e^{\frac{1}{2x}(1+2x)}}{x}$$

$$= \text{Lim}_{x\to 0}(-e)\frac{\left(e^{\frac{\ln(1+2x)}{2x}-1} - 1\right)}{x}$$

$$= \text{Lim}_{x\to 0}(-e)\frac{\ln(1+2x) - 2x}{2x^2}$$

$$= (-e)\times(-1)\frac{4}{2\times 2} = e$$

Q. 2 Consider the line L passing through the points $(1, 2, 3)$ and $(2, 3, 5)$. The distance of the point $\left(\frac{11}{3}, \frac{11}{3}, \frac{19}{3}\right)$ from the line L along the line $\frac{3x-11}{2} = \frac{3y-11}{1} = \frac{3z-19}{2}$ is equal to :

Option 1:
3

Option 2:
5

Option 3:
4

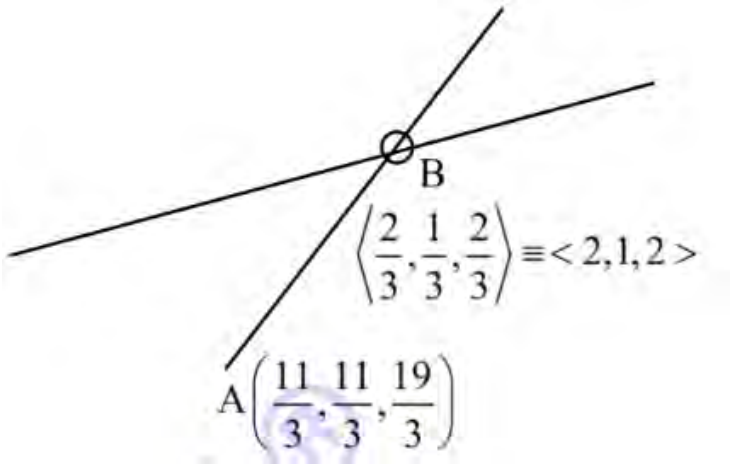
Option 4:
6

Correct Answer:
3

Solution:

$$\frac{x-1}{2-1} = \frac{y-2}{3-2} = \frac{z-3}{5-3}$$

$$\Rightarrow \frac{x-1}{1} = \frac{y-2}{1} = \frac{z-3}{2} = \lambda$$



$B(1+\lambda, 2+\lambda, 3+2\lambda)$

$\text{D.R. of AB} =< \frac{3\lambda-8}{3}, \frac{3\lambda-5}{3}, \frac{6\lambda-10}{3} >$

$B\left(\frac{5}{3}, \frac{8}{3}, \frac{13}{3}\right) \frac{3\lambda-8}{3\lambda-5} = \frac{2}{1} \Rightarrow 3\lambda - 8 = 6\lambda - 10$

$3\lambda = 2$

$\lambda = \frac{2}{3}$

$AB = \frac{\sqrt{36+9+36}}{3} = \frac{9}{3} = 3$

Q. 3

Let $\int_0^x \sqrt{1-(y'(t))^2}dt = \int_0^x y(t)dt, 0 \le x \le 3, y \ge 0$
$y(0) = 0$. Then at $x = 2, y'' + y + 1$ is equal to :

Option 1:
1

Option 2:
2

Option 3:
$\sqrt{2}$

Option 4:
1 / 2

Correct Answer:
1

Solution:

$$\sqrt{1-(y'(x))^2}=y(x)$$

$$1-\left(\frac{dy}{dx}\right)^2=y^2$$

$$\left(\frac{dy}{dx}\right)^2=1-y^2$$

$$\frac{dy}{\sqrt{1-y^2}}=dx \text{ OR } \frac{dy}{\sqrt{1-y^2}}=-dx$$

$$\Rightarrow \sin^{-1}y=x+c, \sin^{-1}y=-x+c$$

$$x=0, y=0 \Rightarrow c=0$$

$$\sin^{-1}y=x, \text{ as } y\geq 0$$

$$\sin x=y$$

$$\Rightarrow \frac{dy}{dx}=\cos x$$

$$\frac{d^2y}{dx^2}=-\sin x$$

$$\Rightarrow -\sin x+\sin x+1=1$$

Q. 4 Let z be a complex number such that the real part of $\frac{z-2i}{z+2i}$ is zero. Then, the maximum value of $|z-(6+8i)|$ is equal to :

Option 1:
12

Option 2:
∞

Option 3:
10

Option 4:
8

Correct Answer:
12

Solution:

$$\frac{z-2i}{z+2i}+\frac{\bar{z}+2i}{\bar{z}-2i}=0$$

$$z\bar{z}-2i\bar{z}-2iz+4(-1)$$
$$+z\bar{z}+2zi+2\bar{z}i+4(-1)=0$$
$$\Rightarrow 2|z|^2=8\Rightarrow |z|=2$$
$$|z-(6+8i)|_{\text{maximum}}=10+2=12$$

Q. 5 The area (in square units) of the region enclosed by the ellipse $x^2+3y^2=18$ in the first quadrant below the line $y=x$ is :

Option 1:
$\sqrt{3}\pi+\frac{3}{4}$

Option 2:
$\sqrt{3}\pi$

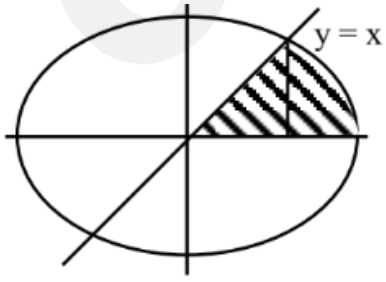
Option 3:
$\sqrt{3}\pi-\frac{3}{4}$

Option 4:
$\sqrt{3}\pi+1$

Correct Answer:
$\sqrt{3}\pi$

Solution:

$\frac{x^2}{18}+\frac{y^2}{6}=1$



$$\frac{x^2}{18}+\frac{3x^2}{18}=1\Rightarrow 4x^2=18\Rightarrow x^2=\frac{9}{2}$$

$$\int_{\frac{3}{\sqrt{2}}}^{3\sqrt{2}}\frac{\sqrt{18-x^2}}{\sqrt{3}}dx$$

$$=\frac{1}{\sqrt{3}}\left(\frac{x\sqrt{18-x^2}}{2}+\frac{18}{2}\sin^{-1}\frac{x}{3\sqrt{2}}\right)_{\frac{3}{\sqrt{2}}}^{3\sqrt{2}}$$

$$=\frac{1}{\sqrt{3}}\left(9\times\frac{\pi}{2}-\frac{3}{2\sqrt{2}}\times\frac{3\sqrt{3}}{\sqrt{2}}-9\times\frac{\pi}{6}\right)$$

Required Area

$$=\frac{1}{2}\times\frac{9}{2}+\left(\frac{18\pi}{6}-\frac{9\sqrt{3}}{4}\right)\frac{1}{\sqrt{3}}$$

$$=\sqrt{3}\pi$$

Q. 6 Let the foci of a hyperbola H coincide with the foci of the ellipse $E:\frac{(x-1)^2}{100}+\frac{(y-1)^2}{75}=1$ and the eccentricity of the hyperbola H be the reciprocal of the eccentricity of the ellipse E. If the length of the transverse axis of H is α and the length of its conjugate axis is β, then $3\alpha^2+2\beta^2$ is equal to :

Option 1:
242

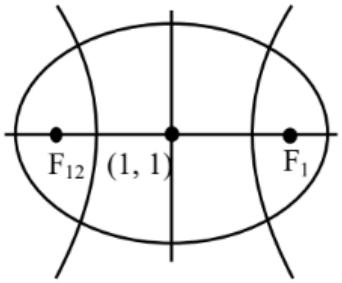
Option 2:
225

Option 3:
237

Option 4:
205

Correct Answer:
225

Solution:



$$e_1 = \sqrt{1 - \frac{75}{100}} = \frac{5}{10} = \frac{1}{2}$$

$$e_2 = 2$$

$$F_1(6, 1), F_2(-4, 1)$$

$$2ae_2 = 10 \Rightarrow a = \frac{5}{2} \Rightarrow 2a = 5$$

$$\Rightarrow \alpha = 5$$

$$4 = 1 + \frac{b^2}{a^2} \Rightarrow b^2 = 3a^2$$

$$b = \sqrt{3} \times \frac{5}{2}$$

$$\beta = 5\sqrt{3}$$

$$3\alpha^2 + 2\beta^2 = 3 \times 25 + 2 \times 25 \times 3$$

$$= 225$$

Q. 7 Two vertices of a triangle ABC are $A(3, -1)$ and $B(-2, 3)$, and its orthocentre is $P(1, 1)$. If the coordinates of the point C are (α, β) and the centre of the circle circumscribing the triangle PAB is (h, k), then the value of $(\alpha + \beta) + 2(h + k)$ equals :

Option 1:
51

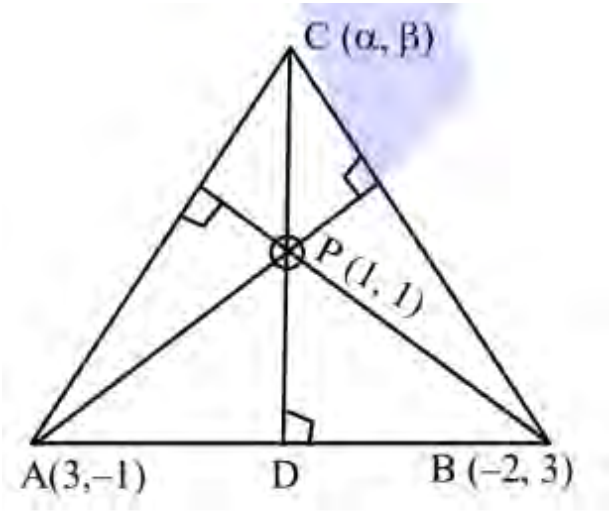
Option 2:
81

Option 3:
5

Option 4:
15

Correct Answer:
5

Solution:



$\mathrm{M_{AB}} = \frac{4}{-5} \Rightarrow \mathrm{M_{DP}} = \frac{5}{4}$

Equation of PC is $\mathrm{y} - 1 = \frac{5}{4}(\mathrm{x} - 1)$

$\mathrm{M_{AP}} = \frac{2}{-2} = -1 \Rightarrow \mathrm{M_{BC}} = +1$

Equation of BC is $\mathrm{y} - 3 = (\mathrm{x} + 2)$

On solving (1) and (2)

$x + 4 = \frac{5}{4}(x - 1) \Rightarrow 4x + 16 = 5x - 5 \Rightarrow \alpha = 21$

$\Rightarrow \beta = y = x + 5 = 26$

$\alpha + \beta = 47$

Equation of $\perp$ bisector of AP

$\mathrm{y} - 0 = (\mathrm{x} - 2)$

Equation of $\perp$ bisector of AB

$y-1=\frac{5}{4}\left(x-\frac{1}{2}\right)$

On solving (3) \& (4)

$$(\mathrm{x}-3)4=5\mathrm{x}-\frac{5}{2}$$

$$\mathrm{x}=\frac{-19}{2}=\mathrm{h}$$

$$\mathrm{y}=\frac{-23}{2}=\mathrm{k}$$

$$\Rightarrow 2(\mathrm{h}+\mathrm{k})=-42$$

Q. 8 If the variance of the frequency distribution is 160 , then the value of $\mathrm{c} \in \mathrm{N}$ is

x	c	2c	3c	4c	5c	6c
f	2	1	1	1	1	1

Option 1:
5

Option 2:
8

Option 3:
7

Option 4:
6

Correct Answer:
7

Solution:

x	c	2c	3c	4c	5c	6c
f	2	1	1	1	1	1

$\bar{x} = \frac{(2+2+3+4+5+6)C}{7} = \frac{22C}{7}$

$$\text{Var}(x) = \frac{c^2\left(2+2^2+3^2+4^2+5^2+6^2\right)}{7} - \left(\frac{22c}{7}\right)^2$$

$$= \frac{92c^2}{7} - c^2 \times \frac{484}{49}$$

$$= \frac{(644-484)c^2}{49} = \frac{160c^2}{49}$$

$$160 = \frac{160 \times c^2}{10} \Rightarrow c = 7$$

Q. 9 Let the range of the function $f(x) = \frac{1}{2+\sin 3x+\cos 3x}, x \in \text{IR}$ be $[a, b]$.

If α and β are respectively the A.M. and the G.M. of a and b, then $\frac{\alpha}{\beta}$ is equal to :

Option 1:
$\sqrt{2}$

Option 2:
2

Option 3:
$\sqrt{\pi}$

Option 4:
π

Correct Answer:
$\sqrt{2}$

Solution:

$f(x)\frac{1}{2+\sin 3x+\cos 3x}$

$\left[\frac{1}{2+\sqrt{2}},\frac{1}{2-\sqrt{2}}\right]$

$\frac{\alpha}{\beta}=\frac{a+b}{2\sqrt{ab}}=\frac{1}{2}\left(\sqrt{\frac{a}{b}}+\sqrt{\frac{b}{a}}\right)$

$=\frac{1}{2}\left(\sqrt{\frac{2-\sqrt{2}}{2+\sqrt{2}}}+\sqrt{\frac{2+\sqrt{2}}{2-\sqrt{2}}}\right)$

$=\frac{(2-\sqrt{2})+(2+\sqrt{2})}{2\times\sqrt{2}}=\sqrt{2}$

Q. 10 Between the following two statements :

Statement-I : Let $\vec{a}=\hat{i}+2\hat{j}-3\hat{k}$ and $\vec{b}=2\hat{i}+\hat{j}-\hat{k}$. Then the vector $\vec{r}$ satisfying $\vec{a}\times\vec{r}=\vec{a}\times\vec{b}$ and $\vec{a}\cdot\vec{r}=0$ is of magnitude $\sqrt{10}$.

Statement-II : In a triangle ABC, $\cos 2A+\cos 2B+\cos 2C\geq-\frac{3}{2}$

Option 1:
Both Statement-I and Statement-II are incorrect

Option 2:
Statement-I is incorrect but Statement-II is correct

Option 3:
Both Statement-I and Statement-II are correct

Option 4:
Statement-I is correct but Statement-II is incorrect

Correct Answer:
Statement-I is incorrect but Statement-II is correct

Solution:

$\bar{a} = \hat{i} + 2\hat{j} - 3\hat{k}$

$\bar{a} = 2\hat{i} + \hat{j} - \hat{k}$

$\bar{a} \times \bar{r} = \bar{a} \times \bar{b}; \bar{a} \cdot \bar{r} = 0$

$\Rightarrow \bar{a} \times (\bar{r} - \bar{b}) = \bar{0}$

$\Rightarrow \bar{a} = \lambda(\bar{r} - \bar{b})$

$\bar{a} \cdot \bar{a} = \lambda(\bar{a} \cdot \bar{r} - \bar{a} \cdot \bar{b})$

$14 = -7\lambda \Rightarrow \lambda = -2$

$\frac{-\bar{a}}{2} = \bar{r} - \bar{b} \Rightarrow \bar{r} = \bar{b} - \frac{\bar{a}}{2}$

$= \frac{2\bar{b} - \bar{a}}{2} = \frac{3\hat{i} + \hat{k}}{2}$

Statement (I) is incorrect

$\cos 2\,\mathrm{A} + \cos 2\,\mathrm{B} + \cos 2\mathrm{c} \geq -\frac{3}{2}$

$2\,\mathrm{A} + 2\,\mathrm{B} + 2\mathrm{C} = 2\pi$

$\cos 2\,\mathrm{A} + \cos 2\,\mathrm{B} + \cos 2\mathrm{C}$

$= -1 - 4\cos\mathrm{A} \cdot \cos\mathrm{B} \cdot \cos\mathrm{C}$

$\geq -1 - 4 \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$

$= -\frac{3}{2}$

Statement (II) is correct.

Q. 11 $\lim_{x \to \frac{\pi}{2}} \left(\frac{\int_{x^3}^{(\pi/2)^3} \left(\sin\left(2t^{1/3}\right) + \cos\left(t^{1/3}\right)\right) dt}{\left(x - \frac{\pi}{2}\right)^2} \right)$ is equal to :

Option 1:

$\frac{9\pi^2}{8}$

Option 2:

$\frac{11\pi^2}{10}$

Option 3:

$\frac{3\pi^2}{2}$

Option 4:
$\frac{5\pi^2}{9}$

Correct Answer:
$\frac{9\pi^2}{8}$

Solution:

$$\lim_{x\to\frac{\pi}{2}} \frac{0-\{\sin(2x)+\cos(x)\}\cdot 3x^2}{2\left(x-\frac{\pi}{2}\right)}$$

$$= \lim_{x\to\frac{\pi}{2}} \frac{-\{2\sin x\cos x+\cos x\}3x^2}{2\left(x-\frac{\pi}{2}\right)}$$

$$= \lim_{x\to\frac{\pi}{2}} \left\{\frac{2\sin x\sin\left(\frac{\pi}{2}-x\right)}{2\left(x-\frac{\pi}{2}\right)}+\frac{\sin\left(\frac{\pi}{2}-x\right)}{2\left(\frac{\pi}{2}-x\right)}\right\}3x^2$$

$$= \left(1(1)+\frac{1}{2}\right)3\left(\frac{\pi}{2}\right)^2$$

$$= \frac{9\pi^2}{8}$$

Q. 12 The sum of the coefficient of $x^{2/3}$ and $x^{-2/5}$ in the binomial expansion of $\left(x^{2/3}+\frac{1}{2}x^{-2/5}\right)^9$ is :

Option 1:
$21/4$

Option 2:
$69/16$

Option 3:
$63/16$

Option 4:
$19/4$

Correct Answer:
$21/4$

Solution:

$$T_{r+1} = {}^{9}C_r \left(x^{2/3}\right)^{9-r} \left(\frac{x^{-2/5}}{2}\right)^{r}$$

$$= {}^{9}C_r \left(\frac{1}{2}\right)^{r} (r)^{\left(6-\frac{2r2r}{3}\right)}$$

for coefficient of $x^{2/3}$, put $6 - \frac{2r}{3} - \frac{2r}{5} = \frac{2}{3}$

$\Rightarrow r = 5$

$\therefore$ Coefficient of $x^{2/3}$ is $= {}^{9}C_5 \left(\frac{1}{5}\right)^{5}$

For coefficient of $x^{-2/5}$, put $6 - \frac{2r}{3} - \frac{2r}{5} = -\frac{2}{5}$

$\Rightarrow r = 6$

Coefficient of $x^{-2/5}$ is ${}^{9}C_6 \left(\frac{1}{2}\right)^{6}$

$$\text{Sum} = {}^{9}C_5 \left(\frac{1}{2}\right)^{5} + {}^{9}C_6 \left(\frac{1}{2}\right)^{6} = \frac{21}{4}$$

Q. 13 Let $B = \begin{bmatrix} 1 & 3 \\ 1 & 5 \end{bmatrix}$ and A be a 2×2 matrix such that $AB^{-1} = A^{-1}$. If $BCB^{-1} = A$ and $C^4 + \alpha C^2 + \beta I = O$, then $2\beta - \alpha$ is equal to :

Option 1:
16

Option 2:
2

Option 3:
8

Option 4:
10

Correct Answer:
10

Solution:

$$BCB^{-1} = A$$
$$\Rightarrow (BCB^{-1})(BCB^{-1}) = A.A$$
$$\Rightarrow BCICB^{-1} = A^2$$
$$\Rightarrow BC^2B^{-1} = A^2$$
$$\Rightarrow B^{-1}(BC^2B^{-1})B = B^{-1}(A.A)B$$

From equation (1)
$$C^2 = A^{-1} \cdot A \cdot B$$
$$C^2 = B$$

Also $AB^{-1} = A^{-1}$
$$\Rightarrow AB^{-1} \cdot A = A^{-1}A = I$$
$$\Rightarrow A^{-1}(AB^{-1}A) = A^{-1}I$$
$$B^{-1}A = A^{-1}$$

Now characteristics equation of C^2 is
$$|C_2 - \lambda I| = 0$$
$$|B - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 3 \\ 1 & 5-\lambda \end{vmatrix} = 0$$
$$\Rightarrow (1-\lambda)(5-1) - 3 = 0 \Rightarrow (\lambda^2 - 6\lambda + 5) - 3 = 0$$
$$\Rightarrow \lambda^2 - 6\lambda + 2 = 0$$
$$\Rightarrow \beta^2 - 6B + 2I = 0$$
$$\Rightarrow C^4 - 6C^2 + 2I = 0$$
$$\alpha = -6$$
$$\beta = 2$$
$$\therefore 2\beta - \alpha = 4 + 6 = 10$$

Q. 14 If $\log_e y = 3\sin^{-1}x$, then $(1-x)^2y'' - xy'$ at $x = \frac{1}{2}$ is equal to:

Option 1:
$9e^{\pi/6}$

Option 2:
$3e^{\pi/6}$

Option 3:
$3e^{\pi/2}$

Option 4:
$9e^{\pi/2}$

Correct Answer:
$9e^{\pi/2}$

Solution:

$\ln(y) = 3\sin^{-1}x$

$$\frac{1}{y}\cdot y' = 3\left(\frac{1}{\sqrt{1-x^2}}\right)$$

$$\Rightarrow y' = \frac{3y}{\sqrt{1-x^2}} \text{ at } x = \frac{1}{2}$$

$$\Rightarrow y' = \frac{3e^{3\left(\frac{\pi}{6}\right)}}{\frac{\sqrt{3}}{2}} = 2\sqrt{3}e^{\frac{\pi}{2}}$$

$$\Rightarrow y'' = 3\left(\frac{\sqrt{1-x^2}y' - y\frac{1}{2\sqrt{1-x^2}}(-2x)}{(1-x^2)}\right)$$

$$\Rightarrow (1-x^2)y'' = 3\left(3y + \frac{xy}{\sqrt{1-x^2}}\right)$$

$$\downarrow \text{ at } x = \frac{1}{2}, y = e^{3\sin^{-1}\left(\frac{1}{2}\right)} = e^{3\left(\frac{\pi}{6}\right)} = e^{\frac{\pi}{2}}$$

$$(1-x^2)y''_{\text{ at } x=\frac{1}{2}} = 3\left(3e^{\frac{\pi}{2}} + \frac{\frac{1}{2}\left(e^{\frac{\pi}{2}}\right)}{\frac{\sqrt{3}}{2}}\right)$$

$$= 3e^{\frac{\pi}{2}}\left(3 + \frac{1}{\sqrt{3}}\right)$$

$$(1-x^2)y'' - xy|_{\text{at } x=\frac{1}{2}}$$

$$= 3e^{\frac{\pi}{2}}\left(3 + \frac{1}{\sqrt{3}}\right) - \frac{1}{2}\left(2\sqrt{3}e^{\frac{\pi}{2}}\right) = 9e^{\frac{\pi}{2}}$$

Q. 15 The integral $\int_{1/4}^{3/4} \cos\left(2\cot^{-1}\sqrt{\frac{1-x}{1+x}}\right)dx$ is equal

Option 1:
-1/2

Option 2:
1/4

Option 3:
1/2

Option 4:
-1/4

Correct Answer:
-1/4

Solution:

$$I=\int_{1/4}^{3/4}\cos\left(2\cot^{-1}\left(\sqrt{\frac{1-x}{1+x}}\right)dx\right)$$

$$\int_{1/4}^{3/4}\cos\left(2\left(\tan^{-1}\sqrt{\frac{1+x}{1+x}}\right)\right)dx$$

$$\int_{1/4}^{3/4}\frac{1-\tan^2\left(\tan^{-1}\sqrt{\frac{1+x}{1-x}}\right)}{1+\tan^2\left(\tan^{-1}\sqrt{\frac{1+x}{1-x}}\right)}dx$$

$$=\int_{1/4}^{3/4}\frac{1-\left(\frac{1+x}{1-x}\right)}{1+\left(\frac{1+x}{1-x}\right)}dx=\int_{1/4}^{3/4}\frac{-2x}{2}dx$$

$$=\int_{1/4}^{3/4}(-x)dx=-\left(\frac{x^2}{2}\right)_{1/4}^{3/4}$$

$$=-\frac{1}{2}\left[\frac{9}{16}-\frac{1}{16}\right]$$

$$=-\frac{1}{4}$$

Q. 16 Let a, ar, ar^2, ...be an infinite G.P. If $\sum_{n=0}^{\infty}a^n=57$ and $\sum_{n=0}^{\infty}a^3r^{3n}=9747$, then $a+18r$ is equal to :

Option 1:
27

Option 2:
46

Option 3:
38

Option 4:
31

Correct Answer:
31

Solution:

$$\sum_{n=0}^{\infty} a^n = 57$$

$$a + ar + a^2 + \infty = 57$$

$$\frac{\mathrm{a}}{1-\mathrm{r}} = 57$$

$$\sum_{n=0}^{\infty} a^3 r^{3n} = 9747$$

$$\mathrm{a}^3 + \mathrm{a}^3 \cdot \mathrm{r}^3 + \mathrm{a}^3 \cdot \mathrm{r}^6 + \ldots\ldots\ldots\ldots \infty = 9746$$

$$\frac{\mathrm{a}^3}{1-\mathrm{r}^3} = 9746$$

$$\frac{(\mathrm{I})^3}{(\mathrm{II})} \Rightarrow \frac{\frac{\mathrm{a}^3}{(1-\mathrm{r})^3}}{\frac{\mathrm{a}^3}{1-\mathrm{r}^3}} = \frac{57^3}{9717} = 19$$

On solving, $\mathrm{r} = \frac{2}{3}$ and $\mathrm{r} = \frac{3}{2}$ (rejected)

$$a = 19$$

$$\therefore a + 18r = 19 + 18 \times \frac{2}{3} = 31$$

Q. 17 If an unbiased dice is rolled thrice, then the probability of getting a greater number in the i^{th} roll than the number obtained in the $(i-1)^{\mathrm{th}}$ roll, $\mathrm{i}=2,3$, is equal to :

Option 1:
3/54

Option 2:
2/54

Option 3:
5/54

Option 4:
1/54

Correct Answer:
5/54

Solution:
Favourable cases $= {}^{6}\mathrm{C}_{3}$
Total out comes $= 6^{3}$
Probability of getting greater number than previous one $= \frac{{}^{6}\mathrm{C}_{3}}{\mathrm{r}^{3}} = \frac{20}{216} = \frac{5}{54}$

Q. 18 The value of the integral $\int_{-1}^{2} \log_{e}\left(x+\sqrt{x^{2}+1}\right)dx$

Option 1:
$\sqrt{5}-\sqrt{2}+\log_{\mathrm{e}}\left(\frac{9+4\sqrt{5}}{1+\sqrt{2}}\right)$

Option 2:
$\sqrt{2}-\sqrt{5}+\log_{\mathrm{e}}\left(\frac{9+4\sqrt{5}}{1+\sqrt{2}}\right)$

Option 3:

$\sqrt{5}-\sqrt{2}+\log_e\left(\frac{7+4\sqrt{5}}{1+\sqrt{2}}\right)$

Option 4:

$\sqrt{2}-\sqrt{5}+\log_e\left(\frac{7+4\sqrt{5}}{1+\sqrt{2}}\right)$

Correct Answer:

$\sqrt{2}-\sqrt{5}+\log_e\left(\frac{9+4\sqrt{5}}{1+\sqrt{2}}\right)$

Solution:

$$I=\int_{-1}^{2}1\log_e\left(x+\sqrt{x^2+1}\right)dx$$

$$=x\log_e\left(x+\sqrt{x^2+1}\right)-\int_{-1}^{2}\left(\frac{1+\frac{x}{\sqrt{x^2+1}}}{x+\sqrt{x^2+1}}\right)dx$$

$$=x\log_e\left(x+\sqrt{x^2+1}\right)-\int_{-1}^{2}\frac{x}{\sqrt{x^2+1}}dx$$

$$=x\log_e\left(x+\sqrt{x^2+1}\right)-\sqrt{x^2+1}\Big|_{-1}^{2}$$

$$=\left(2\log_e(2+\sqrt{5})-\sqrt{5}\right)$$

$$-\left(-\log_e(-1+\sqrt{2})-\sqrt{2}\right)$$

$$=\log_e(2+\sqrt{5})^2-\sqrt{5}+\log_e(\sqrt{2}-1)+\sqrt{2}$$

$$=\log_e(2+\sqrt{5})^2-\sqrt{5}+\log_e(\sqrt{2}-1)+\sqrt{2}$$

$$=\sqrt{2}-\sqrt{5}+\log_e\left(\frac{(2+\sqrt{5})^2}{\sqrt{2+1}}\right)$$

$$=\sqrt{2}-\sqrt{5}+\log_e\left(\frac{9+4\sqrt{5}}{\sqrt{2+1}}\right)$$

Q. 19 Let $\alpha,\beta;\alpha>\beta$, be the roots of the equation $x^2-\sqrt{2}x-\sqrt{3}=0$. Let $P_n=\alpha^n-\beta^n, n\in N$. Then
$(11\sqrt{3}-10\sqrt{2})\mathrm{P}_{10}+(11\sqrt{2}+10)\mathrm{P}_{11}-11\mathrm{P}_{12}$ is equal to :

Option 1:
$10\sqrt{2}\mathrm{P}_9$

Option 2:
$10\sqrt{3}P_9$

Option 3:
$11\sqrt{2}\mathrm{P}_9$

Option 4:
$11\sqrt{3}P_9$

Correct Answer:
$10\sqrt{3}P_9$

Solution:
$x^2-\sqrt{2x}-\sqrt{3}=0\left\langle\begin{matrix}\alpha\\\beta\end{matrix}\right.$

$\alpha^{n+2}-\sqrt{2}\alpha^{n+1}-\sqrt{3}\alpha^n=0$

and $\beta^{n+2}-\sqrt{2}\beta^{n+1}-\sqrt{3}\beta^n=0$

Subtracting

$\left(\alpha^{\mathrm{n}+2}-\beta^{\mathrm{n}+2}\right)-\sqrt{2}\left(\alpha^{\mathrm{n}+1}-\beta^{\mathrm{n}+1}\right)-\sqrt{3}\left(\alpha^{\mathrm{n}}-\beta^{\mathrm{n}}\right)=0$

$\Rightarrow \mathrm{P}_{\mathrm{n}+2}-\sqrt{2}P_{\mathrm{n}+1}-\sqrt{3}\mathrm{P}_{\mathrm{n}}=0$

Put $\mathrm{n}=10$

$\mathrm{P}_{12}-\sqrt{2}\mathrm{P}_{11}-\sqrt{3}\mathrm{P}_{10}=0$

$\mathrm{n}=9$

$\mathrm{P}_{11}-\sqrt{2}\mathrm{P}_{10}-\sqrt{3}\mathrm{P}_9=0$

$11\left(\sqrt{3}\cdot\mathrm{P}_{10}+\sqrt{2}\mathrm{P}_{11}-\mathrm{P}_{11}\right)-10\left(\sqrt{2}\mathrm{P}_{10}-\mathrm{P}_{11}\right)$

$=0-10\left(-\sqrt{3}\mathrm{P}_9\right)=10\sqrt{3}\mathrm{P}_9$

Q. 20 Let $\vec{a}=2\hat{i}+\alpha\hat{j}+\hat{k},\quad \vec{b}=-\hat{i}+\hat{k},\quad \vec{c}=\beta\hat{j}-\hat{k}$, where α and β are integers and $\alpha\beta=-6$. Let the values of the ordered pair (α,β) for which the area of the parallelogram of diagonals $\vec{a}+\vec{b}$ and $\vec{b}+\vec{c}$ is $\frac{\sqrt{21}}{2}$, be (α_1,β_1) and (α_2,β_2).Then $\alpha_1^2+\beta_1^2-\alpha_2\beta_2$ is equal to

Option 1:
17

Option 2:
24

Option 3:
21

Option 4:
19

Correct Answer:
19

Solution:

$$\text{Area of parallelogram} = \frac{1}{2}\ \vec{d}_1 \times \vec{d}_2$$

$$A = \frac{1}{2}|(\vec{a}+\vec{b}) \times (\vec{b}+\vec{c})| = \frac{\sqrt{21}}{2}$$

$$\text{so, } \vec{a}+\vec{b} = \hat{i} + \alpha\hat{j} + 2\hat{k}$$

$$\vec{b}+\vec{c} = -\hat{i} + \beta\hat{j}$$

$$(\vec{a}+\vec{b}) \times (\vec{b}+\vec{c}) = \begin{matrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & \alpha & 2 \\ -1 & \beta & 0 \end{matrix}$$

$$= \hat{i}(-2\beta) - \hat{j}(2) + \hat{k}(\beta+\alpha)$$

$$|(\vec{a}+\vec{b}) \times (\vec{b}+\vec{c})| = \sqrt{4\beta^2 + 4 + (\alpha+\beta)^2} = \sqrt{21}$$

$$4\beta^2 + 4 + \alpha^2 + \beta^2 + 2\alpha\beta = 21$$

$$\alpha^2 + 5\beta^2 - 12 = 17$$

$$\alpha^2 + 5\beta^2 = 29$$

$$\text{and } \alpha\beta = -6$$

and given $\alpha_i\beta$ are integers

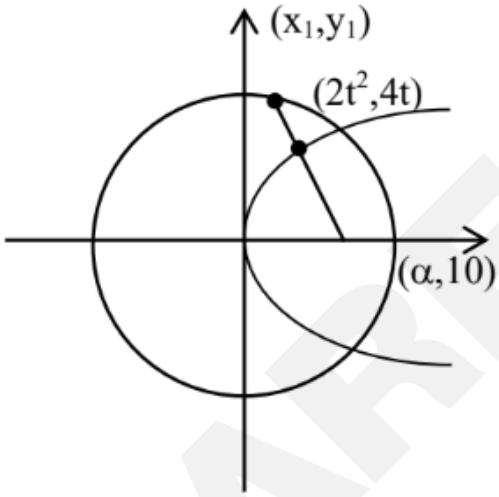
so,

$\alpha = -3, \beta = 2$
or
$\alpha = 3, \quad \beta = -2$
$(\alpha_1, \beta_1) = (-3, 2)$
$(\alpha_2, \beta_2) = (3, -2)$
$\alpha_1^2 + \beta_1^2 - \alpha_2\beta_2 = 9 + 4 + 6 = 19$

Q. 21 Consider the circle $\mathrm{C} : \mathrm{x}^2 + \mathrm{y}^2 = 4$ and the parabola $P : y^2 = 8x$. If the set of all values of α, for which three chords of the circle C on three distinct lines passing through the point $(\alpha, 0)$ are bisected by the parabola P is the interval (p, q), then $(2q - p)^2$ is equal to

Correct Answer:
80

Solution:



$T = S_1$
$x_1 + yy_1 = x_1^2 + y_1^2$
$\alpha x_1 = x_1^2 + y_1^2$
$\alpha\left(2t^2\right) = 4t^4 + 16t^2$
$\alpha = 2t^2 + 8$
$\frac{\alpha - 8}{2} = t^2$

Also, $4t^4 + 16t^2 - 4 < 0$

$t^2 = -2 + \sqrt{5}$

$\alpha = 4 + 2\sqrt{5}$

$\therefore \alpha \in (8, 4 + 2\sqrt{5})$

$\therefore (2q - p)^2 = 80$

Q. 22 Let the set of all values of p, for which $f(x) = \left(p^2 - 6p + 8\right)\left(\sin^2 2x - \cos^2 2x\right) + 2(2 - p)x + 7$ does not have any critical point, be the interval (a, b). Then $16ab$ is equal to

Correct Answer:

252

Solution:

$f(x) = -\left(p^2 - 6p + 8\right)\cos 4n + 2(2 - p)n + 7$

$f^1(x) = +4\left(p^2 - 6p + 8\right)\sin 4x + (4 - 2p) \neq 0$

$$\sin 4x \neq \frac{2p - 4}{4(p - 4)(p - 2)}$$

$$\sin 4x \neq \frac{2(p - 2)}{4(p - 4)(p - 2)}$$

$p \neq 2$

$$\sin 4x \neq \frac{1}{2(p - 4)}$$

$$\Rightarrow \frac{1}{2(p - 4)} > 1$$

on solving we get

$$\therefore p \in \left(\frac{7}{2}, \frac{9}{2}\right)$$

$$\text{Hence } a = \frac{7}{2},\ b = \frac{9}{2}$$

$\therefore 16ab = 252$

Q. 23 For a differentiable function $f : IR \to IR$, suppose $f'(x) = 3f(x) + \alpha$, where $\alpha \in \mathrm{IR}$, $f(0) = 1$ and $\lim_{x \to -\infty} f(x) = 7$. Then $9f(-\log_e 3)$ is equal to

Correct Answer:
61

Solution:

$$\frac{dy}{dx} - 3y = \alpha$$

$$\text{If } = e^{\int -3dx} = e^{-3x}$$

$$\therefore y - e^{-3x} = \int e^{-3x} \cdot \alpha dx$$

$$ye^{-3x} = \frac{\alpha e^{-3x}}{-3} + c$$

$$\left(*e^{3x}\right)$$

$$y = \frac{\alpha}{-3} + C \cdot e^{3x}$$

$$\text{on substituting } x = 0, y = 1$$

$$x \to -\infty, y = 7$$

we get $y = 7 - 6e^{3x}$

$$9f(-\log_e 3) = 61$$

Q. 24 The number of integers, between 100 and 1000 having the sum of their digits equals to 14 , is

Correct Answer:
70

Solution:

$$\mathrm{N} = \mathrm{abc}$$

(i) All distinct digits

$$a + b + c = 14$$

$$a \geq 1$$

$$b, c \in \{0 \text{ to } 9\}$$

by hit & trial : 8 cases

$(6,5,3)$ $(8,6,0)$ $(9,4,1)$

$(7,6,1)$ $(8,5,1)$ $(9,3,2)$

$(7,5,2)$ $(8,4,2)$

$(7,4,3)$ $(9,5,0)$

(ii) 2 same, 1 diff $a = b; c$

$2a + c = 14$

by values :

$(3,8)$ $(4,6)$ $(5,4)$ $(6,2)$ $(7,0)$ $\Big\} \frac{3!}{2!} \times 5 - 1$

$= 14$ cases

(iii) all same :

$3a = 14$

$a = \frac{14}{3} \times$ rejected

0 cases

Hence, Total cases :

$8 \times 3! + 2 \times (4) + 14$

$= 48 + 22$

$= 70$

Q. 25 Let $\mathrm{A} = \{(\mathrm{x}, \mathrm{y}) : 2\mathrm{x} + 3\mathrm{y} = 23, \mathrm{x}, \mathrm{y} \in \mathrm{N}\}$ and $B = \{x : (x, y) \in A\}$. Then the number of one-one functions from A to B is equal to

Correct Answer:

24

Solution:

$2x + 3y = 23$

$x = 1$	$y = 7$
$x = 4$	$y = 5$
$x = 7$	$y = 3$
$x = 10$	$y = 1$

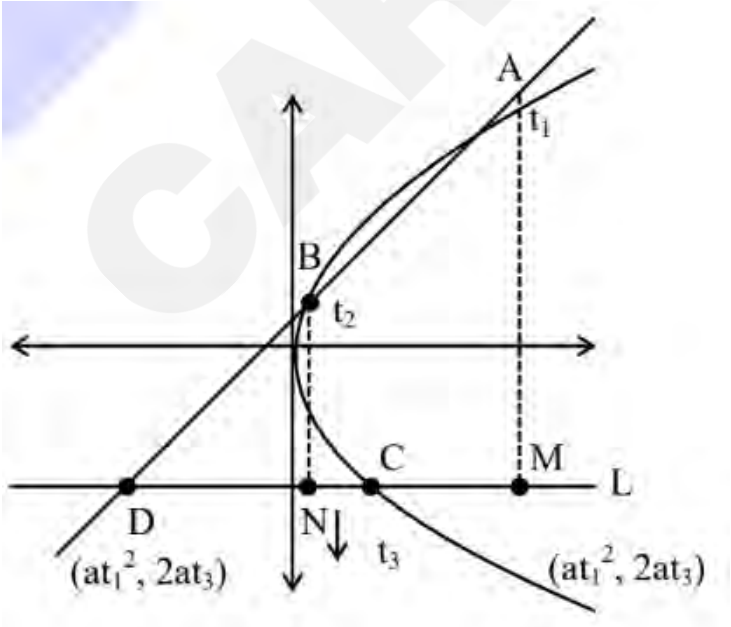
A	B
$(1, 7)$	1
$(4, 5)$	4
$(7, 3)$	7
$(10, 1)$	10

The number of one-one functions from A to B is equal to 4 !

Q. 26 Let A, B and C be three points on the parabola $y^2 = 6x$ and let the line segment AB meet the line L through C parallel to the x-axis at the point D. Let M and N respectively be the feet of the perpendiculars from A and B on L.
Then $\left(\frac{\mathrm{AM} \cdot \mathrm{BN}}{\mathrm{CD}}\right)^2$ is equal to

Correct Answer:
36

Solution:



$$m_{AB} = m_{AD}$$

$$\Rightarrow \quad \frac{2}{t_1 + t_2} = \frac{2a(t_1 - t_3)}{at_1^2 - \alpha}$$

$$\Rightarrow \quad at_1^2 - \alpha = a\{t_1^2 - t_1t_3 + t_1t_2 - t_2t_3\}$$

$$\Rightarrow \quad \alpha = a(t_1t_3 + t_2t_3 - t_1t_2)$$

$$AM = |2a(t_1 - t_3)|, BN = |2a(t_2 - t_3)|,$$

$$CD = at_3^2 - \alpha$$

$$CD = at_3^2 - a(t_1t_3 + t_2t_3 - t_1t_2)$$

$$= a\ t_3^2 - t_1t_3 - t_2t_3 + t_1t_2$$

$$= a|t_3(t_3 - t_1) - t_2(t_3 - t_1)|$$

$$CD = a|(t_3 - t_2)(t_3 - t_1)|$$

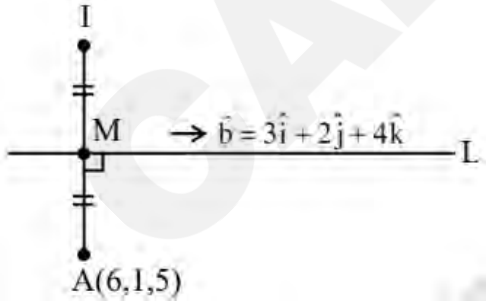
$$\left(\frac{AM \cdot BN}{CD}\right)^2 = \left\{\frac{2a(t_1 - t_3) \cdot 2a(t_2 - t_3)}{a(t_3 - t_2)(t_3 - t_1)}\right\}^2$$

$$16a^2 = 16 \times \frac{9}{4} = 36$$

Q. 27 The square of the distance of the image of the point $(6, 1, 5)$ in the line $\frac{x-1}{3} = \frac{y}{2} = \frac{z-2}{4}$, from the origin is

Correct Answer:
62

Solution:



Let $M(3\lambda + 1, 2\lambda, 4\lambda + 2)$

$$\overrightarrow{AM} \cdot \vec{b} = 0$$

$$\Rightarrow \quad 9\lambda - 15 + 4\lambda - 2 + 16\lambda - 12 = 0$$

$$\Rightarrow \quad 29\lambda = 29$$

$$\Rightarrow \quad \lambda = 1$$

$$M(4, 2, 6), I = (2, 3, 7)$$

Required Distance $= \sqrt{4+9+49} = \sqrt{62}$

Ans. 62

Q. 28 If $\left(\frac{1}{\alpha+1}+\frac{1}{\alpha+2}+\ldots+\frac{1}{\alpha+1012}\right)$
$-\left(\frac{1}{2\cdot 1}+\frac{1}{4\cdot 3}+\frac{1}{6\cdot 5}+\ldots+\frac{1}{2024\cdot 2023}\right)$
$=\frac{1}{2024}$, then α is equal to-

Correct Answer:

1101

Solution:

$$\left(\frac{1}{\alpha+1}+\frac{1}{\alpha+2}+\ldots+\frac{1}{\alpha+2012}\right)$$
$$-\left\{\left(\frac{1}{1}-\frac{1}{2}\right)+\left(\frac{1}{3}-\frac{1}{4}\right)+\ldots+\left(\frac{1}{2023}-\frac{1}{2024}\right)\right\}=\frac{1}{2024}$$
$$\Rightarrow\left(\frac{1}{\alpha+1}+\frac{1}{\alpha+2}+\ldots+\frac{1}{\alpha+2012}\right)$$
$$-\left\{\left(\frac{1}{1}+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}\right)+\ldots+\frac{1}{2023}\right.$$
$$\left.-\frac{1}{2024}-2\left(\frac{1}{2}+\frac{1}{4}+\ldots+\frac{1}{2022}\right)\right\}=\frac{1}{2024}$$
$$\Rightarrow\left(\frac{1}{\alpha+1}+\frac{1}{\alpha+2}+\ldots+\frac{1}{\alpha+2012}\right)$$
$$-\left(\frac{1}{1}+\frac{1}{2}+\ldots+\frac{1}{2023}\right)$$
$$+\frac{1}{2024}+\left(\frac{1}{1}+\frac{1}{2}+\ldots+\frac{1}{1011}\right)=\frac{1}{2024}$$

$$\Rightarrow\left(\frac{1}{\alpha+1}+\frac{1}{\alpha+2}+\ldots+\frac{1}{\alpha+2012}\right)$$
$$-\left(\frac{1}{1}+\frac{1}{2}+\ldots+\frac{1}{2023}\right)$$
$$+\frac{1}{2024}+\left(\frac{1}{1}+\frac{1}{2}+\ldots+\frac{1}{1011}\right)=\frac{1}{2024}$$
$$\Rightarrow\frac{1}{\alpha+1}+\frac{1}{\alpha+2}+\ldots+\frac{1}{\alpha+2012}$$
$$=\frac{1}{1012}+\frac{1}{1013}+\ldots+\frac{1}{2023}$$
$$\Rightarrow\alpha=1011$$

Q. 29 Let the inverse trigonometric functions take principal values. The number of real solutions of the equation $2\sin^{-1}x+3\cos^{-1}x=\frac{2\pi}{5}$, is

Correct Answer:
0

Solution:

$$2\sin^{-1}x+3\cos^{-1}x=\frac{2\pi}{5}$$
$$\Rightarrow\quad \pi+\cos^{-1}x=\frac{2\pi}{5}$$
$$\Rightarrow\quad \cos^{-1}x=\frac{-3\pi}{5}$$

Not possible
Ans. 0

Q. 30 Consider the matrices : $\mathrm{A}=\begin{bmatrix}2 & -5\\ 3 & \mathrm{m}\end{bmatrix}, \mathrm{B}=\begin{bmatrix}2\\ \mathrm{m}\end{bmatrix}$ and $X=\begin{bmatrix}x\\ y\end{bmatrix}$. Let the set of all m, for which system of equations $\mathrm{AX}=\mathrm{B}$ has a negative solut (i.e., $x<0$ and $y<0$), be the interval (a,b).

Then $8\int\!\!\int^{\mathrm{b}}|\mathrm{A}|\mathrm{dm}$ is equal to

Correct Answer:
450

Solution:

$A = \begin{pmatrix} 2 & -5 \\ 3 & m \end{pmatrix}, B = \begin{pmatrix} 20 \\ m \end{pmatrix}$

$X = \begin{pmatrix} x \\ y \end{pmatrix}$

$2x - 5y = 20$

$3x + my = m$

$\Rightarrow y = \dfrac{2m - 60}{2m + 15}$

$y < 0 \Rightarrow m \in \left(\dfrac{-15}{2}, 30\right)$

$x = \dfrac{25m}{2m + 15}$

$x < 0 \Rightarrow m \in \left(\dfrac{-15}{2}, 0\right)$

$\Rightarrow \mathrm{m} \in \left(\dfrac{-15}{2}, 0\right)$

$|\mathrm{A}| = 2\,\mathrm{m} + 15$

Now,

$8\int_{\frac{-15}{2}}^{0} (2\,\mathrm{m} + 15)\mathrm{dm} = 8\{\,\mathrm{m}^2 + 15\,\mathrm{m}\}_{\frac{-15}{2}}^{0}$

$\Rightarrow 8\left\{-\left(\dfrac{225}{4} - \dfrac{225}{2}\right)\right\}$

$= 8 \times \dfrac{225}{4} = 450$